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HIGH-POWER CONVERTERS AND AC DRIVES

Second Edition

Bin Wu
Mehdi Narimani

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About the Authors

Bin Wu graduated from Donghua University, Shanghai, China in 1978, and received his M.A.Sc. and Ph.D. degrees in electrical and computer engineering from the University of Toronto, Canada in 1989 and 1993, respectively. He joined Ryerson University in 1993, where he is currently a Professor and Senior NSERC/Rockwell Automation Industrial Research Chair in Power Electronics and Electric Drives. Dr. Wu has published more than 400 technical papers, authored/coauthored several Wiley-IEEE Press books, and holds more than 30 granted/pending US/European patents in the area of power electronics, medium-voltage drives, and renewable energy systems.

Dr. Wu received the Gold Medal of the Governor General of Canada in 1993, Premier's Research Excellence Award in 2001, NSERC Synergy Award for Innovation in 2002, Ryerson Distinguished Scholar Award in 2003, Ryerson YSGS Outstanding Contribution to Graduate Education Award, and Professional Engineers Ontario (PEO) Engineering Excellence Medal in 2014. He is a fellow of Institute of Electrical and Electronics Engineers (IEEE), Engineering Institute of Canada (EIC), and Canadian Academy of Engineering (CAE).

Mehdi Narimani received his Ph.D. degree from University of Western Ontario, Ontario, Canada in 2012 and received his B.S. and M.S. degrees from Isfahan University of Technology (IUT), Isfahan, Iran in 1999 and 2002, respectively. He is currently assistant professor at the Department of Electrical and Computer Engineering at McMaster University, Hamilton, Ontario, Canada. Prior joining McMaster University, Dr. Narimani was a Power Electronics Engineer at Rockwell Automation Canada, in Cambridge, Ontario. He also worked as a faculty member of Isfahan University of Technology from 2002 to 2009 where he was involved in design and implementation of several industrial projects.

Dr. Narimani has published more than 55 journal and conference proceeding papers, and holds more than four issued/pending US/European patents. His current research interests include power conversion, high power converters, control of power electronics converters, and renewable energy Systems. Dr. Narimani is a senior member of Institute of Electrical and Electronics Engineers (IEEE).

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Preface and Acknowledgments

There have been a number of new developments in high-power medium-voltage (MV) drive technology since 2006 when the first edition of this book was published. The second edition of the book incorporates these developments by adding three new chapters and revising two existing chapters.

The new chapters include Chapter 15 Control of Synchronous Motor Drives, where various control schemes for the synchronous motor drives are presented; Chapter 16 Matrix Converter Fed Drives, where multi-modular cascaded matrix converters developed for the MV drive are analyzed, and Chapter 17 Transformerless MV Drives, in which the technologies for the elimination of the isolation transformer in the MV drive are elaborated. Two chapters are extensively revised, including Chapter 9 Other Multilevel Voltage Source Inverters and Chapter 12 Voltage Source Inverter Fed Drives, where a number of newly developed converter topologies and drive configurations have been added.

The second edition of the book contains 6 parts with 17 chapters. Part 1 (Introduction) is composed of two chapters. Chapter 1 provides an overview of high-power converters, drive configurations, and typical applications. Chapter 2 introduces high-power semiconductor devices.

Part 2 (Multipulse Diode and SCR Rectifiers) contains three chapters on multipulse rectifiers, which are widely used in high-power drives as front-end converters. Chapter 3 deals with multipulse diode rectifiers, Chapter 4 addresses multipulse SCR rectifiers, and Chapter 5 introduces phase-shifting transformers used in the multipulse rectifiers.

Part 3 (Multilevel Voltage Source Converters) is composed of four chapters dealing with a variety of high-power voltage source converters. Chapter 6 introduces modulation techniques for a two-level inverter, which provides a basis for developing modulation schemes for multilevel inverters. Chapter 7 focuses on cascaded H-bridge inverters. Chapter 8 presents a detailed analysis on neutral point clamped multilevel inverters. Chapter 9 introduces other multilevel converter topologies that are recently developed for the MV drive.

Part 4 (PWM Current Source Converters) has two chapters for high-power PWM current source converters. Chapter 10 focuses on the switching schemes for the current source inverters whereas Chapter 11 puts more emphasis on power factor and active damping controls for the current source rectifiers.

Part 5 (High-Power AC Drives) consists of four chapters on high-power drive systems. Chapter 12 presents configurations of voltage source inverter fed MV drives

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while Chapter 13 is on current source inverter based drives. Chapter 14 presents advance control schemes for induction motor MV drives, including field-oriented control and direct torque control. Chapter 15 deals with advanced control schemes for synchronous motor MV drives such as maximum torque per amp control and direct torque control.

Part 6 (Special Topics on MV Drives) has two chapters on the state-of-the-art MV drives. Chapter 16 focuses on multi-modular cascaded matrix converter topologies and matrix converter fed drives. Chapter 17 presents transformerless MV drive configurations for both current and voltage source inverter fed drives.

The second edition of the book presents the latest technology in the field, provides design guidance with tables, charts, and graphs, addresses practical problems and their mitigation methods, and illustrates important concepts with computer simulations and experiments. It can serve as a reference for academic researchers, practicing engineers, and other professionals. This book also provides adequate technical background and can be adopted as a textbook for a graduate-level course in power electronics and ac drives.

Finally, we would like to express our deep gratitude to our colleagues at Rockwell Automation Canada, particularly, Dr. Navid Zargari, for more than 20 years of research collaboration in developing advanced MV drive technologies. We are grateful to our postdoctoral fellows and graduate students in the Laboratory for Electric Drive Applications and Research (LEDAR) at Ryerson University for their assistance in preparing the manuscript of this book. In particular, we would like to thank Drs. Jiacheng Wang and Kai Tian for their great help in preparing Chapters 16 and 17. Our special thanks go to Wiley/IEEE press editor, Ms. Mary Hatcher, for her precious help and support. We also express our sincere appreciation to the Wiley Editorial Program Assistants, Mr. Brady Chin and Ms. Divya Narayanan for their kind help and assistance.

Bin Wu
Mehdi Narimani
Toronto

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List of Abbreviations

ABB	Asea–Brown–Boveri
AFE	Active front end
ANPC	Active neutral point clamped
APOD	Alternative phase opposite disposition
CHB	Cascade H-bridge
CM	Common mode
CMC	Cascaded matrix converter
CMV	Common mode voltage
CSC	Current source converter
CSI	Current source inverter
CSR	Current source rectifier
C-SVM	Conventional space vector modulation
DCC	Diode clamped converter
DF	Distortion factor
DFE	Diode front end
DM	Differential mode
DPF	Displacement power factor
DSP	Digital signal processor
DTC	Direct torque control
emf	Electromotive force
EMI	Electromagnetic interference
ETO	Emitter turn-off thyristor
FC	Flying capacitor
FOC	Field-oriented control
FPGA	Field programmable gate array
GAN	Gallium nitride
GCT	Gate commutated thyristor (also known as integrated gate commutated thyristor)
GTO	Gate turn-off thyristor
HPF	High pass filter
HVDC	High-voltage dc current
IEEE	Institute of Electrical and Electronics Engineers
IEGT	Injection enhanced gate transistor
IGBT	Insulated gate bipolar transistor

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IM	Induction motor
IPD	In-phase disposition
LCI	Load commutated inverter
LPF	Low pass filter
MCT	MOS-controlled thyristor
MC	Matrix converter
MMC	Modular multilevel converter
MOSFET	Metal-oxide semiconductor field-effect transistor
MTPA	Maximum torque per ampere
MV	Medium voltage (2.3KV to 13.8 KV)
NPC	Neutral point clamped
NPP	Neutral point piloted
NNPC	Nested neutral point clamped
PCBB	Power converter building block
PF	Power factor (DF × DPF)
PFC	Power factor compensator
PI	Proportional and integral
PLL	Phase-locked loop
PM	Permanent magnet
PMSM	Permanent magnet synchronous motor
POD	Phase opposite disposition
PS-SPWM	Phase-shifted sinusoidal pulse width modulation
PWM	Pulse width modulation
pu	Per unit
RCM	Reduction common mode
rms	Root mean square
rpm	Revolutions per minute
SCR	Silicon-controlled rectifier (thyristor)
SHE	Selective harmonic elimination
Si	Silicon
SiC	Silicon carbide
SIT	Static induction thyristor
SM	Synchronous motor
SPWM	Sinusoidal pulse width modulation
STATCOM	Static synchronous compensator
SVM	Space vector modulation
THD	Total harmonic distortion
TPWM	Trapezoidal pulse width modulation
VBC	Voltage balancing control
VOC	Voltage-oriented control
VSC	Voltage source converter
VSI	Voltage source inverter
VSR	Voltage source rectifier
VZD	Voltage zero crossing detector
WRSM	Wound-rotor synchronous motor
ZDC	Zero d-axis current

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Part One

Introduction

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Chapter 1

Introduction

1.1 OVERVIEW OF HIGH-POWER DRIVES

The development of high-power converters and medium voltage (MV) drives started in the mid 1980s when 4500 V gate turn off (GTO) thyristors became commercially available [1]. The GTO was the standard for the MV drive until the advent of high-power insulated gate bipolar transistors (IGBTs) and gate commutated thyristors (GCTs) in the 1990s [2, 3]. These switching devices have rapidly progressed into the main areas of high-power electronics due to their superior switching characteristics, reduced power losses, and ease of gate control.

The MV drives cover power ratings from 0.4 to 40 MW at the medium voltage level of 2.3–13.8 kV. The power rating can be extended to 100 MW, where synchronous motor drives with load commutated inverters (LCIs) are often used [4]. However, the majority of the installed MV drives are in the 1–4 MW range with voltage ratings from 3.3 to 6.6 kV as illustrated in Fig. 1.1-1.

The high-power MV drives have found widespread applications in industry. They can be used for pipeline pumps in the petrochemical industry [5], fans in the cement industry [6], pumps in water pumping stations [7], traction applications in the transportation industry [8], steel rolling mills in the metals industry [9], and other applications [10, 11]. A summary of the MV drive applications is given in the appendix of this chapter [12].

Market research has shown that around 85% of the MV drive applications are for pumps, fans, compressors, and conveyors [13]. The technical requirements for these drives are relatively simple and can be accomplished by a standard MV drive. As shown in Fig. 1.1-2, only 15% of the total installed drives are non-standard drives.

One of the major markets for the MV drive is for retrofit applications. Although with the advancements of high-power converter technology, the variable-speed MV drives have been widely accepted in industry over the last three decades, many of the MV motors still operate in the field at a fixed speed. When large fans, pumps, or compressors are driven by a fixed-speed motor, the control of air or liquid flow is

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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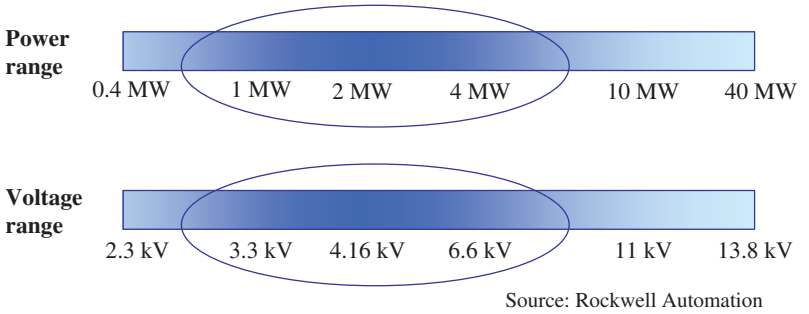

Figure 1.1-1 Voltage and power ranges of the MV drive.

normally achieved by mechanical methods, such as throttling control, inlet dampers, and flow control valves, resulting in a substantial amount of energy loss. The installation of the MV drive can lead to significant savings on energy cost. It was reported that the use of the variable-speed MV drive resulted in a payback time of the investment from 1 to 2 $\frac{1}{2}$ years [7].

The use of the MV drive can also increase productivity in some applications. A case was reported from a cement plant where the speed of a large fan was made adjustable by an MV drive [11]. The collected dust on the fan blades operated at a fixed speed had to be cleaned regularly, leading to a significant downtime per year for maintenance. With variable-speed operation, the blades only had to be cleaned at the standstill of the production once a year. The increase in productivity together with the energy savings resulted in a payback time of the investment within 6 months.

Figure 1.1-3 shows a general block diagram of the MV drive. Depending on the system requirements and the type of the converters employed, the line- and motor-side filters are optional. A phase-shifting transformer with multiple secondary windings is often used mainly for the reduction of line current distortion.

The rectifier converts the utility supply voltage to a dc voltage with a fixed or adjustable magnitude. The commonly used rectifier topologies include multipulse diode rectifiers, multipulse SCR rectifiers, or pulse-width-modulated (PWM) rectifiers. The dc filter can simply be a capacitor that provides a stiff dc voltage in voltage source drives or an inductor that smoothes the dc current in current source drives.

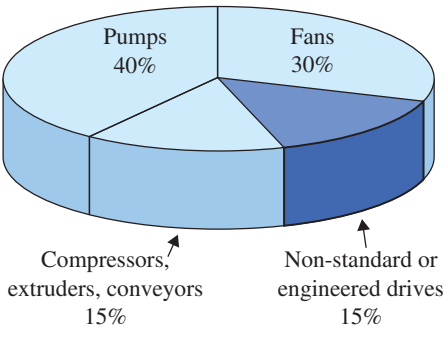

Figure 1.1-2 MV drive market survey.

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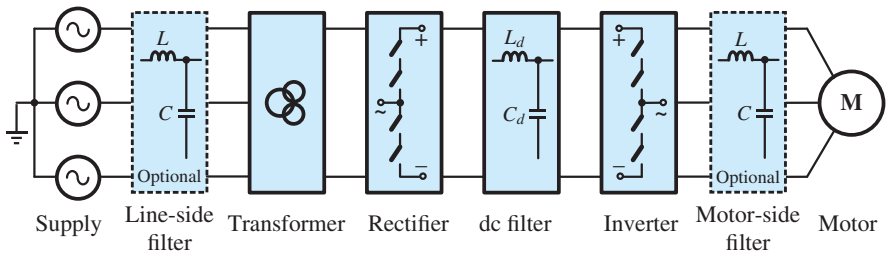

Figure 1.1-3 General block diagram of the MV drive.

The inverter can be generally classified into voltage source inverter (VSI) and current source inverter (CSI). The VSI converts the dc voltage to a three-phase ac voltage with adjustable magnitude and frequency whereas the CSI converts the dc current to an adjustable three-phase ac current. A variety of inverter topologies have been developed for the MV drive, most of which will be analyzed in this book.

1.2 TECHNICAL REQUIREMENTS AND CHALLENGES

The technical requirements and challenges for the MV drive differ in many aspects from those for the low voltage (≤ 600 V) ac drives. Some of them that must be addressed in the MV drive may not even be an issue for the low voltage drives. These requirements and challenges can be generally divided into four groups: the requirements related to the power quality of line-side converters, the challenges associated with the design of motor-side converters, the constraints of the switching devices, and the drive system requirements.

1.2.1 Line-Side Requirements

(a) Line Current Distortion The rectifier normally produces distorted line currents and also causes notches in voltage waveforms. The distorted current and voltage waveforms can cause numerous problems such as nuisance tripping of computer controlled industrial processes, overheating of transformers, equipment failure, computer data loss, and malfunction of communications equipment. Nuisance tripping of industrial assembly lines often leads to expensive downtime and ruined product. There exist certain guidelines for harmonic regulation, such as European Standard IEC1000 and IEEE Standard 519-2014 [14]. The rectifier used in the MV drive should comply with these guidelines.

(b) Input Power Factor High input power factor is a general requirement for all electric equipment. This requirement is especially important for the MV drive due to its high power rating.

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(c) LC Resonance Suppression For the MV drives using line-side capacitors for current THD reduction or power factor compensation, the capacitors form LC resonant circuits with the line inductance of the system. The LC resonant modes may be excited by the harmonic voltages in the utility supply or harmonic currents produced by the rectifier. Since the utility supply at the medium voltage level normally has very little line resistance, the lightly damped LC resonances may cause severe oscillations or over-voltages that may destroy the switching devices and other components in the rectifier circuits. The LC resonance issue should be addressed when the drive system is designed.

1.2.2 Motor-Side Challenges

(a) dv/dt and Wave Reflections Fast switching speed of the semiconductor devices results in high *dv/dt* at the rising and falling edges of the inverter output voltage waveform. Depending on the magnitude of the inverter dc bus voltage and speed of the switching device, the *dv/dt* can well exceed $10,000\,\mathrm{V}/\mu\mathrm{s}$. The high *dv/dt* in the inverter output voltage can cause premature failure of the motor winding insulation due to partial discharges. It induces rotor shaft voltages through stray capacitances between the stator and rotor. The shaft voltage produces a current flowing into the shaft bearing, leading to early bearing failure. The high *dv/dt* also causes electromagnetic emission in the cables connecting the motor to the inverter, affecting the operation of nearby sensitive electronic equipment.

To make the matter worse, the high *dv/dt* may cause voltage doubling effect at the rising and falling edges of the motor voltage waveform due to wave reflections in long cables. The reflections are caused by the mismatch between the wave impedance of the cable and the impedances at its inverter and motor ends, and can double the voltage on the motor terminals at each switching transient if the cable length exceeds a certain limit. The critical cable length for $500\,\mathrm{V}/\mu\mathrm{s}$ is in the 100 m range, for $1000\,\mathrm{V}/\mu\mathrm{s}$ in the 50 m range and for $10,000\,\mathrm{V}/\mu\mathrm{s}$ in the 5 m range [15].

(b) Common-Mode Voltage Stress The switching action of the rectifier and inverter normally generates common-mode voltages [16]. The common-mode voltages are essentially zero-sequence voltages superimposed with switching noise. If not mitigated, they will appear on the neutral of the motor with respect to ground, which should be zero when the motor is powered by a three-phase balanced utility supply. Further, the motor line-to-ground voltage, which should be equal to the motor line-to-neutral (phase) voltage, can be substantially increased due to the common-mode voltages, leading to the premature failure of the motor winding insulation system. As a consequence, the motor life expectancy is shortened.

It is worth noting that the common-mode voltages are generated by the rectification and inversion process of the converters. This phenomenon is different from the high *dv/dt* caused by the switching transients of the high-speed switches. It should be further noted that the common-mode voltage issue is often ignored in the low voltage drives. This is partially due to the conservative design of the insulation system for low

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voltage motors. In the MV drives, the motor should not subject to any common-mode voltages. Otherwise, the replacement of the damaged motor would be very costly in addition to the loss of production.

(c) Motor Derating High-power inverters often generate a large amount of current and voltage harmonics. These harmonics cause additional power losses in the motor winding and magnetic core. As a consequence, the motor is derated and cannot operate at its full capacity.

(d) LC Resonances For the MV drives with a motor-side filter capacitor, the capacitor forms an LC resonant circuit with the motor inductances. The resonant mode of the LC circuit may be excited by the harmonic voltages or currents produced by the inverter. Although the motor winding resistances may provide some damping, this problem should be addressed at the design stage of the drive.

1.2.3 Switching Device Constraints

(a) Device Switching Frequency The device switching loss accounts for a significant amount of the total power loss in the MV drive. The switching loss minimization can lead to a reduction in the operating cost when the drive is commissioned. The physical size and manufacturing cost of the drive can also be reduced due to the reduced cooling requirements for the switching devices. The other reason for limiting the switching frequency is related to the device thermal resistance that may prevent efficient heat transfer from the device to its heatsink. In practice, the device switching frequency is normally limited to around 200 Hz for GTOs and 500 Hz for IGBTs and GCTs.

The reduction of switching frequency generally causes an increase in harmonic distortion of the line- and motor-side waveforms of the drive. Efforts should be made to minimize the waveform distortion with limited switching frequencies.

(b) Series Connection Switching devices in the MV drive are often connected in series for medium voltage operation. Since the series connected devices and their gate drivers may not have identical static and dynamic characteristics, they may not equally share the total voltage in the blocking mode or during switching transients. A reliable voltage equalization scheme should be implemented to protect the switching devices and enhance the system reliability.

1.2.4 Drive System Requirements

The general requirements for the MV drive system include high efficiency, low manufacturing cost, small physical size, high reliability, effective fault protection, easy

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installation, self-commissioning, and minimum downtime for repairs. Some of the application-specific requirements include high dynamic performance, regenerative braking capability, and four-quadrant operation.

1.3 CONVERTER CONFIGURATIONS

Multipulse rectifiers are often employed in the MV drive to meet the line-side harmonic requirements. Figure 1.3-1 illustrates a block diagram of 12-, 18-, and 24-pulse rectifiers. Each multipulse rectifier is essentially composed of a phase-shifting transformer with multiple secondary windings feeding a set of identical six-pulse rectifiers.

Both diode and SCR devices can be used as switching devices. The multipulse diode rectifiers are suitable for VSI fed drives while the SCR rectifiers are normally for CSI drives. Depending on the inverter configuration, the outputs of the six-pulse rectifiers can be either connected in series to form a single dc supply or connected directly to a multilevel inverter that requires isolated dc supplies. In addition to the diode and SCR rectifiers, PWM rectifiers using IGBT or GCT devices can also be employed, where the rectifier usually has the same topology as the inverter.

To meet the motor-side challenges, a variety of inverter topologies can be adopted for the MV drive. Figure 1.3-2 illustrates per-phase diagram of commonly used three-phase multilevel VSI topologies, which include a conventional two-level (2L) VSI, a flying-capacitor(FC) inverter, a neutral point clamped (NPC) inverter, a cascaded H-bridge (CHB) inverter, and a modular multilevel inverter (MMC). For the 2L-VSI, FC, and NPC inverter topologies, either IGBT or GCT devices can be employed, while for the CHB and MMC inverters, IGBT devices are exclusively used.

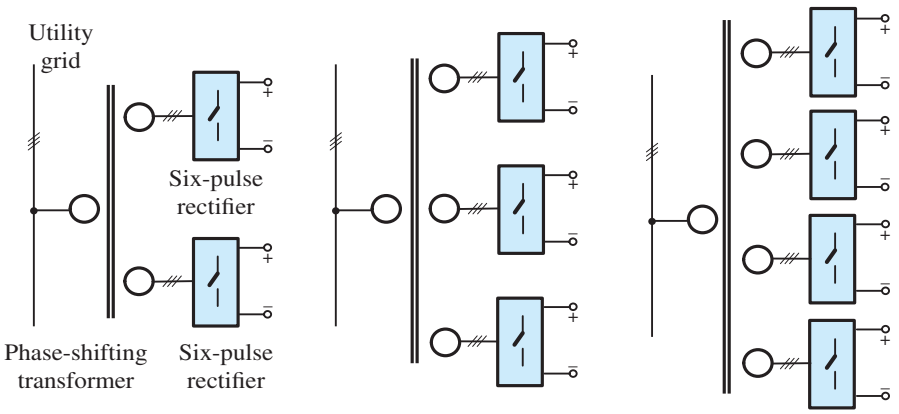

Figure 1.3-1 Multipulse diode/SCR rectifiers.

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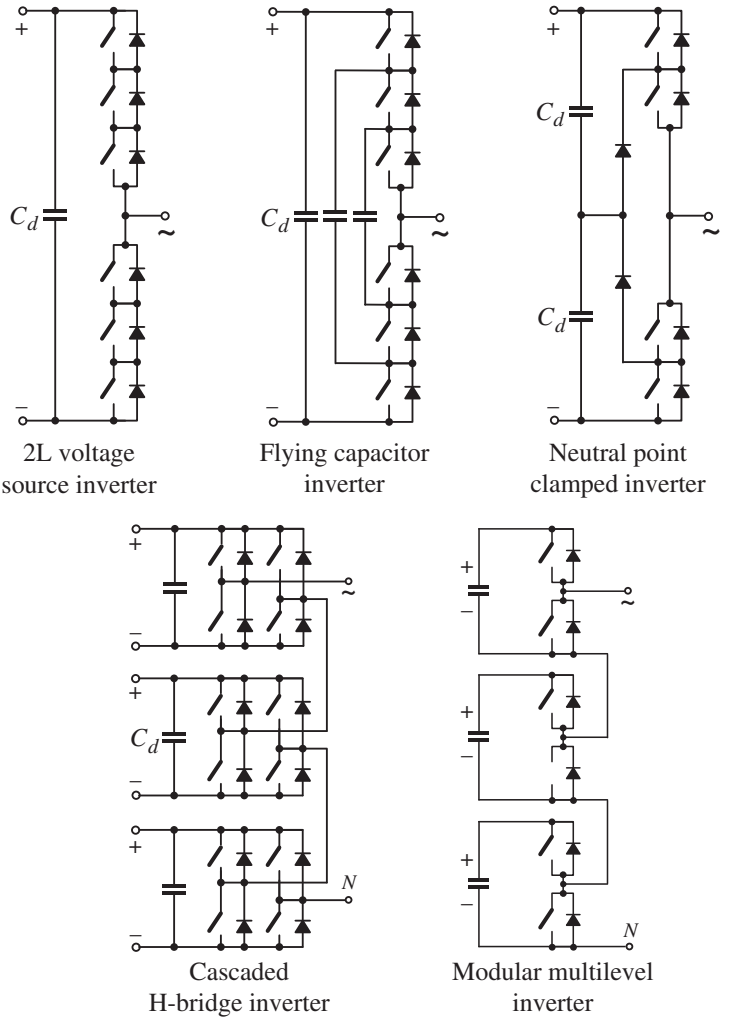

Figure 1.3-2 Per-phase diagram of VSI topologies.

CSI technology has been widely accepted in the drive industry. Figure 1.3-3 shows the per-phase diagram of the CSI topologies for the MV drive. The SCR-based LCI is specially suitable for very large synchronous motor drives while the PWM CSI is a preferred choice for most industrial applications. The parallel PWM CSI is composed of two or more single-bridge inverters connected in parallel for super-high-power applications. Symmetrical GCTs are normally used in the PWM CSIs.

A relatively new power converter topology, cascaded matrix converter (CMC) as shown in Fig. 1.3-4, has been developed for used in commercial MV drives. Unlike the VSI and CSI topologies, the CMC converter performs direct ac–ac conversion

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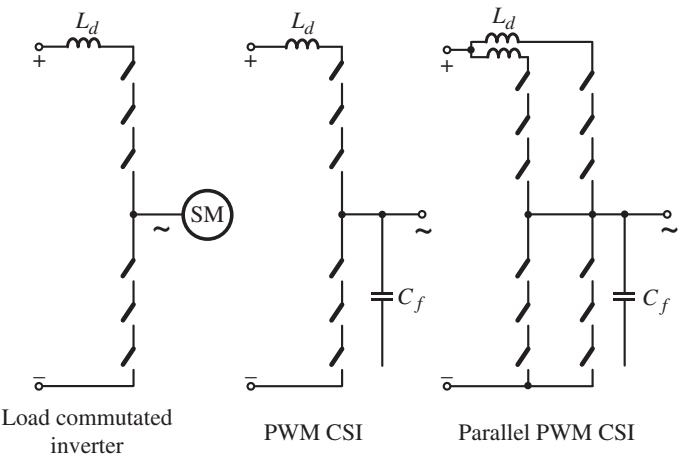

Figure 1.3-3 Per-phase diagram of CSI topologies.

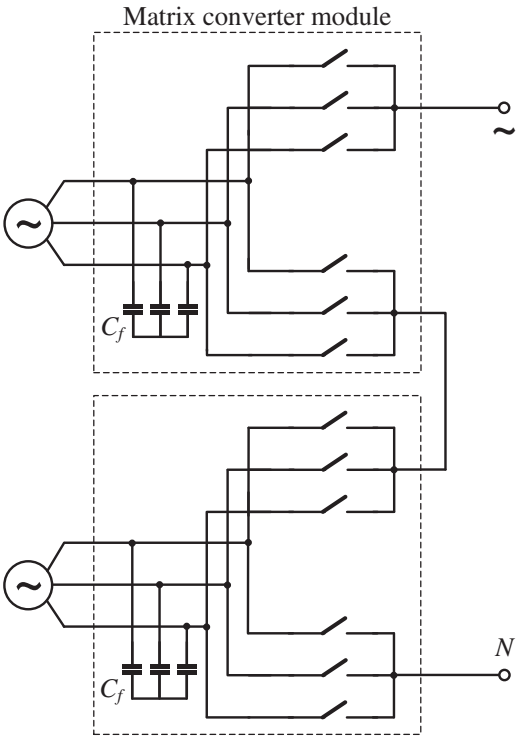

Figure 1.3-4 Per-phase diagram of cascaded matrix converter topology.

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without the need of a dc link. The CMC topology is composed of a number of CMC modules per phase to produce high-quality output voltage waveforms, but requires an isolated three-phase power supply for each of the MC modules. IGBT devices are exclusively used in this topology.

1.4 INDUSTRIAL MV DRIVES

A number of MV drive products are available in the market today. These drives come with different designs using various power converter topologies and control schemes. Each design offers some unique features but also has some limitations. The diversified offering promotes the advancement in the drive technology and the market competition as well. A few examples of the MV industrial drives are as follows.

Figure 1.4-1 illustrates the picture of an MV drive rated at 4.16 kV and 2.0 MW. The drive is composed of a 12-pulse diode rectifier as a front end and a three-level NPC inverter using GCT devices. The drive's digital controller is installed in the left cabinet. The cabinet in the center houses the diode rectifier and air-cooling system of the drive. The inverter and its output filters are mounted in the right cabinet. The phase-shifting transformer for the rectifier is normally installed outside the drive cabinets.

Figure 1.4-1 GCT-based three-level NPC inverter fed MV drive. Courtesy of ABB (ACS1000).

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Figure 1.4-2 IGBT-based three-level NPC inverter fed MV drive. Courtesy of Siemens (SIMOVERT MV).

Figure 1.4-2 shows an MV drive using an IGBT-based three-level NPC inverter. The IGBT–heatsink assemblies in the central cabinet are constructed in a modular fashion for easy assembly and replacement. The front end converter is a standard 12-pulse diode rectifier for line current harmonic reduction. The phase-shifting transformer for the rectifier is not included in the drive cabinet.

Figure 1.4-3 shows a 6.6 kV cascaded H-bridge inverter fed MV drive with a power rating from 0.2 to 3.72 MW. The CHB inverter has 18 IGBT power cells and is installed in the middle cabinet. The inverter line-to-line voltage is composed of 25 small voltage steps, which makes the inverter output voltage waveform nearly sinusoidal. A phase-shifting transformer with 18 secondary windings is in the left cabinet. The dominant switching harmonics produced by the power cells are cancelled by the phase-shifting transformer, which makes its primary current nearly sinusoidal. The digital controller for the drive is mounted in the right cabinet.

Figure 1.4-4 shows a CSI fed MV drive with a power range from 2.3 to 7 MW. The drive comprises two identical PWM GCT current source converters, one for the rectifier and the other for the inverter. The converters are installed in the second cabinet from the left. The dc inductor required by the current source drive is mounted in the fourth cabinet. The fifth (most right) cabinet contains drive's liquid-cooling

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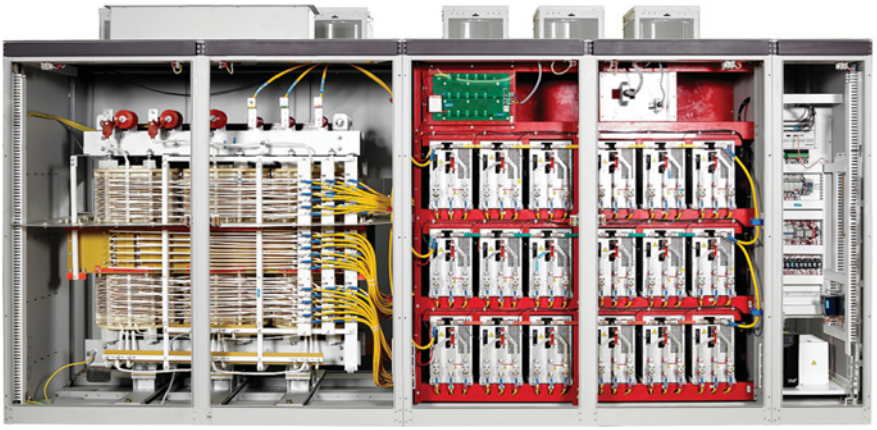

Figure 1.4-3 IGBT cascaded H-bridge inverter fed MV drive. Courtesy of Rockwell Automation (PowerFlex 6000).

system. With the use of a special integrated dc inductor having both differential- and common-mode inductances, the drive does not require an isolation transformer for the common-mode voltage mitigation, leading to a significant reduction in manufacturing cost.

Table 1.4-1 provides a summary of the MV drive products offered by major drive manufacturers in the world, where the inverter configuration, switching device, and power range of the drive are listed.

Figure 1.4-4 CSI fed MV drive using symmetrical GCTs. Courtesy of Rockwell Automation (PowerFlex 7000).

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Table 1.4-1 Examples of the MV Drive Products Marketed by Major Drive Manufacturers

Inverter Configuration	Switching Device	Power Range (MVA)	Manufacturer
Two-level voltage source inverter (VSI)	IGBT	1.4–7.2	Alstom (VDM5000)
Three-level neutral point clamped (NPC) inverter	GCT	0.3–5 3–36	ABB (ACS1000) (ACS6000)
	IGBT	3–21	GE Power Conversion (MV7000)
	IGBT	0.6–7.2	Siemens (SIMOVERT-MV)
Multilevel cascaded H-bridge (CHB) inverter	IGBT	0.2–13	θ Harvest (HARSVERT A/S/VA)
		0.3–60	Siemens (Perfect Harmony) (GH180)
		0.31–16.7	Hitachi (HIVECOL-HVI)
		0.32–5.6	Rockwell Automation (PowerFlex 6000)
Multilevel NPC/H-bridge inverter	IGBT	0.4–4.8	Toshiba (TOSVERT 300 MV)
		0.2–3.75	Yaskawa (MV1000)
Flying-capacitor (FC) inverter	IGBT	0.5–9	Alstom (VDM6000 Symphony)
PWM current source inverter (CSI)	Symmetrical GCT	0.2–25	Rockwell Automation (PowerFlex 7000)
Load commutated inverter (LCI)	SCR	>10	Siemens (SIMOVERT S)
		>10	ABB (LCI)
		>10	Alstom (ALSPA SD7000)

1.5 SUMMARY

This chapter provides an overview of high-power converters and medium voltage (MV) drives, including market analysis, drive system configurations, power converter topologies, drive product analysis, and major manufacturers. The technical requirements and challenges for the MV drive are also summarized. These requirements

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and challenges will be addressed in the subsequent chapters, where various power converters and MV drive systems are analyzed.

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APPENDIX

A Summary of MV Drive Applications

Industry	Application Examples
Petrochemical	Pipeline pumps, gas compressors, brine pumps, mixers/extruders, electrical submersible pumps, induced draft fans, boiler feed water pumps, water injection pumps
Cement	Kiln induced draft fans, forced draft fans, baghouse fans, preheat tower fans, raw mill induced draft fans, kiln gas fans, cooler exhaust fans, separator fans
Mining and Metals	Slurry pumps, ventilation fans, descaling pumps, tandem belt conveyors, baghouse fans, cyclone feed pumps, crushers, rolling mills, hoists, coilers, winders
Water / Wastewater	Raw sewage pumps, bio-roughing tower pumps, treatment pumps, freshwater pumps, storm water pumps
Transportation	Propulsion for naval vessels, shuttle tankers, icebreakers, cruisers. Traction drives for locomotives, light-track trains
Electric Power	Feed water pumps, induced draft fans, forced draft fans, effluent pumps, compressors
Forest Products	Induced draft fans, boiler feed water pumps, pulpers, refiners, kiln drives, line shafts
Miscellaneous	Wind tunnels, agitators, test stands, rubber mixers

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Chapter 2

High-Power Semiconductor Devices

2.1 INTRODUCTION

The development of semiconductor switching devices is essentially a search for the ideal switch. The effort has been made to reduce device power losses, increase switching frequencies, and simplify gate drive circuits. The evolution of the switching devices leads the pace of high-power converter development, and in the meantime the wide application of the high-power converters in industry drives the semiconductor technology toward higher power ratings with improved reliability and reduced cost.

There are two major types of high-power switching devices for use in various converters: the thyristor- and transistor-based devices. The former includes silicon controlled rectifier (SCR), gate turn-off (GTO) thyristor, and gate commutated thyristor (GCT) while the latter embraces insulated gate bipolar transistor (IGBT) and injection enhanced gate transistor (IEGT). Other devices such as power MOSFET, emitter turn-off (ETO) thyristor, MOS controlled thyristor (MCT), and static induction thyristor (SIT) have not gained significant importance in high-power applications.

Figure 2.1-1 shows the voltage and current ratings of major switching devices commercially available for high-power converters [1]. Semiconductor manufacturers can offer SCRs rated at 12 kV/1.5 kA or 1.8 kV/6.1 kA. The GCT devices can reach the voltage and current ratings of 6 kV and 6 kA. The ratings of IGBT devices are relatively low, but can reach as high as 6.5 kV/0.75 kA or 1.7 kV/3.6 kA.

In this chapter, the characteristics of commonly used high-power semiconductor devices are introduced, the static and dynamic voltage equalization techniques for series connected devices are discussed, and the performance of these devices is compared.

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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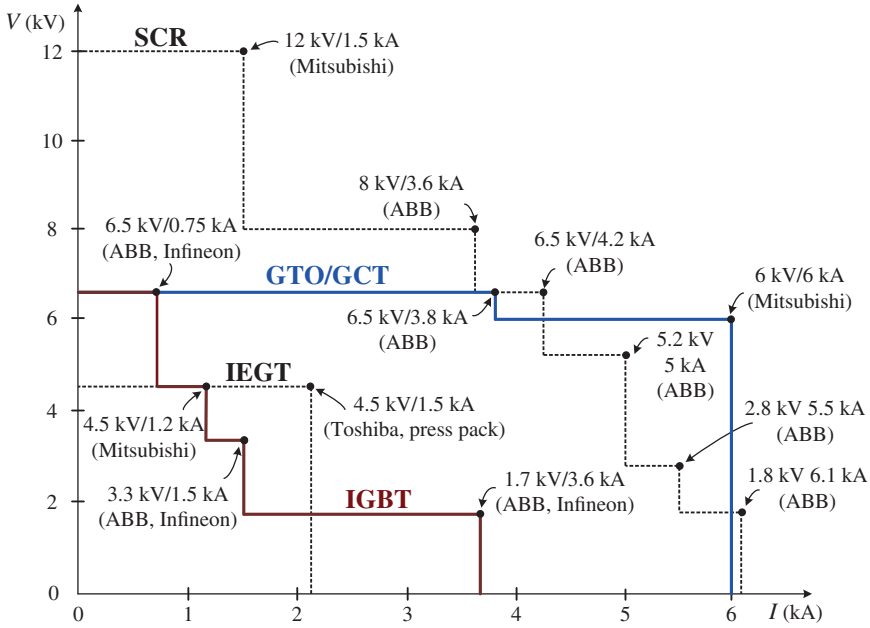

Figure 2.1-1 Voltage and current ratings of high-power semiconductor devices.

2.2 HIGH-POWER SWITCHING DEVICES

2.2.1 Diodes

High-power diodes can be generally classified into two types: the general-purpose type for use in uncontrolled line-frequency rectifiers and the fast recovery type used in voltage source converters as a freewheeling diode. These diodes are commercially available with two packaging techniques: press-pack and module diodes as shown in Fig. 2.2-1.

The device-heatsink assemblies for press-pack and module diodes are shown in Fig. 2.2-2. The press-pack diode features double-sided cooling with low thermal stress. For medium voltage applications where a number of diodes may be connected in series, the diodes and their heatsinks can be assembled with just two bolts, leading to high power density and low assembly costs. This is one of the reasons for the continued popularity of press-pack semiconductors in the medium voltage drives. The modular diode has an insulated baseplate with single-sided cooling, where a number of diodes can be mounted onto a single piece of heatsink.

2.2.2 Silicon Controlled Rectifier (SCR)

The SCR is a thyristor-based device with three terminals: gate, anode, and cathode. It can be turned on by applying a pulse of positive gate current with a short duration

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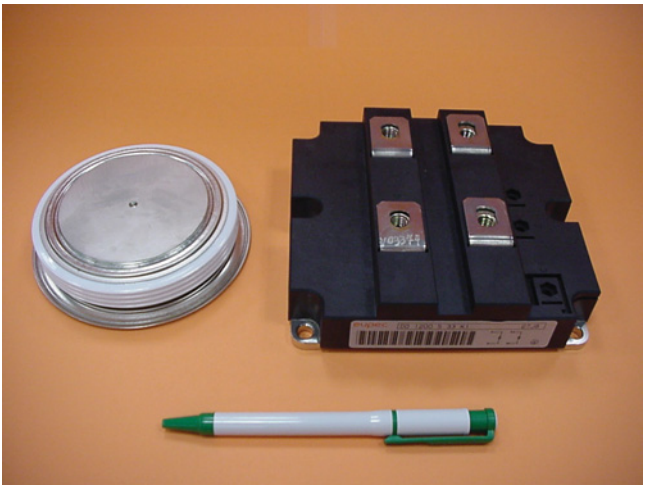


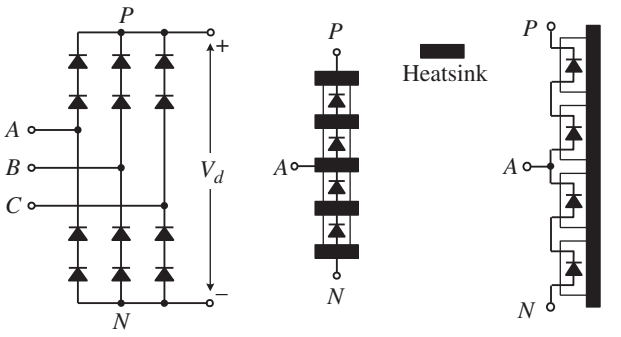
Figure 2.2-1 4.5 kV/0.8 kA press-pack and 1.7 kV/1.2 kA module diodes.

provided that it is forward biased. Once the SCR is turned on, it is latched on. The device can be turned off by applying a negative anode current produced by its power circuit.

The SCR device can be used in phase-controlled rectifiers for PWM current source inverter fed drives or load commutated inverters for synchronous motor drives. Prior to the advent of self-extinguishable devices such as GTO and IGBT, the SCR was also used in forced commutated voltage source inverters.

The majority of high-power SCRs are of press-pack type as shown in Fig. 2.2-3. The SCR modules with an insulated baseplate are more popular for low and medium power applications.

Figure 2.2-4 shows the switching characteristics of the SCR device and typical waveforms for gate current i_G, anode current i_T, and anode–cathode voltage v_T. The



(a) Diode Rectifier (b) Press pack (c) Module

Figure 2.2-2 Device-heatsink assemblies for press-pack and module diodes.

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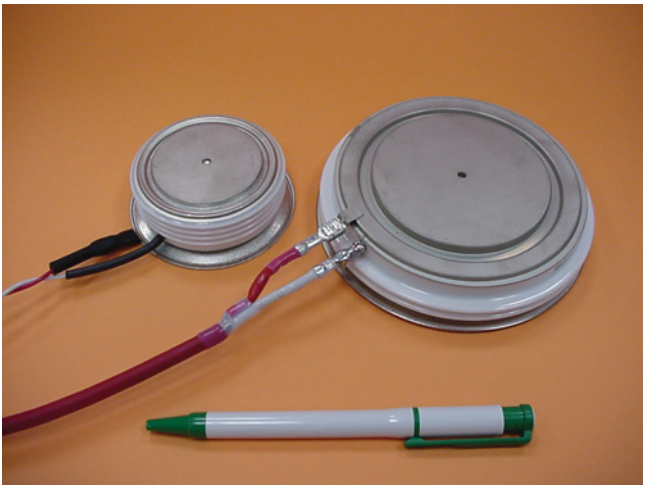

Figure 2.2-3 4.5 kV/0.8 kA and 4.5 kV/1.5 kA SCRs.

turn-on process is initiated by applying a positive gate current i_G to the SCR gate. The turn-on behavior is defined by delay time t_d, rise time t_r, and turn-on time t_{gt}.

The turn-off process is initiated by applying a negative current to the switch at time instant t_1, at which the anode current i_T starts to fall. The negative current is produced by the utility voltage when the SCR is used in a rectifier or by the load voltage in a load commutated inverter. The turn-off transient is characterized by reverse recovery time t_{rr}, peak reverse recovery current I_{rr}, reverse recovery charge Q_{rr}, and turn-off time t_q.

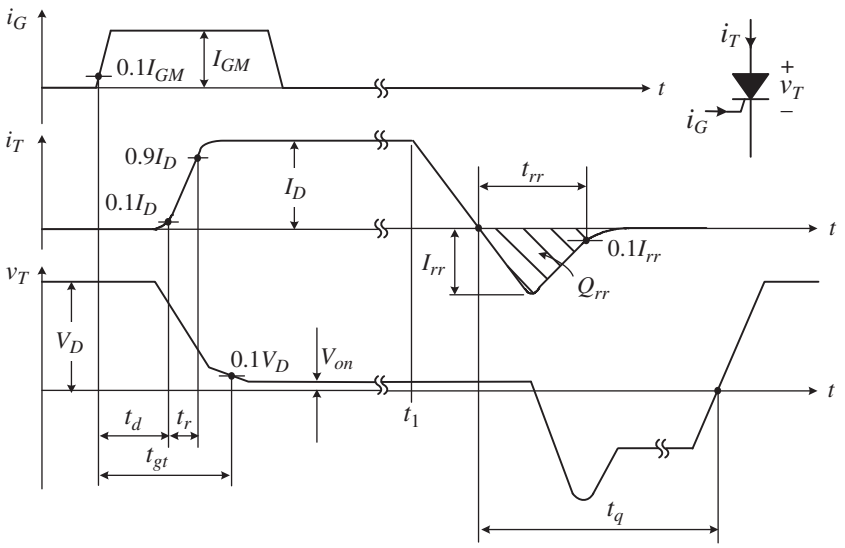

Figure 2.2-4 SCR switching characteristics.

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Table 2.2-1 Main Specifications of a 12 kV/1.5 kA SCR

Maximum Ratings	V_{DRM}	V_{RRM}	I_{TAVM}	I_{TRMS}	—
	12,000 V	12,000 V	1500 A	2360 A	—
Switching Characteristics	**Turn-on Time**	**Turn-off Time**	$\boldsymbol{di_T/dt}$	$\boldsymbol{dv_T/dt}$	$\boldsymbol{Q_{rr}}$
	$t_{gt} = 14\ \mu s$	$t_q = 1200\ \mu s$	100 A/μs	2000 V/μs	7000 μC
Part number: FT1500AU-240 (Mitsubishi)					

Table 2.2-1 lists the main specifications of a 12 kV/1.5 kA SCR device, where V_{DRM} is the maximum repetitive peak off-state voltage, V_{RRM} is the maximum repetitive peak reverse voltage, I_{TAVM} is the maximum average on-state current, and I_{TRMS} is the maximum rms on-state current. The turn-on time t_{gt} is 14 μs and the turn-off time t_q is 1200 μs. The rates of anode current rise di_T/dt at turn-on and device voltage rise dv_T/dt at turn-off are important parameters for converter design. To ensure a proper and reliable operation, the maximum limits for the di_T/dt and dv_T/dt must not be exceeded. The reverse recovery charge Q_{rr} is normally a function of reverse recovery time t_{rr} and reverse recovery current I_{rr}. To reduce the power loss at turn-off, the SCR with a low value of Q_{rr} is preferred.

2.2.3 Gate Turn-Off (GTO) Thyristor

The GTO thyristor is a self-extinguishable device that can be turned off by a negative gate current. The GTOs are normally of press-pack design as shown in Fig. 2.2-5, and the modular design is not commercially available. Several manufacturers offer GTOs up to a rated voltage of 6 kV with a rated current of 6 kA.

Figure 2.2-5 4.5 kV/0.8 kA and 4.5 kV/1.5 kA GTO devices.

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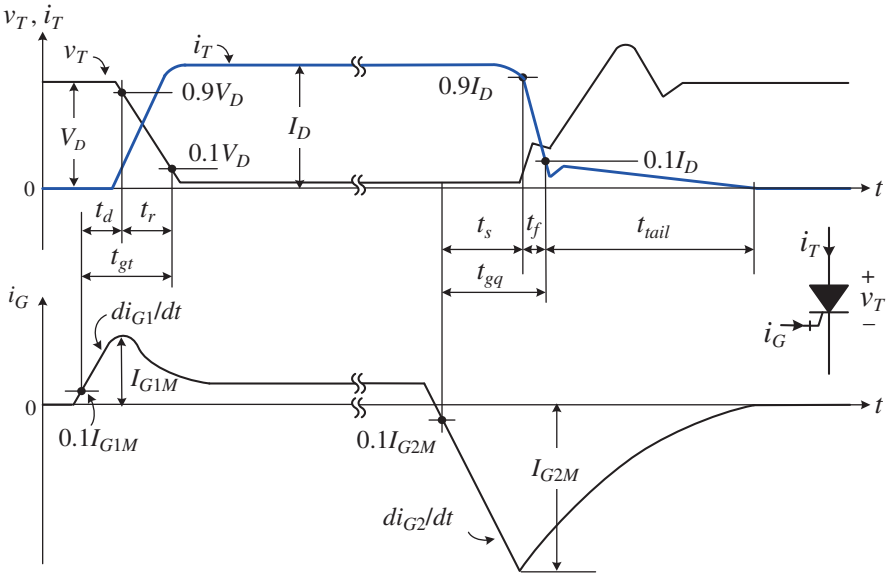

Figure 2.2-6 GTO switching characteristics.

The GTO can be fabricated with symmetrical or asymmetrical structures. The symmetric GTO has reverse voltage blocking capability, making it suitable for current source converters. Its maximum repetitive peak off-state voltage V_{DRM} is approximately equal to its maximum repetitive peak reverse voltage V_{RRM}. The asymmetric GTO is generally used in voltage source converters where the reverse voltage blocking capability is not required. The value of V_{RRM} is typically around 20 V, much lower than V_{DRM}.

The switching characteristics of the GTO thyristor are shown in Fig. 2.2-6, where i_T and v_T are the anode current and anode–cathode voltage, respectively. The GTO turn-on behavior is measured by delay time t_d and rise time t_r. The turn-off transient is characterized by storage time t_s, fall time t_f, and tail time t_{tail}. Some manufacturers provide only turn-on time t_{gt} ($t_{gt} = t_d + t_r$) and turn-off time t_{gq} ($t_{gq} = t_s + t_f$) in their datasheets. The GTO is turned on by a pulse of positive gate current of a few hundred milliamps. Its turn-off process is initiated by a negative gate current. To ensure a reliable turn-off, the rate of change of the negative gate current di_{G2}/dt must meet with the specification set by the device manufacturer.

Table 2.2-2 gives the main specifications of a 4500 V/4000 A asymmetrical GTO device, where V_{DRM}, V_{RRM}, I_{TAVM}, and I_{TRMS} have the same definitions as those for the SCR device. It is worth noting that the current rating of a 4000 A GTO is defined by I_{TGQM}, which is the maximum repetitive controllable on-state current, not by the average current I_{TAVM}. The turn-on delay time t_d and rise time t_r are 2.5 μs and 5.0 μs while the storage time t_s and fall time t_f at turn-off are 25 μs and 3 μs, respectively. The maximum rates of rise of the anode current, gate current, and device voltage are also given in the table.

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Table 2.2-2 Main Specifications of a 4.5 kV/4 kA Asymmetrical GTO

Maximum Rating	V_{DRM}	V_{RRM}	I_{TGQM}	I_{TAVM}	I_{TRMS}	—
	4500 V	17 V	4000 A	1000 A	1570 A	—
Switching Characteristics	**Turn-on Switching**	**Turn-off Switching**	$\boldsymbol{di_T/dt}$	$\boldsymbol{dv_T/dt}$	$\boldsymbol{di_{G1}/dt}$	$\boldsymbol{di_{G2}/dt}$
	$t_d = 2.5\ \mu s$ $t_r = 5.0\ \mu s$	$t_s = 25.0\ \mu s$ $t_f = 3.0\ \mu s$	500 A/μs	1000 V/μs	40 A/μs	40 A/μs
On-state Voltage	$V_{T(on-state)} = 4.4$ V at $I_T = 4000$ A					
Part number: 5SGA 40L4501 (ABB)						

The GTO thyristor features high on-state current density and high blocking voltage. However, the GTO device has a number of drawbacks, including (a) bulky and expensive snubber circuits due to low dv_T/dt, (b) high switching and snubber losses due to slow switching speeds, and (c) complex gate driver. It also needs a turn-on snubber to limit di_T/dt.

2.2.4 Gate Commutated Thyristor (GCT)

The GCT, also known as integrated gate commutated thyristor (IGCT), is developed from the GTO structure [2, 3]. Over the past two decades, the industry has seen the GTO thyristor being replaced by the GCT device. The GCT has become the device of choice for medium voltage drives due to its features such as snubberless operation and low switching loss.

The key GCT technologies include significant improvements in silicon wafer, gate driver, and device packaging. The GCT wafer is much thinner than GTO's, leading to a reduction in on-state power loss. As shown in Fig. 2.2-7, a special gate driver with ring-gate packaging provides an extremely low gate inductance (typically <5 nH) that allows the GCT to operate without snubber circuits. The rate of gate current change at turn-off is normally greater than 3000 A/μs instead of around 40 A/μs for the GTO device. Since the gate driver is an integral part of the GCT, the user only needs to provide the gate driver with a 20–30 V dc power supply and connect the driver to the system controller through two fiber-optic cables for on/off control and device fault diagnostics.

Several manufacturers offer GCT devices with ratings up to 6 kV/6 kA. Ten kilovolt GCTs are technically possible and the development of this technology depends on the market needs [4].

The GCT devices can be classified into asymmetrical, reverse-conducting, and symmetrical types as shown in Table 2.2-3. The asymmetric GCT is generally used in voltage source converters where the reverse voltage blocking capability is not required. The reverse-conducting GCT integrates the freewheeling diode into one

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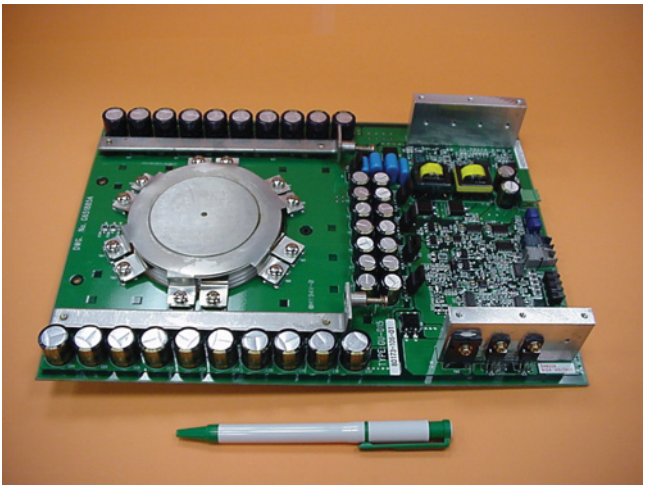

Figure 2.2-7 6.5 kV/1.5 kA symmetrical GCT.

package, resulting in a reduced assembly cost. The symmetric GCT is normally for use in current source converters.

Figure 2.2-8 shows the typical switching characteristics of the GCT device, where the delay time t_d, rise time t_r, storage time t_s, and fall time t_f are defined in the same way as those for the GTO. Note that some semiconductor manufacturers may define the switch times differently or use different symbols. The waveform for the gate current i_G is given as well, where the rate of gate current change di_{G2}/dt at turn-off is substantially higher than that for the GTO.

Table 2.2-4 gives the main specifications of a 6000 V/6000 A asymmetrical GCT, where the maximum repetitive controllable on-state current I_{TQRM} is 6000 A. The turn-on and turn-off times are much faster than those for the GTO. In particular, the

Table 2.2-3 GCT Device Classification

Type	Anti-parallel Diode	Blocking Voltage	Example (6000 V GCT)	Applications
Asymmetrical GCT	Excluded	$V_{RRM} \ll V_{DRM}$	$V_{DRM} = 6000$ V $V_{RRM} = 22$ V	For use in voltage source converters with anti-parallel diodes
Reverse-conducting GCT	Included	$V_{RRM} \approx 0$	$V_{DRM} = 6000$ V	For use in voltage source converters
Symmetrical GCT (reverse blocking)	Not required	$V_{RRM} \approx V_{DRM}$	$V_{DRM} = 6000$ V $V_{RRM} = 6500$ V	For use in current source converters

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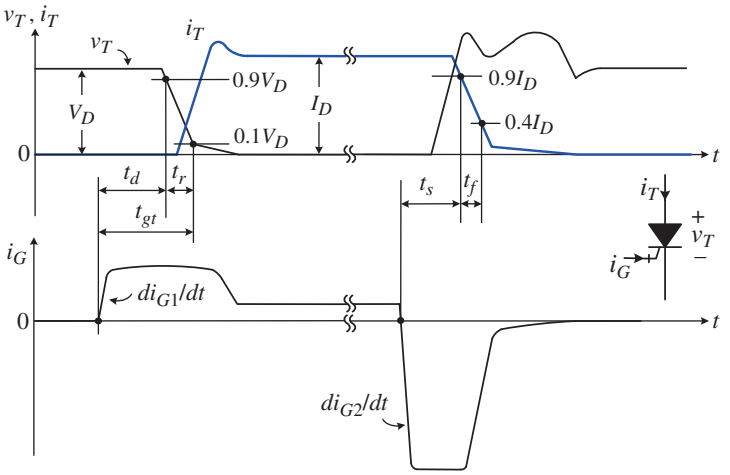

Figure 2.2-8 GCT switching characteristics.

storage time t_s is only 3 μs in comparison with 25 μs for the 4000 A GTO device in Table 2.2-2. The maximum dv_T/dt can be as high as 3000 V/μs. The maximum rate of gate current change, di_{G2}/dt, can be as high as 10,000 A/μs, which helps to reduce the switching time at turn-off. The on-state voltage at $I_T = 6000$ A is only 4 V in comparison with 4.4 V at $I_T = 4000$ A for the GTO device.

The GCT device normally requires a turn-on snubber since the *di/dt* capability of the device is only around 1000 A/μs. Fig. 2.2-9a shows a typical turn-on snubber circuit for voltage source converters [5]. The snubber inductor L_s limits the rate of anode current rise at the moment when one of the six GCTs is gated on. The energy trapped in the inductor is partially dissipated on the snubber resistor R_s. All six GCTs in the converter can share one snubber circuit. In current source converters, the snubber circuit takes a different form as shown in Fig. 2.2-9b, where a *di/dt* limiting

Table 2.2-4 Main Specifications of a 6 kV/6 kA Asymmetrical GCT

Maximum Rating	V_{DRM}	V_{RRM}	I_{TQRM}	I_{TAVM}	I_{TRMS}	—
	6000 V	22 V	6000 A	2000 A	3100 A	—
Switching Characteristics	**Turn-on Switching**	**Turn-off Switching**	$\boldsymbol{di_T/dt}$	$\boldsymbol{dv_T/dt}$	$\boldsymbol{di_{G1}/dt}$	$\boldsymbol{di_{G2}/dt}$
	$t_d < 1.0$ μs $t_r < 2.0$ μs	$t_s < 3.0$ μs t_f—N/A	1000 A/μs	3000 V/μs	200 A/μs	10,000 A/μs
On-state Voltage	$V_{T(on-state)} < 4$ V at $I_T = 6000$ A					
Part number: FGC6000AX120DS (Mitsubishi)						

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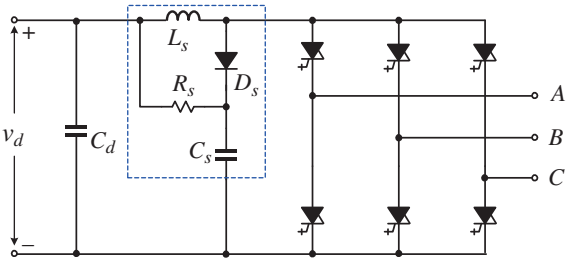
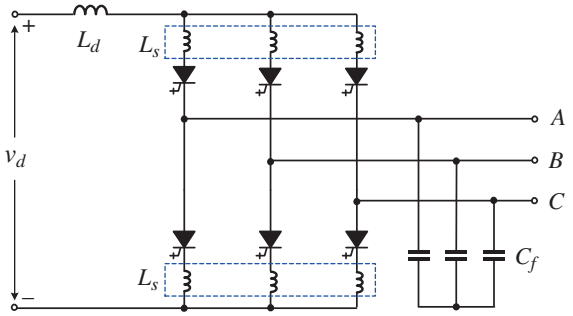


(a) Turn-on snubber in a VSI using reverse-conducting GCTs



(b) Turn-on snubber in a CSI using symmetrical GCTs

Figure 2.2-9 Turn-on *di/dt* snubber for GCTs.

inductor of a few microhenries is required in each of the converter legs, but no other passive components are needed.

2.2.5 Insulated Gate Bipolar Transistor (IGBT)

The IGBT is a voltage controlled device. It can be switched on with a +15 V gate voltage and turned off when the gate voltage is zero. In practice, a negative gate voltage of a few volts is applied during the device off period to increase its noise immunity. The IGBT does not require any gate current when it is fully turned on or off. However, it does need a peak gate current of a few Amperes during switching transients due to the gate-emitter capacitance.

The majority of high-power IGBTs are of modular design as shown in Fig. 2.2-10. Press-pack IGBTs are also available on the market for assembly cost reduction and efficient cooling, but the selection of such devices is limited.

The typical switching characteristics of the IGBT device are shown in Fig. 2.2-11, where the turn-on delay time t_{don}, rise time t_r, turn-off delay time t_{doff}, and fall time t_f are defined. The waveforms for gate driver output voltage v_G, gate-emitter voltage v_{GE}, and collector current i_C are also given. The voltage v_{GE} is equal to v_G after the IGBT is fully turned on or off. These two voltages, however, are not the same during switching transients due to the gate-emitter capacitance. The gate resistor R_G

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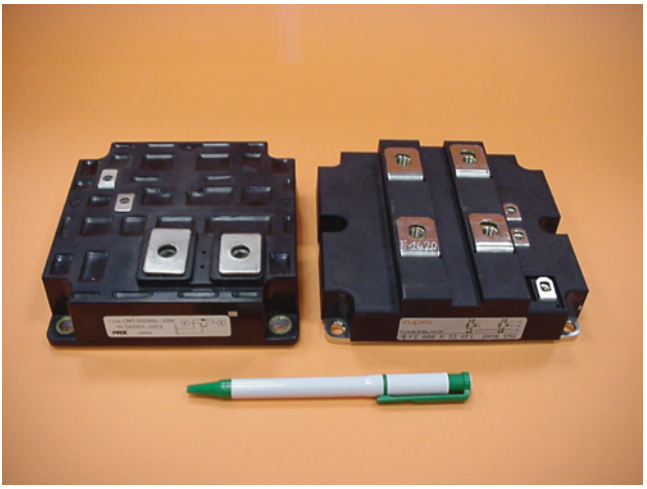

Figure 2.2-10 1.7 kV/1.2 kA and 3.3 kV/1.2 kA IGBT modules.

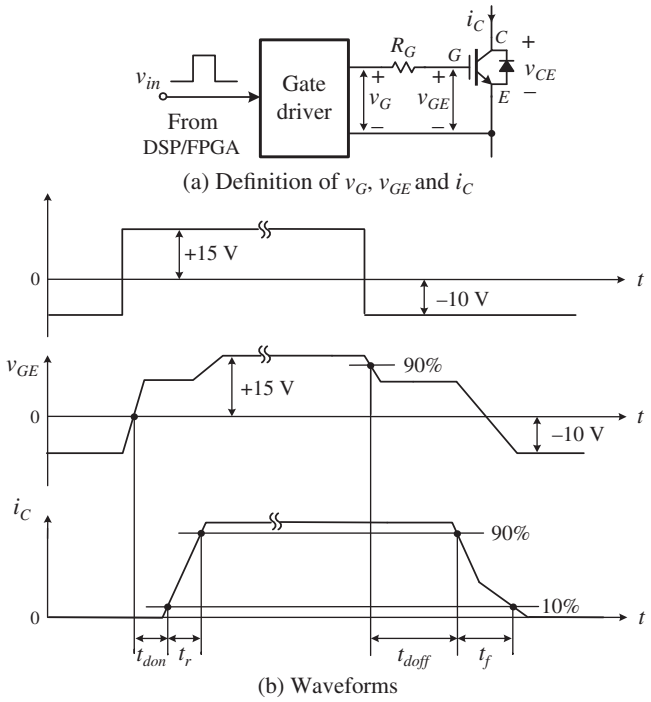

Figure 2.2-11 IGBT switching characteristics.

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Table 2.2-5 Main Specifications of a 3.3 kV/1.2 kA IGBT

Maximum Rating	V_{CE}	I_C	I_{CM}	—
	3300 V	1200 A	2400 A	—
Switching Characteristics	t_{don}	t_r	t_{doff}	t_f
	0.35 μs	0.27 μs	1.7 μs	0.2 μs
Saturation Voltage	$I_{CE\,sat}$ = 4.3 V at I_C = 1200 A			
Part number: FZ1200 R33 KF2 (Eupec)				

is normally required to adjust the device switching speed and to limit the transient gate current.

Table 2.2-5 gives the main specifications of a 3.3 kV/1.2 kA IGBT, where V_{CE} is the rated collector–emitter voltage, I_C is the rated dc collector current, and I_{CM} is the maximum repetitive peak collector current. The IGBT has superior switching characteristics. It can be turned on within 1 μs and turned off within 2 μs.

The IGBT device features simple gate driver, snubberless operation, high switching speed, and modular design with insulated baseplate. More importantly, the IGBT can operate in the active region. Its collector current can be controlled by the gate voltage, providing an effective means for reliable short-circuit protection and active control of *dv/dt* and overvoltage at turn off.

The construction of a medium voltage converter with series connected IGBTs should consider a number of issues such as efficient cooling arrangements, optimal dc bus-bar design and stray capacitance of baseplates to ground. In contrast, press-pack IGBTs allow direct series connection, where the mounting and cooling techniques developed for press-pack thyristors can be utilized.

2.2.6 Other Switching Devices

There are a number of other semiconductor devices, including power MOSFET, ETO thyristor [6], MCT, SIT, and IEGT [7]. However, they have not gained significant importance in high-power applications.

A new type of emerging power semiconductor devices is the silicon carbide (SiC), which is a compound semiconductor composed of silicon and carbon. The SiC power devices are able to operate at higher switching speed and higher temperature with lower switching and conduction losses in comparison to the silicon devices such as IGBTs [1, 8]. With the technological advancements, the power rating of the SiC devices has been significantly increased. It was reported that a hybrid SiC module of 1700 V/1200 A was developed in 2013 [9] and an inverter using 3.3 kV/1500 A SiC power modules was developed for traction applications in 2015 [10]. It is expected that the SiC devices with higher voltage and current ratings will become commercially available with reduced cost in the coming years, which provides a great potential to be used high-power MV drives for reduced power losses and improved system performance. This is an active research area that the researchers around the world and major MV drive manufacturers are moving into.

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2.3 OPERATION OF SERIES CONNECTED DEVICES

In medium voltage drives, switching devices are normally connected in series. It is not necessary to parallel the devices since the current capacity of a single device is usually sufficient. For instance, in a 6.6 kV 10 MW drive the rated motor current is only 880 A in comparison with the current rating of a 6000 A GCT or 3600 A IGBT.

Since the series connected devices and their gate drivers may not have exactly the same static and dynamic characteristics, they may not equally share the total voltage in the blocking mode or during switching transients. The main task for the series connected switches is to ensure equal voltage sharing under both static and dynamic conditions.

2.3.1 Main Causes of Voltage Unbalance

The static voltage unbalance is mainly caused by the difference in the off-state leakage current I_{lk} of series connected switches. Further, the leakage current is a function of device junction temperature and operating voltage. The causes of the dynamic voltage unbalance can be divided into two groups: (a) unbalance due to the difference in device switching behavior, and (b) unbalance caused by the difference in gate signal delays between the system controller and the switches. Table 2.3-1 summarizes the main causes of unequal voltage distribution, where Δ represents the discrepancies between series connected devices.

2.3.2 Voltage Equalization for GCTs

(a) Static Voltage Equalization Figure 2.3-1a shows a commonly used method for static voltage equalization, where each switch is protected by a parallel

Table 2.3-1 Main Causes of Unequal Voltage Distribution Between Series Connected Devices

Type	Causes of Voltage Unbalance	
Static voltage unbalance	ΔI_{lk}—Device off-state leakage current ΔT_j—Junction temperature	
Dynamic voltage unbalance	Device	Δt_{don}—Turn-on delay time Δt_{doff}—Turn-off delay time (IGBT) Δt_s—Storage time (GCT) ΔQ_{rr}—Reverse recovery charge ΔT_j—Junction temperature
	Gate driver	Δt_{GDon}—Gate driver turn-on delay time Δt_{GDoff}—Gate driver turn-off delay time ΔL_{wire}—Wiring inductance between the gate driver output and the device gate

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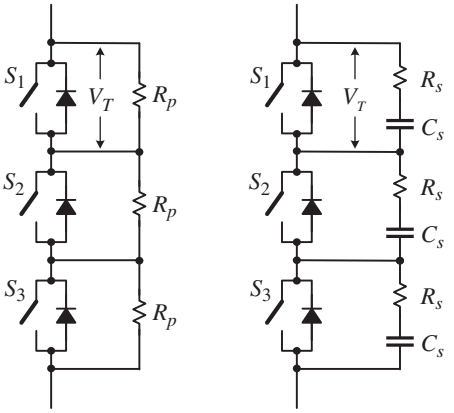

(a) Static voltage equalization (b) Dynamic voltage equalization

Figure 2.3-1 Passive voltage equalization techniques.

resistor R_p. Its resistance can be determined by an empirical equation

$$R_p = \frac{\Delta V_T}{\Delta I_{lk}} \tag{2.3-1}$$

where ΔV_T is the desired maximum voltage discrepancy between the series switches and ΔI_{lk} is the allowable tolerance for the off-state leakage current. Equation (2.3-1) is valid for both asymmetrical and symmetrical GCTs and the value of R_p is normally between 20 and 100 kΩ [11].

(b) Dynamic Voltage Equalization For the dynamic voltage equalization, three modes of GCT operation need to be considered:

- Turn-on transient
- Turn-off transient by gate commutation
- Turn-off transient by natural commutation (for symmetrical GCT only)

The first two operating modes are for asymmetrical and reverse-conducting GCTs used in voltage source converters while all three operating modes are applicable to symmetrical GCTs in current source converters.

To ensure equal dynamic voltage sharing, the following techniques can be employed:

- Use devices of one production lot to minimize Δt_{don}, Δt_s, and ΔQ_{rr}.
- Match the device switching characteristics to minimize Δt_{don}, Δt_s, and ΔQ_{rr}.
- Make the device cooling condition identical to minimize ΔT_j.

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- Design symmetrical gate drivers to minimize Δt_{GDon} and Δt_{GDoff}.
- Place the gate drivers symmetrically to minimize ΔL_{wire}.

The implementation of the above-mentioned techniques can help reduce the device voltage unbalance during switching transients, but does not guarantee a satisfactory result. The series connected switches are often protected by RC snubber circuits shown in Fig. 2.3-1b.

The snubber capacitor C_s should be sized to minimize the effect of the inconsistency in delay times on the GCT voltage equalization. Since the turn-on delay time t_{don} is normally much shorter than the storage time t_s at turn-off, the requirements for turn-off dominate. The value of C_s can be found from an empirical equation

$$C_s = \frac{\Delta t_{delay} \times I_{T\max}}{\Delta V_{T\max}} \tag{2.3-2}$$

where Δt_{delay} is the maximum tolerance in the total turn-off delay time including t_s and the delay time caused by gate drivers, $I_{T\max}$ is the maximum anode current to be commutated and $\Delta V_{T\max}$ is the maximum allowed voltage deviation between the series switches [11].

The GCTs used in the current source converter may be turned off by natural commutation, where the device is commutated by a negative anode current produced by its power circuit. The commutation process is similar to that of an SCR device. The dominant factor affecting the dynamic voltage unbalance in this case is the discrepancy in the GCT reverse recovery charge ΔQ_{rr}. This adds another criterion for choosing the capacitance value:

$$C_s = \frac{\Delta Q_{rr}}{\Delta V_{T\max}} \tag{2.3-3}$$

The value of C_s is normally in the range of 0.1–1 μF for the GCT devices, much lower than that for the GTOs. The snubber resistance R_s should be sized such that (a) it should be small enough to allow fast charging and discharging of the snubber capacitor to accommodate the short pulse-widths of the PWM operation, and (b) it should be large enough to limit the discharging current that flows through the GCT at turn-on. A good compromise should be made.

2.3.3 Voltage Equalization for IGBTs

The static and dynamic voltage equalization techniques for the GCT devices can be equally applied to the IGBTs. In addition, an active overvoltage clamping scheme can be implemented to limit the collector–emitter voltage during switching transients. This scheme is invalid for the GCTs due to the latching mechanism of the thyristor structure.

Figure 2.3-2 illustrates the principle of an active overvoltage clamping scheme [12]. The collector–emitter voltage v_{CE} of each IGBT is detected and compared with

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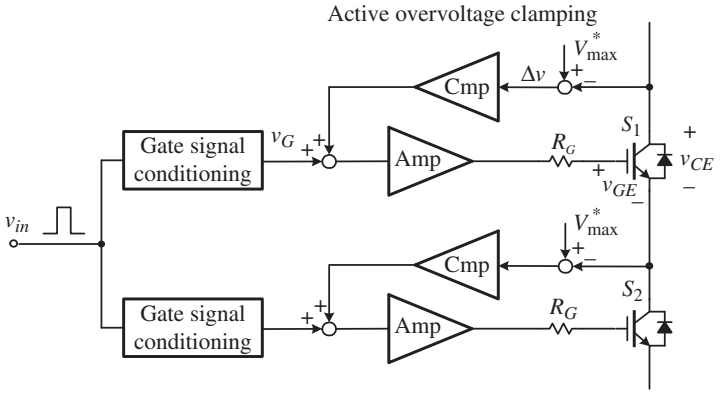

Figure 2.3-2 Principle of active overvoltage clamping for series connected IGBTs.

a reference voltage V^*_{max} that is the maximum allowed voltage for the device. The difference Δv is sent to a comparator. If the detected v_{CE} is lower than V^*_{max} at turn-off, the output of the comparator is zero and the operation of the device is not affected. At the moment that v_{CE} tends to exceed V^*_{max}, $|\Delta v|$ is added to the gate signal v_G, forcing v_{CE} to decrease. Through the feedback control in the IGBT active region, v_{CE} will not exceed the value set by V_{max} during switching transients, effectively protecting the device from overvoltage. However, this is achieved at the expense of an increase in the device switching loss.

2.4 SUMMARY

The chapter focuses on commonly used high-power semiconductor devices including SCRs, GTOs, GCTs, and IGBTs for medium voltage (MV) drives. Their switching characteristics are introduced and main specifications are discussed. Since these devices are often connected in series in the MV drives, the static and dynamic voltage equalization techniques are elaborated. To summarize, a qualitative comparison for the GTO, GCT, and IGBT devices is given below.

Item	GTO	GCT	IGBT
Maximum voltage and current ratings	High	High	Low
Packaging	Press pack	Press pack	Module or press pack
Switching speed	Slow	Moderate	Fast
Turn-on (*di/dt*) snubber	Required	Required	Not required
Turn-off (*dv/dt*) snubber	Required	Not required	Not required
Active overvoltage clamping	No	No	Yes

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Active *di/dt* and *dv/dt* control	No	No	Yes
Active short-circuit protection	No	No	Yes
On-state loss	Low	Low	High
Switching loss	High	Medium	Low
Behavior after destruction	Short circuited	Short circuited	Open circuited
Gate driver	Complex, separate	Complex, integrated	Simple, compact
Gate driver power consumption	High	Medium	Low

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Part Two

Multipulse Diode and SCR Rectifiers

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Chapter 3

Multipulse Diode Rectifiers

3.1 INTRODUCTION

In an effort to comply with the stringent harmonic requirements set by North American and European standards such as IEEE Standard 519-2014 [1], major high-power drive manufacturers around the world are increasingly using multipulse diode rectifiers in their drives as front-end converters [2–5]. The rectifiers can be configured as 12-, 18-, and 24-pulse rectifiers, powered by a phase-shifting transformer with a number of secondary windings. Each secondary winding feeds a **six-pulse diode rectifier**. The dc output of the six-pulse rectifiers is connected to a voltage source inverter.

The main feature of the multipulse rectifier lies in its ability to reduce the line current harmonic distortion. This is achieved by the phase-shifting transformer, through which some of the low-order harmonic currents generated by the six-pulse rectifiers are canceled in the transformer primary winding. In general, the higher the number of rectifier pulses, the lower the line current distortion is. The rectifiers with more than 30 pulses are seldom used in practice mainly due to the increased transformer costs and limited performance improvements.

The multipulse rectifier has a number of other features. It normally does not require any LC filters or power factor compensators, which leads to the elimination of possible LC resonances. The use of the phase-shifting transformer provides an effective means to block common-mode voltages generated by the rectifier and inverter in medium voltage drives, which would otherwise appear on motor terminals, leading to a premature failure of winding insulation if not mitigated [6, 7].

The multipulse diode rectifiers can be classified into two types:

- **Series-type multipulse rectifiers**, where all the six-pulse rectifiers are connected in series on their dc side. In the medium voltage drives, the series-type rectifier can be used as a front end for the inverter that requires a single dc supply such as three-level neutral-point clamped (NPC) inverter and multilevel flying-capacitor inverter [2–3].

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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- **Separate-type multipulse rectifiers**, where each of the six-pulse rectifiers feeds a separate dc load. This type of rectifier is suitable for use in a multilevel cascaded H-bridge inverter that requires multiple units of isolated dc supplies [4, 5].

This chapter starts with an introduction to the six-pulse diode rectifier, followed by a detailed discussion on the series- and separate-type multipulse rectifiers. The rectifier input power factor and line current THD are investigated, and results are summarized in a graphical format. The phase-shifting transformer required by the multipulse rectifiers will be dealt with in Chapter 5.

3.2 SIX-PULSE DIODE RECTIFIER

3.2.1 Introduction

The circuit diagram of a simplified six-pulse diode rectifier is shown in Fig. 3.2-1, where v_a, v_b, and v_c are the phase voltages of the utility supply. For medium voltage applications, each diode in the rectifier may be replaced by two or more diodes in series. To simplify the analysis, all the diodes are assumed to be ideal (no power losses or on-state voltage drop).

Figure 3.2-2 shows a set of voltage and current waveforms of the rectifier. The phase voltages of the utility supply are defined by

$$\begin{aligned} v_a &= \sqrt{2}\, V_{PH} \sin(\omega t) \\ v_b &= \sqrt{2}\, V_{PH} \sin(\omega t - 2\pi/3) \\ v_c &= \sqrt{2}\, V_{PH} \sin(\omega t - 4\pi/3) \end{aligned} \tag{3.2-1}$$

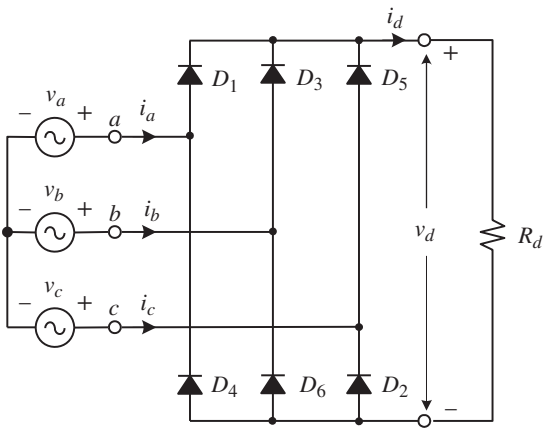

Figure 3.2-1 Six-pulse diode rectifier with a resistive load.

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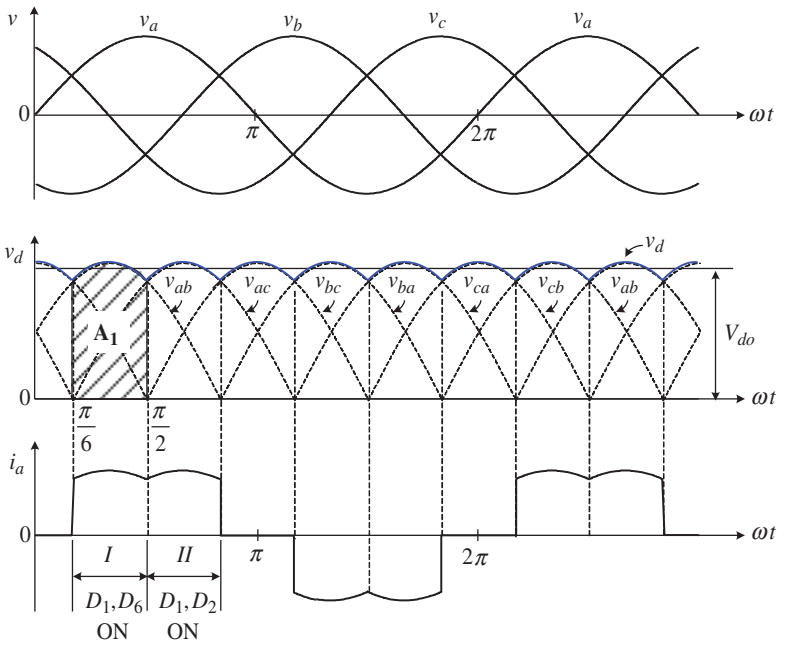

Figure 3.2-2 Waveforms of the six-pulse diode rectifier with a resistive load.

where V_{PH} is the rms value of the phase voltage and ω is the angular frequency of the utility supply, given by $\omega = 2\pi f$. The line-to-line voltage v_{ab} can be calculated by

$$v_{ab} = v_a - v_b = \sqrt{2}\, V_{LL} \sin(\omega t + \pi/6) \tag{3.2-2}$$

where V_{LL} is the rms value of the line-to-line voltage relating to the phase voltage by $V_{LL} = \sqrt{3}\, V_{PH}$.

The waveform of the line current i_a has two humps per half cycle of the supply frequency. During interval I, v_{ab} is higher than the other two line-to-line voltages. Diodes D_1 and D_6 are forward biased and thus turned on. The dc voltage v_d is equal to v_{ab} and the line current $i_a = v_{ab}/R_d$. During interval II, D_1 and D_2 conduct, and $i_a = v_{ac}/R_d$. Similarly, the waveform of i_a in the negative half cycle can be obtained. The other two line currents, i_b and i_c, have the same waveform as i_a, but lag i_a by $2\pi/3$ and $4\pi/3$, respectively.

The dc voltage v_d contains six pulses (humps) per cycle of the supply frequency. The rectifier is, therefore, commonly known as six-pulse rectifier. The average value of the dc voltage can be calculated by

$$V_{do} = \frac{\text{Area } \mathbf{A_1}}{\pi/3} = \frac{1}{\pi/3} \int_{\pi/6}^{\pi/2} \sqrt{2}\, V_{LL} \sin(\omega t + \pi/6)\, d(\omega t) = \frac{3\sqrt{2}}{\pi} V_{LL} \approx 1.35\, V_{LL} \tag{3.2-3}$$

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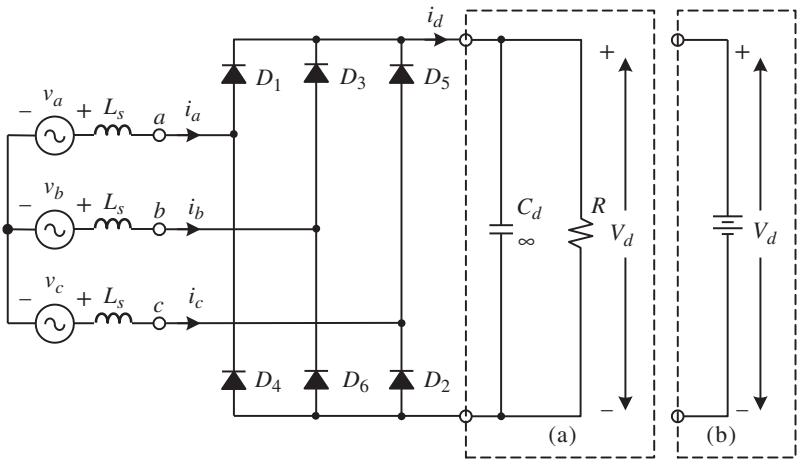

Figure 3.2-3 Six-pulse diode rectifier with a capacitive load.

3.2.2 Capacitive Load

Figure 3.2-3a shows the circuit diagram of the six-pulse rectifier with a capacitive load, where L_s represents the total line inductance between the utility supply and the rectifier, including the equivalent inductance of the supply, the leakage inductances of isolation transformer if any, and the inductance of a three-phase reactor that is often added to the system in practice for the reduction of the line current THD. The dc filter capacitor C_d is assumed to be sufficiently large such that the dc voltage is ripple free. Under this assumption, the capacitor and dc load can be replaced by a dc voltage source V_d as shown in Fig. 3.2-3b. The value of V_d varies slightly with the loading conditions. When the rectifier is lightly loaded, V_d is close to the peak of the ac line-to-line voltage and the dc current i_d may be discontinuous. With the increase in dc current, the voltage across the line inductance L_s increases, causing a reduction in V_d. When the dc current increases to a certain level, it becomes continuous and thus the rectifier operates in a continuous current mode.

(a) Discontinuous Current Operation Figure 3.2-4 illustrates the voltage and current waveforms of the rectifier operating under the light load conditions. Each of the three-phase line currents, i_a, i_b, and i_c, contains two pulses per half cycle of the supply frequency. The rectifier operates in a discontinuous current mode since the dc current i_d falls to zero six times per cycle of the supply frequency.

For the convenience of discussion, Fig. 3.2-5 shows the expanded voltage and current waveforms of the rectifier. During the interval $\theta_1 \leq \omega t < \theta_2$, the line-to-line voltage v_{ab} is higher than the dc voltage V_d, which turns D_1 and D_6 on. The dc current i_d increases from zero and energy is stored in L_s. At θ_2, v_{ab} is equal to V_d, the voltage across the line inductance L_s becomes zero, and i_d reaches its peak value I_p. When $\omega t > \theta_2$, v_{ab} is lower than V_d, and L_s starts to release its stored energy to the load

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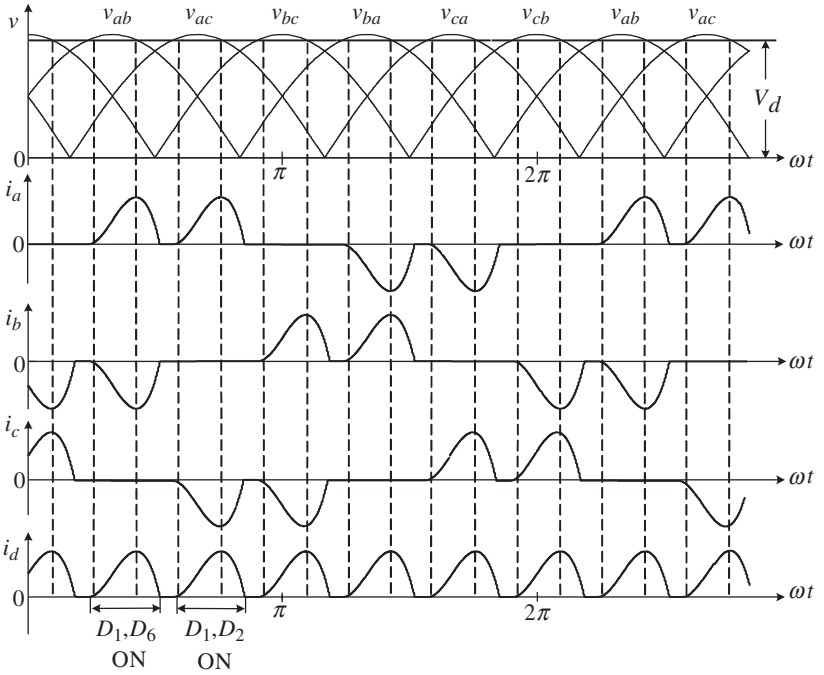

Figure 3.2-4 Discontinuous current operation of the six-pulse diode rectifier.

through D_1 and D_6. Both diodes remain on until $\omega t = \theta_3$, at which i_d falls to zero and the energy stored in L_s is completely discharged. During the interval $\theta_4 \leq \omega t < \theta_5$, v_{ac} is higher than V_d, D_1 is turned on again together with D_2. Obviously, each diode conducts twice per cycle of the supply frequency. The diode conduction angle is given by

$$\theta_c = 2(\theta_3 - \theta_1) \quad 0 \leq \quad \theta_c \leq 2\pi/3 \tag{3.2-4}$$

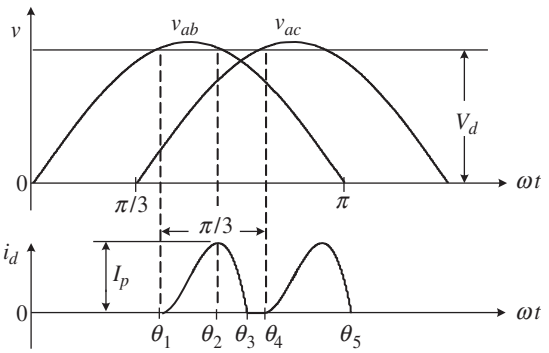

Figure 3.2-5 Details of the dc current waveform in the six-pulse diode rectifier.

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At θ_1 and θ_2, the line-to-line voltage v_{ab} is equal to V_d, from which

$$\theta_1 = \sin^{-1}\left(\frac{V_d}{\sqrt{2}\,V_{LL}}\right) \tag{3.2-5}$$

and

$$\theta_2 = \pi - \theta_1 \tag{3.2-6}$$

The total voltage across the two line inductances in phases a and b when D_1 and D_6 are on can be expressed as

$$2L_s\frac{d\,i_d}{dt} = v_{ab} - V_d \quad \text{for} \quad \theta_1 \le \omega t < \theta_3 \tag{3.2-7}$$

from which

$$\begin{aligned} i_d(\theta) &= \frac{1}{2\omega\,L_s}\int_{\theta_1}^{\theta}(\sqrt{2}\,V_{LL}\sin(\omega t) - V_d)\,d\,(\omega t) \\ &= \frac{1}{2\omega\,L_s}(\sqrt{2}\,V_{LL}(\cos\theta_1 - \cos\theta) + V_d(\theta_1 - \theta)) \end{aligned} \tag{3.2-8}$$

The peak dc current can be calculated by substituting θ_2 into (3.2-8)

$$\hat{I}_d = \frac{1}{2\omega\,L_s}(\sqrt{2}\,V_{LL}(\cos\theta_1 - \cos\theta_2) + V_d(\theta_1 - \theta_2)) \tag{3.2-9}$$

The average dc current can be calculated by

$$I_d = \frac{1}{\pi/3}\int_{\theta_1}^{\theta_3} i_d(\theta)\,d(\theta) \tag{3.2-10}$$

Substituting the condition of $i_d(\theta_3) = 0$ into (3.2-8) yields

$$\frac{V_d}{\sqrt{2}\,V_{LL}} = \frac{\cos\theta_3 - \cos\theta_1}{\theta_1 - \theta_3} \tag{3.2-11}$$

from which θ_3 can be calculated for a given V_{LL} and V_d. It is interesting to note that the angles θ_1, θ_2, and θ_3 are a function of V_{LL} and V_d only, irrelevant to the line inductance L_s.

(b) Continuous Current Operation As discussed earlier, the dc voltage V_d of the rectifier decreases with the increase in dc load current. The decrease in V_d makes θ_3 and θ_4 in Fig. 3.2-5 move toward each other. At the moment θ_3 and θ_4 start to overlap, the dc current i_d becomes continuous.

Figure 3.2-6 shows the waveforms of the rectifier operating with a continuous dc current. During interval *I*, a positive i_a keeps D_1 conducting while a negative i_c keeps D_2 on. The dc current is given by $i_d = i_a = -i_c$.

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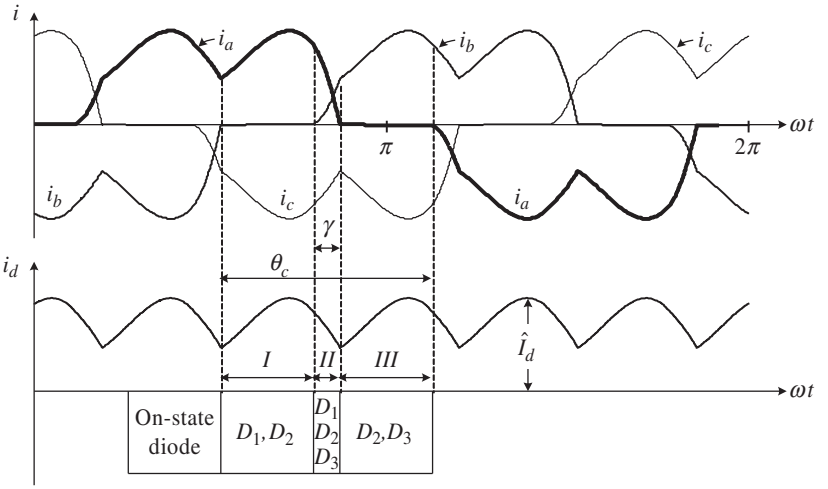

Figure 3.2-6 Current waveforms of the six-pulse diode rectifier operating in a continuous current mode.

Interval *II* is the commutation period, during which the current flowing in D_1 is commutated to D_3. The commutation is initiated by a forward biased voltage on D_3, which turns it on. Due to the presence of the line inductance L_s, the commutation process cannot complete instantly. It takes a finite moment for current i_b in D_3 to build up and current i_a in D_1 to fall. During this period, three diodes (D_1, D_2, and D_3) conduct simultaneously, and the dc current $i_d = i_a + i_b = -i_c$. The commutation period ends at the end of interval *II*, at which i_a decreases to zero and D_1 is turned off.

During interval *III*, diodes D_2 and D_3 are on, and the dc current is $i_d = i_b = -i_c$. The diode conduction angle θ_c is $2\pi/3 + \gamma$, where γ is the commutation interval. Compared with discontinuous current operation, the rectifier operating under a continuous current mode draws a line current with less harmonic distortion. The details of the line current distortion are discussed in the following sections.

3.2.3 Definition of THD and PF

Assume that the phase voltage v_a of the utility supply is sinusoidal

$$v_a = \sqrt{2}\, V_a \sin \omega t \tag{3.2-12}$$

The line current i_a drawn by a rectifier is generally periodical but non-sinusoidal. The line current can be expressed by a Fourier series

$$i_a = \sum_{n=1,2,3,\ldots}^{\infty} \sqrt{2}\, I_{an} (\sin(\omega_n t) - \phi_n) \tag{3.2-13}$$

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where n is the harmonic order, I_{an} and ω_n are the rms value and angular frequency of the nth harmonic current, and ϕ_n is the phase displacement between V_a and I_{an}, respectively.

The rms value of the distorted line current i_a can be calculated by

$$I_a = \left(\frac{1}{2\pi} \int_0^{2\pi} (i_a)^2 d(\omega t) \right)^{1/2} = \left(\sum_{n=1,2,3,\ldots}^{\infty} I_{an}^2 \right)^{1/2} \tag{3.2-14}$$

The total harmonic distortion is defined by

$$\text{THD} = \frac{\sqrt{I_a^2 - I_{a1}^2}}{I_{a1}} \tag{3.2-15}$$

where I_{a1} is the rms value of the fundamental current. The per-phase average power delivered from the supply to the rectifier is

$$P = \frac{1}{2\pi} \int_0^{2\pi} v_a \times i_a \, d(\omega t) \tag{3.2-16}$$

Substituting (3.2-12) and (3.2-13) into (3.2-16) yields

$$P = V_a I_{a1} \cos \phi_1 \tag{3.2-17}$$

where ϕ_1 is the phase displacement between V_a and I_{a1}. The per-phase apparent power is given by

$$S = V_a I_a \tag{3.2-18}$$

The input power factor is defined as

$$\text{PF} = \frac{P}{S} = \frac{I_{a1}}{I_a} \cos \phi_1 = \text{DF} \times \text{DPF} \tag{3.2-19}$$

where DF is the distortion factor and DPF is the displacement power factor, given by

$$\begin{cases} \text{DF} = I_{a1}/I_a \\ \text{DPF} = \cos \phi_1 \end{cases} \tag{3.2-20}$$

For a given THD and DPF, the power factor can also be calculated by

$$\text{PF} = \frac{\text{DPF}}{\sqrt{1 + \text{THD}^2}} \tag{3.2-21}$$

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3.2.4 Per Unit System

It is convenient to use per unit system for the analysis of power converter systems. Assume that the converter system under investigation is three-phase balanced with a rated apparent power S_R and rated line-to-line voltage V_{LL}. The base voltage, which is the rated phase voltage of the system, is given by

$$V_B = \frac{V_{LL}}{\sqrt{3}} \tag{3.2-22}$$

The base current and impedance are then defined as

$$I_B = \frac{S_R}{3V_B} \quad \text{and} \quad Z_B = \frac{V_B}{I_B} \tag{3.2-23}$$

The base frequency is

$$\omega_B = 2\pi f \tag{3.2-24}$$

where f is the nominal frequency of the utility supply or the rated output frequency of an inverter. The base inductance and capacitance can be found from

$$L_B = \frac{Z_B}{\omega_B} \quad \text{and} \quad C_B = \frac{1}{\omega_B Z_B} \tag{3.2-25}$$

Consider a three-phase diode rectifier rated at 4160 V, 60 Hz, and 2 MVA. The base current I_B of the rectifier is 277.6 A and the base inductance L_B is 22.9 mH. Assuming that the rectifier has a line inductance of 2.29 mH per phase and draws a line current of 138.8 A, the corresponding per unit value for the inductance and current is 0.1 per unit (pu) and 0.5 pu, respectively.

3.2.5 THD and PF of Six-Pulse Diode Rectifier

Two typical waveforms of the line current drawn by the six-pulse diode rectifier are shown in Fig. 3.2-7. When the rectifier operates under the light load conditions with the fundamental line current of $I_{a1} = 0.2$ pu, the line current waveform is somewhat spiky. The line current waveform contains two separate pulses per half cycle of the supply frequency, which makes the dc current discontinuous. With the rectifier operating at the rated condition ($I_{a1} = 1.0$ pu), the two current pulses are partially overlapped, which leads to a continuous dc current.

The harmonic spectrum of the line current waveform is shown in Fig. 3.2-7c. The line current i_a does not contain any even-order harmonics since the current waveform is of **half-wave symmetry**, defined by $f(\omega t) = -f(\omega t + \pi)$. It does not contain any triplen (**zero sequence**) harmonic currents either due to a balanced three-phase system. The dominant harmonics, such as the 5th and 7th, usually have a much

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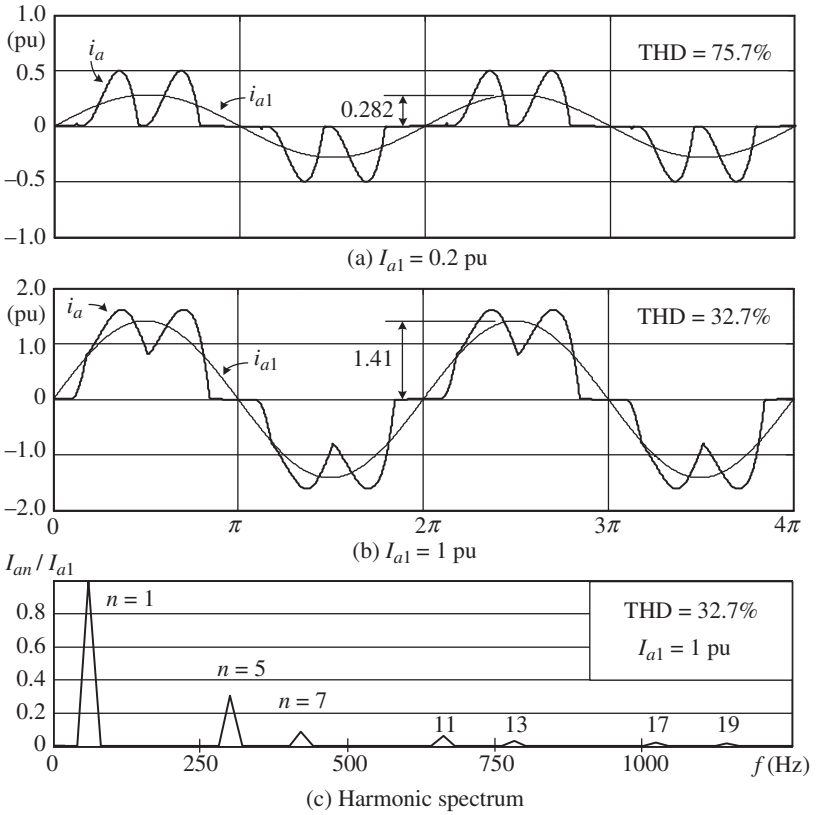

Figure 3.2-7 Line current waveform and harmonic content of the six-pulse diode rectifier (L_s = 0.05 pu).

higher magnitude than other harmonics. The line current THD is a function of the fundamental current I_{a1}, which is 75.7% at I_{a1} = 0.2 pu and 32.7% at I_{a1} = 1.0 pu.

The THD and PF curves of the six-pulse diode rectifier are shown in Fig. 3.2-8, where the fundamental current I_{a1} varies from 0.1 to 1 pu and the line inductance L_s changes from 0.05 pu to 0.15 pu. With the increase of the line current, its THD decreases while the overall power factor (PF) of the rectifier increases. The increase in PF is mainly due to the THD reduction which improves the distortion power factor of the rectifier. For a given value of THD and PF, the displacement power factor DPF can be found from

$$\text{DPF} = \cos\phi_1 = \text{PF}\sqrt{1 + \text{THD}^2} \tag{3.2-26}$$

As mentioned earlier, the low-order dominant harmonics in the line current are of high magnitude. An effective approach to reducing the line current THD is, therefore, to remove these dominant harmonics from the system. This can be achieved by using multipulse rectifiers.

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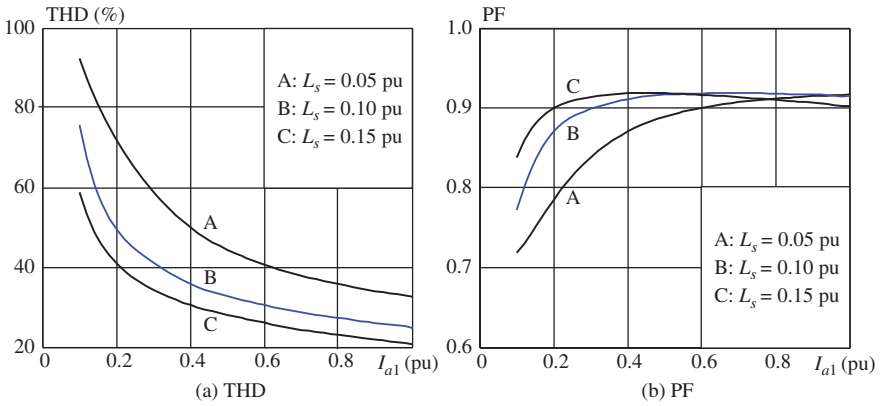

Figure 3.2-8 Calculated THD and PF of the six-pulse diode rectifier.

3.3 SERIES-TYPE MULTIPULSE DIODE RECTIFIERS

In this section, the configuration of 12-, 18-, and 24-pulse series-type rectifiers is introduced. The THD and PF performance of these rectifiers are investigated through simulation and experiments.

3.3.1 12-Pulse Series-Type Diode Rectifier

(a) Rectifier Configuration Figure 3.3-1 shows the typical configuration of a 12-pulse series-type diode rectifier. There are two identical six-pulse diode rectifiers powered by a phase-shifting transformer with two secondary windings. The dc outputs of the six-pulse rectifiers are connected in series. To eliminate low-order harmonics in the line current i_A, the line-to-line voltage v_{ab} of the wye-connected secondary winding is in phase with the primary voltage v_{AB} while the delta-connected secondary winging voltage $v_{\tilde{a}\tilde{b}}$ leads v_{AB} by

$$\delta = \angle v_{\tilde{a}\tilde{b}} - \angle v_{AB} = 30° \tag{3.3-1}$$

The rms line-to-line voltage of each secondary winding is

$$V_{ab} = V_{\tilde{a}\tilde{b}} = V_{AB}/2 \tag{3.3-2}$$

from which the turns ratio of the transformer can be determined by

$$\frac{N_1}{N_2} = 2 \quad \text{and} \quad \frac{N_1}{N_3} = \frac{2}{\sqrt{3}} \tag{3.3-3}$$

The inductance L_s represents the total line inductance between the utility supply and the transformer, and L_{lk} is the total leakage inductance of the transformer referred

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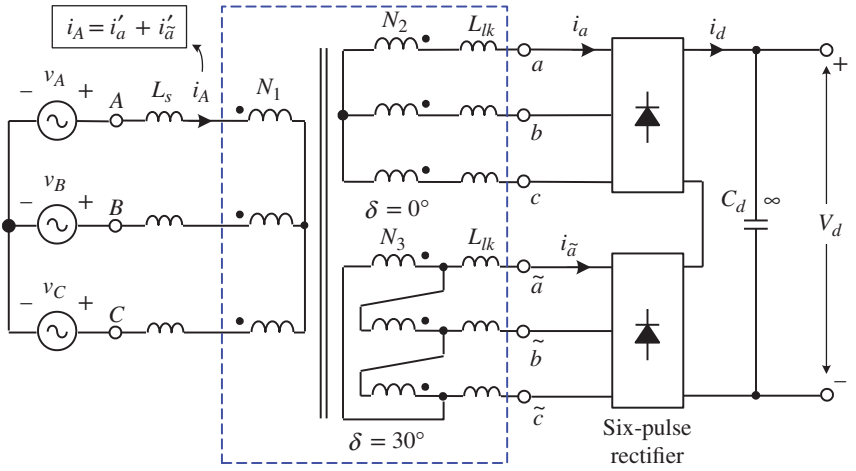
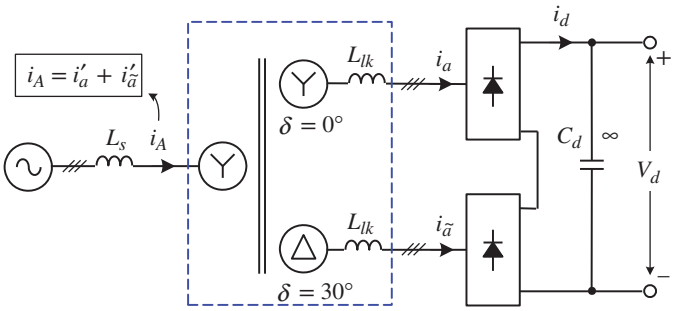


(a) 12-pulse diode rectifier



(b) Simplified diagram

Figure 3.3-1 12-pulse series-type diode rectifier.

to the secondary side. The dc filter capacitor C_d is assumed to be sufficiently large such that the dc voltage V_d is ripple free.

Figure 3.3-1b shows the simplified diagram of the 12-pulse diode rectifier. The transformer winding is represented by the sign "Y" or "Δ" enclosed by a circle, where "Y" denotes a three-phase wye-connected winding while "Δ" represents a delta-connected winding.

(b) Current Waveforms Figure 3.3-2 shows a set of simulated current waveforms of the rectifier operating under the rated conditions. The line inductance L_s is assumed to be zero, and the total leakage inductance L_{lk} is 0.05 pu, which is a typical value for a phase-shifting transformer.

The dc current i_d is continuous, containing 12 pulses per cycle of the supply frequency. At any time instant (excluding commutation intervals), the dc current i_d flows through four diodes simultaneously, two in the top six-pulse rectifier and

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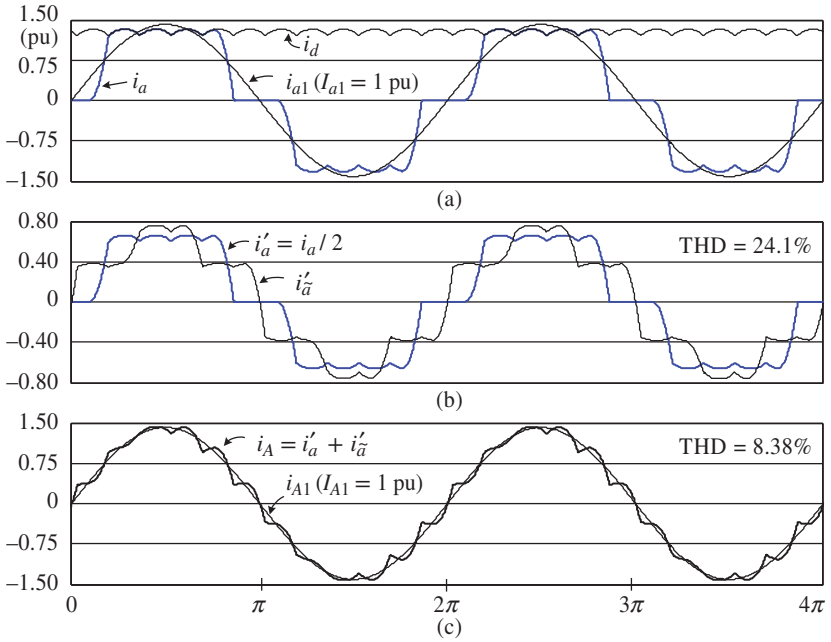

Figure 3.3-2 Current waveforms in the 12-pulse series-type rectifier ($L_s = 0$, $L_{lk} = 0.05$ pu, and $I_{A1} = 1.0$ pu).

two in the bottom rectifier. The dc current ripple is relatively low due to the series connection of the two six-pulse rectifiers, where the leakage inductances of the secondary windings can be considered in series.

The waveform of the line current i_a in the wye-connected secondary winding looks like a trapezoidal wave with four humps on the top. The waveform of $i_{\tilde{a}}$ in the delta-connected winding is identical to $i_a$except for a 30° phase displacement and therefore not shown in the figure.

The currents i'_a and $i'_{\tilde{a}}$ in Fig. 3.3-2b are the secondary line currents i_a and $i_{\tilde{a}}$ referred to the primary side. Since both primary and top secondary windings are connected in wye, the waveform of the referred current i'_a is identical to that of i_a except that its magnitude is halved due to the turns ratio of the two windings. When $i_{\tilde{a}}$ is referred to the primary side, the referred current $i'_{\tilde{a}}$ does not keep the same waveform as $i_{\tilde{a}}$. The changes in waveform are caused by the phase displacement of the harmonic currents when they are referred from the delta-connected secondary winding to the wye-connected primary winding. It is the phase displacement that makes certain harmonics, such as the 5th and 7th, in $i'_{\tilde{a}}$ out of phase with those in i'_a. As a result, these harmonic currents are canceled in the transformer primary winding and do not appear in the primary line current, given by

$$i_A = i'_a + i'_{\tilde{a}} \tag{3.3-4}$$

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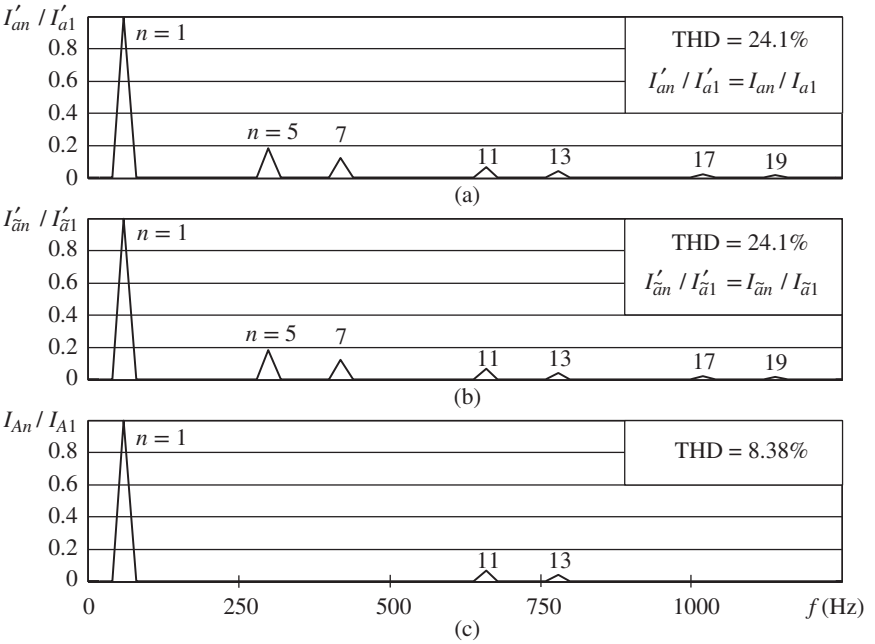

Figure 3.3-3 Harmonic spectrum of the current waveforms in Fig. 3.3-2.

Figure 3.3-3 shows the harmonic spectrum of the rectifier currents in Fig. 3.3-2, where I'_{an}, $I'_{\tilde{a}n}$, and I_{An} are the nth order harmonic currents (rms) in i'_a, $i'_{\tilde{a}}$, and i_A, respectively. The harmonic content of the referred currents i'_a and $i'_{\tilde{a}}$ is identical although their waveforms are quite different. This is understandable since the harmonic content should not alter when a secondary current is referred to the primary side. The magnitude of the 5th and 7th harmonics is 18.6% and 12.4%, respectively, which are much higher than other harmonics. The THD of the primary line current i_A is only 8.38% in comparison to 24.1% of the secondary line current i_a. The substantial reduction in THD is owing to the elimination of dominant harmonics by the phase-shifting transformer. The principle of harmonic cancellation by the phase-shifting transformer will be discussed in Chapter 5.

Figure 3.3-4 shows the waveforms measured from a 12-pulse diode rectifier operating under the rated conditions. The phase-shifting transformer has a total leakage inductance of 0.045 pu and a voltage ratio of $V_{AB}/V_{ab} = V_{AB}/V_{\tilde{a}\tilde{b}} = 2.05$. The line inductance L_s between the utility supply and the transformer is negligible due to the high capacity of the supply and low power rating of the rectifier.

The measured secondary currents, i_a and $i_{\tilde{a}}$, are of a quasi-trapezoidal wave with a 30° phase displacement. The harmonic spectrum in Fig. 3.3-4b indicates that $i_a$and $i_{\tilde{a}}$ contain the 5th and 7th harmonics, but these harmonics are canceled by the transformer and do not appear in i_A. It should be pointed out that the amplitudes of

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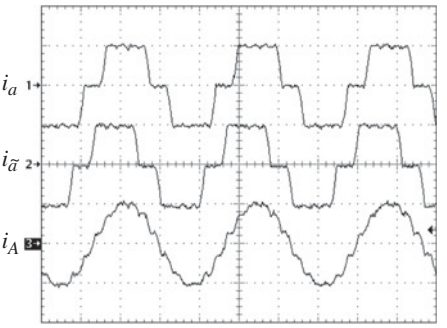
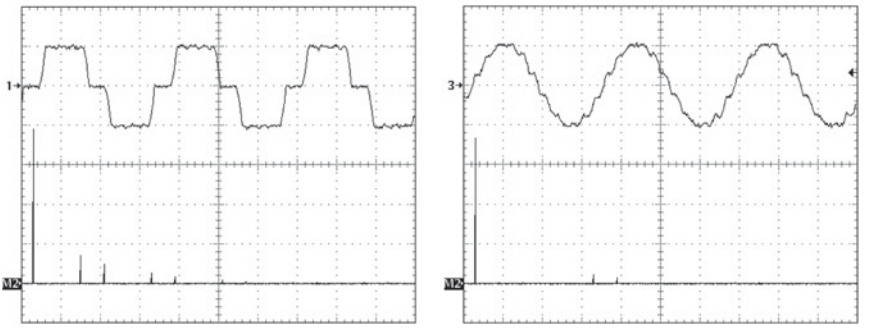


(a) Currents: $\sqrt{2}$ pu/div, 5 ms/div



(b) Spectrum: $\sqrt{2}/5$ pu/div, 200 Hz/div

Figure 3.3-4 Measured waveforms and harmonic spectrum of the 12-pulse series-type rectifier.

the fundamental component in i_a and i_A are not exactly the same. They would be identical if the voltage ratio of the transformer were equal to 2 instead of 2.05.

(c) THD and Power Factor Figure 3.3-5 shows the calculated line current THD and input power factor versus the fundamental line current I_{A1}. The leakage inductance L_{lk} is typically 0.05 pu while the line inductance L_s usually varies with the capacity and operating conditions of the power system. To investigate the effect of L_s, three typical values, $L_s = 0$, 0.05, and 0.1 pu, are selected. The THD of the line current i_A decreases with the increase of I_{A1} and L_s. Compared with the six-pulse rectifier, the 12-pulse rectifier can achieve a substantial reduction in the line current THD. Its input power factor PF is also improved thanks to the lower line current THD and higher displacement power factor.

Generally speaking, the THD of the line current in the 12-pulse rectifier does not meet the harmonic requirements set by IEEE Standard 519-2014 except when it operates under the rated conditions with $L_s \geq 1.0$ pu. In practice, a high value of $L_s$can be obtained by adding a three-phase line reactor or designing the phase-shifting transformer with a high leakage inductance.

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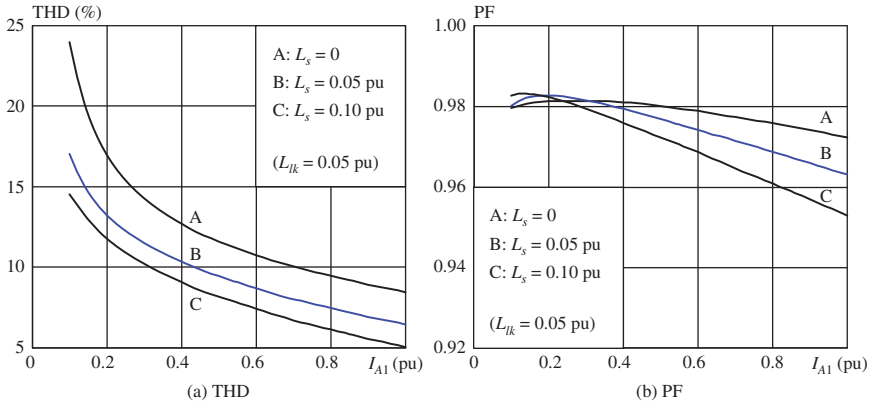

Figure 3.3-5 Line current THD and PF of the 12-pulse series-type diode rectifier.

3.3.2 18-Pulse Series-Type Rectifier

(a) Rectifier Configuration The block diagram of an 18-pulse series-type diode rectifier is shown in Fig. 3.3-6. The rectifier has three units of identical six-pulse diode rectifiers fed by a phase-shifting transformer. The sign "Z" enclosed by a circle represents a three-phase zigzag-connected winding, which provides a required phase displacement δ between the primary and secondary line-to-line voltages. The detailed analysis of the zigzag transformer is given in Chapter 5.

The 18-pulse rectifier is able to eliminate four dominant harmonics (the 5th, 7th, 11th, and 13th). This can be achieved by employing a phase-shifting transformer with a 20° phase displacement between any two adjacent secondary windings. The typical values of δ are 20°, 0°, and −20° for the top, middle, and bottom secondary windings, respectively. Other arrangements for δ are possible, such as $\delta = 0°$, 20°, and 40°. The turns ratio of the transformer for the 18-pulse rectifier is usually selected

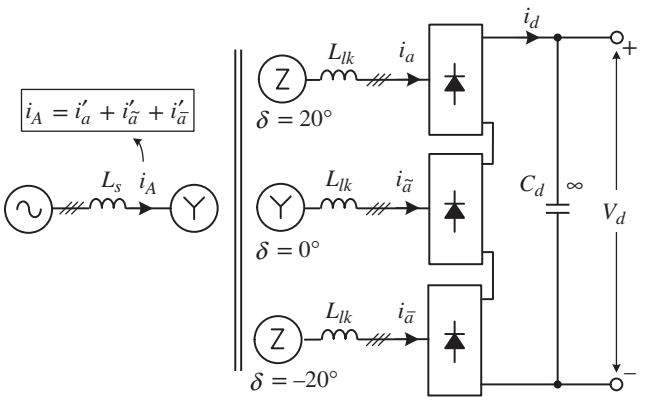

Figure 3.3-6 18-pulse series-type diode rectifier.

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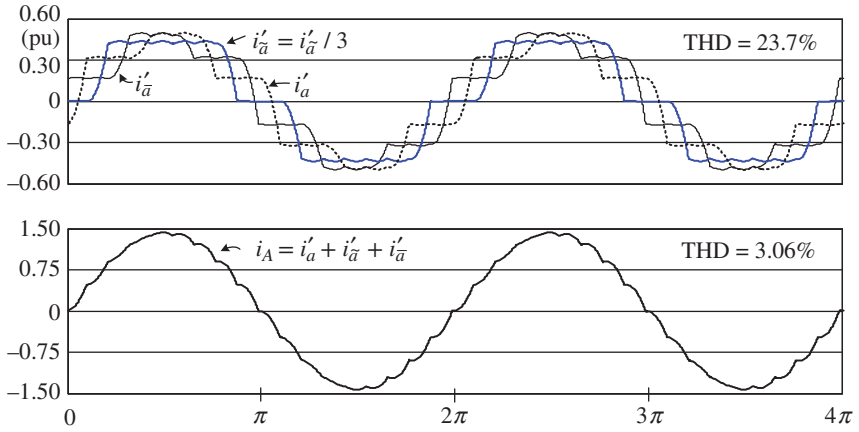

(a) Waveforms

Harmonics (n)	5	7	11	13	17	19	23	25	THD (%)
I'_{an} / I'_{a1} (%)	18.5	12.2	6.11	4.35	2.14	2.05	0.98	0.84	23.7
I_{An} / I_{A1} (%)	0	0	0	0	2.14	2.05	0	0	3.06

(b) Harmonic content

Figure 3.3-7 Current waveforms in the 18-pulse series-type rectifier ($L_s = 0$, $L_{lk} = 0.05$ pu, and $I_{A1} = 1.0$ pu).

such that the line-to-line voltage of each secondary winding is one-third that of the primary winding.

(b) Waveforms Assume that the 18-pulse diode rectifier in Fig. 3.3-6 operates under the rated conditions with $L_s = 0$ and $L_{lk} = 0.05$ pu. A set of simulated waveforms for the rectifier are shown in Fig. 3.3-7, where i'_a, $i'_{\tilde{a}}$, and $i'_{\bar{a}}$ are the primary current components referred from the secondary side of the transformer. The waveforms of these currents are all different due to the phase displacement of harmonic currents when they are transferred from the secondary to the primary winding. The waveforms of the secondary currents i_a, $i_{\tilde{a}}$, and $i_{\bar{a}}$ are not given in the figure, but they can be easily obtained by referencing to the $i'_{\tilde{a}}$ waveform. The harmonic content of the primary and secondary line currents are given in Fig. 3.3-7b. The secondary line current has a THD of 23.7% while the THD of the primary line current is only 3.06% due to the elimination of four dominant harmonics.

The waveforms measured from an 18-pulse diode rectifier under the rated operating conditions are shown in Fig. 3.3-8. The phase-shifting transformer has a leakage inductance of 0.05 pu and a voltage ratio of $V_{AB}/V_{ab} = V_{AB}/V_{\tilde{a}\tilde{b}} = V_{AB}/V_{\bar{a}\bar{b}} = 2.95$. The waveforms of the secondary line currents i_a, $i_{\tilde{a}}$, and $i_{\bar{a}}$ has a 20° phase

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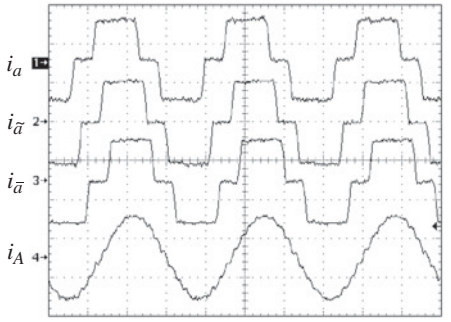
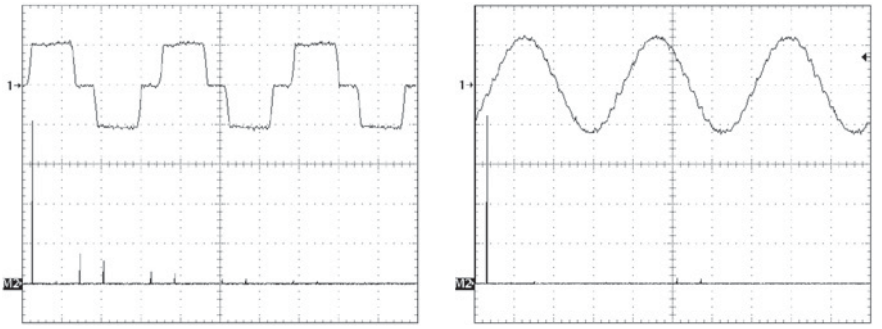
displacement between each other. The harmonic spectrum illustrates that the primary line current i_A does not contain the 5th, 7th, 11th, or 13th harmonics, and therefore is nearly sinusoidal.

(c) Line Current THD and Input Power Factor Figure 3.3-9 shows the calculated primary line current THD and input power factor of the 18-pulse diode rectifier. Compared with the 12-pulse rectifier, the 18-pulse rectifier has lower line current THD and better power factor. For instance, when the 18-pulse operates under the rated load conditions ($I_{A1} = 1$ pu) with $L_s = 0.05$ pu, the THD of i_A is reduced from 6.4% of the 12-pulse rectifier to 2.3% and the power factor is slightly increased as well.



(a) Currents: $\sqrt{2}$ pu/div, 5 ms/div



(b) Spectrum: $\sqrt{2}/5$ pu/div, 200 Hz/div

Figure 3.3-8 Waveforms and harmonic spectrum measured from an 18-pulse series-type rectifier.

3.3.3 24-Pulse Series-Type Rectifier

The configuration of a 24-pulse series-type diode rectifier is shown in Fig. 3.3-10, where a phase-shifting transformer is used to power four sets of six-pulse diode rectifiers. To eliminate six dominant current harmonics (the 5th, 7th, 11th, 13th,

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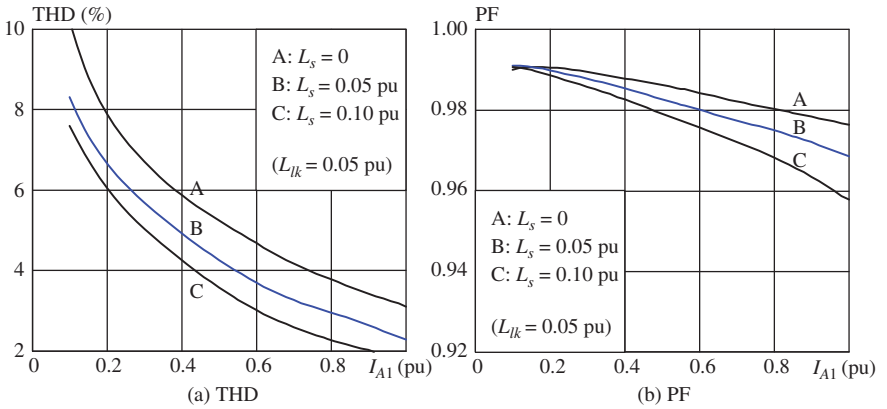

Figure 3.3-9 THD and PF of the 18-pulse series-type diode rectifier.

17th, and 19th), the transformer should be arranged such that there is a 15° phase displacement between the voltages of any two adjacent secondary windings. The line-to-line voltage of each secondary winding is usually one-fourth that of the primary winding.

Figure 3.3-11 shows the current waveforms in the 24-pulse diode rectifier operating under rated conditions, where i'_a, $i'_{\tilde{a}}$, $i'_{\bar{a}}$, and $i'_{\bar{\tilde{a}}}$ are the primary currents referred from the secondary side of the transformer. Each of these currents has a THD of 24%. The primary line current i_A is virtually a sinusoid with only 1.49% total harmonic distortion.

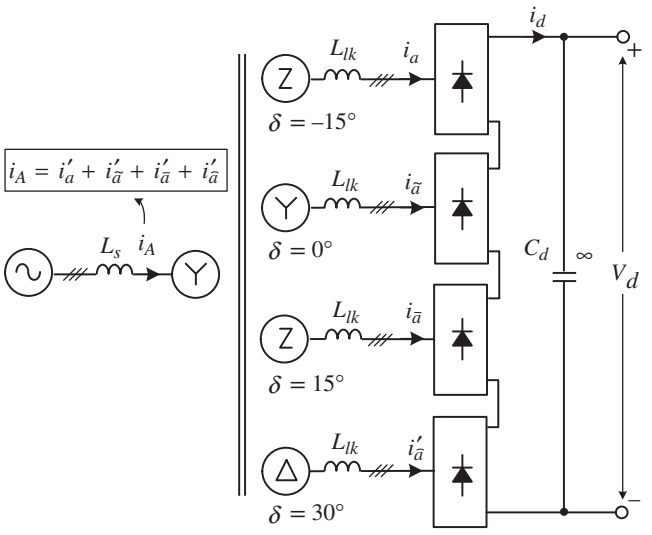

Figure 3.3-10 24-pulse series-type diode rectifier.

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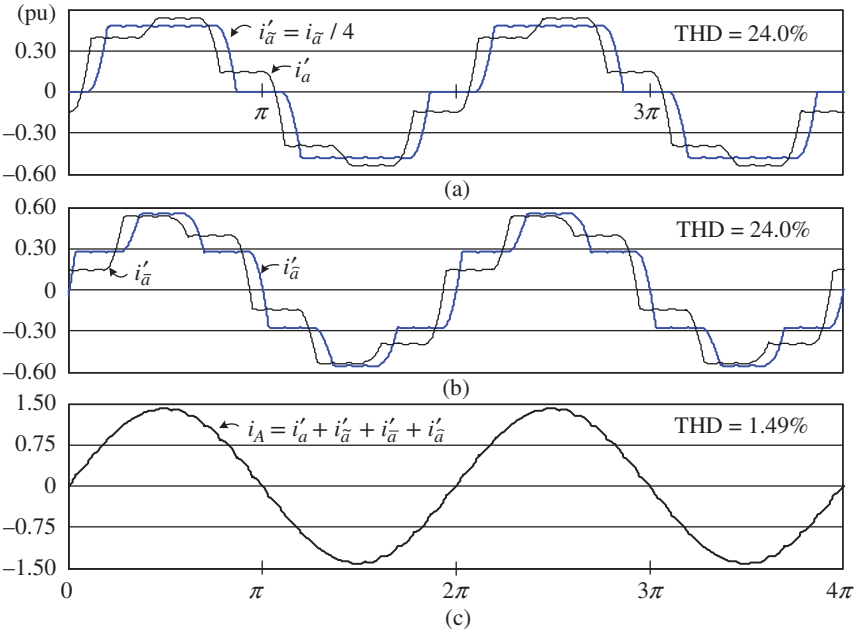

Figure 3.3-11 Current waveforms in the 24-pulse series-type rectifier ($L_s = 0, L_{lk} = 0.05$ pu, and $I_{A1} = 1.0$ pu).

The calculated THD of the line current i_A and input power factor of the 24-pulse rectifier is shown in Fig. 3.3-12. It can be observed that the rectifier has an excellent THD profile, which meets the harmonic requirements specified by IEEE Standard 519-2014.

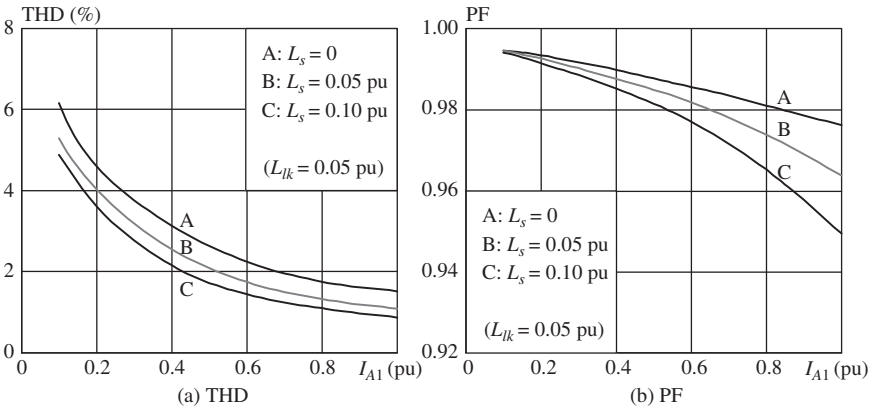

Figure 3.3-12 THD and PF of the 24-pulse series-type diode rectifier.

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3.4 SEPARATE-TYPE MULTIPULSE DIODE RECTIFIERS

In the previous section, we have discussed series-type multipulse diode rectifiers, where the dc outputs of all the six-pulse diode rectifiers are connected in series. This section focuses on the separate-type multipulse rectifiers, where each of its six-pulse rectifiers feeds a separate dc load.

3.4.1 12-Pulse Separate-Type Diode Rectifier

The block diagram of a 12-pulse separate-type diode rectifier is shown in Fig. 3.4-1. The rectifier configuration is essentially the same as that of the 12-pulse series-type rectifier except that two separate dc loads are employed instead of a single dc load.

Figure 3.4-2 illustrates an application example of the 12-pulse separate-type rectifier as a front end for a cascaded H-bridge multilevel inverter fed drive. The phase-shifting transformer has six secondary windings, of which three are wye-connected with $\delta = 0$ and the other three are delta-connected having a δ of 30°. Each of the secondary windings feeds a six-pulse diode rectifier. Since all wye-connected secondary windings are identical and so are the delta-connected windings, it is essentially a 12-pulse transformer. The six-pulse rectifiers provide isolated dc voltages to H-bridge inverters, whose outputs are connected in cascade, providing a three-phase ac voltage to the motor.

Figure 3.4-3 shows the current waveforms of the 12-pulse separate-type rectifier operating with a rated line current. The waveform of the secondary line current i_a is similar to that of the "stand-alone" six-pulse rectifier. This is due to the fact that with the line inductance L_s assumed to be zero, the 12-pulse rectifier is essentially composed of two units of stand-alone six-pulse rectifiers. The dc currents in the two rectifiers, i_d and $i_{\tilde{d}}$, contain a higher ripple component compared with that in the 12-pulse series-type rectifier where the dc load sees the leakage inductances of the two secondary windings in series.

The currents i'_a and $i'_{\tilde{a}}$ are the secondary line currents i_a and $i_{\tilde{a}}$ referred to the primary side. For the reasons discussed earlier, the 5th and 7th harmonic currents in

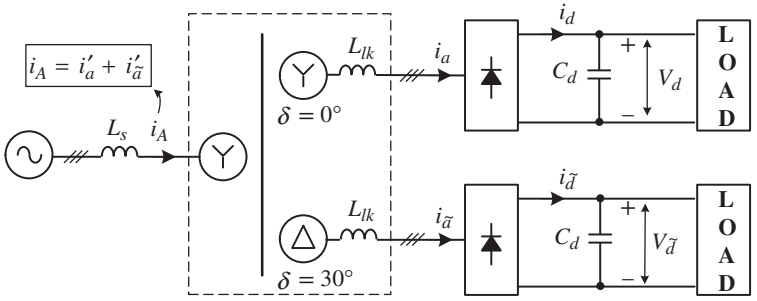

Figure 3.4-1 12-pulse separate-type diode rectifier.

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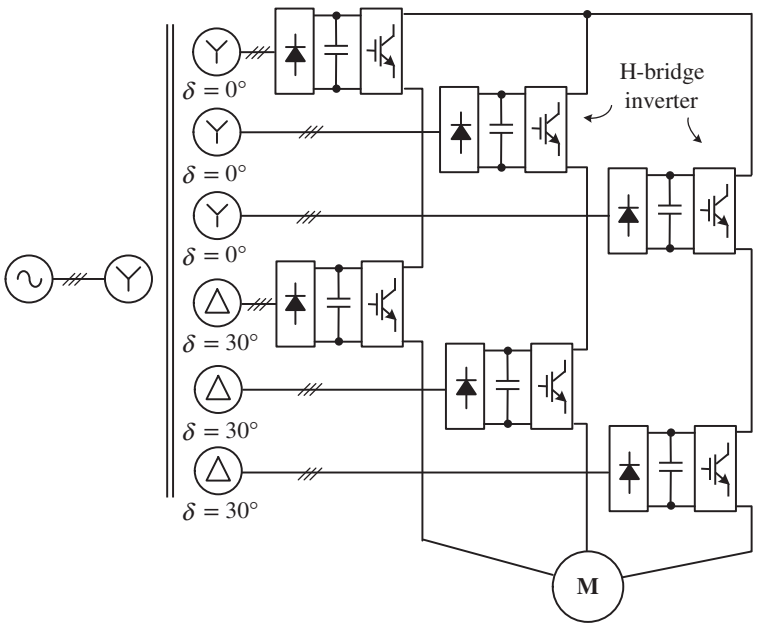

Figure 3.4-2 Application of the 12-pulse separate-type diode rectifier in a cascaded H-bridge multilevel inverter fed drive.

i'_a and $i'_{\tilde{a}}$ are out of phase and therefore are cancelled in the primary winding of the transformer.

It is interesting to note that although the waveforms of i_a and $i_{\tilde{a}}$ differ significantly from those in 12-pulse series-type rectifier, the primary line current i_A in both rectifiers has a similar waveform, and so is its THD. This is mainly due to the 12-pulse configuration, where the two most detrimental harmonics, the 5th and 7th, are eliminated. The remaining harmonics have less influence on the line current waveform and its THD.

Figure 3.4-4 shows the waveforms measured from a 12-pulse separate-type rectifier operating under the rated conditions. The phase-shifting transformer has a leakage inductance of 0.045 pu and a voltage ratio of $V_{AB}/V_{ab} = V_{AB}/V_{\tilde{a}\tilde{b}} = 2.05$. The waveform of the secondary line currents, i_a and $i_{\tilde{a}}$, has two humps per half cycle while the primary current i_A is close to sinusoidal due to the elimination of the 5th and 7th harmonics as shown in Fig. 3.4-4b.

The THD of the line current i_A in the 12-pulse separate-type rectifier shown in Fig. 3.4-5 is somewhat lower than that of the series type. This is mainly due to the differences in harmonic distribution. The secondary line currents in the separate-type rectifier contain higher 5th and 7th harmonics but lower 11th and 13th harmonics than those in the series type. When they are reflected to the primary side, the 5th and 7th harmonics are cancelled, and thus the lower magnitude of the 11th and 13th harmonics makes a reduction in the line current THD.

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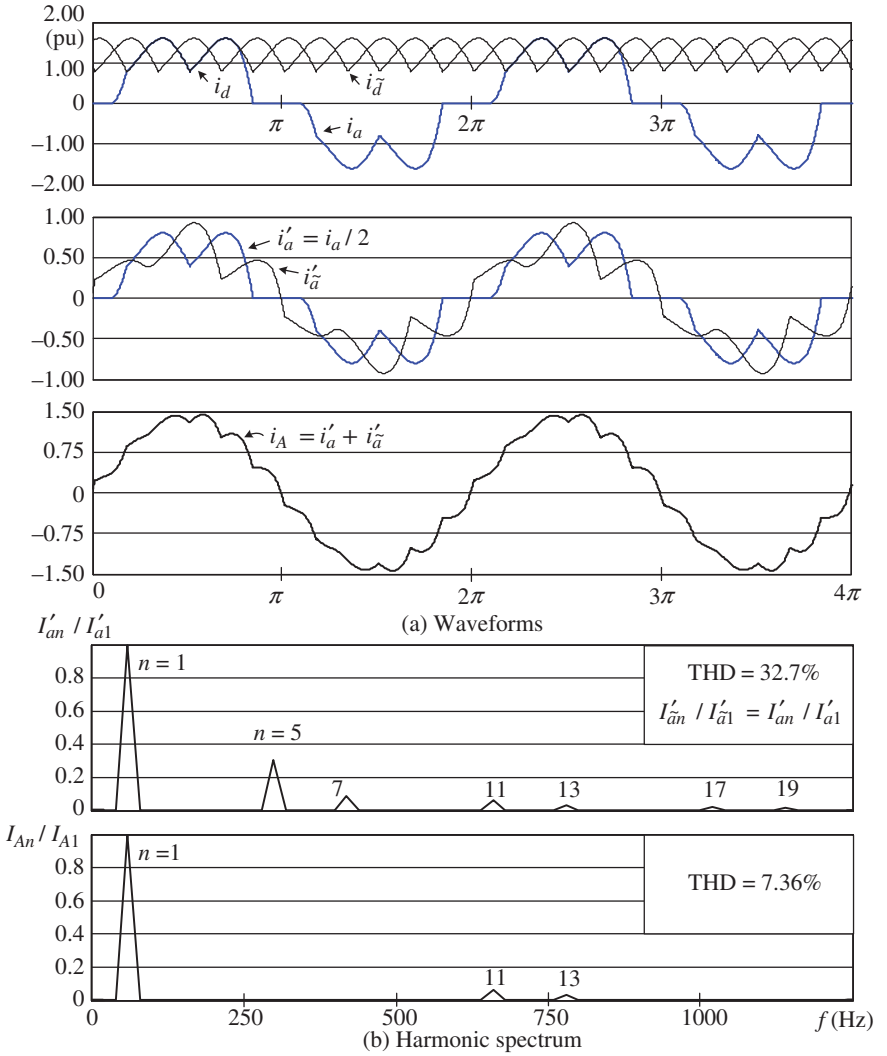

Figure 3.4-3 Current waveforms in the 12-pulse separate-type rectifier ($L_s = 0, L_{lk} = 0.05$ pu, and $I_{A1} = 1.0$ pu).

The power factor profile is also different from that of the 12-pulse series-type rectifier. A notch occurs approximately at $I_{A1} = 0.22$ pu, which signifies a boundary between continuous and discontinuous current operation of the rectifier. The discontinuous current operation normally does not occur in the series-type rectifiers since the dc load sees the leakage inductances of the secondary windings in series, making the dc current continuous over almost the full operation range. The power factor of the separate type is slightly lower than the series type. This is mainly due to the dc load connection, which affects the equivalent inductance seen by the utility supply.

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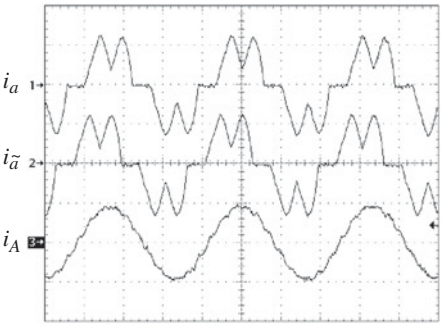
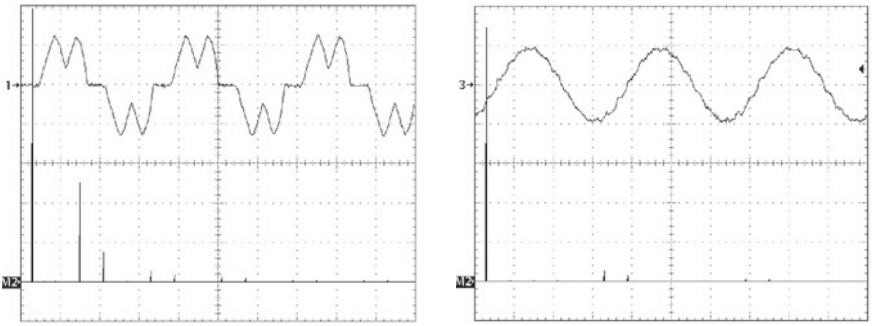


(a) Currents: $\sqrt{2}$ pu/div, 5 ms/div



(b) Spectrum: $\sqrt{2}/10$ pu/div, 200 Hz/div

Figure 3.4-4 Waveforms and harmonic spectrum measured from a 12-pulse separate-type rectifier.

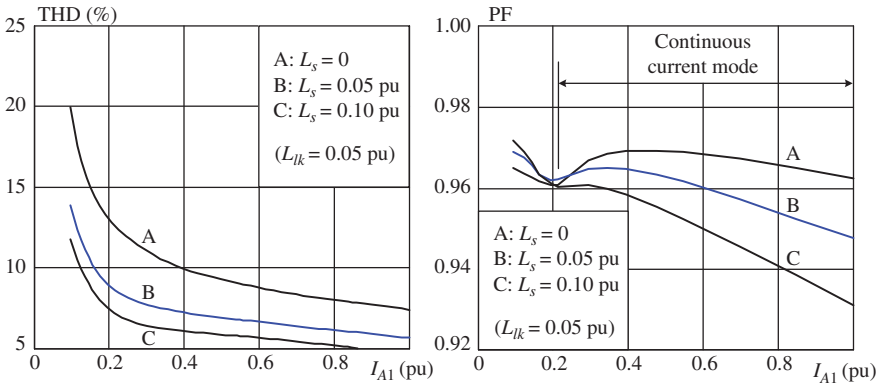


Figure 3.4-5 THD and PF of the 12-pulse separate-type diode rectifier.

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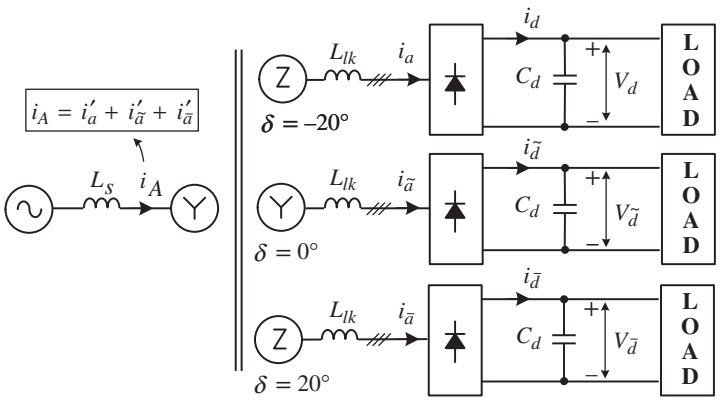

Figure 3.4-6 18-pulse separate-type diode rectifier.

3.4.2 18- and 24-Pulse Separate-Type Diode Rectifiers

The configuration of the 18-pulse separate-type diode rectifier is shown in Fig. 3.4-6. It is essentially the same as that of 18-pulse series-type except the dc side connection.

The waveforms of the 18-pulse separate-type rectifier are shown in Fig. 3.4-7. Due to the elimination of four dominant harmonics, the line current i_A is close to sinusoidal with THD of 3.05%.

Figure 3.4-8 shows the line current THD and input power factor of the 18- and 24-pulse separate-type rectifiers, respectively. In general, the THD profile of the separate-type rectifiers is somewhat better and the power factor profile is slightly worse than their series-type counterparts.

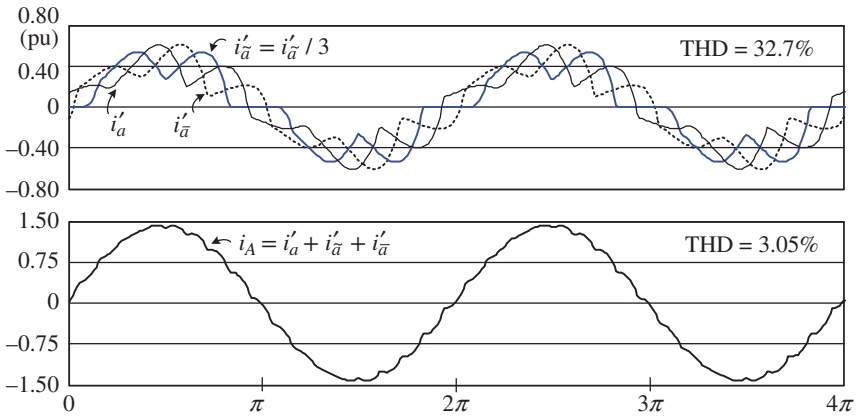

Figure 3.4-7 Current waveforms in the 18-pulse separate-type rectifier ($L_s = 0, L_{lk} = 0.05$ pu, and $I_{A1} = 1.0$ pu).

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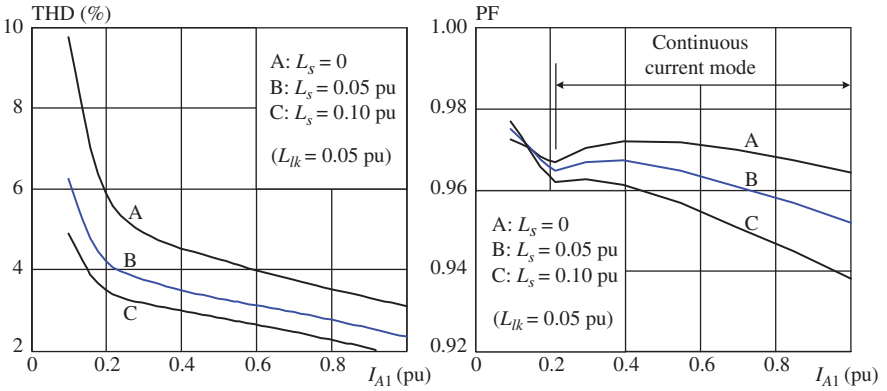
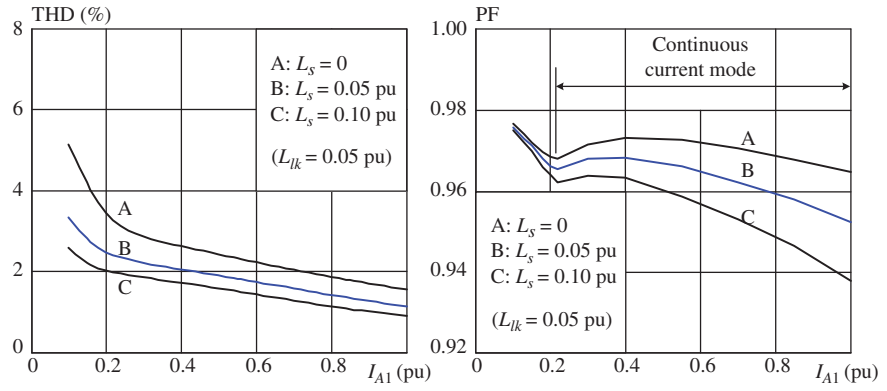


(a) 18-pulse rectifier



(b) 24-pulse rectifier

Figure 3.4-8 THD and PF of the 18- and 24-pulse separate-type diode rectifiers.

3.5 SUMMARY

This chapter provides a comprehensive analysis on the multipulse diode rectifiers widely used in high-power medium voltage drives as front-end converters. The main issues discussed in the chapter are summarized below.

- **Systematic analysis of 12-, 18-, and 24-pulse diode rectifiers.** The line current THD and input power factor of the multipulse rectifiers are analyzed. The line current THD of the 12-pulse diode rectifiers normally do not satisfy the harmonic requirements specified by IEEE Standard 519-2014. The 18-pulse rectifiers have a better harmonic profile while the 24-pulse rectifiers provide excellent harmonic performance. The input power factor of the multipulse rectifiers is also analyzed. Rectifiers with more than 30 pulses are seldom used in practice mainly due to the increased transformer costs and limited performance improvements.

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- **Comparison between series- and separate-type rectifiers.** The multipulse rectifiers can be classified into series and separate types for various multilevel voltage source inverters. In general, the line current THD profile of the separate-type rectifiers is a little better and the input power factor is a little worse than their series-type counterparts.

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Chapter 4

Multipulse SCR Rectifiers

4.1 INTRODUCTION

The multipulse diode rectifiers presented in the preceding chapter are normally used in voltage source inverter (VSI) fed drives while the multipulse SCR rectifiers to be discussed in this chapter can be used in current source inverter (CSI) based drives. The SCR rectifier provides an adjustable dc current for the CSI which converts the dc current to a three-phase PWM ac current with variable magnitude and frequency.

This chapter starts with an overview of six-pulse SCR rectifier, which is the building block for the multipulse SCR rectifiers, followed by an analysis of 12-, 18-, and 24-pulse rectifiers. The line current THD and input power factor of these rectifiers are investigated, and the results are summarized in a graphic format.

4.2 SIX-PULSE SCR RECTIFIER

Figure 4.2-1 shows a simplified circuit diagram for the six-pulse SCR rectifier, where RC snubber circuits for the SCR devices are omitted. The line inductance L_s represents the total inductance between the utility supply and the rectifier, including the equivalent inductance of the supply, the total leakage inductance of isolation transformer, if any, and the inductance of a three-phase line reactor that is often added to the system for the reduction of line current THD. On the dc side of the rectifier, a dc choke L_d is used to smooth the dc current. The choke is normally constructed with a single magnetic core and two coils, one coil in the positive dc bus and the other in the negative bus. Such an arrangement is preferable in medium voltage drives since it helps to reduce the common-mode voltage imposed on the motor without increasing the manufacturing cost of the choke [1, 2].

To simplify the analysis, it is assumed that the inductance of the dc choke L_d is sufficiently high such that the dc current I_d is ripple free. The dc choke and the load can then be replaced by an adjustable dc current source as shown in Fig. 4.2-1b.

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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4.2.1 Idealized Six-Pulse Rectifier

Let us consider an idealized six-pulse SCR rectifier, where the line inductance L_s in Fig. 4.2-1 is assumed to be zero. Fig. 4.2-2 shows typical waveforms of the rectifier, where v_a, v_b, and v_c are the phase voltages of the utility supply, i_{g1} to i_{g6} are the gate signals for SCR switches S_1 to S_6, and α is the **firing angle** of the SCRs, respectively.

During interval I ($\frac{\pi}{6} + \alpha \le \omega t < \frac{\pi}{2} + \alpha$), v_a is higher than the other two phase voltages (v_b and v_c), making S_1 forward biased. When S_1 is fired at $\omega t = \pi/6 + \alpha$ by its gate signal i_{g1}, it is turned on. The positive dc bus voltage v_P with respect to ground G is equal to v_a. Assuming that S_6 was conducting prior to the turn-on of S_1, it continues to conduct until the end of interval I, during which the negative bus voltage v_N is equal to v_b. The dc output voltage can be found from $v_d = v_P - v_N = v_{ab}$. The dc current I_d flows from v_a to v_b through S_1, the load, and S_6. The three-phase line currents are $i_a = I_d, i_b = -I_d$, and $i_c = 0$ as shown in Fig. 4.2-2.

During interval II ($\frac{\pi}{2} + \alpha \le \omega t < \frac{5\pi}{6} + \alpha$), v_c is lower than the other two phase voltages (v_a and v_b), making S_2 forward biased. At the moment the gate signal i_{g2} arrives, S_2 is switched on. The conduction of S_2 makes S_6 reverse biased, forcing it to turn off. The dc current I_d is then commutated from S_6 to S_2, which leads to $i_b = 0$ and $i_c = -I_d$. The commutation process in this case completes instantly due to the absence of the line inductance. The dc output voltage is given by $v_d = v_P - v_N = v_{ac}$. Following the same procedure, all the current and voltage waveforms in other intervals can be obtained.

The average dc output voltage can be determined by

$$V_d = \frac{\text{Area } \mathbf{A_1}}{\pi/3} = \frac{1}{\pi/3}\int_{\pi/6+\alpha}^{\pi/2+\alpha} v_{ab}\,d(\omega t) = \frac{3\sqrt{2}}{\pi}V_{LL}\cos\alpha = 1.35V_{LL}\cos\alpha \tag{4.2-1}$$

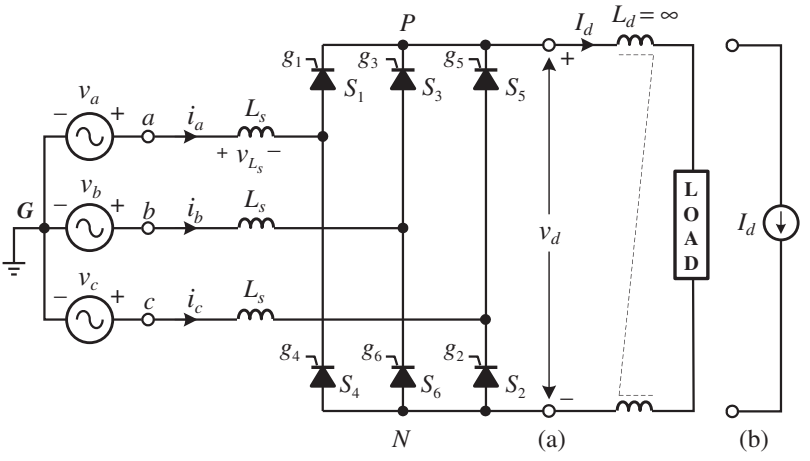

Figure 4.2-1 Simplified circuit diagram of a six-pulse SCR rectifier.

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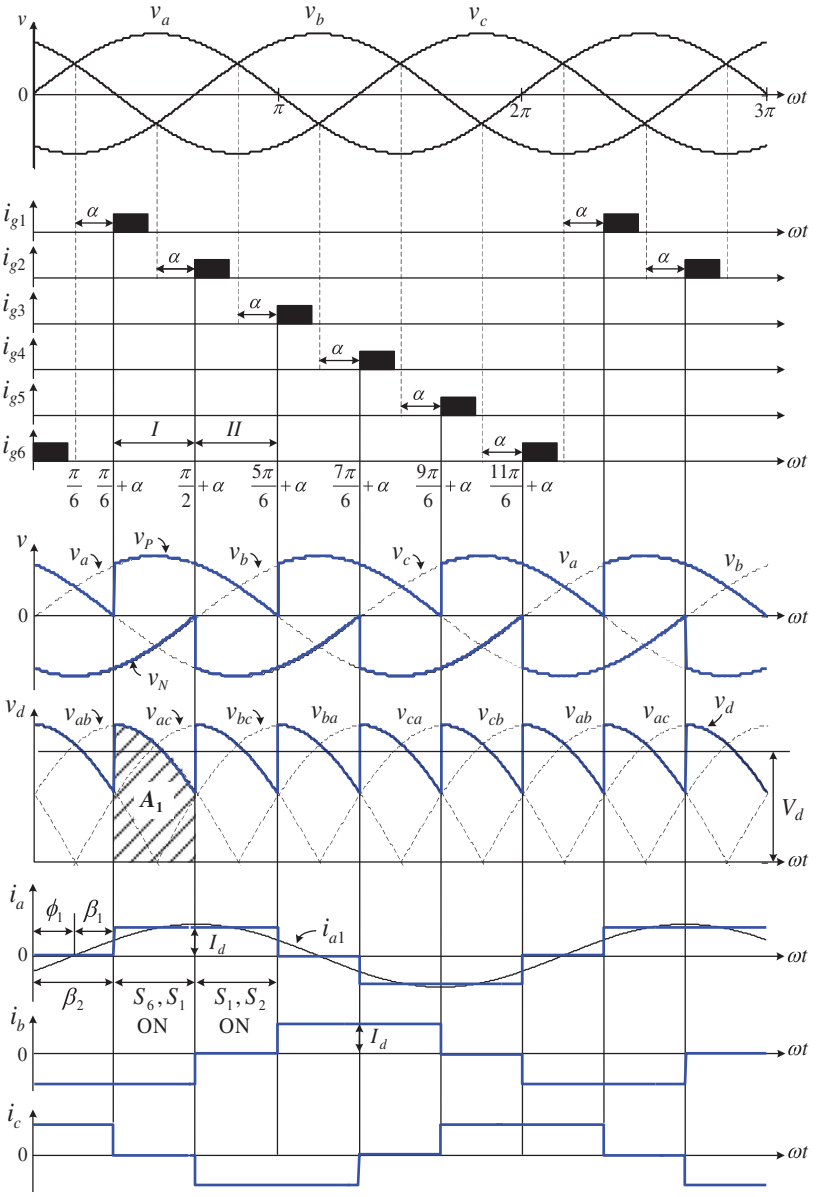

Figure 4.2-2 Waveforms of the idealized six-pulse SCR rectifier operating at $\alpha = 30°$.

where $v_{ab} = \sqrt{2}V_{LL}\sin(\omega t + \pi/6)$. The equation illustrates that the rectifier dc output voltage V_d is positive when the firing angle α is less than $\pi/2$ and becomes negative for an α greater than $\pi/2$. However, the dc current I_d is always positive, irrelevant to the polarity of the dc output voltage.

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When the rectifier produces a positive dc voltage, the power is delivered from the supply to the load. With a negative dc voltage, the rectifier operates in an **inverting mode**, and the power is fed from the load back to the supply. This often takes place in a CSI drive during rapid speed deceleration where the kinetic energy of the rotor and its mechanical load is converted to the electrical energy by the inverter and then sent back to the power supply by the SCR rectifier for fast **dynamic braking**. The power flow in the SCR rectifier is, therefore, **bidirectional**, which also enables the CSI drive to operate in four quadrants, an important feature provided by the SCR rectifier.

The line current i_a in Fig. 4.2-2 can be expressed in a Fourier series as

$$\begin{aligned} i_a = \frac{2\sqrt{3}}{\pi} I_d \Big(& \sin(\omega t - \varphi_1) - \frac{1}{5}\sin 5(\omega t - \varphi_1) - \frac{1}{7}\sin 7(\omega t - \varphi_1) \\ & + \frac{1}{11}\sin 11(\omega t - \varphi_1) + \frac{1}{13}\sin 13(\omega t - \varphi_1) \\ & - \frac{1}{17}\sin 17(\omega t - \varphi_1) - \frac{1}{19}\sin 19(\omega t - \varphi_1) + \ldots . \Big) \end{aligned} \tag{4.2-2}$$

where φ_1 is the phase angle between the supply voltage v_a and the fundamental-frequency line current i_{a1}.

The rms value of i_a can be calculated by

$$\begin{aligned} I_a &= \left(\frac{1}{2\pi} \int_0^{2\pi} (i_a)^2 d(\omega t) \right)^{1/2} \\ &= \left(\frac{1}{2\pi} \left(\int_{\frac{\pi}{6}+\alpha}^{\frac{5\pi}{6}+\alpha} (I_d)^2 d(\omega t) + \int_{\frac{7\pi}{6}+\alpha}^{\frac{11\pi}{6}+\alpha} (-I_d)^2 d(\omega t) \right) \right)^{1/2} \\ &= \sqrt{\frac{2}{3}}\, I_d = 0.816\, I_d \end{aligned} \tag{4.2-3}$$

from which the total harmonic distortion for the line current i_a is

$$\text{THD} = \frac{\sqrt{I_a^2 - I_{a1}^2}}{I_{a1}} = \frac{\sqrt{(0.816\, I_d)^2 - (0.78\, I_d)^2}}{0.78\, I_d} = 0.311 \tag{4.2-4}$$

where I_{a1} is the rms value of i_{a1}.

To find the displacement power factor (DPF), we can refer to β_1 and β_2 in Fig. 4.2-2. Since β_1 is fixed to $\pi/6$ and β_2 is equal to $\pi/6 + \alpha$, the DPF angle is

$$\varphi_1 = \beta_2 - \beta_1 = \alpha$$

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from which

$$\text{DPF} = \cos\varphi_1 = \cos\alpha \tag{4.2-5}$$

The overall power factor for the six-pulse SCR rectifier can be obtained from

$$\text{PF} = \text{DPF} \times \text{DF} = \frac{\cos\varphi_1}{\sqrt{1 + \text{THD}^2}} = 0.955\cos\alpha \tag{4.2-6}$$

where DF is the distortion factor defined in Chapter 3.

Figure 4.2-3 shows the voltage waveforms of the rectifier with various firing angles. The average dc output voltage V_d is positive at $\alpha = 45°$, falls to zero at

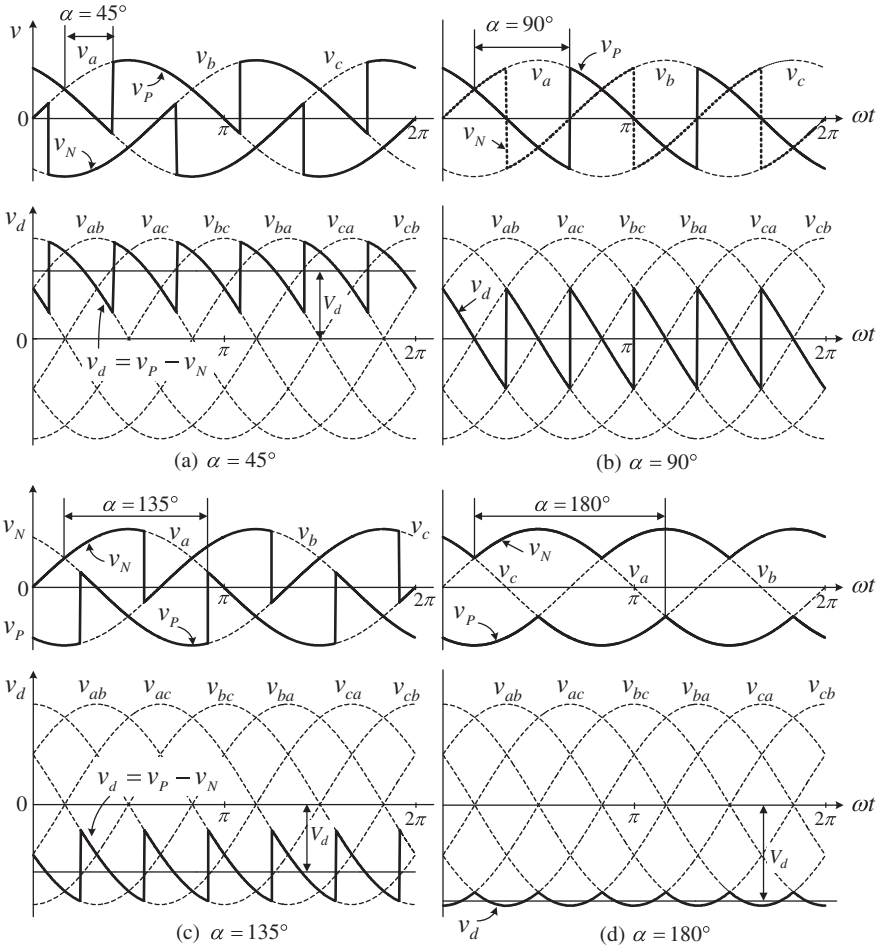

Figure 4.2-3 Voltage waveforms of the idealized six-pulse SCR rectifier operating at various firing angles.

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$\alpha = 90°$, and becomes negative when $\alpha = 135°$. It reaches its maximum negative value at $\alpha = 180°$. In a practical rectifier where the line inductance L_s is present, the firing angle α should be less than 180° to prevent SCR commutation failure [3].

4.2.2 Effect of Line Inductance

With the presence of the line inductance L_s, the commutation of the SCR devices will not complete instantly. Consider a case where the dc output current I_d is commutated from S_5 to S_1 as shown in Fig. 4.2-4. Assuming that S_5 and S_6 are conducting prior to the turn-on of S_1, the dc current flows through both devices. The commutation process is initiated by turning S_1 on at α. At the moment the incoming device S_1 is gated on, its current i_a starts to rise from zero, but cannot jump to I_d instantly due to the line inductance L_s. In the meantime, the current i_c in the outgoing device S_5 starts to decrease since $i_c = I_d - i_a$. As a result, three SCR devices, S_1, S_5, and S_6, conduct simultaneously. The commutation completes at the end of the commutation interval γ, at which the current i_a in S_1 reaches I_d whereas the current i_c in S_5 falls to zero.

The commutation causes a reduction in the average dc voltage V_d. Since both S_1 and S_5 conduct simultaneously during the γ interval, the positive bus voltage v_P with respect to ground G can be expressed as

$$v_P = -L_s \frac{di_a}{dt} + v_a = -L_s \frac{di_c}{dt} + v_c \tag{4.2-7}$$

from which

$$v_P = \frac{v_a + v_c}{2} - \frac{L_s}{2}\left(\frac{di_a}{dt} + \frac{di_c}{dt}\right) \tag{4.2-8}$$

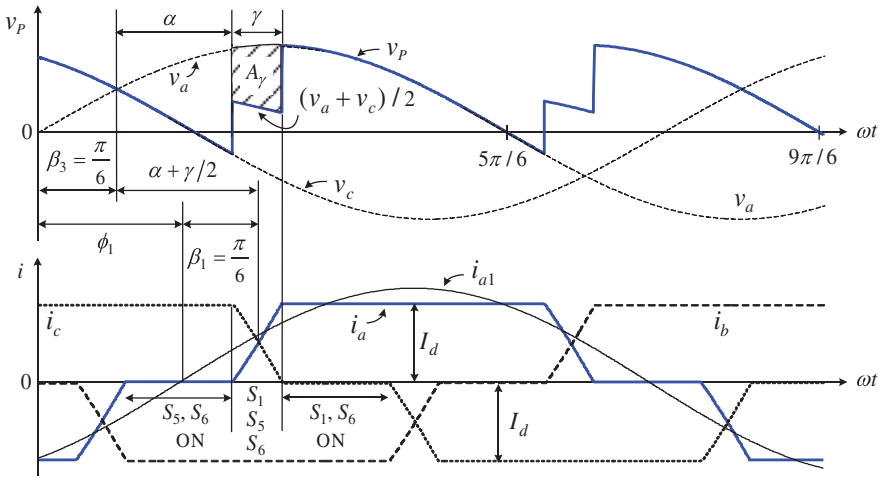

Figure 4.2-4 Voltage and current waveforms during commutation ($\alpha = 45°$).

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Since $i_a + i_c = I_d = \text{constant}$, we have

$$\frac{di_a}{dt} + \frac{di_c}{dt} = 0 \tag{4.2-9}$$

Substituting (4.2-9) into (4.2-8) leads to

$$v_P = \frac{v_a + v_c}{2} \tag{4.2-10}$$

The waveform of v_P during the γ interval is also shown in Fig. 4.2-4. The shaded area A_γ, representing the amount of voltage reduction caused by the commutation, can be found from

$$A_\gamma = \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{6}+\alpha+\gamma} (v_a - v_P) d(\omega t) \tag{4.2-11}$$

where

$$v_a - v_P = L_s(di_a/dt) \tag{4.2-12}$$

Substituting (4.2-12) into (4.2-11) yields

$$A_\gamma = \int_0^{I_d} \omega L_s di_a = \omega L_s I_d \tag{4.2-13}$$

The average dc voltage loss ΔV can then be calculated by

$$\Delta V = \frac{A_\gamma}{\pi/3} = \frac{3\omega L_s}{\pi} I_d \tag{4.2-14}$$

Taking the effect of the line inductance L_s into account, the average dc output voltage of the six-pulse SCR rectifier is

$$V_d = 1.35\, V_{LL} \cos\alpha - \frac{3\omega L_s}{\pi} I_d \tag{4.2-15}$$

The commutation angle γ can be derived from (4.2-11)

$$\gamma = \cos^{-1}\left(\cos\alpha - \frac{\sqrt{2}\omega L_s}{V_{LL}} I_d\right) - \alpha \tag{4.2-16}$$

Figure 4.2-5 shows the relationship between the commutation angle γ and the firing angle α. For a given α, the lower the value for L_s and I_d, the smaller the commutation angle γ is.

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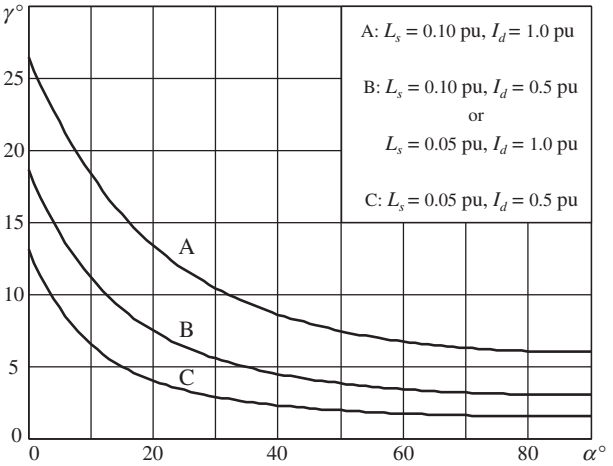

Figure 4.2-5 Commutation angle γ versus firing angle α.

The input power factor is affected by the line inductance L_s as well. Assuming that i_a and i_c shown in Fig. 4.2-4 varies linearly over time during the commutation interval, β_1 is equal to $\pi/6$. The DPF angle φ_1 can be calculated by

$$\varphi_1 = \beta_3 + (\alpha + \gamma/2) - \beta_1 = \alpha + \gamma/2 \tag{4.2-17}$$

from which

$$\text{DPF} = \cos\varphi_1 = \cos(\alpha + \gamma/2) \tag{4.2-18}$$

The overall power factor of the rectifier can be determined by

$$\text{PF} = \text{DPF} \times \text{DF} = \frac{\cos(\alpha + \gamma/2)}{\sqrt{1 + \text{THD}^2}} \tag{4.2-19}$$

4.2.3 Power Factor and THD

Figure 4.2-6 shows the simulated waveforms for the line current i_a when the rectifier operates with the rated line current ($I_{a1} = 1$ pu). The line inductance L_s is assumed to be 0.05 pu and the firing angle α is 0° in Fig.4.2-6a and 30° in Fig.4.2-6b, respectively. It is interesting to note that waveform of i_a during the γ interval varies with α. It rises

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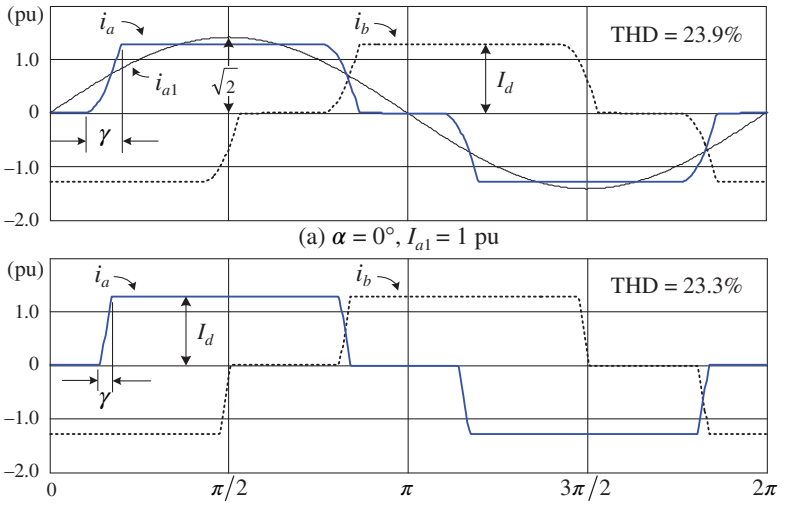

Harmonics (n)	5	7	11	13	17	19	23	25	THD (%)
I_{an} / I_{a1} (%) $\alpha = 0°$	18.6	12.4	6.32	4.58	2.40	1.73	1.02	0.87	23.9
I_{an} / I_{a1} (%) $\alpha = 30°$	19.7	14.1	8.58	7.27	5.16	4.62	3.43	3.16	28.3

(c) Harmonic content

Figure 4.2-6 Line current waveforms in the six-pulse SCR rectifier with $L_s = 0.05$ pu.

nonlinearly when $\alpha = 0°$ and looks somewhat like linear for $\alpha = 30°$. This is because the line current i_a is a function of α during commutation, given by

$$i_a = \frac{V_{LL}}{\sqrt{2}\omega L_s}(\cos\alpha - \cos(\omega t + \alpha)) \quad 0 \le \omega t \le \gamma \tag{4.2-20}$$

Figure 4.2-6c shows the line current harmonic content for the six-pulse SCR rectifier. Its THD is more than 20%, which is not acceptable in practice, especially when the rectifier is for high power applications.

Figure 4.2-7 shows the line current THD versus I_{a1} with L_s and α as parameters. The THD reduces with the increase of I_{a1} and L_s as shown in Fig. 4.2-7a. It also decreases with the firing angle α as illustrated in Fig. 4.2-7b.

Figure 4.2-8 shows the input power factor profile of the six-pulse SCR rectifier as a function of I_{a1} and α. The power factor varies slightly with the line current I_{a1}. However, it reduces substantially with large values of α. This is, in fact, the main drawback of the SCR rectifier.

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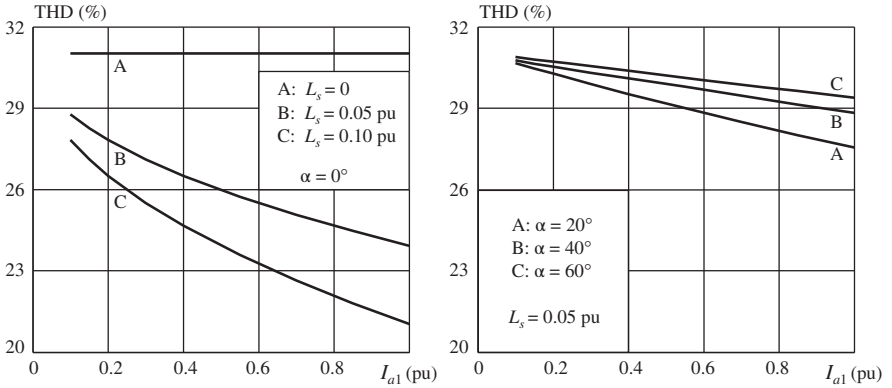

(a) Fixed α and variable L_s (b) Fixed L_s and variable α

Figure 4.2-7 Line current THD of the six-pulse SCR rectifier.

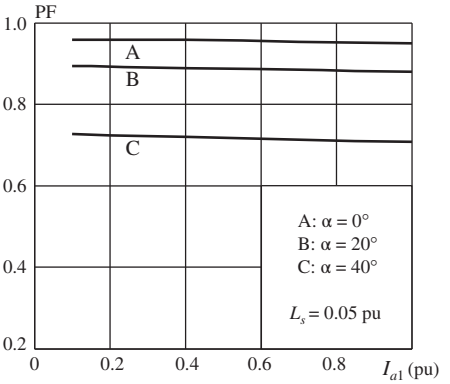

Figure 4.2-8 Power factor of the six-pulse SCR rectifier.

4.3 12-PULSE SCR RECTIFIER

The block diagram of a 12-pulse SCR rectifier is shown in Fig. 4.3-1. It is composed of a phase-shifting transformer and two identical six-pulse SCR rectifiers. The transformer has two secondary windings, one connected in wye and the other in delta. The line-to-line voltage of the secondary windings is normally half of its primary line-to-line voltage. The dc outputs of the two SCR rectifiers are connected in series for a single dc load. The dc choke L_d is assumed to be sufficiently large and the resultant dc current I_d is ripple free.

As shown in Fig. 4.3-2, the 12-pulse SCR rectifier can be used as a front end for a CSI fed drive. The inverter converts the dc current I_d to a three-phase PWM current i_w. The magnitude of i_w is proportional to I_d, and thus can be adjusted by the rectifier through firing angle control. The details of the CSI drive will be discussed in the later chapters.

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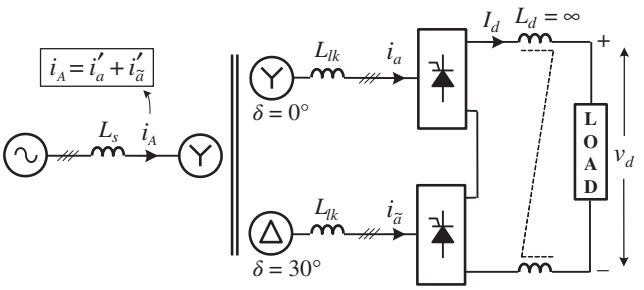

Figure 4.3-1 Block diagram of a 12-pulse SCR rectifier.

4.3.1 Idealized 12-Pulse Rectifier

Consider an idealized 12-pulse rectifier where the line inductance L_s and the total leakage inductance L_{lk} of the transformer are assumed to be zero. The current waveforms in the rectifier are shown in Fig. 4.3-3, where i_a and $i_{\tilde{a}}$ are the secondary line currents, i'_a and $i'_{\tilde{a}}$ are the primary currents referred from the secondary side, and i_A is the primary line current given by $i_A = i'_a + i'_{\tilde{a}}$, respectively.

The secondary line current i_a can be expressed as

$$i_a = \frac{2\sqrt{3}}{\pi} I_d \left(\sin \omega t - \frac{1}{5} \sin 5\omega t - \frac{1}{7} \sin 7\omega t + \frac{1}{11} \sin 11\omega t + \frac{1}{13} \sin 13\omega t - \frac{1}{17} \sin 17\omega t - \frac{1}{19} \sin 19\omega t + \cdots . \right) \tag{4.3-1}$$

where $\omega = 2\pi f_1$ is the angular frequency of the supply voltage. Since the waveform of i_a is of half-wave symmetry, it does not contain any even-order harmonics. In addition, i_a does not contain any triplen harmonics either due to the balanced three-phase system.

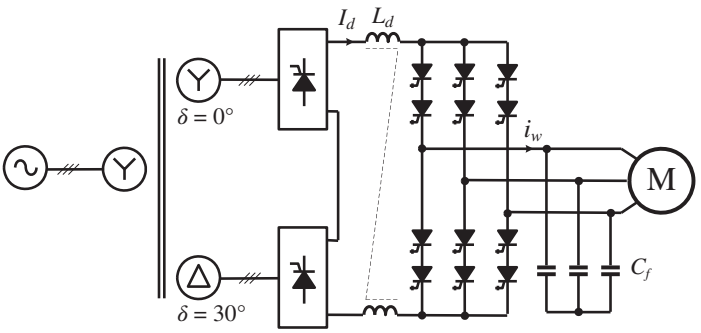

Figure 4.3-2 A CSI fed drive using the 12-pulse SCR rectifier as a front end.

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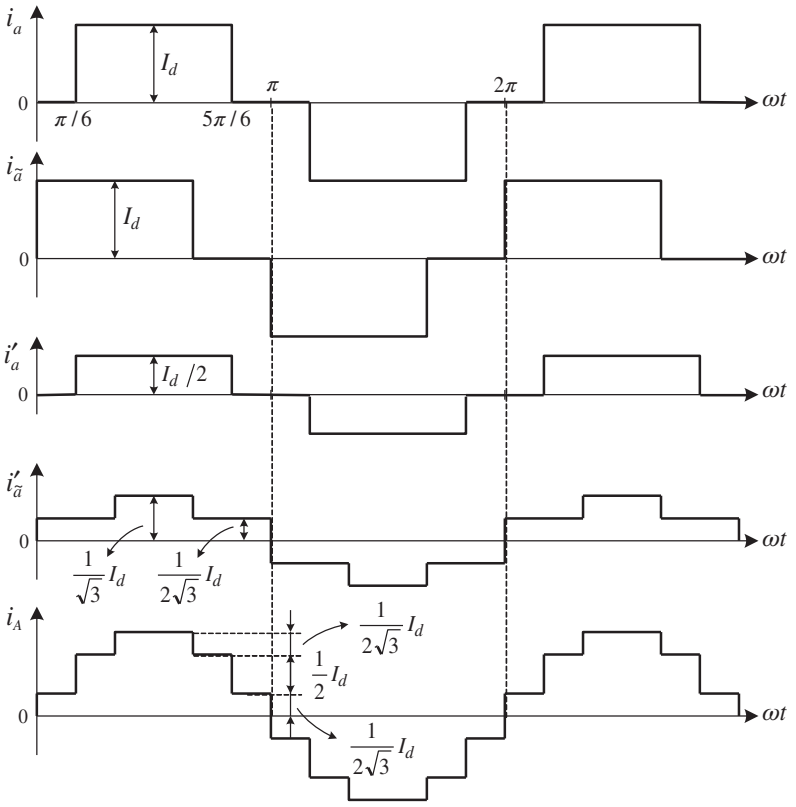

Figure 4.3-3 Current waveforms of the 12-pulse SCR rectifier ($L_s = L_{lK} = 0$).

The other secondary current $i_{\tilde{a}}$ leads i_a by 30°, and its Fourier expression is

$$i_{\tilde{a}} = \frac{2\sqrt{3}}{\pi} I_d \left(\sin(\omega t + 30°) - \frac{1}{5}\sin 5(\omega t + 30°) - \frac{1}{7}\sin 7(\omega t + 30°) + \frac{1}{11}\sin 11(\omega t + 30°) + \frac{1}{13}\sin 13(\omega t + 30°) - \frac{1}{17}\sin 17(\omega t + 30°) - \frac{1}{19}\sin 19(\omega t + 30°) + \cdots . \right) \tag{4.3-2}$$

The waveform for the referred current i'_a in Fig. 4.3-3 is identical to i_a except that its magnitude is halved due to the turns ratio of the Y/Y-connected windings. The current i'_a can be expressed in Fourier series as

$$i'_a = \frac{\sqrt{3}}{\pi} I_d \left\{ \sin \omega t - \frac{1}{5}\sin 5\omega t - \frac{1}{7}\sin 7\omega t + \frac{1}{11}\sin 11\omega t + \frac{1}{13}\sin 13\omega t - \frac{1}{17}\sin 17\omega t - \frac{1}{19}\sin 19\omega t + \cdots . \right\} \tag{4.3-3}$$

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When the current $i_{\tilde{a}}$ is referred to the primary side, the phase angles of some harmonic currents are altered due to the Y/Δ-connected windings. As a result, the referred current $i'_{\tilde{a}}$ does not keep the same wave shape as $i_{\tilde{a}}$. The Fourier expression for $i'_{\tilde{a}}$ is

$$i'_{\tilde{a}} = \frac{\sqrt{3}}{\pi} I_d \left\{ \sin \omega t + \frac{1}{5} \sin 5\omega t + \frac{1}{7} \sin 7\omega t + \frac{1}{11} \sin 11\omega t + \frac{1}{13} \sin 13\omega t + \frac{1}{17} \sin 17\omega t + \frac{1}{19} \sin 19\omega t + \cdots . \right\} \tag{4.3-4}$$

The line current i_A can be found from

$$i_A = i'_a + i'_{\tilde{a}} = \frac{2\sqrt{3}}{\pi} I_d \left\{ \sin \omega t + \frac{1}{11} \sin 11\omega t + \frac{1}{13} \sin 13\omega t + \frac{1}{23} \sin 23\omega t + \frac{1}{25} \sin 25\omega t + \cdots . \right\} \tag{4.3-5}$$

where the two dominant current harmonics, the 5th and 7th, are cancelled in addition to the 17th and 19th.

The THD of the secondary and primary line currents i_a and i_A can be determined by

$$\text{THD}\,(i_a) \frac{\sqrt{I_a^2 - I_{a1}^2}}{I_{a1}} = \frac{\left(I_{a5}^2 + I_{a7}^2 + I_{a11}^2 + I_{a13}^2 + \cdots\right)^{1/2}}{I_{a1}} = 31.1\% \tag{4.3-6}$$

and

$$\text{THD}\,(i_A) \frac{\sqrt{I_A^2 - I_{A1}^2}}{I_{A1}} = \frac{\left(I_{A11}^2 + I_{A13}^2 + I_{A23}^2 + I_{A25}^2 + \ldots\right)^{1/2}}{I_{A1}} = 15.3\% \tag{4.3-7}$$

The THD of the primary line current i_A in the idealized 12-pulse rectifier is reduced approximately by 50% compared with that of the secondary line current i_a.

4.3.2 Effect of Line and Leakage Inductances

Figure 4.3-4 shows typical current waveforms for the 12-pulse rectifier taking into account the transformer leakage inductance L_{lk}. The rectifier operates under the condition of $\alpha = 0°$, $I_{A1} = 1$ pu, $L_s = 0$, and $L_{lk} = 0.05$ pu. The waveform for the secondary line current i_a is close to a trapezoid and contains the 5th and 7th harmonics with a magnitude of 18.8% and 12.7%, respectively. However, these two harmonics are cancelled by the phase-shifting transformer, and thus do not appear in the primary line current i_A. Due to the effect of the leakage inductance, the THD of i_A is reduced from 15.3% in the idealized rectifier to 8.61%.

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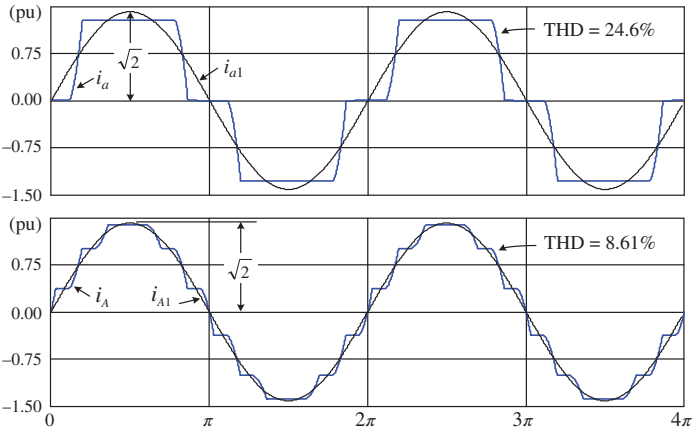


(a) Waveforms

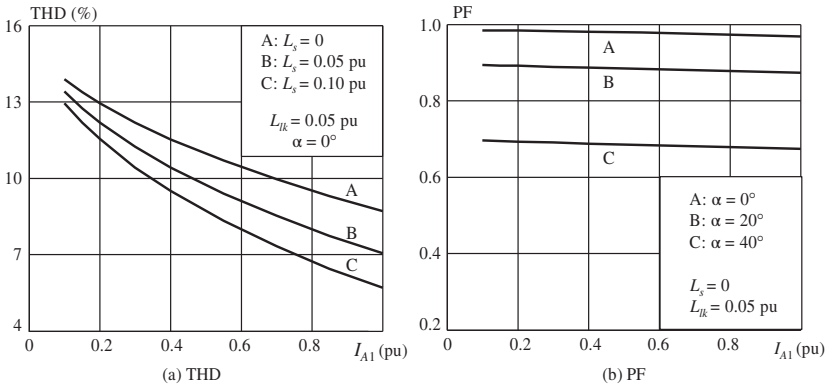
Harmonics (n)	5	7	11	13	17	19	23	25	THD (%)
I_{an} / I_{a1} (%)	18.8	12.7	6.78	5.05	2.77	2.01	1.01	0.75	24.6
I_{An} / I_{A1} (%)	0	0	6.78	5.05	0	0	1.01	0.75	8.61

(b) Harmonic content

Figure 4.3-4 Typical current waveforms and harmonic contents of the 12-pulse SCR rectifier with $L_s = 0$ and $L_{lk} = 0.05$ pu.

4.3.3 THD and PF

The THD of the primary line current i_A as a function of I_{A1} and L_s is illustrated in Fig. 4.3-5a. Compared with the six-pulse SCR rectifier, the 12-pulse rectifier has a much better THD profile. However, it generally does not meet the harmonic guidelines set by IEEE Standard 519-2014 [4]. The input power factor of the rectifier varies greatly with the firing angle α as shown in Fig. 4.3-5b.



(a) THD (b) PF

Figure 4.3-5 Primary line current THD and input PF of the 12-pulse SCR rectifier.

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4.4 18- AND 24-PULSE SCR RECTIFIERS

The block diagram of an 18-pulse SCR rectifier is depicted in Fig. 4.4-1. Similar to the 18-pulse diode rectifiers, the rectifier employs a phase-shifting transformer

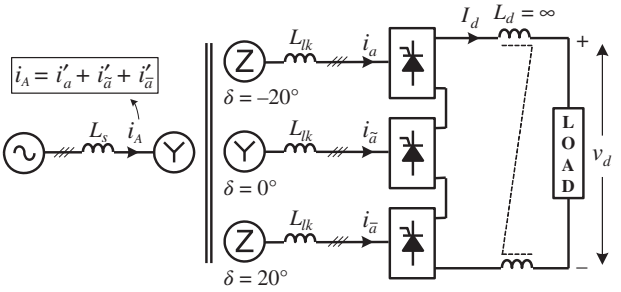
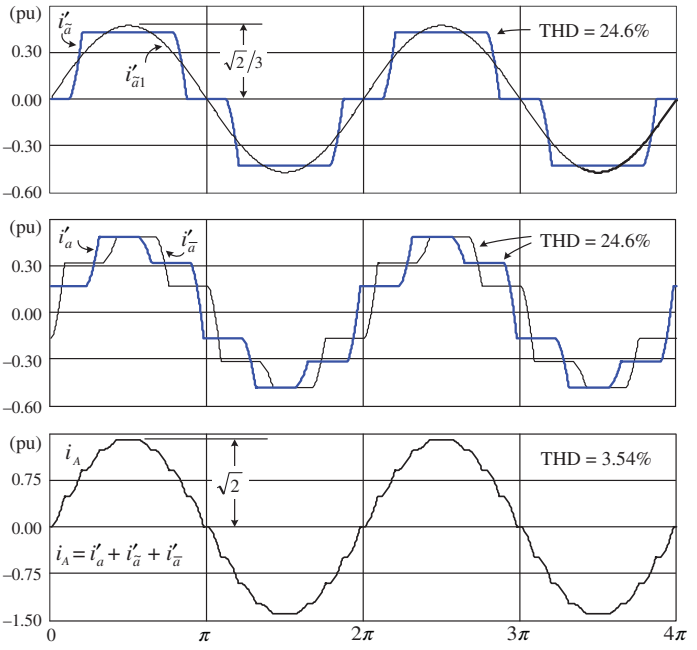


Figure 4.4-1 Block diagram of an 18-pulse SCR rectifier.



(a) Waveforms

Harmonics (n)	5	7	11	13	17	19	23	25	THD (%)
I_{an}/I_{a1} (%)	18.8	12.7	6.78	5.05	2.77	2.01	1.01	0.75	24.6
I_{An}/I_{A1} (%)	0	0	0	0	2.77	2.01	0	0	3.54

(b) Harmonic content

Figure 4.4-2 Current waveforms and harmonic contents of the 18-pulse SCR rectifier with $L_s = 0$ and $L_{lk} = 0.05$ pu.

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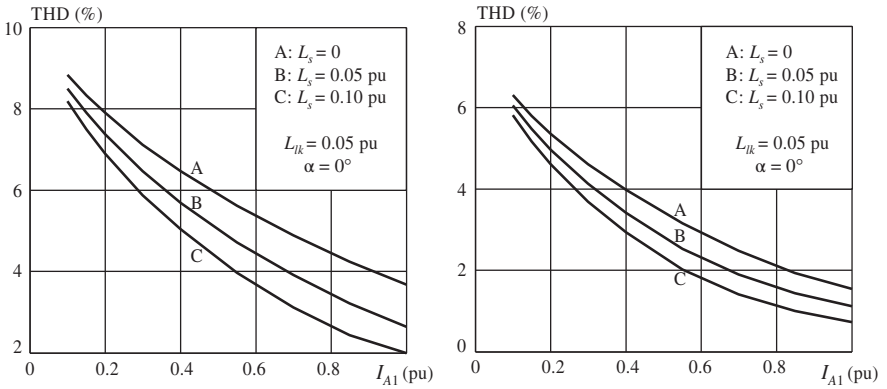

Figure 4.4-3 THD of the primary line current i_A in the 18- and 24-pulse SCR rectifiers.

with three secondary windings feeding three identical six-pulse SCR rectifiers. The configuration of the 24-pulse SCR rectifier can be easily derived and thus is not shown.

Figure 4.4-2 shows the typical current waveforms of the 18-pulse SCR rectifier operating under the condition of $\alpha = 0°$, $I_{A1} = 1$ pu, $L_s = 0$, and $L_{lk} = 0.05$ pu, where i'_a, $i'_{\bar{a}}$, and $i'_{\tilde{a}}$ are the primary currents referred from the transformer secondary side. All these currents have the same THD of 24.6% although their waveforms are all different. The primary line current i_A does not contain the 5th, 7th, 11th, or 13th harmonics, resulting in a nearly sinusoidal waveform with a THD of only 3.54%.

Figure 4.4-3 shows the primary line current THD for the 18- and 24-pulse SCR rectifiers versus I_{A1} with the line inductance L_s as a parameter. As expected, the 18-pulse rectifier has better line current THD profile than the 12-pulse SCR rectifier while the 24-pulse rectifier is superior to the 18-pulse rectifier. The input power factor of the 18- and 24-pulse rectifiers is similar to that of the 12-pulse and therefore is not presented.

4.5 SUMMARY

In this chapter, the operation of the six-pulse SCR rectifier is introduced and its performance is analyzed. The six-pulse rectifier is the building block for the multipulse SCR rectifiers, and therefore is discussed in detail. The line current THD of the 12-pulse SCR rectifier normally does not satisfy the harmonic guidelines set by IEEE Standard 519-2014. The 18-pulse SCR rectifier has a better line current harmonic profile while the 24-pulse rectifier provides a superior harmonic performance. The input power factor of the SCR rectifiers varies with the firing angle, which is the major disadvantage of the rectifiers.

The multipulse SCR rectifier is naturally suited for use in medium voltage CSI fed drives. It was extensively used in the CSI MV drives as a front-end converter

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due to its good harmonic performance and electric isolation provided by the phase-shifting transformer. However, the SCR rectifier has been replaced by PWM GCT current source rectifiers in most applications for higher input power factor and better dynamic performance.

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Chapter 5

Phase-Shifting Transformers

5.1 INTRODUCTION

The phase-shifting transformer is an indispensable device in multipulse diode/SCR rectifiers. It provides three main functions: (a) a required phase displacement between the primary and secondary line-to-line voltages for harmonic cancellation, (b) a proper secondary voltage, and (c) an electric isolation between the rectifier and the utility supply. According to the winding arrangements, the transformers can be classified into Y/Z and Δ/Z configurations, where the primary winding can be connected in wye (Y) or delta (Δ) while the secondary windings are normally in zigzag (Z) connection. Both configurations can be equally used in the multipulse rectifiers.

In this chapter, a number of issues concerning the phase-shifting transformer are addressed, including the configuration of the transformer, the design of turns ratios, and the principle of harmonic current cancellation.

5.2 Y/Z PHASE-SHIFTING TRANSFORMERS

Depending on winding connections, the line-to-line voltage of the transformer secondary winding may lead or lag its primary voltage by a phase angle δ. The Y/Z-1 transformers to be presented below provide a leading phase angle while the Y/Z-2 transformers generate a lagging angle.

5.2.1 Y/Z-1 Transformers

Figure 5.2-1 shows a Y/Z-1 phase-shifting transformer and its phasor diagram. The primary winding is connected in wye with N_1 turns per phase. The secondary winding is composed of two sets of coils having N_2 and N_3 turns per phase. The N_2 coils are connected in delta and then in series with the N_3 coils. Such an arrangement is

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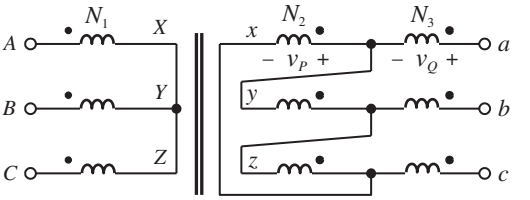
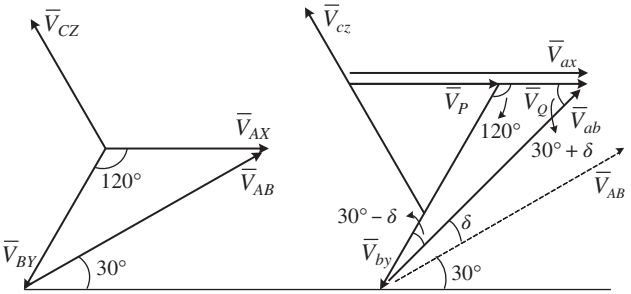


(a) Connection diagram



(b) Phasor diagram

Figure 5.2-1 Y/Z-1 phase-shifting transformer.

known as **zigzag** or **extended-delta** connection. As shown in its phasor diagram, the transformer can produce a phase-shifting angle δ, defined by

$$\delta = \angle \overline{V}_{ab} - \angle \overline{V}_{AB} \tag{5.2-1}$$

where $\overline{V}_{AB}$ and $\overline{V}_{ab}$ are the phasors for the primary and secondary line-to-line voltages v_{AB} and v_{ab}, respectively. To determine the turns ratio for the transformer, let us consider a triangle composed of $\overline{V}_Q, \overline{V}_{by}$, and $\overline{V}_{ab}$ in the phasor diagram, from which

$$\frac{V_Q}{\sin(30° - \delta)} = \frac{V_{by}}{\sin(30° + \delta)} \qquad 0° \le \delta \le 30° \tag{5.2-2}$$

where V_Q is the rms voltage across the N_3 coil and V_{by} is the rms phase voltage between notes b and y. Since V_{by} is equal to V_{ax} in a balanced three-phase system, (5.2-2) can be rewritten as

$$\frac{V_Q}{V_{ax}} = \frac{\sin(30° - \delta)}{\sin(30° + \delta)} \tag{5.2-3}$$

from which the turns ratio of the secondary coils is

$$\frac{N_3}{N_2 + N_3} = \frac{V_Q}{V_{ax}} = \frac{\sin(30° - \delta)}{\sin(30° + \delta)} \tag{5.2-4}$$

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For a given value of δ, the ratio of N_3 to $(N_2 + N_3)$ can be determined. Similarly, the following relationship can be derived

$$\frac{V_{ab}}{\sin 120^\circ} = \frac{V_{by}}{\sin(30^\circ + \delta)} \tag{5.2-5}$$

from which

$$V_{ax} = V_{by} = \frac{2}{\sqrt{3}} \sin(30^\circ + \delta) V_{ab} \tag{5.2-6}$$

The turns ratio of the transformer is defined by

$$\frac{N_1}{N_2 + N_3} = \frac{V_{AX}}{V_{ax}} \tag{5.2-7}$$

Substituting (5.2-6) into (5.2-7) yields

$$\frac{N_1}{N_2 + N_3} = \frac{1}{2\sin(30^\circ + \delta)} \frac{V_{AB}}{V_{ab}} \tag{5.2-8}$$

where $V_{AB} = \sqrt{3}\, V_{AX}$.

Let us now examine two extreme cases. Assuming that N_2 is reduced to zero, the secondary winding in Fig. 5.2-1 becomes wye connected, and thus V_{ab} is in phase with V_{AB}, leading to $\delta = 0^\circ$. Alternatively, if $N_3 = 0$, the secondary winding becomes delta connected, resulting in $\delta = 30^\circ$. Therefore, the phase-shifting angle δ for the Y/Z-1 transformer is in the range of 0–30°.

5.2.2 Y/Z-2 Transformers

The configuration of a Y/Z-2 phase-shifting transformer is shown in Fig. 5.2-2, where the primary winding remains the same as that in the Y/Z-1 transformer while the secondary delta-connected coils are connected in a reverse order. Following the same procedure presented earlier, the transformer turns ratio can be found from

$$\begin{cases} \dfrac{N_3}{N_2 + N_3} = \dfrac{\sin(30^\circ - |\delta|)}{\sin(30^\circ + |\delta|)} \\ \dfrac{N_1}{N_2 + N_3} = \dfrac{1}{2\sin(30^\circ + |\delta|)} \dfrac{V_{AB}}{V_{ab}} \end{cases} \tag{5.2-9}$$

The phase angle δ has a negative value for the Y/Z-2 transformer, indicating that V_{ab} lags V_{AB} by $|\delta|$ as shown in Fig. 5.2-2b.

Table 5.2-1 gives the typical value of δ and turns ratio of the Y/Z transformers for use in multipulse rectifiers. The voltage ratio V_{AB}/V_{ab} is normally equal to 2, 3, and 4 for the 12-, 18-, and 24-pulse rectifiers, respectively.

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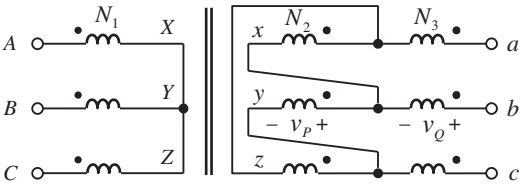
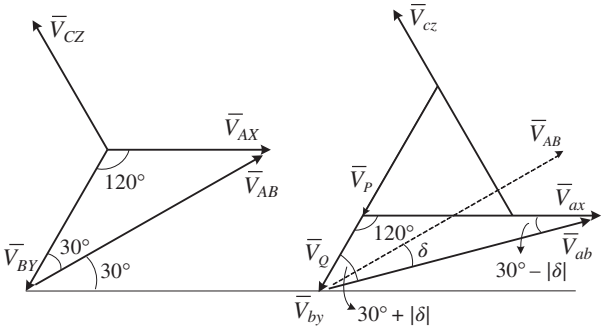


(a) Connection diagram



(b) Phasor diagram

Figure 5.2-2 Y/Z-2 phase-shifting transformer.

Table 5.2-1 Turns Ratio for Y/Z Transformers

$\delta(\angle\overline{V}_{ab} - \angle\overline{V}_{AB})$		$\frac{N_3}{N_2 + N_3}$	$\frac{N_1}{N_2 + N_3}$	Applications
Y/Z-1	Y/Z-2			
0°	0°	1.0	$1.0\ \frac{V_{AB}}{V_{ab}}$	12-, 18-, and 24-pulse rectifiers
15°	−15°	0.366	$0.707\ \frac{V_{AB}}{V_{ab}}$	24-pulse rectifiers
20°	−20°	0.227	$0.653\ \frac{V_{AB}}{V_{ab}}$	18-pulse rectifiers
30°	−30°	0	$0.577\ \frac{V_{AB}}{V_{ab}}$	12- and 24-pulse rectifiers
$\frac{V_{AB}}{V_{ab}}$ = 2, 3, and 4 for 12-, 18-, and 24-pulse rectifiers, respectively				

5.3 Δ/Z TRANSFORMERS

Figure 5.3-1 shows two typical configurations for Δ/Z phase-shifting transformers, where the primary winding is connected in delta and the secondary winding is zigzag connected. The phasor diagram for the $\Delta/Z - 1$ transformer is given in Fig. 5.3-1c, in

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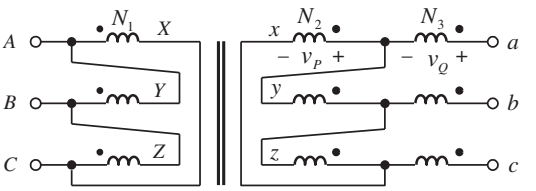
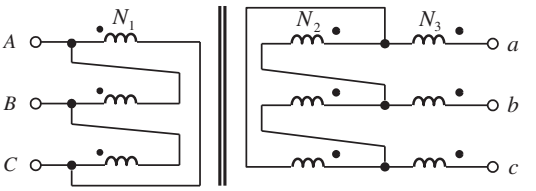
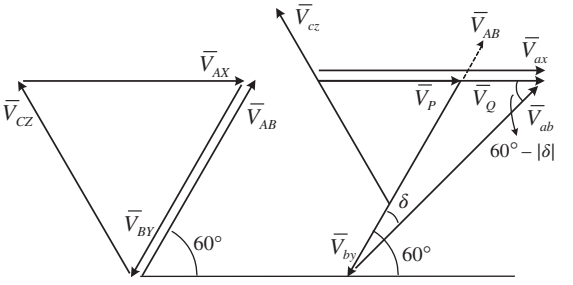


(a) Δ / Z – 1 transformer



(b) Δ / Z – 2 transformer



(c) Phasor diagram for Δ /Z – 1 transformer

Figure 5.3-1 Δ/Z phase-shifting transformers.

which the secondary voltage V_{ab} lags the primary voltage V_{AB} by $|\delta|$. The turns ratio of the $\Delta/Z-1$ transformer is given by

$$\begin{cases} \dfrac{N_3}{N_2+N_3} = \dfrac{V_Q}{V_{ax}} = \dfrac{\sin(|\delta|)}{\sin(60^\circ - |\delta|)} \\ \dfrac{N_1}{N_2+N_3} = \dfrac{V_{AX}}{V_{ax}} = \dfrac{\sqrt{3}}{2\sin(60^\circ - |\delta|)} \dfrac{V_{AB}}{V_{ab}} \end{cases} \qquad -30^\circ \le \delta \le 0^\circ \quad (5.3\text{-}1)$$

Table 5.3-1 illustrates the relationship between the phase-shifting angle δ and the turns ratio of the Δ/Z transformers for multipulse rectifiers. The phase angle δ varies from 0° to − 30° for the $\Delta/Z-1$ transformer and from − 30° to − 60° for the $\Delta/Z-2$ transformer.

Figure 5.3-2 shows a few examples of the phase-shifting transformers for use in multipulse diode/SCR rectifiers. The transformer for the 12-pulse rectifiers has two secondary windings with a 30° phase shift between them. The 18-pulse rectifiers require a transformer with three secondary windings having a 20° phase displacement

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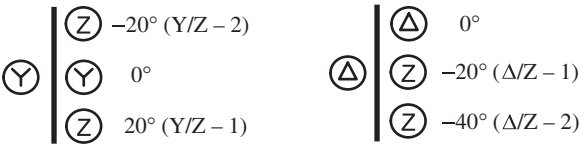
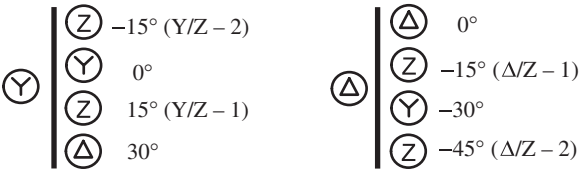
Table 5.3-1 Turns Ratio for Δ/Z Transformers

Zigzag Transformer	$\delta(\angle\overline{V}_{ab} - \angle\overline{V}_{AB})$	$\frac{N_3}{N_2 + N_3}$	$\frac{N_1}{N_2 + N_3}$	Applications
Δ/Z-1	0°	0	$1.0\frac{V_{AB}}{V_{ab}}$	12-, 18-, and 24-pulse rectifiers
	−15°	0.366	$1.225\frac{V_{AB}}{V_{ab}}$	24-pulse rectifiers
	−20°	0.532	$1.347\frac{V_{AB}}{V_{ab}}$	18-pulse rectifiers
	−30°	1.0	$1.732\frac{V_{AB}}{V_{ab}}$	12- and 24-pulse rectifiers
Δ/Z-2	−40°	0.532	$1.347\frac{V_{AB}}{V_{ab}}$	18-pulse rectifiers
	−45°	0.366	$1.225\frac{V_{AB}}{V_{ab}}$	24-pulse rectifiers
	−60°	0	$1.0\frac{V_{AB}}{V_{ab}}$	18-pulse rectifiers
$\frac{V_{AB}}{V_{ab}}$ = 2, 3, and 4 for 12-, 18-, and 24-pulse rectifiers, respectively				

(a) For 12-pulse rectifiers



(b) For 18-pulse rectifiers



(c) For 24-pulse rectifiers

Figure 5.3-2 Examples of phase-shifting transformers for multipulse rectifiers.

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among each other. The transformer used in the 24-pulse rectifiers has four secondary windings with a 15° phase shift between any two adjacent windings.

5.4 HARMONIC CURRENT CANCELLATION

5.4.1 Phase Displacement of Harmonic Currents

The main purpose of this section is to investigate the phase displacement of harmonic currents when they are referred from the secondary to the primary side of a phase-shifting transformer. It is the phase displacement that makes it possible to cancel certain harmonic currents generated by a three-phase nonlinear load.

Figure 5.4-1 shows a Δ/Y transformer feeding a nonlinear load. Assume that the voltage ratio V_{AB}/V_{ab} of the transformer is unity with a turns ratio of $N_1/N_2 = \sqrt{3}$. The transformer has a phase angle of $\delta = \angle\overline{V}_{ab} - \angle\overline{V}_{AB} = -30°$. For a three-phase balanced system, the line currents of the nonlinear load can be expressed as

$$\begin{cases} i_a = \displaystyle\sum_{n=1,5,7,11,\ldots}^{\infty} \hat{I}_n \sin(n\omega t) \\ i_b = \displaystyle\sum_{n=1,5,7,11,\ldots}^{\infty} \hat{I}_n \sin(n(\omega t - 120°)) \\ i_c = \displaystyle\sum_{n=1,5,7,11,\ldots}^{\infty} \hat{I}_n \sin(n(\omega t - 240°)) \end{cases} \tag{5.4-1}$$

where $\hat{I}_n$ is the peak value of the nth order harmonic current. When i_a and i_b are referred to the primary side, the referred currents i'_{ap} and i'_{bp} in the primary winding

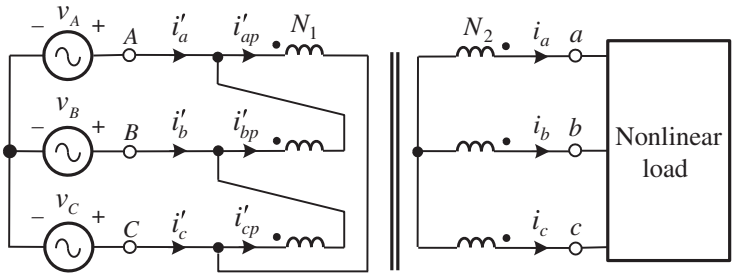

Figure 5.4-1 Investigation of harmonic currents in the primary and secondary windings.

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can be described by

$$\begin{cases} i'_{ap} = i_a \dfrac{N_2}{N_1} = \dfrac{1}{\sqrt{3}}(\hat{I}_1 \sin(\omega t) + \hat{I}_5 \sin(5\omega t) + \hat{I}_7 \sin(7\omega t) + \hat{I}_{11} \sin(11\omega t) + \ldots) \\ i'_{bp} = i_b \dfrac{N_2}{N_1} = \dfrac{1}{\sqrt{3}}(\hat{I}_1 \sin(\omega t - 120°) + \hat{I}_5 \sin(5\omega t - 240°) + \hat{I}_7 \sin(7\omega t - 120°) \\ \qquad + \hat{I}_{11} \sin(11\omega t - 240°) + \cdots) \end{cases} \tag{5.4-2}$$

from which the primary line current can be found from

$$\begin{aligned} i'_a = i'_{ap} - i'_{bp} &= \hat{I}_1 \sin(\omega t + 30°) + \hat{I}_5 \sin(5\omega t - 30°) + \hat{I}_7 \sin(7\omega t + 30°) \\ &\quad + \hat{I}_{11} \sin(11\omega t - 30°) + \ldots \\ &= \sum_{n=1,7,13,\ldots}^{\infty} \hat{I}_n \sin(n\omega t - \delta) + \sum_{n=5,11,17,\ldots}^{\infty} \hat{I}_n \sin(n\omega t + \delta) \end{aligned} \tag{5.4-3}$$

The first $\sum$ on the right-hand side of (5.4-3) includes all the harmonic currents of **positive sequence** ($n = 1, 7, 13, \ldots$) while the second $\sum$ represents all the **negative sequence** harmonics ($n = 5, 11, 17, \ldots$).

Comparing the primary line current i'_a in (5.4-3) with the secondary line current i_a in (5.4-1), we have

$$\begin{cases} \angle i'_{an} = \angle i_{an} - \delta \quad \text{for} \quad n = 1, 7, 13, 19, \ldots \quad \text{(positive sequence harmonics)} \\ \angle i'_{an} = \angle i_{an} + \delta \quad \text{for} \quad n = 5, 11, 17, 23, \ldots \quad \text{(negative sequence harmonics)} \end{cases} \tag{5.4-4}$$

where $\angle i'_{an}$ and $\angle i'_{an}$ are the phase angles of nth order harmonic currents i'_{an} and i'_{an}, respectively. Equation 5.4-4 describes the relationship between the phase angles of the harmonic currents when referred from the secondary to the primary of the phase-shifting transformer. It can be proven that equation 5.4-4 is valid for any δ values.

5.4.2 Harmonic Cancellation

To illustrate how the harmonic currents are cancelled by a phase-shifting transformer, let us examine a 12-pulse rectifier shown in Fig. 5.4-2. The phase-shifting angle δ of the wye- and delta-connected secondary windings is zero and 30°, respectively. The

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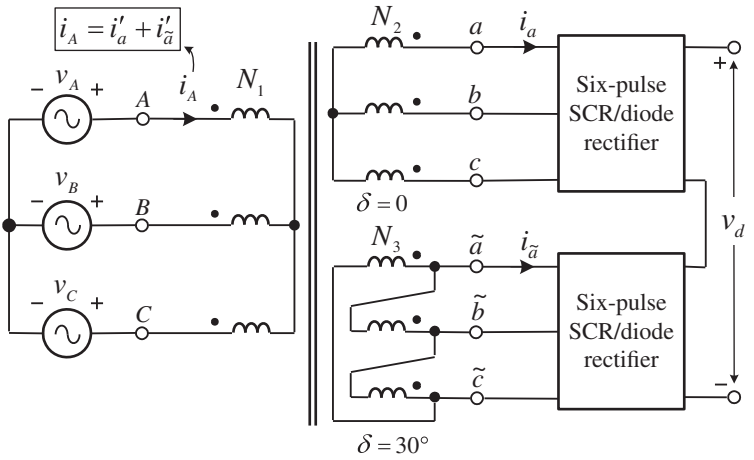

Figure 5.4-2 An example of harmonic current cancellation.

voltage ratio is $V_{AB}/V_{ab} = V_{AB}/V_{\tilde{a}\tilde{b}} = 2$. The line currents in the secondary windings can be expressed as

$$\begin{cases} i_a = \displaystyle\sum_{n=1,5,7,11,13,\ldots}^{\infty} \hat{I}_n \sin(n\omega t) \cdots \\ i_{\tilde{a}} = \displaystyle\sum_{n=1,5,7,11,13,\ldots}^{\infty} \hat{I}_n \sin\left(n(\omega t + \delta)\right) \end{cases} \tag{5.4-5}$$

When i_a is referred to the primary side, the phase angle of all the harmonic currents remains unchanged due to the Y/Y connection. The referred current i'_a is then given by

$$i'_a = \frac{1}{2}(\hat{I}_1 \sin(\omega t) + \hat{I}_5 \sin(5\omega t) + \hat{I}_7 \sin(7\omega t) + \hat{I}_{11} \sin(11\omega t) + \hat{I}_{13} \sin(13\omega t) + \cdots) \tag{5.4-6}$$

To transfer $i_{\tilde{a}}$ to the primary side, we can make use of (5.4-4), from which

$$\begin{aligned} i'_{\tilde{a}} &= \frac{1}{2}\left(\sum_{n=1,7,13,\ldots}^{\infty} \hat{I}_n \sin\left(n(\omega t + \delta) - \delta\right) + \sum_{n=5,11,17,\ldots}^{\infty} \hat{I}_n \sin\left(n(\omega t + \delta) + \delta\right)\right) \\ &= \frac{1}{2}(\hat{I}_1 \sin(\omega t) - \hat{I}_5 \sin(5\omega t) - \hat{I}_7 \sin(7\omega t) + \hat{I}_{11} \sin(11\omega t) + \hat{I}_{13} \sin(13\omega t) - \ldots) \\ &\quad \text{for } \delta = 30° \end{aligned} \tag{5.4-7}$$

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The primary line current i_A can then be found from

$$i_A = i'_a + i'_{\bar{a}} = \hat{I}_1 \sin \omega t + \hat{I}_{11} \sin 11\omega t + \hat{I}_{13} \sin 13\omega + \hat{I}_{23} \sin 23\omega + \cdots . \quad (5.4\text{-}8)$$

where the 5th, 7th, 17th, and 19th harmonic currents in i_a and i'_a are 180° out of phase, and therefore cancelled.

5.5 SUMMARY

To reduce the line current THD in high-power rectifiers, multipulse diode/SCR rectifiers powered by phase-shifting transformers are often employed. In this chapter, the typical configurations of the phase-shifting transformers for 12-, 18-, and 24-pulse rectifiers are presented. The structure and phasor diagrams of the transformers are discussed. To assist the transformer design, the relationship between the required phase-shifting angle and transformer turns ratio is tabulated. The principle of harmonic current cancellation by the phase-shifting transformers is also demonstrated.

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Part Three

Multilevel Voltage Source Converters

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Chapter 6

Two-Level Voltage Source Inverter

6.1 INTRODUCTION

The primary function of a voltage source inverter (VSI) is to convert a fixed dc voltage to a three-phase ac voltage with variable magnitude and frequency. A simplified circuit diagram for a two-level VSI for high-power medium voltage applications is shown in Fig. 6.1-1. The inverter is composed of six group of active switches, S_1–S_6, with a free-wheeling diode in parallel with each switch. Depending on the dc operating voltage of the inverter, each switch group consists of two or more IGBT or GCT switching devices connected in series.

This chapter focuses on **pulse width modulation** (PWM) schemes for the high-power two-level inverter, where the device switching frequency is normally below 1 kHz. A carrier based sinusoidal PWM (SPWM) scheme is reviewed, followed by a detailed analysis of **space vector modulation** (SVM) algorithms. The conventional SVM scheme usually generates both even- and odd-order harmonic voltages. The mechanism of even-order harmonic generation is analyzed, and a modified SVM scheme for even-order harmonic elimination is presented.

6.2 SINUSOIDAL PWM

6.2.1 Modulation Scheme

The principle of the SPWM scheme for the two-level inverter is illustrated in Fig. 6.2-1, where v_{mA}, v_{mB}, and v_{mC} are the three-phase sinusoidal **modulating**

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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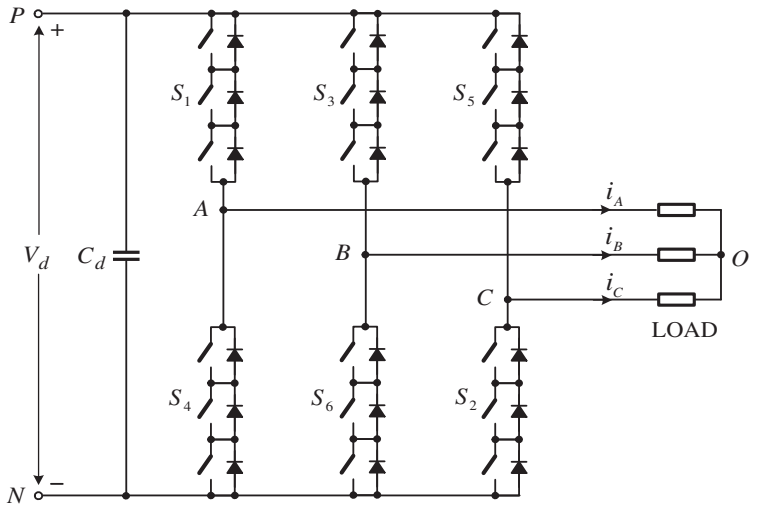

Figure 6.1-1 Simplified two-level inverter for high-power applications.

waves and v_{cr} is the triangular **carrier wave**. The fundamental-frequency component in the inverter output voltage can be controlled by **amplitude modulation index**

$$m_a = \frac{\hat{V}_m}{\hat{V}_{cr}} \tag{6.2-1}$$

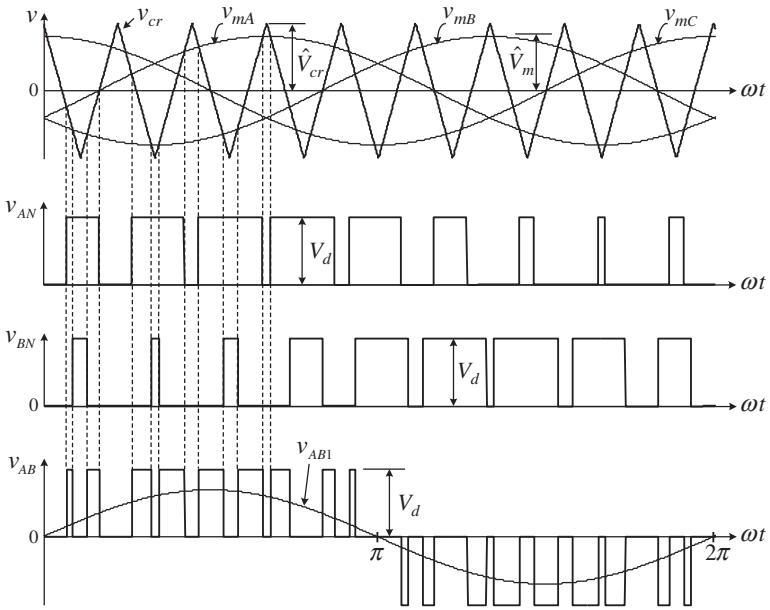

Figure 6.2-1 Sinusoidal pulse width modulation (SPWM).

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where $\hat{V}_m$ and $\hat{V}_{cr}$ are the peak values of the modulating and carrier waves, respectively. The amplitude modulation index m_a is usually adjusted by varying $\hat{V}_m$ while keeping $\hat{V}_{cr}$ fixed. The **frequency modulation index** is defined by

$$m_f = \frac{f_{cr}}{f_m} \tag{6.2-2}$$

where f_m and f_{cr} are the frequencies of the modulating and carrier waves, respectively.

The operation of switches S_1–S_6 is determined by comparing the modulating waves with the carrier wave. When $v_{mA} > v_{cr}$, the upper switch S_1 in inverter leg A is turned on. The lower switch S_4 operates in a complementary manner and thus is switched off. The resultant **inverter terminal voltage** v_{AN}, which is the voltage at the phase A terminal with respect to the negative dc bus N, is equal to the dc voltage V_d. When $v_{mA} < v_{cr}$, S_4 is on and S_1 is off, leading to $v_{AN} = 0$ as shown in Fig. 6.2-1. Since the waveform of v_{AN} has only two levels, V_d and 0, the inverter is known as a **two-level inverter**. It should be noted that to avoid possible short circuit during switching transients of the upper and lower devices in an inverter leg, a **blanking time** should be implemented, during which both switches are turned off.

The inverter line-to-line voltage v_{AB} can be determined by $v_{AB} = v_{AN} - v_{BN}$. The waveform of its fundamental-frequency component v_{AB1} is also given in the figure. The magnitude and frequency of v_{AB1} can be independently controlled by m_a and f_m, respectively.

The **switching frequency** of the active switches in the two-level inverter can be found from $f_{sw} = f_{cr} = f_m \times m_f$. For instance, v_{AN} in Fig. 6.2-1 contains nine pulses per cycle of the fundamental frequency. Each pulse is produced by turning S_1 on and off once. With the fundamental frequency of 60 Hz, the resultant switching frequency for S_1 is $f_{sw} = 60 \times 9 = 540$ Hz, which is also the carrier frequency f_{cr}. It is worth noting that the device switching frequency may not always be equal to the carrier frequency in multilevel inverters. This issue will be addressed in the later chapters.

When the carrier wave is synchronized with the modulating wave (m_f is an integer), the modulation scheme is known as **synchronous PWM** in contrast to **asynchronous PWM** whose carrier frequency f_{cr} is usually fixed and independent of f_m. The asynchronous PWM features a fixed switching frequency and easy implementation with analog circuits. However, it may generate **non-characteristic harmonics**, whose frequency is not a multiple of the fundamental frequency. The synchronous PWM scheme is more suitable for implementation with a digital processor.

6.2.2 Harmonic Content

Figure 6.2-2 shows a set of simulated waveforms for the two-level inverter, where v_{AB} is the inverter line-to-line voltage, v_{AO} is the **load phase voltage** and i_A is the load current. The inverter operates under the condition of $m_a = 0.8$, $m_f = 15$, $f_m = 60$ Hz,

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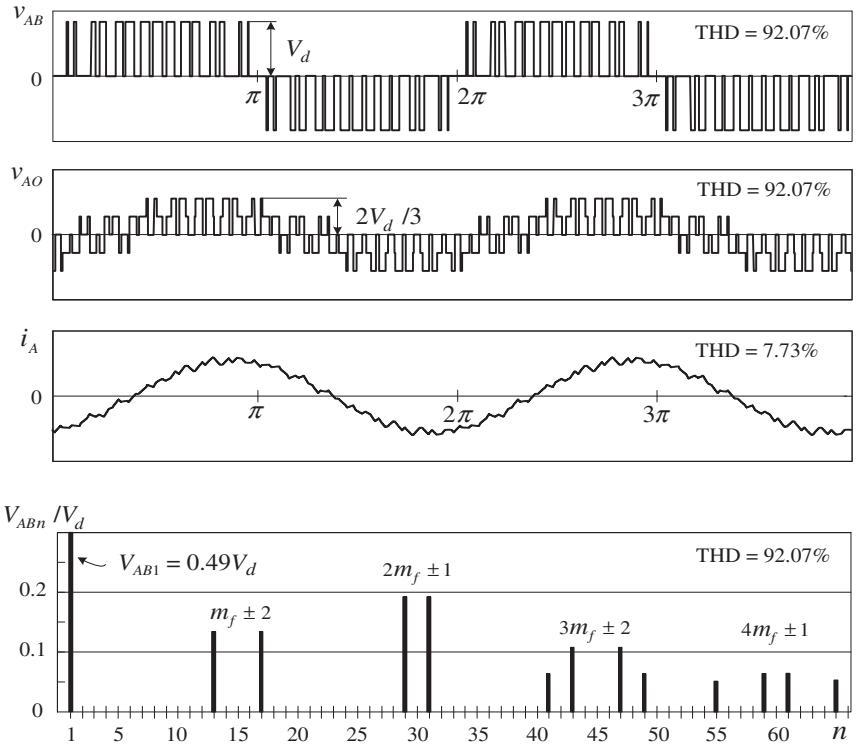

Figure 6.2-2 Simulated waveforms for the two-level inverter operating at $m_a = 0.8$, $m_f = 15$ and $f_m = 60$ Hz, $f_{sw} = 900$ Hz.

and $f_{sw} = 900$ Hz with a rated three-phase inductive load. The load power factor is 0.9 per phase. It can be observed that

- All the harmonics in v_{AB} with the order lower than $(m_f - 2)$ are eliminated.
- The harmonics are centered around m_f and its multiples such as $2m_f$ and $3m_f$.

The above statements are valid for $m_f \geq 9$ provided that m_f is a multiple of 3 [1].

The waveform of the load current i_A is close to sinusoidal with a THD of 7.73%. The low amount of harmonic distortion is due to the elimination of low-order harmonics by the modulation scheme and the filtering effect of the load inductance.

Figure 6.2-3 shows the harmonic content of the inverter line-to-line voltage v_{AB} normalized to its dc voltage V_d as a function of m_a, where V_{ABn} is the nth order harmonic voltage (rms). The fundamental-frequency component V_{AB1} increases linearly with m_a, whose maximum value can be found from

$$V_{AB1,\max} = 0.612\, V_d \quad \text{for} \quad m_a = 1 \tag{6.2-3}$$

The THD curve for v_{AB} is also given in the figure.

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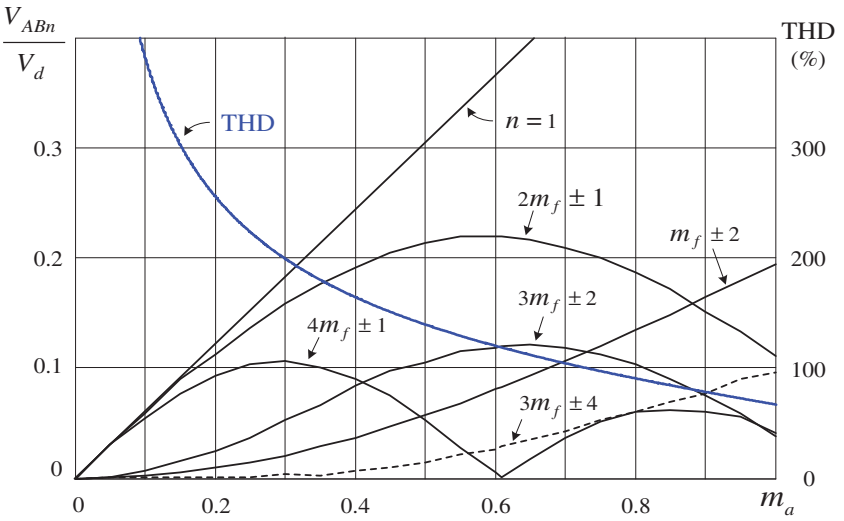

Figure 6.2-3 Harmonic content of v_{AB} in Fig. 6.2-2.

6.2.3 Over-Modulation

Over-modulation occurs when the amplitude modulation index m_a is greater than unity. Figure 6.2-4 shows such a case with $m_a = 2$. The over-modulation causes a reduction in number of pulses in the line-to-line voltage waveform, leading to the emergence of low-order harmonics such as the 5th and 11th. However, the fundamental voltage V_{AB1} is boosted to $0.744V_d$, which represents a 22% increase in comparison with $0.612V_d$ at $m_a = 1$. With m_a further increased to 3.24, v_{AB} becomes a square wave, whose fundamental voltage is $V_{AB1} = 0.78V_d$, which is the highest possible value produced by the two-level VSI. The over-modulation is seldom used in practice due to the difficulties to filter out the low-order harmonics and the nonlinear relationship between V_{AB1} and m_a.

6.2.4 Third Harmonic Injection PWM

The inverter fundamental voltage V_{AB1} can also be increased by adding a third harmonic component to the three-phase sinusoidal modulating wave without causing over-modulation. This modulation technique is known as **third harmonic injection** PWM.

Figure 6.2-5 illustrates the principle of this PWM scheme, where the modulating wave v_{mA} is composed of a fundamental component v_{m1} and a third harmonic component v_{m3}, making v_{mA} somewhat flattened on the top. As a result, the peak fundamental component $\hat{V}_{m1}$ can be higher than the peak triangular carrier wave $\hat{V}_{cr}$, which boosts the fundamental voltage V_{AB1}. In the meantime the peak modulating wave $\hat{V}_{mA}$ can be

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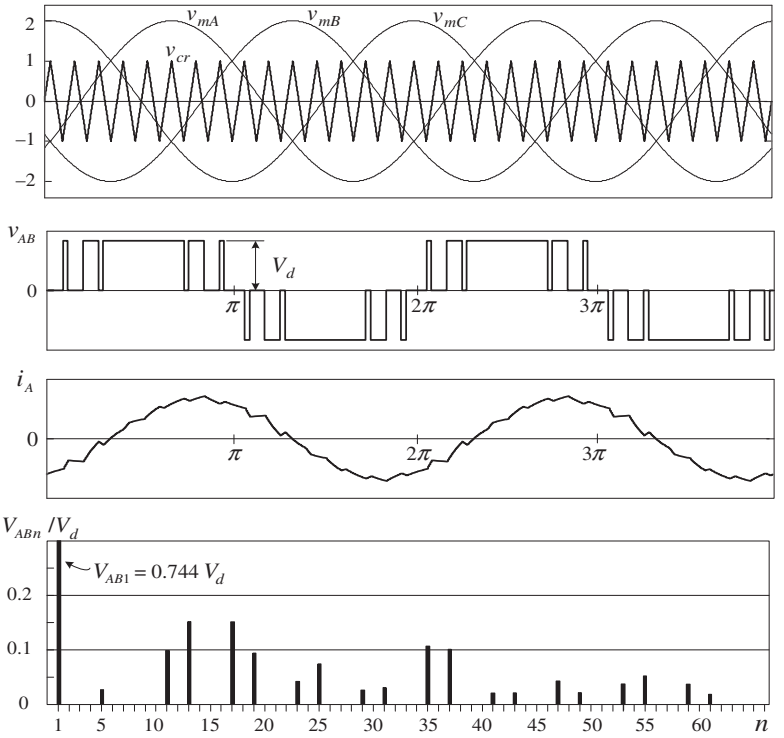

Figure 6.2-4 Over-modulation ($m_a = 2.0$, $m_f = 15$, and $f_m = 60$ Hz).

kept lower than $\hat{V}_{cr}$, avoiding the problems caused by over-modulation. The maximum amount of V_{AB1} that can be increased by this scheme is 15.5% [2, 3].

The injected third harmonic component v_{m3} will not increase the harmonic distortion for v_{AB}. Although it appears in each of the inverter terminal voltages v_{AN}, v_{BN}, and v_{CN}, the third-order harmonic voltage does not exist in the line-to-line voltage v_{AB}. This is because the line-to-line voltage is given by $v_{AB} = v_{AN} - v_{BN}$, where the

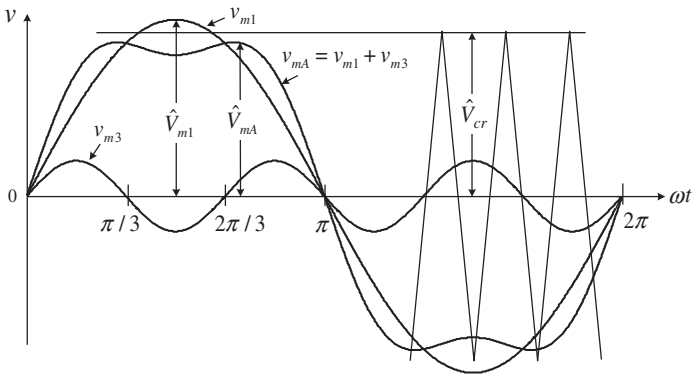

Figure 6.2-5 Modulating wave v_{mA} with 3rd harmonic injection.

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third-order harmonic in v_{AN} and v_{BN} is of zero sequence with the same magnitude and phase displacement, and thus cancels each other.

6.3 SPACE VECTOR MODULATION

SVM is one of the preferred real-time modulation techniques and is widely used for digital control of VSIs [3, 4]. This section presents the principle and implementation of the SVM for the two-level inverter.

6.3.1 Switching States

The operating status of the switches in the two-level inverter in Fig. 6.1-1 can be represented by **switching states**. As indicated in Table 6.3-1, switching state "P" denotes that the upper switch in an inverter leg is on and the inverter terminal voltage (v_{AN}, v_{BN}, or v_{CN}) is positive ($+V_d$) while "O" indicates that the inverter terminal voltage is zero due to the conduction of the lower switch.

There are eight possible combinations of switching states in the two-level inverter as listed in Table 6.3-2. The switching state [POO], for example, corresponds to the conduction of S_1, S_6, and S_2 in the inverter legs A, B, and C, respectively. Among the eight switching states, [PPP] and [OOO] are **zero states** and the others are **active states**.

6.3.2 Space Vectors

The active and zero switching states can be represented by active and zero **space vectors**, respectively. A typical space vector diagram for the two-level inverter is shown in Fig. 6.3-1, where the six **active vectors** $\vec{V}_1$ to $\vec{V}_6$ form a regular hexagon with six equal sectors (I–VI). The **zero vector** $\vec{V}_0$ lies on the center of the hexagon.

To derive the relationship between the space vectors and switching states, refer to the two-level inverter in Fig. 6.1-1. Assuming that the operation of the inverter is three-phase balanced, we have

$$v_{AO}(t) + v_{BO}(t) + v_{CO}(t) = 0 \tag{6.3-1}$$

where v_{AO}, v_{BO}, and v_{CO} are the instantaneous load phase voltages. From mathematical point of view, one of the phase voltages is redundant since given any two phase

Table 6.3-1 Definition of Switching States

	Leg A			Leg B			Leg C		
Switching State	S_1	S_4	v_{AN}	S_3	S_6	v_{BN}	S_5	S_2	v_{CN}
P	On	Off	V_d	On	Off	V_d	On	Off	V_d
O	Off	On	0	Off	On	0	Off	On	0

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Table 6.3-2 Space Vectors, Switching States, and On-State Switches

Space Vector		Switching State (Three Phases)	On-State Switch	Vector Definition
Zero Vector	$\vec{V}_0$	PPP	S_1, S_3, S_5	$\vec{V}_0 = 0$
		OOO	S_4, S_6, S_2	
Active Vector	$\vec{V}_1$	POO	S_1, S_6, S_2	$\vec{V}_1 = \frac{2}{3} V_d e^{j0}$
	$\vec{V}_2$	PPO	S_1, S_3, S_2	$\vec{V}_2 = \frac{2}{3} V_d e^{j\frac{\pi}{3}}$
	$\vec{V}_3$	OPO	S_4, S_3, S_2	$\vec{V}_3 = \frac{2}{3} V_d e^{j\frac{2\pi}{3}}$
	$\vec{V}_4$	OPP	S_4, S_3, S_5	$\vec{V}_4 = \frac{2}{3} V_d e^{j\frac{3\pi}{3}}$
	$\vec{V}_5$	OOP	S_4, S_6, S_5	$\vec{V}_5 = \frac{2}{3} V_d e^{j\frac{4\pi}{3}}$
	$\vec{V}_6$	POP	S_1, S_6, S_5	$\vec{V}_6 = \frac{2}{3} V_d e^{j\frac{5\pi}{3}}$

voltages, the third one can be readily calculated. Therefore, it is possible to transform the three-phase variables to equivalent two-phase variables [5]:

$$\begin{bmatrix} v_\alpha(t) \\ v_\beta(t) \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_{AO}(t) \\ v_{BO}(t) \\ v_{CO}(t) \end{bmatrix} \tag{6.3-2}$$

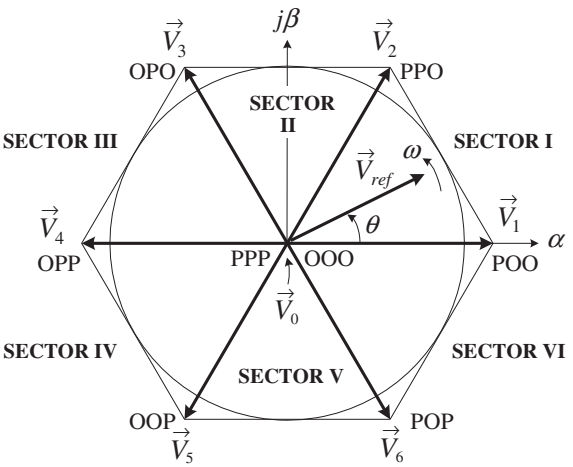

Figure 6.3-1 Space vector diagram for the two-level inverter.

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The coefficient $2/3$ is somewhat arbitrarily chosen. The commonly used value is $2/3$ or $\sqrt{2/3}$. The main advantage of using $2/3$ is that the magnitude of the two-phase voltages will be equal to that of the three-phase voltages after the transformation. A space vector can be generally expressed in terms of the two-phase voltages in the α–β plane

$$\vec{V}(t) = v_\alpha(t) + j\, v_\beta(t) \tag{6.3-3}$$

Substituting (6.3-2) into (6.3-3), we have

$$\vec{V}(t) = \frac{2}{3}\left[v_{AO}(t)\, e^{j0} + v_{BO}(t)\, e^{j2\pi/3} + v_{CO}(t)\, e^{j4\pi/3}\right] \tag{6.3-4}$$

where $e^{jx} = \cos x + j \sin x$ and $x = 0, 2\pi/3, or\, 4\pi/3$. For active switching state [POO], the generated load phase voltages are

$$v_{AO}(t) = \frac{2}{3} V_d, \quad v_{BO}(t) = -\frac{1}{3} V_d, \quad \text{and} \quad v_{CO}(t) = -\frac{1}{3} V_d \tag{6.3-5}$$

The corresponding space vector, denoted as $\vec{V}_1$, can be obtained by substituting (6.3-5) into (6.3-4):

$$\vec{V}_1 = \frac{2}{3} V_d\, e^{j0} \tag{6.3-6}$$

Following the same procedure, all six active vectors can be derived

$$\vec{V}_k = \frac{2}{3} V_d\, e^{j(k-1)\frac{\pi}{3}} \quad k = 1, 2, ..., 6 \tag{6.3-7}$$

The zero vector $\vec{V}_0$ has two switching states [PPP] and [OOO], one of which seems redundant. As will be seen later, the **redundant switching state** can be utilized to minimize the switching frequency of the inverter or perform other useful functions. The relationship between the space vectors and their corresponding switching states is given in Table 6.3-2

Note that the zero and active vectors do not move in space, and thus are referred to as **stationary vectors**. On the contrary, the **reference vector** $\vec{V}_{ref}$ in Fig. 6.3-1 rotates in space at an angular velocity

$$\omega = 2\pi f_1 \tag{6.3-8}$$

where f_1 is the fundamental frequency of the inverter output voltage. The angular displacement between $\vec{V}_{ref}$ and the α-axis of the α–β plane can be obtained by

$$\theta(t) = \int_0^t \omega(t)\, dt + \theta(0) \tag{6.3-9}$$

For a given magnitude (length) and position, $\vec{V}_{ref}$ can be synthesized by three nearby stationary vectors, based on which the switching states of the inverter can be selected and gate signals for the active switches can be generated. When $\vec{V}_{ref}$ passes

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through sectors one by one, different sets of switches will be turned on or off. As a result, when $\vec{V}_{ref}$ rotates one revolution in space, the inverter output voltage varies one cycle over time. The inverter output frequency corresponds to the rotating speed of $\vec{V}_{ref}$ while its output voltage can be adjusted by the magnitude of $\vec{V}_{ref}$.

6.3.3 Dwell Time Calculation

As mentioned earlier, the reference $\vec{V}_{ref}$ can be synthesized by three stationary vectors. The dwell time for the stationary vectors essentially represents the duty-cycle time (on-state or off-state time) of the chosen switches during a **sampling period** T_s. The dwell time calculation is based on "**volt-second balancing"** principle, that is, the product of the reference voltage $\vec{V}_{ref}$ and sampling period T_s equals the sum of the voltage multiplied by the time interval of chosen space vectors.

Assuming that the sampling period T_s is sufficiently small, the reference vector $\vec{V}_{ref}$ can be considered constant during T_s. Under this assumption, $\vec{V}_{ref}$ can be approximated by two adjacent active vectors and one zero vector. For example, when $\vec{V}_{ref}$ falls into sector I as shown in Fig. 6.3-2, it can be synthesized by $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_0$. The volt-second balancing equation is

$$\begin{cases} \vec{V}_{ref}\, T_s = \vec{V}_1\, T_a + \vec{V}_2\, T_b + \vec{V}_0\, T_0 \\ T_s = T_a + T_b + T_0 \end{cases} \tag{6.3-10}$$

where T_a, T_b, and T_0 are the **dwell times** for the vectors $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_0$, respectively. The space vectors in (6.3-10) can be expressed as

$$\vec{V}_{ref} = V_{ref}\, e^{j\theta},\ \vec{V}_1 = \frac{2}{3} V_d,\ \vec{V}_2 = \frac{2}{3} V_d\, e^{j\frac{\pi}{3}},\ \text{and } \vec{V}_0 = 0 \tag{6.3-11}$$

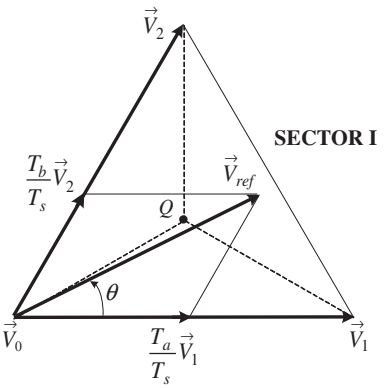

Figure 6.3-2 $\vec{V}_{ref}$ synthesized by $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_0$.

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Substituting (6.3-11) into (6.3-10) and then splitting the resultant equation into the real (α-axis) and imaginary (β-axis) components in the α–β plane, we have

$$\begin{cases} \text{Re:} & V_{ref}(\cos\theta)\,T_s = \dfrac{2}{3}\,V_d\,T_a + \dfrac{1}{3}\,V_d\,T_b \\ \text{Im:} & V_{ref}(\sin\theta)\,T_s = \dfrac{1}{\sqrt{3}}V_d\,T_b \end{cases} \tag{6.3-12}$$

Solving (6.3-12) together with $T_s = T_a + T_b + T_0$ yields

$$\begin{cases} T_a = \dfrac{\sqrt{3}\,T_s\,V_{ref}}{V_d}\sin\left(\dfrac{\pi}{3} - \theta\right) \\ T_b = \dfrac{\sqrt{3}\,T_s\,V_{ref}}{V_d}\sin\theta \qquad \text{for } 0 \le \theta < \pi/3 \\ T_0 = T_s - T_a - T_b \end{cases} \tag{6.3-13}$$

To visualize the relationship between the location of $\vec{V}_{ref}$ and the dwell times, let us examine some special cases. If $\vec{V}_{ref}$ lies exactly in the middle between $\vec{V}_1$ and $\vec{V}_2$ (i.e., $\theta = \pi/6$), the dwell time T_a for $\vec{V}_1$ will be equal to T_b for $\vec{V}_2$. When $\vec{V}_{ref}$ is closer to $\vec{V}_2$ than $\vec{V}_1$, T_b will be greater than T_a. If $\vec{V}_{ref}$ is coincident with $\vec{V}_2$, T_a will be zero. With the head of $\vec{V}_{ref}$ located right on the central point Q, $T_a = T_b = T_0$. The relationship between the $\vec{V}_{ref}$ location and dwell times is summarized in Table 6.3-3.

Note that although equation 6.3-13 is derived when $\vec{V}_{ref}$ is in sector I, it can also be used when $\vec{V}_{ref}$ is in other sectors provided that a multiple of $\pi/3$ is subtracted from the actual angular displacement θ such that the modified angle θ' falls into the range between zero and $\pi/3$ for use in the equation, that is,

$$\theta' = \theta - (k-1)\,\pi/3 \text{ for } 0 \le \theta' < \pi/3 \tag{6.3-14}$$

where $k = 1, 2, \ldots, 6$ for sectors I, II, …, VI, respectively. For example, when $\vec{V}_{ref}$ is in sector II, the calculated dwell times T_a, T_b, and T_0 based on (6.3-13) and (6.3-14) are for vectors $\vec{V}_2$, $\vec{V}_3$, and $\vec{V}_0$, respectively.

Table 6.3-3 $\vec{V}_{ref}$ Location and Dwell Times

$\vec{V}_{ref}$ Location	$\theta = 0$	$0 < \theta < \dfrac{\pi}{6}$	$\theta = \dfrac{\pi}{6}$	$\dfrac{\pi}{6} < \theta < \dfrac{\pi}{3}$	$\theta = \dfrac{\pi}{3}$
Dwell Times	$T_a > 0$ $T_b = 0$	$T_a > T_b$	$T_a = T_b$	$T_a < T_b$	$T_a = 0$ $T_b > 0$

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6.3.4 Modulation Index

Equation 6.3-13 can be also expressed in terms of **modulation index** m_a

$$\begin{cases} T_a = T_s m_a \sin\left(\frac{\pi}{3} - \theta\right) \\ T_b = T_s m_a \sin\theta \\ T_0 = T_s - T_a - T_b \end{cases} \tag{6.3-15}$$

where

$$m_a = \frac{\sqrt{3}\, V_{ref}}{V_d} \tag{6.3-16}$$

The maximum magnitude of the reference vector, $V_{ref,\max}$, corresponds to the radius of the largest circle that can be inscribed within the hexagon as shown in Fig. 6.3-1. Since the hexagon is formed by six active vectors having a length of $2\,V_d/3$, $V_{ref,\max}$ can be found from

$$V_{ref,\max} = \frac{2}{3} V_d \times \frac{\sqrt{3}}{2} = \frac{V_d}{\sqrt{3}} \tag{6.3-17}$$

Substituting (6.3-17) into (6.3-16) gives the maximum modulation index

$$m_{a,\max} = 1$$

from which the modulation index for the SVM scheme is in the range of

$$0 \le m_a \le 1 \tag{6.3-18}$$

The maximum fundamental line-to-line voltage (rms) produced by the SVM scheme can be calculated by

$$V_{\max,SVM} = \sqrt{3}\left(V_{ref,\max}/\sqrt{2}\right) = 0.707 V_d \tag{6.3-19}$$

where $V_{ref,\max}/\sqrt{2}$ is the maximum rms value of the fundamental phase voltage of the inverter.

With the inverter controlled by the SPWM scheme, the maximum fundamental line-to-line voltage is

$$V_{\max,SPWM} = 0.612\, V_d \tag{6.3-20}$$

from which

$$\frac{V_{\max,SVM}}{V_{\max,SPWM}} = 1.155 \tag{6.3-21}$$

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Equation (6.3-21) indicates that for a given dc bus voltage the maximum inverter line-to-line voltage generated by the SVM scheme is 15.5% higher than that by the SPWM scheme. However, the use of third harmonic injection SPWM scheme can also boost the inverter output voltage by 15.5%. Therefore, the two schemes have essentially the same **dc bus voltage utilization**.

6.3.5 Switching Sequence

With the space vectors selected and their dwell times calculated, the next step is to arrange switching sequence. In general, the switching sequence design for a given $\vec{V}_{ref}$ is not unique, but it should satisfy the following two requirements for the minimization of the device switching frequency:

a. The transition from one switching state to the next involves only two switches in the same inverter leg, one being switched on and the other switched off.

b. The transition for $\vec{V}_{ref}$ moving from one sector in the space vector diagram to the next requires no or minimum number of switchings.

Figure 6.3-3 shows a typical **seven-segment** switching sequence and inverter output voltage waveforms for $\vec{V}_{ref}$ in sector I, where $\vec{V}_{ref}$ is synthesized by $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_0$. The sampling period T_s is divided into seven segments for the selected vectors. It can be observed that

- The dwell times for the seven segments add up to the sampling period ($T_s = T_a + T_b + T_0$).

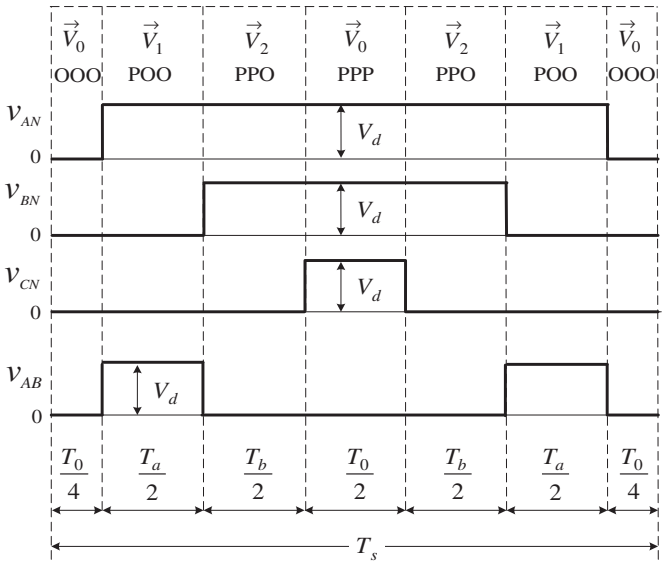

Figure 6.3-3 Seven-segment switching sequence for $\vec{V}_{ref}$ in sector I.

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- The design requirement (*a*) is satisfied. For instance, the transition from [OOO] to [POO] is accomplished by turning S_1 on and S_4 off, which involves only two switches.
- The redundant switching states for $\vec{V}_0$ are utilized to reduce the number of switchings per sampling period. For the $T_0/2$ segment in the center of the sampling period, the switching state [PPP] is selected while for the $T_0/4$ segments on both sides, the state [OOO] is used.
- Each of the switches in the inverter turns on and off once per sampling period. The **switching frequency** f_{sw} of the devices is thus equal to the **sampling frequency** f_{sp}, that is, $f_{sw} = f_{sp} = 1/T_s$.

Let us now examine a case given in Fig. 6.3-4, where the vectors $\vec{V}_1$ and $\vec{V}_2$ in Fig. 6.3-3 are swapped. Some switching state transitions, such as the transition from [OOO] to [PPO], are accomplished by turning on and off four switches in two inverter legs simultaneously. As a consequence, the total number of switchings during the sampling period increases from six in the previous case to ten. Obviously, this switching sequence does not satisfy the design requirement and thus should not be adopted.

It is interesting to note that the waveforms of v_{AB} in Figs. 6.3-3 and 6.3-4 produced by two different switching sequences seem different, but they are essentially the same. If these two waveforms are drawn for two or more consecutive sampling periods, we will notice that they are identical except for a small time delay ($T_s/2$). Since T_s is much shorter than the period of the inverter fundamental frequency, the effect caused by the time delay is negligible.

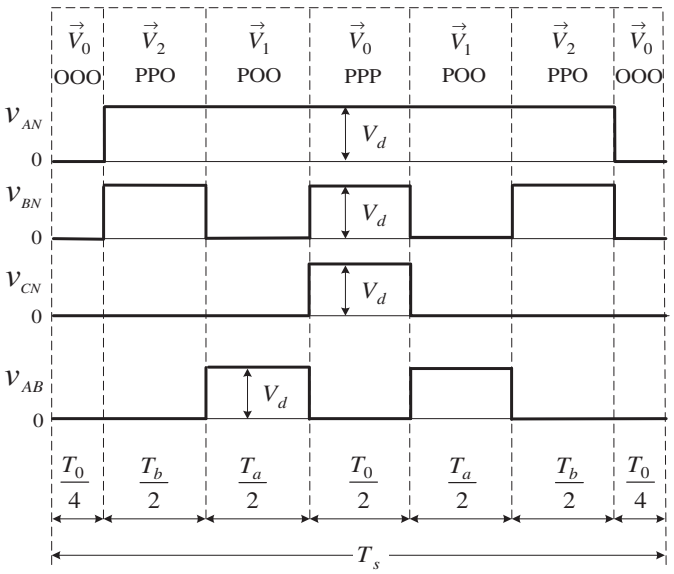

Figure 6.3-4 Undesirable seven-segment switching sequence.

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Table 6.3-4 Seven-Segment Switching Sequence

Sector	Switching Segment						
	1	2	3	4	5	6	7
I	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_2$	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_1$	$\vec{V}_0$
	OOO	POO	PPO	PPP	PPO	POO	OOO
II	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_2$	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_3$	$\vec{V}_0$
	OOO	OPO	PPO	PPP	PPO	OPO	OOO
III	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_4$	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_3$	$\vec{V}_0$
	OOO	OPO	OPP	PPP	OPP	OPO	OOO
IV	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_4$	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_5$	$\vec{V}_0$
	OOO	OOP	OPP	PPP	OPP	OOP	OOO
V	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_6$	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_5$	$\vec{V}_0$
	OOO	OOP	POP	PPP	POP	OOP	OOO
VI	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_6$	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_1$	$\vec{V}_0$
	OOO	POO	POP	PPP	POP	POO	OOO

Table 6.3-4 gives the seven-segment switching sequences for $\vec{V}_{ref}$ residing in all six sectors. Note that all the switching sequences start and end with switching state [OOO], which indicates that the transition for $\vec{V}_{ref}$ moving from one sector to the next does not require any switchings. The switching sequence design requirement (b) is satisfied.

6.3.6 Spectrum Analysis

The simulated waveforms for the inverter output voltages and load current are shown in Fig. 6.3-5. The inverter operates under the condition of $f_1 = 60$ Hz, $T_s = 1/720$ s, $f_{sw} = 720$ Hz, and $m_a = 0.8$ with a rated three-phase inductive load. The load power factor is 0.9 per phase. It can be observed that the waveform of the inverter line-to-line voltage v_{AB} is not **half-wave symmetrical**, that is, $v_{AB}(\omega t) \neq -v_{AB}(\omega t + \pi)$. Therefore, it contains even-order harmonics, such as 2nd, 4th 8th, and 10th, in addition to odd-order harmonics. The THD of v_{AB} and i_A is 80.2% and 8.37%, respectively.

Figure 6.3-6 shows waveforms measured from a laboratory two-level inverter operating under the same conditions as those given in Fig. 6.3-5. The top and bottom traces in Fig. 6.3-6a are the inverter line-to-line voltage v_{AB} and load phase voltage v_{AO}, and the spectrum of v_{AB} is given in Fig. 6.3-6b. The experimental results match with the simulation very well.

Figure 6.3-7 shows the harmonic content of v_{AB} for the inverter operating at $f_1 = 60$ Hz and $f_{sw} = 720$ Hz. Although the low-order harmonics, such as 2nd, 4th

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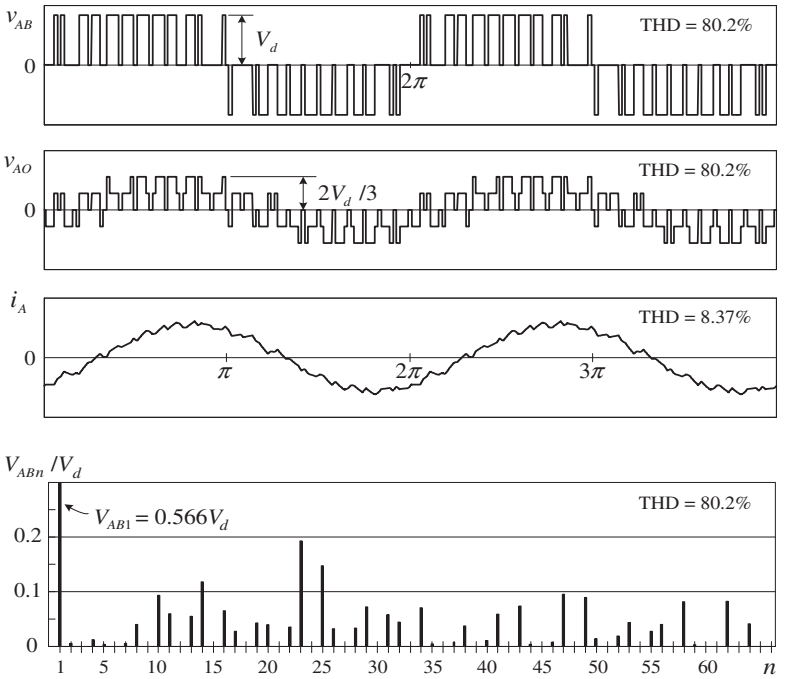
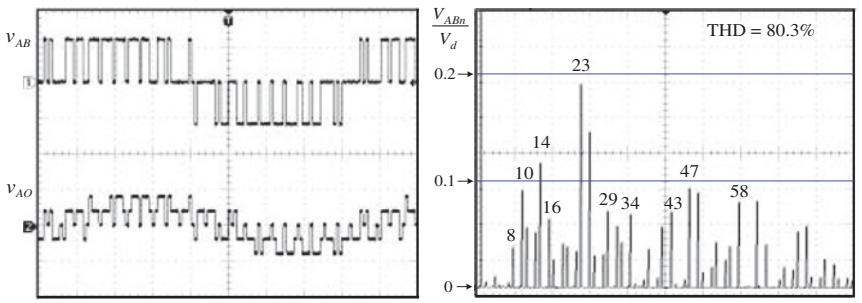


Figure 6.3-5 Inverter output waveforms produced by SVM scheme with $f_1 = 60$ Hz, $f_{sw} = 720$ Hz, and $m_a = 0.8$.

5th, and 7th, are not eliminated, they have very low magnitudes. The maximum fundamental line-to-line voltage (rms) occurs at $m_a = 1$ and can be found from

$$V_{AB1,\max} = 0.707V_d \text{ for } m_a = 1 \tag{6.3-22}$$

which is around 15.5% higher than that given in (6.2-3) for the SPWM scheme without using the third harmonic injection technique.



(a) Waveforms 2 ms/div (b) Spectrum (500 Hz/div)

Figure 6.3-6 Measured inverter voltage waveforms and harmonic spectrum for the verification of simulated waveforms in Fig. 6.3.5.

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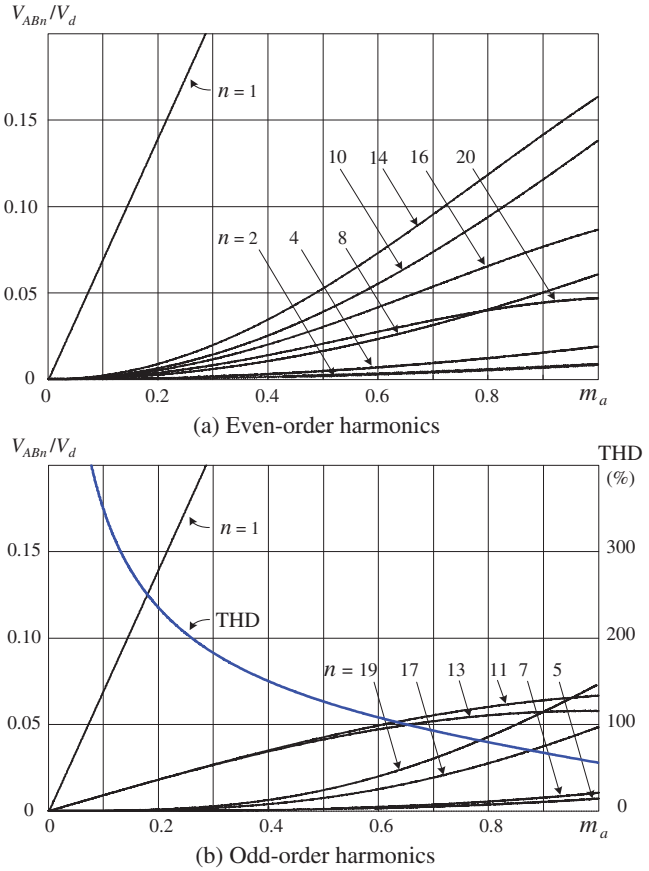

Figure 6.3-7 Harmonic content of v_{AB} with $f_1 = 60$ Hz and $f_{sw} = 720$ Hz.

6.3.7 Even-Order Harmonic Elimination

As indicated earlier, the line-to-line voltage waveform produced by the SVM inverter contains even-order harmonics. In the inverter fed medium voltage drives, these harmonics may not have a significant impact on the operation of the motor. However, when the two-level converter is used as a rectifier, its line current THD should comply with harmonic standards such as IEEE 519-2014 [6]. Since most standards have more stringent requirements on even-order harmonics than odd ones, this section presents a modified SVM scheme with even-order harmonic elimination.

To investigate the mechanism of even-order harmonic generation, consider a case where the reference vector $\vec{V}_{ref}$ falls into sector IV. Based on the switching sequence given in Table 6.3-4, the waveform of inverter line-to-line voltage v_{AB} in a sampling period is illustrated in Fig. 6.3-8a. The waveform does not have a mirror image (not symmetrical about the horizontal axis) with that in Fig. 6.3-3, where $\vec{V}_{ref}$ is in a

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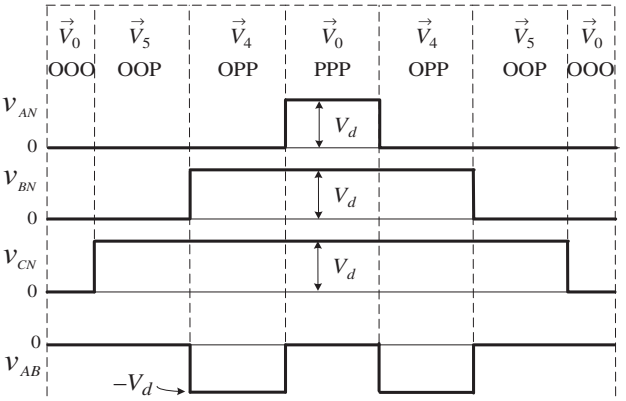
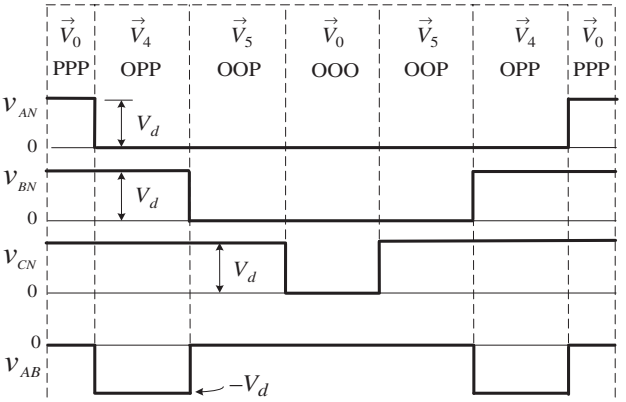


(a) Type-A sequence (starts and ends with [OOO])



(b) Type-B sequence (starts and ends with [PPP])

Figure 6.3-8 Two valid switching sequences for $\vec{V}_{ref}$ in sector IV.

sector 180° apart from sector IV. This implies that the waveform generated by the SVM scheme is not half-wave symmetrical, leading to the generation of even-order harmonics.

Let us now consider type-B switching sequence shown in Fig. 6.3-8b, which is also a valid switching sequence that satisfies the design requirement (*a*) stated earlier. By comparing the waveform of v_{AB} with that in Fig. 6.3-3, it is clear that the use of this switching sequence would lead to $v_{AB}(\omega t) = -v_{AB}(\omega t + \pi)$. As a result, the waveform of v_{AB} would not contain any even-order harmonics.

Examining the two switching sequences in Fig. 6.3-8, we can find out that the type-A sequence starts and ends with [OOO] while the type-B sequence commences and finishes with [PPP]. The waveforms of v_{AB} generated by both sequences seem

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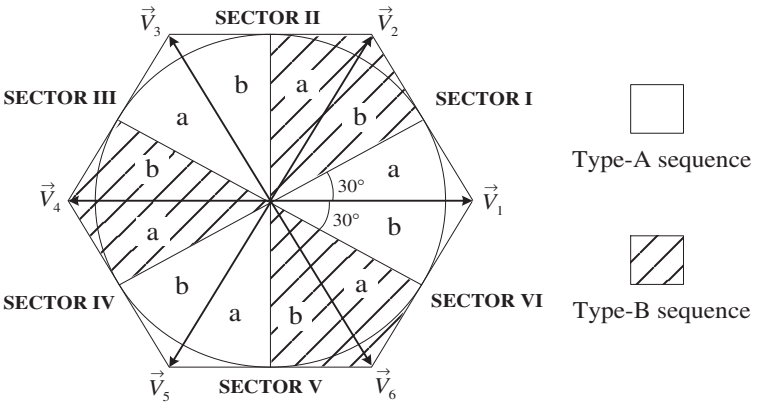

Figure 6.3-9 Alternative use of two switching sequences for even-order harmonic elimination.

different. However, they are essentially the same except for a small time delay ($T_s/2$) if these waveforms are drawn for two or more consecutive sampling periods.

To make the three-phase line-to-line voltage half-wave symmetrical, type A and B switching sequences can be alternatively used. In addition, each sector in the space vector diagram is divided into two regions as shown in Fig. 6.3-9. Type-A sequence is used in the non-shaded regions while type-B sequence is employed in the shaded regions. The detailed switching sequence arrangements are given in Table 6.3-5.

It can be observed from the table that the transition for $\vec{V}_{ref}$ moving from region a to b causes additional switchings. This implies that the even-order harmonic elimination is achieved at the expense of an increase in switching frequency. The amount of switching frequency increase can be determined by

$$\Delta f_{sw} = 3f_1 \tag{6.3-23}$$

where f_1 is the fundamental frequency of the inverter output voltage.

The inverter output waveforms measured form a laboratory two-level inverter with modified SVM scheme are shown in Fig. 6.3-10. The inverter operates under the condition of $f_1 = 60$ Hz, $T_s = 1/720$ s, and $m_a = 0.8$. The waveforms of the inverter line-to-line voltage v_{AB} and load phase voltage v_{AO} are of half-wave symmetry, containing no even-order harmonics. Compared with the harmonic spectrum given in Fig. 6.3-6, the magnitude of the 5th and 7th harmonics in v_{AB} is increased while the THD essentially remains the same.

6.3.8 Discontinuous Space Vector Modulation

As pointed out earlier, the switching sequence design is not unique for a given set of stationary vectors and dwell times. Figure 6.3-11 shows two **five-segment** switching sequences and generated inverter terminal voltages for $\vec{V}_{ref}$ in sector I. For type-A

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Table 6.3-5 Switching Sequence of the Modified SVM for Even-Order Harmonic Elimination

Sector	Switching Sequence						
I-a	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_2$	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_1$	$\vec{V}_0$
	OOO	POO	PPO	PPP	PPO	POO	OOO
I-b	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_1$	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_2$	$\vec{V}_0$
	PPP	PPO	POO	OOO	POO	PPO	PPP
II-a	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_3$	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_2$	$\vec{V}_0$
	PPP	PPO	OPO	OOO	OPO	PPO	PPP
II-b	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_2$	$\vec{V}_0$	$\vec{V}_2$	$\vec{V}_3$	$\vec{V}_0$
	OOO	OPO	PPO	PPP	PPO	OPO	OOO
III-a	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_4$	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_3$	$\vec{V}_0$
	OOO	OPO	OPP	PPP	OPP	OPO	OOO
III-b	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_3$	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_4$	$\vec{V}_0$
	PPP	OPP	OPO	OOO	OPO	OPP	PPP
IV-a	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_5$	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_4$	$\vec{V}_0$
	PPP	OPP	OOP	OOO	OOP	OPP	PPP
IV-b	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_4$	$\vec{V}_0$	$\vec{V}_4$	$\vec{V}_5$	$\vec{V}_0$
	OOO	OOP	OPP	PPP	OPP	OOP	OOO
V-a	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_6$	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_5$	$\vec{V}_0$
	OOO	OOP	POP	PPP	POP	OOP	OOO
V-b	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_5$	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_6$	$\vec{V}_0$
	PPP	POP	OOP	OOO	OOP	POP	PPP
VI-a	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_1$	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_6$	$\vec{V}_0$
	PPP	POP	POO	OOO	POO	POP	PPP
VI-b	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_6$	$\vec{V}_0$	$\vec{V}_6$	$\vec{V}_1$	$\vec{V}_0$
	OOO	POO	POP	PPP	POP	POO	OOO

sequence, the zero switching sate [OOO] is assigned for $\vec{V}_0$ while type-B sequence utilizes [PPP] for $\vec{V}_0$.

In the five-segment sequence, one of the three inverter output terminals is clamped either to the positive or negative dc bus without any switchings during the sampling period T_s. Further, the switching sequence can be arranged such that the switching in an inverter leg is continuously suppressed for a period of $2\pi/3$ per cycle of the fundamental frequency. For instance, the inverter terminal voltage v_{CN} can be clamped

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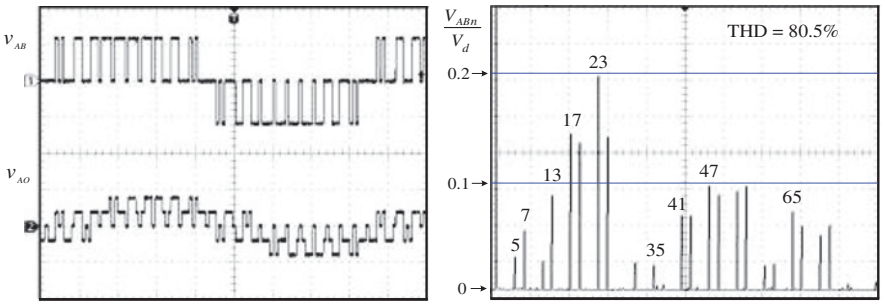


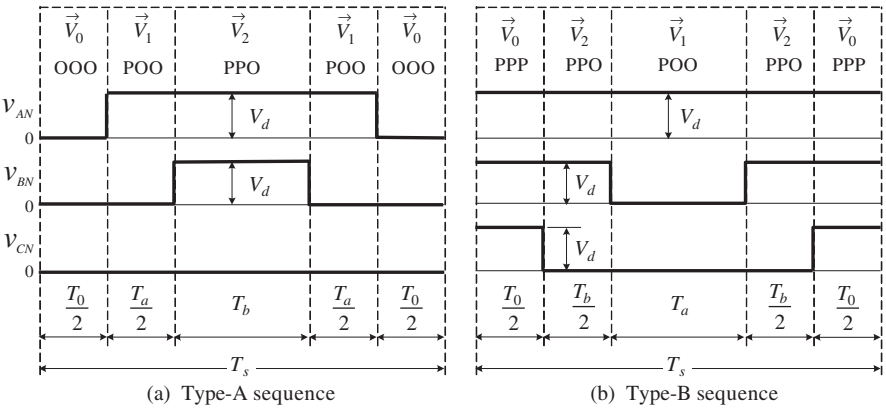
(a) Waveforms 2 ms/div (b) Spectrum (500 Hz/div)

Figure 6.3-10 Measured waveforms produced by the modified SVM with even-order harmonic elimination ($f_1 = 60$ Hz, $T_s = 1/720$ s, and $m_a = 0.8$).

to the negative dc bus continuously in sectors I and II as shown in Table 6.3-6. Due to the switching discontinuity, the five-segment scheme is also known as **discontinuous space vector modulation** [4].

The use of type-A sequence alone will make the conduction angle of the lower switch in an inverter leg longer than that of the upper switch, causing unbalanced power and thermal distributions. The problem can be mitigated by swapping the two types of the switching sequences periodically. The switching frequency of the inverter will increase accordingly.

Figure 6.3-12 shows the simulated waveforms for v_{AB} and i_A when the inverter operates at $f_1 = 60$ Hz, $f_{sw} = 600$ Hz, $T_s = 1/900$ s, and $m_a = 0.8$ with a rated three-phase inductive load. The load power factor is 0.9 per phase. Since the gate signals for S_1, S_3, and S_5 are suppressed continuously for a period of $2\pi/3$ per cycle of the fundamental frequency, the switching frequency of the five-segment sequence is reduced by $1/3$ compared with the seven-segment sequence with the same sampling period. The waveform of v_{AB} is not half-wave symmetrical, containing large amount



(a) Type-A sequence (b) Type-B sequence

Figure 6.3-11 Five-segment switching sequence.

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Table 6.3-6 Five-Segment Switching Sequence

Sector	Switching Sequence (Type A)					
I	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_2$	$\vec{V}_1$	$\vec{V}_0$	$v_{CN} = 0$
	OOO	POO	PPO	POO	OOO	
II	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_2$	$\vec{V}_3$	$\vec{V}_0$	$v_{CN} = 0$
	OOO	OPO	PPO	OPO	OOO	
III	$\vec{V}_0$	$\vec{V}_3$	$\vec{V}_4$	$\vec{V}_3$	$\vec{V}_0$	$v_{AN} = 0$
	OOO	OPO	OPP	OPO	OOO	
IV	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_4$	$\vec{V}_5$	$\vec{V}_0$	$v_{AN} = 0$
	OOO	OOP	OPP	OOP	OOO	
V	$\vec{V}_0$	$\vec{V}_5$	$\vec{V}_6$	$\vec{V}_5$	$\vec{V}_0$	$v_{BN} = 0$
	OOO	OOP	POP	OOP	OOO	
VI	$\vec{V}_0$	$\vec{V}_1$	$\vec{V}_6$	$\vec{V}_1$	$\vec{V}_0$	$v_{BN} = 0$
	OOO	POO	POP	POO	OOO	

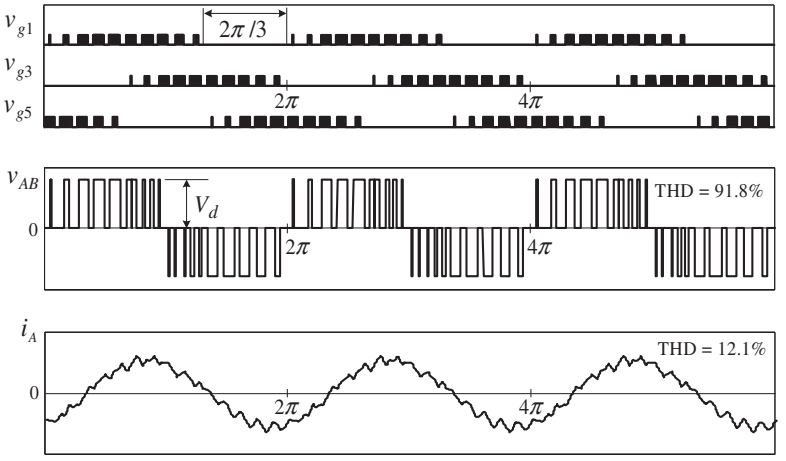

Figure 6.3-12 Waveforms produced by five-segment SVM with $f_1 = 60$ Hz, $f_{sw} = 600$ Hz, $T_s = 1/900$ s, and $m_a = 0.8$.

of even-order harmonics. The THD of v_{AB} and i_A are 91.8% and 12.1%, respectively, which are higher than those in the seven-segment sequence. This is mainly caused by the reduction of switching frequencies.

6.4 SUMMARY

This chapter focuses on PWM schemes for the two-level VSI. The switching frequency of the inverter is usually limited to a few hundred hertz for high-power

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medium voltage (MV) drives. A carrier based SPWM scheme is reviewed, followed by a detailed analysis on SVM algorithms, including derivation of space vectors, calculation of dwell times, design of switching sequence, and analysis of harmonic spectrum and THD.

The SVM schemes usually generate both odd- and even-order harmonics in the inverter output voltages. The even-order harmonics may not have a significant impact on the operation of the motor. However, they are strictly regulated by harmonic guidelines such as IEEE Standard 519-2014 when the two-level converter is used as a rectifier in the MV drive. Since the two-level voltage source rectifier is not separately discussed in the book, the mechanism of even-order harmonic generation is analyzed and a modified SVM scheme is presented.

The two-level inverter has a number of features, including simple converter topology and modular structure for the switching devices. However, it produces high dv/dt and THD in its output voltage, and therefore often requires a large size LC filter installed at its output terminals. Other advantages and drawbacks of the two-level inverter for use in the MV drive will be elaborated in Chapter 12.

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Chapter 7

Cascaded H-Bridge Multilevel Inverters

7.1 INTRODUCTION

Cascaded H-bridge (CHB) multilevel inverter is one of the popular converter topologies used in high-power medium voltage (MV) drives [1–3]. It is composed of multiple units of single-phase H-bridge power cells. The H-bridge cells are normally connected in cascade on their ac side to achieve medium voltage operation and low harmonic distortion. In practice, the number of power cells in a CHB inverter is mainly determined by its operating voltage and manufacturing cost. For instance, in the MV drives with a rated line-to-line voltage of 3300 V, a seven-level inverter can be used, where the CHB inverter has a total of 12 power cells using 600 V class components [1]. The use of identical power cells leads to a modular structure, which is an effective means for cost reduction.

The CHB multilevel inverter requires a number of isolated dc supplies, each of which feeds an H-bridge power cell. The dc supplies are normally obtained from multipulse diode rectifiers presented in Chapter 3. For the seven- and nine-level inverters, 18- and 24-pulse diode rectifiers can be employed, respectively, to achieve low line-current harmonic distortion and high input power factor.

In this chapter, the single-phase H-bridge power cell, which is the building block for the CHB inverter, is reviewed. Various inverter topologies are introduced. Two carrier-based PWM schemes, phase-shifted and level-shifted modulations, are analyzed and their performance is compared. A staircase modulation with selective harmonic elimination is also presented.

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7.2 H-BRIDGE INVERTER

Figure 7.2-1 shows a simplified circuit diagram of a single-phase H-bridge inverter. It is composed of two **inverter legs** with two IGBT devices in each leg. The inverter dc bus voltage V_d is usually fixed while its ac output voltage v_{AB} can be adjusted by either bipolar or unipolar modulation schemes.

7.2.1 Bipolar Pulse Width Modulation

Figure 7.2-2 shows a set of typical waveforms of the H-bridge inverter with bipolar modulation, where v_m is the sinusoidal modulating wave, v_{cr} is the triangular carrier wave, and v_{g1} and v_{g3} are the gate signals for the upper switches S_1 and S_3, respectively. The upper and the lower switches in the same inverter leg operate in a complementary manner with one switch turned on and the other turned off. Thus, we only need to consider two independent gate signals, v_{g1} and v_{g3}, which are generated by comparing v_m with v_{cr}. Following the same procedures given in Chapter 6, the waveforms of the inverter terminal voltages v_{AN} and v_{BN} can be derived, from which the inverter output voltage can be found from $v_{AB} = v_{AN} - v_{BN}$. Since the waveform of v_{AB} switches between the positive and negative dc voltages $\pm V_d$, this scheme is known as **bipolar modulation** [4].

The harmonic spectrum of the inverter output voltage v_{AB} normalized to its dc voltage V_d is shown in Fig. 7.2-2b, where V_{ABn} is the rms value of the nth order harmonic voltage. The harmonics appear as sidebands centered around the frequency modulation index m_f and its multiples such as $2m_f$ and $3m_f$. The voltage harmonics with the order lower than $(m_f - 2)$ are either eliminated or negligibly small. The switching frequency of the IGBT device, referred to as **device switching frequency** $f_{sw,dev}$ is equal to the carrier frequency f_{cr}.

Figure 7.2-3 shows the harmonic content of v_{AB} versus the amplitude modulation index m_a. The fundamental voltage V_{AB1} (rms) increases linearly with m_a. The

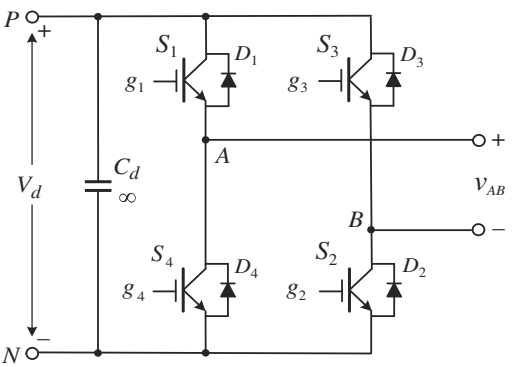

Figure 7.2-1 Single-phase H-bridge inverter.

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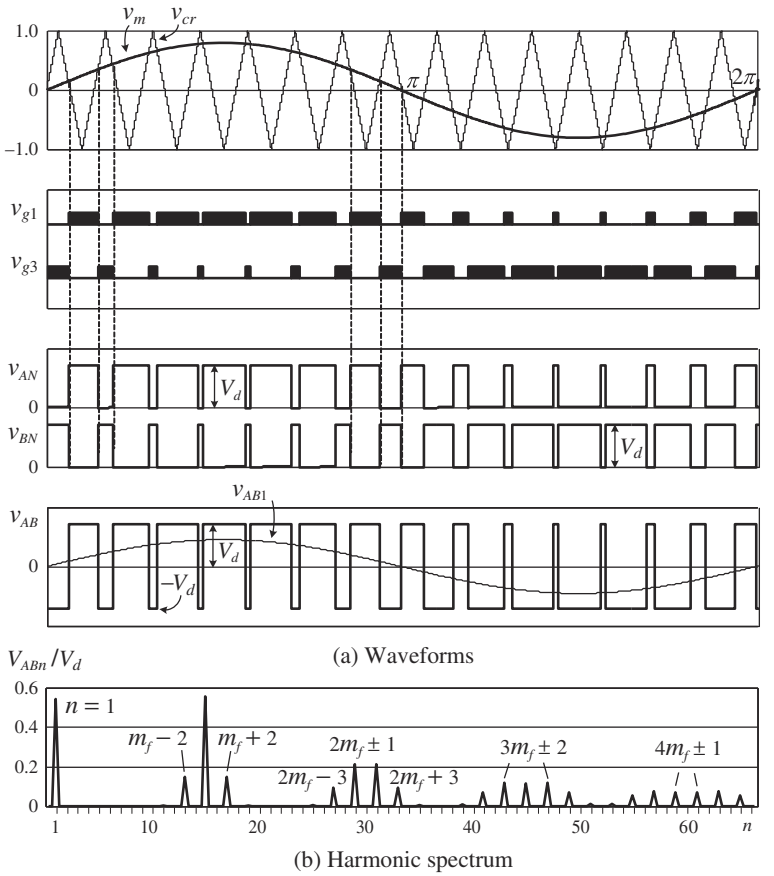

Figure 7.2-2 Bipolar PWM for the H-bridge inverter operating at $m_f = 15$, $m_a = 0.8$, $f_m = 60$ Hz, and $f_{cr} = 900$ Hz.

dominant harmonic m_f has a high magnitude, which is even higher than V_{AB1} for $m_a < 0.8$. This harmonic along with its sidebands can be eliminated by the unipolar pulse width modulation scheme.

7.2.2 Unipolar Pulse Width Modulation

The unipolar modulation normally requires two sinusoidal modulating waves, v_m and v_{m-}, which are of the same magnitude and frequency but 180° out of phase as shown in Fig. 7.2-4. The two modulating waves are compared with a common triangular carrier wave v_{cr}, generating two gating signals, v_{g1} and v_{g3}, for the upper switches, S_1 and S_3, respectively. It can be observed that the two upper switches do not commutate simultaneously, which is distinguished from the bipolar PWM where all four devices are commutated at the same time. The inverter output voltage v_{AB} switches either

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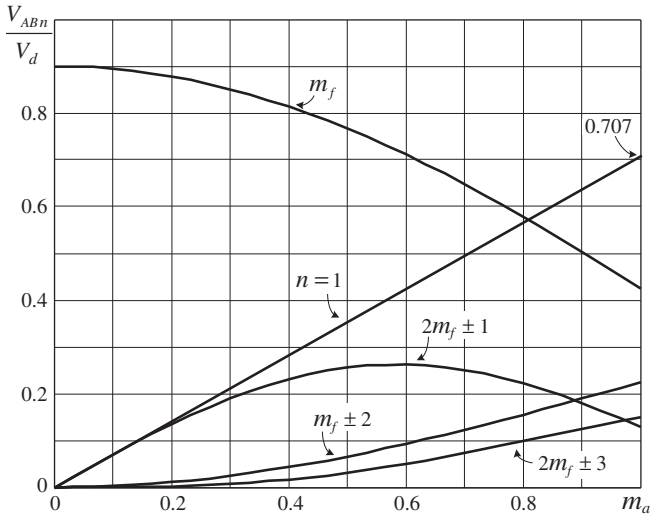

Figure 7.2-3 Harmonic content of v_{AB} produced by the H-bridge inverter with bipolar PWM.

between zero and $+V_d$ during the positive half cycle or between zero and $-V_d$ during the negative half cycle of the fundamental frequency. Thus, this scheme is known as **unipolar modulation** [4].

Figure 7.2-4b shows the harmonic spectrum of the inverter output voltage v_{AB}. The harmonics appear as sidebands centered around $2m_f$ and $4m_f$. The low-order harmonics generated by the bipolar modulation, such as m_f and $m_f \pm 2$, are eliminated by the unipolar modulation. The dominant harmonics are distributed around $2m_f$, and their frequencies are in the neighborhood of 1800 Hz. This is essentially the equivalent **inverter switching frequency** $f_{sw,inv}$, which is also the switching frequency seen by the load. Compared with the device switching frequency of 900 Hz, the inverter switching frequency is doubled. This phenomenon can also be explained from another perspective. The H-bridge inverter has two complementary switch pairs switching at 900 Hz. But the two pairs normally switch at different time instants, leading to $f_{sw,inv} = 2f_{sw,dev}$.

It is interesting to note that the dominant harmonics, $2m_f \pm 1$ and $2m_f \pm 3$, produced by the unipolar modulation have exactly the same magnitude as those generated by bipolar modulation. As a result, Fig. 7.2-3 can also be used to determine the magnitude of these harmonics at various m_a.

The unipolar modulation can also be implemented by using only one modulating wave v_m but two phase-shifted carrier waves, v_{cr} and v_{cr-}, as shown in Fig. 7.2-5. The two carrier waves are of same amplitude and frequency, but 180° out of phase. Switch S_1 is turned on by v_{g1} when $v_m > v_{cr}$ whereas S_3 is on when $v_m < v_{cr-}$. The waveform of v_{AB} is identical to that shown in Fig. 7.2-4. This modulation technique is often used in the CHB multilevel inverters.

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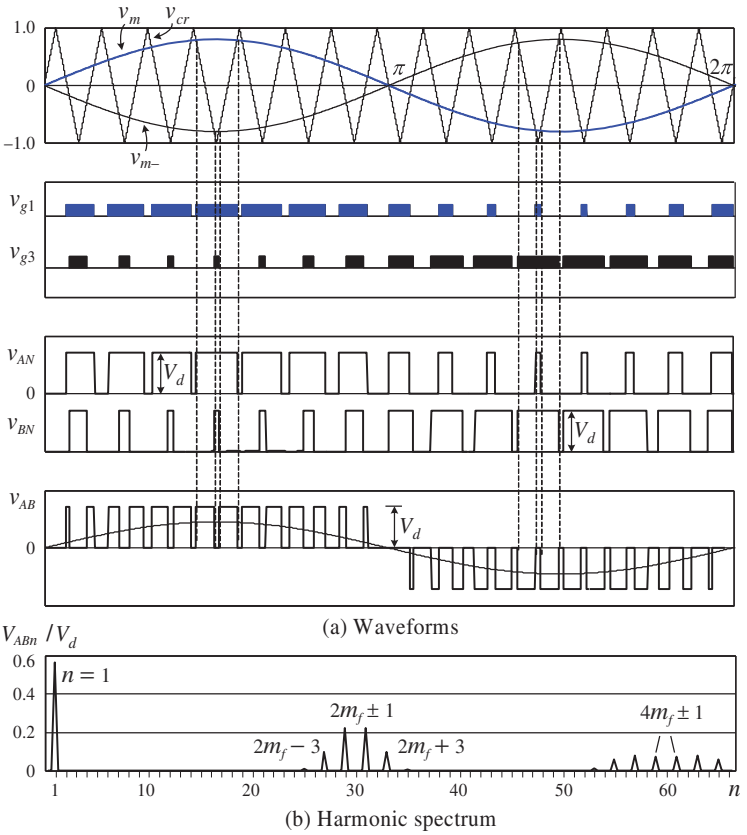

Figure 7.2-4 Unipolar PWM with two phase-shifted modulating waves ($m_f = 15$, $m_a = 0.8$, $f_m = 60$ Hz, and $f_{cr} = 900$ Hz).

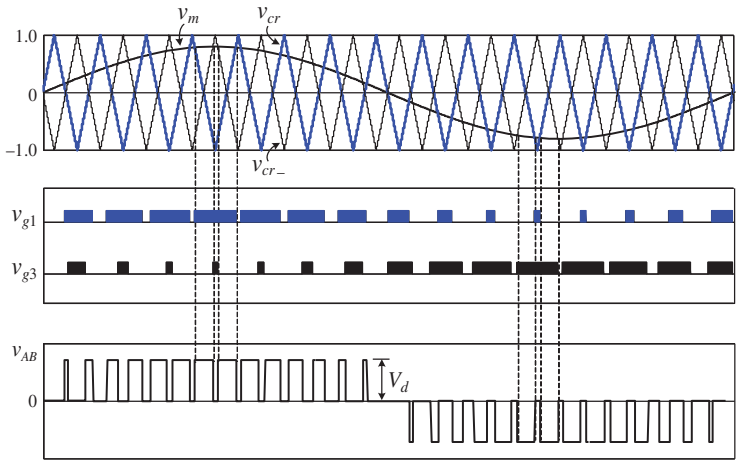

Figure 7.2-5 Unipolar PWM with two phase-shifted carriers ($m_f = 15$, $m_a = 0.8$, $f_m = 60$ Hz, and $f_{cr} = 900$ Hz).

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7.3 MULTILEVEL INVERTER TOPOLOGIES

7.3.1 CHB Inverter with Equal DC Voltage

As the name suggests, the CHB multilevel inverter uses multiple units of H-bridge power cells connected in a series chain to produce high ac voltages. A typical configuration of a five-level CHB inverter is shown in Fig. 7.3-1, where each phase leg consists of two H-bridge cells powered by two isolated dc supplies of equal voltage E. The dc supplies are normally obtained by multipulse diode rectifiers discussed in Chapter 3.

The CHB inverter in Fig. 7.3-1 can produce a phase voltage with five voltage levels. When switches S_{11}, S_{21}, S_{12}, and S_{22} conduct, the output voltage of the H-bridge cells $H1$ and $H2$ is $v_{H1} = v_{H2} = E$, and the resultant **inverter phase voltage** is $v_{AN} = v_{H1} + v_{H2} = 2E$, which is the voltage at the inverter terminal A with respect to the inverter neutral N. Similarly, with S_{31}, S_{41}, S_{32}, and S_{42} switched on, $v_{AN} = -2E$. The other three voltage levels are E, 0, and $-E$, which correspond to various switching states summarized in Table 7.3-1. It is worth noting that the inverter phase voltage v_{AN} may not necessarily equal the **load phase voltage** v_{AO}, which is the voltage at node A with respect to the load neutral O.

It can be observed from Table 7.3-1 that some voltage levels can be obtained by more than one switching state. The voltage level E, for instance, can be produced by four sets of different (redundant) switching states. The switching state redundancy is a common phenomenon in multilevel converters. It provides a great flexibility for switching pattern design, especially for space vector modulation schemes.

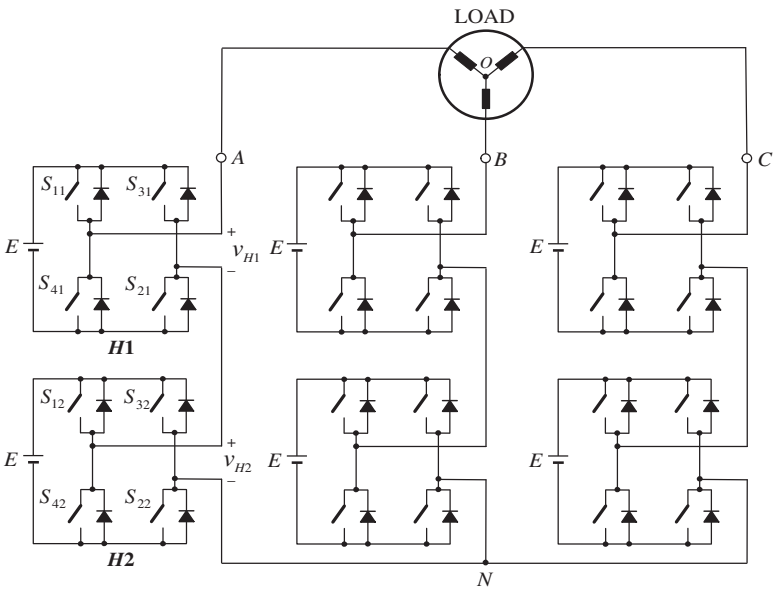

Figure 7.3-1 Five-level cascaded H-bridge inverter.

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Table 7.3-1 Voltage Level and Switching State of the Five-Level CHB Inverter

Output Voltage v_{AN}	Switching State				v_{H1}	v_{H2}
	S_{11}	S_{31}	S_{12}	S_{32}		
$2E$	1	0	1	0	E	E
E	1	0	1	1	E	0
	1	0	0	0	E	0
	1	1	1	0	0	E
	0	0	1	0	0	E
0	0	0	0	0	0	0
	0	0	1	1	0	0
	1	1	0	0	0	0
	1	1	1	1	0	0
	1	0	0	1	E	$-E$
	0	1	1	0	$-E$	E
$-E$	0	1	1	1	$-E$	0
	0	1	0	0	$-E$	0
	1	1	0	1	0	$-E$
	0	0	0	1	0	$-E$
$-2E$	0	1	0	1	$-E$	$-E$

The number of voltage levels in a CHB inverter can be found from

$$m = (2H + 1) \tag{7.3-1}$$

where H is the number of H-bridge cells per phase leg. The voltage level m is always an odd number for the CHB inverter while in other multilevel topologies such as diode-clamped inverters, it can be either an even or odd number.

The CHB inverter introduced above can be extended to any number of voltage levels. The per-phase diagram of seven- and nine-level inverters are depicted in Fig. 7.3-2, where the seven-level inverter has three H-bridge cells in cascade while the nine-level has four cells in series. The total number of active switches (IGBTs) used in the CHB inverters can be calculated by

$$N_{sw} = 6\,(m - 1) \tag{7.3-2}$$

7.3.2 H-Bridges with Unequal DC Voltages

The dc supply voltages of the H-bridge power cells introduced in the previous section are all the same. Alternatively, different dc voltages may be selected for the power

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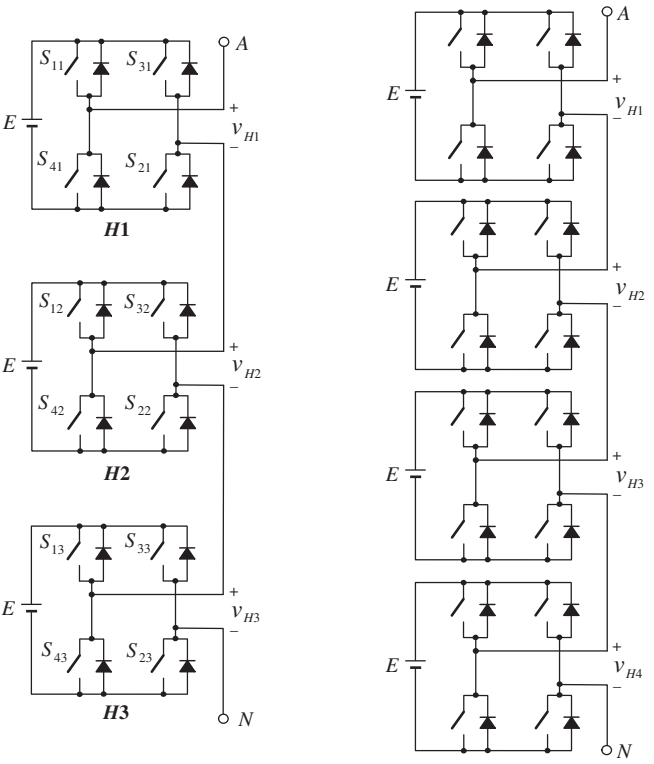

(a) Seven-level inverter (b) Nine-level inverter

Figure 7.3-2 Per-phase diagram of seven- and nine-level CHB inverters.

cells. With unequal dc voltages, the number of voltage levels can be increased without necessarily increasing the number of H-bridge cells in cascade. This allows more voltage steps in the inverter output voltage waveform for a given number of power cells [5, 6].

Figure 7.3-3 shows two inverter topologies, where the dc voltages for the H-bridge cells are not equal. In the seven-level topology, the dc voltages for $H1$ and $H2$ are E and $2E$, respectively. The two-cell inverter leg is able to produce seven voltage levels: $3E, 2E, E, 0, -E, -2E$, and $-3E$. The relationship between the voltage levels and their corresponding switching states is summarized in Table 7.3-2. In the nine-level topology, the dc voltage of $H2$ is three times that of $H1$. All the nine voltage levels can obtained by replacing the $H2$ output voltage of $v_{H2} = \pm 2E$ in Table 7.3-2 with $v_{H2} = \pm 3E$ and then calculating the inverter phase voltage v_{AN}.

There are some drawbacks associated with the CHB inverter using unequal dc voltages. The merits of the modular structure are essentially lost. In addition, switching pattern design becomes much more difficult due to the reduction in redundant switching states [5]. Therefore, this inverter topology has limited industrial applications.

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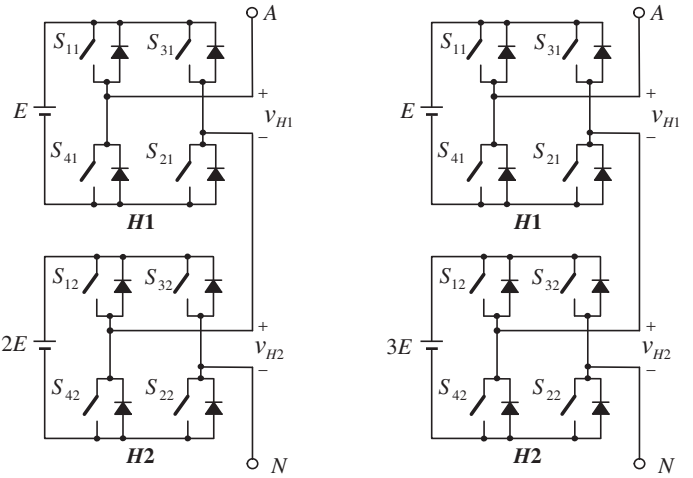

(a) Two-cell seven-level topology

(b) Two-cell nine-level topology

Figure 7.3-3 Per-phase diagram of CHB inverters with unequal dc voltages.

Table 7.3-2 Voltage Level and Switching State of the Two-Cell Seven-Level CHB Inverter with Unequal dc Voltages

Output Voltage v_{AN}	Switching State				v_{H1}	v_{H2}
	S_{11}	S_{31}	S_{12}	S_{32}		
$3E$	1	0	1	0	E	$2E$
$2E$	1	1	1	0	0	$2E$
	0	0	1	0	0	$2E$
E	1	0	1	1	E	0
	1	0	0	0	E	0
	0	1	1	0	$-E$	$2E$
0	0	0	0	0	0	0
	0	0	1	1	0	0
	1	1	0	0	0	0
	1	1	1	1	0	0
$-E$	1	0	0	1	E	$-2E$
	0	1	1	1	$-E$	0
	0	1	0	0	$-E$	0
$-2E$	1	1	0	1	0	$-2E$
	0	0	0	1	0	$-2E$
$-3E$	0	1	0	1	$-E$	$-2E$

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7.4 CARRIER-BASED PWM SCHEMES

The carrier-based modulation schemes for multilevel inverters can be generally classified into two categories: **phase-shifted** and **level-shifted modulations**. Both modulation schemes can be applied to the CHB inverters.

7.4.1 Phase-Shifted Multicarrier Modulation

In general, a multilevel inverter with m voltage levels requires $(m - 1)$ triangular carriers. In the phase-shifted multicarrier modulation, all the triangular carriers have the same frequency and the same peak-to-peak amplitude, but there is a phase shift between any two adjacent carrier waves, given by

$$\varphi_{cr} = 360^\circ/(m - 1) \tag{7.4-1}$$

The modulating signal is usually a three-phase sinusoidal wave with adjustable amplitude and frequency. The gate signals are generated by comparing the modulating wave with the carrier waves.

Figure 7.4-1 shows the principle of the phase-shifted modulation for a seven-level CHB inverter, where six triangular carriers are required with a 60° phase displacement

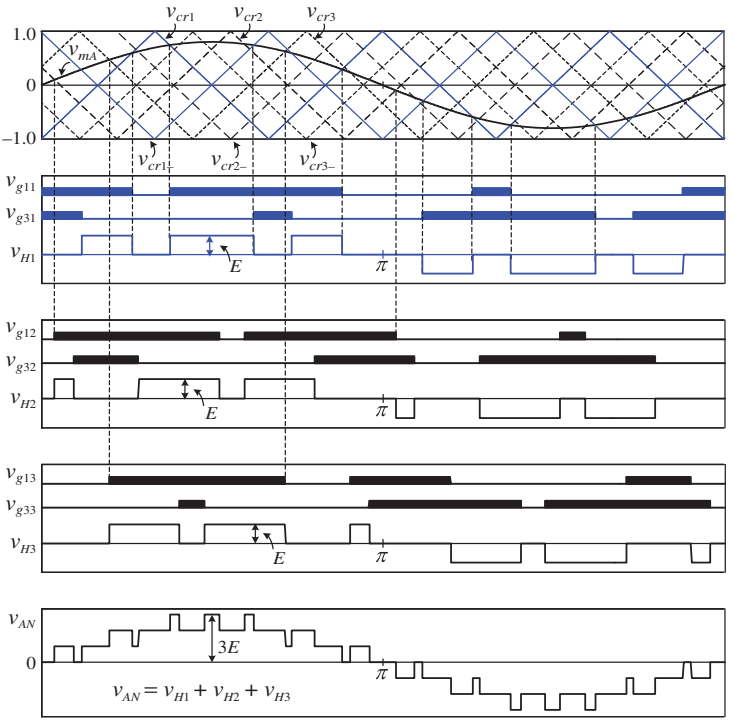

Figure 7.4-1 Phase-shifted PWM for seven-level CHB inverters ($m_f = 3$, $m_a = 0.8$, $f_m = 60$ Hz, and $f_{cr} = 180$ Hz).

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between any two adjacent carriers. Of the three-phase sinusoidal modulating waves, only the phase A modulating wave v_{mA} is plotted for simplicity. The carriers v_{cr1}, v_{cr2}, and v_{cr3} are used to generate gatings for the upper switches S_{11}, S_{12}, and S_{13} in the left legs of power cells $H1, H2$, and $H3$ in Fig. 7.3-2a, respectively. The other three carriers, v_{cr1-}, v_{cr2-}, and v_{cr3-}, which are 180° out of phase with v_{cr1}, v_{cr2}, and v_{cr3}, respectively, produce the gatings for the upper switches S_{31}, S_{32}, and S_{33} in the right legs of the H-bridge cells. The gate signals for all the lower switches in the H-bridge legs are not shown since these switches operate in a complementary manner with respect to their corresponding upper switches.

The PWM scheme discussed above is essentially the unipolar modulation. As shown in Fig. 7.4-1, the gatings for the upper switches S_{11} and S_{31} in $H1$ are generated by comparing v_{cr1} and v_{cr1-} with v_{mA}. The $H1$ output voltage v_{H1} is switched either between zero and E during the positive half cycle or between zero and $-E$ during the negative half cycle of the fundamental frequency. The frequency modulation index in this example is $m_f = f_{cr}/f_m = 3$ and the amplitude modulation index is $m_a = \hat{V}_{mA}/\hat{V}_{cr} = 0.8$, where f_{cr} and f_m are the frequencies of the carrier and modulating waves, and $\hat{V}_{mA}$ and $\hat{V}_{cr}$ are the peak amplitudes of v_{mA} and v_{cr}, respectively.

The inverter phase voltage can be found from

$$v_{AN} = v_{H1} + v_{H2} + v_{H3} \tag{7.4-2}$$

where v_{H1}, v_{H2}, and v_{H3} are the output voltages of cells $H1, H2$, and $H3$, respectively. It is clear that the inverter phase voltage waveform is formed by seven voltage steps: $+3E$, $2E$, E, 0, $-E$, $-2E$, and $-3E$.

Figure 7.4-2 shows the simulated voltage waveforms and their harmonic content of the seven-level inverter operating under the condition of $f_m = 60$ Hz, $m_f = 10$, and $m_a = 1.0$. The device switching frequency can be calculated by $f_{sw,dev} = f_{cr} = f_m \times m_f = 600$ Hz, which is a typical value for the switching devices in high-power converters. The waveforms of v_{H1}, v_{H2}, and v_{H3} are almost identical except a small phase displacement caused by the phase-shifted carriers.

The waveform of v_{AN} is composed of seven voltage levels with a peak value of $3E$. Since the IGBTs in the different H-bridges do not switch simultaneously, the magnitude of voltage step change during switching is only E. This leads to a low dv/dt and reduced electromagnetic interference (EMI). The line-to-line voltage v_{AB} has 13 voltage levels with an amplitude of $6E$.

The harmonic spectrum for the waveforms of v_{H1}, v_{AN}, and v_{AB} is shown in Fig. 7.4-2b. The harmonics in v_{H1} appear as sidebands centered around $2m_f$ and its multiples such as $4m_f$ and $6m_f$. The harmonic content of v_{H2} and v_{H3} is identical to that of v_{H1}, and thus not given in the figure. The inverter phase voltage v_{AN} does not contain any harmonics of the order lower than $4m_f$, which leads to a significant reduction in THD. The THD for v_{AN} is only 18.8% in comparison to 53.9% for v_{H1}. It can be observed that v_{AN} contains triplen harmonics such as $(6m_f \pm 3)$ and $(6m_f \pm 9)$. However, these harmonics do not appear in the line-to-line voltage v_{AB} due to the balanced three-phase system, resulting in a further reduction in THD to 15.5%.

As stated earlier, the frequency of the dominant harmonic in the inverter output voltage represents the inverter switching frequency $f_{sw,inv}$. Since the dominant

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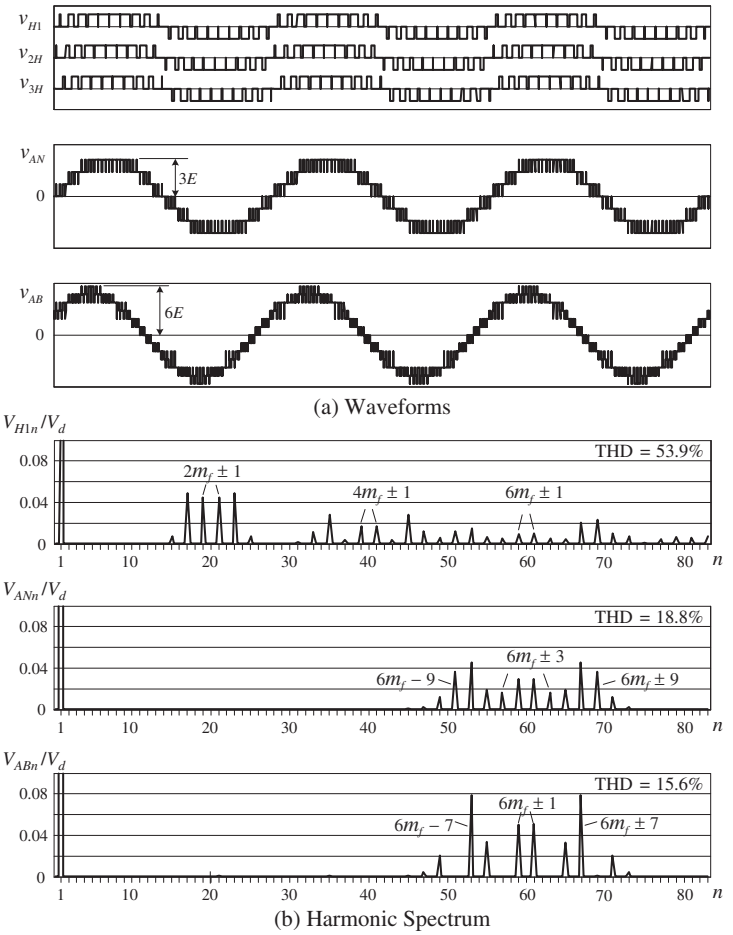

Figure 7.4-2 Simulated waveforms for a seven-level CHB inverter with phase-shifted PWM ($m_f = 10$, $m_a = 1.0$, $f_m = 60$ Hz, and $f_{cr} = 600$ Hz).

harmonics in v_{AN} and v_{AB} in Fig. 7.4-2 are distributed around $6m_f$, the inverter switching frequency can be found from $f_{sw,inv} = 6m_f \times f_m = 6f_{sw,dev}$, which is six times the device switching frequency. This is a desirable feature attained by the multilevel inverter since a high value of $f_{sw,inv}$ allows more harmonics in v_{AB} to be eliminated while a low value of $f_{sw,dev}$ helps to reduce device switching losses. In general, the switching frequency of the inverter using the phase-shifted modulation is related to the device switching frequency by

$$f_{sw,inv} = 2Hf_{sw,dev} = (m-1)f_{sw,dev} \tag{7.4-3}$$

The harmonic content of v_{AB} versus the modulation index m_a is shown in Fig. 7.4-3. Since the high-order harmonic components can be easily attenuated by filters or load inductances, only the dominant harmonics centered around $6m_f$ are plotted.

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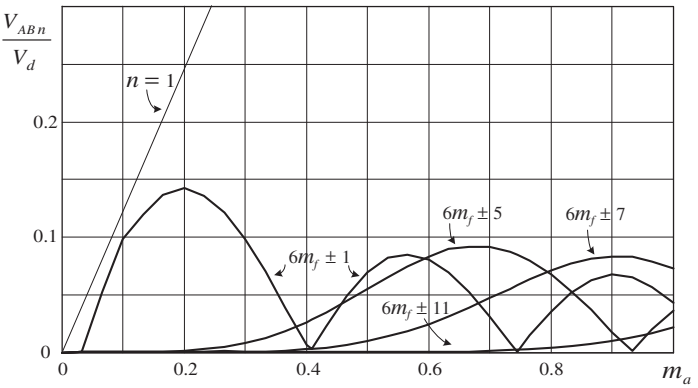

Figure 7.4-3 Harmonic content of v_{AB} produced by a seven-level CHB inverter with phase-shifted PWM.

The nth order harmonic voltage V_{ABn} (rms) is normalized with respect to the total dc voltage

$$V_d = \frac{m-1}{2}E \tag{7.4-4}$$

For a seven-level inverter, $V_d = 3E$. The maximum fundamental-frequency voltage can be found from

$$V_{AB1,\max} = 1.224V_d = 0.612(m-1)E \quad \text{for} \quad m_a = 1 \tag{7.4-5}$$

As discussed in Chapter 6, the maximum voltage $V_{AB1,\max}$ can be boosted by 15.5% by the **third harmonic injection method**. This technique can also be applied to the phase- and level-shifted modulation schemes for the CHB inverters.

7.4.2 Level-Shifted Multicarrier Modulation

Similar to the phase-shifted modulation, an m-level CHB inverter using level-shifted multicarrier modulation scheme requires (m-1) triangular carriers, all having the same frequency and amplitude. The (m-1) triangular carriers are vertically disposed such that the bands they occupy are contiguous. The frequency modulation index is given by $m_f = f_{cr}/f_m$, which remains the same as that for the phase-shifted modulation scheme whereas the amplitude modulation index is defined as

$$m_a = \frac{\hat{V}_m}{\hat{V}_{cr}(m-1)} \quad \text{for} \quad 0 \le m_a \le 1 \tag{7.4-6}$$

where $\hat{V}_m$ is the peak amplitude of the modulating wave v_m and $\hat{V}_{cr}$ is the peak amplitude of each carrier wave.

Figure 7.4-4 shows three schemes for the level-shifted multicarrier modulation: (a) in-phase disposition (IPD), where all carriers are in phase; (b) alternative phase

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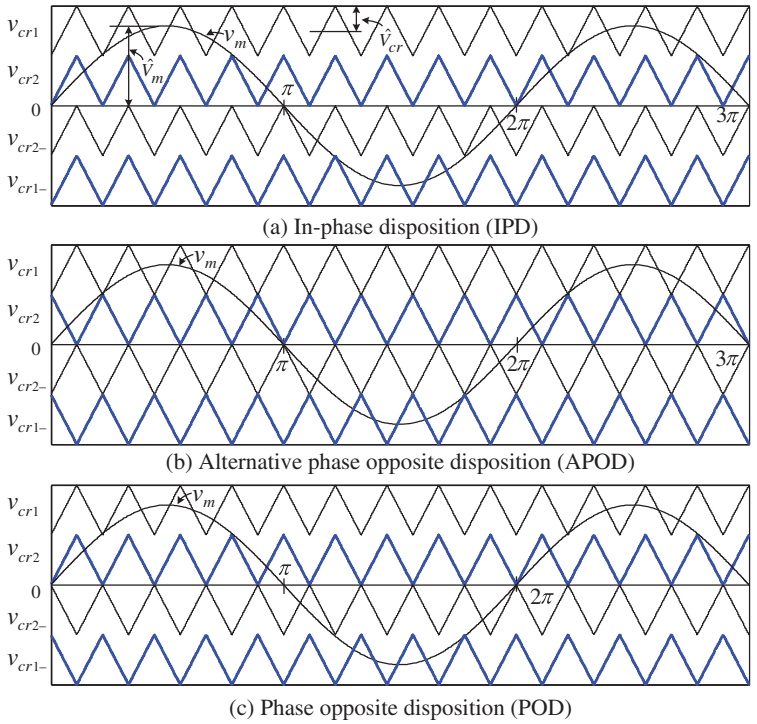

(a) In-phase disposition (IPD)

(b) Alternative phase opposite disposition (APOD)

(c) Phase opposite disposition (POD)

Figure 7.4-4 Level-shifted multicarrier modulation for five-level inverters.

opposite disposition (APOD), where all carriers are alternatively in opposite disposition; and (c) phase opposite disposition (POD), where all carriers above the zero reference are in phase but in opposition with those below the zero reference. In what follows, only IPD modulation scheme is discussed since it provides the best harmonic profile of all three modulation schemes [7].

Figure 7.4-5 shows the principle of the IPD modulation for a seven-level CHB inverter operating under the condition of $m_f = 15$, $m_a = 0.8$, $f_m = 60$ Hz, and $f_{cr} = f_m \times m_f = 900$ Hz. The uppermost and lowermost carrier pair, v_{cr1} and v_{cr1-}, are used to generate the gatings for switches S_{11} and S_{31} in power cell $H1$ of Fig. 7.3-2a. The innermost carrier pair, v_{cr3} and v_{cr3-}, generate gatings for S_{13} and S_{33} in $H3$. The remaining carrier pair, v_{cr2} and v_{cr2-}, are for S_{12} and S_{32} in $H2$. For the carriers above the zero reference (v_{cr1}, v_{cr2}, and v_{cr3}), the switches S_{11}, S_{12}, and S_{13} are turned on when the phase A modulating signal v_{mA} is higher than the corresponding carriers. For the carriers below the zero reference (v_{cr1-}, v_{cr2-}, and v_{cr3-}), S_{31}, S_{32} and S_{33} are switched on when v_{mA} is lower than the carrier waves. The gate signals for the lower switches are complementary to their corresponding upper switches, and thus not shown for simplicity. The resultant H-bridge output voltage waveforms v_{H1}, v_{H2}, and v_{H3} are all unipolar as shown in Fig. 7.4-5. The inverter phase voltage waveform v_{AN} is formed with seven voltage levels.

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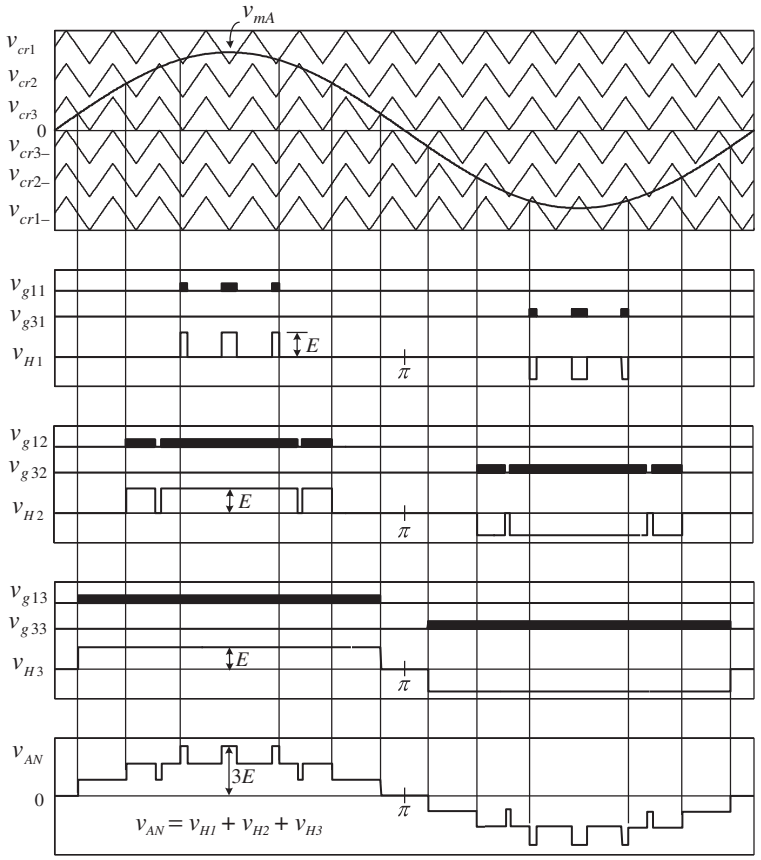

Figure 7.4-5 Level-shifted PWM for a seven-level CHB inverter ($m_f = 15$, $m_a = 0.8$, $f_m =$ 60 Hz, and $f_{cr} = 900$ Hz).

In the phase-shifted modulation, the device switching frequency is equal to the carrier frequency. This relationship, however, is no longer held true for the IPD modulation. For example, with the carrier frequency of 900 Hz in Fig. 7.4-5, the switching frequency of the devices in $H1$ is only 180 Hz, which is obtained by the number of gating pulses per cycle multiplied by the frequency of the modulating wave (60 Hz). Further, the switching frequency is not the same for all the devices. The switches in $H3$ are turned on and off only once per cycle, which translates into a switching frequency of 60 Hz. In general, the switching frequency of the inverter using the level-shifted modulation is equal to the carrier frequency, that is,

$$f_{sw,inv} = f_{cr} \tag{7.4-7}$$

from which the average device switching frequency is

$$f_{sw,dev} = f_{cr}/(m-1) \tag{7.4-8}$$

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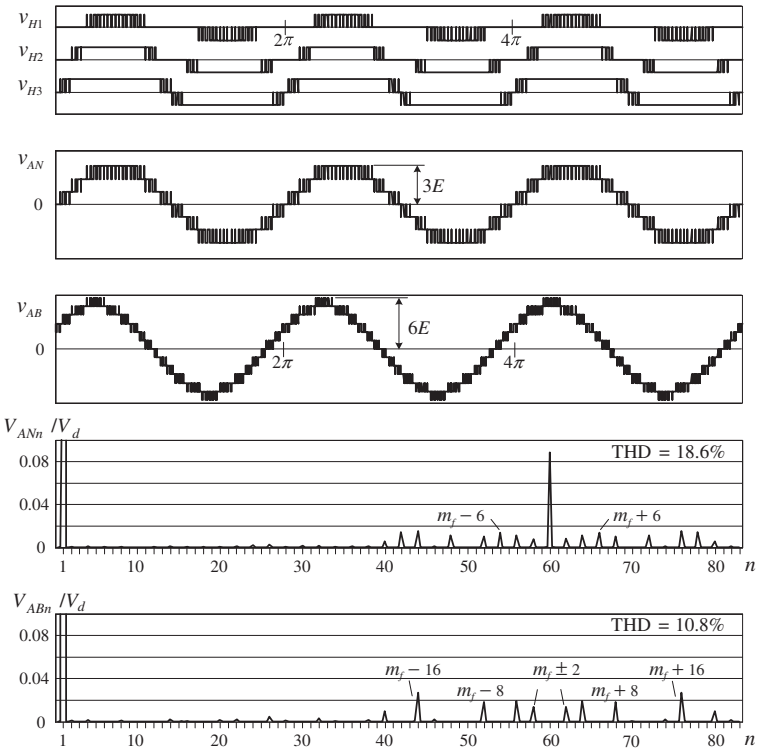

Figure 7.4-6 Simulated waveforms for a seven-level CHB inverter with IPD modulation ($m_f = 60$, $m_a = 1.0$, $f_m = 60$ Hz, $f_{cr} = 3600$ Hz, and $f_{sw,dev} = 600$ Hz).

In addition to the unequal device switching frequencies, the conduction time of the devices is not evenly distributed either. For example, the device S_{11} in $H1$ conduct much less time than S_{13} in $H3$ per cycle of the fundamental frequency. To evenly distribute the switching and conduction losses, the switching pattern should rotate among the H-bridge cells.

Figure 7.4-6 shows the simulated waveforms for a seven-level inverter operating under the condition of $m_f = 60$, $m_a = 1.0$, $f_m = 60$ Hz, and $f_{cr} = 3600$ Hz. Although the carrier frequency of 3600 Hz seems high for high-power converters, the average device switching frequency is only 600 Hz. The output voltages of the H-bridge cells, v_{H1}, v_{H2}, and v_{H3}, are all different, signifying that the IGBTs operate at different switching frequencies with various conduction times.

Similar to the voltage waveforms produced by the phase-shifted modulation, the inverter phase voltage v_{AN} is composed of seven voltage levels while the line-to-line voltage v_{AB} has thirteen voltage levels. The dominant harmonics in v_{AN} and v_{AB} appear as sidebands centered around m_f. The phase voltage contains triplen harmonics, such as m_f and $m_f \pm 6$, with m_f being a dominant harmonic. Since these harmonics do not appear in the line-to-line voltage, the THD of v_{AB} is only 10.8% in comparison

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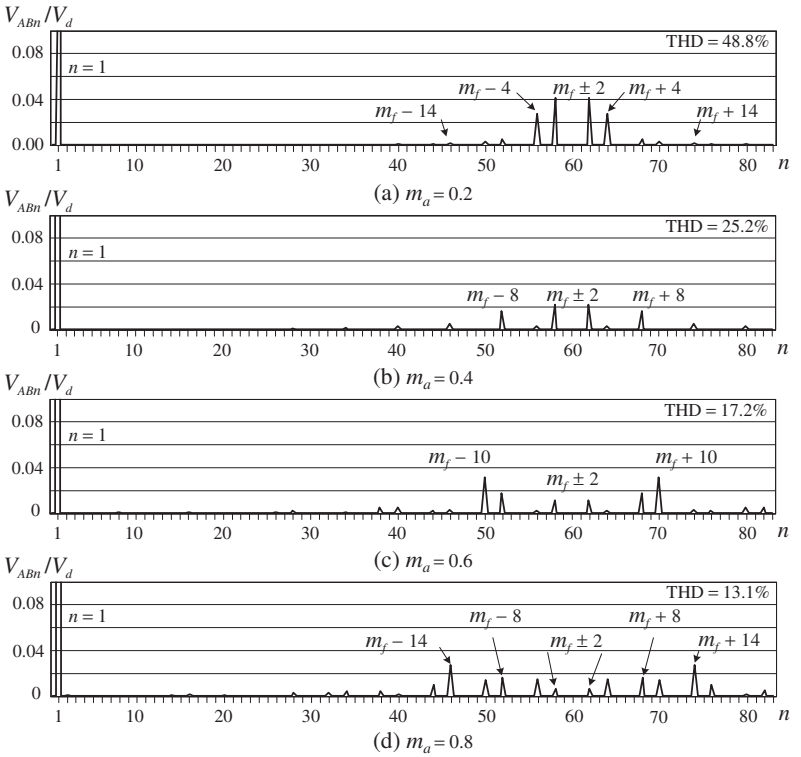

Figure 7.4-7 Harmonic content of v_{AB} produced by a seven-level CHB inverter with IPD modulation ($m_f = 60$, $f_m = 60$ Hz, $f_{cr} = 3600$ Hz, and $f_{sw,dev} = 600$ Hz).

to 18.6% for v_{AN}. The spectra of v_{AB} at other modulation indices m_a are shown in Fig. 7.4-7. The THD of v_{AB} decreases from 48.8% at $m_a = 0.2$ to 13.1% at $m_a = 0.8$.

The waveforms for v_{AN} and v_{AB} measured from a laboratory seven-level CHB inverter are illustrated in Fig. 7.4-8. The inverter operates under the condition of $m_f = 60$, $m_a = 1.0$, $f_m = 60$ Hz, and $f_{cr} = 3600$ Hz. The measured waveforms and their harmonic spectra are consistent with the simulation results shown in Fig. 7.4-6.

7.4.3 Comparison Between Phase- and Level-Shifted PWM Schemes

To compare the performance of phase- and level-shifted modulation schemes, it is assumed that the average switching frequency of the solid-state devices is the same for both schemes. Figure 7.4-9 shows the output voltage waveforms of a seven-level inverter operating with $f_{sw,dev} = 600$ Hz and $m_a = 0.2$, at which the differences between the two modulation schemes can be easily distinguished.

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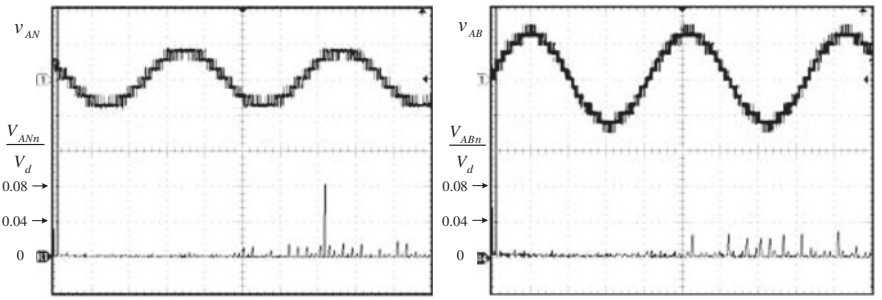

Figure 7.4-8 Waveforms measured from a laboratory seven-level CHB inverter with IPD modulation ($m_f = 60$, $m_a = 1.0$, $f_m = 60$ Hz, $f_{cr} = 3600$ Hz, and $f_{sw,dev} = 600$ Hz).

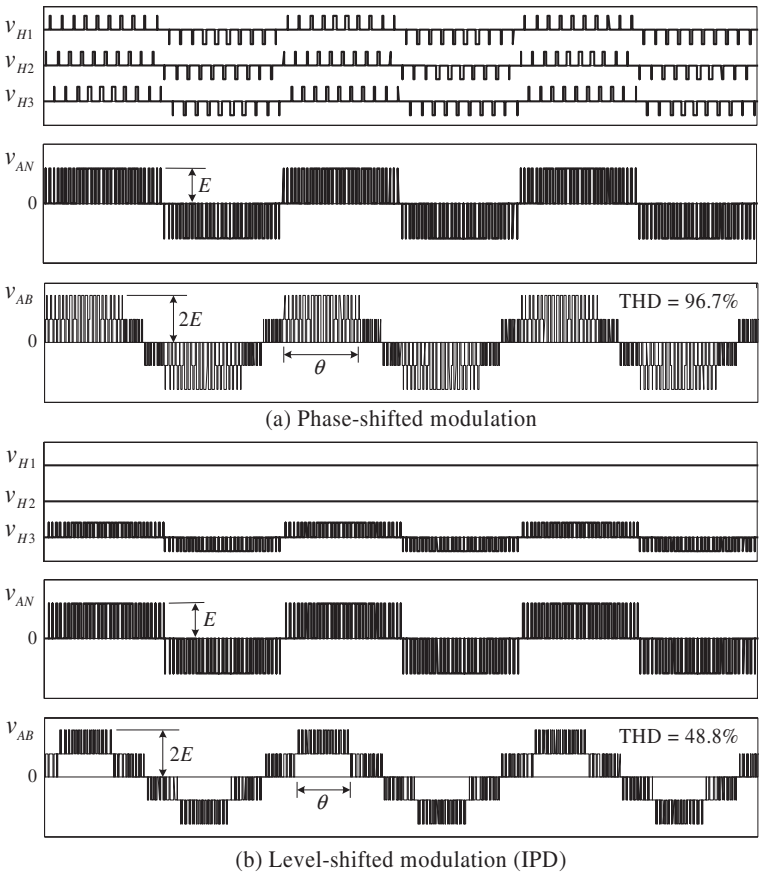

Figure 7.4-9 Output voltage waveforms of the seven-level inverter operating at a low modulation index.

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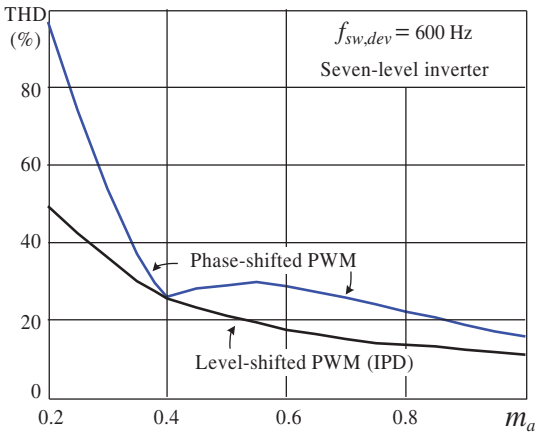

Figure 7.4-10 THD profile of v_{AB} produced by the seven-level CHB inverter with phase- and level-shifted modulation schemes.

The H-bridge output voltages, v_{H1}, v_{H2}, and v_{H3}, produced by the phase-shifted modulation are almost identical except for a small phase displacement among them. All the devices operate at the same switching frequency and conduction time. However, v_{H1} and v_{H2} produced by the level-shifted modulation are equal to zero and thus no switchings occur in power cells $H1$ and $H2$. The devices in $H3$ switch at the carrier frequency of 3600 Hz. To evenly distribute the switching and conduction losses, the switching pattern for the devices in the H-bridge cells should rotate.

The inverter phase voltage v_{AN} produced by both modulation schemes looks similar. It contains only three voltage levels instead of seven due to the low modulation index. The voltage levels of the inverter line-to-line voltage v_{AB} are reduced accordingly. Further, the THD of v_{AB} produced by the phase-shifted modulation is 96.7%, much higher than 48.8% for the level-shifted modulation. This is mainly caused by the waveform differences in the center portion of the positive and negative half cycles of v_{AB}.

Figure 7.4-10 shows the THD profile of the line-to-line voltage v_{AB} modulated by the phase- and level-shifted schemes. A summary of the carrier-based modulation schemes for the CHB multilevel inverters is given in Table 7.4-1.

Table 7.4-1 Comparison Between the Phase- and Level-Shifted PWM Schemes

Comparison	Phase-Shifted Modulation	Level-Shifted Modulation (IPD)
Device switching frequency	Same for all devices	Different
Device conduction period	Same for all devices	Different
Rotating of switching patterns	Not required	Required
Line-to-line voltage THD	Good	Better

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7.5 STAIRCASE MODULATION

The staircase modulation can be easily implemented for the CHB inverter due to its unique structure [8, 9]. The principle of this modulation scheme is illustrated in Fig. 7.5-1, where v_{H1}, v_{H2}, and v_{H3} are the output voltages of the H-bridge cells in a seven-level inverter shown in Fig. 7.3-2a. The inverter phase voltage v_{AN} is formed by a seven-level staircase.

The waveform of v_{AN} can be expressed in terms of Fourier series as

$$v_{AN} = \frac{4E}{\pi} \sum_{n=1,3,5...}^{\infty} \frac{1}{n} \{\cos(n\theta_1) + \cos(n\theta_2) + \cos(n\theta_3)\} \sin(n\omega t) \quad \text{for}$$
$$0 \le \theta_3 < \theta_2 < \theta_1 \le \pi/2 \tag{7.5-1}$$

where n is the harmonic order, and θ_1, θ_2, and θ_3 are the independent switching angles. The coefficient $4E/\pi$ represents the peak value of the maximum fundamental voltage $\hat{V}_{H,\max}$ of an H-bridge cell, which occurs when the switching angle θ_1 of v_{H1}, for example, reduces to zero.

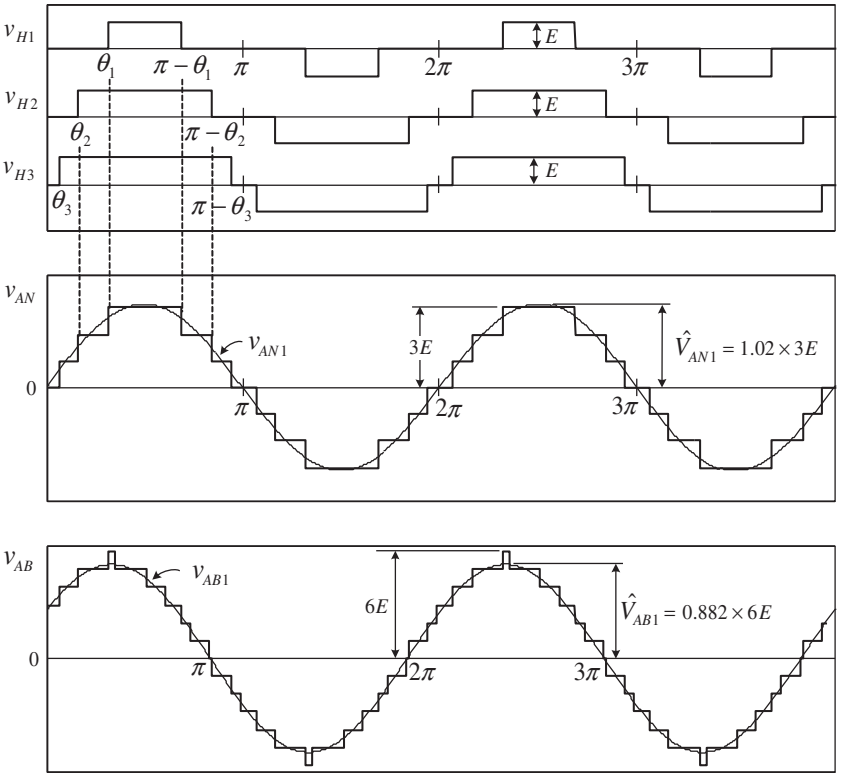

Figure 7.5-1 Staircase modulation with 5th and 7th harmonic elimination ($m_a = 0.8$).

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The three independent angles can be used to eliminate two harmonics in v_{AN} and also provide an adjustable modulation index, defined by

$$m_a = \frac{\hat{V}_{AN1}}{H \times \hat{V}_{H,\max}} = \frac{\hat{V}_{AN1}}{H \times 4E/\pi} \tag{7.5-2}$$

where $\hat{V}_{AN1}$ is the peak value of the fundamental inverter phase voltage v_{AN1} and H is the number of H-bridge cells per phase.

For the seven-level CHB inverter with 5th and 7th harmonic elimination, the following equations can be formulated

$$\begin{cases} \cos(\theta_1) + \cos(\theta_2) + \cos(\theta_3) = 3m_a \\ \cos(5\theta_1) + \cos(5\theta_2) + \cos(5\theta_3) = 0 \\ \cos(7\theta_1) + \cos(7\theta_2) + \cos(7\theta_3) = 0 \end{cases} \tag{7.5-3}$$

from which

$$\theta_1 = 57.106°,\ \theta_2 = 28.717°, \quad \text{and} \quad \theta_3 = 11.504° \quad \text{for} \quad m_a = 0.8 \tag{7.5-4}$$

The inverter output voltage waveforms based on (7.5-4) are shown in Fig. 7.5-1, and their spectrum is illustrated in Fig. 7.5-2. The waveform of v_{AN} does not contain the 5th or 7th harmonics and its THD is 12.5%. The inverter line-to-line voltage v_{AB} does not have any triplen harmonics such as 3rd, 9th, and 15th, resulting in a further reduction in THD.

The staircase modulation scheme is simple to implement. All the switching angles can be calculated off-line and then stored in a look-up table for digital implementation. Compared with the carrier-based PWM schemes, the staircase modulation features low switching losses since all the IGBTs operate at the fundamental frequency.

It is worth noting that the equations such as (7.5-3) for the switching angle calculation are nonlinear and transcendental, and thus may not always have a valid

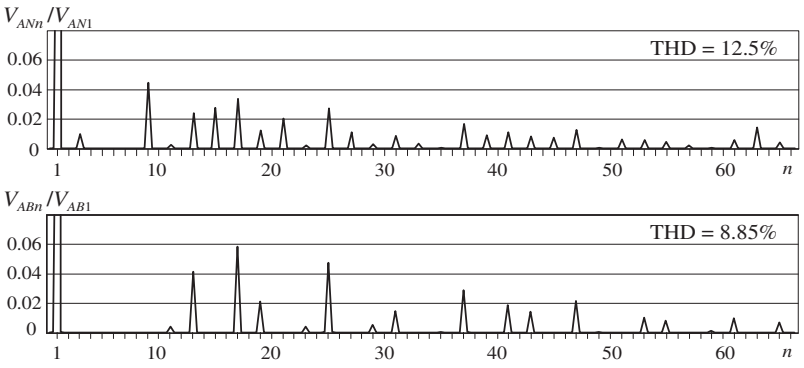

Figure 7.5-2 Harmonic spectrum for the waveforms of v_{AN} and v_{AB} in Fig. 7.5-1

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solution over the full range of m_a [10]. When it happens, the switching angles should be calculated to minimize the magnitude of those harmonics that cannot be eliminated.

7.6 SUMMARY

This chapter focuses on the configurations and modulation schemes for the CHB multilevel inverters. The inverter is mainly composed of a number of identical CHB power cells. In practice, the number of H-bridge cells in a CHB inverter is primarily determined by the inverter operating voltage, harmonic requirements, and manufacturing cost. The CHB inverter with 7–11 voltage levels has been increasingly used in high-power medium (MV) voltage drives, where the low-voltage (1200 V or 1700 V) IGBTs are normally used as switching devices.

Two multi-carrier-based PWM schemes, the phase- and level-shifted modulations, are presented. Various aspects associated with the modulation schemes for the CHB multilevel inverters are discussed, which include gate signal arrangements, spectrum analyses, and THD profiles. The performance of the modulation schemes is compared. Another commonly used modulation technique, space vector modulation, is not discussed in this chapter. The reader can refer to Chapters 6 and 8 for details.

The CHB multilevel inverter has a number of features and drawbacks, including:

- Modular structure. The multilevel inverter is composed of multiple units of identical H-bridge power cells, which leads to a reduction in manufacturing cost.
- Lower voltage THD and *dv/dt*. The inverter output voltage waveform is formed by several voltage levels with small voltage steps. Compared with a two-level inverter, the CHB multilevel inverter can produce an output voltage with much lower THD and *dv/dt.*
- High voltage operation without switching devices in series. The H-bridge power cells are connected in cascade to produce high ac voltages. The problems associated with equal voltage sharing for series-connected devices are avoided.
- Large number of isolated dc supplies. The dc supplies for the CHB inverter are usually obtained from a multipulse diode rectifier employing an expensive phase-shifting transformer.
- High component count. The CHB inverter uses a large number of IGBTs. A nine-level CHB inverter requires 48 IGBTs with the same number of gate drivers.

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Chapter 8

Diode-Clamped Multilevel Inverters

8.1 INTRODUCTION

The diode-clamped multilevel inverter employs clamping diodes and cascaded dc capacitors to produce ac voltage waveforms with multiple levels. The inverter can be generally configured as a three-, four-, or five-level topology, but only the three-level inverter, often known as **neutral-point clamped** (NPC) inverter, has found practical application in high-power medium voltage (MV) drives [1–3]. The main features of the NPC inverter include reduced *dv/dt* and THD in its ac output voltages in comparison to the two-level inverter discussed earlier. More importantly, the inverter can be used in the MV drive without switching devices in series. For instance, the NPC inverter using 6000 V devices is suitable for the drives rated at 4160 V.

In this chapter, various aspects of the three-level (3L) NPC inverter are discussed, including the inverter topology, operating principle, and device commutation. A conventional **space vector modulation** (SVM) scheme for the NPC inverter is discussed in detail. To eliminate the even-order harmonics produced by the SVM, a modified modulation scheme is presented. The dc input voltage of the inverter is normally split by two cascaded dc capacitors, providing a floating neutral point. The control of the neutral-point voltage deviation is also elaborated. Finally, the operation of four- and five-level diode-clamped inverters with carrier-based modulation techniques is introduced.

8.2 THREE-LEVEL INVERTER

8.2.1 Converter Configuration

Figure 8.2-1 shows the simplified circuit diagram of a three-level NPC inverter. The inverter leg A is composed of four active switches S_1 to S_4 with four anti-parallel

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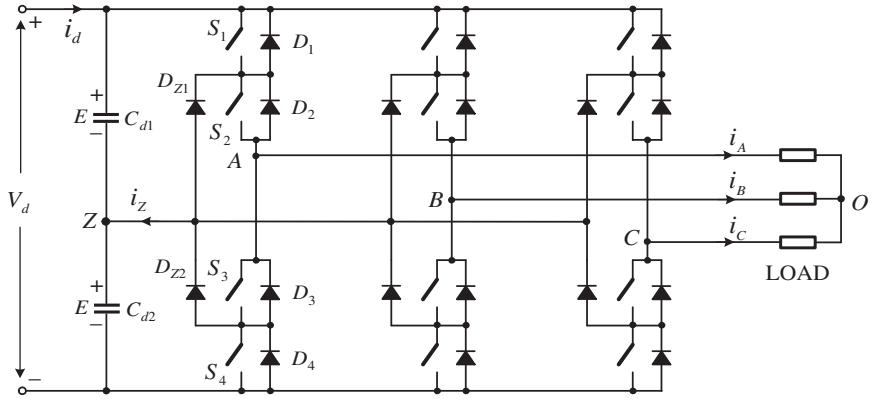

Figure 8.2-1 Three-level NPC inverter.

diodes D_1 to D_4. In practice, either IGBT or GCT can be employed as a switching device.

On the dc side of the inverter, the dc bus capacitor is split into two, providing a **neutral point** Z. The diodes connected to the neutral point, D_{Z1} and D_{Z2}, are the **clamping diodes.** When switches S_2 and S_3 are turned on, the inverter output terminal A is connected to the neutral point through one of the clamping diodes. The voltage across each of the dc capacitors is E, which is normally equal to half of the total dc voltage V_d. With a finite value for C_{d1} and C_{d2}, the capacitors can be charged or discharged by **neutral current** i_Z, causing **neutral-point voltage deviation**. This issue will be further discussed in the later sections.

8.2.2 Switching State

The operating status of the switches in the NPC inverter can be represented by switching states shown in Table 8.2-1. Switching state "P" denotes that the upper two switches in leg A are on and the **inverter terminal voltage** v_{AZ}, which is the voltage at terminal A with respect to the neutral point Z, is $+E$, whereas "N" indicates that the lower two switches conduct, leading to $v_{AZ} = -E$.

Table 8.2-1 Definition of Switching States

Switching State	Device Switching Status (Phase A)				Inverter Terminal Voltage V_{AZ}
	S_1	S_2	S_3	S_4	
P	On	On	Off	Off	E
O	Off	On	On	Off	0
N	Off	Off	On	On	$-E$

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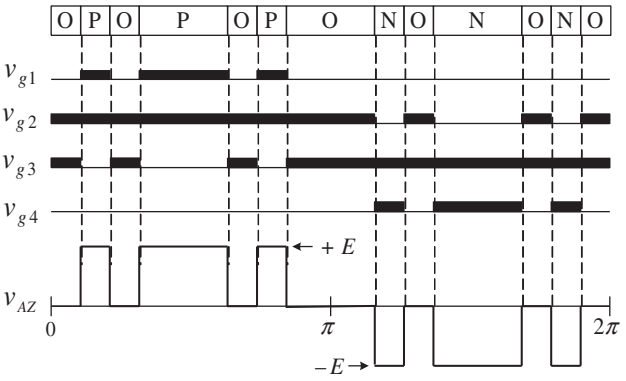

Figure 8.2-2 Switching states, gate signals, and inverter terminal voltage v_{AZ}.

Switching state "O" signifies that the inner two switches S_2 and S_3 are on and v_{AZ} is clamped to zero through the clamping diodes. Depending on the direction of load current i_A, one of the two clamping diodes is turned on. For instance, a positive load current ($i_A > 0$) forces D_{Z1} to turn on, and the terminal A is connected to the neutral point Z through the conduction of D_{Z1} and S_2.

It can be observed from Table 8.2-1 that switches S_1 and S_3 operate in a complementary manner. With one switched on, the other must be off. Similarly, S_2 and S_4 are a complementary pair as well.

Figure 8.2-2 shows an example of switching state and gate signal arrangements, where v_{g1} to v_{g4} are the gate signals for S_1 to S_4, respectively. The gate signals can be generated by carrier-based modulation, SVM, or selective harmonic elimination schemes. The waveform for v_{AZ} has three voltage levels, $+E$, 0, and $-E$, based on which the inverter is referred to as a three-level inverter.

Figure 8.2-3 shows how the line-to-line voltage waveform is obtained. The inverter terminal voltages v_{AZ}, v_{BZ}, and v_{CZ} are three-phase balanced with a phase shift of $2\pi/3$ between each other. The line-to-line voltage v_{AB} can be found from $v_{AB} = v_{AZ} - v_{BZ}$, which contains five voltage levels ($+2E$, $+E$, 0, $-E$, and $-2E$).

8.2.3 Commutation

To investigate the commutation of switching devices in the NPC inverter, consider a transition from switching state [O] to [P] by turning S_3 off and turning S_1 on. Figure 8.2-4a shows the gate signals v_{g1} to v_{g4} for switches S_1 to S_4, respectively. Similar to the gating arrangement in the two-level inverter, a blanking time of δ is required for the complementary switch pair $S_1$and S_3.

Figures 8.2-4b and 8.2-4c show the circuit diagram of the inverter leg A during commutation, where each of the active switches has a parallel resistor for static voltage sharing. According to the direction of the phase A load current i_A, the following two cases are investigated.

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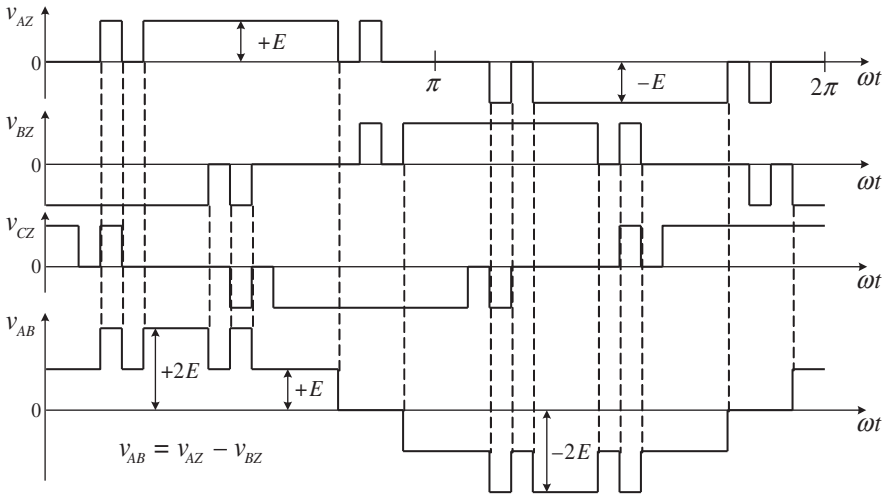

Figure 8.2-3 Inverter terminal and line-to-line voltage waveforms.

Case 1: Commutation with $i_A > 0$. The commutation process is illustrated in Fig. 8.2-4b. It is assumed that (a) the load current i_A is constant during the commutation due to the inductive load, (b) the dc bus capacitors C_{d1} and C_{d2} are sufficiently large such that the voltage across each capacitor is kept at E, and (c) all the switches are ideal. In switching state [O], switches S_1 and S_4 are switched off while S_2 and S_3 conduct. The clamping diode D_{Z1} is turned on by the positive load current ($i_A > 0$). The voltages across the on-state switches S_2 and S_3 are given by $v_{S2} = v_{S3} = 0$ while the voltage on each of the off-state switches S_1 and S_4 is equal to E.

During the δ interval, S_3 is being turned off. The paths of i_A remain unchanged. When S_3 is completely switched off, the voltages across S_3 and S_4 become $v_{S3} = v_{S4} = E/2$ due to the static voltage sharing resistors R_3 and R_4.

In switching state [P], the top switch S_1 is gated on ($v_{S1} = 0$). The clamping diode D_{Z1} is reverse biased and thus turned off. The load current $i_A$is commutated from D_{Z1} to S_1. Since both S_3 and S_4 have already been in the off-state, the voltage across these two switches is equally divided by R_3 and R_4, leading to $v_{S3} = v_{S4} = E$.

Case 2: Commutation with $i_A < 0$. The commutation process with $i_A < 0$ is illustrated in Fig. 8.2-4c. In switching state [O], S_2 and S_3 conduct, and the clamping diode D_{Z2} is turned on by the negative load current i_A. The voltage across the off-state switches S_1 and S_4 is $v_{S1} = v_{S4} = E$.

During the δ interval, S_3 is being turned off. Since the inductive load current $i_A$cannot change its direction instantly, it forces diodes D_1 and D_2 to turn on, resulting in $v_{S1} = v_{S2} = 0$. The load current is commutated from S_3 to the diodes. During the S_3 turn-off transient, the voltage across S_4 will not be higher than E due to the clamping diode D_{Z2}, and will not be lower than E either since the equivalent resistance of

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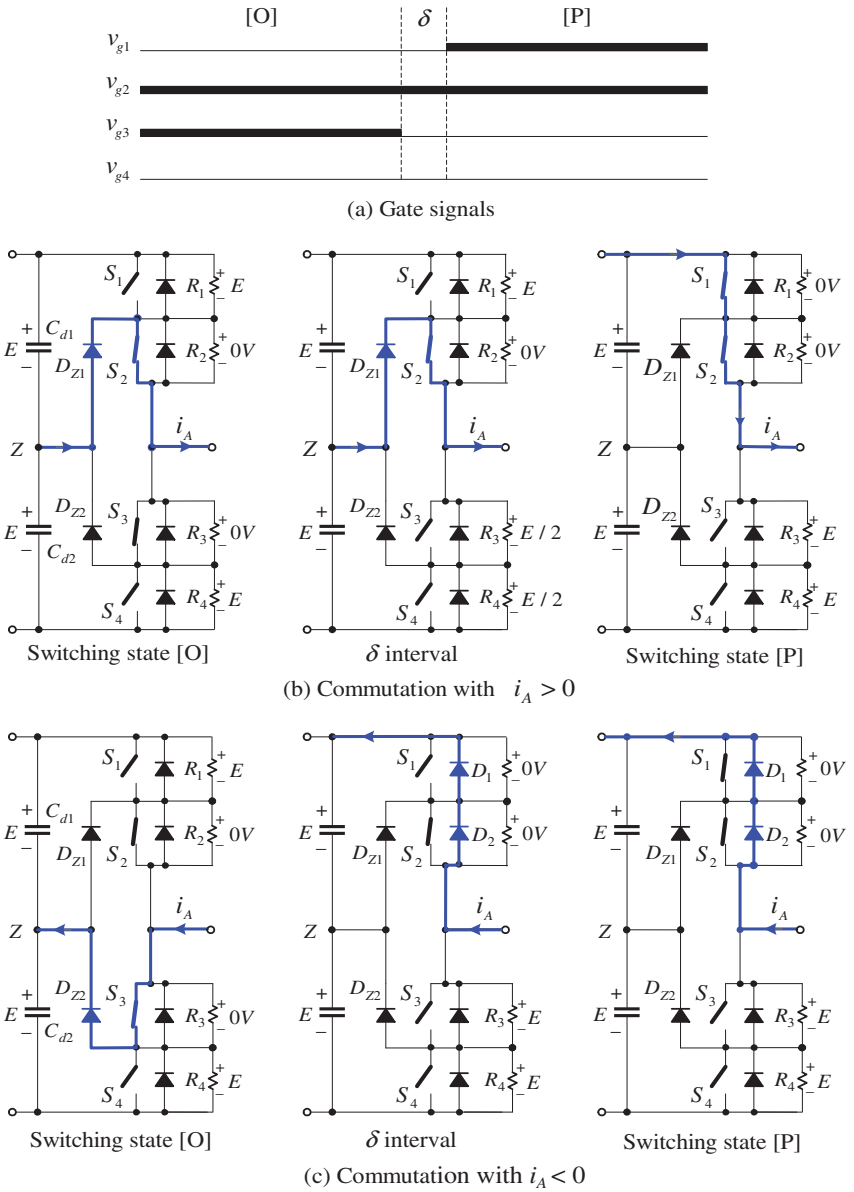

Figure 8.2-4 Commutation during a transition from switching state [O] to [P].

S_3 during turn-off is always lower than the off-state resistance of S_4. Therefore, v_{S3} increases from zero to E while v_{S4} is kept at E.

In switching state [P], the turn-on of S_1 does not affect the operation of the circuit. Although S_1 and S_2 are switched on, they do not carry the load current due to the conduction of D_1 and D_2.

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It can be concluded that all the switching devices in the NPC inverter withstand only half of the dc bus voltage during the commutation from switching state [O] to [P]. Similarly, the same conclusion can be drawn for the commutation from [P] to [O], [N] to [O], or vice versa. Therefore, the NPC inverter does not have dynamic voltage sharing problem.

It should be pointed out that the switching between [P] and [N] is prohibited for two reasons: (a) it involves all four switches in an inverter leg, two being turned on and the other two being commutated off, during which the dynamic voltage on each switch may not be kept same; and (b) the switching loss is doubled.

It is worth noting that the static voltage sharing resistors R_1 to R_4 may be omitted if the leakage current of the top and bottom switches (S_1 and S_4) in each inverter leg is selected to be lower than that of the inner two switches (S_2 and S_3). In doing so, the voltages across the top and bottom switches, which tend to be higher than those of the inner switches, are clamped to E by the clamping diodes in steady state. As a result, the voltage on each of the inner two switches is also equal to E, and the static voltage equalization is achieved.

To summarize, the three-level NPC inverter offers the following features:

- No dynamic voltage sharing problem. Each of the switches in the NPC inverter withstands only half of the total dc voltage during commutation.
- Static voltage equalization without using additional components. The static voltage equalization can be achieved when the leakage current of the top and bottom switches in an inverter leg is selected to be lower than that of the inner switches.
- Low THD and *dv/dt*. The waveform of the line-to-line voltages is composed of five voltage levels, which leads to lower THD and *dv/dt* in comparison to the two-level inverter operating at the same voltage rating and device switching frequency.

However, the NPC inverter has some drawbacks such as additional clamping diodes, complicated PWM switching pattern design and possible deviation of neutral-point voltage. The latter two are dealt with in the following sections.

8.3 SPACE VECTOR MODULATION

Various SVM schemes have been proposed for the three-level NPC inverter [4–7]. This section presents a "conventional" SVM scheme for the NPC inverter, followed by a modified SVM scheme for even-order harmonic elimination [8].

8.3.1 Stationary Space Vectors

As indicated earlier, the operation of each inverter phase leg can be represented by three switching states [P], [O], and [N]. Taking all three phases into account,

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Table 8.3-1 Voltage Vectors and Switching States

Space Vector		Switching State		Vector Classification	Vector Magnitude
$\vec{V}_0$		PPP, OOO, NNN		Zero Vector	0
$\vec{V}_1$		**P-type**	**N-type**	Small vector	$\frac{1}{3}\ V_d$
	$\vec{V}_{1P}$	POO			
	$\vec{V}_{1N}$		ONN		
$\vec{V}_2$	$\vec{V}_{2P}$	PPO			
	$\vec{V}_{2N}$		OON		
$\vec{V}_3$	$\vec{V}_{3P}$	OPO			
	$\vec{V}_{3N}$		NON		
$\vec{V}_4$	$\vec{V}_{4P}$	OPP			
	$\vec{V}_{4N}$		NOO		
$\vec{V}_5$	$\vec{V}_{5P}$	OOP			
	$\vec{V}_{5N}$		NNO		
$\vec{V}_6$	$\vec{V}_{6P}$	POP			
	$\vec{V}_{6N}$		ONO		
$\vec{V}_7$		PON		Medium vector	$\frac{\sqrt{3}}{3}V_d$
$\vec{V}_8$		OPN			
$\vec{V}_9$		NPO			
$\vec{V}_{10}$		NOP			
$\vec{V}_{11}$		ONP			
$\vec{V}_{12}$		PNO			
$\vec{V}_{13}$		PNN		Large vector	$\frac{2}{3}V_d$
$\vec{V}_{14}$		PPN			
$\vec{V}_{15}$		NPN			
$\vec{V}_{16}$		NPP			
$\vec{V}_{17}$		NNP			
$\vec{V}_{18}$		PNP			

the inverter has a total of 27 valid combinations of switching states. As listed in Table 8.3-1, these three-phase switching states are represented by three letters in square brackets for the inverter phases *A*, *B,* and *C*.

To find the relationship between the switching states and their corresponding space voltage vectors, we can follow the same procedures presented in Chapter 6. The 27 switching states listed in the table correspond to 19 voltage vectors whose space

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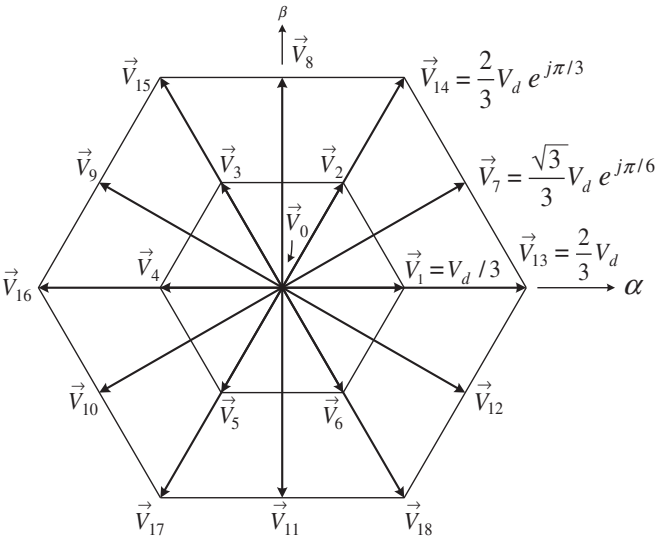

Figure 8.3-1 Space vector diagram of the NPC inverter.

vector diagram is given in Fig. 8.3-1. Based on their magnitude (length), the voltage vectors can be divided into four groups:

- **Zero vector** ($\vec{V}_0$), representing three switching states [PPP], [OOO], and [NNN]. The magnitude of $\vec{V}_0$ is zero;
- **Small vectors** ($\vec{V}_1$ to $\vec{V}_6$), all having a magnitude of $V_d/3$. Each small vector has two switching states, one containing [P] and the other containing [N], and therefore can be further classified into a P- or N-type small vector;
- **Medium vectors** ($\vec{V}_7$ to $\vec{V}_{12}$), whose magnitude is $\sqrt{3}V_d/3$; and
- **Large vectors** ($\vec{V}_{13}$ to $\vec{V}_{18}$), all having a magnitude of $2V_d/3$.

8.3.2 Dwell Time Calculation

To facilitate the dwell time calculation, the space vector diagram of Fig. 8.3-1 can be divided into six triangular **sectors**(I to VI), each of which can be further divided into four triangular **regions**(1 to 4) as illustrated in Fig. 8.3-2. The switching states of all the vectors are also shown in the figure.

Similar to the SVM algorithm for the two-level inverter, the SVM for the NPC inverter is also based on "volt-second balancing" principle, that is, the product of the reference voltage $\vec{V}_{ref}$ and sampling period T_s equals the sum of the voltage multiplied by the time interval of chosen space vectors. In the NPC inverter, the reference vector $\vec{V}_{ref}$ can be synthesized by three nearest stationary vectors. For instance, when $\vec{V}_{ref}$

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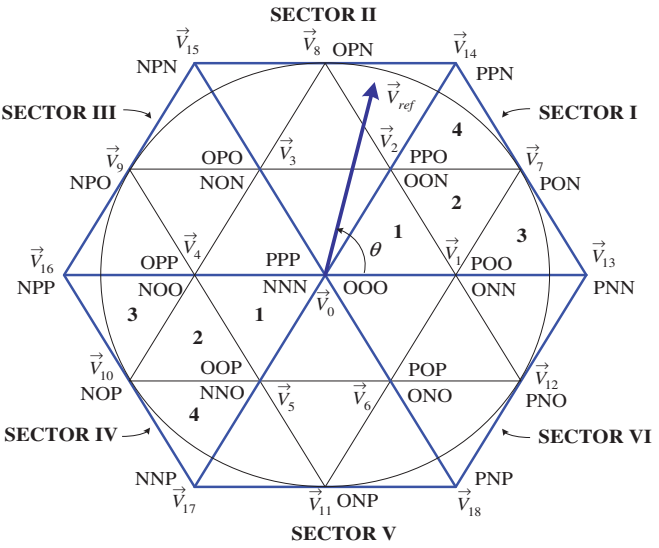

Figure 8.3-2 Division of sectors and regions.

falls into region 2 of sector I as shown in Fig. 8.3-3, the three nearest vectors are $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_7$, from which

$$\begin{cases} \vec{V}_1\, T_a + \vec{V}_7\, T_b + \vec{V}_2\, T_c = \vec{V}_{ref}\, T_s \\ T_a + T_b + T_c = T_s \end{cases} \tag{8.3-1}$$

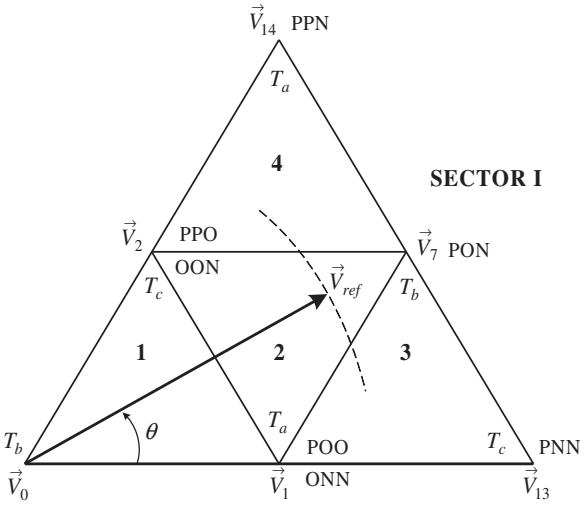

Figure 8.3-3 Voltage vectors and their dwell times.

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where T_a, T_b, and T_c are the dwell times for $\vec{V}_1, \vec{V}_7$, and $\vec{V}_2$, respectively. Note that $\vec{V}_{ref}$ can also be synthesized by other space vectors instead of the "nearest three." However, it will cause higher harmonic distortion in the inverter output voltage, which is undesirable in most cases.

The voltage vectors $\vec{V}_1, \vec{V}_2, \vec{V}_7$, and $\vec{V}_{ref}$ in Fig. 8.3-3 can be expressed as

$$\vec{V}_1 = \frac{1}{3}V_d, \vec{V}_2 = \frac{1}{3}V_d e^{j\frac{\pi}{3}}, \vec{V}_7 = \frac{\sqrt{3}}{3}V_d\, e^{j\frac{\pi}{6}}, \text{ and } \vec{V}_{ref} = V_{ref}\, e^{j\theta} \tag{8.3-2}$$

Substituting (8.3-2) into (8.3-1) yields

$$\frac{1}{3}\,V_d\,T_a + \frac{\sqrt{3}}{3}\,V_d\,e^{j\frac{\pi}{6}}\,T_b + \frac{1}{3}\,V_d\,e^{j\frac{\pi}{3}}\,T_c = V_{ref}\,e^{j\theta}\,T_s \tag{8.3-3}$$

from which

$$\begin{aligned}\frac{1}{3}\,V_d\,T_a + \frac{\sqrt{3}}{3}\,V_d\left(\cos\frac{\pi}{6} + j\sin\frac{\pi}{6}\right)\,T_b + \frac{1}{3}\,V_d\left(\cos\frac{\pi}{3} + j\sin\frac{\pi}{3}\right)\,T_c \\ = V_{ref}\,(\cos\theta + j\sin\theta)\,T_s\end{aligned} \tag{8.3-4}$$

Splitting (8.3-4) into the real and imaginary parts, we have

$$\begin{cases}\text{Re:} \quad T_a + \dfrac{3}{2}\,T_b + \dfrac{1}{2}\,T_c = 3\dfrac{V_{ref}}{V_d}(\cos\theta)\,T_s \\ \text{Im:} \quad \dfrac{\sqrt{3}}{2}\,T_b + \dfrac{\sqrt{3}}{2}\,T_c = 3\dfrac{V_{ref}}{V_d}(\sin\theta)\,T_s\end{cases} \tag{8.3-5}$$

Solve (8.3-5) together with $T_s = T_a + T_b + T_c$ for dwell times

$$\begin{cases}T_a = T_s[1 - 2\,m_a\sin\theta] \\ T_b = T_s\left[2\,m_a\sin\left(\dfrac{\pi}{3} + \theta\right) - 1\right] \quad \text{for} \quad 0 \le \theta < \pi/3 \\ T_c = T_s\left[1 - 2\,m_a\sin\left(\dfrac{\pi}{3} - \theta\right)\right]\end{cases} \tag{8.3-6}$$

Where m_a is the **modulation index**, defined by

$$m_a = \sqrt{3}\,\frac{V_{ref}}{V_d} \tag{8.3-7}$$

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Table 8.3-2 Dwell Time Calculation for $\vec{V}_{ref}$ in Sector I

Region	T_a		T_b		T_c	
1	$\vec{V}_1$	$T_s[2m_a \sin(\frac{\pi}{3} - \theta)]$	$\vec{V}_0$	$T_s[1 - 2m_a \sin(\frac{\pi}{3} + \theta)]$	$\vec{V}_2$	$T_s[2m_a \sin\theta]$
2	$\vec{V}_1$	$T_s[1 - 2m_a \sin\theta]$	$\vec{V}_7$	$T_s[2m_a \sin(\frac{\pi}{3} + \theta) - 1]$	$\vec{V}_2$	$T_s[1 - 2m_a \sin(\frac{\pi}{3} - \theta)]$
3	$\vec{V}_1$	$T_s[2 - 2m_a \sin(\frac{\pi}{3} + \theta)]$	$\vec{V}_7$	$T_s[2m_a \sin\theta]$	$\vec{V}_{13}$	$T_s[2m_a \sin(\frac{\pi}{3} - \theta) - 1]$
4	$\vec{V}_{14}$	$T_s[2m_a \sin\theta - 1]$	$\vec{V}_7$	$T_s[2m_a \sin(\frac{\pi}{3} - \theta)]$	$\vec{V}_2$	$T_s[2 - 2m_a \sin(\frac{\pi}{3} + \theta)]$

The maximum length of the reference vector $\vec{V}_{ref}$ corresponds to the radius of the largest circle that can be inscribed within the hexagon of Fig. 8.3-2, which happens to be the length of the medium voltage vectors

$$V_{ref,\max} = \sqrt{3}\, V_d/3$$

Substituting $V_{ref,\max}$ into (8.3-7) yields the maximum modulation index

$$m_{a,\max} = \sqrt{3}\,\frac{V_{ref,\max}}{V_d} = 1 \tag{8.3-8}$$

from which the range of m_a is

$$0 \le m_a \le 1 \tag{8.3-9}$$

Table 8.3-2 gives the equations for the calculation of dwell times for $\vec{V}_{ref}$ in sector I.

The equations in Table 8.3-2 can also be used to calculate the dwell times when $\vec{V}_{ref}$ is in other sectors (II to VI) provided that a multiple of $\pi/3$ is subtracted from the actual angular displacement θ such that the modified angle falls into the range between zero and $\pi/3$ for use in the equations. The reader can refer to Chapter 6 for details.

8.3.3 Relationship Between $\vec{V}_{ref}$ Location and Dwell Times

To demonstrate the relationship between the $\vec{V}_{ref}$ location and dwell times, consider an example shown in Fig. 8.3-4. Assuming that the head of $\vec{V}_{ref}$ points to the center Q of region 4, the dwell times for the nearest three vectors $\vec{V}_2$, $\vec{V}_7$, and $\vec{V}_{14}$ should be identical since the distance from Q to these vectors is the same. This can be verified

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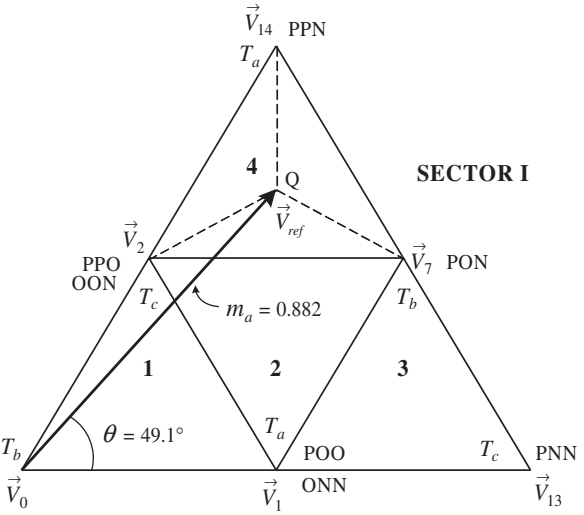

Figure 8.3-4 An example to demonstrate the relationship between the location of $\vec{V}_{ref}$ and dwell times.

by substituting $m_a = 0.882$ and $\theta = 49.1°$ into the equations in Table 8.3-2, from which the calculated dwell times are $T_a = T_b = T_c = 0.333T_s$.

With $\vec{V}_{ref}$ moving toward $\vec{V}_2$ from Q along the dashed line, the influence of $\vec{V}_2$ on $\vec{V}_{ref}$ becomes stronger, which translates into a longer dwell time for $\vec{V}_2$. When $\vec{V}_{ref}$ is identical to $\vec{V}_2$, the dwell time T_c for $\vec{V}_2$ reaches its maximum value ($T_c = T_s$) while T_b and T_c for $\vec{V}_2$ and $\vec{V}_7$ diminish to zero.

8.3.4 Switching Sequence Design

The neutral-point voltage v_Z, which is defined as the voltage between the neutral point Z and the negative dc bus, normally varies with the switching state of the NPC inverter. When designing the switching sequence, we should minimize the effect of the switching state on neutral-point voltage deviation. Taking into account the two requirements presented in Chapter 6 for the two-level inverter, the overall requirements for switching sequence design in the NPC inverter are

- **a.** The transition from one switching state to the next involves only two switches in the same inverter leg, one being switched on and the other switched off.
- **b.** The transition for $\vec{V}_{ref}$ moving from one sector (or region) to the next requires no or minimum number of switchings.
- **c.** The effect of switching state on the neutral-point voltage deviation is minimized.

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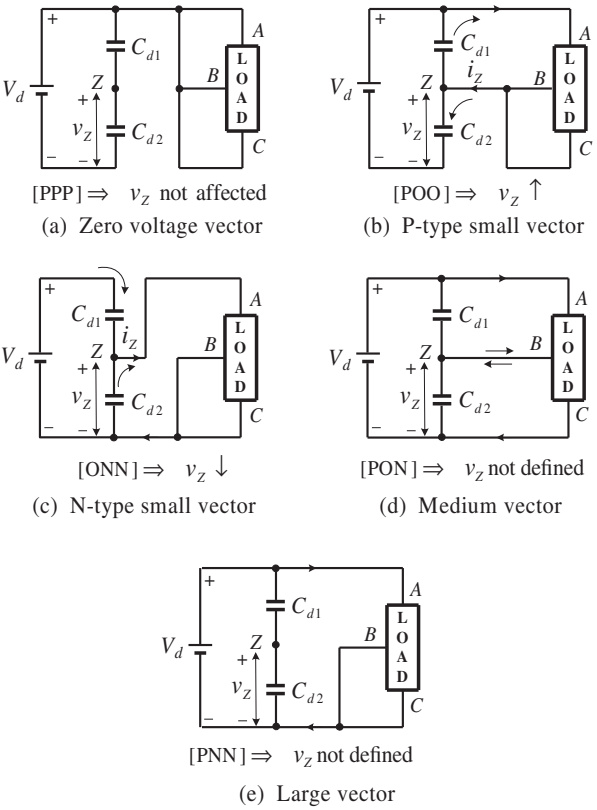

Figure 8.3-5 Effect of switching states on neutral-point voltage deviation.

(a) Effect of Switching States on Neutral-Point Voltage Deviation The effect of switching states on neutral voltage deviation is illustrated in Fig. 8.3-5. When the inverter operates with switching state [PPP] of zero vector $\vec{V}_0$, the upper two switches in each of the three inverter legs are turned on, connecting the inverter terminals *A, B,* and *C* to the positive dc bus as shown in Fig. 8.3-5a. Since the neutral point *Z* is left unconnected, this switching state does not affect v_Z. Similarly, the other two zero switching states, [OOO] and [NNN], do not cause v_Z to shift either.

Figure 8.3-5b shows the inverter operation with P-type switching state [POO] of small vector $\vec{V}_1$. Since the three-phase load is connected between the positive dc bus and neutral point *Z*, the neutral current i_Z flows into *Z*, causing v_Z to increase. On the contrary, the N-type switching state [ONN] of $\vec{V}_1$ makes v_Z to decrease as shown in Fig. 8.3-5c.

The medium voltage vectors also affect the neutral-point voltage. For medium vector $\vec{V}_7$ with switching state [PON] in Fig. 8.3-5d, load terminals *A*, *B,* and *C* are connected to the positive bus, the neutral point and the negative bus, respectively.

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Depending on the inverter operating conditions, the neutral-point voltage v_Z may rise or drop.

Considering a large vector $\vec{V}_{13}$ with switching state [PNN] shown in Fig. 8.3-5e, the load terminals are connected between the positive and negative dc buses. The neutral point Z is left unconnected, and thus the neutral voltage is not affected.

It can be summarized that

- Zero vector $\vec{V}_0$ does not affect the neutral-point voltage v_Z;
- Small vectors $\vec{V}_1$ to $\vec{V}_6$ have a dominant influence on v_Z. A P-type small vector makes v_Z rise while an N-type small vector causes v_Z to decline;
- Medium vectors $\vec{V}_7$ to $\vec{V}_{12}$ also affect v_Z, but the direction of voltage deviation is undefined; and
- The large vectors $\vec{V}_{13}$ to $\vec{V}_{18}$ do not play a role in neutral-point voltage deviation.

Note that the above summary is made under the assumption that the inverter is in normal (motoring) operating mode. The effect of the regenerative operation on neutral-point voltage shift will be addressed later.

(b) Switching Sequence with Minimal Neutral-Point Voltage Deviation As mentioned earlier, a P-type small vector causes the neutral-point voltage v_Z to rise while an N-type small vector makes v_Z fall. To minimize the neutral-point voltage deviation, the dwell time of a given small vector can be equally distributed between the P- and N-type switching states over a sampling period. According to the triangular region that the reference vector $\vec{V}_{ref}$ lies in, the following two cases are investigated.

Case 1: One Small Vector Among Three Selected Vectors. When the reference vector $\vec{V}_{ref}$ is in region 3 or 4 of sector I in Fig. 8.3-3, only one of the three selected vectors is the small vector. Assuming that $\vec{V}_{ref}$ falls into region 4, it can be synthesized by $\vec{V}_2$, $\vec{V}_7$, and $\vec{V}_{14}$. The small vector $\vec{V}_2$ has two switching states [PPO] and [OON]. To minimize the neutral voltage deviation, the dwell time for $\vec{V}_2$ should be equally distributed between the P- and N-type states. Fig. 8.3-6 shows a typical seven-segment switching sequence for the NPC inverter, from which we can observe that

- The dwell times for the seven segments add up to the sampling period ($T_s = T_a + T_b + T_c$).
- The design requirement *a*) is satisfied. For instance, the transition from [OON] to [PON] is accomplished by turning S_1 on and switching S_3 off, which involves only two switches.
- The dwell time T_c for $\vec{V}_2$ is equally divided between the P- and N-type switching states, which satisfies the design requirement (*c*).

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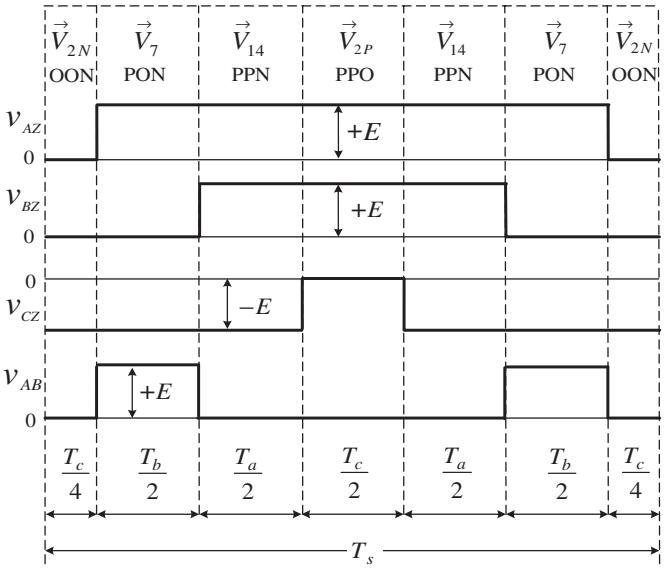

Figure 8.3-6 Seven-segment switching sequence for $\vec{V}_{ref}$ in sector I-4.

- Among the four switching devices in an inverter leg, only two are tuned on and off once per sampling period. Assuming that the transition for $\vec{V}_{ref}$ moving from one sector (or region) to the next does not involve any switchings, the **device switching frequency** $f_{sw,dev}$ is equal to half of the sampling frequency f_{sp}, that is,

$$f_{sw,dev} = f_{sp}/2 = 1/(2T_s) \tag{8.3-10}$$

Case 2: Two Small Vectors Among Three Selected Vectors. When $\vec{V}_{ref}$ is in region 1 or 2 of sector I in Fig. 8.3-3, two of the three selected vectors are small vectors. To reduce the neutral voltage deviation, each of the two regions is further divided into two sub-regions as shown in Fig. 8.3-7. Assuming that $\vec{V}_{ref}$ lies in region 2a, it can be approximated by $\vec{V}_1$, $\vec{V}_2$ and $\vec{V}_7$. Since $\vec{V}_{ref}$ is closer to $\vec{V}_1$ than $\vec{V}_2$, the corresponding dwell time T_a for $\vec{V}_1$ is longer than T_c for $\vec{V}_2$. The vector $\vec{V}_1$ is referred to as **dominant small vector**, whose dwell time is equally divided between $\vec{V}_{1P}$ and $\vec{V}_{1N}$ as shown in Table 8.3-3.

Based on the above discussions, all the switching sequences in sectors I and II are summarized in Table 8.3-4. It can be observed that (1) when $\vec{V}_{ref}$ crosses the border between sectors I and II, the transition does not involve any switchings; and (2) an extra switching takes place when $\vec{V}_{ref}$ moves from region a to b within a sector. The graphical representation is illustrated in Fig. 8.3-8, where the large and small circles are the steady-state trajectories of $\vec{V}_{ref}$ and the dots represent the locations at which

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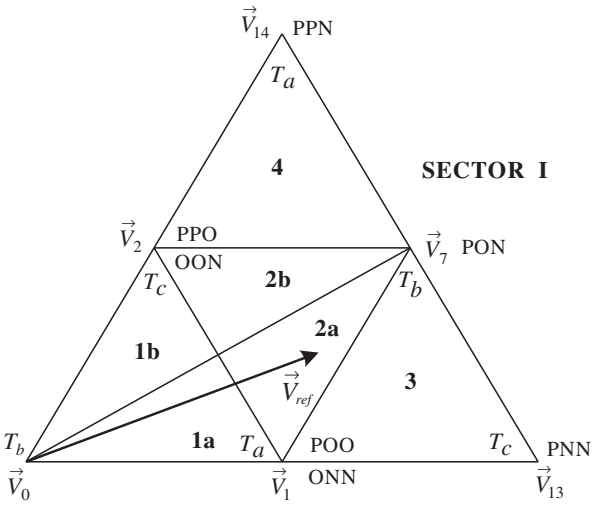

Figure 8.3-7 Division of six regions of sector I for the minimization of neutral-point voltage deviation.

an extra switching takes place. Since each of these extra switchings involves only two devices (out of 12) and there are only six extra switchings per cycle of fundamental frequency, the average switching frequency of the device is increased to

$$f_{sw,dev} = f_{sp}/2 + f_1/2 \tag{8.3-11}$$

8.3.5 Inverter Output Waveforms and Harmonic Content

Figure 8.3-9 shows the simulated waveforms for the NPC inverter operating at $f_1 = 60\,\text{Hz}$, $T_s = 1/1080\,\text{s}$, $f_{sw,dev} = 1080/2 + 60/2 = 570\,\text{Hz}$ and $m_a = 0.8$.The inverter is loaded with a three-phase inductive load with a power factor of 0.9. The gate signals v_{g1} and v_{g4} are for the switches S_1 and S_4 of the inverter circuit in Fig. 8.2-1. Since the inner switches S_2 and S_3 operate complementarily with S_4 and S_1, their gatings are not shown.

Table 8.3-3 Seven-Segment Switching Sequence for $\vec{V}_{ref}$ in Sector I-2a

Segment	1st	2nd	3rd	4th	5th	6th	7th
Voltage Vector	$\vec{V}_{1N}$	$\vec{V}_{2N}$	$\vec{V}_7$	$\vec{V}_{1P}$	$\vec{V}_7$	$\vec{V}_{2N}$	$\vec{V}_{1N}$
Switching State	[ONN]	[OON]	[PON]	[POO]	[PON]	[OON]	[ONN]
Dwell Time	$\frac{T_a}{4}$	$\frac{T_c}{2}$	$\frac{T_b}{2}$	$\frac{T_a}{2}$	$\frac{T_b}{2}$	$\frac{T_c}{2}$	$\frac{T_a}{4}$

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Table 8.3-4 Seven-Segment Switching Sequence

Sector I												
Sgmt	**1a**		**1b**		**2a**		**2b**		**3**		**4**	
1st	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON
2nd	$\vec{V}_{2N}$	OON	$\vec{V}_{0}$	OOO	$\vec{V}_{2N}$	OON	$\vec{V}_{7}$	PON	$\vec{V}_{13}$	PNN	$\vec{V}_{7}$	PON
3rd	$\vec{V}_{0}$	OOO	$\vec{V}_{1P}$	POO	$\vec{V}_{7}$	PON	$\vec{V}_{1P}$	POO	$\vec{V}_{7}$	PON	$\vec{V}_{14}$	PPN
4th	$\vec{V}_{1P}$	POO	$\vec{V}_{2P}$	PPO	$\vec{V}_{1P}$	POO	$\vec{V}_{2P}$	PPO	$\vec{V}_{1P}$	POO	$\vec{V}_{2P}$	PPO
5th	$\vec{V}_{0}$	OOO	$\vec{V}_{1P}$	POO	$\vec{V}_{7}$	PON	$\vec{V}_{1P}$	POO	$\vec{V}_{7}$	PON	$\vec{V}_{14}$	PPN
6th	$\vec{V}_{2N}$	OON	$\vec{V}_{0}$	OOO	$\vec{V}_{2N}$	OON	$\vec{V}_{7}$	PON	$\vec{V}_{13}$	PNN	$\vec{V}_{7}$	PON
7th	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON	$\vec{V}_{1N}$	ONN	$\vec{V}_{2N}$	OON
Sector II												
Sgmt	**1a**		**1b**		**2a**		**2b**		**3**		**4**	
1st	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON
2nd	$\vec{V}_{0}$	OOO	$\vec{V}_{2N}$	OON	$\vec{V}_{8}$	OPN	$\vec{V}_{2N}$	OON	$\vec{V}_{8}$	OPN	$\vec{V}_{15}$	NPN
3rd	$\vec{V}_{3P}$	OPO	$\vec{V}_{0}$	OOO	$\vec{V}_{3P}$	OPO	$\vec{V}_{8}$	OPN	$\vec{V}_{14}$	PPN	$\vec{V}_{8}$	OPN
4th	$\vec{V}_{2P}$	PPO	$\vec{V}_{3P}$	OPO	$\vec{V}_{2P}$	PPO	$\vec{V}_{3P}$	OPO	$\vec{V}_{2P}$	PPO	$\vec{V}_{3P}$	OPO
5th	$\vec{V}_{3P}$	OPO	$\vec{V}_{0}$	OOO	$\vec{V}_{3P}$	OPO	$\vec{V}_{8}$	OPN	$\vec{V}_{14}$	PPN	$\vec{V}_{8}$	OPN
6th	$\vec{V}_{0}$	OOO	$\vec{V}_{2N}$	OON	$\vec{V}_{8}$	OPN	$\vec{V}_{2N}$	OON	$\vec{V}_{8}$	OPN	$\vec{V}_{15}$	NPN
7th	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON	$\vec{V}_{2N}$	OON	$\vec{V}_{3N}$	NON

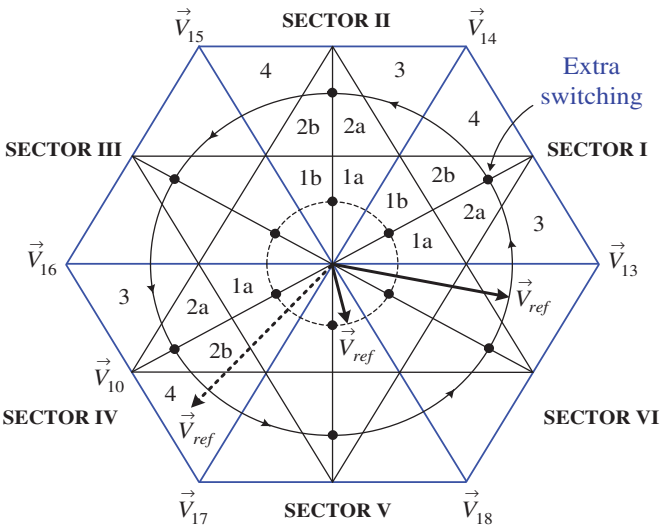

Figure 8.3-8 Graphical representation of extra switchings when $\vec{V}_{ref}$ moves from region a to b.

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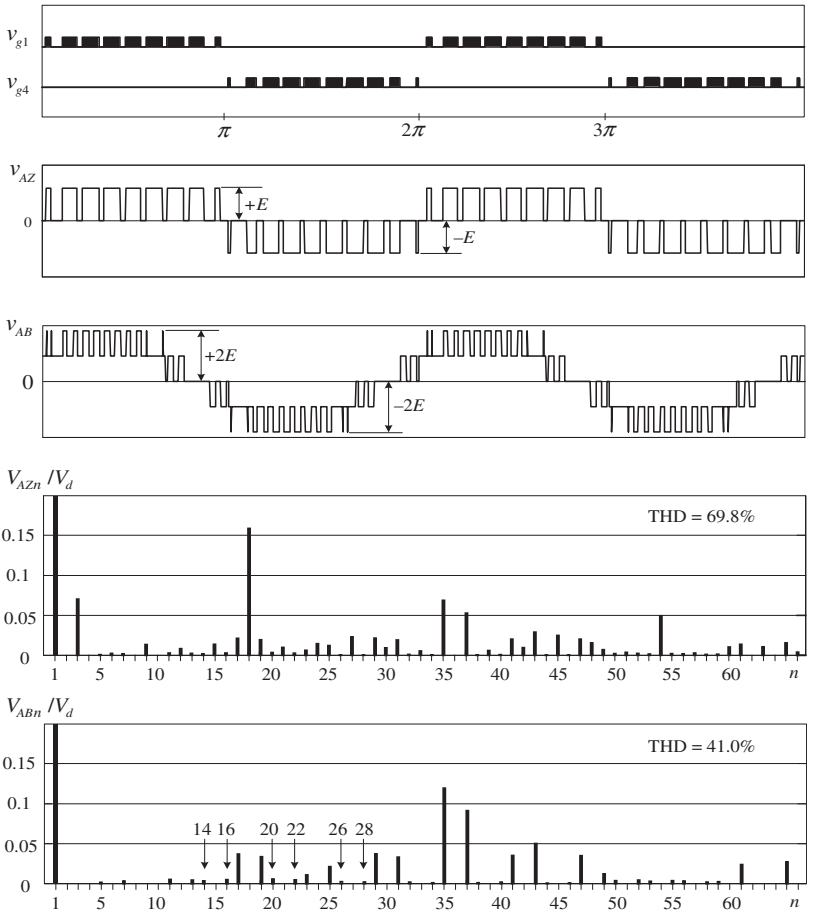

Figure 8.3-9 Simulated voltage waveforms of the NPC inverter ($f_1 = 60$ Hz, $T_s = 1/1080$ s, $f_{sw,dev} = 570$ Hz, and $m_a = 0.8$).

The waveform of the inverter terminal voltage v_{AZ} is composed of three voltage levels while the inverter line-to-line voltage v_{AB} has five voltage levels. The waveform for v_{AZ} contains triplen harmonics with the 3rd and 18th being dominant. Since the triplen harmonics are of zero sequence, they do not appear in the line-to-line voltage v_{AB}. However, v_{AB} contains even-order harmonics such as the 14th and 16th in addition to odd-order harmonics. This is due to the fact that waveform of v_{AB} produced by the SVM scheme is not half-wave symmetrical.

The dominant harmonics in v_{AB} are the 17th and 19th, centered around the 18th harmonic whose frequency is 1080 Hz. As discussed in Chapter 6, this frequency can be considered as the equivalent **inverter switching frequency** $f_{sw,inv}$, which is approximately twice the device switching frequency $f_{sw,dev}$.

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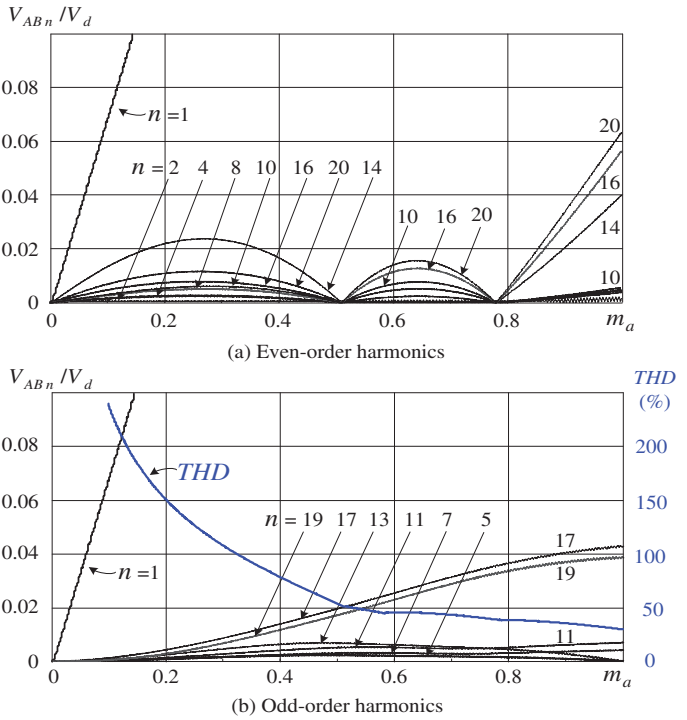

Figure 8.3-10 Harmonic content and THD of the inverter line-to-line voltage v_{AB} ($f_1 = 60$ Hz, $T_s = 1/1080$ s, and $f_{sw,dev} = 570$ Hz).

The harmonic content and THD of v_{AB} versus m_a are illustrated in Fig. 8.3-10, where V_{ABn} is the rms value of the n^{th}order harmonic voltage. The waveform of v_{AB} contains all the low-order harmonics except for triplen harmonics. The magnitude of most even-order harmonics peaks at $m_a = 1$. The maximum rms fundamental voltage occurs at $m_a = 1$, at which

$$V_{AB1,\max} = 0.707 V_d \tag{8.3-12}$$

Figure 8.3-11 shows waveforms measured from a laboratory three-level NPC inverter operating at $f_1 = 60$ Hz, $T_s = 1/1080$ s, and $f_{sw,dev} = 570$ Hz with the modulation index m_a equal to 0.8 and 0.9, respectively. The measured waveforms and their spectrum at $m_a = 0.8$ correlate closely with the simulated results in Fig. 8.3-9. It can also be observed that the magnitude of the even-order harmonics at $m_a = 0.9$ is much higher than that for $m_a = 0.8$, which is consistent with the harmonic content illustrated in Fig. 8.3-10.

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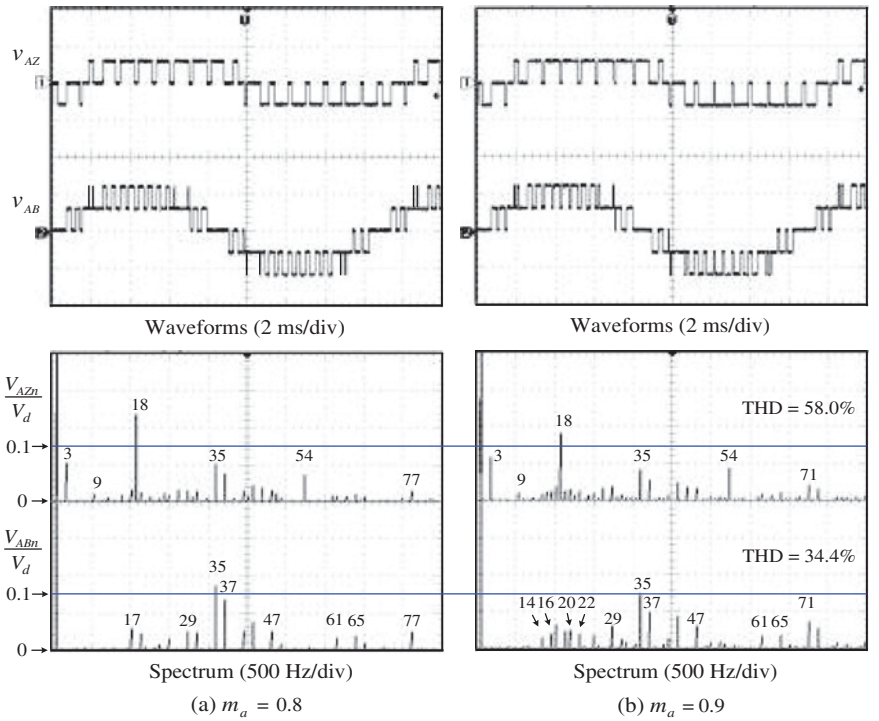

Figure 8.3-11 Measured waveforms and their harmonic spectra ($f_1 = 60\,\text{Hz}$, $T_s = 1/1080\,\text{s}$, and $f_{sw,dev} = 570\,\text{Hz}$).

8.3.6 Even-Order Harmonic Elimination

The mechanism of even-order harmonic generation and the reasons for its elimination have been discussed in Chapter 6 for the two-level inverter. They can be equally applied to the three-level NPC inverter, and therefore are not repeated here.

Figure 8.3-12 shows two valid switching sequences for $\vec{V}_{ref}$ in sector IV-4 of the space vector diagram in Fig. 8.3-8. It can be observed that type-A sequence starts with an N-type small vector while type-B sequence commences with a P-type small vector. Although the waveforms of v_{AZ}, v_{BZ}, v_{CZ}, and v_{AB} in Figs. 8.3-12a and 8.3-12b seem quite different, they are essentially the same except a small amount of time delay ($T_s/2$), which can be clearly observed if these waveforms are drawn for two or more consecutive sampling periods.

In the conventional SVM scheme for the NPC inverter, only the type-A switching sequence is employed. To eliminate the even-order harmonics in v_{AB}, type A and B switching sequences can be alternatively used as illustrated in Fig. 8.3-13. The reader can refer to Chapter 6 for the principle of even-order harmonic elimination. A complete set of the switching sequences for the modified SVM scheme is given in the appendix of this chapter. Compared with the conventional SVM, the modified

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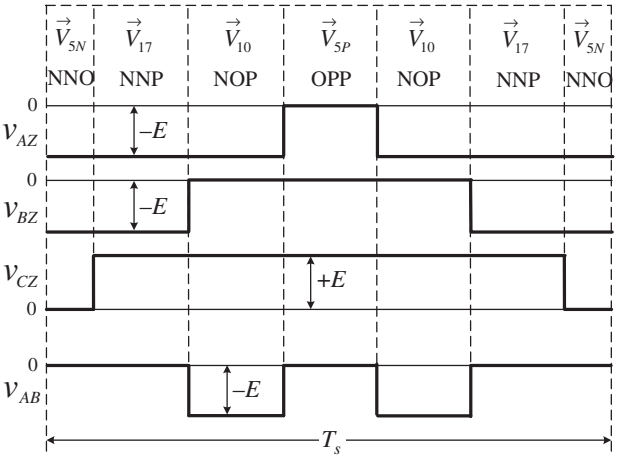
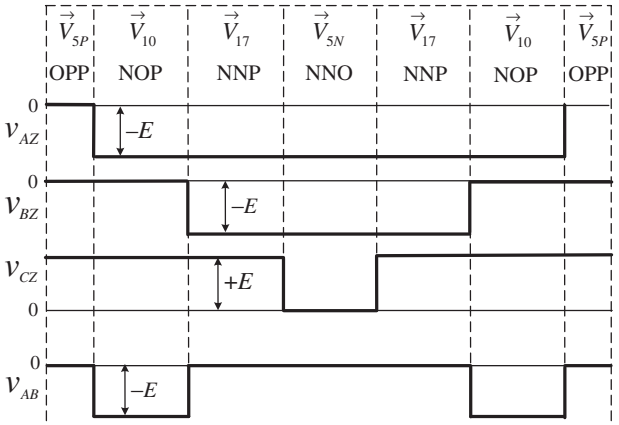


(a) Type-A sequence
(starts with an N-type small vector)



(b) Type-B sequence
(starts with a P-type small vector)

Figure 8.3-12 Two valid switching sequences for $\vec{V}_{ref}$ in sector IV-4.

scheme causes a slight increase in the device switching frequency. The amount of increase is given by $\Delta f_{sw} = f_1/2$, from which

$$f_{sw,dev} = f_{sp}/2 + f_1 \tag{8.3-13}$$

Figure 8.3-14 shows waveforms measured from a laboratory NPC inverter with the modified SVM scheme. The inverter output voltage waveforms of v_{AZ} and v_{AB} are of half-wave symmetry, leading to the elimination of even-order harmonics. It is interesting to note that although the harmonic spectrum of v_{AB} differs from that in Fig. 8.3-11, its THD essentially remains unchanged.

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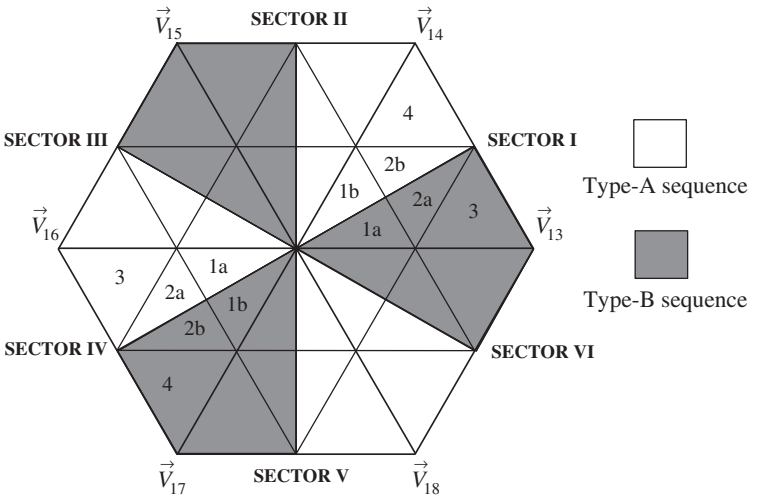

Figure 8.3-13 Alternative use of type A and B switching sequences for even-order harmonic elimination.

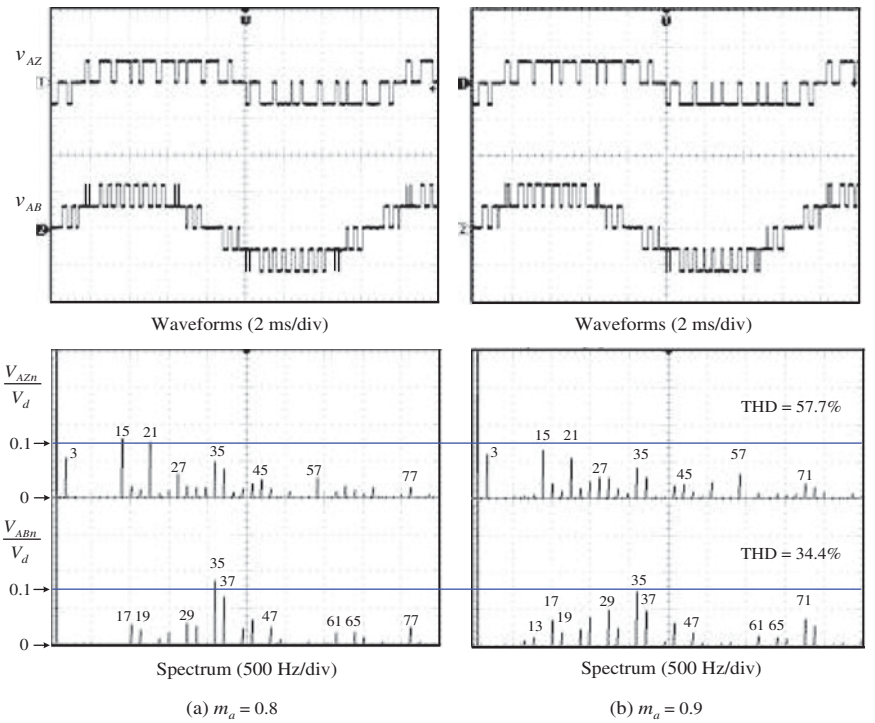

Figure 8.3-14 Measured waveforms produced by the modified SVM with even-order harmonic elimination ($f_1 = 60\,\text{Hz}$, $T_s = 1/1080\,\text{s}$, $f_{sw,dev} = 600\,\text{Hz}$).

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8.4 NEUTRAL-POINT VOLTAGE CONTROL

As indicated earlier, the neutral-point voltage v_Z varies with the operating condition of the NPC inverter. If the neutral-point voltage deviates too far, an uneven voltage distribution takes place, which may lead to premature failure of the switching devices and cause an increase in the THD of the inverter output voltage.

8.4.1 Causes of Neutral-Point Voltage Deviation

In addition to the influence of small and medium voltage vectors, the neutral-point voltage may also be affected by a number of other factors, including

- Unbalanced dc capacitors due to manufacturing tolerances
- Inconsistency in switching device characteristics
- Unbalanced three-phase operation

To minimize the neutral-point voltage shift, a feedback control scheme can be implemented, where the neutral-point voltage is detected and then controlled [7, 9, 10].

8.4.2 Effect of Motoring and Regenerative Operation

When the NPC inverter is used in the MV drives, the operating mode of the drive may also influence the neutral-point voltage. Fig. 8.4-1 shows the effect of motoring and regenerative operations of the drive on neutral-point voltage shift. When the drive is in the motoring mode as shown in Fig. 8.4-1a where the dc current i_d flows from the dc source to the inverter, the P-type state [POO] of small vector $\vec{V}_1$ causes the neutral-point voltage v_Z to rise while the N-type state [ONN] makes v_Z to decline. An opposite action takes place in the regenerative mode in which the dc current reverses its direction as shown in Fig. 8.4-1b. This phenomenon should be taken into account when designing the feedback control for v_Z.

8.4.3 Feedback Control of Neutral-Point Voltage

The neutral-point voltage v_Z can be controlled by adjusting the time distribution between the P- and N-type states of a small voltage vector. There always exists a small voltage vector in each switching sequence, whose dwell time is divided into two sub-periods, one for its P-type and the other for its N-type switching state. For instance, the dwell time T_a for $\vec{V}_{1P}$ and $\vec{V}_{1N}$, which is 50/50 split in Table 8.3-3, can be redistributed as

$$T_a = T_{aP} + T_{aN} \tag{8.4-1}$$

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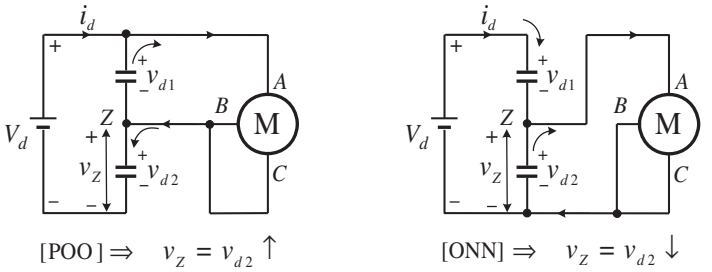
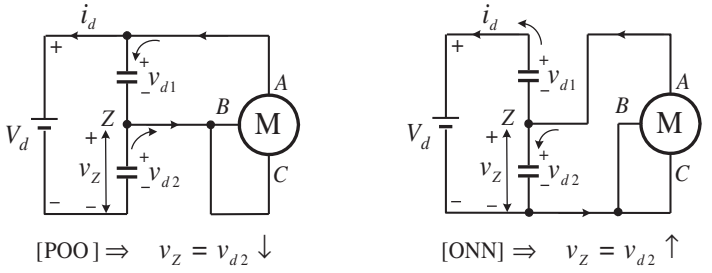


(a) Motoring operation



(b) Regenerative operation

Figure 8.4-1 Effect of drive operating modes on neutral-point voltage deviation.

where T_{aP} and T_{aN} are given by

$$\begin{cases} T_{aP} = \dfrac{T_a}{2}(1 + \Delta t) \\ T_{aN} = \dfrac{T_a}{2}(1 - \Delta t) \end{cases} \quad \text{for } -1 \le \Delta t \le 1 \tag{8.4-2}$$

The deviation of the neutral-point voltage can be minimized by adjusting the incremental time interval Δt in (8.4-2) according to the detected dc capacitor voltages v_{d1} and v_{d2}. For instance, if $(v_{d1} - v_{d2})$ is greater than the maximum allowed dc voltage deviation ΔV_d for some reasons, we can increase T_{aP} and decrease T_{aN} by Δt ($\Delta t > 0$) simultaneously for the drive in a motoring mode. A reverse action ($\Delta t < 0$) should be taken when the drive is in a regenerative mode. The relationship between the capacitor voltages and the incremental time interval Δt is summarized in Table 8.4-1.

Figure 8.4-2 shows the simulated waveforms of the two dc capacitor voltages v_{d1} and v_{d2} with an equal initial value of 2800 V. To make v_{d1} and v_{d2} unbalanced on purpose, a resistor is connected in parallel with the bottom capacitor C_{d2}. Without the neutral-point voltage control, the capacitor C_{d2} discharges through the resistor and its voltage v_{d2} decreases over time while the voltage on C_{d1} increases. At $t = 0.026$ s,

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Table 8.4-1 Relationship Between Capacitor Voltages and Incremental Time Interval Δt

Neutral Point Deviation Level	Motoring Mode $i_d > 0$	Regenerating Mode $i_d < 0$
$(v_{d1} - v_{d2}) > \Delta V_d$	$\Delta t > 0$	$\Delta t < 0$
$(v_{d2} - v_{d1}) > \Delta V_d$	$\Delta t < 0$	$\Delta t > 0$
$\lvert v_{d1} - v_{d2} \rvert < \Delta V_d$	$\Delta t = 0$	$\Delta t = 0$
ΔV_d—maximum allowed dc voltage deviation ($\Delta V_d > 0$).		

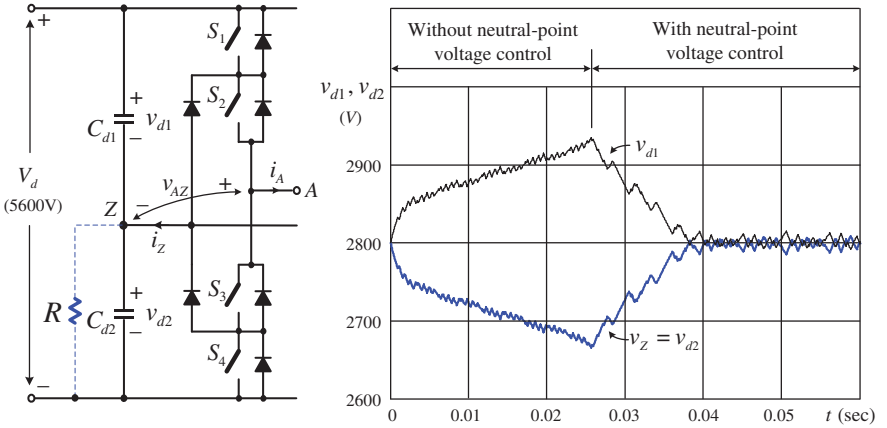

Figure 8.4-2 Neutral-point voltage control by SVM scheme.

the neutral-point voltage control is activated, making dc capacitor voltages balanced at $t = 0.04$ s.

8.5 CARRIER-BASED PWM SCHEME AND NEUTRAL-POINT VOLTAGE CONTROL

The carrier-based modulation schemes presented in Chapter 7 can also be used for the NPC inverters. Figure 8.5-1 illustrates the simulated waveforms of the three-level NPC inverter modulated by a level-shifted IPD modulation scheme. The three-level inverter requires two carriers v_{cr1} and v_{cr2}, which are compared with modulating wave v_m to generate gate signals v_{g1} and v_{g2} for the top two switches S_1 and S_2 in the inverter shown in Fig. 8.4-2a. The waveforms of the inverter phase voltage v_{Az} and line-to-line voltage v_{AB} are also illustrated in the figure. Since the carrier-based modulation schemes have been extensively discussed in Chapter 7, the details of the waveform generation are not provided here.

The main idea for the control of the neutral-point voltage v_z of the 3L-NPC inverter with the carrier-based PWM scheme is to add a dc offset voltage v_{ofs} to the

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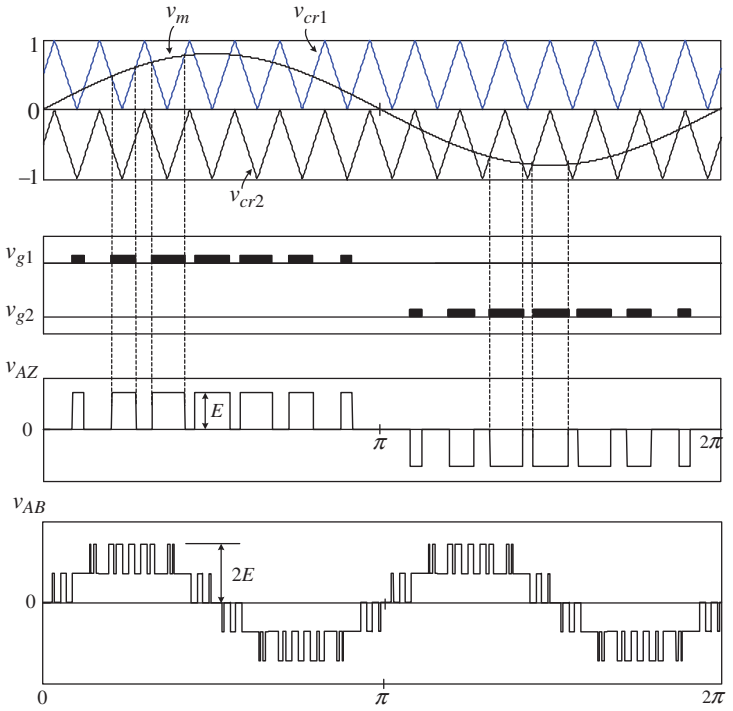

Figure 8.5-1 Waveforms in the 3L-NPC inverter using IPD modulation ($f_1 = 60$ Hz, $f_{sw,dev} = 450$ Hz, $m_a = 0.9$, and $m_f = 15$).

inverter phase voltage v_{AZ}. If the offset voltage is positive, the bottom capacitor C_{d2} in Fig. 8.4-2a will be charged and the top capacitor C_{d1} will be discharged, and vice versa [15].

Figure 8.5-2 shows the block diagram of the voltage balancing control based on the carrier-based PWM scheme. The reference for the capacitor voltage deviation Δv_z^* is set to zero. This reference is compared with the measured voltage deviation Δv_z, which is obtained by $\Delta v_z = v_{d1} - v_{d2}$, and its difference is sent to a PI controller. The PI controller generates the dc offset voltage v_{ofs}, which is added to the modulating

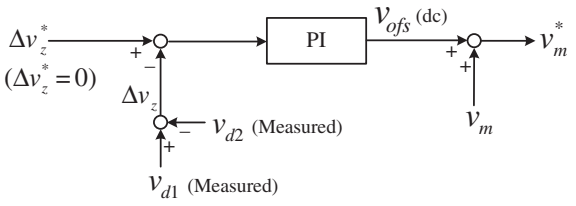

Figure 8.5-2 Neutral-point voltage control for the 3L-NPC inverter.

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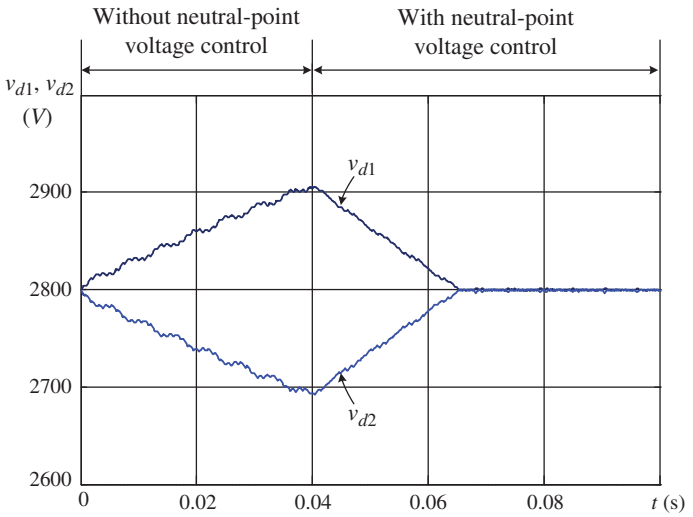

Figure 8.5-3 Neutral-point voltage control by a carrier-based PWM scheme.

wave v_m to produce the reference modulating wave v_m^*. The reference modulating wave is then compared with the carrier wave to generate gate signals for the inverter to make the capacitor voltage balanced. With the PI controller, the measured voltage deviation Δv_z will be kept at zero when the inverter operates in steady state.

To verify the neutral-point voltage control scheme discussed above, the simulated waveforms of the dc capacitor voltages are illustrated in Fig. 8.5-3. The two capacitors have an equal initial value of 2800 V. To make v_{d1} and v_{d2} unbalanced on purpose, a resistor is connected in parallel with the bottom dc capacitor C_{d2} as shown in Fig. 8.4-2a. Without the neutral-point voltage control, the capacitor C_{d2} discharges through the resistor and its voltage v_{d2} decreases over time while the voltage on C_{d1} increases. At $t = 0.04$ s, the neutral-point voltage control is activated, making dc capacitor voltages balanced at $t = 0.065$ s.

8.6 OTHER SPACE VECTOR MODULATION ALGORITHMS

In addition to the SVM schemes presented in the previous section, other SVM algorithms have been proposed for the NPC inverter [11–14]. Two of them are briefly introduced here.

8.6.1 Discontinuous Space Vector Modulation

The principle of the discontinuous (five-segment) SVM scheme presented in Chapter 6 for the two-level inverter can also be applied to the NPC inverter. The five-segment switching sequence can be arranged such that switching for the devices in one of the

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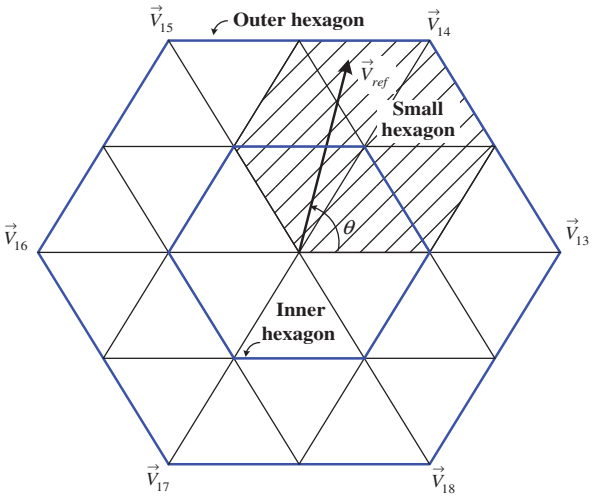

Figure 8.6-1 Space vector diagram of the three-level NPC inverter using two-level SVM algorithm.

three inverter legs is avoided for a period of $\pi/3$ during the positive half cycle and another $\pi/3$ during the negative half cycle of the fundamental frequency. If the $\pi/3$ no-switching period is centered on the positive or negative peaks of the load current, the switching loss can be reduced. The reader can refer to References [11] and [12] for the details.

8.6.2 SVM Based on Two-Level Algorithm

The space vector diagram for the NPC inverter has an **outer hexagon** containing all 24 triangular regions and an **inner hexagon** with six triangular regions as shown in Fig. 8.6-1. The space vector diagram can be decomposed into six **small hexagons**, each of which centers at the six apexes of the inner hexagon [13]. Each of these small hexagons is composed of six triangular regions.

The position of the reference vector $\vec{V}_{ref}$ in the NPC space vector diagram determines which of the six small hexagons is selected. The selected hexagon is then shifted toward the center of the inner hexagon for the dwell time calculation and switching sequence design. Accordingly, $\vec{V}_{ref}$ should be also referred to the new coordinate system. In doing so, the SVM algorithm for the NPC inverter is simplified and can be performed in the same manner as for the two-level inverter.

8.7 HIGH-LEVEL DIODE-CLAMPED INVERTERS

To increase the inverter voltage rating and improve its waveform quality, high-level diode-clamped inverters can be employed. This section presents four- and five-level diode-clamped inverters.

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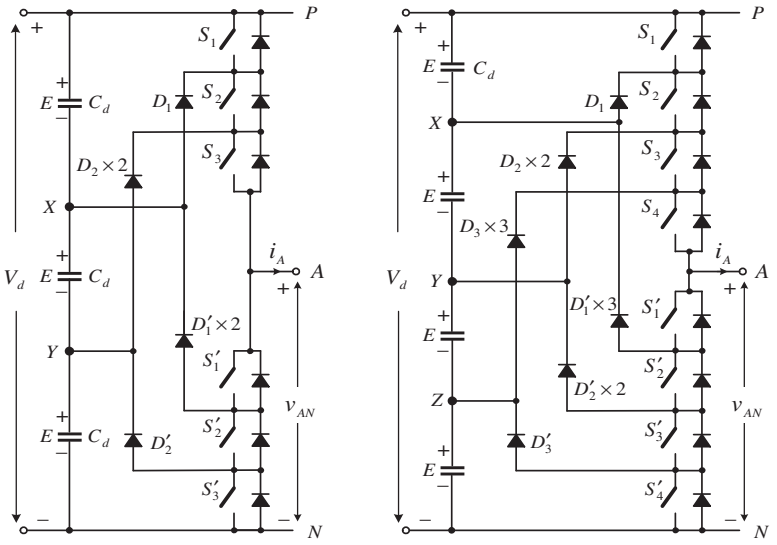
8.7.1 Four- and Five-Level Diode-Clamped Inverters

Figure 8.7-1a shows the simplified per-phase diagram of the four-level diode-clamped inverter. The inverter is composed of six active switches and a number of clamping diodes per phase. The dc capacitors C_d are shared by all three phases. It is assumed in the following analysis that the voltage across each capacitor is E and the total dc voltage V_d is equally divided by the capacitors ($V_d = 3E$).

The switch operating status and the inverter terminal voltage v_{AN} of the four-level inverter are summarized in Table 8.7-1, where "1" signifies that an active switch is turned on while "0" indicates the switch is off. When the top three switches in leg A are on ($S_1 = S_2 = S_3 =$ "1"), v_{AN} is $3E$ whereas the conduction of the bottom three switches makes v_{AN} to be zero. When the inverter terminal A is connected to node X or Y of the capacitor circuit through the conduction of the middle three switches and clamping diodes, v_{AN} will be equal to $2E$ or E. Clearly, the waveform of v_{AN} is composed of four voltage levels: $3E$, $2E$, E, and 0. It can also be observed from the table that (a) in the four-level inverter, three switches conduct at any time instant, and (b) switch pairs (S_1, S'_1), (S_2, S'_2), and (S_3, S'_3) operate in a complementary manner.

It should be pointed out that the clamping diodes may withstand different reverse blocking voltages. For instance, when the inverter operates with $S_1 = S_2 = S_3 =$ "1", the anode of the clamping diodes D_1 and D_2 in Fig. 8.7-1a is connected to the positive dc bus. The voltage applied to D_1 and D_2 is then E and $2E$, respectively. In practice, the voltage rating for all the clamping diodes is normally selected to be the same



(a) Four level ($m = 4$) (b) Five level ($m = 5$)

Figure 8.7-1 Per-phase diagram of four- and five-level diode-clamped inverters.

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Table 8.7-1 Switch Status and Inverter Terminal Voltage v_{AN}

Switch Status						v_{AN}
Four-level Inverter						
S_1	S_2	S_3	S'_1	S'_2	S'_3	
1	1	1	0	0	0	***3E***
0	1	1	1	0	0	***2E***
0	0	1	1	1	0	***E***
0	0	0	1	1	1	**0**
Five-level Inverter						v_{AN}
S_1	S_2	S_3	S_4	S'_1	S'_2	S'_3	S'_4	
1	1	1	1	0	0	0	0	***4E***
0	1	1	1	1	0	0	0	***3E***
0	0	1	1	1	1	0	0	***2E***
0	0	0	1	1	1	1	0	***E***
0	0	0	0	1	1	1	1	**0**

as the active switches. As a result, two diodes should be in series for D_2 (denoted by$D_2 \times 2$ in the figure).

The per-phase circuit diagram of the five-level diode-clamped inverter is shown in Fig. 8.7-1b, and the relationship between the switch status and v_{AN} for the five-level inverter is also given in Table 8.7-1. With various combinations of switch operating status, the waveform of v_{AN} contains five voltage levels: $4E$, $3E$, $2E$, E, and 0.

Table 8.7-2 lists the component count for the multilevel diode-clamped inverters. Assuming that all the active switches and clamping diodes have the same voltage rating, the rated inverter output voltage is proportional to the number of active switches. This suggests that if the number of the switches is doubled, the maximum inverter output voltage increases twofold, so does its output power. However, the number of clamping diodes increases dramatically with the voltage level. For

Table 8.7-2 Component Count of Diode-Clamped Multilevel Inverters

Voltage Level	Active Switches	Clamping Diodes[a]	dc capacitors
m	6(m-1)	3(m-1)(m-2)	(m-1)
3	12	6	2
4	18	18	3
5	24	36	4
6	30	60	5

[a] All diodes and active switches have the same voltage rating.

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example, the three-level inverter only requires six clamping diodes while the five-level inverter needs 36 clamping diodes. This is, in fact, one of the main reasons why the four- and five-level inverters are seldom found in industrial applications.

8.7.2 Carrier-Based PWM for High-Level Diode-Clamped Inverters

The carrier-based modulation schemes presented in Chapter 7 for cascaded H-bridge multilevel inverters can also be used for the diode-clamped inverters. Figure 8.7-2 illustrates the simulated waveforms of a four-level inverter modulated by an in-phase disposition (IPD) modulation scheme. The four-level inverter requires three carriers v_{cr1}, v_{cr2}, and v_{cr3}, which are disposed vertically, but all in phase. The amplitude modulation index m_a is equal to 0.9 and frequency modulation index m_f is 15. The definition of the modulation indices is given in Chapter 7.

The gate signals v_{g1}, v_{g2}, and v_{g3} for the top three switches S_1, S_2, and S_3 in Fig. 8.7-1a are generated at the intersections of the carrier waves and phase A modulation wave v_{mA}, respectively. The gatings for the bottom three devices S_1', S_2', and S_3' are complementary to v_{g1}, v_{g2} , and v_{g3}, and therefore are not shown.

The inverter operates at $f_1 = 60\,\text{Hz}$ and $f_{sw,dev} = 300\,\text{Hz}$, and feeds a three-phase inductive load with a power factor of 0.9. The inverter terminal voltage v_{AN} has four voltage levels and its line-to-line voltage v_{AB} contains seven voltage levels. The load current i_A is close to a sinusoid having a THD of only 2.53%. The waveform of v_{AB} contains low-order harmonics, such as the 5th and 7th, but their magnitudes are relatively low.

The harmonic content of v_{AB} is shown in Fig. 8.7-3. Although the four-level inverter with the IPD scheme generates low-order harmonics, the overall harmonic profile is quite good. At $m_a = 1$, the rms fundamental line-to-line voltage is $V_{AB1} = 0.612\,V_d$, which can be further boosted by 15.5% to $0.707V_d$ using the third harmonic injection technique presented in Chapter 6. It is worth mentioning that v_{AB} is composed of three, five, and seven voltage levels for $0 \le m_a < 0.33$, $0.33 \le m_a < 0.74$, and $0.74 \le m_a \le 1.0$, respectively.

Figure 8.7-4 shows the waveforms of the four-level inverter modulated by an alternative phase opposite disposition (APOD) scheme, where all carriers are alternatively in opposite disposition. The inverter operates under the same conditions as those in the previous case. The THD of v_{AB} and i_A is 37.3% and 4.85%, respectively, much higher than those generated by the IPD modulation. The waveform of v_{AB} contains two pairs of dominant harmonics, (11th, 13th) and (17th, 19th), with relatively high magnitudes as shown in Figs. 8.7-4 and 8.7-5, but the 5th and 7th harmonics are eliminated.

In summary, the IPD modulation produces better harmonic profile than the APOD modulation, which is consistent with the conclusion made in Chapter 7. It should be noted that the phase-shifted modulation schemes cannot be utilized for the diode-clamped multilevel inverters.

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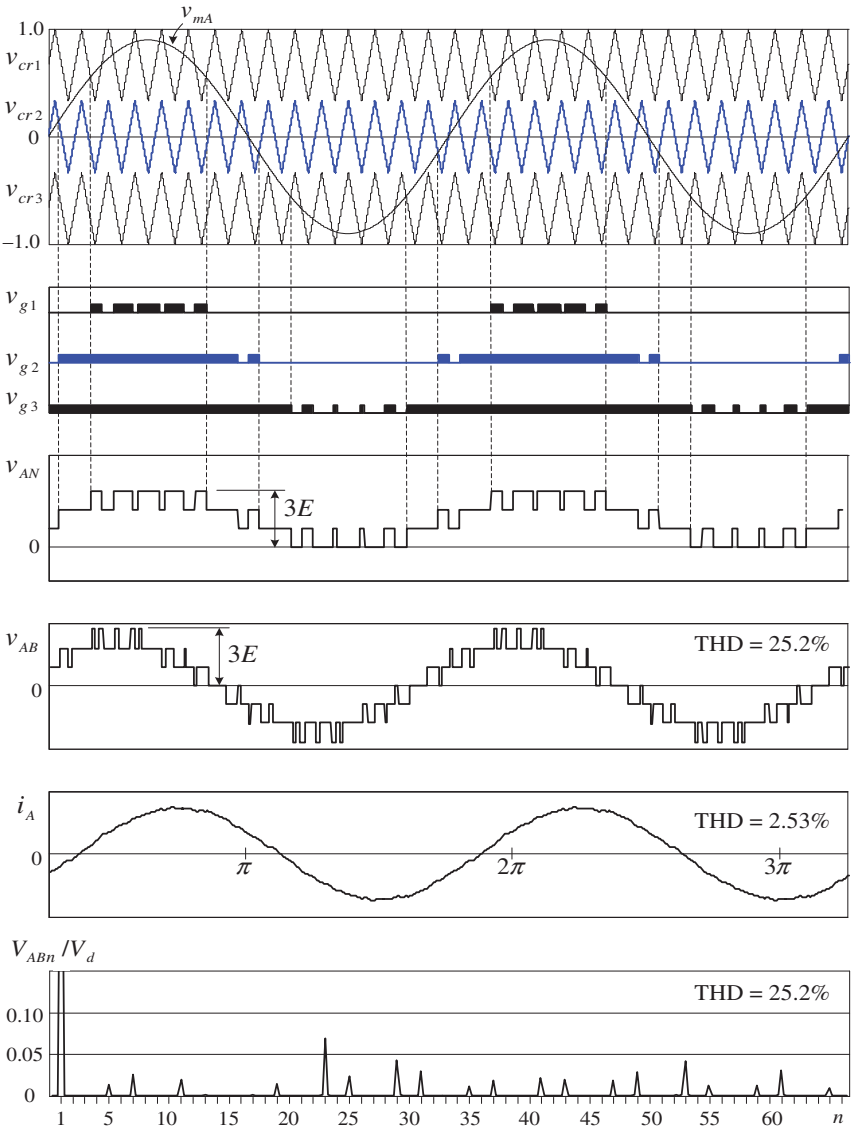

Figure 8.7-2 Simulated waveforms in the four-level inverter using IPD modulation (f_1 = 60 Hz, $f_{sw,dev}$ = 300 Hz, m_a = 0.9, and m_f = 15).

8.8 NPC/H-BRIDGE INVERTER

The NPC/H-bridge inverter is developed from the three-level NPC inverter topology. This inverter has some unique features that have promoted its application in the MV drive industry [16, 17].

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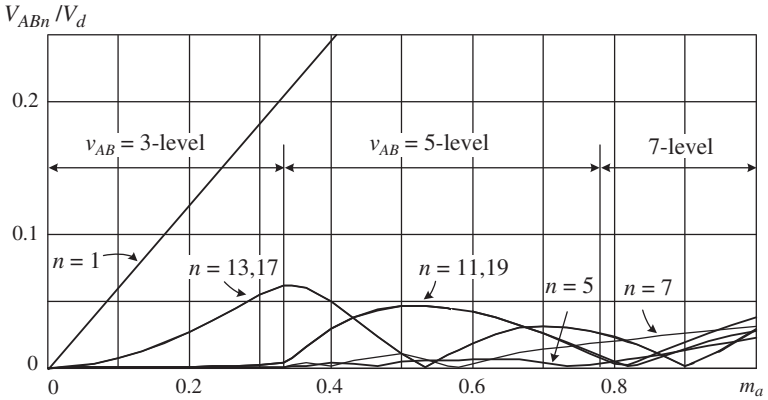

Figure 8.7-3 Harmonic content of v_{AB} in the four-level inverter ($f_1 = 60\,\text{Hz}, f_{sw,dev} = 300\,\text{Hz}$, and $m_f = 15$).

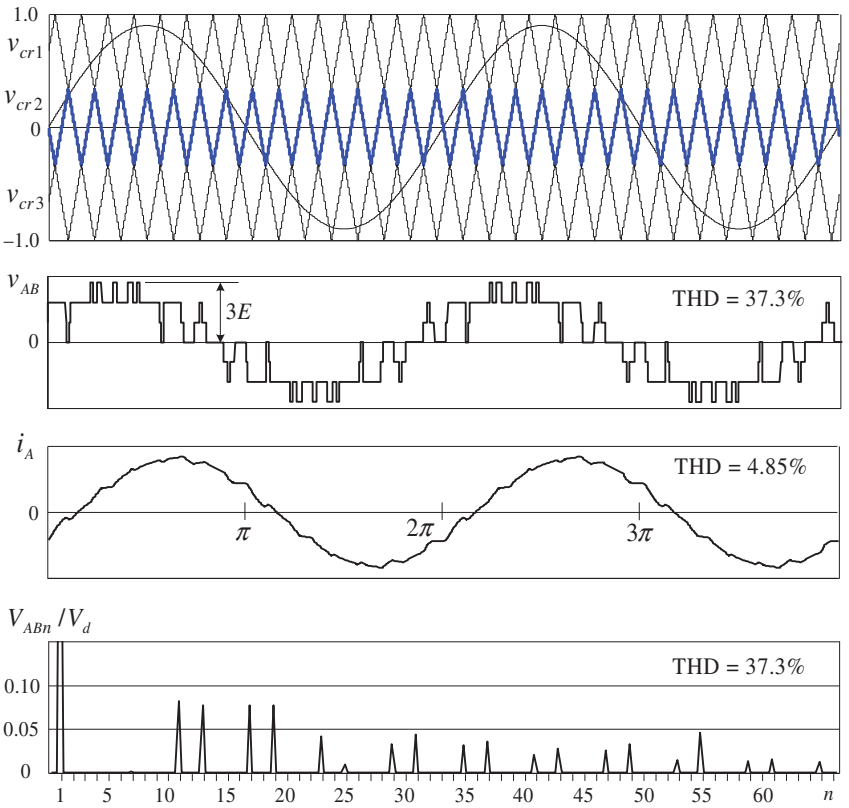

Figure 8.7-4 Simulated waveforms in the four-level inverter using APOD modulation ($f_1 = 60\,\text{Hz}, f_{sw,dev} = 300\,\text{Hz}, m_a = 0.9$, and $m_f = 15$).

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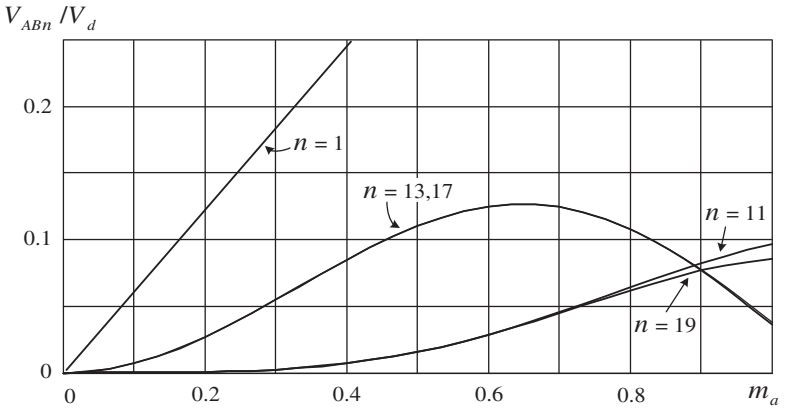

Figure 8.7-5 Harmonic content of v_{AB} in the four-level inverter ($f_1 = 60$ Hz, $f_{sw,dev} = 300$ Hz, and $m_f = 15$).

8.8.1 Inverter Topology

The output voltage and power of a three-level NPC inverter can be doubled by using 24 active switches, every two of which are connected in series. The NPC/H-bridge inverter shown in Fig. 8.8-1 also uses 24 active switches to achieve the same voltage and power ratings as the 24-switch NPC inverter. Each of the inverter phases is composed of two NPC legs in an H-bridge form.

The NPC/H-bridge inverter has some advantages over the three-level NPC inverter. The **inverter phase voltages**, v_{AN}, v_{BN}, and v_{CN}, contain five voltage levels instead of three levels for the NPC inverter, leading to a lower *dv/dt* and THD. The inverter does not have any switching devices in series, which eliminates the device dynamic

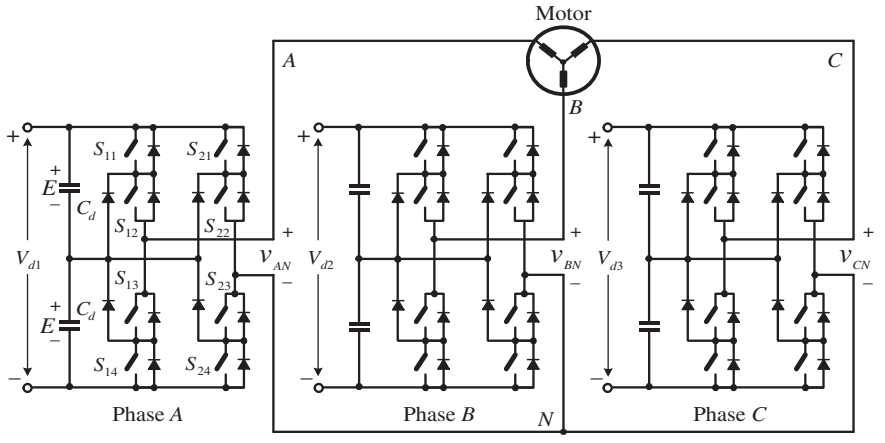

Figure 8.8-1 Five-level NPC/H-bridge inverter.

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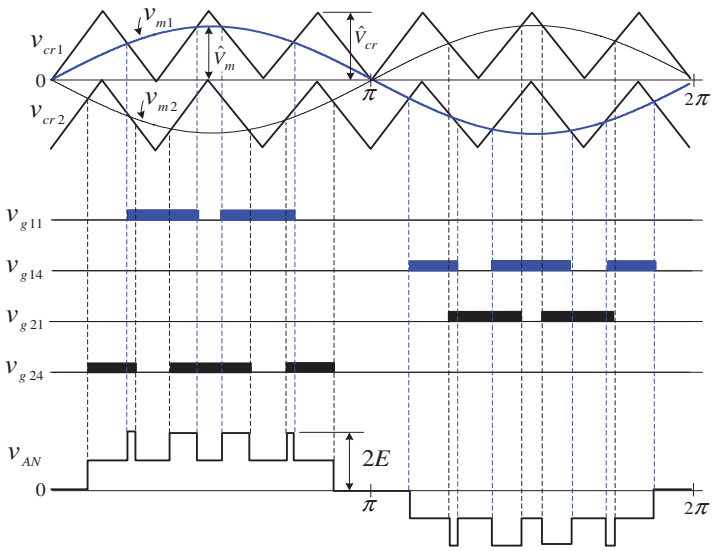

Figure 8.8-2 Modified IPD modulation for the five-level NPC/H-bridge inverter ($m_f = 6$).

and static voltage sharing problems. However, the inverter requires three isolated dc supplies, which increases the complexity and cost of the dc supply system.

8.8.2 Modulation Scheme

Figure 8.8-2 shows a modified IPD modulation scheme for the five-level NPC/H-bridge inverter, where only the waveforms for the inverter phase *A* are given. The two modulating waves, v_{m1} and v_{m2}, have the same frequency and amplitude, but are 180° out of phase. Similar to the IPD modulation presented in Chapter 7, the triangular carriers v_{cr1} and v_{cr2} are in phase but vertically disposed. The frequency modulation index is defined by $m_f = f_{cr}/f_m$, where f_m and f_{cr} are the frequencies of the modulating and carrier waves. The amplitude modulation index is given by $m_a = \hat{V}_m/\hat{V}_{cr}$, where $\hat{V}_m$ and $\hat{V}_{cr}$ are the peak amplitudes of the modulating and carrier waves, respectively.

The eight switches in inverter phase *A* constitute four complementary switch pairs: (S_{11}, S_{13}), (S_{12}, S_{14}), (S_{21}, S_{23}), and (S_{22}, S_{24}). Therefore, the modulation scheme needs to generate only four independent gate signals for the top and bottom four switches. The modulating wave v_{m1} is used to generate the gatings v_{g11} and v_{g14} for $S_{11}$and S_{14}, where v_{g11} is generated during the positive half cycle of v_{m1} while v_{g14} is produced during the negative half cycle of v_{m1}. The gate signals for S_{21} and S_{24} are arranged in a similar manner. The waveform for the inverter phase voltage v_{AN} is formed with five voltage levels.

Unlike the IPD modulation for three-level NPC inverter where the conduction angle of the switching devices over a fundamental-frequency cycle is not the same, the gating arrangements for the five-level NPC/H-bridge inverter ensure an equal

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conduction angle for all the switches ($m_f > 6$). This facilitates inverter thermal design and device selection as well. The phase-shifted multicarrier modulation discussed in Chapter 7 does not work for the NPC/H-bridge inverter.

There are three neutral points in the inverter, whose potential may need a tight control to avoid any voltage deviation. However, for the drive system using a multipulse diode rectifier as a front end, the rectifier can be designed such that its midpoints can be directly connected to the neutral points of the inverter. In doing so, the inverter neutral voltages are fixed by the rectifier and thus do not vary with the inverter operating conditions.

8.8.3 Waveforms and Harmonic Content

Figure 8.8-3 shows the simulated waveform for the phase voltage v_{AN} of the NPC/H-bridge inverter and its harmonic content. The inverter operates under the condition of $f_m = 60$ Hz, $m_f = 18$, and $m_a = 0.9$. The device switching frequency can be found from $f_{sw,dev} = f_m \times m_f/2 = 540$ Hz. The waveform of v_{AN} is composed of five voltage levels, whose harmonics appear as sidebands centered around $2m_f$ and its multiples such as $4m_f$. The phase voltage v_{AN} does not contain any harmonics with the order lower than the 27th, but has triplen harmonics such as $(2m_f \pm 3)$ and $(4m_f \pm 3)$.

The simulated waveform for the inverter line-to-line voltage v_{AB} is illustrated in Fig. 8.8-4. It contains nine voltage levels. The triplen harmonics in v_{AN} do not appear in v_{AB} due to the three-phase balanced system, resulting in a reduction of THD from 33.1–28.4%.

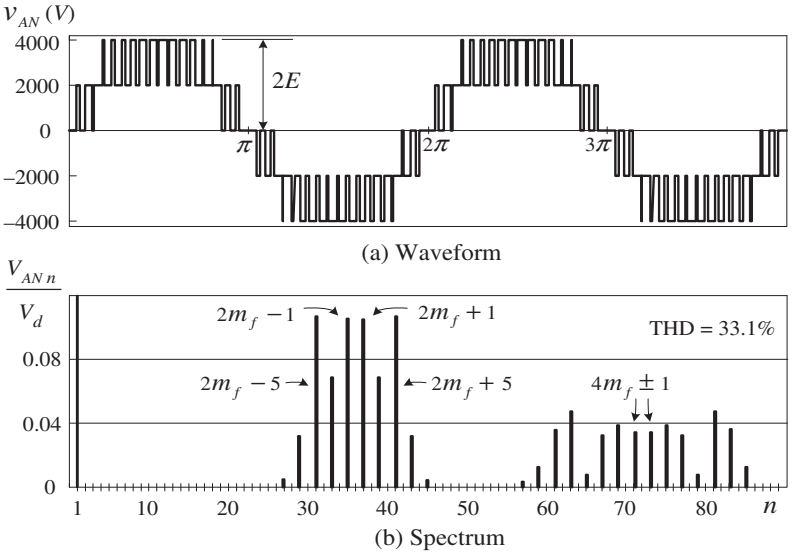

Figure 8.8-3 Waveform and spectrum of the inverter phase voltage ($f_m = 60$ Hz, $f_{sw,dev} = 540$ Hz, $m_f = 18$, and $m_a = 0.9$).

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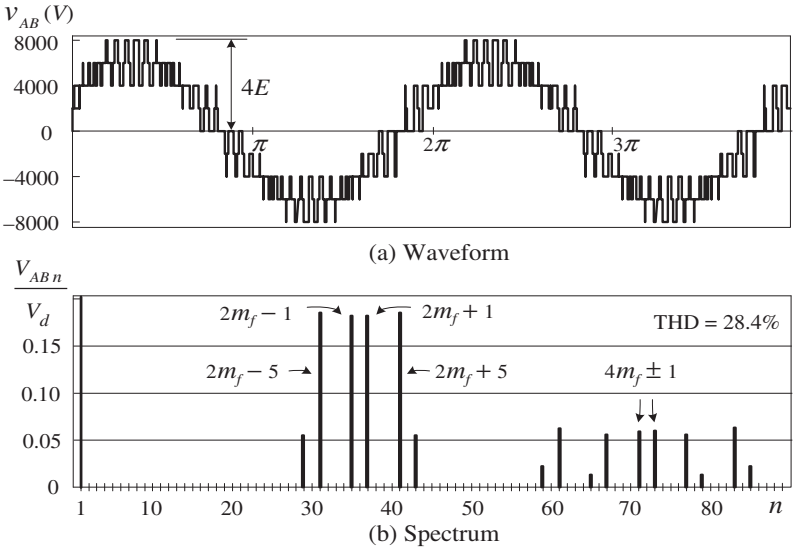

Figure 8.8-4 Waveform and spectrum of the inverter line-to-line voltage ($f_m = 60$ Hz, $f_{sw,dev} = 540$ Hz, $m_f = 18$, and $m_a = 0.9$).

The frequency of the dominant harmonics in inverter output voltages represents the equivalent **inverter switching frequency** $f_{sw,inv}$. Since the dominant harmonics in v_{AN} and v_{AB} are distributed around $2m_f$, the inverter switching frequency can be calculated by $f_{sw,inv} = f_m \times 2m_f = 4f_{sw,dev}$, four times as high as the device switching frequency.

The harmonic content of v_{AB} versus modulation index m_a is shown in Fig. 8.8-5. Since the high-order harmonic components can be easily attenuated by

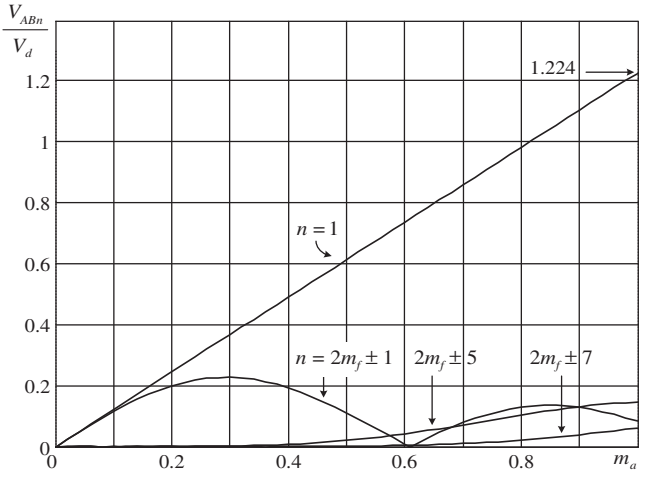

Figure 8.8-5 Harmonic content of the five-level NPC/H-bridge inverter ($m_f = 18$).

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filters or load inductances, only the dominant harmonics centered around $2m_f$ are plotted. The nth order harmonic voltage V_{ABn} (rms) is normalized with respect to the dc voltage V_d. The maximum fundamental-frequency voltage at $m_a = 1$ is given by $V_{AB1,\max} = 1.224V_d$, which is two times that of the three-level NPC inverter.

8.9 SUMMARY

This chapter provides a comprehensive analysis on the three-level diode-clamped inverter, also known as NPC inverter. A number of issues are investigated, including the inverter configuration, operating principle, SVM techniques, and neutral-point voltage control. The emphasis of the chapter is on the SVM schemes, where the conventional SVM algorithm and its modified version for even-order harmonic elimination are discussed in detail. The harmonic profile and THD of the inverter output voltage are evaluated. Important concepts are illustrated with simulations and experiments.

In addition to the three-level inverter, four- and five-level neutral-point inverters are also introduced. These inverters are seldom employed in practice mainly due to their high component count and complexity in dc capacitor voltage balancing control.

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APPENDIX

Seven-Segment Switching Sequence for the Three-Level NPC Inverter with Even-Order Harmonic Elimination

Sector I											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]
$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{13}$	[PNN]	$\vec{V}_{14}$	[PPN]
$\vec{V}_{1N}$	[ONN]	$\vec{V}_{2P}$	[PPO]	$\vec{V}_{1N}$	[ONN]	$\vec{V}_{2P}$	[PPO]	$\vec{V}_{1N}$	[ONN]	$\vec{V}_{2P}$	[PPO]
$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{13}$	[PNN]	$\vec{V}_{14}$	[PPN]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]	$\vec{V}_7$	[PON]
$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{2N}$	[OON]
Sector II											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]
$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{14}$	[PPN]	$\vec{V}_{15}$	[NPN]
$\vec{V}_{2P}$	[PPO]	$\vec{V}_{3N}$	[NON]	$\vec{V}_{2P}$	[PPO]	$\vec{V}_{3N}$	[NON]	$\vec{V}_{2P}$	[PPO]	$\vec{V}_{3N}$	[NON]
$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{14}$	[PPN]	$\vec{V}_{15}$	[NPN]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]	$\vec{V}_8$	[OPN]
$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{2N}$	[OON]	$\vec{V}_{3P}$	[OPO]
Sector III											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]
$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{15}$	[NPN]	$\vec{V}_{16}$	[NPP]
$\vec{V}_{3N}$	[NON]	$\vec{V}_{4P}$	[OPP]	$\vec{V}_{3N}$	[NON]	$\vec{V}_{4P}$	[OPP]	$\vec{V}_{3N}$	[NON]	$\vec{V}_{4P}$	[OPP]
$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{15}$	[NPN]	$\vec{V}_{16}$	[NPP]
$\vec{V}_0$	[OOO]	$\vec{V}_0$	[OOO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]	$\vec{V}_9$	[NPO]
$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{3P}$	[OPO]	$\vec{V}_{4N}$	[NOO]

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Sector IV											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]
$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{16}$	[NPP]	$\vec{V}_{17}$	[NNP]
$\vec{V}_{4P}$	[OPP]	$\vec{V}_{5N}$	[NNO]	$\vec{V}_{4P}$	[OPP]	$\vec{V}_{5N}$	[NNO]	$\vec{V}_{4P}$	[OPP]	$\vec{V}_{5N}$	[NNO]
$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{16}$	[NPP]	$\vec{V}_{17}$	[NNP]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]	$\vec{V}_{10}$	[NOP]
$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{4N}$	[NOO]	$\vec{V}_{5P}$	[OOP]
Sector V											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]
$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{17}$	[NNP]	$\vec{V}_{18}$	[PNP]
$\vec{V}_{5N}$	[NNO]	$\vec{V}_{6P}$	[POP]	$\vec{V}_{5N}$	[NNO]	$\vec{V}_{6P}$	[POP]	$\vec{V}_{5N}$	[NNO]	$\vec{V}_{6P}$	[POP]
$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{17}$	[NNP]	$\vec{V}_{18}$	[PNP]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]	$\vec{V}_{11}$	[ONP]
$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{5P}$	[OOP]	$\vec{V}_{6N}$	[ONO]
Sector VI											
1a		**1b**		**2a**		**2b**		**3**		**4**	
$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]
$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{18}$	[PNP]	$\vec{V}_{13}$	[PNN]
$\vec{V}_{6P}$	[POP]	$\vec{V}_{1N}$	[ONN]	$\vec{V}_{6P}$	[POP]	$\vec{V}_{1N}$	[ONN]	$\vec{V}_{6P}$	[POP]	$\vec{V}_{1N}$	[ONN]
$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{18}$	[PNP]	$\vec{V}_{13}$	[PNN]
$\vec{V}_{0}$	[OOO]	$\vec{V}_{0}$	[OOO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]	$\vec{V}_{12}$	[PNO]
$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]	$\vec{V}_{6N}$	[ONO]	$\vec{V}_{1P}$	[POO]

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Chapter 9

Other Multilevel Voltage Source Inverters

9.1 INTRODUCTION

The multilevel CHB and NPC inverters presented in the preceding chapters are among the first group of the multilevel inverters used in commercial MV drives. With the advancements in power converter technologies, a number of advanced multilevel inverter topologies have been developed and some of them have been successfully used in the MV drives. These topologies include active neutral-point clamped (ANPC) inverters, neutral-point piloted (NPP) inverters, nested neutral-point clamped (NNPC) inverters, and modular multilevel converters (MMCs).

In this chapter, a classic multilevel flying-capacitor (FC) inverter is introduced first since some of the advanced inverter topologies are derived from this topology. The chapter will then focus on the analysis of advanced inverter topologies, including multilevel ANPC, NPP, NNPC, and MMC inverters. The operating principle of these inverters is introduced, and their merits and drawbacks are discussed.

9.2 MULTILEVEL FLYING-CAPACITOR INVERTER

9.2.1 Inverter Configuration

Figure 9.2-1 shows a typical configuration of a five-level flying-capacitor (5L-FC) inverter [1, 2]. This inverter is evolved from the two-level inverter by adding dc capacitors to the cascaded switches. Each of the inverter legs has three flying capacitors with voltage ratings of $3E$, $2E$, and E, where E is equal to one-fourth of the dc bus voltage V_d ($E = V_d / 4$). There are four complementary switch pairs in each inverter leg. For example, the switch pairs in leg A are (S_1, S'_1), (S_2, S'_2), (S_3, S'_3),

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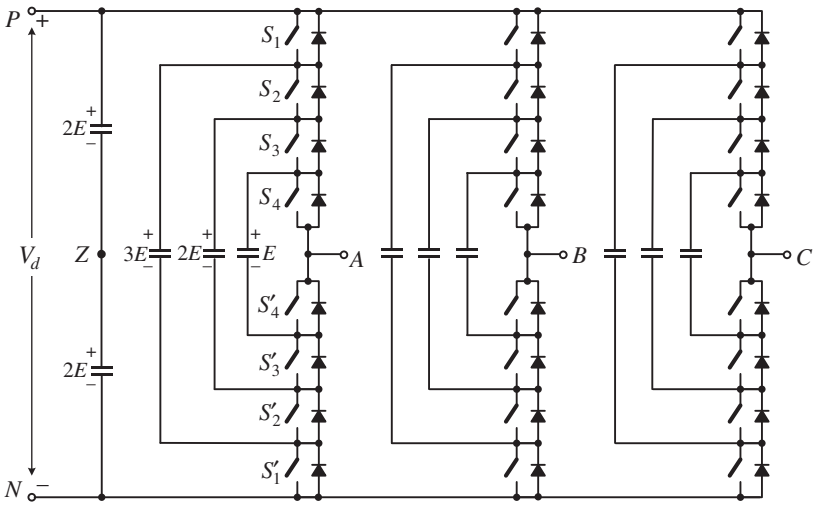

Figure 9.2-1 Five-level flying-capacitor inverter.

and (S_4, S'_4). Therefore, only four independent gate signals are required for the eight switches in an inverter leg.

The flying-capacitor inverter of Fig. 9.2-1 can produce an inverter phase voltage with five voltage levels. For example, when switches S_1, S_2, S_3, and S_4 conduct, the inverter phase voltage v_{AZ} is $2E$, which is the voltage at the inverter output terminal A with respect to the neutral point (midpoint) of the dc bus, Z. Similarly, when S_1, S_2, and S_3 are switched on, $v_{AZ} = E$. Table 9.2-1 lists all the voltage levels and their corresponding switching states. Some voltage levels can be produced by more than one switching state. For instance, voltage level 0 can be produced by six sets of different switch combinations. The redundancy of the switching states is common in multilevel inverters, which provides great flexibility for the switching pattern design.

9.2.2 Modulation Schemes

Both phase- and level-shifted modulation schemes presented in Chapter 7 can be used for the multilevel FC inverters. Figure 9.2-2 shows simulated waveforms for the inverter phase voltage v_{AZ} and line-to-line voltage v_{AB} of the 5L-FC inverter operating at the fundamental frequency of 60 Hz with a phase-shifted modulation scheme. The amplitude modulation index m_a is 0.8, frequency modulation index m_f is 12, device switching frequency $f_{sw,dev}$ is 720 Hz, and the equivalent switching frequency of the inverter $f_{sw,inv}$ is 2880 Hz, which is four times the device switching frequency due to the multilevel output voltages. The inverter equivalent switching frequency can also be observed from the harmonic spectrum of the inverter line-to-line voltage v_{AB}, where the dominant switching harmonics are around the 48th harmonic, whose frequency is 60 Hz $\times$ 48 = 2880 Hz.

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Table 9.2-1 Switching States and Voltage Levels of a 5L-FC Inverter

Switching States (Phase A)				Inverter Phase Voltage v_{AZ}
S_1	S_2	S_3	S_4	
1	1	1	1	$2E$
1	1	1	0	E
0	1	1	1	
1	0	1	1	
1	1	0	1	
1	1	0	0	0
0	0	1	1	
1	0	0	1	
0	1	1	0	
1	0	1	0	
0	1	0	1	
1	0	0	0	$-E$
0	1	0	0	
0	0	1	0	
0	0	0	1	
0	0	0	0	$-2E$

The multilevel flying-capacitor inverter has some features, including

- Modular structure for the switching devices. The inverter topology is derived from the two-level inverter, and it carries the same features as the two-level inverter such as modular structure for the switches, where each switch module include a gate driver, short circuit protection, over-voltage protection, device fault diagnosis, and bypass circuit. The modular structure helps to reduce the manufacturing cost of the inverter since the switch modules can be in mass production for the inverters and drives with different voltage ratings.
- Reduced *dv/dt* and THD. The multilevel inverter produces multi-step voltage waveforms as shown in Fig. 9.2-2, which leads to a reduced *dv/dt* and THD of the inverter output voltages.

However, the flying-capacitor inverter has some drawbacks, including

- A large number of dc capacitors with pre-charging circuits. The inverter requires several banks of dc flying capacitors with different voltage ratings, each of which needs a separate pre-charging circuit.

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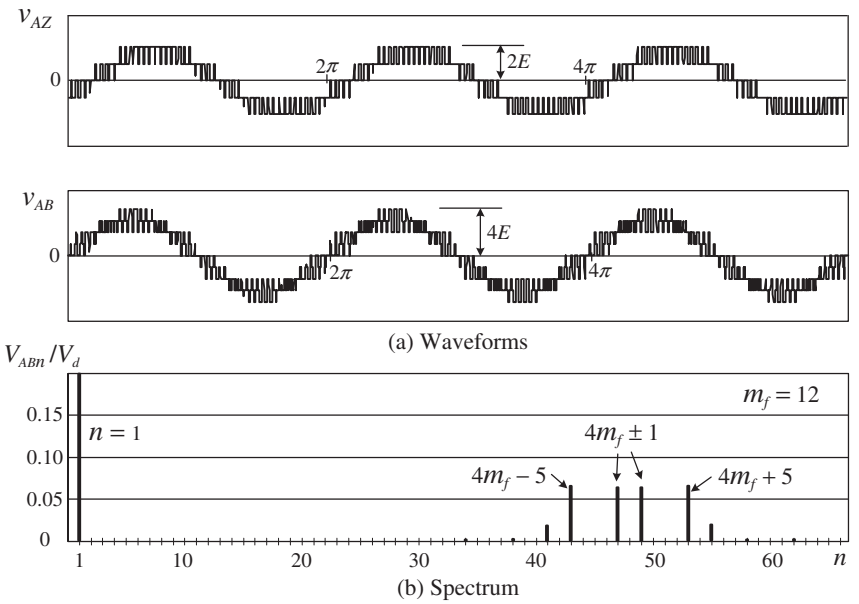

Figure 9.2-2 Simulated waveforms and spectrum of a 5L-FC inverter.

- Complex flying capacitor voltage control. The flying capacitor voltages in the inverter normally vary with the inverter operating conditions. To avoid the problems caused by the dc voltage deviation, the voltages on the flying capacitors should be tightly controlled, which increases the complexity of the control scheme.

Due to the above mentioned drawbacks, the practical use of the flying-capacitor inverter in the MV drives seems limited. However, this inverter topology can be used in deriving other advanced converter topologies, such as the NNPC and MMC converters.

9.3 ACTIVE NEUTRAL-POINT CLAMPED INVERTER

9.3.1 Inverter Configuration

The three-level neutral-point clamped (3L-NPC) inverter presented in Chapter 8 has been widely used in high-power MV drives. However, this popular inverter topology has an issue with unequal power losses among the switching devices, which leads to an uneven semiconductor-junction temperature distribution. This causes unequal utilization of the switching devices in the inverter. For a given power rating of the inverter, the switching devices that have lower junction temperatures are not fully utilized [3–5].

To solve the problem in the 3L-NPC inverter, a three-level ANPC inverter shown in Fig. 9.3-1 was proposed [3]. This inverter topology is derived from the 3L-NPC

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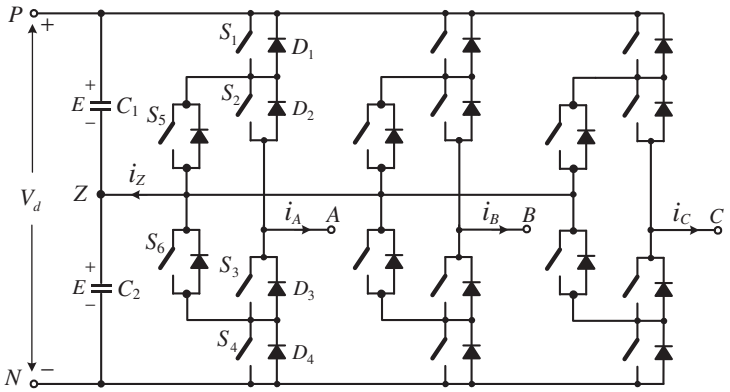

Figure 9.3-1 Three-level ANPC inverter.

inverter, where the clamping diodes are replaced by active clamping switches, such as S_5 and S_6 in the inverter leg A. The active clamping switches can control the power loss distribution and thus the junction temperature among the switches in each of the three inverter legs. Therefore, the ANPC topology overcomes the limitation of the NPC inverter, and thus enables higher output power assuming that the same switching devices are used in both inverters. However, the use of active clamping switches increases the cost and complexity of the overall converter system.

9.3.2 Switching States

The operating status of the switches in the 3L-ANPC inverter can be represented by the switching states shown in Table 9.3-1. Switching state [P] denotes that the upper two switches, S_1 and S_2, in the inverter leg A are on, and the inverter phase voltage v_{AZ}, which is the voltage at the inverter output terminal A with respect to the neutral point Z, is $+E$ ($E = V_d/2$). Switch S_6 is also turned on during the [P] state for equal sharing of dc bus voltage between S_3 and S_4 since the total dc bus voltage

Table 9.3-1 Switching States for a 3L-ANPC Inverter

Switching State		Switching States (Phase A)						Inverter Phase Voltage v_{AZ}
		S_1	S_2	S_3	S_4	S_5	S_6	
[P]		1	1	0	0	0	1	E
[O]	[OU_1]	0	1	0	0	1	0	0
	[OU_2]	0	1	0	1	1	0	
	[OL_1]	0	0	1	0	0	1	
	[OL_2]	1	0	1	0	0	1	
[N]		0	0	1	1	1	0	$-E$

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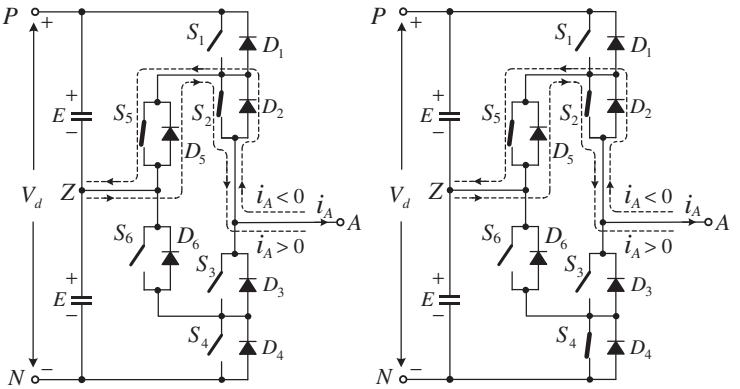

(a) [OU_1] (S_2, S_5 on) (b) [OU_2] (S_2, S_5, S_4 on)

Figure 9.3-2 Current paths of the switching states [OU_1] and [OU_2] in a 3L-ANPC inverter.

V_d is applied to these two devices. Switching state [N] indicates that the lower two switches, S_3 and S_4, conduct, leading to $v_{AZ} = -E$. Similarly, during the [N] state, S_5 is turned on for equal sharing of the dc bus voltage between S_1 and S_2.

The zero state [O] has four possible switching states: [OU_1], [OU_2], [OL_1], and [OL_2]. For switching state [OU_1], switches S_2 and S_5 are turned on, the inverter phase-A output current i_A flows through the upper path of the clamping circuits as shown in Fig. 9.3-2a. When $i_A > 0$, the output current i_A flows from the neutral point Z to the output terminal A through D_5 and S_2, whereas when $i_A < 0$, it flows from the terminal A to the neutral point Z through D_2 and S_5. In both cases, the inverter phase voltage v_{AZ} is zero.

For switching state [OU_2], S_4 is turned on in addition to the turn-on of S_2 and S_5 as shown in Fig. 9.3-2b. The operation of S_2 and S_5 remains the same as that of [OU_1], which is not affected by the turn-on of S_4. However, the turn-on of S_4 is lossless since no current flows through it when it is turned on. The turn-on of S_4 is for the preparation of the next switching state. If the next switching state is [N], S_4 should be switched on according to Table 9.3-1, but this switch has already been turned on during the [OU_2] state, which leads to the elimination of the turn-on losses for S_4 during the commutation from [OU_2] to [P]. On the other hand, if the next switching state after the zero state is [P], during which S_1 and S_2 will be turned on, there is no point to turn S_4 on during the state [OU_2]. In this case, switching state [OU_1] can be used instead of [OU_2].

For switching states [OL_1] and [OL_2], the inverter phase current i_A flows through the lower path of the clamping circuits composed of S_3 and S_6. The operation of [OL_1] and [OL_2] is similar to that of [OU_1] and [OU_2], and therefore is not repeated.

9.3.3 Principle of Switch Power Loss Distribution

In the conventional 3L-NPC inverter, there is only one zero state [O] as shown in Table 8.2-1. The current paths between the neutral point Z and the inverter terminals at

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the zero state [O] are determined by the direction of the inverter output current. When inverter output current is positive, it flows through the upper path of the clamping circuits whereas with a negative inverter output current, it flows through the lower path as shown in Fig. 8.2-4. As a result, there is no freedom available to select the upper or lower path for the redistribution of the switch power losses.

In the 3L-ANPC inverter, there are four zero switching states. Selection of a proper zero state can make the inverter output current flow through the upper or lower path of the clamping circuits, which in turn controls the switching and conduction losses of the switching devices. The following example illustrates how the switching losses of an outer switch, such as S_1, is transferred to the inner switch, such as S_2, during the commutation from state [P] to zero state [O].

Let us first consider the commutation from [P] to [OU_1] in the inverter leg A with a positive phase current i_A as shown in Fig. 9.3-3. It is assumed that the inverter operates with a full modulation index of 1.0 and is loaded with an inductive load having a power factor close to unity. Before the commutation, S_1 and S_2 are turned on, and the phase current i_A flows through S_1 and S_2 as shown in Fig. 9.3-3a. S_6 is also turned on, but does not carry current. The turn-on of S_6 is to ensure an equal sharing of the dc bus voltage between S_3 and S_4. During the commutation from [P] to [OU_1], S_1 and S_6 are turned off, and S_5 is turned on, the inverter output current i_A is commutated from S_1 to S_5 (D_5) and flows through the upper path of the clamping circuits as shown in Fig 9.3-3b. The conduction losses for S_1 and S_2 are almost the same since the [P] state has a much longer duration than the [O] state due to the high modulation index of 1.0. In addition to the conduction losses, S_1 has a significant turn-off switching loss, whereas S_2 is kept conducting without any switching losses during the commutation. Thus, S_1 has more power losses and is more stressed than S_2.

To reduce the power losses on S_1, commutation from [P] to [OL_2] can be selected as shown in Fig. 9.3-3c. During the commutation, S_2 is turned off, S_3 is turned on, and S_1 and S_6 remain on. The inverter output current i_A is commutated from S_2 to S_3 and flows through the lower path of the clamping circuits. Thus, S_2 has a significant turn-off switching loss while S_1 is kept conducting without any switching losses during the commutation. The switching loss of S_1 during the commutation from [P] to [OU_1] is thus transferred to S_2 when [OL_2] is selected, which makes the power losses between S_1 and S_2 more evenly distributed.

In summary, the use of the active clamping switches in the ANPC inverter can redistribute the power losses among the switching devices, which can in turn make the junction temperature of the switches more evenly distributed.

9.3.4 Modulation Schemes and Device Power Loss Distribution

Both carrier-based sinusoidal PWM and SVM schemes can be used for the ANPC inverter, but the gate signals generated by these modulation schemes should be

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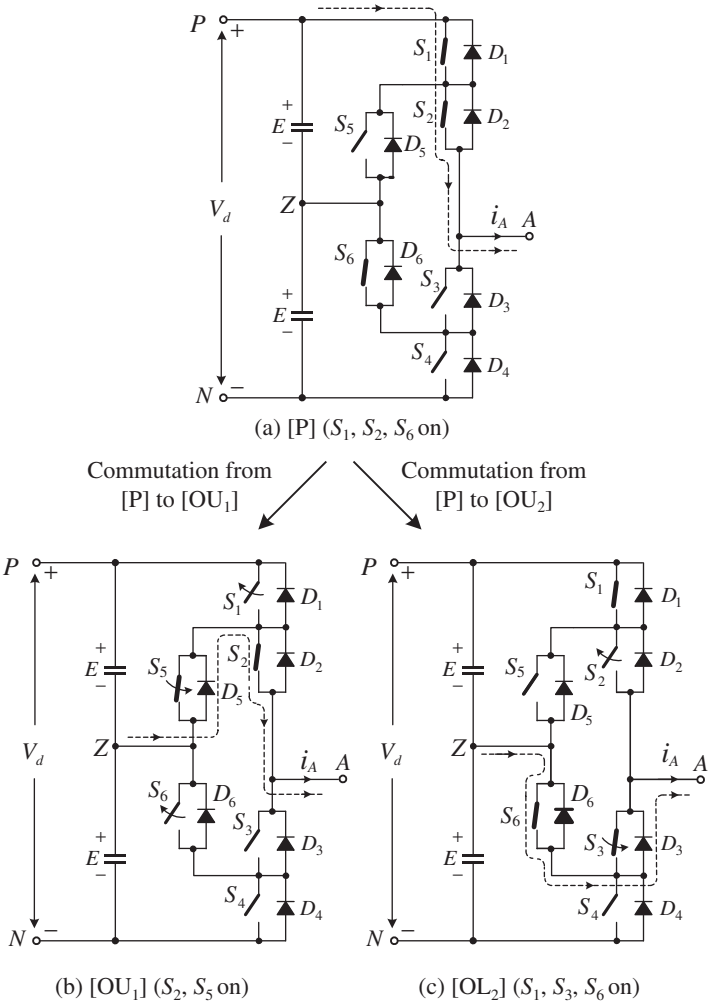

Figure 9.3-3 Commutations from [P] to [O] in a 3L-ANPC inverter with $i_A > 0$.

modified to redistribute the power losses of the switches to achieve even junction temperature distribution for the switches in the inverter.

Figure 9.3-4 shows the principle of a carrier-based in-phase-disposition (IPD) modulation scheme for the 3L-ANPC inverter. The inverter requires two carriers v_{cr1} and v_{cr2}, which are in phase and disposed vertically. The sinusoidal modulating wave v_m is compared with the two triangular carriers v_{cr1} and v_{cr2}. The gate signals v_{g1} and v_{g2} for the switches S_1 and S_2 are generated at the intersections of the modulating and carrier waves. Using the switching states defined in Table 9.3-1, a reference PWM waveform v_{pwm}, which is also a desired waveform for the inverter phase voltage v_{AZ}, is generated.

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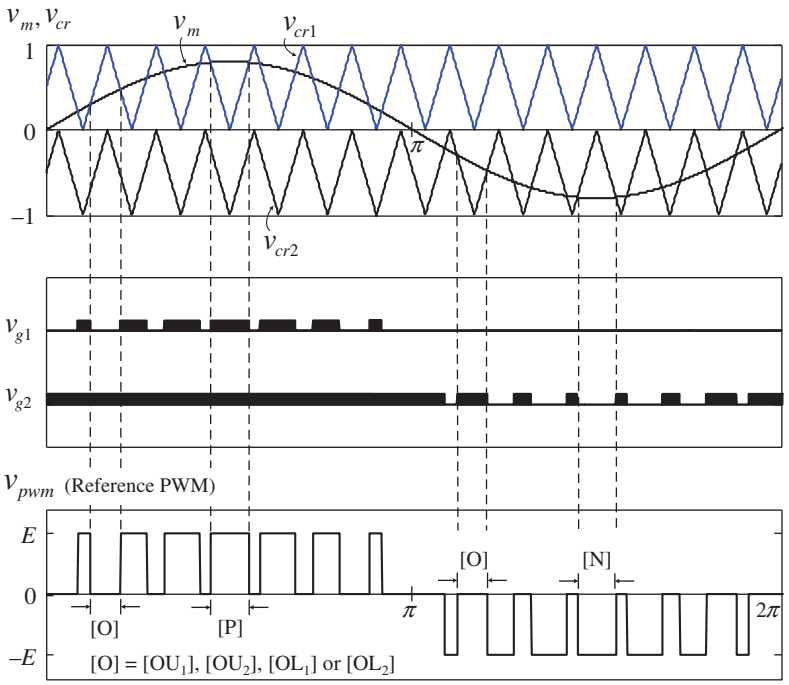

Figure 9.3-4 Principle of the carrier-based IPD modulation for a 3L-ANPC inverter.

The reference PWM waveform v_{pwm} in Fig. 9.3-4 has three switching states [N], [P], and [O]. The zero state has four redundant states: [OU$_1$], [OU$_2$], [OL$_1$], and [OL$_2$], which can be selected to redistribute the power losses of the switches to achieve even switch junction temperature distribution. As a result, the gate signals for the switching devices during the zero state [O] should be rearranged according to the selection of redundant switching states, but the gate signals for the [N] and [P] states remain unchanged.

Figure 9.3-5 shows the block diagram for the switch power loss distribution and junction temperature control. The main idea is to estimate the power losses and junction temperature of all the switches in the inverter and then select a proper commutation and zero state to transfer the power losses from the hottest switches to the cooler switches [3].

To calculate the power losses of the switches in real time, an accurate switch loss model is needed with measured dc bus voltage V_d and inverter three-phase currents i_A, i_B, and i_C. The switch junction temperature is determined based on the calculated switch power losses and an accurate converter thermal model. Then a proper zero state, [OU$_1$], [OU$_2$], [OL$_1$], or [OL$_2$], is selected to redistribute the power losses among the relevant switches that can make the junction temperature of the switches more even.

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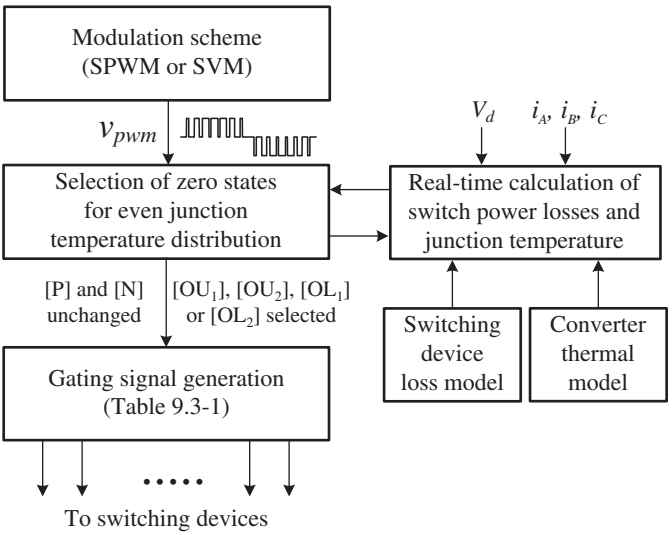

Figure 9.3-5 Even distribution of switch power loss and junction temperature.

The even distribution of the junction temperature among switching devices enables the ANPC inverter to have a higher power in comparison to the conventional NPC inverter under the assumption that the same switching devices are used in both inverters. However, this is achieved at the expense of higher cost due to the use of active clamping switches and more complex control schemes.

9.3.5 Five-Level ANPC Inverter

In addition to the 3L-ANPC inverter discussed above, a five-level (5L) ANPC inverter was proposed for the MV drives [6]. Fig. 9.3-6 shows the phase-*A* circuit of the 5L-ANPC inverter, where a flying-capacitor power cell is added to the 3L-ANPC inverter. The power cell is composed of two switches, S_7 and S_8, and a flying capacitor C_3. To achieve five voltage levels, the voltage on the flying capacitor should be kept at a quarter of the dc bus voltage ($V_d/4 = E/2$), through which two additional voltage levels are added to the 3L-ANPC inverter.

Table 9.3-2 shows the switching states of the 5L-ANPC inverter [6]. There are five pairs of complementary switches, (S_1, S_4), (S_1, S_5), (S_4, S_6), (S_2, S_3), and (S_7, S_8), and eight switching states [P], [P$_1$], [P$_2$], [O$_1$], [O$_2$], [N$_1$], [N$_2$], and [N], which correspond to five voltage levels in the inverter phase voltage: E, $E/2$, 0, $-E/2$, and $-E$, respectively.

For switching state [P], switches S_1, S_2, and S_7 in leg *A* are on, and the inverter phase voltage v_{AZ} is E. Switch S_6 is also turned on during the [P] state for equal dc voltage sharing between S_3 and S_4 since a dc voltage of ($V_d - E/2$) is applied to these two devices. For state [N], switches S_3, S_4, and S_8 conduct, which leads to $v_{AZ} = -E$. Similarly, during the [N] state, S_5 is turned on for equal dc voltage sharing between

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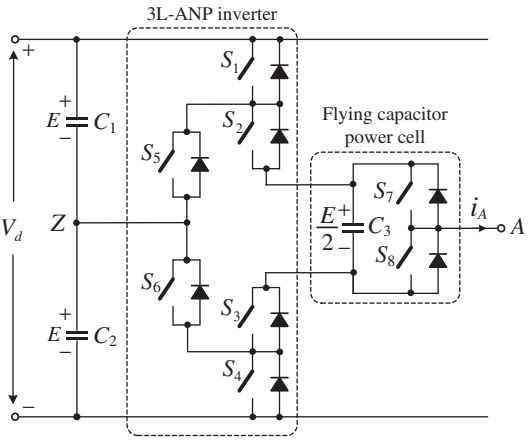

Figure 9.3-6 Diagram of a five-level ANPC inverter (Phase *A*).

S_1 and S_2. States [P] and [N] do not have an impact on the voltage of flying-capacitor C_3.

For switching state [P_1], switches S_1, S_2, and S_8 are turned on according to Table 9.3-2. Switch S_6 is also turned on for equal dc voltage sharing between S_3 and S_4. When the phase current i_A is positive ($i_A > 0$), the current flows from the positive dc bus through C_3 to the output terminal *A*, which charges C_3 as shown in Fig. 9.3-7a. When $i_A < 0$, the current flows from the terminal *A* through C_3 to the positive dc bus, which discharges C_3. In both cases, the inverter phase voltage v_{AZ} is the same, which is $E/2$.

Table 9.3-2 Switching States and Inverter Phase Voltage of a 5L-ANPC Inverter

Switching State	Switching States (Phase *A*)								Flying-Capacitor Voltage (C_3 in Phase *A*)		Inverter Phase Voltage v_{AZ}
	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	$i_A > 0$	$i_A < 0$	
[P]	1	1	0	0	0	1	1	0	—	—	E
[P_1]	1	1	0	0	0	1	0	1	↑	↓	$E/2$
[P_2]	1	0	1	0	0	1	1	0	↓	↑	
[O_1]	1	0	1	0	0	1	0	1	—	—	0
[O_2]	0	1	0	1	1	0	1	0	—	—	
[N_1]	0	1	0	1	1	0	0	1	↑	↓	$-E/2$
[N_2]	0	0	1	1	1	0	1	0	↓	↑	
[N]	0	0	1	1	1	0	0	1	—	—	$-E$

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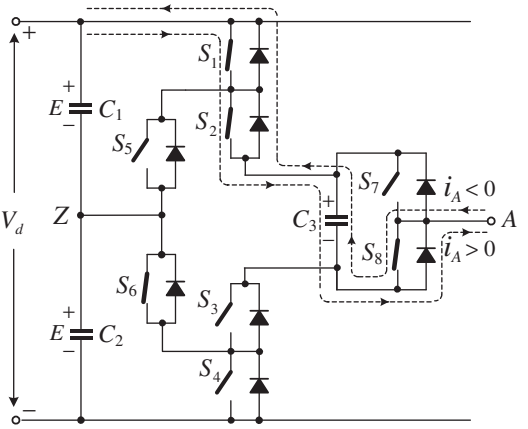
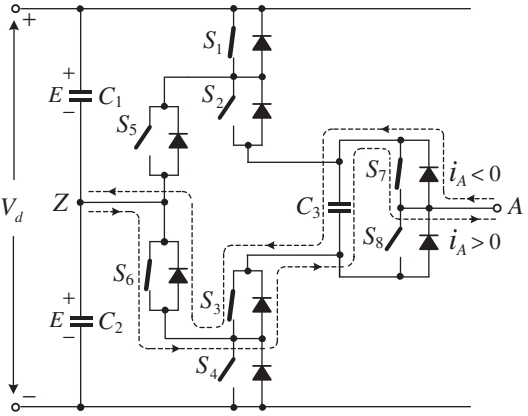


(a) State [P_1]; $v_{AZ} = E/2$



(b) State [P_2]; $v_{AZ} = E/2$

Figure 9.3-7 Switching states [P_1, P_2] and current paths in a 5L-ANPC inverter.

During switching state [P_2] with a positive output current ($i_A > 0$), the current flows from the neutral point Z through the switche S_6, diode of S_3, C_3, and S_7 to the output terminal A, and the flying capacitor is discharged as shown in Fig. 9.3-7b. When $i_A < 0$, the current flows from the terminal A through C_3 to the neutral point Z, and the capacitor is charged. In both cases, the inverter phase voltage v_{AZ} is $E/2$.

Similarly, the phase-A output voltage of $-E/2$ can be produced by either [N_1] or [N_2], which can be selected to adjust the voltage on the flying-capacitor C_3. The zero output voltage can be generated by [O_1] or [O_2], but they do not affect the flying-capacitor voltage. The impact of the switching states on the flying-capacitor voltage is summarized in Table 9.3-2.

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To evenly distribute the power losses among the switching devices, the redundant switching states, [P_1], [P_2], [N_1], and [N_2], can be selected to make the inverter output current i_A flows through the different paths as shown in Fig. 9.3-7. The principle of even power loss distribution has been discussed in the preceding section, and therefore is not repeated here. The redundancy of the switching states is utilized to keep the voltage on the flying capacitor around its nominal value of $E/2$ in addition to the even power loss distribution among the switching devices.

The 5L-ANPC inverter can produce higher number of voltage levels in comparison to the 3L-ANPC inverter, and thus the dv/dt and harmonic distortion of the inverter output voltages are reduced. However, the voltage stress on the switches in the 5L-ANPC inverter is not identical. The voltage on S_1, S_4, S_5, and S_6 is half of the dc bus voltage, whereas the voltage on S_2, S_3, S_7, and S_8 is a quarter of the dc bus voltage. In practical applications, where the same switching devices are used in the inverter, each of the switches S_1, S_4, S_5, and S_6 needs two devices in series. The 5L-ANPC inverter has been used in commercial MV drives with voltage and power ratings up to 6.9 kV and 2.0 MW, respectively [7].

9.4 NEUTRAL-POINT PILOTED INVERTER

9.4.1 Inverter Configuration

In the MV drive industry, another multilevel topology, known as the neutral-point piloted (NPP) inverter, has been used in the drives [8]. Fig. 9.4-1 illustrates a three-level NPP inverter topology, which is derived from the two-level inverter, where the output of each inverter leg is connected to the neutral point Z of the dc bus capacitors through a bidirectional switch [9].

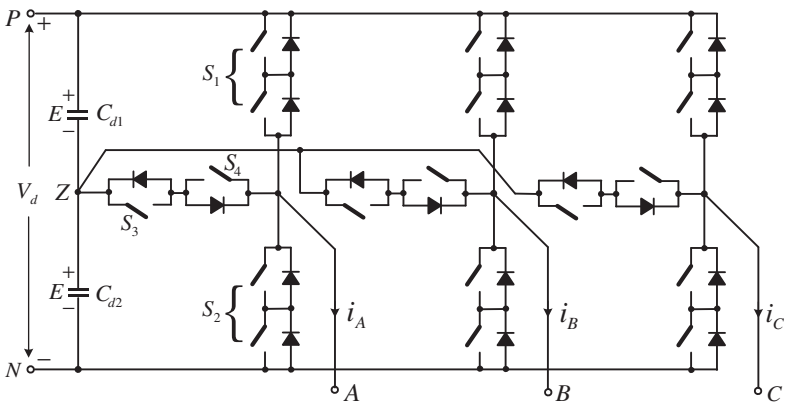

Figure 9.4-1 Three-level neutral-point piloted (NPP) inverter.

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Table 9.4-1 Switching States for a Three-Level NPP Inverter

Switching State	Switching States (Phase A)				Inverter Phase Voltage v_{AZ}
	S_1	S_2	S_3	S_4	
[P]	1	0	1	0	E
[O]	0	0	1	1	0
[N]	0	1	0	1	$-E$

The bidirectional switch provides a controllable path for the current between the neutral point Z and inverter output terminal A. The bidirectional switch is composed of two anti-series switches, such as S_3 and S_4, connected between the neutral point Z and the inverter terminal A. The main switch S_1 on the positive dc bus and S_2 on the negative dc bus are normally composed of two or more switches in series to achieve medium voltage operation.

9.4.2 Switching States

The operating status of the switches in the NPP inverter is shown in Table 9.4-1. Switching state [P] denotes that the upper switch S_1 in the inverter leg A is on, and the inverter phase voltage v_{AZ}, which is the voltage at the inverter output terminal A with respect to the neutral point Z of the dc bus is E, where E is equal to half of the dc bus voltage V_d. Switching state [N] indicates that the lower switch S_2 conducts, leading to $v_{AZ} = -E$. For zero state [O], S_3 and S_4 are on, and $v_{AZ} = 0$. In the NPP inverter, the three levels of the inverter phase voltage, E, 0, and $-E$, are produced by three switching states, [P], [O], and [N], respectively. Therefore there are no redundant switching states.

Figure 9.4-2 shows the current paths of the 3L-NPP inverter operating with different switching states. Fig. 9.4-2a shows the current paths of the inverter during the [P] state, where the current flows through S_1 from the positive dc bus to the inverter terminal A when the inverter output current i_A is positive ($i_A > 0$). When i_A is negative, the current flow through the diode of S_1 to the positive dc bus. For zero state [O], the bidirectional switch is on and the current in the switch is bidirectional, depending on the polarity of i_A as shown in Fig. 9.4-2b. The current paths during the [N] state are illustrated in Fig. 9.4-2c.

It is noted that during the [P] state, S_3 is also turned on, which is a lossless turn-on since no current flows through this switch during the turn-on process. The turn-on of S_3 does not affect the operation of the [P] state. The main purpose of turning S_3 on during the [P] state is to reduce the switching losses during the commutation between the states [P] and [O]. For example, during the commutation from the [P] state to the [O] state, the main switch S_1 is switched off and both S_3 and S_4 need to be turned on, but S_3 has already been on during the [P] state, and its turn-on losses are eliminated. Similarly, during the [N] state, S_4 should be turned on to reduce the switching losses during the commutation between the [N] and [O] states.

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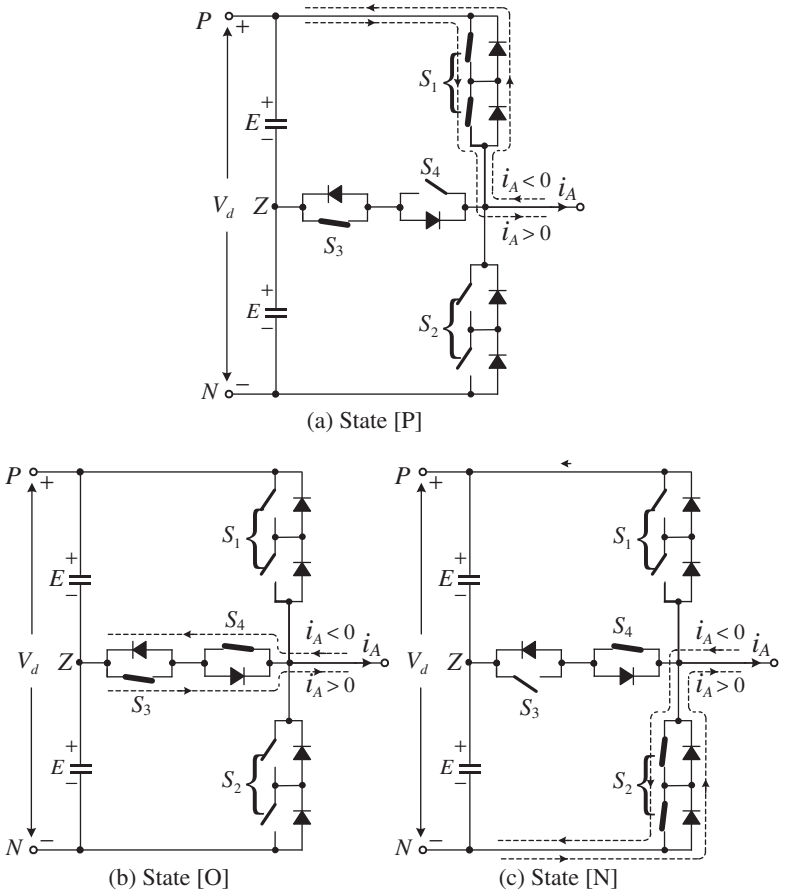

Figure 9.4-2 Switching states and current paths in a 3L-NPP inverter.

9.4.3 Modulation Scheme and Neutral Point Voltage Control

Both carrier-based SPWM and SVM schemes can be used for the NPP inverter. The principle and implementation of these modulation schemes are the same as those for the NPC and ANPC inverters. Therefore, the modulation schemes for the 3L-NPP inverter are not discussed here.

Similar to the 3L-NPC inverter, the neutral point voltage v_Z of the 3L-NPP inverter varies with the operating conditions. The principle of the neutral-point voltage control for the NPP inverter is the same as that of the NPC inverter. Thus the control scheme presented in Chapter 8 for the NPC inverter can be equally applied to the NPP inverter, and therefore not repeated here.

The NPP inverter topology is derived from the two-level inverter by adding bidirectional switches between the inverter output terminals and the neutral point of the dc

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bus. Thus, this inverter can produce the three-level voltage waveforms with reduced *dv/dt* and THD. However, the NPP inverter has some limitations:

- Depending on the voltage ratings of the inverter, two or more switches need to be connected in series in each of the switch positions in the inverter. Thus, reliable static and dynamic voltage sharing methods should be implemented for the series-connected switches, which increases the complexity and reduces the reliability of the inverter.
- The dc bus capacitor voltages should be balanced, which increases the complexity of the control scheme.

9.5 NESTED NEUTRAL-POINT CLAMPED INVERTER

9.5.1 Inverter Configuration

A new multilevel topology, known as nested neutral point clamped (NNPC) inverter, has been proposed in the recent years for use in the MV drive [10]. Figure 9.5-1 shows a four-level topology for the NNPC inverter. This topology is a combination of a flying-capacitor (FC) topology composed of outer devices S_1, S_2, S_5, S_6, C_1, and C_2 in the inverter leg A and an NPC topology composed of inner devices S_3, S_4, D_1, D_2, S_2, and S_5, in the same inverter leg. The switches S_2 and S_5 are shared by both topologies. The inner NPC and outer FC topologies are nested into one topology, and therefore it is called the nested NPC inverter. The voltage rating of each device in the inverter is $2E/3$, where E is half of the dc bus voltage V_d, except for the voltage on the dc bus capacitors, C_{d1} and C_{d2}, which is equal to E.

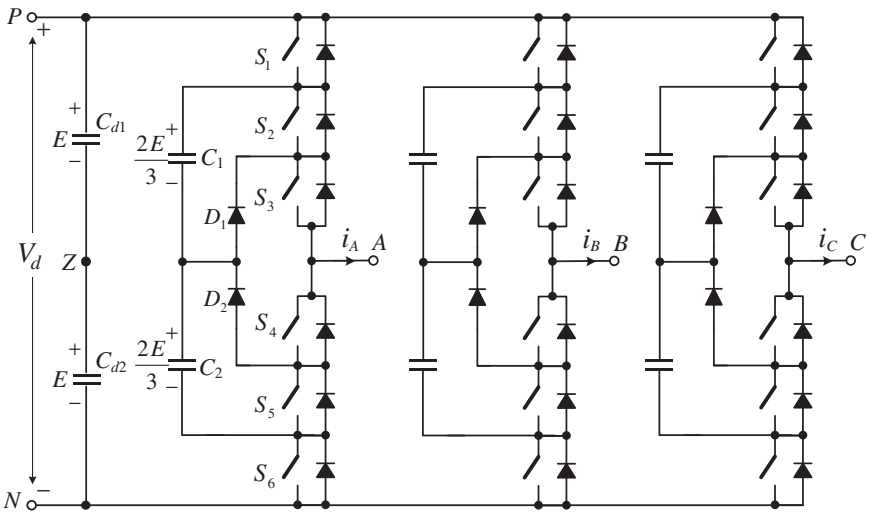

Figure 9.5-1 Four-level nested neutral point clamped (NNPC) inverter.

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Different from the NPC inverter, the clamping diodes D_1 and D_2 in the inverter leg A are not connected to the dc bus midpoint Z. Instead, they are connected to the midpoint of the two flying-capacitors C_1 and C_2. It is noted that the operation of the NNPC inverter does not require the dc bus midpoint Z provided by dc capacitors C_{d1} and C_{d2}. A medium-voltage film capacitor directly connected across the positive and negative dc buses is preferred. The dc bus midpoint Z in Fig. 9.5-1 is provided to facilitate the analysis and discussion of the inverter.

Compared with the 4L-NPC inverter topology shown in Fig. 8.7-1a, where it requires a total of 39 components, including 18 switches, 18 clamping diodes, and three dc capacitors, the 4L-NNPC inverter requires only 31 components, including 18 switches, 6 clamping diodes, 6 flying capacitors, and one dc bus capacitor. Therefore, the 4L-NNPC inverter topology has less component count than the 4L-NPC inverter. However, there are six flying capacitors in the 4L-NNPC inverter and the voltages on these capacitors need to be controlled, whereas in the 4L-NPC inverter only two neutral point voltages provided by three cascaded dc bus capacitors need to be controlled.

9.5.2 Switching States

To achieve four voltage levels in the NNPC inverter shown in Fig. 9.5-1, the voltage on each of the flying capacitors, such as C_1 and C_2 in the inverter leg A, should be kept at $2E/3$. Table 9.5-1 shows the switching states of the switches in the inverter leg A of the 4L-NNPC topology. There are three pairs of complementary switches (S_1, S_6), (S_2, S_4), and (S_3, S_5), and six switching states [P], [P_1], [P_2], [N_2], [N_1], and [N], which correspond to four voltage levels in the inverter phase voltage: E, $E/3$, $-E/3$, and $-E$, respectively.

For switching state [P], the upper three switches, S_1, S_2, and S_3, in leg A are on, and the inverter phase voltage v_{AZ}, which is the voltage between the inverter output terminal A and the dc bus neutral point Z, is E. For state [N], the lower three switches, S_4, S_5, and S_6, conduct, which leads to $v_{AZ} = -E$.

Table 9.5-1 Switching States of a 4L-NNPC Inverter

Switching State	Switching States (Phase A)						Inverter Phase Voltage v_{AZ}
	S_1	S_2	S_3	S_4	S_5	S_6	
[P]	1	1	1	0	0	0	E
[P_1]	1	0	1	1	0	0	$E/3$
[P_2]	0	1	1	0	0	1	
[N_2]	1	0	0	1	1	0	$-E/3$
[N_1]	0	0	1	1	0	1	
[N]	0	0	0	1	1	1	$-E$

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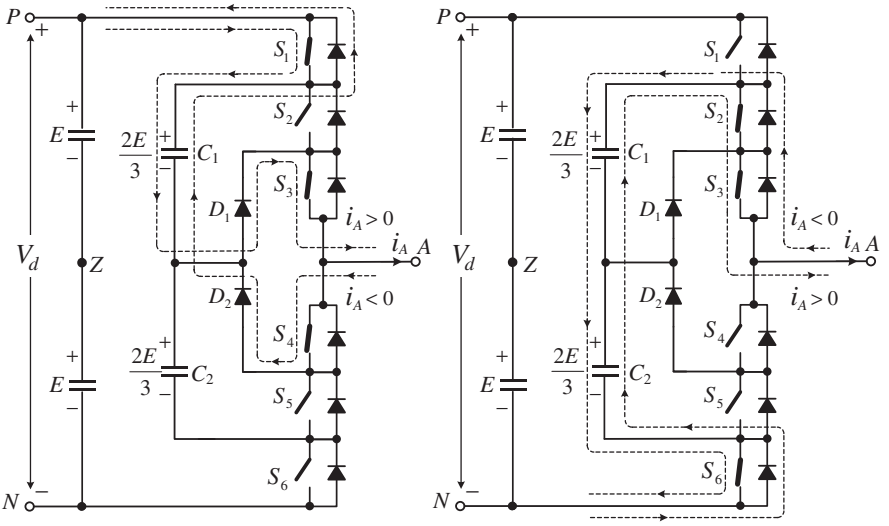

Figure 9.5-2 Switching states [P_1, P_2] and current paths in a 4L-NNPC inverter.

For switching state [P_1], switches S_1, S_3, and S_4 are turned on according to Table 9.5-1, and the current paths for the inverter phase current i_A are shown in Fig. 9.5-2a. When the phase current i_A is positive ($i_A > 0$), the current flows from the positive dc bus through S_1, C_1, D_1, and S_3 to the output terminal A. When $i_A < 0$, the current flows from the terminal A through S_4, D_2, C_1, and the diode of S_1 to the positive dc bus. In both cases, the inverter phase voltage v_{AZ} is the same, given by $E - 2E/3 = E/3$.

During switching state [P_2], switches S_2, S_3, and S_6 are turned on, and the current paths are shown in Fig. 9.5-2b. When the output current i_A is positive, the current flows from the negative dc bus N through the diode of S_6, C_2, C_1, S_2, and S_3 to the output terminal A. When $i_A < 0$, the current flows from the terminal A through the diodes of S_3 and S_2, C_1, C_2, and S_6 to the negative dc bus. In both cases, the inverter phase voltage v_{AZ} is given by $4E/3 - E = E/3$. Both switching states [P_1] and [P_2] produce the same voltage of $E/3$ at the inverter output. The redundancy of the switching states provides an effective means for the flying-capacitor voltage control, which will be discussed in the following section. Similarly, switching states [N_1] and [N_2] produce an output voltage of $-E/3$ at the inverter terminal A with respect to the dc bus midpoint Z.

9.5.3 Principle of Flying-Capacitor Voltage Control

The flying capacitors in the 4L-NNPC inverter, such as C_1 and C_2 in the inverter leg A, should be kept at $2E/3$ by a capacitor voltage control scheme to achieve four voltage levels at the inverter output. This can be accomplished by making use of

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Table 9.5-2 Impact of Switching States on Flying-Capacitor Voltages

Switching State	Flying-Capacitor Voltage (Phase A)		Inverter Phase Voltage v_{AZ}
	v_{C1}	v_{C2}	
[P]	No impact	No impact	E
[P_1]	$i_A > 0$: ↑ $i_A < 0$: ↓	No impact	$E/3$
[P_2]	$i_A > 0$: ↓ $i_A < 0$: ↑	$i_A > 0$: ↓ $i_A < 0$: ↑	
[N_2]	$i_A > 0$: ↑ $i_A < 0$: ↓	$i_A > 0$: ↑ $i_A < 0$: ↓	$-E/3$
[N_1]	No impact	$i_A > 0$: ↓ $i_A < 0$: ↑	
[N]	No impact	No impact	$-E$

the redundancy of the switching states [P_1], [P_2], [N_1], and [N_2] discussed earlier. Table 9.5-2 shows the impact of the switching states on the flying-capacitor voltages v_{C1} and v_{C2} in the 4L-NNPC inverter.

Switching states [P] and [N] do not have an impact on the flying-capacitor voltage since the inverter output current i_A does not flow through C_1 or C_2 during these two states. The impact of [P_1] on the flying-capacitor voltages can be observed from Fig. 9.5-2a. The state [P_1] makes v_{C1} increase when $i_A > 0$, and makes v_{C1} decrease when $i_A < 0$, but it does not affect the voltage on C_2.

The impact of state [P_2] on the flying capacitors can be found from Fig. 9.5-2b and is summarized in Table 9.5-2. Also, the impact of [N_1] and [N_2] on the flying-capacitor voltages is summarized in the table [11].

To control the flying-capacitor voltage tightly, the voltage on each of the flying capacitors needs to be measured. The deviation of the measured capacitor voltage from its nominal value of $2E/3$ is defined by

$$\Delta v_{Ci} = v_{Ci} - \frac{2E}{3} = v_{Ci} - \frac{V_d}{3} \quad i = 1, 2 \quad \text{for inverter phase } A \tag{9.5-1}$$

A capacitor voltage balancing control (VBC) scheme is required to ensure that the voltage deviation $|\Delta v_{ci}|$ is within a given range. Table 9.5-3 illustrates the mechanism of flying-capacitor VBC [12], where the inputs of the VBC scheme are the inverter phase voltage levels $E/3$ and $-E/3$, the capacitor voltage deviations Δv_{C1} and Δv_{C2}, and the polarity of the inverter output current i_A. The inverter phase voltage levels of $+E$ and $-E$, which correspond to the switching states [P] and [N], are not used since these two switching states have no impact on the capacitor voltages as shown in Table 9.5-2.

Based on its input, the VBC scheme selects a proper switching state from [P_1], [P_2], [N_1], and [N_2] to reduce $|\Delta v_{ci}|$ through charging or discharging the flying-capacitors C_1 and C_2. For example, at the voltage level of $E/3$, $\Delta v_{C1} < 0$, and $i_A \geq$

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Table 9.5-3 Mechanism of Capacitor Voltage Balancing Control (VBC)

Input Variables of VBC			Selection of Switching State	v_{C1}
Voltage level of v_{AZ}	Δv_{C1}	i_A		
$E/3$	<0	≥0	$[P_1]$	↑
		<0	$[P_2]$	↑
	≥0	≥0	$[P_2]$	↓
		<0	$[P_1]$	↓
	Δv_{C2}	i_A		v_{C2}
$-E/3$	<0	≥0	$[N_2]$	↑
		<0	$[N_1]$	↑
	≥0	≥0	$[N_1]$	↓
		<0	$[N_2]$	↓

0, the capacitor voltage v_{C1} is lower than its nominal value of $2E/3$. Switching state $[P_1]$ is selected from Table 9.5-3, which will make v_{C1} increase.

9.5.4 Modulation Schemes with Capacitor Voltage Balancing Control

Both carrier-based PWM and SVM schemes can be used for the NNPC inverter, but the gate signals generated by these modulation schemes need to be modified for the flying-capacitorVBC.

Figure 9.5-3 shows an example of a carrier-based IPD modulation scheme for the 4L-NNPC inverter. The four-level inverter requires three carriers v_{cr1}, v_{cr2}, and v_{cr3}, which are compared with the sinusoidal modulating wave v_m. The gate signals v_{g1}, v_{g2}, and v_{g3} for the switches S_1, S_2, and S_3 are generated at the intersections of the modulating and carrier waves. Using the switching states defined in Table 9.5-1, a reference PWM waveform v_{pwm}, which is also a desired waveform for the inverter phase voltage v_{AZ}, is generated.

The reference PWM waveform v_{pwm} in Fig. 9.5-3 has four voltage levels, E, $E/3$, $-E/3$, and $-E$. As discussed earlier, the flying-capacitor voltages cannot be controlled by switching states [P] or [N] at the voltage levels of E or $-E$. The voltage level of $E/3$ can be produced by switching states $[P_1]$ or $[P_2]$, which can be selected to control the voltage on the flying capacitors according to Table 9.5-2. Similarly, when the reference PWM voltage level is at $-E/3$, switching states $[N_1]$ or $[N_2]$ can be selected to control the flying-capacitor voltages.

Figure 9.5-4 shows a block diagram of the flying-capacitor VBC scheme for the 4L-NNPC inverters. A carrier-based sinusoidal PWM or a space vector modulation scheme can be employed to produce a reference PWM waveform v_{pwm}. To ensure

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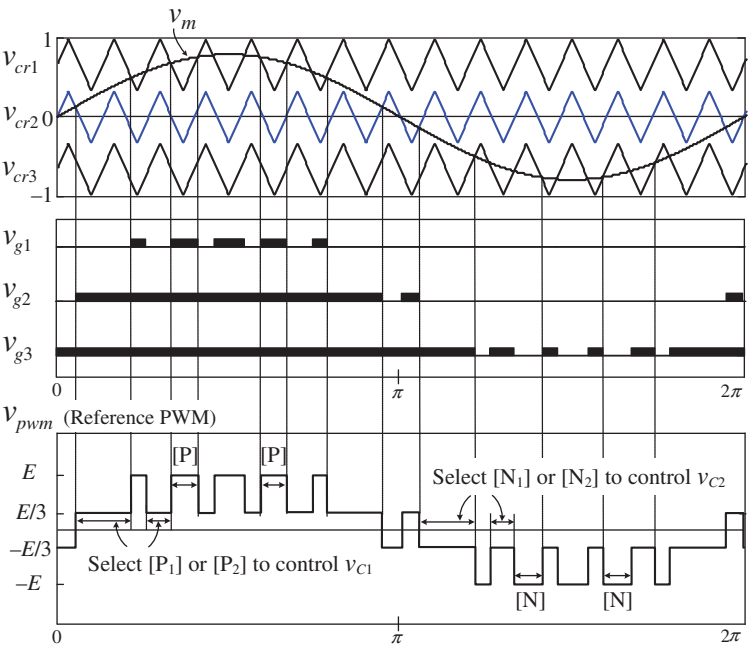

Figure 9.5-3 Principle of the carrier-based IPD modulation for a 4L-NNPC inverter.

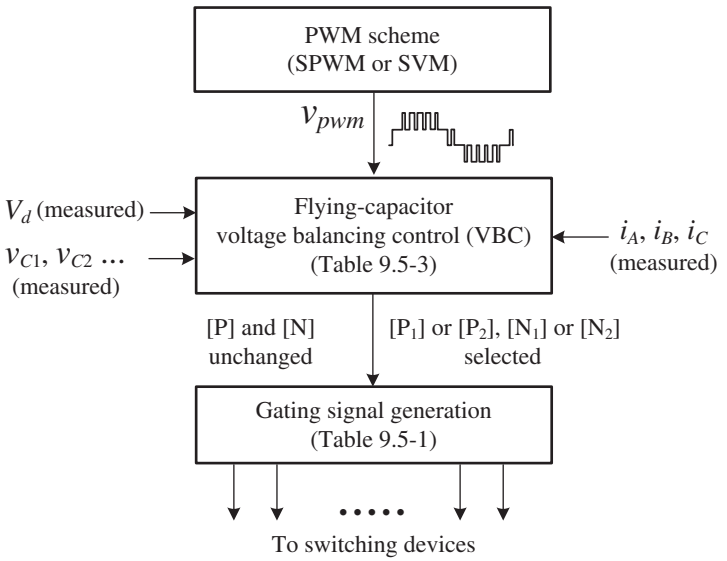

Figure 9.5-4 Block diagram of flying-capacitor voltage control.

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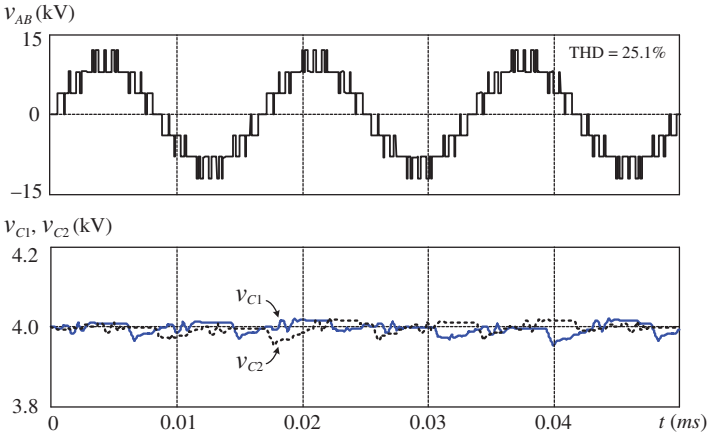

Figure 9.5-5 Simulated waveforms for a 4L-NNPC inverter.

that the voltage on the flying capacitors is kept around its nominal value of $2E/3$, the capacitor voltages, such as v_{C1} and v_{C2} in the inverter leg A, and the dc bus voltage V_d are measured. The polarity of the inverter output currents, i_A, i_B, and i_C, is also detected.

The main function of the VBC block is to select a proper switching state from $[P_1]$, $[P_2]$, $[N_1]$, and $[N_2]$ according to Table 9.5-3 for the capacitor voltage control, whereas the [P] and [N] states remain unchanged. In doing so, the inverter can produce the desired output waveform defined by v_{pwm} while keeping the capacitor voltage under tight control. The switching states are then sent to the gate signal generation block to generate the gate signals for the switching devices in the inverter.

Figure 9.5-5 shows the simulated waveforms for a 7.2 kV drive using a 4L-NNPC inverter. The drive operates at the fundamental frequency of 60 Hz, line-to-line voltage of 7.2 kV, dc bus voltage of 12 kV, average switching frequency of 400 Hz, and the flying capacitor of 4 pu. The waveform of the line-to-line voltage v_{AB} has seven voltage levels with a THD of 25.1%. The voltages on the two flying capacitors in the inverter leg A, v_{C1} and v_{C2}, are controlled by the VBC scheme, and kept around its nominal value of 4 kV with small ripples.

Figure 9.5-6 shows measured waveforms from a laboratory 4L-NNPC inverter operating at the fundamental frequency of 60 Hz, switching frequency of 400 Hz, line-to-line voltage of 208 V, dc bus voltage of 300 V, load power factor of 0.95, and flying capacitor of 4 pu. The inverter line-to-line voltage waveform is close to the simulated one shown in Fig. 9.5-5, and inverter output currents are nearly sinusoidal due to the load inductance. The voltages on the flying capacitors are kept by the controller around the nominal value of 100 V and the voltage ripple is less than 5%.

The 4L-NNPC inverter presented in this section for the MV drive has the following features:

- The inverter has one more voltage level than the three-level NPC, ANPC, and NPP inverters, and therefore can produce higher output voltages. The

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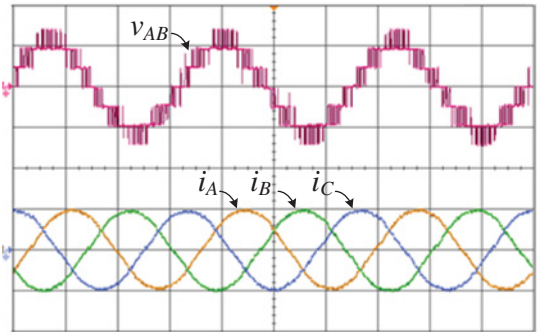
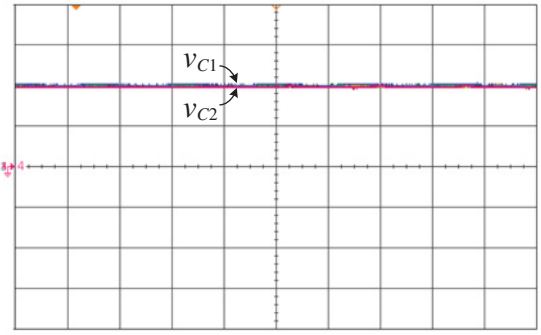


(a) v_{AB}: 200 V/div; i_A, i_B, i_C: 20 A/div; 5 ms/div



(b) v_{C1}, v_{C2}: 50 V/div; 5 ms/div

Figure 9.5-6 Measured waveforms from a 4L-NNPC lab prototyping inverter.

4L-NNPC inverter is suitable for the MV drives with the voltage rating up to 7.2 kV without switches in series.

- All the switches in the inverter subject to the same voltage stress.
- The inverter requires fewer components than the 4L-NPC inverter.

9.5.5 High-Level NNPC Inverters

The 4L-NNPC inverter discussed above can be extended to five or six voltage levels. Fig. 9.5-7a shows the simplified per-phase diagram of the 5L-NNPC inverter. The inverter is composed of eight active switches with the same voltage stress each, two clamping diodes and two flying capacitors per phase. All the components in the inverter have the same nominal voltage of $E/2$ (V_d /4). Table 9.5-4 shows the switching states of the 5L-NNPC inverter. There are three pairs of complementary switches (S_1, S_6), (S_2, S_4), and (S_3, S_5) and the waveform of the inverter terminal voltage v_{AZ} is composed of five voltage levels: E, $E/2$, 0, $-E/2$, and $-E$.

The per-phase circuit diagram of a 6L-NNPC inverter is shown in Fig. 9.5-7b. The inverter is composed of 10 active switches with the same voltage stress, which

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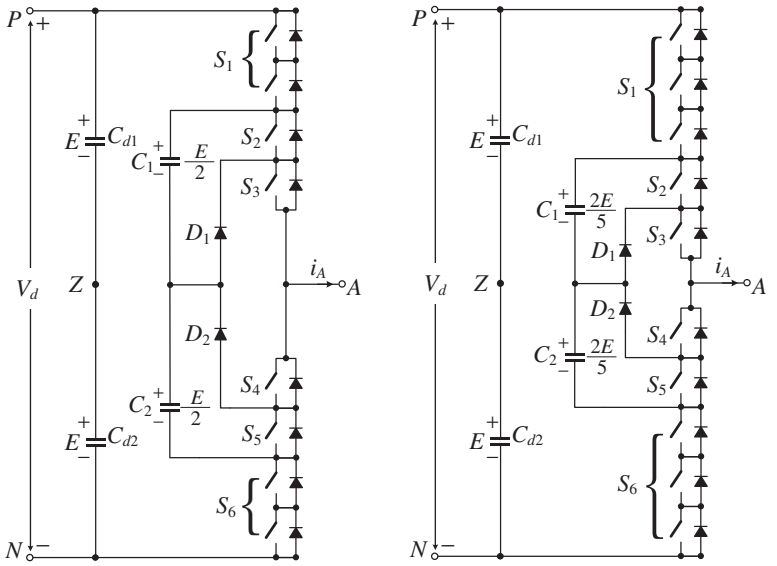

(a) Five level (b) Six level

Figure 9.5-7 Per-phase diagram of 5L- and 6L-NNPC inverters.

Table 9.5-4 Switching States and Inverter Output Voltage in 5L- and 6L-NNPC Inverters

Switching States (Phase *A*)						Inverter Phase Voltage v_{AZ}
5L-NNPC Inverter						
S_1	S_2	S_3	S_4	S_5	S_6	
1	1	1	0	0	0	E
1	0	1	1	0	0	$E/2$
0	1	1	0	0	1	0
1	0	0	1	1	0	
0	0	1	1	0	1	$-E/2$
0	0	0	1	1	1	$-E$
6L-NNPC Inverter						Inverter Phase Voltage v_{AZ}
1	1	1	0	0	0	E
1	0	1	1	0	0	$3E/5$
0	1	1	0	0	1	$E/5$
1	0	0	1	1	0	$-E/5$
0	0	1	1	0	1	$-3E/5$
0	0	0	1	1	1	$-E$

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is one-fifth of the dc bus voltage, two clamping diodes, and two flying capacitors per phase. To produce six voltage levels in v_{AZ}, the flying-capacitor voltage should be kept at its nominal value of $2E/5$ (V_d /5). The relationship between the switching states and v_{AZ} for the inverter is also given in Table 9.5-4. The waveform of v_{AZ} contains six voltage levels: E, $3E/5$, $E/5$, $-E/5$, $-3E/5$, and $-E$.

The 5L- and 6L-NNPC inverters produce the output voltage waveforms with lower dv/dt and THD than the 4L-NNPC inverter, and they can produce higher output voltages. However, some of the switches in the inverter, such as such as S_1 and S_6 in Fig. 9.5-7, withstand higher voltages than the other switches. Therefore these switches requires two or more devices in series. Static and dynamic voltage sharing methods need to be implemented for the series-connected switches.

9.6 MODULAR MULTILEVEL CONVERTER

9.6.1 Inverter Configuration

The modular multilevel converter (MMC), which was originally developed for high voltage dc (HVDC) transmission systems, has gained more attention for use in the medium-voltage drives [13]. Figure 9.6-1 shows a typical configuration of the MMC, where a number of submodules (SM) are connected in cascade in each of the six converter arms. The submodules can have a number of different designs, but the half-bridge submodule shown in Fig. 9.6-1b is widely used.

Unlike the cascaded H-bridge (CHB) inverter, where an isolated dc power supply is required for each of the cascaded power cells, the submodules in the MMC do not need isolated dc supplies. Instead, a flying dc capacitor, such as C_1 in the figure, is required in each of the submodules. Therefore, the MMC can be directly connected to a common dc bus with a dc voltage of V_d.

Each of the converter arms has an inductor L, which is used to limit the inrush current during the startup, circulating current in the steady-state operation, and surge current during the fault.

The midpoint of the dc bus Z is formed by two dc capacitors, C_{d1} and C_{d2}. However, in the practical converters, these capacitors are not required since each of the submodules has its own dc flying capacitor. The dc bus midpoint Z is not required either, but it is provided to facilitate the discussion and analysis of the converter.

The power rating of the MV drive using the MMC topology with half-bridge submodules can reach 16 MVA with the output voltage up to 7.2 kV [14]. The power and voltage ratings of the drive can be easily extended due to the modular structure of the converter.

The half-bridge submodule in the converter is composed of two power switches, S_1 and S_1', and a flying capacitor C_1 as shown in Fig. 9.6-1b. The two switches operate in a complementary manner: when one is turned on, the other must be turned off. The nominal voltage on the capacitor is E, which is equal to V_d/m, where m is the number of submodules per arm. When switch S_1 is turned on ($S_1 = 1$), the submodule output voltage v_{H1} is equal to the capacitor voltage E, which is called "On state." When S_1

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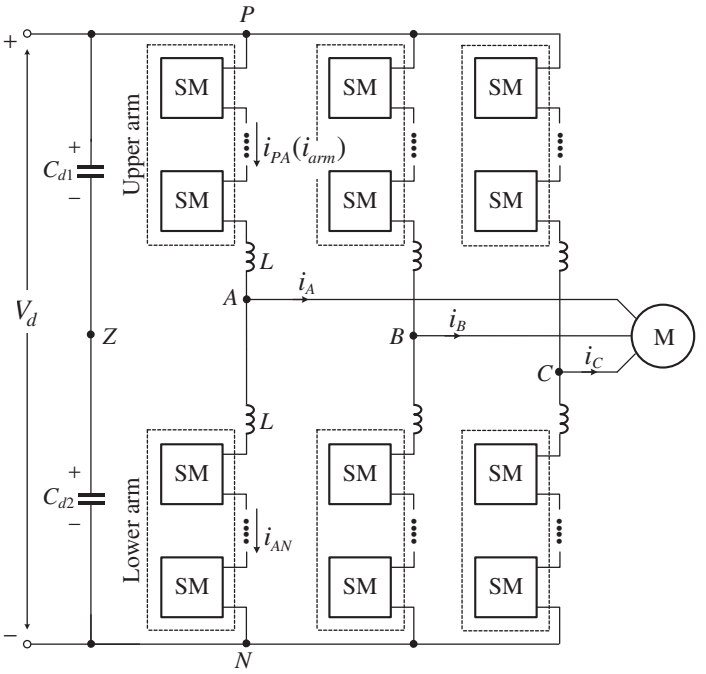
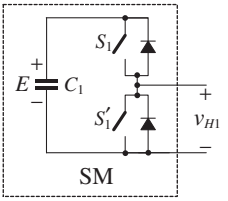


(a) Converter configuration



Switching states

S_1	S'_1	v_{H1}	SM status
1	0	E	On state
0	1	0	Bypass state

(b) Half-bridge submodule (SM)

Figure 9.6-1 Modular multilevel converter for MV drives.

is off ($S_1 = 0$), the submodule output voltage is equal to zero, which is referred to as "Bypass state" since the submodule is bypassed.

In addition to the half-bridge submodule, other types of the submodules can be used in the MMCs [15–17], three of which are shown in Fig. 9.6-2. The full-bridge submodule is formed by four switches and a flying capacitor. This submodule can generate three voltage levels: 0, E, and $-E$. The flying-capacitor and three-level submodules (Figs. 9.6-2b and 9.6-2c) are also composed of four switches each, but require two dc flying capacitors. They can generate three voltage levels: 0, E, and $2E$.

Among the four submodules mentioned above, the half-bridge submodule has the simplest structure with the least component count. It has the lowest conduction losses as well since only one device carries the current during the operation. For the full-bridge submodule, the number of switches is doubled and two devices carry current

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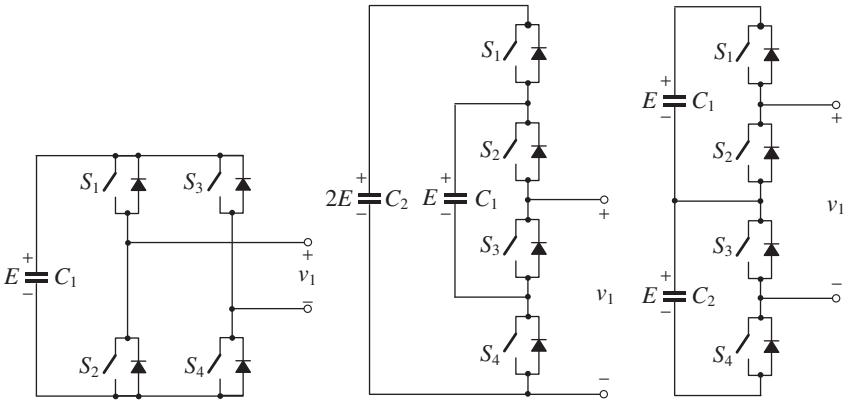

(a) Full-bridge SM (b) Flying-capacitor SM (c) Three-level SM

Figure 9.6-2 Examples of other submodules for the MMC.

during the operation, which increases the cost and power losses. However, it can output a negative voltage ($v_1 = -E$) that can help to limit the current during the fault. For the flying-capacitor submodule, it generates three voltage levels, which provides a design option for the reduction of the number of cascaded submodules for a given MMC output voltage. However, the flying capacitors may have different nominal voltages, which increase the complexity for the control and design. The flying-capacitor module has the same conduction losses as the full-bridge submodule. The three-level submodule has the same characteristics as the flying-capacitor submodule, but the two dc capacitors have the same nominal voltage.

9.6.2 Switching States and Arm Voltage

To investigate the relationship between the switching states and arm voltages, let us consider an upper arm circuit shown in Fig. 9.6-3, where three half-bridge submodules are in cascade. It is assumed in the following analysis that the voltage on the flying capacitors is kept by the controller at its nominal value of E, and the arm inductor L has little impact on the steady-state analysis, and it is, therefore, neglected.

Table 9.6-1 illustrates the relationship among the switching state of the switches, S_1, S_2, and S_3, the output voltage of the submodules, v_{H1}, v_{H2}, and v_{H3}, and the arm voltage v_{PA}, which is the voltage at the positive dc bus P with respect to the converter phase-A output. Switches S_1', S_2', and S_3' in Fig. 9.6-3 are complementary to S_1, S_2, and S_3, and therefore their switching states are not listed in the table. The arm voltage v_{PA} has four voltage levels: 0, E, $2E$, and $3E$. The voltage level E can be produced by one of the three switching states [100], [010], and [001]. Similarly, the voltage level $2E$ can be generated by any of the three switching states [110], [101], and [011]. It will be seen later that the redundancy of the switching states can be utilized to balance the voltages on the submodule flying capacitors.

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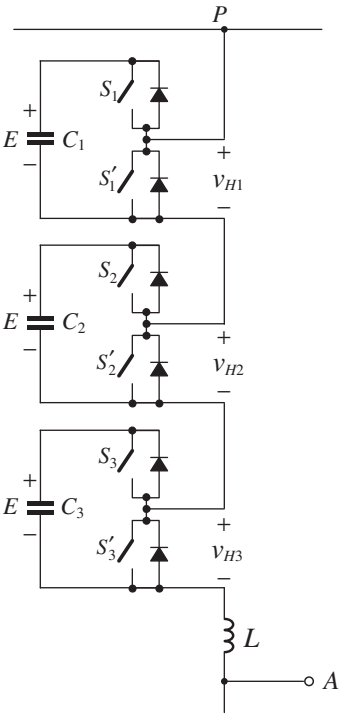

Figure 9.6-3 Configuration of an arm with half-bridge submodules in cascade.

9.6.3 Modulation Scheme

Both carrier-based SPWM and SVM schemes can be used for the MMC converter. As an example, the phase-shifted modulation for the MMC with three half-bridge submodules per arm is shown in Fig. 9.6-4. The converter requires two modulating waves v_m and v_{m-} and three carriers, v_{cr1}, v_{cr2}, and v_{cr3}. All carrier waves are identical except a phase displacement of 60° between any two adjacent carriers.

Table 9.6-1 Switching States and Arm Voltages

S_1	S_2	S_3	v_{H1}	v_{H2}	v_{H3}	v_{PA}
0	0	0	0	0	0	0
1	0	0	E	0	0	E
0	1	0	0	E	0	E
0	0	1	0	0	E	E
1	1	0	E	E	0	$2E$
1	0	1	E	0	E	$2E$
0	1	1	0	E	E	$2E$
1	1	1	E	E	E	$3E$

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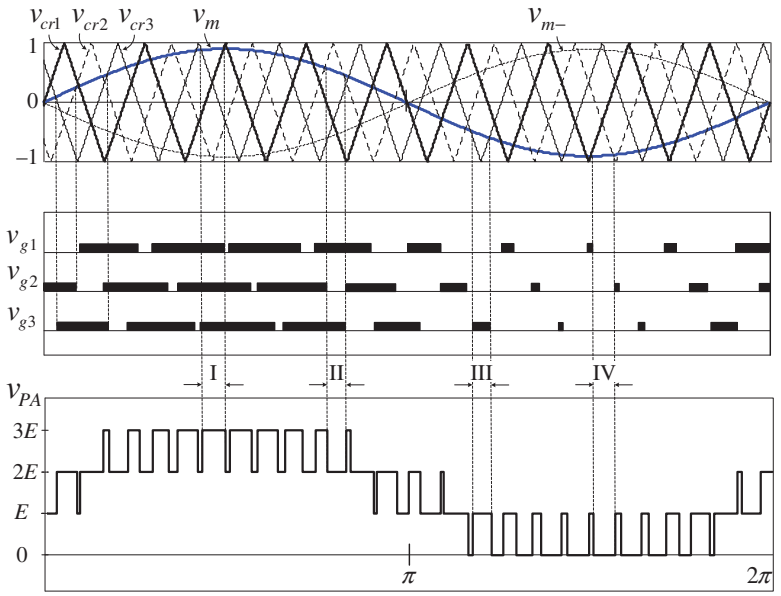

Figure 9.6-4 Principle of phase-shifted modulation for the MMC with three SMs per arm.

The gate signals, v_{g1}, v_{g2}, and v_{g3} for the switches in the upper arm, S_1, S_2, and S_3, are generated at the intersections of the modulating wave v_m and carrier waves v_{cr1}, v_{cr2}, and v_{cr3}, whereas the gate signals for the switches in the lower arm are generated (not shown) by comparing the modulating wave v_{m-}, which has the same magnitude and frequency as v_m but 180° out of phase, with the carriers v_{cr1}, v_{cr2}, and v_{cr3}.

The upper arm voltage v_{PA} is produced by the gate signals for the upper arm switches. For example, during period I, all three switches are gated on, v_{PA} can be determined according to Table 9.6-1, which is equal to $3E$. During period II, S_1 and S_3 are turned on, and $v_{PA} = 2E$. For period III, S_3 is conducting, $v_{PA} = E$. When none of the three switches is gated on during period IV, $v_{PA} = 0$.

Note that in the above analysis the arm voltage $2E$ is produced by turning S_1 and S_3 on, but this voltage can also be generated by turning on S_1 and S_2 or S_2 and S_3 according to Table 9.6-1. Similarly, the voltage level E can be generated by three different switching states. The redundancy of the switching states provides a degree of freedom to achieve additional control objectives, such as voltage balancing for the flying capacitors in the MMC.

Figure 9.6-5 shows the output voltage waveforms of the MMC with three submodules per arm. The converter output frequency and the device switching frequency is 60 Hz and 420 Hz, respectively. The phase-shifted modulation is used with an amplitude modulation m_a of 0.9. Both upper arm voltage v_{PA} and the lower arm voltage v_{AN} have four voltage levels. The converter phase-A output voltage v_{AZ}, which is the

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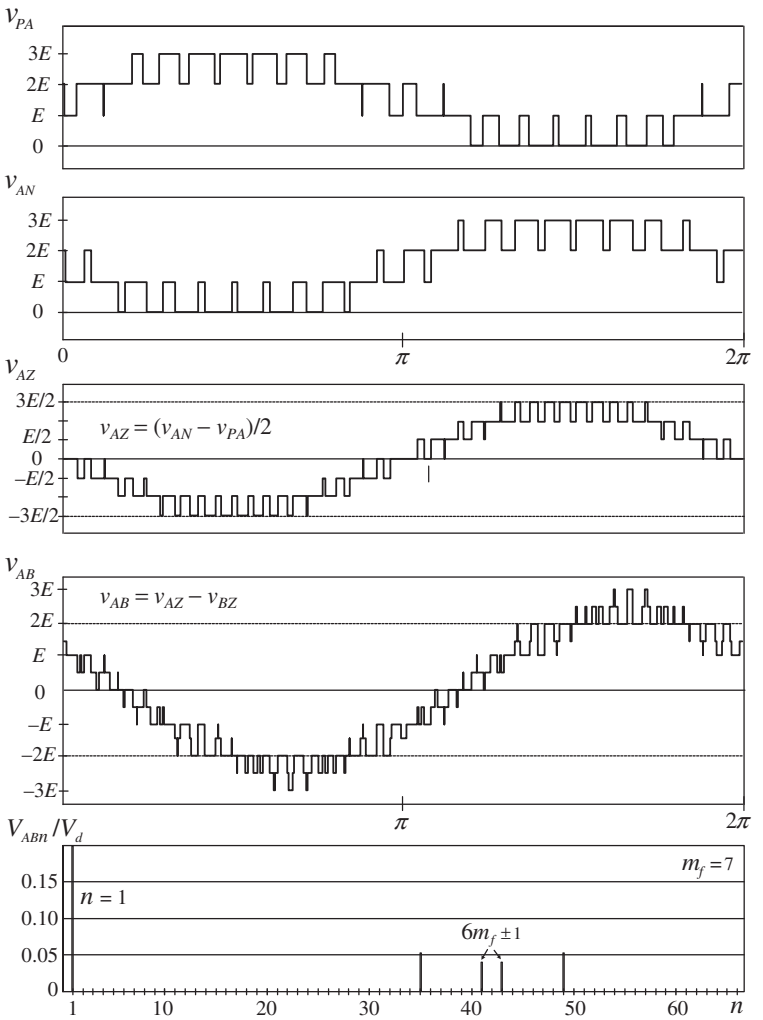

Figure 9.6-5 Output voltage waveforms of the MMC with three SMs per arm.

voltage at the output terminal A with respect to the midpoint of the dc bus Z, can be found from

$$v_{AZ} = \begin{cases} +E - v_{PA} \\ -E + v_{AN} \end{cases} \tag{9.6-1}$$

from which

$$v_{AZ} = \frac{1}{2}(v_{AN} - v_{PA}) \tag{9.6-2}$$

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The waveform of phase-*A* output voltage v_{AZ} has seven voltage levels, while the line-to-line voltage v_{AB}, which can be obtained by $v_{AB} = v_{Az} - v_{BZ}$, is composed of 13 voltage levels. The large number of voltage levels leads to a much lower dv/dt and THD in comparison to the two-level voltage source converters.

The harmonic spectrum of v_{AB} is also shown in the figure, where the dominant harmonics are centered around the 42nd harmonic, whose frequency is 2520 Hz. This can be considered as the equivalent switching frequency of the converter, which is six times the device switching frequency. This is mainly due to the fact that the switches in the three cascaded submodules in an arm are turned on or off at different times.

9.6.4 Voltage Balancing of Flying Capacitors in MMCs

In a MMC with m half-bridge submodules per arm, the total dc bus voltage V_d should be equally distributed among the submodule flying capacitors and maintained around its nominal value of E ($E = V_d/m$). There are a number of proposed methods for the capacitor VBC [15–20], most of which use logic/sorting algorithms to make the capacitor voltage balanced. In this section, a simple capacitor voltage balancing method, which is also based on a logic/sorting algorithm, is presented [15–16].

Figure 9.6-6 shows how the operating modes of a submodule affect the voltage on the flying capacitors. When the submodule is in the On state, the flying capacitor C_1 will be charged if the arm current i_{arm} is positive ($i_{arm} > 0$), whereas it will be

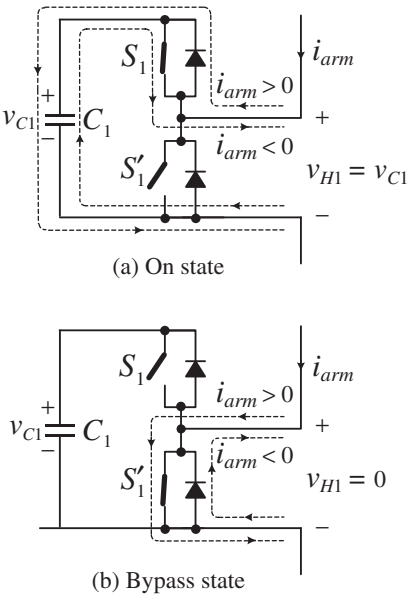

Figure 9.6-6 Operating modes of the half-bridge submodule.

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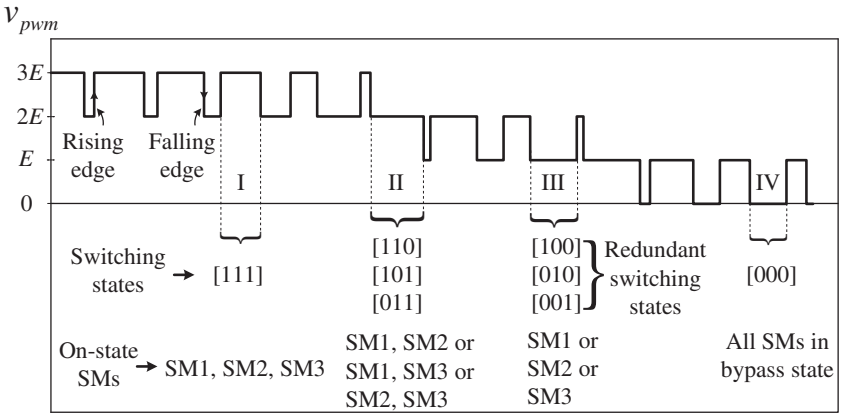

Figure 9.6-7 Example of using redundant switching states for capacitor voltage balancing control.

discharged with a negative arm current ($i_{arm} < 0$). When the submodule is in the Bypass state, the flying capacitor voltage is kept unchanged as shown in Fig. 9.6-6b.

Figure 9.6-7 shows an example of using redundant switching states for the capacitor VBC in an MMC with three submodules per arm, where v_{pwm} is the reference PWM voltage. The upper arm voltage v_{PA} or lower arm voltage v_{AN} generated by a modulation scheme can be used as the reference voltage v_{pwm}. To illustrate the concept of submodule capacitor voltage control, let us consider four typical periods in the reference PWM voltage v_{pwm}:

- Period I with $v_{pwm} = 3E$: Only one switching state [111] can be used to produce $3E$ according to Table 9.6-1. No redundant switching states can be utilized to adjust the capacitor voltage. The three submodules in the arm are all in the On state.
- Period II with $v_{pwm} = 2E$: Any of the three switching states, [110], [101], and [011], can be used to produce $2E$, which corresponds to three pairs of on-state submodules: (SM1, SM2), (SM1, SM3), and (SM2, SM3). Based on measured capacitor voltages and arm current polarity, one pair of the on-state submodules can be selected to charge the capacitor with the lowest voltage or discharge the capacitor with the highest voltage.
- Period III with $v_{pwm} = E$: Any of the three switching states, [100], [010], and [001], can be used to produce E, which corresponds to three on-state submodules: SM1, SM2, SM3. One of three submodules can be selected to charge or discharge the capacitor with the lowest or highest voltage, respectively.
- Period IV with $v_{pwm} = 0$: All three submodules are in the Bypass state, and the submodule capacitor voltage is not affected.

The above analysis indicates that the submodule capacitor voltage can be controlled by the redundant switching states at the reference PWM voltage levels of E

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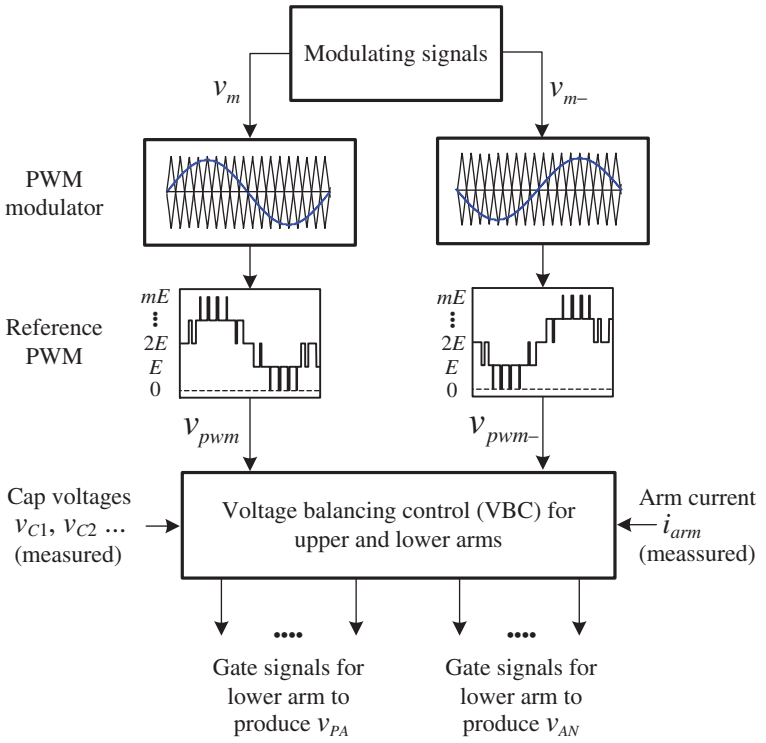

Figure 9.6-8 Block diagram of capacitor voltage balancing control.

and $2E$. For the MMCs with m submodules per arm, the submodule capacitor voltage can be controlled at the reference PWM voltage levels of $E, 2E \ldots (m-1)E$. When v_{pwm} is at the voltage levels of zero or mE, the submodule capacitor voltage cannot be adjusted.

Figure 9.6-8 shows a general block diagram for the control of submodule capacitor voltages. The phase-shifted modulation scheme is used as an example. Based on the two modulation signals, v_m and v_{m-}, two reference PWM voltages, v_{pwm} and v_{pwm-}, are generated to control the voltages on capacitors in the upper and lower arms, respectively. Based on the measured capacitor voltages and arm current direction, the VBC block generates the gate signals for the switches in the upper and lower arms, which produce the desired waveforms for the arm voltages v_{PA} and v_{AN}.

Figure 9.6-9 shows the detailed algorithm for the VBC block in Fig. 9.6-8. In this algorithm, the rising and falling edges of the reference PWM waveform v_{pwm} are detected. The rising edge indicates that during the upcoming switching period the voltage level of v_{pwm} will be increased by an E, which implies that one of the submodules in the arm should be changed from the Bypass state to On state as shown in Fig. 9.6-7. The falling edge indicates that during the upcoming switching period the v_{pwm} voltage level will be decreased by an E, which implies that one of the submodules in the arm should be changed from the On state to Bypass state.

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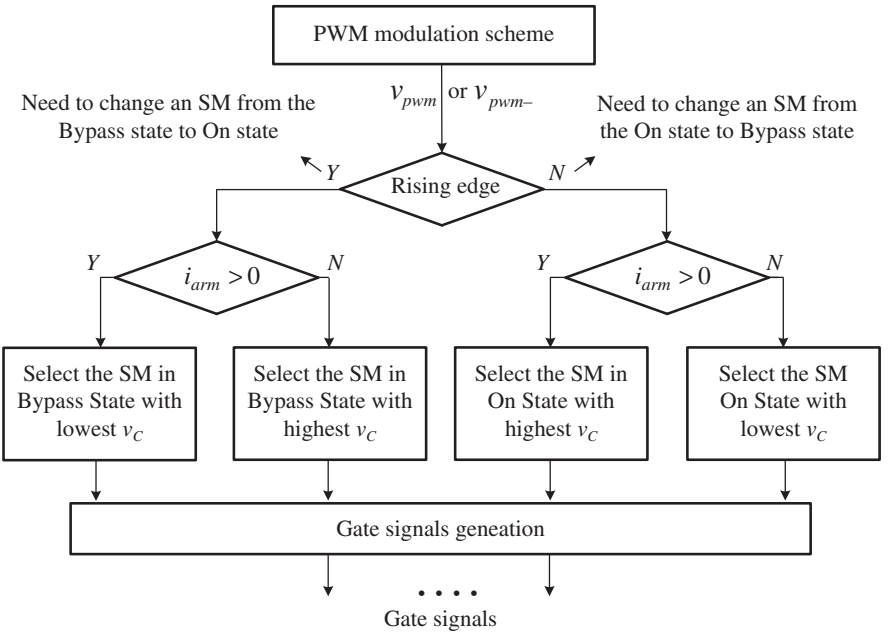

Figure 9.6-9 Flowchart of capacitor voltage balancing control algorithm.

Based on the detected rising/falling edges in v_{pwm}, the measured polarity of the arm current i_{arm}, and the submodule capacitor voltages, the submodule with the lowest capacitor voltage should be selected to charge the capacitor during the upcoming switching period or the submodule with the highest capacitor voltage should be selected to discharge the capacitor.

For example, when a rising edge in v_{pwm} is detected, a submodule in the arm should be changed from the Bypass state to On state (refer to Fig. 9.6-9). If the arm current is positive ($i_{arm} > 0$), a submodule in the Bypass state with the lowest capacitor voltage will be selected to turn on. The capacitor in this submodule will be then charged during the upcoming switching period as shown in Fig. 9.6-6a. Similarly, if the arm current is negative ($i_{arm} < 0$), a submodule in the Bypass state with the highest capacitor voltage will be selected to turn on. The capacitor in the submodule will be discharged during the upcoming switching period.

When a falling edge in v_{pwm} is detected, a submodule in the arm should be changed from the On state to Bypass state. If the arm current is positive ($i_{arm} > 0$), a submodule in the On state with the highest capacitor voltage will be selected to bypass. Otherwise, this capacitor will be charged again during the upcoming switching period, which should be avoided. Similarly, if the arm current is negative ($i_{arm} < 0$), a submodule in the On state with the lowest capacitor voltage will be selected to bypass. Otherwise, this capacitor will be kept discharging during the upcoming switching period.

Figure 9.6-10 shows the simulated waveforms for the MMC with four submodules per arm. The operating condition and converter parameters are given in Table 9.6-2.

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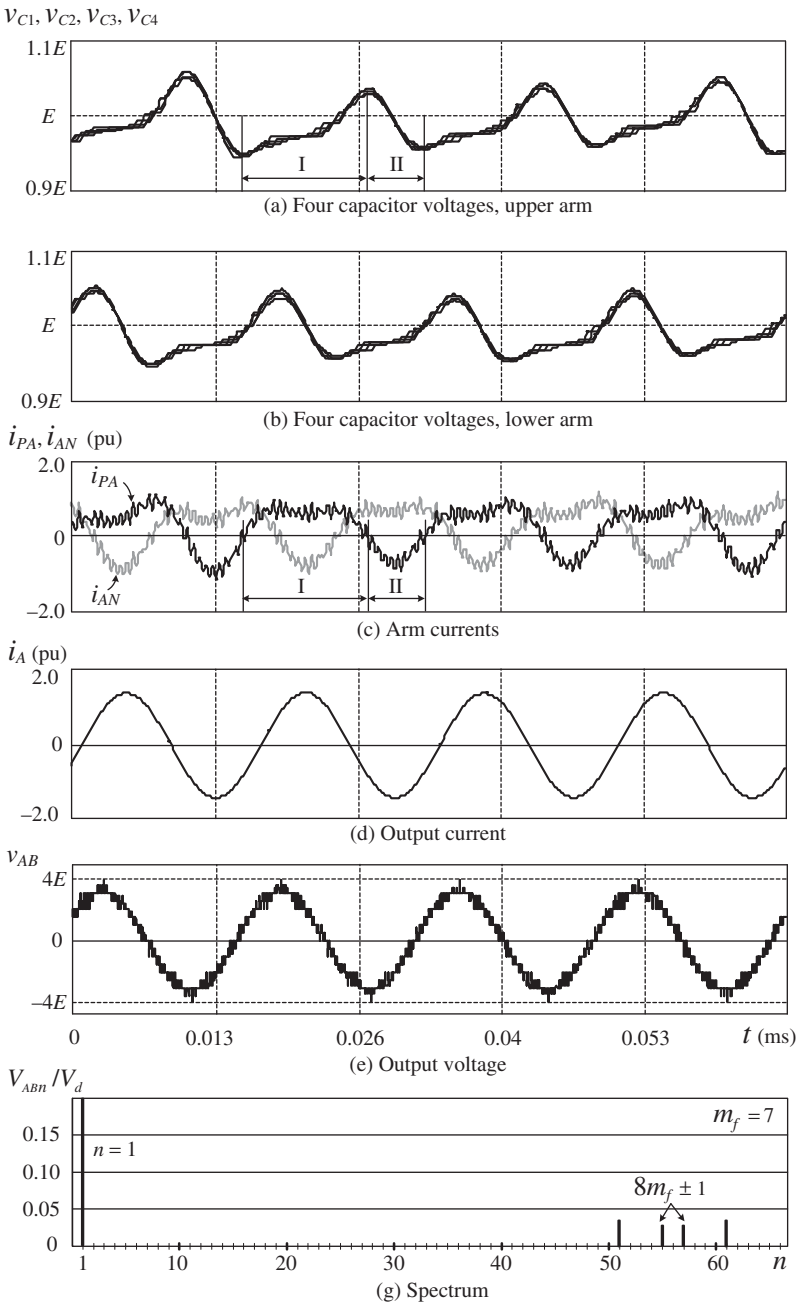

Figure 9.6-10 Simulated waveforms for an MMC with capacitor voltage balancing control.

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Table 9.6-2 Simulation Parameters for an MMC with Four SMs Per Arm

Output voltage V_{AB} (rms)	4000 V
Output current I_A (rms)	160 A (1.0 pu)
Output frequency f_o	60 Hz
DC bus voltage V_d	7045 V (3.05 pu)
Type of submodule	Half-bridge
Number of submodules per arm	4
Flying capacitor	2.8 mF (15 pu)
Arm inductor	3.0 mH (0.08 pu)
Nominal SM capacitor voltage E	$V_d/4 = 1761$ V
Switching frequency	420 Hz ($m_f = 7$)
Amplitude modulation index	0.9
Load power factor	0.9

The voltage waveforms on the four capacitors in the upper arm and another four capacitors in the lower arm are shown in Figs. 9.6-10a and 9.6-10b, respectively. The capacitor voltages fluctuate around the nominal value of $E = V_d/m$, which is 1761 V as given in the table. The magnitude of the fluctuation depends on the capacitor value and the output frequency of the converter. In this study, the capacitor value is 15 pu and the converter output frequency is 60 Hz.

It can be observed that when the converter output voltage v_{AB} varies one cycle, the capacitor voltage ripple also varies one cycle. It implies that the frequency of the dominant harmonic in the voltage ripple is the same as the converter output frequency of 60 Hz, not the switching frequency of 420 Hz. This is the reason why a relatively large size flying capacitor (15 pu in this study) is required in the MMCs.

The waveforms of the upper and lower arm currents, i_{PA} and i_{AN} are shown in Fig. 9.6-10c. Each of the arm current contains a dc component, fundamental frequency component of 60 Hz, and switching harmonics. The converter phase-*A* output current is obtained by $i_A = i_{PA} - i_{AN}$.

The line-to-line voltage is composed of 17 small voltage steps, which makes the waveform close to sinusoidal. The dominant harmonics are centered around the 56th harmonic, whose frequency is 3360 Hz, which is 8 times of the device switching frequency of 420 Hz. If the converter has a motor load, its stator current will be virtually sinusoidal since the high-order switching harmonics in the stator current will be greatly attenuated by the motor leakage inductances.

9.6.5 Capacitor Voltage Ripples and Circulating Currents

When the MMCs are used in the MV drive, the voltage ripple on the flying capacitors will be excessively high when the motor operates at the low speeds. The main cause for the capacitor voltage ripples is partially explained in the preceding section when the converter operates at the output frequency of 60 Hz.

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The capacitor voltage ripple will be substantially increased as the converter output frequency decreases, for example, to a few hertz. The cause of the excessive capacitor voltage ripple can be further investigated by examining the waveforms in Fig. 9.6-10. During period I, the upper arm current i_{PA} is positive, which makes the capacitor in the upper arm continuously charged (refer to Fig. 9.6-6a). As a result, the capacitor voltage keeps increasing as shown in Fig. 9.6-10a. Similarly, during period II, i_{PA} is negative, which makes the capacitor continuously discharged.

The above analysis reveals that as long as the arm current is positive ($i_{arm} > 0$), the capacitors in the arm will keep charging, and their voltages will keep increasing, whereas with a negative arm current, the capacitor voltage will keep decreasing. When the converter operates at low frequencies, the positive and negative periods of the arm current will be extensively long, leading to excessive voltage ripples on the flying capacitors.

To solve the issue with the excessive capacitor voltage ripples at low output frequencies, the size of the flying capacitors should be increased significantly. However, this is not a practical solution due to the cost associated with the large number of large size flying capacitors in the MMC.

This problem can be mitigated by injecting a relatively high frequency common-mode (CM) voltage signal to the modulating signals of a PWM scheme [20]. In doing so, the flying capacitors will be charged and discharged at high rates determined by the frequency of the injected CM voltage, which leads to the reduction in the capacitor voltage ripples. However, this method generates additional CM voltage stress on the stator winding. If not mitigated, it will cause premature failure of the stator winding insulation. The CM voltage issue in the MV drive is further analyzed in detail in Chapter 17, and therefore are not discussed here.

Another technical issue in the MMCs is the circulating currents that flow among the three legs of the converter. The circulating currents are mainly caused by the differences of the submodule capacitor voltages in the upper and lower arms of the three converter legs that are connected in parallel on a common dc bus. Since the circulating currents flow internally among the three converter legs, they do not affect the waveforms of the converter output voltage or current. However, they increase the conduction losses of the switches, which reduces the efficiency of the converter. The circulating currents can be reduced to a certain extent by increasing the arm inductance [13], but this is at the expense of increased converter cost. They can also be minimized by the closed loop control schemes [17].

To summarize, the MMC presented in this section has a number of features. Its cascaded modular structure offers high modularity and scalability with low-voltage cost-effective submodules, which helps to reduce the manufacturing cost of the converter. With a number of submodules in cascade per arm, the converter output voltage waveform is of high quality with low THD and dv/dt. The converter submodules do not require isolated dc supplies, which are normally provided by the secondary windings of a transformer. Instead, the converter can operate with a single dc supply (V_d), which makes it possible to develop a transformerless drive.

However, there are a number of disadvantages associated with the converter. The converter requires a large number of flying capacitors. The value of these capacitors

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is relatively high since they are charged and discharged at the fundamental frequency of the converter, not at the switching frequency. Each of the submodule capacitor voltages needs to be measured and controlled, which increases the complexity of the control scheme. The capacitor voltage ripples will be excessively high at low converter output frequencies, and mitigation methods should be provided for the drive to operate at low speeds. Despite its disadvantages, the MMC has been successfully used in the commercial MV drives [14, 21].

9.7 SUMMARY

This chapter presents the multilevel inverter topologies that are not covered in the earlier chapters. These inverter topologies include the flying-capacitor (FC) inverters, ANPC inverters, NPP inverters, NNPC inverters, and MMCs. The operating principle of these inverters is explained, their modulation schemes are introduced, and the dc capacitor voltage balancing control is elaborated.

The FC inverter may have limited practical applications due to the large number of flying capacitors that need isolated pre-charging circuits and complex voltage balancing control, but this topology can be used to derive other multilevel topologies. The ANPC topology is evolved from the NPC inverter and has a unique feature for even distribution of the power losses among the switches. This feature makes each of the switches in the inverter fully utilized for a higher output power. The NNPC inverter is developed by combining the FC and NPC topologies, which leads to less components compared with the NPC inverters with the same voltage levels. The MMC topology has a modular structure. With a number of low-voltage submodules in cascade, the converter can operate in a wide voltage range. It can be used in the MV drives with the voltage ratings from 2.3 to 13.8 kV.

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Part Four

PWM Current Source Converters

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Chapter 10

PWM Current Source Inverters

10.1 INTRODUCTION

The inverters used in medium voltage (MV) drives can be generally classified into voltage source inverter (VSI) and current source inverter (CSI). The VSI produces a defined three-phase PWM voltage waveform for the load while the CSI outputs a defined PWM current waveform. The PWM CSI features simple converter topology, motor-friendly waveforms, and reliable short-circuit protection, and therefore it is one of the preferred converter topologies for the MV drive [1].

Two types of CSIs are commonly used in the MV drive: PWM inverters and load-commutated inverter (LCI). The PWM inverter uses switching devices with self-extinguishable capability. Prior to the advent of GCT devices in the late 1990s, GTOs were dominantly used in the CSI fed drives [2, 3]. The LCI employs SCR thyristors whose commutation is assisted by the load with a leading power factor. The LCI topology is particularly suitable for very large synchronous motor drives with a power rating up to 100 MW [4].

This chapter mainly deals with the PWM CSI. Various modulation techniques for the inverter are discussed, which include trapezoidal pulse width modulation (TPWM), selective harmonic elimination (SHE), and space vector modulation (SVM). These modulation schemes are developed for high-power inverters operating with a switching frequency of around 500 Hz. The operating principle of the modulation schemes is elaborated and their harmonic performance is analyzed. A relatively new CSI topology using parallel inverters is introduced. An SVM-based dc current balance control algorithm is developed for the parallel inverters. The chapter ends with an introduction to the Load-commutated inverter.

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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10.2 PWM CURRENT SOURCE INVERTER

An idealized PWM CSI is shown in Fig. 10.2-1. The inverter is composed of six GCT devices, each of which can be replaced with two or more devices in series for medium voltage operation. The GCT devices used in the CSI are of symmetrical type with a reverse voltage blocking capability. The inverter produces a defined PWM output current i_w. On the dc side of the inverter is an ideal dc current source I_d. In practice, I_d can be obtained by a current source rectifier (CSR).

The CSI normally requires a three-phase capacitor C_f at its output to assist the commutation of the switching devices. For instance, at the turn-off of switch S_1, the inverter PWM current i_w falls to zero within a very short period of time. The capacitor provides a current path for the energy trapped in the phase-*A* load inductance. Otherwise, a high voltage spike would be induced, causing damages to the switching devices. The capacitor also acts as a harmonic filter, improving the load current and voltage waveforms. The value of capacitor is in the range of 0.3–0.6 pu for the MV drive with a switching frequency of around 200 Hz [5]. The capacitor value can be reduced accordingly with the increase of the switching frequency.

The dc current source I_d can be realized by an SCR or PWM current source rectifier with a dc current feedback control as shown in Fig. 10.2-2. To make the dc current I_d smooth and continuous, a dc choke L_d is an indispensable device for the CSR. Through the feedback control the magnitude of I_d is kept at a value set by its reference I_d^*. The size of the dc choke is normally in the range of 0.5–0.8 pu.

The PWM CSI has the following characteristics:

- Simple converter topology. The GCT devices used in the inverter are of symmetrical type, which do not require anti-parallel freewheeling diodes.
- Motor-friendly waveforms. The CSI produces a three-phase PWM current instead of PWM voltage as in the VSI. With the filter capacitor installed at the inverter output, the load current and voltage waveforms are close to sinusoidal. The high *dv/dt* problem associated with the VSI does not exist in the CSI.

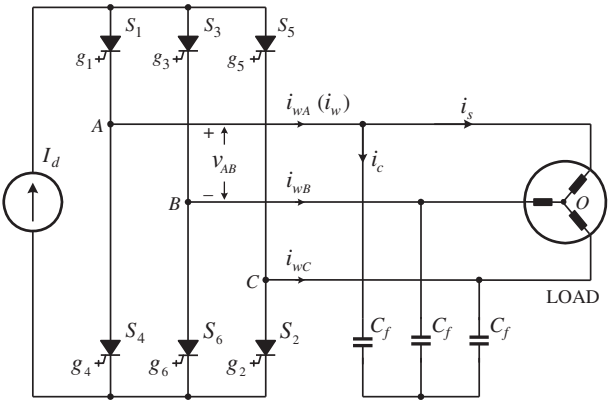

Figure 10.2-1 PWM GCT current source inverter.

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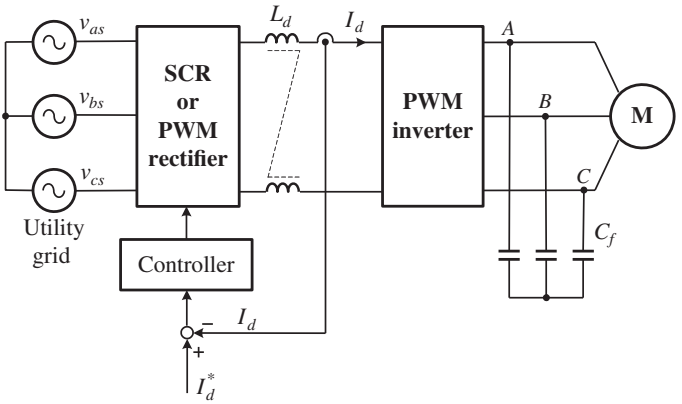

Figure 10.2-2 Realization of a dc current source I_d.

- Reliable short-circuit protection. In case of a short circuit at the inverter output terminals, the rate of rise of the dc current is limited by the dc choke, allowing sufficient time for the protection circuit to function.
- Limited dynamic performance. The dc current cannot be changed instantaneously during transients, which reduces the system dynamic performance.

10.2.1 Trapezoidal Modulation

The switching pattern design for the CSI should generally satisfy two conditions: (a) the dc current I_d should be continuous, and (b) the inverter PWM current i_w should be defined. The two conditions can be translated into a **switching constraint**: at any instant of time (excluding commutation intervals) there are only two switches conducting, one in the top half of the bridge and the other in the bottom half. With only one switch turned on, the continuity of the dc current is lost. A very high voltage will be induced by the dc choke, causing damage to the switching devices. If more than two devices are on simultaneously, the PWM current i_w is not defined by the switching pattern. For instance, with S_1, S_2, and S_3 conducting at the same time, the currents in S_1 and S_3, which are the PWM currents in the inverter phases A and B, are load dependent although the sum of the two currents is equal to I_d.

Figure 10.2-3 shows the principle of TPWM, where v_m is a trapezoidal modulating wave and v_{cr} is a triangular carrier wave. The **amplitude modulation index** is defined by

$$m_a = \frac{\hat{V}_m}{\hat{V}_{cr}} \tag{10.2-1}$$

where $\hat{V}_m$ and $\hat{V}_{cr}$ are the peak values of the modulating and carrier waves, respectively. Similar to the carrier-based PWM schemes for VSIs, the gate signal v_{g1} for

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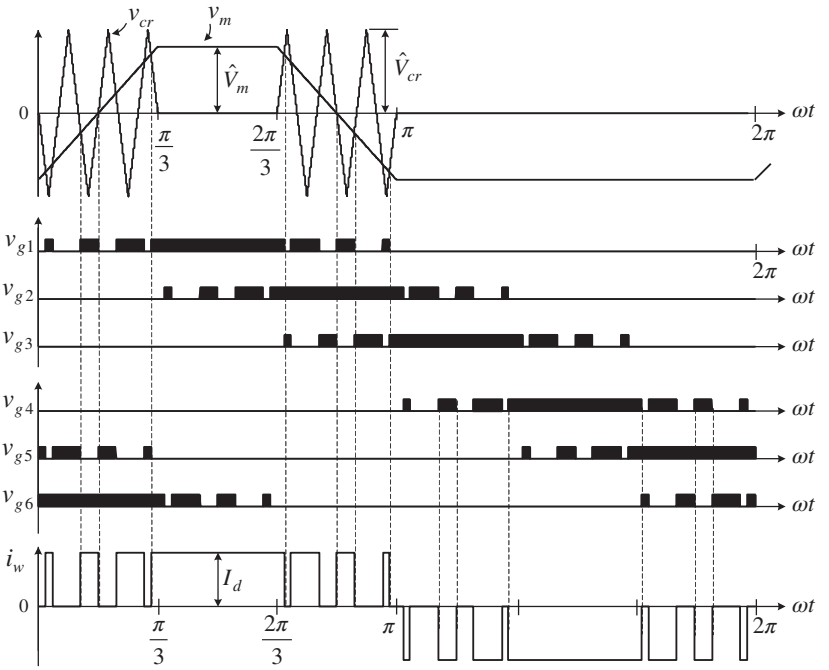

Figure 10.2-3 Trapezoidal pulse width modulation.

switch S_1 is generated by comparing v_m with v_{cr}. However, the trapezoidal modulation does not generate gatings in the center $\pi/3$ interval of the positive half cycle or in the negative half cycle of the inverter fundamental frequency. Such an arrangement leads to the satisfaction of the switching constraint for the CSI. It can be observed from the gate signals that only two GCTs conduct at any time, resulting in a defined i_w. The magnitude of i_w is set by the dc current I_d.

The switching frequency of the devices can be calculated by

$$f_{sw} = f_1 \times N_p \tag{10.2-2}$$

where f_1 is the fundamental frequency and N_p is the number of pulses per half cycle of i_w.

Figure 10.2-4a shows the spectrum of the inverter PWM current i_w with $N_p = 13$ and $m_a = 0.85$, where I_{wn} is the rms value of the nth order harmonic current in i_w and $I_{w1,\max}$ is the maximum rms fundamental-frequency current given by

$$I_{w1,\max} = 0.74 I_d \quad \text{for} \quad m_a = 1 \tag{10.2-3}$$

The PWM current i_w does not contain any even-order harmonics since its waveform is of half-wave symmetry. The TPWM scheme produces two pairs of dominant

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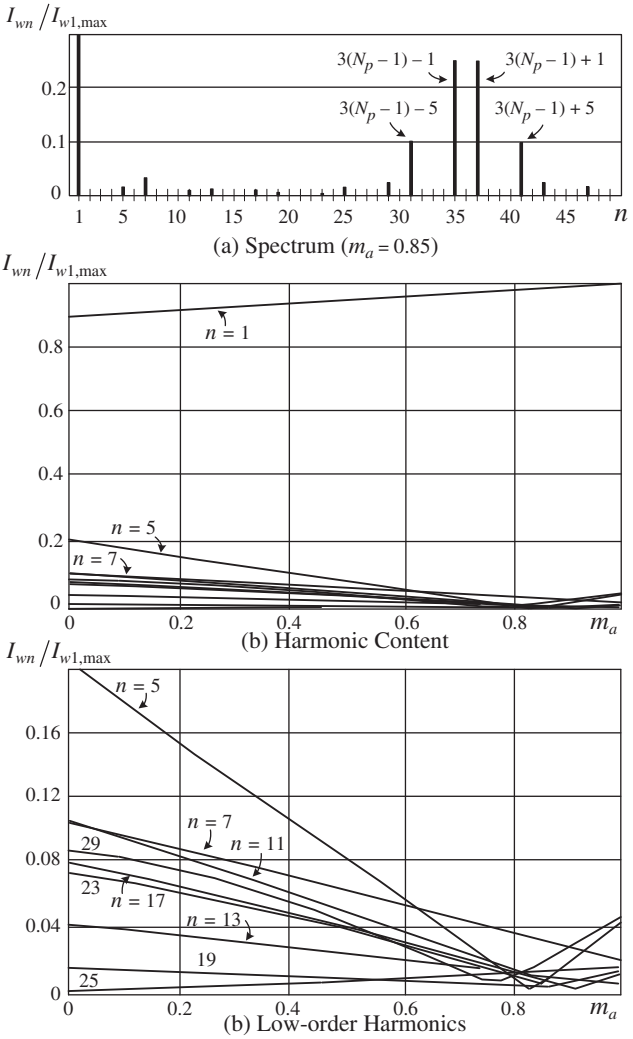

Figure 10.2-4 Harmonic content of i_w produced by TPWM with $N_p = 13$.

harmonics at $n = 3(N_p - 1) \pm 1$ and $n = 3(N_p - 1) \pm 5$, which are the 31st, 35th, 37th, and 41st in this case.

The harmonic content in i_w is shown in Fig. 10.2-4b. Its fundamental-frequency component I_{w1} does not vary significantly with the modulation index m_a. When m_a varies from zero to its maximum value of 1.0, I_{w1} changes from its minimum value of $0.89I_{w1,\max}$ to $I_{w1,\max}$, presenting only an 11% increase. This is mainly due to the fact that i_w is not modulated in the center $\pi/3$ interval of each half cycle. In practice, the adjustment of I_{w1} is normally accomplished by varying the dc current I_d through the rectifier instead of m_a.

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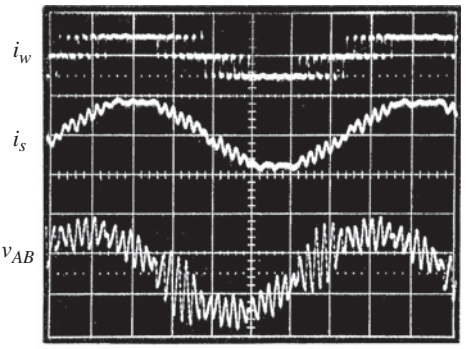
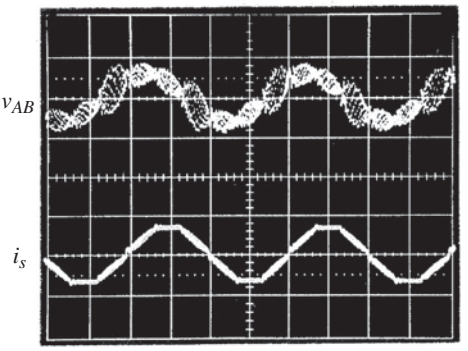


(a) $f_1 = 13.8$ Hz



(b) $f_1 = 5$ Hz

Figure 10.2-5 Waveforms measured from a CSI drive with TPWM scheme (a) $f_1 = 13.8$ Hz, $N_p = 13, f_{sw} = 180$ Hz; and (b) $f_1 = 5$ Hz, $N_p = 31, f_{sw} = 155$ Hz.

Figure 10.2-4c shows the details of the low-order harmonic currents. At the modulation index of 0.85, the magnitudes of most harmonic currents are close to their lowest values, leading to a lowest harmonic distortion. This phenomenon holds true for other values of N_p. Therefore, the modulation index of 0.85 can be selected, at which the THD of i_w is minimized while the fundamental current I_{w1} is close to $I_{w1,\max}$.

Figure 10.2-5 shows the experimental results obtained from a low-power laboratory CSI fed induction motor drive with $m_a = 0.85$ and $C_f = 0.66$ pu. The motor operated at low speeds with a rated stator current. Figure 10.2-5a illustrates the waveforms for the inverter PWM current i_w, stator current i_s, and line-to-line v_{AB} with the fundamental frequency of $f_1 = 13.8$ Hz and switching frequency of $f_{sw} = 180$ Hz. The PWM current i_w contains 13 pulses per half cycle ($N_p = 13$). The waveform for i_s looks like a trapezoid superimposed with some switching noise. The v_{AB} waveform, although contains more harmonics than i_s, is much better than that of the two-level VSI in terms of harmonic distortion and *dv/dt*. The measured waveforms for the inverter operating at $f_1 = 5$ Hz and $f_{sw} = 155$ Hz ($N_p = 31$) are shown in Fig. 10.2-5b.

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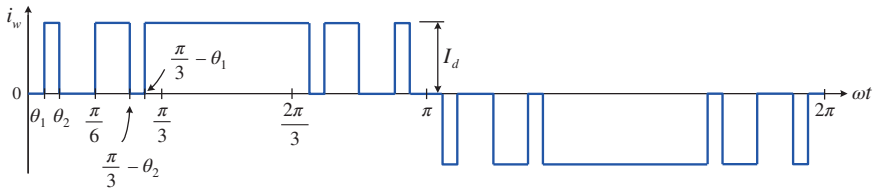

Figure 10.2-6 Selective harmonic elimination (SHE) scheme.

Considering the fact that the switching frequency in these two cases is only 180 Hz and 155 Hz, the waveforms for v_{AB} and i_s are fairly good.

As indicated earlier, the TPWM scheme produces two pairs of dominant harmonics at $n = 3(N_p - 1) \pm 1$ and $n = 3(N_p - 1) \pm 5$. With $N_p = 5$, the dominant harmonics in i_w include the 7th, 11th, 13th, and 17th. These low-order harmonics are difficult to be fully attenuated by the filter capacitor and motor inductance, causing a detrimental effect on motor operation and harmonic power losses. Therefore, the applicability of the trapezoidal modulation is diminished for $N_p \leq 7$.

10.2.2 Selective Harmonic Elimination

SHE is an off-line modulation scheme, which is able to eliminate a number of low-order unwanted harmonics in the inverter PWM current i_w. The switching angles are pre-calculated and then imported into a digital controller for implementation. Fig. 10.2-6 shows a typical SHE waveform that satisfies the switching constraint for the CSI. There are five pulses per half cycle ($N_p = 5$) with five switching angles in the first $\pi/2$ period. However, only two out of the five angles, θ_1 and θ_2, are independent. Given these two angles, all other switching angles can be calculated.

The two switching angles provide two degrees of freedom, which can be used to either eliminate two harmonics in i_w without modulation index control or eliminate one harmonic and provide an adjustable modulation index m_a. The first option is preferred since the adjustment of I_{w1} is normally done by varying I_d through the rectifier. The number of harmonics to be eliminated is then given by

$$k = (N_p - 1)/2$$

The inverter PWM current i_w can generally be expressed as

$$i_w(\omega t) = \sum_{n=1}^{\infty} a_n \sin(n\omega t) \tag{10.2-4}$$

where

$$a_n = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} i_w(\omega t) \sin(n\omega t) d(\omega t) \tag{10.2-5}$$

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The Fourier coefficient a_n can be found from

$$a_n = \frac{4I_{dc}}{\pi} \times \begin{cases} \displaystyle\int_{\theta_1}^{\theta_2} \sin(n\omega t)d(\omega t) + \cdots + \int_{\theta_k}^{\frac{\pi}{6}} \sin(n\omega t)d(\omega t) + \\ \displaystyle\int_{\frac{\pi}{3}-\theta_k}^{\frac{\pi}{3}-\theta_{k-1}} \sin(n\omega t)d(\omega t) + \cdots + \int_{\frac{\pi}{3}-\theta_1}^{\frac{\pi}{2}} \sin(n\omega t)d(\omega t) & k = \text{odd}; \\ \displaystyle\int_{\theta_1}^{\theta_2} \sin(n\omega t)d(\omega t) + \cdots + \int_{\theta_{k-1}}^{\theta_k} \sin(n\omega t)d(\omega t) + \\ \displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}-\theta_k} \sin(n\omega t)d(\omega t) + \cdots + \int_{\frac{\pi}{3}-\theta_1}^{\frac{\pi}{2}} \sin(n\omega t)d(\omega t) & k = \text{even} \end{cases} \tag{10.2-6}$$

from which

$$a_n = \frac{4I_{dc}}{\pi\, n} \times \begin{cases} \cos(n\theta_1) + \cos[n(\pi/3-\theta_1)] - \cos(n\theta_2) - \cos[n(\pi/3-\theta_2)] + \cdots \\ + \cos(n\theta_k) + \cos[n(\pi/3-\theta_k)] - \cos(n\pi/6) & k = \text{odd}; \\ \cos(n\theta_1) + \cos[n(\pi/3-\theta_1)] - \cos(n\theta_2) - \cos[n(\pi/3-\theta_2)] + \cdots \\ - \cos(n\theta_k) - \cos[n(\pi/3-\theta_k)] + \cos(n\pi/6) & k = \text{even} \end{cases} \tag{10.2-7}$$

To eliminate k harmonics, k equations can be formulated by setting $a_n = 0$,

$$F_i(\theta_1, \theta_2, \theta_3, \ldots \theta_k) = 0 \qquad i = 1, 2, \ldots, k \tag{10.2-8}$$

For example, to eliminate the 5th, 7th, and 11th harmonics in i_w, the following three functions can be derived

$$\begin{cases} F_1 = \cos(5\theta_1) + \cos[5(\pi/3-\theta_1)] - \cos(5\theta_2) - \cos[5(\pi/3-\theta_2)] \\ \qquad + \cos(5\theta_3) + \cos[5(\pi/3-\theta_3)] - \cos(5\pi/6) = 0 \\ F_2 = \cos(7\theta_1) + \cos[7(\pi/3-\theta_1)] - \cos(7\theta_2) - \cos[7(\pi/3-\theta_2)] \\ \qquad + \cos(7\theta_3) + \cos[7(\pi/3-\theta_3)] - \cos(7\pi/6) = 0 \\ F_3 = \cos(11\theta_1) + \cos[11(\pi/3-\theta_1)] - \cos(11\theta_2) - \cos[11(\pi/3-\theta_2)] \\ \qquad + \cos(11\theta_3) + \cos[11(\pi/3-\theta_3)] - \cos(11\pi/6) = 0 \end{cases} \tag{10.2-9}$$

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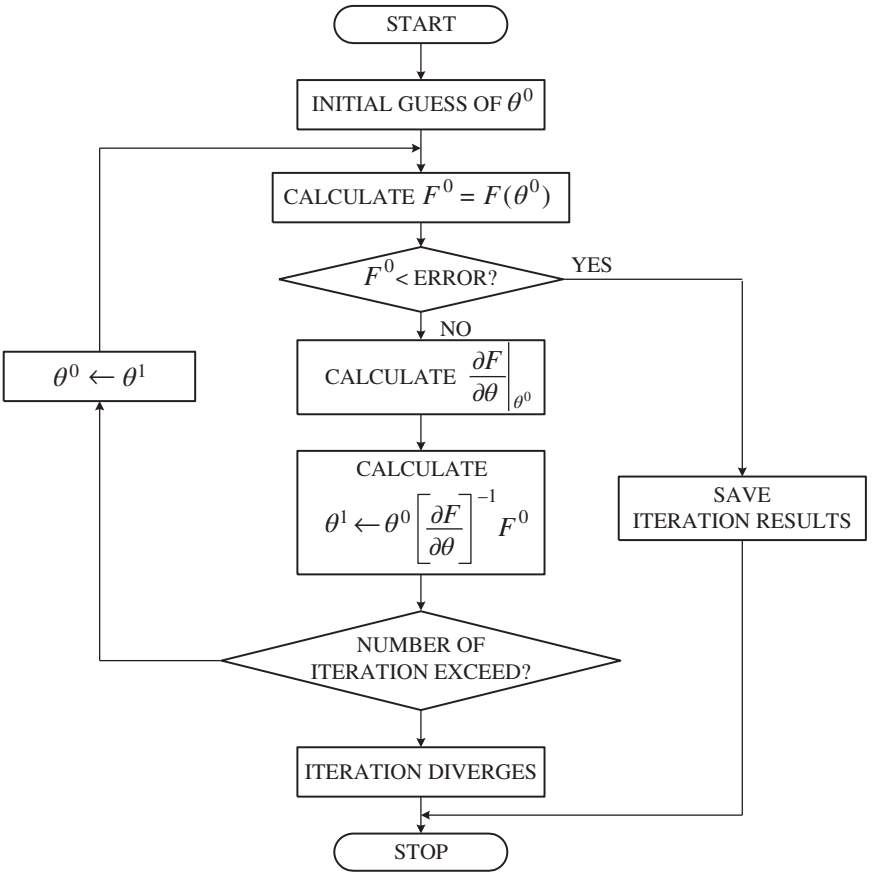

Figure 10.2-7 Flowchart of Newton–Raphson algorithm.

The nonlinear and transcendental equation of (10.2-9) can be solved by a number of numerical methods, one of which is the Newton–Raphson iteration algorithm [6]. The flowchart of this algorithm is shown in Fig. 10.2-7, where θ^0 is the initial guess of the switching angles and $\partial F/\partial\theta$ is the Jacob matrix given by

$$\frac{\partial F}{\partial \theta} = \begin{bmatrix} \frac{\partial F_1}{\partial \theta_1} & \frac{\partial F_1}{\partial \theta_2} & \frac{\partial F_1}{\partial \theta_3} & \cdots & \frac{\partial F_1}{\partial \theta_k} \\ \frac{\partial F_2}{\partial \theta_1} & \frac{\partial F_2}{\partial \theta_2} & \frac{\partial F_2}{\partial \theta_3} & \cdots & \frac{\partial F_2}{\partial \theta_k} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\partial F_k}{\partial \theta_1} & \frac{\partial F_k}{\partial \theta_2} & \frac{\partial F_k}{\partial \theta_3} & \cdots & \frac{\partial F_k}{\partial \theta_k} \end{bmatrix} \tag{10.2-10}$$

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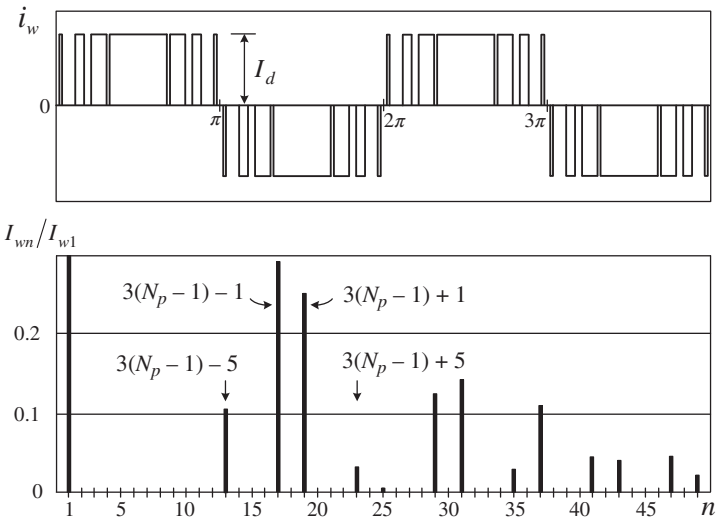

Figure 10.2-8 Waveforms of i_w and its spectrum with 5th, 7th, and 11th harmonic elimination.

Based on the algorithm, the switching angles for the 5th, 7th, and 11th harmonic elimination are $\theta_1 = 2.24°$, $\theta_2 = 5.60°$, and $\theta_3 = 21.26°$. The PWM current i_w and its spectrum are illustrated in Fig. 10.2-8. Similar to the TPWM scheme, the SHE modulation generates two pairs of dominant harmonics at $n = 3(N_p - 1) \pm 1$ and $n = 3(N_p - 1) \pm 5$.

The switching angles for the elimination of up to four harmonics in i_w are given in the appendix of this chapter. It is worth noting that the harmonics with the lowest order normally have the highest priority to be eliminated, but it is not always the case. For example, if an LC resonant mode, caused by the filter capacitor and the load inductance, happens to be the frequency of the 11th harmonics, the 5th and 11th harmonics should be eliminated to avoid any possible resonances instead of the 5th and 7th harmonic elimination. In addition, equation (10.2-8) may not always have a valid solution. For example, no valid solutions can be found to eliminate the 5th, 7th, 11th, 13th, and 17th simultaneously.

Figure 10.2-9 shows a set of voltage and current waveforms measured from a low-horsepower CSI fed induction motor drive operating at various speeds with $C_f =$ 0.66 pu [5]. To keep the motor air-gap flux constant, the magnitude of the stator voltage v_{AB} changes with the inverter fundamental frequency f_1 accordingly while the stator current i_s is kept at its rated value.

Figure 10.2-9a shows the waveforms of the inverter operating at 20 Hz, where three low-order harmonics, the 5th, 7th, and 11th, are eliminated. The resultant switching frequency is only 140 Hz. When the inverter operates at 35 Hz as illustrated in Fig. 10.2-9b with the 5th and 7th harmonics eliminated, its switching frequency is 175 Hz. The waveforms of the inverter working at 60 Hz is shown in Fig. 10.2-9c,

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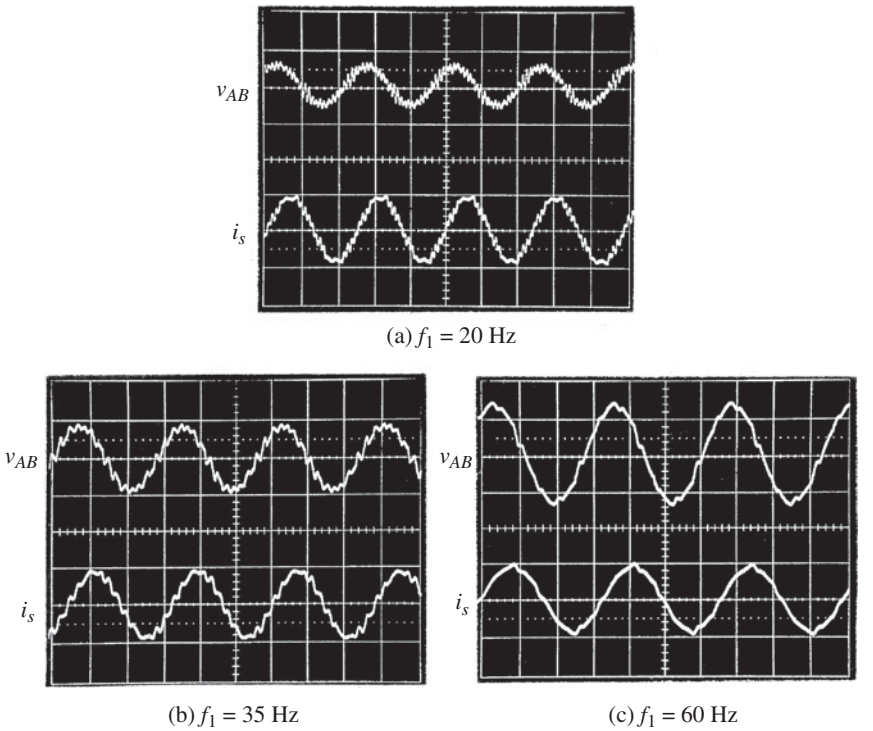

Figure 10.2-9 Waveforms measured from a CSI fed drive using SHE scheme (a) $f_1 = 20$ Hz, $f_{sw} = 140$ Hz; (b) $f_1 = 35$ Hz, $f_{sw} = 175$ Hz; and (c) $f_1 = 60$ Hz, $f_{sw} = 180$ Hz.

where only the 5th harmonic current is eliminated, leading to a switching frequency of 180 Hz.

It can be observed that the CSI produces nearly sinusoidal motor-friendly waveforms even with a very low switching frequency ($\leq$180 Hz). The high *dv/dt* problems associated with the VSIs do not exist in the CSIs.

The SHE and TPWM modulation techniques presented above can be combined for high-power MV drives, where the SHE modulation can be used when the inverter operates at high fundamental frequencies whereas the TPWM scheme can be utilized for the inverter running at low frequencies. This topic will be further discussed in Chapter 13.

10.3 SPACE VECTOR MODULATION

In addition to the TPWM and SHE schemes, the CSI can also be controlled by SVM [7, 8]. In this section, the principle and implementation of the SVM scheme are presented and its performance is compared with the TPWM and SHE modulation techniques.

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Table 10.3-1 Switching States and Space Vectors

Type	Switching State	On-State Switch	Inverter PWM Current			Space Vector
			i_{wA}	i_{wB}	i_{wC}	
Zero States	[14]	S_1, S_4	0	0	0	$\vec{I}_0$
	[36]	S_3, S_6				
	[52]	S_5, S_2				
Active States	[61]	S_6, S_1	I_d	$-I_d$	0	$\vec{I}_1$
	[12]	S_1, S_2	I_d	0	$-I_d$	$\vec{I}_2$
	[23]	S_2, S_3	0	I_d	$-I_d$	$\vec{I}_3$
	[34]	S_3, S_4	$-I_d$	I_d	0	$\vec{I}_4$
	[45]	S_4, S_5	$-I_d$	0	I_d	$\vec{I}_5$
	[56]	S_5, S_6	0	$-I_d$	I_d	$\vec{I}_6$

10.3.1 Switching States

As stated earlier, the PWM switching pattern for the CSI shown in Fig. 10.2-1 must satisfy a constraint, that is, only two switches in the inverter conduct at any time instant, one in the top half of the CSI bridge and the other in the bottom half. Under this constraint, the three-phase inverter has a total of nine switching states as listed in Table 10.3-1. These switching states can be classified as **zero switching states** and **active switching states**.

There are three zero switching states [14], [36], and [52]. The zero state [14] signifies that switches S_1 and S_4 in inverter phase leg A conduct simultaneously and the other four switches in the inverter are off. The dc current source I_d is bypassed, leading to $i_{wA} = i_{wB} = i_{wC} = 0$. This operating mode is often referred to as **bypass operation**.

There exist six active switching states. State [12] indicates that switch S_1 in leg A and S_2 in leg C are on. The dc current I_d flows through S_1, the load, S_2, and then back to the dc source, resulting in $i_{wA} = I_d$ and $i_{wC} = -I_d$. The definition of other five active states is also given in the table.

10.3.2 Space Vectors

The active and zero switching states can be represented by active and zero space vectors, respectively. A typical space vector diagram for the CSI is shown in Fig. 10.3-1, where $\vec{I}_1$ to $\vec{I}_6$ are the **active vectors** and $\vec{I}_0$ is the **zero vector**. The active vectors form a regular hexagon with six equal sectors while the zero vector $\vec{I}_0$ lies at the center of the hexagon.

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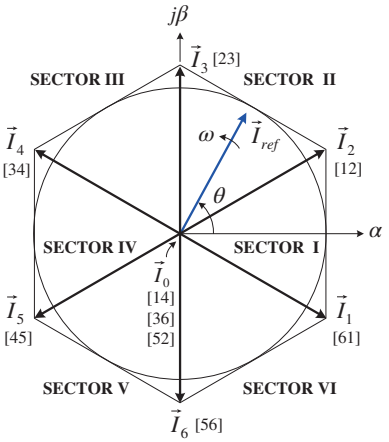

Figure 10.3-1 Space vector diagram for the current source inverter.

To derive the relationship between the space vectors and switching states, we can follow the same procedures given in Chapter 6. Assuming that the operation of the inverter in Fig. 10.2-1 is three-phase balanced, we have

$$i_{wA}(t) + i_{wB}(t) + i_{wC}(t) = 0 \tag{10.3-1}$$

where i_{wA}, i_{wB}, and i_{wC} are the instantaneous PWM output currents in the inverter phases A, B, and C, respectively. The three-phase currents can be transformed into two-phase currents in the α–β plane

$$\begin{bmatrix} i_\alpha(t) \\ i_\beta(t) \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} i_{wA}(t) \\ i_{wB}(t) \\ i_{wC}(t) \end{bmatrix} \tag{10.3-2}$$

A current space vector can be generally expressed in terms of the two-phase currents as

$$\vec{I}(t) = i_\alpha(t) + j i_\beta(t) \tag{10.3-3}$$

Substituting (10.3-2) into (10.3-3), $\vec{I}(t)$ can be express in terms of i_{wA}, i_{wB}, and i_{wC}

$$\vec{I}(t) = \frac{2}{3}\left[i_{wA}(t)e^{j0} + i_{wB}(t)e^{j2\pi/3} + i_{wC}(t)e^{j4\pi/3}\right] \tag{10.3-4}$$

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For the active state [61], S_1 and S_6 are turned on, the inverter PWM currents are

$$i_{wA}(t) = I_d, i_{wB}(t) = -I_d, \quad \text{and} \quad i_{wC}(t) = 0 \tag{10.3-5}$$

Substituting (10.3-5) into (10.3-4) yields

$$\vec{I}_1 = \frac{2}{\sqrt{3}} I_d e^{j(-\pi/6)} \tag{10.3-6}$$

Similarly, the other five active vectors can be derived. The active vectors can be expressed as

$$\vec{I}_k = \frac{2}{\sqrt{3}} I_d \, e^{j\left((k-1)\frac{\pi}{3}-\frac{\pi}{6}\right)} \text{ for } k = 1, 2, \ldots, 6. \tag{10.3-7}$$

Note that the active and zero vectors do not move in space, and thus are referred to as **stationary vectors**. On the contrary, the **current reference vector** $\vec{I}_{ref}$ in Fig. 10.3-1 rotates in space at an angular velocity

$$\omega = 2\pi f_1 \tag{10.3-8}$$

where f_1 is the fundamental frequency of the inverter output current i_w. The angular displacement between $\vec{I}_{ref}$ and the α-axis of the α–β plane can be obtained by

$$\theta(t) = \int_0^t \omega(t)dt + \theta(0) \tag{10.3-9}$$

For a given length and position, $\vec{I}_{ref}$ can be synthesized by three nearby stationary vectors, based on which the switching states of the inverter can be selected and gate signals for the active switches can be generated. When $\vec{I}_{ref}$ passes through sectors one by one, different sets of switches are turned on or off. As a result, when $\vec{I}_{ref}$ rotates one revolution in space, the inverter output current varies one cycle over time. The inverter output frequency corresponds to the rotating speed of $\vec{I}_{ref}$ while its output current can be adjusted by the length of $\vec{I}_{ref}$.

10.3.3 Dwell Time Calculation

As indicated above, the reference $\vec{I}_{ref}$ can be synthesized by three stationary vectors. The dwell time for the stationary vectors essentially represents the duty-cycle time (on-state or off-state time) of the chosen switches during a **sampling period** T_s. The dwell time calculation is based on ampere-second balancing principle, that is, the product of the reference vector $\vec{I}_{ref}$ and sampling period T_s equals the sum of the current vectors multiplied by the time interval of chosen space vectors. Assuming

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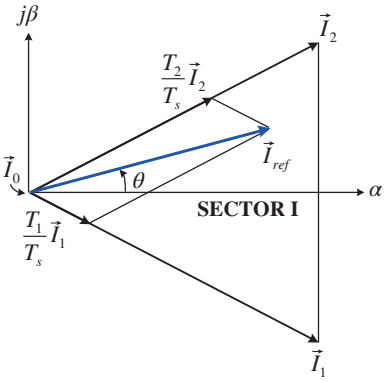

Figure 10.3-2 Synthesis of $\vec{I}_{ref}$ by $\vec{I}_1$, $\vec{I}_2$, and $\vec{I}_0$.

that the sampling period T_s is sufficiently small, the reference vector $\vec{I}_{ref}$ can be considered constant during T_s. Under this assumption, $\vec{I}_{ref}$ can be approximated by two adjacent active vectors and a zero vector. For example, with $\vec{I}_{ref}$ falling into sector I as shown in Fig. 10.3-2, it can be synthesized by $\vec{I}_1$, $\vec{I}_2$, and $\vec{I}_0$. The ampere-second balancing equation is thus given by

$$\begin{cases} \vec{I}_{ref}\, T_s = \vec{I}_1\, T_1 + \vec{I}_2\, T_2 + \vec{I}_0\, T_0 \\ T_s = T_1 + T_2 + T_0 \end{cases} \tag{10.3-10}$$

where T_1, T_2, and T_0 are the dwell times for the vectors $\vec{I}_1$, $\vec{I}_2$, and $\vec{I}_0$, respectively. Substituting

$$\vec{I}_{ref} = I_{ref}\, e^{j\theta},\ \vec{I}_1 = \frac{2}{\sqrt{3}} I_d e^{-j\frac{\pi}{6}},\ \vec{I}_2 = \frac{2}{\sqrt{3}} I_d\, e^{j\frac{\pi}{6}},\ \text{and}\ \vec{I}_0 = 0 \tag{10.3-11}$$

into (10.3-10) and then splitting the resultant equation into the real (α-axis) and imaginary (β-axis) components leads to

$$\begin{cases} \text{Re:} & I_{ref}(\cos\theta)\, T_s = I_d\,(T_1 + T_2) \\ \text{Im:} & I_{ref}(\sin\theta)\, T_s = \dfrac{1}{\sqrt{3}} I_d\,(-T_1 + T_2) \end{cases} \tag{10.3-12}$$

Solving (10.3-12) together with $T_s = T_1 + T_2 + T_0$ gives

$$\begin{cases} T_1 = m_a \sin(\pi/6 - \theta) T_s \\ T_2 = m_a \sin(\pi/6 + \theta) T_s \quad \text{for } -\pi/6 \le \theta < \pi/6 \\ T_0 = T_s - T_1 - T_2 \end{cases} \tag{10.3-13}$$

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where m_a is the modulation index, given by

$$m_a = \frac{I_{ref}}{I_d} = \frac{\hat{I}_{w1}}{I_d} \tag{10.3-14}$$

in which $\hat{I}_{w1}$ is the peak value of the fundamental-frequency component in i_w.

Note that although equation (10.3-13) is derived when $\vec{I}_{ref}$ is in sector I, it can also be used when $\vec{I}_{ref}$ is in other sectors provided that a multiple of $\pi/3$ is subtracted from the actual angular displacement θ such that the modified angle θ' falls into the range of $-\pi/6 \le \theta' < \pi/6$ for use in the equation, that is,

$$\theta' = \theta - (k-1)\,\pi/3 \quad \text{for} \quad -\pi/6 \le \theta' < \pi/6 \tag{10.3-15}$$

where $k = 1, 2, \ldots, 6$ for sectors I, II, …, VI, respectively.

The maximum length of the reference vector, $I_{ref,\max}$, corresponds to the radius of the largest circle that can be inscribed within the hexagon as shown in Fig. 10.3-1. Since the hexagon is formed by the six active vectors having a length of $2\,I_d/\sqrt{3}$, $I_{ref,\max}$ can be found from

$$I_{ref,\max} = \frac{2I_d}{\sqrt{3}} \times \frac{\sqrt{3}}{2} = I_d \tag{10.3-16}$$

Substituting (10.3-16) into (10.3-14) gives the maximum modulation index

$$m_{a,\max} = 1 \tag{10.3-17}$$

from which the modulation index is in the range of

$$0 \le m_a \le 1 \tag{10.3-18}$$

10.3.4 Switching Sequence

Similar to the SVM for the two-level VSI, the switching sequence design for the CSI should satisfy the following two requirements for the minimization of switching frequencies:

(a) The transition from one switching state to the next involves only two switches, one being switched on and the other switched off.

(b) The transition for $\vec{I}_{ref}$ moving from one sector to the next requires minimum number of switchings.

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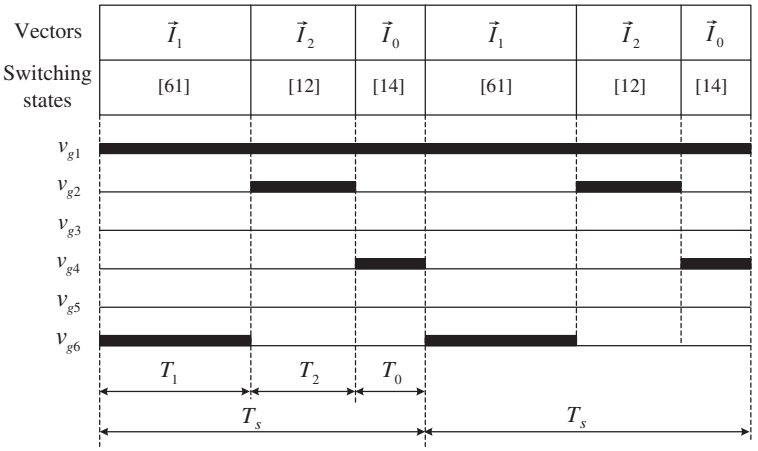

Figure 10.3-3 Switching sequence for $\vec{I}_{ref}$ in sector I.

Figure 10.3-3 shows a typical three-segment sequence for the reference vector $\vec{I}_{ref}$ residing in sector I, where v_{g1} to v_{g6} are the gate signals for switches S_1 to S_6, respectively. The reference vector $\vec{I}_{ref}$ is synthesized by $\vec{I}_1$, $\vec{I}_2$, and $\vec{I}_0$. The sampling period T_s is divided into three segments composed of T_1, T_2, and T_0. The switching states for vectors $\vec{I}_1$ and $\vec{I}_2$ are [61] and [12], and their corresponding on-state switch pairs are (S_6, S_1) and (S_1, S_2). The zero state [14] is selected for $\vec{I}_0$ such that the design requirement (*a*) is satisfied.

Figure 10.3-4 shows the details of the switching sequence and gate signal arrangements over a fundamental-frequency cycle. There are twelve samples per cycle with

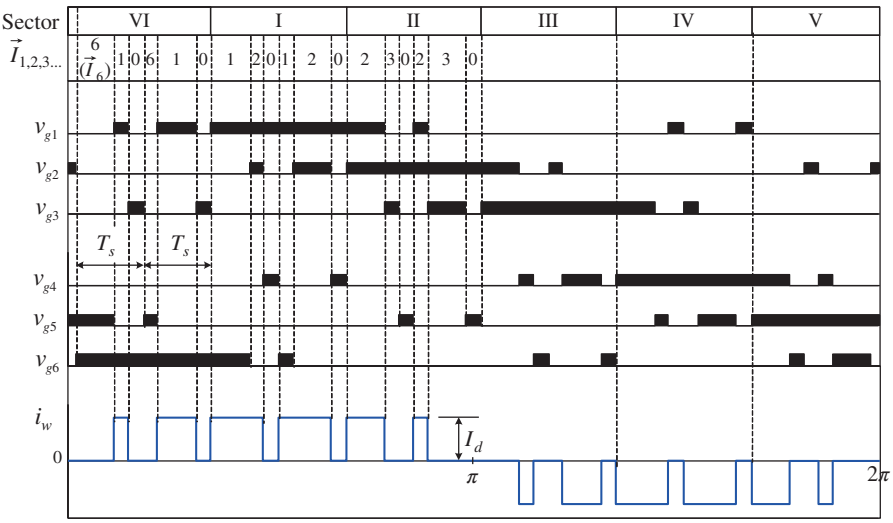

Figure 10.3-4 SVM switching sequence over a fundamental-frequency cycle.

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two samples in each sector. It can be observed that

- At any time instant, only two switches conduct, one in the top half of the bridge and the other in the bottom half.
- By a proper selection of the redundant switching states for $\vec{I}_0$, the requirements for switching sequence design are satisfied. In particular, the transition for $\vec{I}_{ref}$ moving from one sector to the next involves only two switches.
- The dc current I_d is bypassed 12 times per fundamental-frequency cycle by the zero vector. It is the bypass operation that makes the magnitude of the fundamental-frequency current i_{w1} adjustable.
- The inverter PWM current i_w varies one cycle when the reference vector $\vec{I}_{ref}$ passes through all six sectors once.
- The device switching frequency f_{sw} can be calculated by $f_{sw} = f_1 \times N_p$.
- The sampling frequency is $f_{sp} = 1/T_s$, which relates the switching frequency by $f_{sw} = f_{sp}/2$.
- The switching sequence for the SVM scheme is

$$\begin{cases} \vec{I}_k, \vec{I}_{k+1}, \vec{I}_0 & \text{for } k = 1, 2, \ldots, 5 \\ \vec{I}_k, \vec{I}_1, \vec{I}_0 & \text{for } k = 6 \end{cases} \tag{10.3-19}$$

where k represents the sector number.

The simulated waveforms for a 1 MW/4160 V CSI using the SVM is shown in Fig. 10.3-5, where i_w is the inverter PWM current, i_s is the load phase current, and v_{AB} is the inverter line-to-line voltage. The inverter operates at $f_1 = 60\,\text{Hz}$, $f_{sp} = 1080\,\text{Hz}$, and $f_{sw} = 540\,\text{Hz}$ with $m_a = 1$. The filter capacitor C_f is 0.3 pu per phase. The inverter is loaded with a three-phase balanced inductive load having a resistance of 1.0 pu and an inductance of 0.1 pu per phase. The dc current of the inverter is adjusted such that i_{w1} is rated.

The spectrum for i_w, i_s, and v_{AB} are also shown in the figure, where I_{wn} is the rms value of the nth order harmonic current in i_w and $I_{w1,\max}$ is the maximum rms fundamental-frequency current that can be found from (10.3-14) and (10.3-17):

$$I_{w1,\max} = \frac{m_{a,\max} \times I_d}{\sqrt{2}} = 0.707 I_d \quad \text{for} \quad m_{a,\max} = 1 \tag{10.3-20}$$

The PWM current i_w contains no even-order harmonics, and its THD is 45.7%. Similar to the two-level and three-level NPC inverters, the SVM CSI produces low-order harmonics, such as the 5th and 7th. The THD of i_s and v_{AB} is 6.36% and 8.77%, respectively.

10.3.5 Harmonic Content

The harmonic content of the PWM current i_w for the inverter operating at $f_1 = 60\,\text{Hz}$ with $f_{sw} = 540\,\text{Hz}$ and $f_{sw} = 720\,\text{Hz}$ is shown in Figs. 10.3-6a and 10.3-6b,

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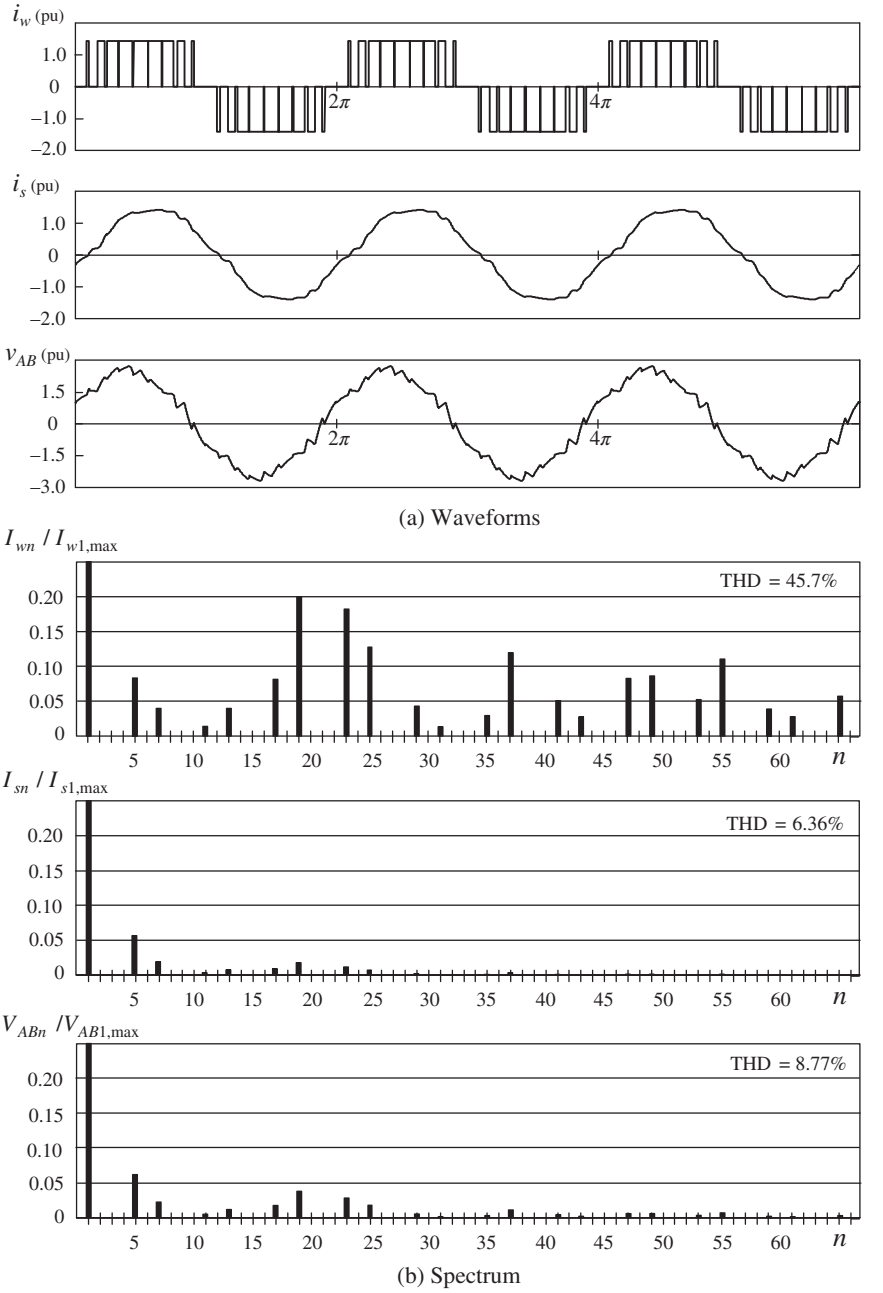

Figure 10.3-5 Waveforms produced by the SVM CSI ($f_1 = 60$ Hz, $f_{sw} = 540$ Hz, $m_a = 1$, and $C_f = 0.3$ pu).

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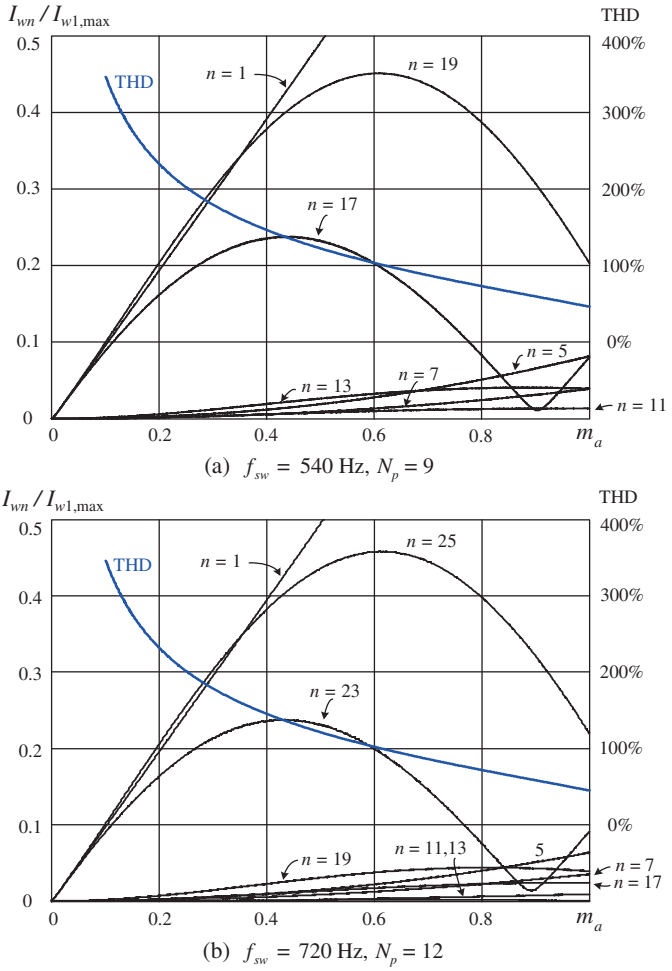

Figure 10.3-6 Harmonic content of i_w produced by the SVM CSI.

respectively. There are a pair of dominant harmonics, whose order can be determined by $n = 2N_p \pm 1$. It is interesting to note that the THD curves for the two cases are almost identical. This is because the magnitudes of the two dominant harmonics in Fig. 10.3-6a are almost identical to those in Fig. 10.3-6b.

10.3.6 SVM Versus TPWM and SHE

Table 10.3-2 provides a brief comparison among the three modulation schemes for the CSI. The main feature of the SVM scheme is fast dynamic response. This is in view of the fact that (a) its modulation index can be adjusted within a sampling period T_s, and (b) the inverter PWM current i_w can be directly controlled by the bypass operation instead of dc current adjustment by the rectifier. Therefore, the SVM scheme is suitable for applications where a fast dynamic response is required.

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Table 10.3-2 Comparison of CSI Modulation Schemes

Item	SVM	TPWM	SHE
DC current utilization $I_{w1,\max}/I_d$	0.707	0.74	0.73–0.78
Dynamic performance	High	Medium	Low
Digital implementation	Real time	Real time or look-up table	Look-up table
Harmonic performance	Adequate	Good	Best
dc current bypass operation	Yes	No	Optional

The SVM scheme has the lowest dc current utilization due to its bypass operation. The SHE scheme features superior harmonic performance. Its dynamic performance can also be improved by allowing dc current bypass operation for the quick adjustment of i_w. The performance of the TPWM modulation is somewhere between the SVM and SHE schemes.

10.4 PARALLEL CURRENT SOURCE INVERTERS

10.4.1 Inverter Topology

To increase the power rating of a CSI fed drive, two or more CSIs can operate in a parallel manner [9, 10]. Figure 10.4-1 shows such a configuration where two inverters are connected in parallel. Each inverter has its own dc choke, but the two inverters share a common filter capacitor C_f at their output.

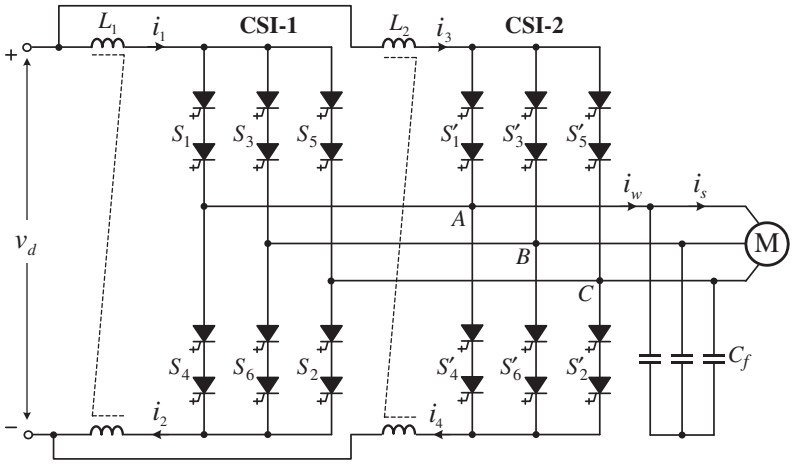

Figure 10.4-1 Parallel current source inverters for high-power MV drives.

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In practice, the parallel operation of the two inverters may cause unbalanced dc currents. The main causes for the unbalanced operation include (a) unequal on-state voltages of the semiconductor devices, which affects dc current balance in steady state; (b) variations in time delay of the gating signals of the two inverters, which affects both transient and steady-state current balance; and (c) manufacturing tolerance in dc choke parameters. In what follows, a SVM algorithm is introduced, which can effectively solve the dc current unbalance problem [9].

10.4.2 Space Vector Modulation for Parallel Inverters

Following the procedure presented in Section 10.3, a space vector diagram composed of 19 current space vectors for the parallel inverters is illustrated in Fig. 10.4-2. These vectors can be divided into four groups according to their length: large, medium, small, and zero vectors. The 19 vectors correspond to 51 switching states given in Table 10.4-1. Each switching state is represented by four digits separated by a semicolon, the first two representing two on-state switches in CSI-1 and the last two denoting two on-state devices in CSI-2, respectively. For instance, switching state [12;16] for medium vector $\vec{I}_7$ indicates that switches S_1, S_2, S'_1, and S'_6 in the two inverters are turned on.

When designing the SVM algorithm for the parallel inverters, the effect of current vectors on the dc currents should be taken into account.

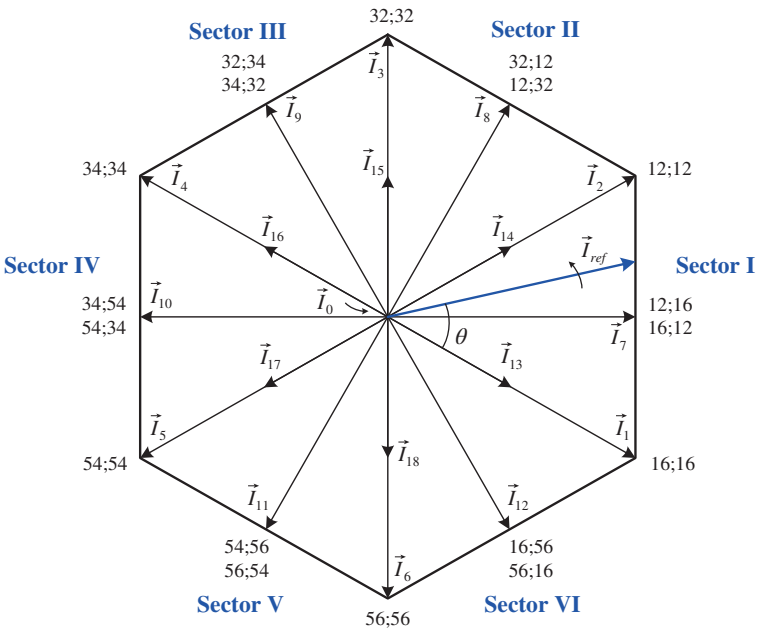

Figure 10.4-2 Space vector diagram for the parallel current source inverters.

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Table 10.4-1 Classification of Space Vectors and Their Switching States

Classification	Current Vector	Switching State
Large vectors	$\vec{I}_1$	16;16
	$\vec{I}_2$	12;12
	$\vec{I}_3$	32;32
	$\vec{I}_4$	34;34
	$\vec{I}_5$	54;54
	$\vec{I}_6$	56;56
Medium vectors	$\vec{I}_7$	12;16, 16;12
	$\vec{I}_8$	32;12, 12;32
	$\vec{I}_9$	32;34, 34;32
	$\vec{I}_{10}$	34;54, 54;34
	$\vec{I}_{11}$	54;56, 56;54
	$\vec{I}_{12}$	16;56, 56;16
Small vectors	$\vec{I}_{13}$	16;14, 14;16, 16;36, 36;16, 56;12, 12;56
	$\vec{I}_{14}$	12;14, 14;12, 12;52, 52;12, 16;32, 32;16
	$\vec{I}_{15}$	32;36, 36;32, 32;52, 52;32, 12;34, 34;12
	$\vec{I}_{16}$	34;36, 36;34, 34;14, 14;34, 32;54, 54;32
	$\vec{I}_{17}$	54;14, 14;54, 54;52, 52;54, 34;56, 56;34
	$\vec{I}_{18}$	56;52, 52;56, 56;36, 36;56, 54;16, 16;54
Zero vectors	$\vec{I}_0$	14;14, 14;36, 14;52, 36;14, 36;52, 36;36, 52;14, 52;36, 52;52, 12;54, 54;12, 32;56, 56;32 34;16, 16;34

- Small and zero current vectors are not allowed to be used since they introduce a bypass operation (shoot through) by turning on the two devices in the same inverter leg simultaneously, resulting in increased switching frequency and energy loss. More importantly, the inverter output current in a practical MV drive is normally adjusted by the dc current instead of bypass operation through the modulation index control.
- Large vectors cannot be used for the dc current balance control. They turn on the devices in the same switch position of the two inverters. For instance, the switching state of $\vec{I}_1$ turns on S_1, S_6, S_1', and S_6' simultaneously, which does not have an effect on the dc currents.
- Only medium vectors can be utilized for the dc current balance control. Making use of the redundant switching states of the medium vectors, the dc currents

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in the two inverters can be controlled independently. The detailed analysis is given in the following section.

10.4.3 Effect of Medium Vectors on DC Currents

Figure 10.4-3 shows dc current paths in the parallel inverters with switching state [16;56] for medium vector $\vec{I}_{12}$. The dc current i_1 flows through S_1, the load (phases A and B), and back to the dc source through S_6 and S'_6. The dc current i_3 flows through S'_5, the load (phases C and B), and back to the dc source also through S_6 and S'_6. Assuming that the two inverters are identical and symmetrical, the dc currents in the negative dc buses are balanced, leading to $i_2 = i_4$.

The currents in the positive dc buses, i_1 and i_3, are affected by the load voltages. Assuming that the load phase voltage v_{AO} happens to be equal to v_{CO}, the two positive dc bus currents in this special case are balanced ($i_1 = i_3$). However, when v_{AO} is greater than v_{CO} ($v_{AO} > v_{CO}$), i_1 will decrease and in the meanwhile i_3 will increase. If $v_{AO} < v_{CO}$, a reverse action will take place for the dc currents.

Let us now consider the other switching state [56;16] of the medium vector $\vec{I}_{12}$. The effect of this switching state on i_1 and i_3 is exactly opposite to that of [16;56]. Table 10.4-2 provides a summary for both cases. It can be observed that for a given load voltage (except for $v_{AO} = v_{CO}$), one switching state can make the dc current increase while the other can make the same dc current decrease.

It should be further noted that the medium vectors in the even sectors (II, IV, and VI) of the space vector diagram can be used to adjust the positive dc bus currents (i_1 and i_3), but they do not have an effect on the negative dc bus currents. On the contrary, the medium vectors in the odd sectors (I, III, and V) can be used to control

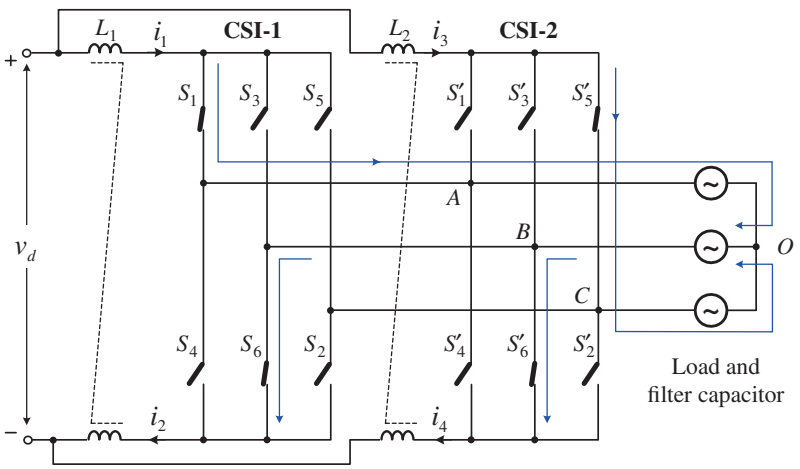

Figure 10.4-3 Current paths in the parallel inverters with switching state [16;56].

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Table 10.4-2 Effect of the Switching States of Medium Vector $\vec{I}_{12}$ on dc Currents

Switching State	Load Voltage	i_1	i_2	i_3	i_4
[16;56] and [56;16]	$v_{AO} = v_{CO}$	×	×	×	×
[16;56]	$v_{AO} > v_{CO}$	↓	×	↑	×
	$v_{AO} < v_{CO}$	↑	×	↓	×
[56;16]	$v_{AO} > v_{CO}$	↑	×	↓	×
	$v_{AO} < v_{CO}$	↓	×	↑	×
Symbol "×": dc currents not affected.					

the negative dc bus currents (i_2 and i_4), but they do not affect the positive dc bus currents. Therefore, the positive and negative dc bus currents can be independently controlled by the two switching states of medium vectors.

10.4.4 DC Current Balance Control

To ensure a balanced operation for the two inverters, all the dc currents should be detected and controlled. The error signals for the detected dc currents are defined by

$$\begin{cases} \Delta i_p = i_1 - i_3 \\ \Delta i_n = i_2 - i_4 \end{cases} \tag{10.4-1}$$

where Δi_p and Δi_n are the current differences in the positive and negative dc buses, which should be zero when the inverters operate under the balanced condition. The error signals are then sent to two PI controllers for the dc current balance control. The output of the PI controllers are used to adjust the dwell time of medium vectors, given by

$$\begin{cases} t_p = K\Delta i_p + \dfrac{1}{\tau}\displaystyle\int \Delta i_p dt \\ t_n = K\Delta i_n + \dfrac{1}{\tau}\displaystyle\int \Delta i_n dt \end{cases} \tag{10.4-2}$$

where t_p and t_n are the dwell times for the medium vectors in the even and odd sectors for the adjustment of positive and negative dc bus currents, and K and τ are the gain and time constant of the PI controllers, respectively.

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Assuming that the reference vector $\vec{I}_{ref}$ is in sector I as shown in Fig. 10.4-2, $\vec{I}_{ref}$ can be synthesized by two large vectors ($\vec{I}_1$ and $\vec{I}_2$) and a medium vector ($\vec{I}_7$), that is,

$$\vec{I}_{ref} T_s = T_1 \vec{I}_1 + T_2 \vec{I}_2 + T_m \vec{I}_7 \tag{10.4-3}$$

where T_1, T_2, and T_m are the dwell times for $\vec{I}_1$, $\vec{I}_2$, and $\vec{I}_7$, and T_s is the sampling period, respectively. To balance the dc currents, the dwell time T_m for the medium vector is adjusted by the output of the PI regulators:

$$T_m = \begin{cases} t_p & \text{for } \vec{I}_{ref} \text{ in sectors II, IV, and VI} \\ t_n & \text{for } \vec{I}_{ref} \text{ in sectors I, III, and V} \end{cases} \tag{10.4-4}$$

The dwell times for the large vectors can be calculated by

$$\begin{cases} T_1 = \dfrac{\sqrt{3} - \tan\theta}{\sqrt{3} + \tan\theta} T_s - \dfrac{1}{2} T_m \\ T_2 = T_s - T_1 - T_m \end{cases} \tag{10.4-5}$$

where θ is the phase displacement between $\vec{I}_1$ and $\vec{I}_{ref}$ as shown in Fig. 10.4-2.

It should be pointed out that the three-phase load voltage should also be detected for the proper selection of the switching state of medium vectors. For the CSI drives, the combined power factor of the motor and the filter capacitor may vary from inductive to capacitive when the motor operates under different conditions. But this will not affect the dc current balance control since the switching states of the medium vectors are selected according to the sign of the measured load voltages, independent of the load power factor.

10.4.5 Experimental Verification

The SVM-based dc current balance control algorithm is implemented on a laboratory 5 hp (4-pole) induction motor drive using parallel CSIs. The drive system operates under a light load condition with a maximum switching frequency of 420 Hz. To make the two inverters unbalanced on purpose, two power diodes are added to the dc circuit of CSI-2, one in the positive dc bus and the other in the negative bus. It is the diode voltage drop that makes the dc currents of the two inverters unbalanced.

Figure 10.4-4 shows the measured dc current waveforms during motor speed acceleration from 90 to 1500 rpm. Without the dc current balance control, the current i_1 in CSI-1 is always higher than i_3 in CSI-2 as shown in Fig. 10.4-4a. When the motor operates at 1500 rpm with an increased dc current, the voltage drop on the diodes increases, resulting in a higher dc current difference. With the current balance

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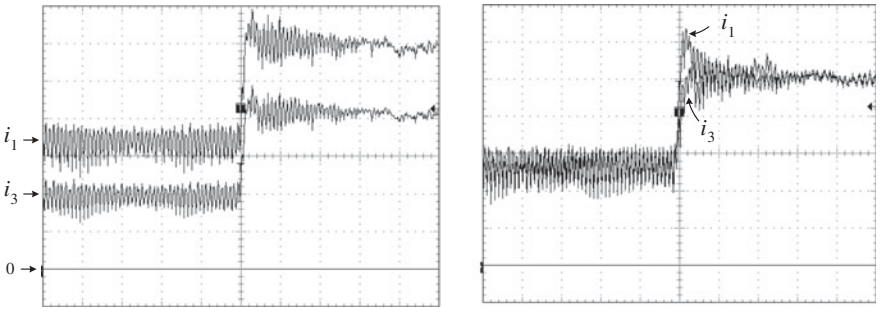
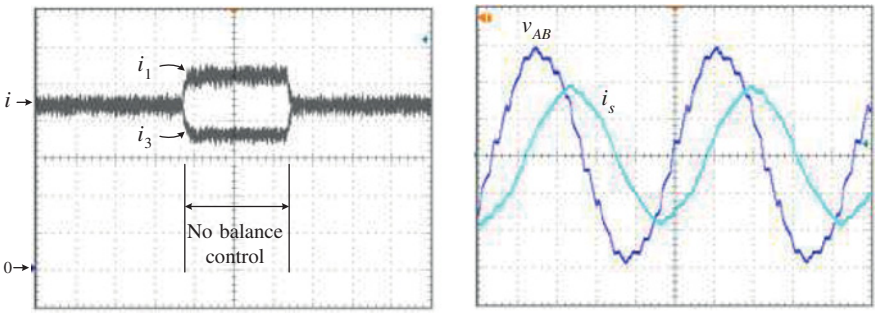


Timebase: 1.0 sec/div
(a) Without dc current balance control

Timebase: 1.0 sec/div
(b) With dc current balance control

Figure 10.4-4 Measured dc current waveforms during motor speed acceleration from 90 to 1500 rpm.



Timebase: 1.0 sec/div
(a) dc current transient response

Timebase: 4 ms/div
(b) Steady state ac waveforms

Figure 10.4-5 Measured inverter dc- and ac-side waveforms.

control activated, both currents are kept almost the same as illustrated in Fig. 10.4-4b during transient and steady-state operations.

Figure 10.4-5 shows the measured waveforms when the drive is running at 1500 rpm. The dc currents i_1 and i_3 in Fig. 10.4-5a are kept balanced by the drive controller except for the middle portion of the current waveforms where the current balance control is temporarily disabled. The steady-state motor voltage and current waveforms are shown in Fig. 10.4-5b, which are close to sinusoidal.

10.5 LOAD-COMMUTATED INVERTER (LCI)

One of the well-known CSI topologies is the LCI [11]. Figure 10.5-1 shows the typical LCI configuration for synchronous motor (SM) drives. On the dc side of the inverter, a dc choke L_d is required to provide a smooth dc current I_d. The inverter employs the SCR thyristor as a switching device instead of the symmetrical GCT for the PWM CSI. The SCRs do not have self turn-off capability, but they can be

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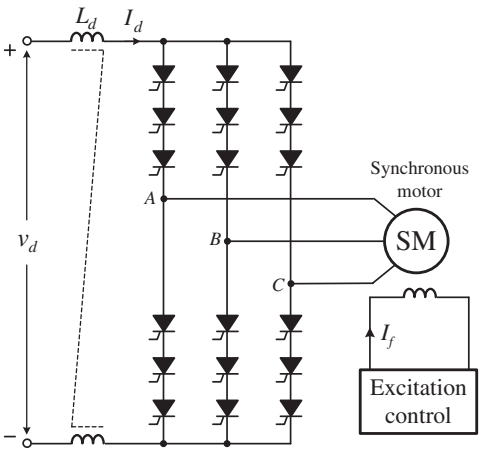

Figure 10.5-1 Load-commutated inverter for synchronous motor drive.

naturally commutated by the load voltage with a leading power factor. The ideal load for the LCI is, therefore, synchronous motors operating at a leading power factor which can be easily achieved by adjusting the excitation current I_f.

The natural commutation of the SCRs is essentially accomplished by a leading EMF induced by the motor operating at certain speeds. At low motor speeds (typically lower than 10% the rated speed), the induced EMF may be too low to commutate the SCRs. In this case, the commutation is usually assisted by the front end SCR rectifier.

The LCI fed motor drive features low manufacturing cost and high efficiency mainly due to the use of low-cost SCR devices and lack of PWM operation. The LCI is a popular solution for very large drives, where the initial investment and operating efficiency are of great importance. A typical example is a 100 MW wind tunnel synchronous motor drive [4], where the efficiency of the power converters including the rectifier and inverter reaches 99%.

The main drawback of the LCI drive is its limited dynamic performance. However, the majority of the LCI drives are for fans, pumps, compressors, and conveyors, where the dynamic response is usually not a critical requirement. In addition, the power losses in the motor are high due to the large amount of harmonics in the stator current [12].

10.6 SUMMARY

This chapter focuses on the PWM CSI technologies for high-power medium voltage drives. The operating principle for the CSI inverters is discussed. Three modulation techniques for the CSI are analyzed, which include TPWM, SHE, and SVM. These modulation techniques are developed for the high-power GCT inverters where the switching frequency of the inverter is normally below 500 Hz. This chapter also

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presents a relatively new CSI topology using parallel inverters. An SVM-based dc current balance control algorithm is developed for the parallel inverters.

The PWM CSI features simple converter topology, motor-friendly waveforms, and reliable short-circuit protection, and therefore it is one of the preferred converter topologies for the MV drive. Prior to the advent of GCTs, the GTO thyristors were dominant for the CSI drives. Although there are still a large number of installed GTO CSI drives in the field, this technology has been replaced by the GCTs-based current source drives since the late 1990s.

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APPENDIX

SHE Switching Angles for Inverter Circuit of Fig. 10.2-1

Harmonics Eliminated	Switching Angles			Harmonics Eliminated	Switching Angles			
	θ_1	θ_2	θ_3		θ_1	θ_2	θ_3	θ_4
5	18.00	—	—	7, 11, 17	11.70	14.12	24.17	—
7	21.43	—	—	7, 13, 17	12.69	14.97	24.16	—
11	24.55	—	—	7, 13, 19	13.49	15.94	24.53	—
13	25.38	—	—	11, 13, 17	14.55	15.97	25.06	—
5, 7	7.93	13.75	—	11, 13, 19	15.24	16.71	25.32	—
5, 11	12.96	19.14	—	13, 17, 19	17.08	18.23	25.84	—
5, 13	14.48	21.12	—	13, 17, 23	18.03	19.22	26.16	—
7, 11	15.23	19.37	—	5, 7, 11, 13	0.00	1.60	15.14	20.26
7, 13	16.58	20.79	—	5, 7, 11, 17	0.07	2.63	16.57	21.80
7, 17	18.49	23.08	—	5, 7, 11, 19	1.11	4.01	18.26	23.60
11, 13	19.00	21.74	—	5, 7, 13, 17	1.50	4.14	16.40	21.12
11, 17	20.51	23.14	—	5, 7, 13, 19	2.56	5.57	17.82	22.33
11, 19	21.10	23.75	—	5, 7, 17, 19	4.59	7.96	17.17	20.55
13, 17	21.19	23.45	—	5, 11, 13, 17	4.16	6.07	16.79	22.04
13, 19	21.71	23.94	—	5, 11, 13, 19	5.13	7.26	17.57	22.72
5, 7, 11	2.24	5.60	21.26	5, 11, 17, 19	6.93	9.15	17.85	22.77
5, 7, 13	4.21	8.04	22.45	5, 13, 17, 19	7.80	9.82	18.01	23.25
5, 7, 17	6.91	11.96	25.57	7, 11, 13, 17	5.42	6.65	18.03	22.17
5, 11, 13	7.81	11.03	22.13	7, 11, 13, 19	6.35	7.69	18.67	22.74
5, 11, 17	10.16	14.02	23.34	7, 11, 17, 19	8.07	9.44	19.09	22.93
5, 13, 17	11.24	14.92	22.98	7, 13, 17, 19	8.88	10.12	19.35	23.22
7, 11, 13	9.51	11.64	23.27	11, 13, 17, 19	10.39	11.14	20.56	23.60

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Chapter 11

PWM Current Source Rectifiers

11.1 INTRODUCTION

With the advent of gate commutated thyristor (GCT) devices in the late 1990s, the PWM current source rectifier (CSR) using symmetrical GCT switches has become a preferred choice for current source fed medium voltage (MV) drives. Compared with the multipulse SCR rectifiers presented in Chapter 4, the PWM rectifier features improved input power factor, reduced line current distortion, and superior dynamic response.

The PWM CSR normally requires a three-phase filter capacitor at its input. The capacitor provides two basic functions: (a) to assist the commutation of switching devices, and (b) to filter out line current harmonics. However, the use of the filter capacitor may cause LC resonances and affect the input power factor of the rectifier as well.

This chapter addresses four main aspects of the CSRs, including converter topologies, PWM schemes, power factor control, and active damping control for LC resonance suppression. The important concepts are elaborated with simulations and experiments.

11.2 SINGLE-BRIDGE CURRENT SOURCE RECTIFIER

11.2.1 Introduction

Figure 11.2-1 shows the circuit diagram of a single-bridge GCT CSR [1–3]. When the rectifier is used in high-power MV drives as a front end, two or more GCTs

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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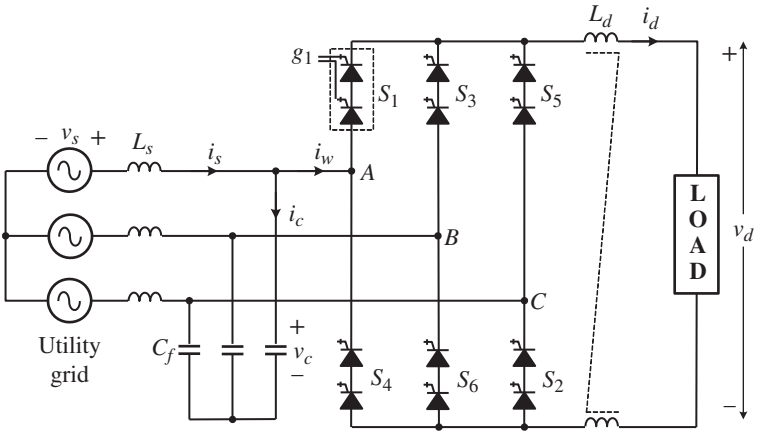

Figure 11.2-1 Single-bridge GCT current source rectifier.

can be connected in series. The **line inductance** L_s on the ac side of the rectifier represents the total inductance between the utility supply and the rectifier, including the equivalent inductance of the supply, leakage inductances of isolation transformer if any, and inductance of the line reactor for line current THD reduction. The line inductance L_s is normally in the range of 0.1–0.15 per unit (pu). The definition of per unit system is given in Chapter 3.

The PWM rectifier requires a **filter capacitor** C_f to assist the commutation of the GCT devices and filter out harmonic currents. The capacitor size is dependent on a number of factors such as the rectifier switching frequency, LC resonant mode, required line current THD, and input power factor. It is normally in the range of 0.3–0.6 pu for high-power PWM rectifiers with a switching frequency of a few hundred hertz.

On the dc side of the rectifier, a **dc choke** L_d is required to smooth the dc current. The choke usually has a magnetic core with two coils, one connected to the positive dc bus and the other to the negative bus. Such an arrangement is preferred in practice for motor common-mode voltage reduction [4]. To limit the dc link current ripple to an acceptable level (<15%), the size of the dc choke is normally in the range of 0.5–0.8 pu.

11.2.2 Selective Harmonic Elimination

As discussed in Chapter 10, the **selective harmonic elimination**(SHE) is considered as an optimum modulation scheme, which provides a superior harmonic profile with a minimum switching frequency. Unlike the SHE modulation schemes for current source inverters where the modulation index is usually fixed to its maximum value, the SHE scheme for the CSR should provide an adjustable modulation index for dc current control in addition to harmonic elimination [5, 6].

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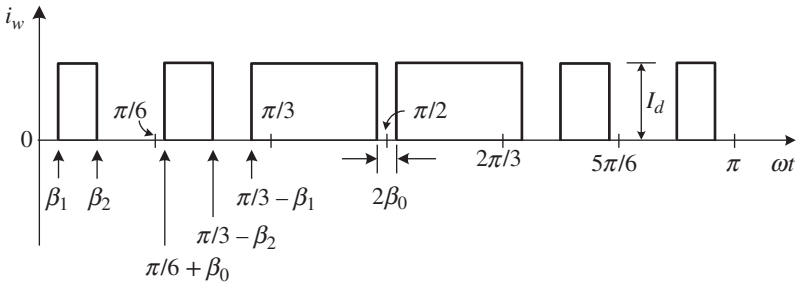

Figure 11.2-2 PWM current waveform (half cycle) with three independent switching angles (β_1, β_2, and β_0).

Figure 11.2-2 shows a typical half-cycle waveform of the rectifier PWM current i_w. The current waveform is designed such that the **switching constraints** for the current source converters discussed in the previous chapter are satisfied. There are six current pulses per half cycle of the fundamental frequency with only three independent **switching angles**, β_1, β_2, and β_0. These angles can be used to eliminate two harmonics and in the meanwhile provide an adjustable modulation index.

The PWM current waveform in Fig. 11.2-2 can be expressed in Fourier series as

$$i_w(\omega t) = \sum_{n=1}^{\infty} a_n \sin(n\omega t) \tag{11.2-1}$$

where

$$\begin{aligned} a_n &= \frac{4}{\pi}\int_0^{\frac{\pi}{2}} i_w(\omega t)\sin(n\omega t)d(\omega t) \\ &= \frac{4I_d}{n\pi}\left\{\cos(n\beta_1) - \cos(n\beta_2) + \cos\left(n\left(\frac{\pi}{6}+\beta_0\right)\right) - \cos\left(n\left(\frac{\pi}{3}-\beta_2\right)\right)\right. \\ &\quad \left. + \cos\left(n\left(\frac{\pi}{3}-\beta_1\right)\right) - \cos\left(n\left(\frac{\pi}{2}-\beta_0\right)\right)\right\} \end{aligned} \tag{11.2-2}$$

To eliminate two dominant harmonics such as the 5th and 7th, we have

$$\begin{aligned} F_1 &= \cos(5\beta_1) - \cos(5\beta_2) + \cos\left(5\left(\frac{\pi}{6}+\beta_0\right)\right) - \cos\left(5\left(\frac{\pi}{3}-\beta_2\right)\right) \\ &\quad + \cos\left(5\left(\frac{\pi}{3}-\beta_1\right)\right) - \cos\left(5\left(\frac{\pi}{2}-\beta_0\right)\right) = 0 \end{aligned} \tag{11.2-3}$$

$$\begin{aligned} F_2 &= \cos(7\beta_1) - \cos(7\beta_2) + \cos\left(7\left(\frac{\pi}{6}+\beta_0\right)\right) - \cos\left(7\left(\frac{\pi}{3}-\beta_2\right)\right) \\ &\quad + \cos\left(7\left(\frac{\pi}{3}-\beta_1\right)\right) - \cos\left(7\left(\frac{\pi}{2}-\beta_0\right)\right) = 0 \end{aligned} \tag{11.2-4}$$

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Table 11.2-1 Switching Angles for the Single-Bridge CSR with 5th and 7th Harmonic Elimination

m_a	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.03
β_1	−13.5°	−11.9°	−10.3°	−8.60°	−6.86°	−5.00°	−3.98°	−0.67°	2.17°	6.24°	7.93°
β_2	14.2°	13.5°	12.7°	12.0°	11.4°	10.8°	10.4°	10.3°	10.8	12.6°	13.8°
β_0	13.6°	12.2°	10.9°	9.5°	8.0°	6.6°	5.1°	3.6°	2.1°	0.5°	0

A third equation can be derived for modulation index adjustment

$$F_3 = \frac{a_1}{I_d} - m_a = \frac{4}{\pi}\left\{\cos\beta_1 - \cos\beta_2 + \cos\left(\frac{\pi}{6} + \beta_0\right) - \cos\left(\frac{\pi}{3} - \beta_2\right) + \cos\left(\frac{\pi}{3} - \beta_1\right) - \cos\left(\frac{\pi}{2} - \beta_0\right)\right\} - m_a = 0 \tag{11.2-5}$$

where m_a is **amplitude modulation index**, given by

$$m_a = \frac{\hat{I}_{w1}}{I_d} \tag{11.2-6}$$

$\hat{I}_{w1}$ in (11.2-6) is the peak fundamental-frequency component of the PWM current and I_d is the average dc current. Using Newton–Raphson iteration algorithm introduced in Chapter 10 or other numerical methods to solve (11.2-3) to (11.2-5) simultaneously, we can obtain all the switching angles at various modulation indices. The calculated results are given in Table 11.2-1. The maximum modulation index is 1.03, at which β_0 becomes zero and the notch in the center of the PWM waveform disappears. The switching angles β_1 and β_2 at $m_a = 1.03$ are identical to those for the current source inverter with 5th and 7th harmonic elimination.

For a given value of the switching angels β_1, β_2, and β_0, the gate signals for the switching devices in the CSR can be arranged. An example is shown in Fig. 11.2-3, where v_{g1} and v_{g4} are the gate signals for S_1 and S_4 in rectifier leg A, respectively.

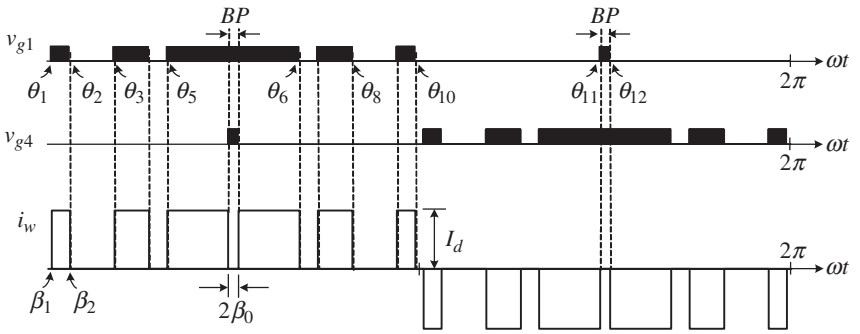

Figure 11.2-3 Gate signal arrangement for the single-bridge current source rectifier ($m_a = 0.9$).

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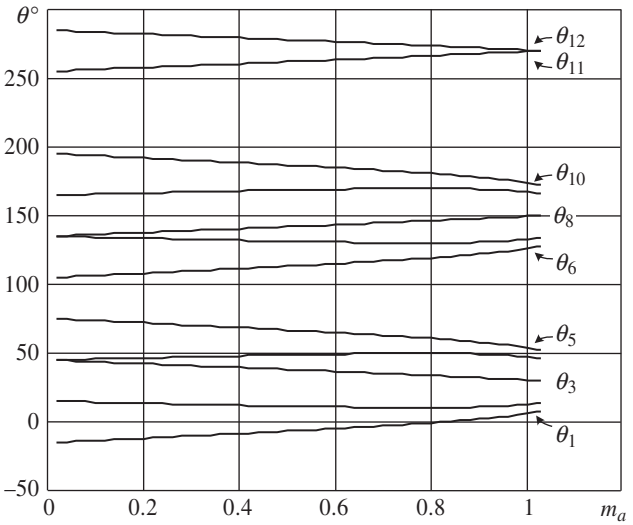

Figure 11.2-4 Gating angles versus modulation index for the single-bridge CSR with 5th and 7th harmonic elimination.

The gate signal v_{g1} is composed of six pulses, of which one is the **bypass pulse**, defined by θ_{11} and θ_{12}. The notch $2\beta_0$ in the center of the i_w waveform is realized by turning on S_1 and S_4 simultaneously. The dc current i_d is then bypassed (shorted) by the rectifier, leading to $i_w = 0$. During the bypass interval, i_d is kept constant by the large dc choke.

Based on the gate signals in Fig. 11.2-3, the **switching frequency** of the GCT device can be calculate by $f_{sw} = f_s \times N_p = 60\,\text{Hz} \times 6 = 360\,\text{Hz}$, where f_s is the supply frequency and N_p is the number of pulses per cycle of the supply frequency. This is the lowest possible switching frequency that can be used to eliminate two harmonics and in the meantime to provide an adjustable modulation index. All the **gating angles**, θ_1 *to* θ_{12}, can be calculated from Table 11.2-1 and are plotted in Fig. 11.2-4. Their numerical values are given in the Appendix of this chapter.

It is interesting to note that when the modulation index m_a is lower than 0.826, gating angle θ_1 becomes negative and θ_{10} is larger than 180°. Figure 11.2-5 shows such a case with $m_a = 0.4$, $\theta_1 = -8.6°$, and $\theta_{10} = 188.6°$. There are four bypass intervals (*BP*1–*BP*4) per cycle. The bypass intervals, *BP*2 and *BP*4, are created by bypass pulses while the other two, *BP*1 and *BP*3, are due to the overlapping of the gate signals.

The harmonic content of the rectifier PWM current i_w is illustrated in Fig. 11.2-6, where I_{w1} and I_{wn} are the rms values of the fundamental frequency and nth order harmonic currents, respectively. The fundamental current I_{w1} increases linearly with m_a. With the 5th and 7th harmonics eliminated, the magnitudes of remaining harmonics are high, especially for the 11th harmonic. However, these

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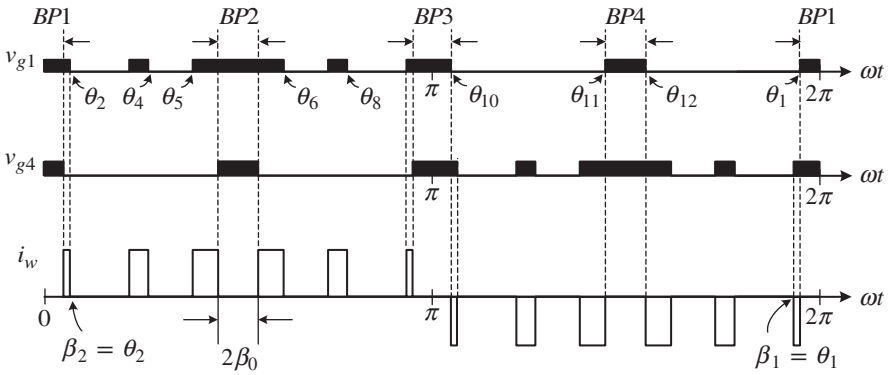

Figure 11.2-5 Four bypass intervals ($m_a < 0.826$).

harmonic currents can be substantially attenuated by the filter capacitor C_f and line inductance L_s.

Figure 11.2-7 shows simulated current and voltage waveforms of the single-bridge rectifier with 5th and 7th harmonic elimination. The rectifier has a line inductance of 0.1 pu and filter capacitor of 0.6 pu. It operates at the rated dc current with a modulation index of 0.9. As shown in the figure, the THD of the PWM i_w is quite high (75%) whereas the THD of the line (source) current i_s is only 6.5% due to the use of SHE modulation and the LC filter.

Figure 11.2-8 shows the current waveforms measured from a laboratory CSR system operating under the condition of $f_s = 60\,\text{Hz}$, $m_a = 0.7$, $N_p = 6$, $L_s = 0.1\,\text{pu}$,

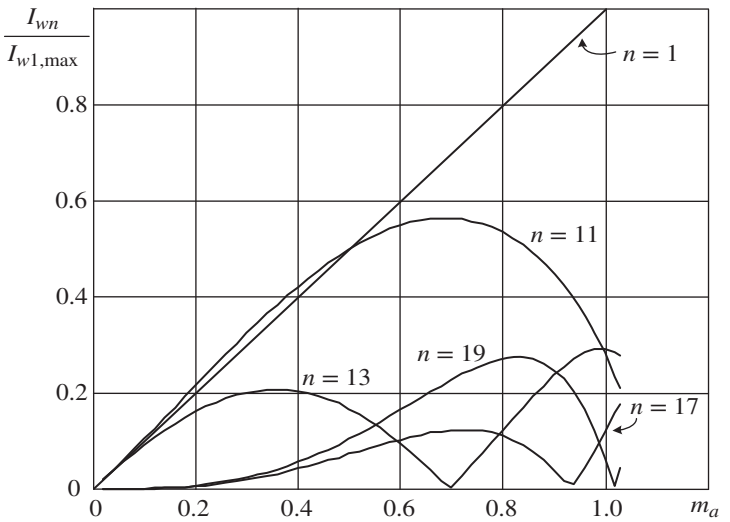

Figure 11.2-6 Harmonic content of I_w in the single-bridge CSR with 5th and 7th harmonic elimination.

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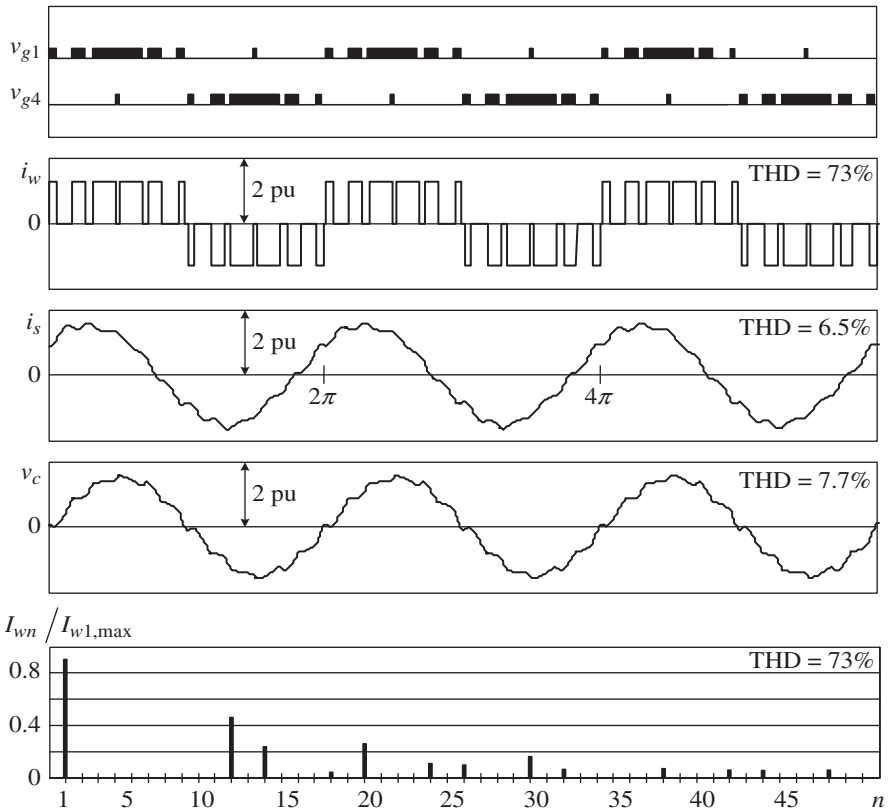

Figure 11.2-7 Simulated waveforms in the single-bridge rectifier ($L_s = 0.1$ pu, $C_f = 0.6$ pu, and $m_a = 0.9$).

and $C_f = 0.66$ pu. The traces from the top to bottom are the switch S_1 current i_{GCT}, the rectifier PWM current i_w, and the line current i_s. With the modulation index lower than 0.826, the dc current i_d should be bypassed by the rectifier four times per cycle of the supply frequency. The bypasses are reflected in the differences between i_{GCT} and positive portion of i_w. If there were no bypass operation, these two waveforms would be identical. The line current i_s is close to sinusoidal due to the SHE modulation and the filter capacitor.

11.2.3 Rectifier DC Output Voltage

The dc output voltage v_d of the rectifier can be adjusted by two methods, modulation index (m_a) control and delay angle (α) control, where α is the phase displacement between the utility supply voltage v_s and the rectifier fundamental current i_{w1}. The operating principle of delay angle control is the same as that in phase-controlled SCR rectifiers.

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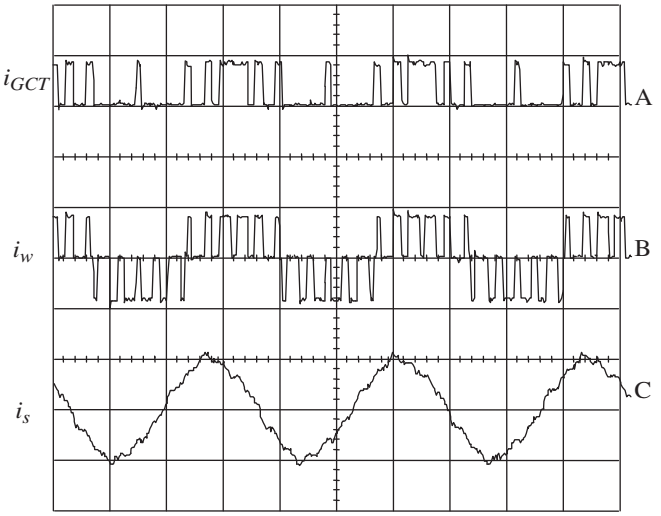

All traces: 0.75 pu/div, 5 ms/div.

Figure 11.2-8 Current waveforms measured from a laboratory single-bridge CSR (f_s = 60 Hz, $m_a = 0.7$, $L_s = 0.1$ pu, and $C_f = 0.66$ pu).

The ac input power can be expressed as

$$P_{ac} = \sqrt{3}\, V_{LL}\, I_{w1}\, \cos\alpha \tag{11.2-7}$$

where V_{LL} is the rms line-to-line voltage of the utility supply. The dc output power is given by

$$P_d = V_d\, I_d \tag{11.2-8}$$

where V_d and I_d are the average dc voltage and current, respectively. Neglecting the power losses in the rectifier, the ac input power should be equal to the dc output power,

$$\sqrt{3}\, V_{LL} I_{w1} \cos\alpha = V_d I_d \tag{11.2-9}$$

from which

$$V_d = \sqrt{3/2}\, V_{LL} m_a \cos\alpha \tag{11.2-10}$$

where $m_a = \hat{I}_{w1}/I_d = \sqrt{2} I_{w1}/I_d$

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Considering the maximum modulation index of $m_{a,\max} = 1.03$ for the single-bridge CSR, the maximum dc output voltage is

$$V_{d,\max} = \sqrt{3/2} V_{LL} m_{a,\max} \cos 0° = 1.26 V_{LL} \quad \text{for } \alpha = 0 \qquad (11.2\text{-}11)$$

Compared with a phase-controlled SCR rectifier where the maximum dc output voltage is 1.35 V_{LL}, the single-bridge PWM rectifier has a dc voltage derating of 7%.

11.2.4 Space Vector Modulation

The SHE modulation is an off-line PWM scheme. All the switching angles are pre-calculated and then stored in a look-up table for digital implementation. Although SHE scheme provides a superior harmonic performance with a minimum switching frequency, it may not be suitable for the CSRs that require an instantaneous adjustment of modulation index.

The space vector modulation (SVM) is an on-line scheme suitable for real-time digital implementation. Compared with the SHE modulation, the SVM provides faster dynamic response and greater flexibility since the modulation index can be adjusted within each sampling period. As an example, the SVM modulation can be used in the active damping control, where the modulation index should be adjusted dynamically to suppress oscillations caused by LC resonances. The SVM scheme discussed in Chapter 10 for the CSI can be directly applied to the CSR, and therefore is not discussed in this chapter.

11.3 DUAL-BRIDGE CURRENT SOURCE RECTIFIER

11.3.1 Introduction

Figure 11.3-1 shows the converter configuration of a dual-bridge CSR [7, 8]. This topology is composed of two identical single-bridge CSRs powered by a **phase-shifting transformer** with two secondary windings, one connected in delta (Δ) and the other in wye (Y). The line-to-line voltage of each secondary winding is normally half of that of the primary winding. The total line inductance L_s is referred to the secondary side of the transformer for the convenience of discussion. The filter capacitor C_f is normally in the range of 0.15–0.3pu, which is smaller than that for the single-bridge CSR.

The dual-bridge rectifier has the following features:

- **Sinusoidal line current**. The transformer is used to cancel the 5th, 7th, 17th, and 19th harmonic currents while the PWM technique is employed to eliminate the 11th and 13th harmonics. As a result, the primary-side line current i_A does not contain any harmonics lower than the 23rd. The other high-order harmonics can be attenuated by the filter capacitor C_f.

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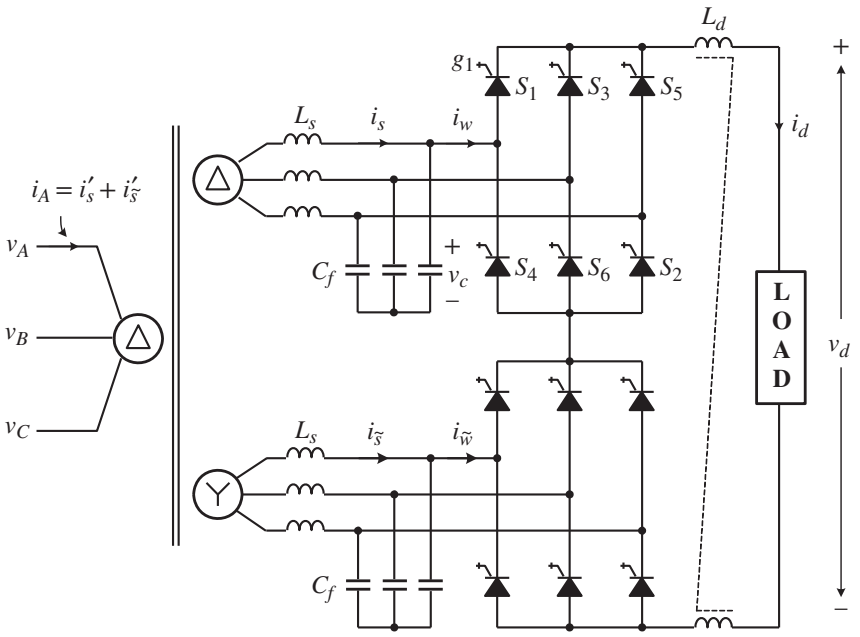

Figure 11.3-1 Dual-bridge GCT current source rectifier.

- **Low switching frequency**. The rectifier can draw sinusoidal input current with a low switching frequency, typically 360 Hz or 420 Hz for the elimination of the 11th and 13th harmonics.
- **Reliable operation**. No GCT devices are connected in series in the dual-bridge CSR, which enhances the reliability of the system.
- **Suitable for retrofit applications**. The dual-bridge rectifier requires a phase-shifting transformer for harmonic cancellation. The transformer can also block common-mode voltages that would otherwise appear on the motor windings, causing premature failure of the winding insulation system. The MV drive with the dual-bridge CSR as a front end is suited for retrofit applications, where a standard ac motor is usually employed.

11.3.2 PWM Schemes

To design the SHE switching pattern for the dual-bridge rectifier, the following requirements should be satisfied:

- To eliminate 11th and 13th harmonics in the PWM current of each bridge;
- To provide a full range of modulation index control; and
- To minimize switching frequency.

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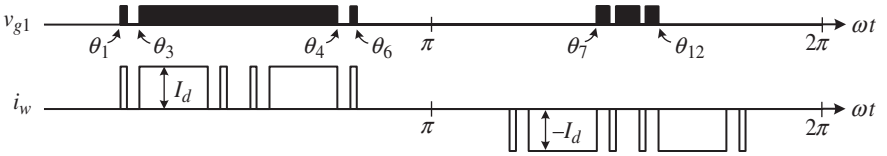
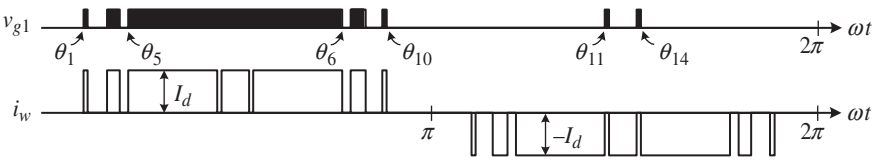


(a) Switching pattern A for a low m_a



(b) Switching pattern B for a high m_a

Figure 11.3-2 Two switching patterns for the dual-bridge CSR with 11th and 13th harmonic elimination.

In addition, the PWM pattern design must satisfy the **switching constraints** for current source converters: only two switching devices should conduct at any time, one connected to the positive dc bus and the other to the negative dc bus.

The switching pattern developed for the single-bridge CSR with one bypass pulse per cycle (Fig. 11.2-3) does not work for the 11th and 13th harmonic elimination. Figure 11.3-2 shows two new switching patterns for the dual-bridge CSR. Pattern A is suited for the rectifier operating at low modulation indices. The gate signal v_{g1} for switch S_1 is composed of six pulses per cycle of the supply frequency, of which three are the bypass pulses specified by θ_7 to θ_{12}. The resultant device switching frequency is 360 Hz, six times the supply frequency of 60 Hz. Pattern B is developed for the rectifier operating at high modulation indices. It can be observed that S_1 switches seven times per cycle, resulting in a switching frequency of 420 Hz.

Figure 11.3-3 illustrates the calculated gating angles versus modulation index m_a for both patterns. Pattern A is valid for $0.02 \leq m_a \leq 0.857$. It is noted that the difference between the gating angles θ_9 and θ_{10} reduces with the increase of m_a. The two angles finally merge at $m_a = 0.857$, beyond which the 11th and 13th harmonics are no longer eliminated. Similarly, pattern B is valid only for $0.84 \leq m_a \leq 1.085$. These two patterns can then be combined to provide a continuous adjustment of m_a over the full operating range of the rectifier.

11.3.3 Harmonic Contents

The harmonic content in the rectifier PWM current i_w is shown in Fig. 11.3-4, where I_{w1} and I_{wn} are the rms values of the fundamental frequency and nth order harmonic currents, respectively. The fundamental current I_{w1} increases linearly with m_a. The

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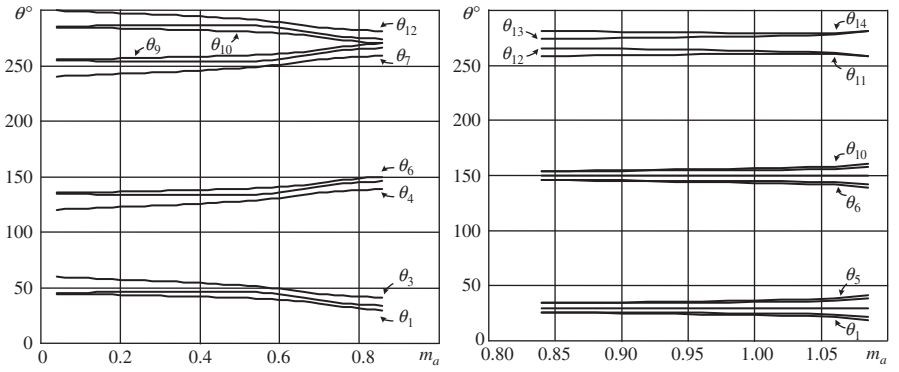

(a) Pattern A: $m_a = 0 - 0.857$ (b) Pattern B: $m_a = 0.84 - 1.085$

Figure 11.3-3 Gating angles θ versus modulation index m_a for the dual-bridge CSR with 11th and 13th harmonic elimination.

current i_w contains the 5th, 7th, 17th, and 19th harmonics, but these harmonics can be eliminated by the phase-shifting transformer (refer to Chapter 5 for details). It is interesting to note that the harmonic curves are smoothly joined for the two patterns.

Figure 11.3-5 shows the current waveforms measured from a laboratory dual bridge CSR system operating at $m_a = 0.5$ of pattern A and $m_a = 1.02$ of pattern B, respectively. The total line inductance L_s is 0.067 pu and the filter capacitance is 0.15 pu.

The traces from top to bottom are the transformer primary line current i_A, the secondary line current i_s, and the rectifier PWM current i_w. The waveform of the primary current i_A is virtually sinusoidal, which is the main feature of the dual-bridge rectifier.

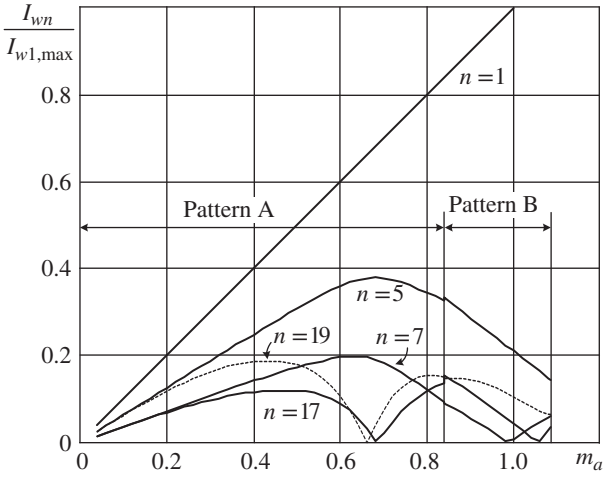

Figure 11.3-4 Harmonic content of i_w produced by combined PWM patterns A and B.

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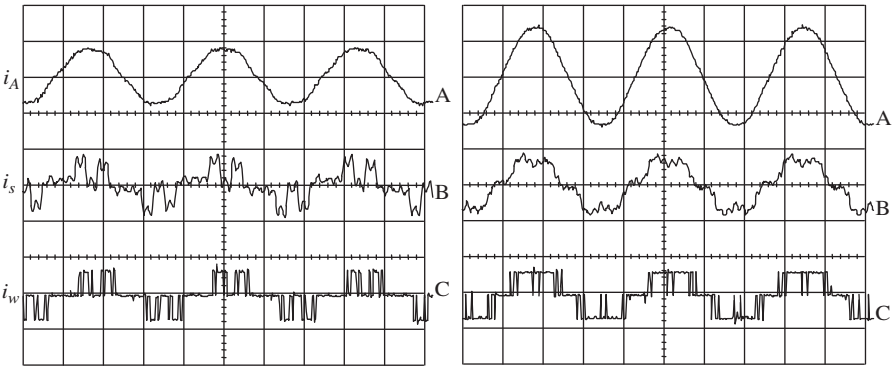

(a) Pattern A: $m_a = 0.5$ (b) Pattern B: $m_a = 1.02$

All traces: 1.0 pu/div, 5 ms/div.

Figure 11.3-5 Current waveforms measured from a laboratory dual-bridge CSR.

11.4 POWER FACTOR CONTROL

11.4.1 Introduction

The PWM CSR requires a three-phase filter capacitor at its input terminals. Its capacitance is typically in a range of 0.3–0.6pu for the single-bridge CSR with a switching frequency of 200 Hz. The capacitor size can be reduced when the CSR operates at higher switching frequencies or the dual-bridge rectifier is used. Neglecting a small voltage drop across the line inductance, the capacitor voltage is approximately equal to the supply voltage, resulting in a constant current flowing through the capacitor regardless of the operating condition of the rectifier.

The rectifier dc output current can be controlled by modulation index m_a. Alternatively, it can also be adjusted by delay angle α in the same manner as that for phase-controlled SCR rectifiers. The delay angle control produces a lagging power factor, which compensates the leading power factor caused by the filter capacitor. By controlling both modulation index and delay angle simultaneously, the rectifier can potentially achieve unity power factor operation while its dc current can also be controlled [9, 10].

11.4.2 Simultaneous α and m_a Control

Figure 11.4-1 shows two phasor diagrams for the single-bridge CSR, where all the voltage and current phasors, such as I_s, I_w, and V_c, represent their fundamental-frequency components only. The subscript "1" designated for the fundamental frequency is omitted for simplicity. The phasor diagram in Fig. 11.4-1b illustrates the operation of the rectifier with modulation index control only. The PWM current I_w is

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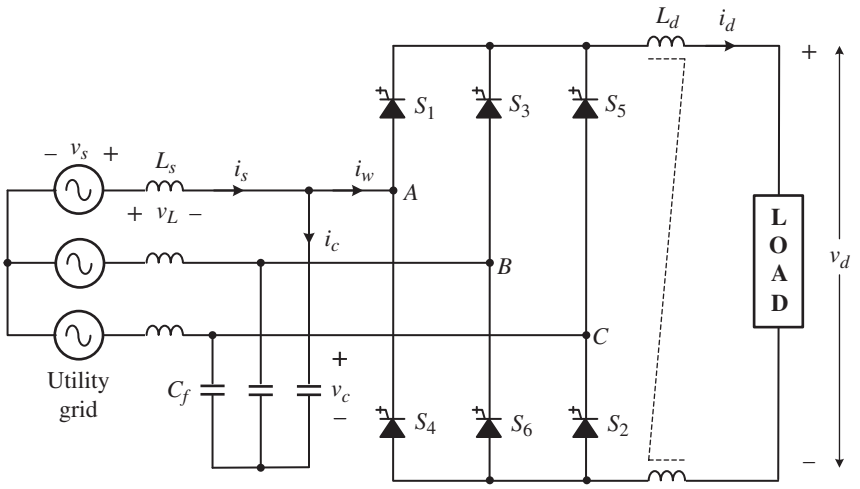
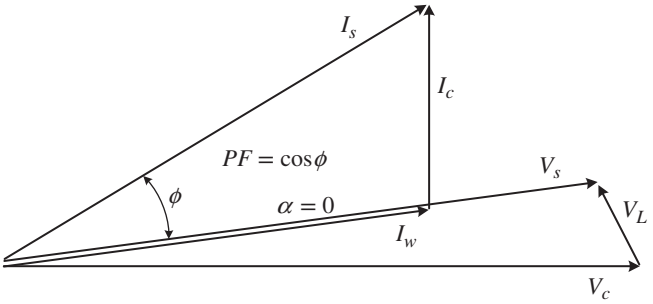
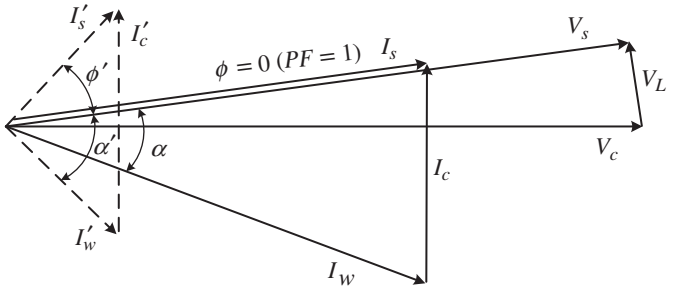


(a) Circuit diagram



(b) Phasor diagram (m_a control only)



(c) Phasor diagram (m_a and α control)

Figure 11.4-1 Single-bridge CSR and its phasor diagram.

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in phase with the supply voltage V_s while the line current I_s leads V_s by the **power factor angle** φ. The input power factor of the rectifier is then given by

$$PF = DF \times \cos\varphi = \cos\varphi \tag{11.4-1}$$

where the distortion power factor DF is assumed to be unity, which is based on the fact that the waveform of the line current i_s is close to sinusoidal. Under this assumption, the control of the power factor in this section is essentially to control the displacement power factor of the rectifier.

The input power factor can be improved by increasing the delay angle α between I_w and V_s as shown in Fig. 11.4-1c and in the meanwhile increasing the modulation index m_a to compensate the dc voltage reduction due to the increase of α. The dc voltage is a function of m_a and α, given by

$$V_d = \sqrt{3/2}\, V_{LL} m_a \cos\alpha \tag{11.4-2}$$

To achieve a unity power factor operation, the delay angle should satisfy

$$\alpha \approx \sin^{-1}\frac{I_c}{I_w} \approx \sin^{-1}\frac{\omega C_f V_s}{m_a I_d} \tag{11.4-3}$$

where the voltage drop on the line inductance is neglected.

However, the unity power factor operation is not always achievable. The phasor diagram composed of I'_w, I'_c, and I'_s in Fig. 11.4-1c illustrates such a case, where the rectifier operates under the light load condition. Obviously, the PWM current I'_w is too small to provide enough lagging current component to cancel the leading capacitor current. Therefore, the power factor control scheme should be designed such that the rectifier will (a) operate at a unity power factor when it is achievable, and (b) produce the highest possible power factor when the unity power factor operation is not achievable. The latter can be realized by

- Increasing the PWM current I_w to its highest value by setting m_a to its maximum value such that I_w is able to produce largest lagging current component; and then
- Adjusting the delay angle α to produce a required dc current set by its reference.

Figure 11.4-2 shows a power factor control scheme derived from the above-discussed principle. There exist two control loops. In the m_a control loop, the supply voltage V_s and line current I_s are detected through a low-pass filter (LPF) and then sent to the power factor angle detector. The detected power factor angle φ is compared with its reference φ^*, which is normally set to zero, demanding a unity power factor operation. The resultant error signal $\Delta\varphi$ is used to control modulation index m_a through a PI regulator.

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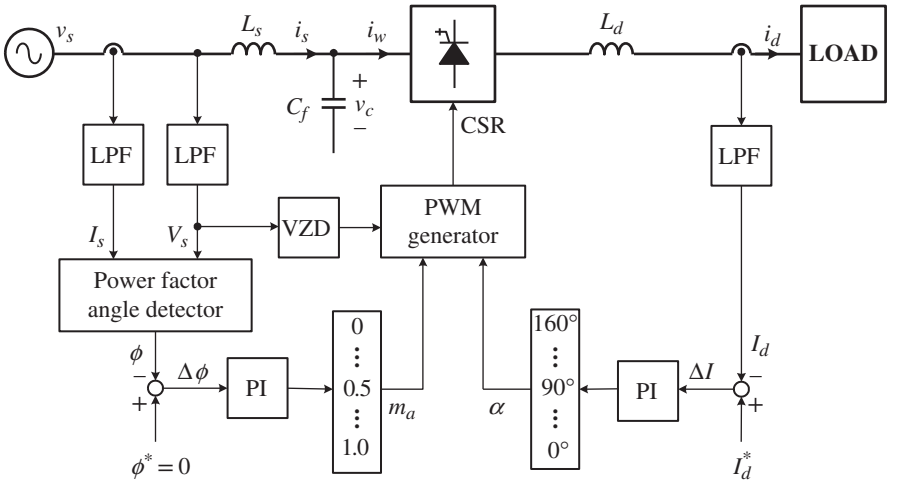

Figure 11.4-2 Block diagram of power factor control scheme.

In the α control loop, the detected dc current I_d is compared with its reference I_d^*. The error signal ΔI is then sent to a PI for delay angle control. The dc current is essentially controlled by both m_a and α. The PWM generator produces the gate signals for the GCTs in the CSR based on calculated modulation index m_a and delay angle α. The voltage zero crossing detector (VZD) provides a reference for the delay angle control.

The control scheme can guarantee that the input power factor of the rectifier is unity when it is achievable. Assuming that the line current I_s leads the supply voltage V_s by an angle φ due to a change in load, an error signal $\Delta\varphi$ is generated. This error signal results in a higher modulation index m_a, which boosts the dc voltage V_d governed by (11.4-2). The increase in V_d makes the dc current I_d rise, which causes the α control loop to respond. The α control loop tries to bring I_d back to the value set by I_d^* by increasing α. The increase in α causes a reduction in φ, which improves the input power factor. This process continues until the unity power factor is reached, at which the power factor angle $\varphi = 0$, the phase displacement error $\Delta\varphi = 0$, the dc current $I_d = I_d^*$, and the rectifier operates at a new operating point.

When the rectifier operates under light load conditions, the unity power factor operation may not be achievable. Similar to the case discussed above, the modulation index m_a keeps increasing due to $\Delta\varphi$, and in the meantime the delay angle α also keeps increasing for the dc current adjustment and power factor improvement. Since $\Delta\varphi$ will not be reduced to zero, the process continues until the PI regulator in the m_a loop is saturated, at which m_a reaches its maximum value $m_{a,\max}$ and the delay angle α also reaches a value that produces the highest possible input power factor while maintaining I_d at its reference value.

Obviously, the transition between the two operation modes, the unity and maximum power factor operations, is smooth and seamless. No extra measures should be taken for the transition.

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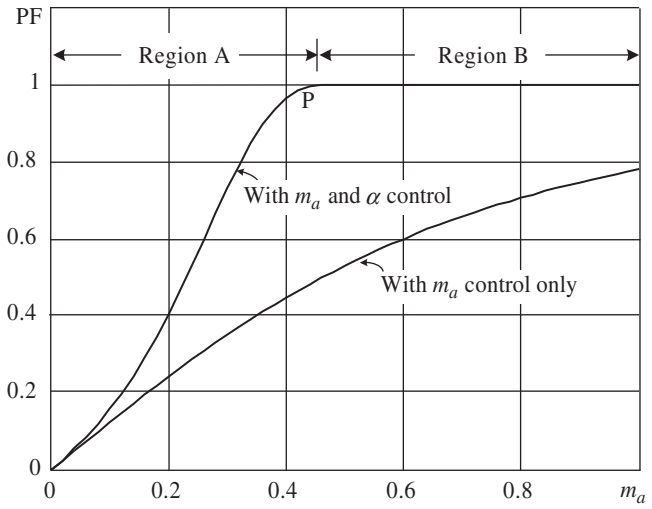

Figure 11.4-3 Rectifier power factor versus modulation index m_a ($V_d = 0.5$ pu and $C_f = 0.4$ pu).

11.4.3 Power Factor Profile

Figure 11.4-3 shows the power factor profile for the rectifier operating at a dc voltage of 0.5 pu and filter capacitor of 0.4 pu. With simultaneous m_a and α control, the rectifier can operate with a unity power factor in Region A. When the unity power factor is not achievable under light load conditions, the controller can produce a highest possible power factor as shown in Region B. The power factor of the rectifier with the m_a control only is also plotted in the figure, where the power factor never reaches unity due to the leading capacitor current. It is clear that the combination of the m_a and α control significantly improves the input power factor of the rectifier.

Figure 11.4-4 shows the waveforms of dc current i_d, delay angle α, modulation index m_a, and PWM current i_w measured from a laboratory single-bridge CSR. The rectifier has a total line inductance of $L_s = 0.1$ pu, filter capacitance of $C_f = 0.6$ pu, and load resistance of 0.64 pu. It operates at a dc current of 0.5 pu with a switching frequency of 360 Hz for the 5th and 7th harmonic elimination. Under this operating condition, the rectifier cannot achieve unity power factor operation since the PWM current i_w is not high enough to fully compensate the leading capacitor current of $I_c = 0.6$ pu. The PI regulator in the m_a control loop is saturated, keeping m_a at its maximum value of 1.03. The delay angle α is adjusted by its PI regulator to 67°, at which the dc current $I_d = I_d^* = 0.5$ pu and the power factor $PF = 0.93$ (leading), which is the maximum achievable value.

When the load current is increased from 0.5 to 1.0 pu by a step increase in I_d^*, the rectifier is able to achieve unity power factor operation since the PWM current i_w is now sufficiently high to compensate the leading capacitor current. The load current i_d rises from 0.5 to 1.0 pu, the delay angle α decreases from its original value of 67°

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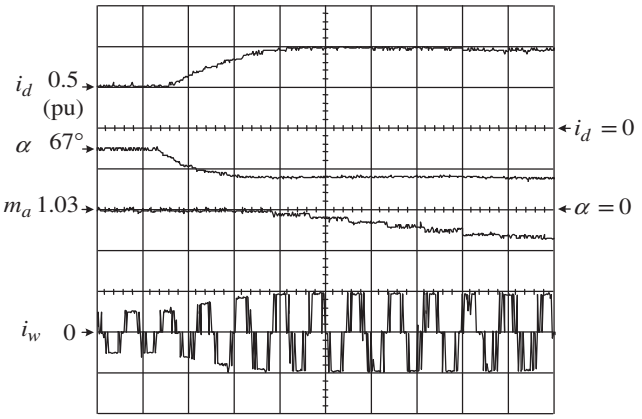
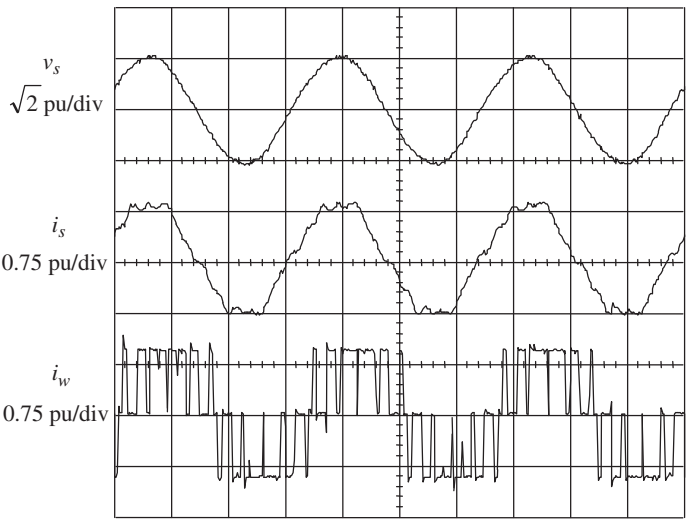


i_d: 0.5 pu/div, α: 40°/div, m_a: 0.1/div, i_w: 1.0 pu/div.
Timebase: 20 ms/div

Figure 11.4-4 Rectifier dynamic response to a step increase in I_d^*, leading to a unity power factor operation.

to 35°, and the modulation index m_a falls from its maximum value to 0.96, at which the rectifier operates at the unity power factor. Fig. 11.4-5 shows the waveforms of the supply voltage v_s, line current i_s, and PWM current i_w when the rectifier reaches the unity power factor operation, where the line current i_s is in phase with the supply voltage v_s.

It is worth noting that the power factor control scheme does not require any system parameters, such as the values of the line inductance or filter capacitor. Variations



Timebase: 5 ms/div

Figure 11.4-5 Waveforms measured from the single-bridge CSR at unity power factor operation.

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in the line inductance due to the power system operation or the changes in filter capacitor size do not affect the process of tracking unity or maximum power factor, which is desirable in practice.

11.5 ACTIVE DAMPING CONTROL

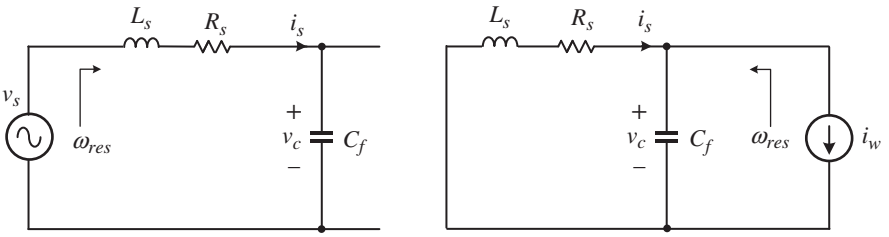
11.5.1 Introduction

The filter capacitor C_f at the input of the CSR forms an LC resonant circuit with the total line inductance L_s of the system. The LC resonant mode may be excited by the harmonic voltages in the utility supply and harmonic currents produced by the rectifier. In this section, the principle of active damping is introduced, based on which a control algorithm is developed. The effectiveness of the active damping control is investigated by simulation and verified by experiments.

11.5.2 Series and Parallel Resonant Modes

The LC resonant circuit in the PWM rectifier has two modes of resonance depending on the source of excitation. Assuming a balanced three-phase system, the LC circuit can be analyzed on a per-phase basis. The **series resonant mode** can be obtained by looking into the LC circuit and rectifier of Fig. 11.4-1 from the utility supply, where the rectifier can be considered as a constant current source and thus can be open circuited as shown in Fig. 11.5-1a. The line inductance L_s is essentially placed in series with the filter capacitor C_f. The LC circuit in high-power rectifiers is lightly damped due to a very low equivalent series resistance R_s, which is the total resistance of the distribution line, line reactor, and isolation transformer if any. The series LC resonance may be excited by voltage harmonics in the utility supply.

The **parallel resonant mode** can be identified by looking into the LC circuit and utility supply from the rectifier, where the supply is essentially a constant voltage source and thus can be short circuited as shown in Fig. 11.5-1b. The parallel resonance may be excited by the current harmonics generated by the PWM rectifier. Neglecting R_s, the resonant mode of the series and parallel resonances can be found from $\omega_{res} = 1/\sqrt{L_s C_f}$. With the filter capacitor C_f in the range of 0.3–0.6 pu and the total



(a) Series resonant mode (b) Parallel resonant mode

Figure 11.5-1 Series and parallel resonant modes in a current source rectifier.

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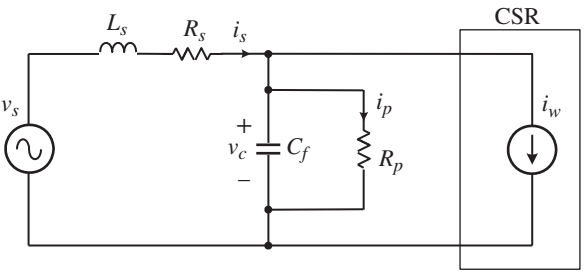
line inductance L_s between 0.1 and 0.15 pu, the resultant LC resonant frequency ω_{res} is in the range of 3.3–5.8 pu.

The LC resonance problem is traditionally addressed by selecting a proper LC resonant frequency using reasonable values for L_s and C_f such that the frequency of the LC resonance is lower than that of the lowest harmonic produced by the rectifier. However, the source inductance may vary with power system operations. It is possible for the source inductance to change such that the LC resonant mode is excited by a low-order harmonic. The use of the active damping control can effectively suppress the LC resonances, and simplifies the design of the LC filter as well.

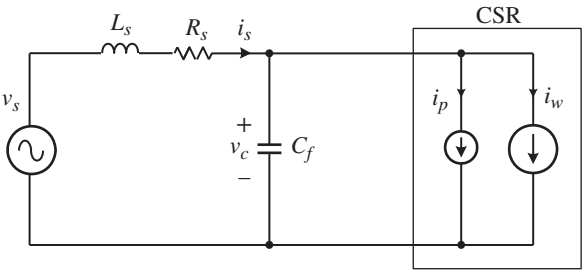
11.5.3 Principle of Active Damping

It is well known that an LC resonance can be suppressed by adding a physical damping resistor to the resonant circuit. For the CSR, the most effective location for the damping resistor R_p is in parallel with the filter capacitor C_f as shown in Fig. 11.5-2a. This location is effective in damping both series and parallel resonances. However, adding a resistor to the system results in additional power losses and thus is not a practical solution.

Active damping uses the rectifier to emulate a damping resistance R_p in the system [11–13]. Fig. 11.5-2b illustrates the principle of one of the active damping schemes



(a) Passive damping



(b) Active damping

Figure 11.5-2 Realization of passive and active damping.

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[11]. The damping current i_p can be generated by the PWM rectifier through the following steps:

- Detect the instantaneous capacitor voltage v_c;
- Calculate the damping current by $i_p = v_c/R_p$; and
- Adjust the modulation index m_a dynamically based on the calculated i_p.

By selecting a proper value for the emulated damping resistance R_p, the LC resonances can be effectively suppressed without causing any additional power losses.

The detected capacitor voltage v_c is composed of the fundamental component and all the harmonic components, so is the calculated damping current i_p. In practice, the active damping control needs to damp out the LC resonances at all frequencies except the fundamental since the fundamental component of i_p will interfere with the control of the rectifier dc output current [11].

Figure 11.5-3 shows a block diagram of the active damping control algorithm. The capacitor voltage v_c is detected and then transformed to the synchronous reference frame. The transformation is performed by the *abc/dq* block based on a reference angle θ that is obtained from a digital phase-locked loop (PLL). The fundamental component of the capacitor voltage v_c in the stationary frame becomes a dc component in the synchronous frame. The dc component is then removed with a high-pass filter (HPF), and the remaining signal represents the capacitor harmonic voltages v_c' in the synchronous frame. This harmonic detection method is insensitive to the variations in the supply frequency, a desirable feature in practice. The HPF is set to a low cut-off frequency, resulting in a minimal phase delay at the resonant frequencies.

The damping current reference i_p' is obtained by dividing v_c' by the desired damping resistance R_p. The reference current i_p' is then converted to the damping modulation index m_a' by normalizing to the dc output current I_d. The damping modulation index m_a' is summed with the modulation index m_a'' for dc current control in the space vector modulator for active damping and dc current control.

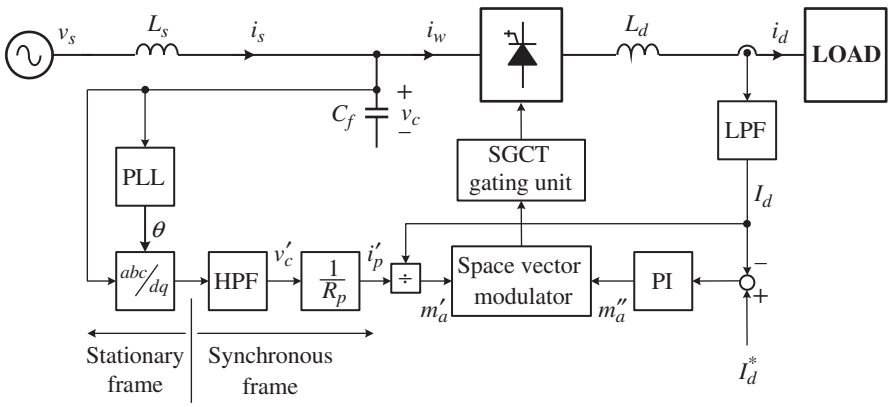

Figure 11.5-3 Block diagram of a vector-controlled CSR with active damping control.

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Table 11.5-1 Rectifier Ratings and Parameters Used in Simulation of a 1 MVA Rectifier

Rated power	1 MVA
Rated voltage (line-to-line, rms)	4160 V
Rated line current (rms)	138 A
Total line inductance L_s	0.17 pu
Filter capacitor C_f	0.3 pu
Equivalent series resistance R_s	0.002 pu
Active damping resistance R_p	1.5 pu
LC resonant frequency ω_{res}	4.43 pu (266 Hz)
Switching frequency f_{sw}	540 Hz

It should be noted that SVM schemes are recommended for the active damping control. The SHE is an off-line scheme and may not be able to provide a fast adjustment of its modulation index. The SVM scheme discussed in Chapter 10 for the current source inverter can be directly applied to the CSR for active damping control.

11.5.4 LC Resonance Suppression

To investigate the performance of the active damping control illustrated in Fig. 11.5-3, computer simulations were carried out [14]. The PWM rectifier under investigation is powered by an ideal three-phase 4160 V utility supply. The parameters of the rectifier are given in Table 11.5-1. The emulated damping resistance R_p used in the controller is 1.5 pu.

Let us consider a case where the dc current reference I_d^* is reduced suddenly from 100% to 20% and the load resistance changes with the reference to maintain a constant load voltage of 50%. This simulates the dc load that the rectifier would see if it were used to feed an inverter-based drive system operating at a constant motor speed with a sudden decrease in load torque.

The simulated waveforms for the capacitor voltage v_c and line current i_s are shown in Fig. 11.5-4. The step change in the reference current forces the dc current control to react as quickly as possible, resulting in a fast transient on the ac side of the rectifier. This transient may excite the natural resonance of the LC circuit. Without active damping control, the line current i_s and capacitor voltage v_c undergo oscillations during the dc side step transient as shown in 11.5-4a. Due to the light damping, the oscillations persist for many cycles of the fundamental frequency. However, when active damping is employed, the oscillations are suppressed within one cycle of the fundamental frequency as shown in Fig. 11.5-4b.

The active damping control is verified by experiments on a laboratory GTO CSR with a digital controller. The parameters used in the experimental system are given in Table 11.5-2. The rectifier is controlled by a SVM scheme with a switching frequency of 540 Hz. The supply voltage contains approximately 1% 5th harmonic voltage. Since the LC resonant frequency of 5.43 pu is close to the frequency of the

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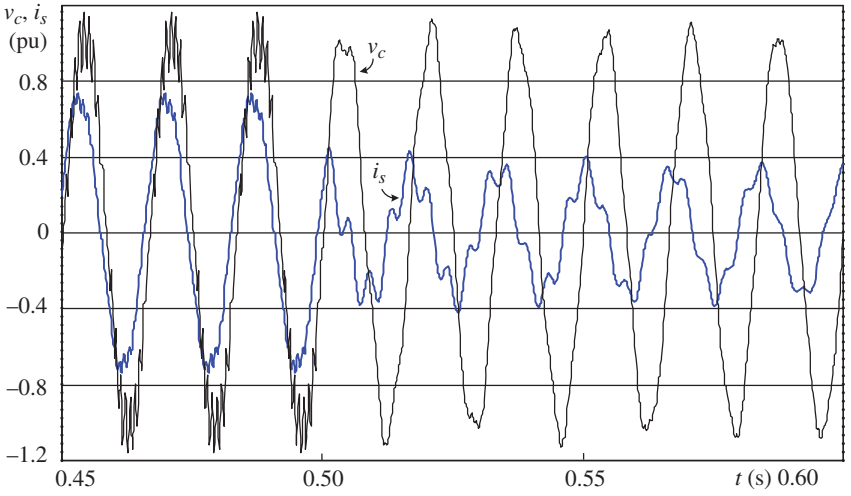
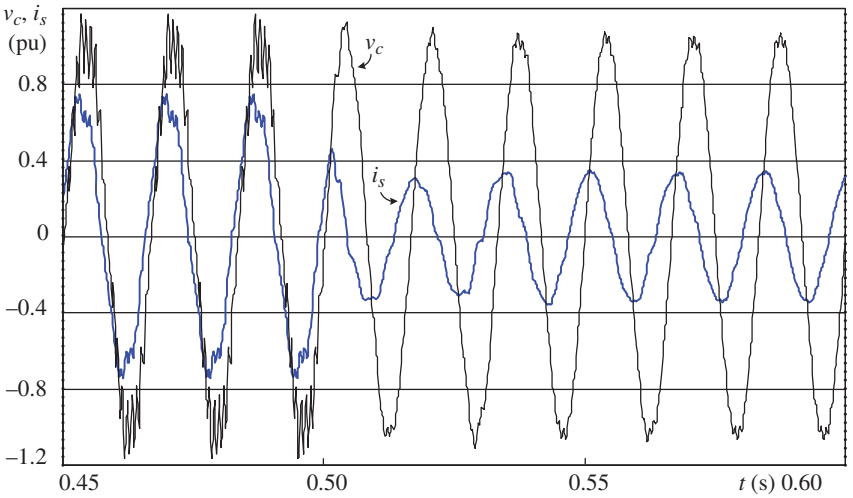


(a) Without active damping control



(b) With active damping control

Figure 11.5-4 Simulated waveforms for 1 MVA rectifier with a step reduction in I_d^*.

Table 11.5-2 Parameters in a Laboratory GTO Current Source Rectifier with Active Damping Control

Total line inductance L_s	0.087 pu
Filter capacitor C_f	0.39 pu
dc current inductance L_d	0.87 pu
dc load resistance R_d	1.2 pu
Active damping resistance R_p	0.96 pu
LC resonant frequency ω_{res}	5.43 pu (326 Hz)

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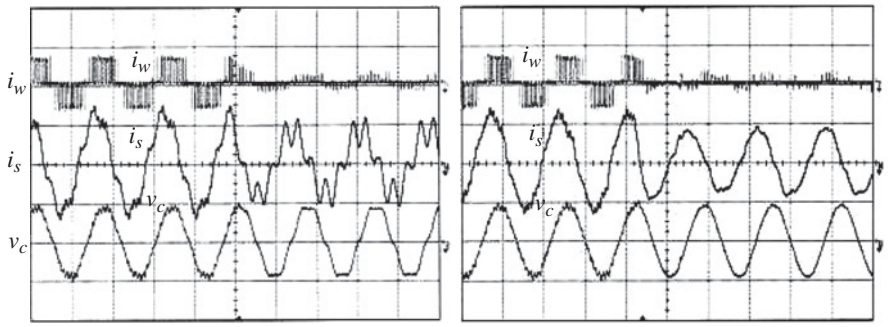
5th harmonic, the LC resonance can be excited by the 5th harmonic voltage in the supply and the 5th harmonic current generated by the rectifier.

Figure 11.5-5 shows the current and voltage waveforms measured from the laboratory rectifier without and with active damping control. The dc current reference is stepped from 75% to 20% of its rated value. The system response in Fig. 11.5-5a exhibits oscillations due to the change in the dc current reference. In particular, the waveform of the line current i_s has two humps per half cycle of the fundamental frequency and therefore contains a substantial amount of 5th harmonic component due to the LC resonance. With the active damping control implemented, the LC resonance is effectively suppressed as shown in Fig. 11.5-5b.

11.5.5 Harmonic Reduction

The active damping control can also reduce the line current distortion during the steady-state operation of the CSRs [15]. To investigate the effect of LC resonances on line current THD, let us consider a 1 MW/4160 V rectifier with $L_s = 0.1$ pu, $C_f = 0.4$ pu, $R_s = 0.005$ pu, $L_d = 1.0$ pu, and $f_{sw} = 540$ Hz (SVM). The PWM current i_w of the rectifier normally contains a few percent of 5th and 7th harmonics produced by the SVM. The natural frequency of the LC circuit is tuned exactly to the 5th harmonic ($\omega_{res} = 1/\sqrt{L_s C_f} = 5.0$ pu) on purpose, which is the worst case and should be avoided in practice. The simulated PWM current i_w, line current i_s, and capacitor phase voltage v_c are shown in Fig. 11.5-6. Since there is no active damping present, the parallel resonant mode of the LC circuit is excited by the 5th harmonic current produced by the PWM rectifier. This is evident in the line current and capacitor voltage waveforms. The total harmonic distortion of the line current i_s is substantial (105%).



(a) Without active damping control (b) With active damping control

i_w: 1.3 pu/div, i_s: 0.5 pu/div, v_c: 1.41 pu/div, Timebase: 10 ms/div.

Figure 11.5-5 Waveforms measured from a laboratory CSR with a step reduction in I_d^* at cursor ▲.

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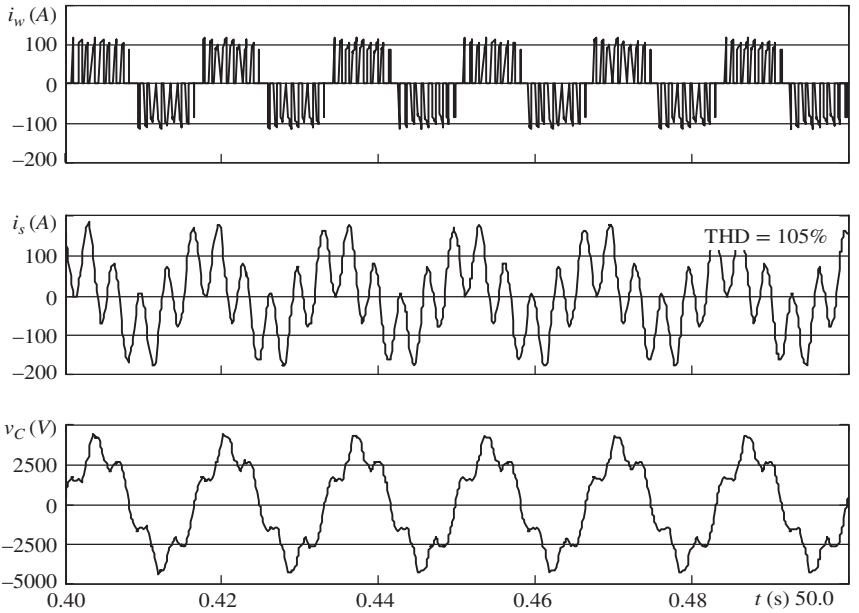

Figure 11.5-6 Simulated waveforms with the LC resonant mode tuned exactly to the 5th harmonic (worst case) without active damping control.

Figure 11.5-7 shows the line current THD profile for the rectifier system as a function of the series resistance R_s and resonant frequency f_{res}. The LC values are calculated using a 4:1 ratio of per unit filter capacitor C_f to total line inductance L_s. The dominant shape of this THD profile is a crest for the resonant mode of the LC filter tuned to the fifth harmonic. The THD of the line current for a resonant frequency f_{res} of 5 pu and a series resistance of 0.5% is 105%. Considering an extreme case where the series resistance is 1.5% (impractically high), the line current THD is still 60%, which is too high to accept.

Figure 11.5-8 shows the effect of the active damping control on line current THD for the rectifier operating under the same conditions as in the previous case. When the LC filter is tuned to the 5th harmonic, an active damping resistance of 1.0 pu results in a line current THD of 5%, which is about 21 times lower than that without active damping control.

The effect of active damping on the system sensitivity to the LC resonant mode is easily seen in the THD profile shown in Fig. 11.5-9. As the strength of active damping increases (i.e., a small value of R_p), the crest in the THD profile flattens. With an active damping resistance of 1 pu, the THD profile is almost a linear function of f_{res}. Active damping resistances below 1 pu are not shown in the figure because of the system instability caused by strong damping, where the significance of THD is lost due to non-periodical waveforms.

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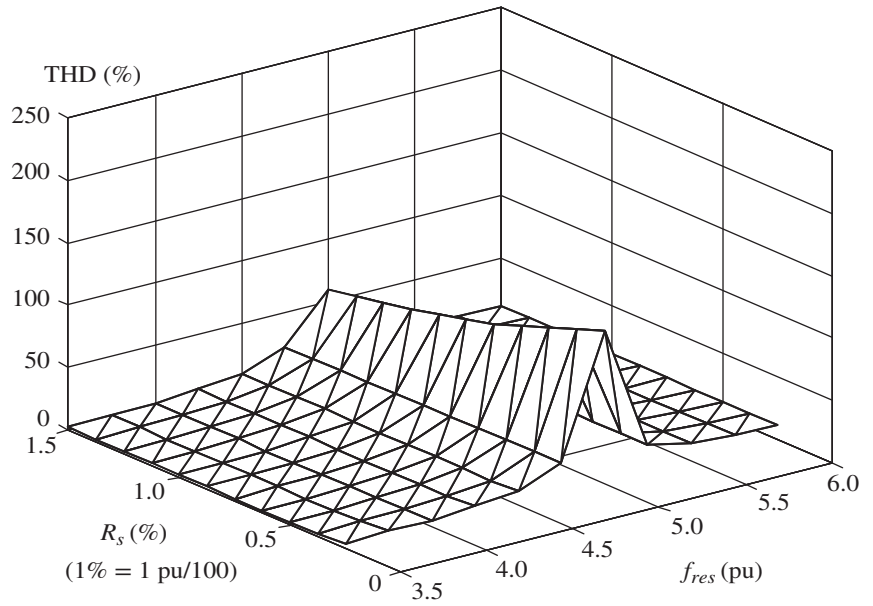

Figure 11.5-7 Line current THD versus series resistance R_s and resonant frequency f_{res} without active damping control.

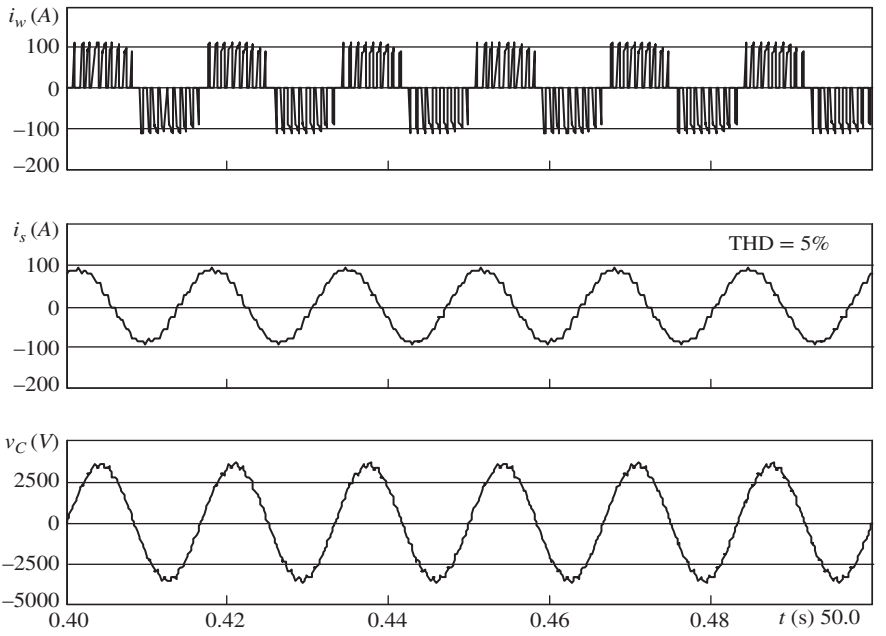

Figure 11.5-8 Simulated waveforms with the LC resonant mode tuned exactly to the 5th harmonic (worst case) with 1.0 pu active damping resistance.

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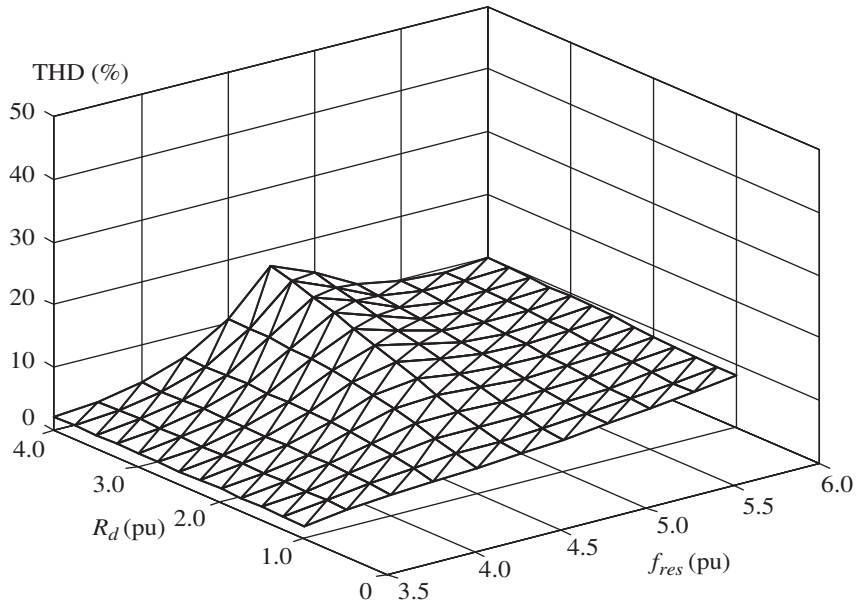

Figure 11.5-9 Line current THD versus damping resistance R_p and resonant frequency f_{res} ($R_s = 0.005$ pu).

11.5.6 Selection of Active Damping Resistance

As stated earlier, a small value of the active damping resistance R_p is preferred for minimizing the line current THD. However, if R_p is too small, the rectifier system may become unstable. The selection of the damping resistance is a complex issue since it is determined by many factors, including LC resonant modes, device switching frequency, dc current and voltage levels, magnitude of dominant harmonics, and value of modulation index. As a rule of thumb, the value of R_p for a high-power rectifier with a switching frequency of a few hundred hertz should not be lower than 1.0 pu.

To optimize the system performance, the damping resistance R_p should not be kept constant over the full operating range of the rectifier. Its minimum value can be determined by the stability requirements and may be used over most of the operating range. With the modulation index approaching to its maximum value, the damping resistance should increase accordingly [15].

11.6 SUMMARY

This chapter addresses four main aspects of the current source rectifiers (CSRs) used in high-power ac drives, including converter topologies, PWM patterns, power factor control, and LC resonance with active damping control. The single-bridge CSR

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topology features simple converter structure and low manufacturing cost while the dual-bridge CSR draws a sinusoidal current from the utility supply. SHE schemes for the two rectifiers are presented. To minimize the switching losses, these schemes are designed with a switching frequency of only 360 or 420 Hz.

The CSR requires a filter capacitor at its input terminals. To compensate the effect of the filter capacitor, a power factor control scheme with simultaneous delay angle and modulation index control is presented. This scheme can make the rectifier operate at the unity power factor or highest achievable power factor over the full operating range. Further, the power factor control algorithm does not require any system parameters, which is desirable in practice.

The filter capacitor and line inductance constitute an LC resonant circuit. The LC resonances may be excited by voltage harmonics in the utility supply or current harmonics produced by the rectifier. An active damping control scheme is elaborated, which can effectively suppress the LC resonances during system transients and also reduce the line current distortion during the steady state operation of the rectifier.

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APPENDIX

Gating Angles for Current Source Rectifiers

Table A-1 Gating Angles for the Single-Bridge CSR with 5th and 7th Harmonic Elimination

m_a	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.03
	Degrees										
θ_1	−13.5	−11.9	−10.3	−8.60	−6.86	−5.00	−3.98	−0.67	2.17	6.24	7.93
θ_2	14.2	13.5	12.7	12.0	11.4	10.8	10.4	10.3	10.8	12.6	13.8
θ_3	43.6	42.2	40.9	39.5	38.0	36.6	35.1	33.6	32.1	30.5	30.0
θ_4	45.8	46.5	47.3	48.0	48.6	49.2	49.6	49.7	49.2	47.3	46.2
θ_5	73.5	71.9	70.3	68.6	66.9	65.0	63.0	60.7	57.8	53.8	52.1
θ_6	106.5	108.1	109.7	111.4	113.1	115.0	117.0	119.3	122.2	126.2	127.9
θ_7	134.2	133.5	132.7	132.0	131.4	130.8	130.4	130.3	130.8	132.6	133.8
θ_8	136.4	137.8	139.1	140.5	142.0	143.4	144.9	146.4	147.9	149.5	150.0
θ_9	165.8	166.5	167.3	168.0	168.6	169.2	169.6	169.7	169.2	167.3	166.2
θ_{10}	193.5	191.9	190.3	188.6	186.9	185.0	183.0	180.7	177.8	173.7	172.1
θ_{11}	256.4	257.8	259.1	260.5	262.0	263.4	264.9	266.4	267.9	269.5	270.0
θ_{12}	283.6	282.2	280.9	279.5	278.0	276.6	275.1	273.6	272.1	270.5	270.0

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Table A-2 Gating Angles for the Dual-Bridge CSR with 11th and 13th Harmonic Elimination

	Pattern A								Pattern B			
m_a	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.857	0.84	0.9	1.0	1.085
	Degrees											
θ_1	43.6	42.8	41.9	41.0	39.5	35.5	31.5	30.0	18.9	19.1	19.4	19.0
θ_2	46.2	46.6	46.8	46.4	44.4	38.7	35.1	34.2	18.6	19.6	21.1	21.7
θ_3	57.4	55.9	54.3	52.2	49.0	44.6	42.2	41.1	30.0	30.0	30.0	30.0
θ_4	122.6	124.1	125.7	127.8	131.0	135.4	137.8	138.9	33.9	34.8	36.4	38.3
θ_5	133.8	133.4	133.2	133.6	135.6	141.3	144.9	145.8	41.1	40.9	40.6	41.0
θ_6	136.4	137.2	138.1	139.0	140.6	144.5	148.5	150.0	138.9	139.1	139.4	139.0
θ_7	242.6	244.1	245.7	247.8	251.0	255.4	257.8	258.9	146.1	145.2	143.6	141.7
θ_8	253.8	253.4	253.2	253.6	255.6	261.3	264.9	265.8	150.0	150.0	150.0	150.0
θ_9	256.4	257.2	258.1	259.0	260.5	264.5	268.5	270.0	161.4	160.4	158.9	158.3
θ_{10}	283.6	282.8	281.9	281.0	279.5	275.5	271.5	270.0	161.1	160.9	160.6	161.0
θ_{11}	286.2	286.6	286.8	286.4	284.4	278.7	275.1	274.2	258.6	259.6	261.1	261.7
θ_{12}	297.4	295.9	294.3	292.2	289.0	284.6	282.2	281.1	266.1	265.2	263.6	261.7
θ_{13}	—	—	—	—	—	—	—	—	273.9	274.8	276.4	278.3
θ_{14}	—	—	—	—	—	—	—	—	281.4	280.4	278.9	278.3

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Part Five

High-Power AC Drives

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Chapter 12

Voltage Source Inverter Fed Drives

12.1 INTRODUCTION

The medium voltage (MV) drives using multilevel voltage source inverters have found wide application in industry. Most of the MV drives are for the general applications, where the dynamic performance of the drive is not a top priority whereas the other MV drives are used in areas where four-quadrant operation and high dynamic performance are required. To meet the industrial needs, the MV drives come with a number of different configurations using various converter topologies, each of which has some unique features. This chapter presents the multilevel VSI fed MV drives marketed by the world leading drive manufacturers. The advantages and limitations of these drives are discussed.

12.2 TWO-LEVEL VSI-BASED MV DRIVES

It is well known that the two-level voltage source inverter is a dominant converter topology for low-voltage (≤ 600 V) drives. This technology has now been extended to the MV drives, which are commercially available for the power rating up to a few megawatts [1].

12.2.1 Power Converter Building Block

Figure 12.2-1a shows a typical two-level inverter topology for the MV drive. There are three switch modules in series per inverter branch. Each switch module is composed of an IGBT device, gate driver, snubber circuit, parallel resistor R_p, and bypass

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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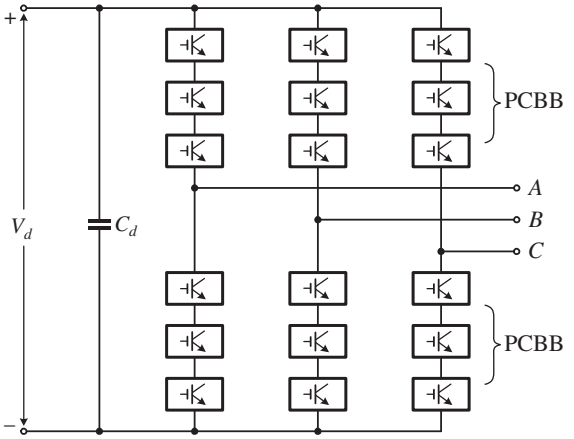
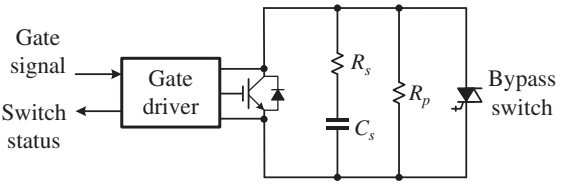


(a) Two-level inverter



(b) Power converter building block (PCBB)

Figure 12.2-1 Two-level medium voltage inverter with PCBBs.

switch as shown in Fig. 12.2-1b. This type of switch module is also known as **power converter building block** (PCBB) [2].

The gate driver receives a gate signal from the digital controller of the drive and generates conditioned gating pulses for the IGBT. It also detects the operating status of the IGBT and sends it back to the controller for fault diagnosis. The gate driver normally communicates with the drive controller through fiber optic cables for electrical isolation and high noise immunity. A number of protection functions can be implemented in the gate driver as well, such as IGBT overvoltage and short-circuit protections. The active overvoltage clamping scheme presented in Chapter 2 for series-connected IGBTs can also be integrated into the gate driver.

Each IGBT switch is protected by an RC snubber network (C_s and R_s) from over-voltages at turn-off. The snubber also facilitates dynamic voltage equalization for the series-connected devices during switching transients. Alternatively, the active overvoltage clamping scheme can be implemented instead of the snubber circuit. However, the use of active overvoltage clamping causes additional switching losses for the IGBTs as discussed in Chapter 2.

The snubber circuit provides an effective means of transferring the switching losses from the IGBT to the snubber resistor, leading to a lower junction temperature rise and better thermal management for the IGBTs. The snubber circuit also helps to reduce the *dv/dt* during the IGBT turn-off transients. The parallel resistor R_p shown

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in Fig. 12.2-1b is for static voltage sharing, and the function of the bypass switch will be discussed later.

12.2.2 Two-Level VSI Drive with Passive Front End

Figure 12.2-2 illustrates a typical configuration for the two-level VSI drive. A 12-pulse diode rectifier is employed as a front end for the reduction of line current harmonic distortion. For applications with more stringent harmonic requirements, the 12-pulse rectifier can be replaced by an 18- or 24-pulse diode rectifier. The detailed analysis of the multipulse diode rectifiers is given in Chapter 3.

The inverter is composed of 24 switch modules with four modules per inverter branch. Using 3300 V IGBTs, the two-level inverter is suitable for 4160 V (line-to-line) ac motors. The dynamic braking circuit in the dc link is optional. The dc capacitors are normally of oil-filled type instead of electrolytic type commonly used in the low-voltage VSI drives since the latter has limited voltage ratings (a few hundred volts each). The two-level inverter usually requires an LC filter at its output. The inverter can be controlled by either carrier-based modulation or space vector modulation scheme presented in Chapter 6.

The two-level voltage source inverter has the following features:

- Modular structure using PCBBs. The IGBT device, gate driver, bypass switch, and snubber circuits are integrated into a single switch module for easy assembly and mass production, leading to a reduction in manufacturing cost. The modular design also facilitates fast replacement of failed modules when the drive operates in the field.

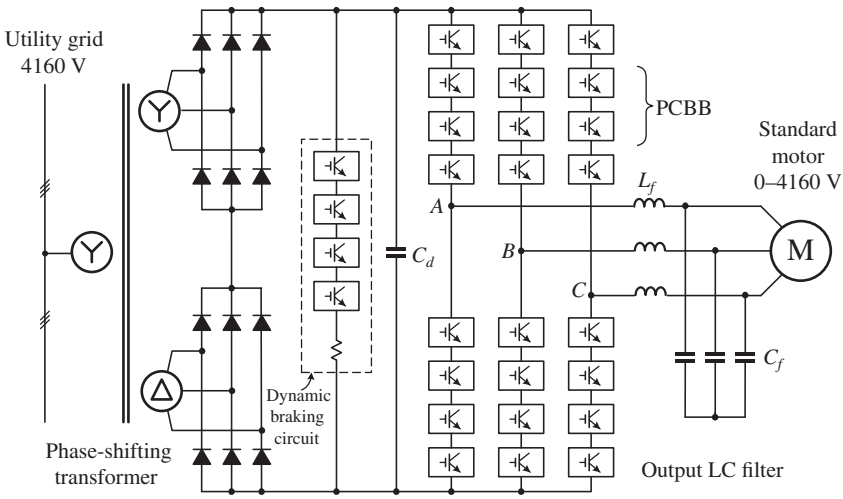

Figure 12.2-2 Typical two-level VSI drive with a passive front end.

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- Simple PWM scheme. The conventional carrier-based sinusoidal modulation or space vector modulation scheme can be implemented for the inverter. Only six gate signals are required for the six groups of synchronous switches. The number of the gate signals does not vary with the number of the switches in series.
- Active overvoltage clamping for series-connected IGBTs. The maximum dynamic voltage on the IGBT device at turn-off can be effectively clamped by the gate driver. The IGBT can be safely protected from over-voltages caused by switching transients.
- $N+1$ provision for high reliability. In applications where high system reliability is required, a redundant switching device ($N+1$) can be added to each of the six inverter branches. When a switch module malfunctions during operation, the defective module can be shorted out by the bypass switch, and the drive is able to operate continuously at full load with a failed module.
- Ease of dc capacitor pre-charging. The dc capacitor in the two-level inverter needs only one pre-charging circuit. This is in contrast to the multilevel inverters where multiple sets of pre-charging circuits are normally required.
- Provision for four-quadrant operation and regenerative braking. The multipulse diode rectifier can be replaced by an active front end with the same configuration as the inverter for four-quadrant operation or regenerative braking.

However, there are some drawbacks associated with the two-level voltage source inverter, including the following:

- High *dv/dt* in the inverter output voltage. Fast switching speed of IGBTs results in high *dv/dt* at the rising and falling edges of the inverter output voltage waveform. The *dv/dt* is particularly high for the two-level inverter employing series-connected IGBTs switching in a synchronous manner. Depending on the magnitude of the dc bus voltage and switching speed of the IGBT, the *dv/dt* can well exceed $10,000\,\mathrm{V}/\mu s$ [3], which causes a number of problems such as premature failure of motor winding insulation, early bearing failure, and wave reflections. More detailed explanation is given in Chapter 1.
- Motor harmonic losses. The two-level inverter usually operates at low switching frequencies, typically around 500 Hz, resulting in high harmonic distortion in the stator voltage and current. The harmonics produce additional power losses in the motor.
- Common-mode voltages. As discussed in Chapter 1, the rectification and inversion process in any converters generates **common-mode voltages** [4]. If not mitigated, these voltages would appear on the motor, causing premature failure of its winding insulation.

The first two problems can be effectively solved by adding a properly designed LC filter between the inverter output and the motor as shown in Fig. 12.2-2. With the use of the filter, the high *dv/dt* in the inverter output voltage is now applied to

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the filter inductor L_f instead of the motor. The insulation of the inductor should be properly designed for the high *dv/dt*. The LC filter is normally installed inside the drive cabinet and connected to the inverter with short cables to avoid wave reflections. The motor voltage and current can be made nearly sinusoidal by the filter, leading to low harmonic losses in the motor.

However, the use of the LC filter causes some practical consequences, including an increase in manufacturing cost, fundamental voltage drops, and circulating current between the filter and dc circuit. It may also cause LC resonances that can be excited by the harmonics in the inverter PWM voltages. The problem can be mitigated at the design stage by placing the LC resonance frequency below the lowest harmonic frequency [3]. The principle of the active damping control presented in the previous chapter can also be used for the suppression of the LC resonances.

The third problem can be effectively mitigated by the phase-shifting transformer in Fig. 12.2-2, which is used to block the common-mode voltages. To ensure that the motor is not subject to any common-mode voltages, the neutral of the filter capacitor C_f is grounded directly or through an RC grounding network. In a three-phase balanced system, the neutral points of the capacitor and stator winding should have the same potential. Grounding one makes the other equivalently grounded.

It is worth mentioning that the use of the phase-shifting transformer does not lead to the elimination of the common-mode voltage. Grounding the capacitor neutral essentially makes the common-mode voltage be transferred from the motor to the transformer [4]. The insulation system of the transformer should, therefore, be properly designed. The MV drive system is suitable for retrofit applications, where standard ac motors (which are not designed to withstand the common-mode voltages) are usually used.

12.3 NEUTRAL POINT CLAMPED (NPC) INVERTER FED DRIVES

The MV drive using three-level neutral point clamped (NPC) inverter technology is marketed by a number of leading drive manufacturers [5–7]. Some manufacturers use GCTs in their drives while the others prefer IGBTs.

12.3.1 GCT-Based NPC Inverter Drives

Figure 12.3-1a shows a typical configuration of a three-level NPC inverter fed drive. A 12-pulse diode rectifier is normally adopted as a front end. The inverter consists of twelve reverse-conducting GCT devices and six clamping diodes. The mechanical assembly for one of the three inverter legs is illustrated in Fig. 12.3-1b, where four GCTs, two diodes, and a number of heatsinks can be assembled with just two bolts, leading to high power density and low package costs.

There are two *di/dt* clamp circuits, each composed of L_s, D_s, R_s, and C_s. One clamp circuit is in the positive dc bus for the switches in the upper half bridge and the other is in the negative dc bus for those in the lower half bridge. With a few

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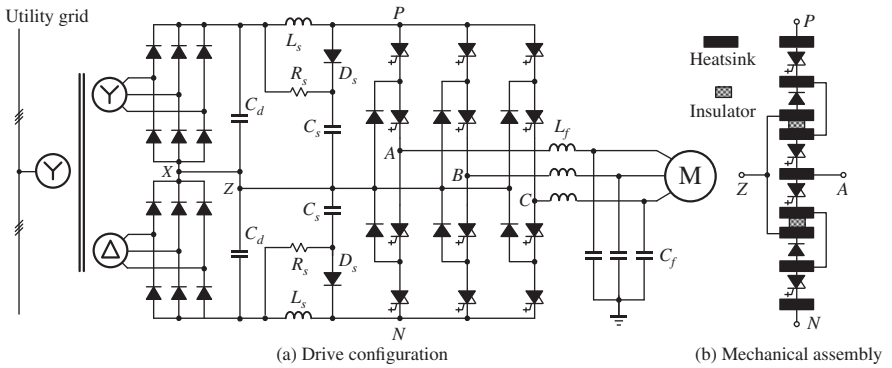

Figure 12.3-1 Typical configuration for a three-level NPC inverter fed drive.

microhenries for the *di/dt* limiting choke L_s, the rate of current rise during GCT turn-on transients can be limited to a certain value, typically below 1000 A/μs.

The neutral point *Z* of the NPC inverter can be connected to the midpoint *X* of the diode rectifier. Such a connection makes the total dc voltage equally divided between the two dc capacitors. The inverter neutral point voltage control in this case is no longer an issue.

Similar to the two-level VSI drive, an LC filter is usually installed at the inverter output terminals for sinusoidal outputs. The filter can also solve the *dv/dt* problems caused by fast switching of the GCT devices.

Figure 12.3-2 illustrates a three-level NPC drive with two more protection schemes added to the drive system of Fig. 12.3-1 [7]. The drive is equipped with protection GCT switches S_d in the dc circuit for fuseless short-circuit protection. The *di/dt* limiting choke L_s limits the rate of rise of the dc current and facilitates a safe shutdown of the drive during a short-circuit fault.

As mentioned earlier, the common-mode voltages produced by the rectifier and inverter are transferred from the motor to the transformer for motor protection by

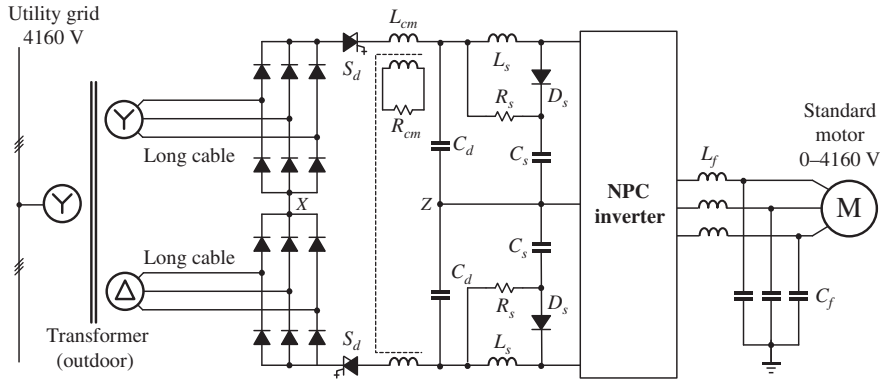

Figure 12.3-2 Three-level NPC drive with a common-mode choke for long transformer cables.

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grounding the neutral point of the filter capacitor C_f. To minimize the effect of the common-mode voltages on the transformer and its cables, a special common-mode choke L_{cm} is added to the dc link for the reduction of peak currents that cause charging and discharging of the capacitance of the cables that connect the transformer secondary windings to the rectifier. The choke has an auxiliary coil, to which a resistor R_{cm} is connected to suppress transient oscillations. With a properly designed L_{cm} and R_{cm}, the cable length can reach 300 m without causing any problems [6]. The transformer can then be placed outside the control room, which reduces the floor space and room cooling requirements as well.

It should be mentioned that the neutral point Z of the NPC inverter and midpoint X of the 12-pulse rectifier should be left unconnected due to the use of S_d and L_{cm}. As a result, the inverter neutral point voltage should be tightly controlled as discussed in Chapter 8.

Table 12.3-1 gives the main specifications for the three-level NPC inverter fed MV drive [5]. Without any GCTs connected in series, the drive is capable of powering ac motors with a rated voltage of 2300 V, 3300 V or 4160 V. For the 4160 V drives, the GCTs rated at 5500 V can be selected. The rated power of the drive is typically in the range of 0.3–5 MW. It can be extended to 10 MW for 6600 V applications, where each switch position in the NPC inverter is replaced by two series-connected GCTs.

The switching frequency of the GCTs is typically around 500 Hz. However, the motor sees an equivalent switching frequency of 1000 Hz due to asynchronous switchings of the GCT devices, leading to a reduction in harmonic distortion and output filter size. This is one of the main features of the three-level NPC inverter fed drive. In addition, the drive can operate at the medium voltages up to 4160 V without GCTs connected in series, which reduces the cost and increases the reliability of the drive due to the low component count.

12.3.2 IGBT-Based NPC Inverter Drives

The configuration of an IGBT-based three-level NPC drive is shown in Fig. 12.3-3. The drive topology is essentially the same as that given in Fig. 12.3-2 except that neither the *di/dt* clamp circuits nor the protection switches are required. This is due to the fact that the rate of rise of IGBT anode current can be effectively controlled and the short-circuit protection can be fully implemented by the IGBT gate driver.

The main specifications of the three-level IGBT drive are given in Table 12.3-2. Depending on applications and customer requirements, the front end can be either the 12- or 24-pulse diode rectifier. A three-level IGBT-based NPC rectifier can also be used for the drives requiring four-quadrant operation or regenerative braking.

The MV drive can operate at the nominal utility/motor voltages of 2300 V, 3300 V, 4160 V, and 6600 V. For the 2300 V applications, the NPC inverter is composed of 12 pieces of 3300 V IGBTs without devices in series. For the drives operating at higher voltages, two series-connected IGBTs can be used in each switch position in the inverter.

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Table 12.3-1 Main Specifications for the Three-Level GCT-Based Drives

Drive System Specifications	Nominal input voltage	2300 V, 3300 V, 4160 V
	Output power rating	400–6700 HP (0.3–5 MW)
	Output voltage rating	0–2300 V, 0–3300 V, 0–4160 V
	Output frequency	0–66 Hz (up to 200 Hz optional)
	Drive system efficiency	Typically >98.0% (including output filter losses but excluding transformer losses)
	Input power factor	>0.95 (displacement power factor >0.97)
	Output waveform	Sinusoidal (with output filter)
	Motor type	Induction or synchronous
	Overload capability	Standard: 10% for 1 minute every 10 minutes Optional: 150% for 1 minute every 10 minutes
	Cooling	Forced air or liquid
	Mean time between failure (MTBF)	>6 years
	Regenerative braking capability	No
Control Specifications	Control scheme	Direct torque control (DTC)
	Dynamic speed error	<0.4% without encoder <0.1% with encoder
	Steady state speed error	<0.5% without encoder <0.01% with encoder
	Torque response time	<10 ms
Power Converter Specifications	Rectifier type	Standard: 12-pulse diode rectifier Optional: 24-pulse diode rectifier
	Inverter type	Three-level NPC inverter
	GCT switching frequency	500 Hz
	Number of GCTs per phase	4
	Number of clamping diodes per phase	2
	Modulation technique	Hysteresis modulation generated by DTC scheme
	Inverter/rectifier switch failure mode	Non-rupture, non-arc

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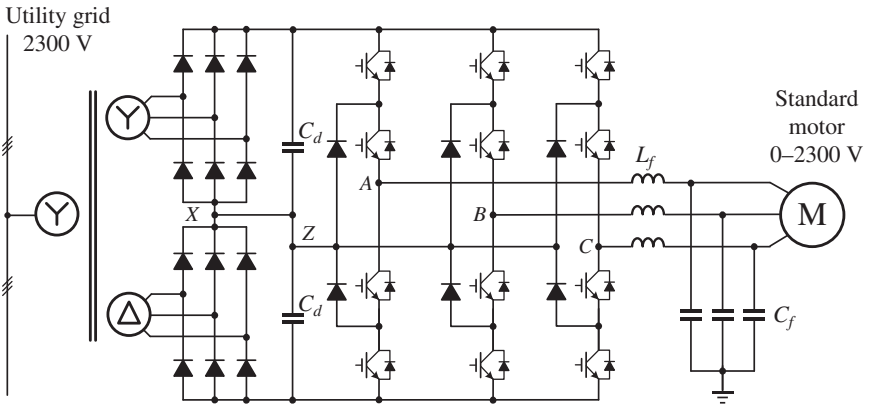

Figure 12.3-3 Three-level IGBT inverter fed drive.

As listed in the table, the maximum power ratings of the drive are 2.4 MW at 2300 V, 3.1 MW at 3300 V, and 4 MW at 4160 V. The power rating can be further increased to 7.2 MW at 4160 V with the two inverters operating in parallel. The operating voltage of the drive can be extended to 6600 V by adding a step-up autotransformer to the output of the 2300 V drive [8]. The leakage inductance of the autotransformer can also serve as the filter inductance, a viable solution for cost reduction.

Table 12.3-2 Main Specifications for the Three-Level IGBT-Based Drives

Rectifier	Standard: 12-pulse diode rectifier Optional: 24-pulse diode rectifier or active front end (PWM IGBT rectifier)
Displacement power factor ($\cos\phi$)	>0.96 (12-pulse diode rectifier)
Nominal utility/motor voltage	2300 V, 3300 V, 4160 V, 6600 V
Output power rating	0.8–2.4 MW@2300 V 1.0–3.1 MW@3300 V 1.3–4.0 MW@4160 V 4.7–7.2 MW@4160 V (parallel converter configuration) 0.6–2.0 MW@6600 V
Output voltage range	0–2300 V, 0–3300 V, 0–4160 V, 0–6600 V
Output frequency	0–100 Hz (standard)
Motor speed range	1:1000 (with encoder)
Drive system efficiency	Typically >98.5% (at rated operating point, excluding transformer losses)

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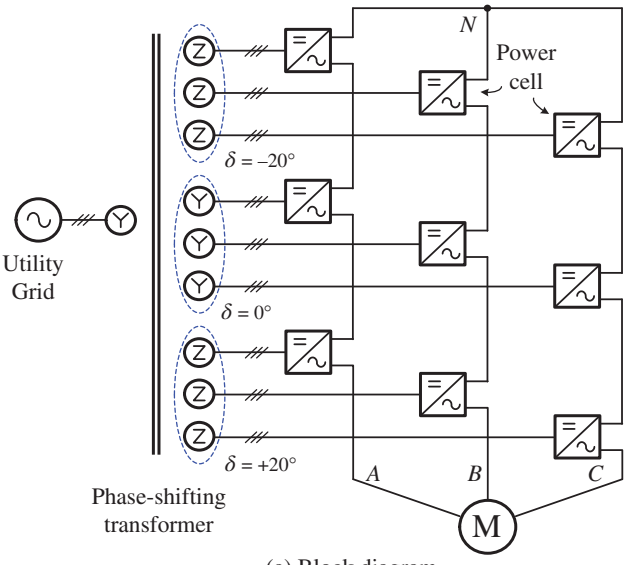
12.4 MULTILEVEL CASCADED H-BRIDGE (CHB) INVERTER FED DRIVES

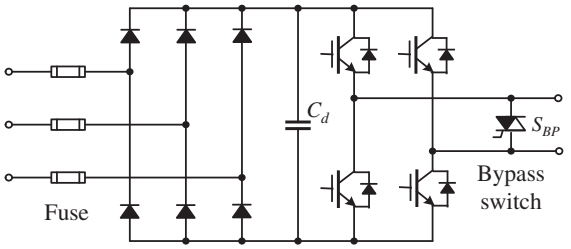
The multilevel cascaded H-bridge (CHB) inverter is one of the popular inverter topologies for the MV drive [9, 10]. Unlike other multilevel inverters where high-voltage IGBTs or GCTs are used, the CHB inverter normally employs low-voltage IGBTs as a switching device in H-bridge power cells. The power cells are then connected in cascade to achieve medium voltage operation.

12.4.1 CHB Inverter Fed Drives for 2300 V/4160 V Motors

The CHB inverters can be configured with different voltage levels. A seven-level CHB inverter fed MV drive is illustrated in Fig. 12.4-1. The phase-shifting transformer is



(a) Block diagram



(b) Power cell

Figure 12.4-1 Seven-level CHB drive with an 18-pulse diode rectifier.

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an indispensable device for the CHB inverter. It provides three main functions: (a) isolated power supplies for the power cells, (b) line current THD reduction, and (c) isolation between the utility and the converter for common-mode voltage mitigation.

The phase-shifting transformer has three groups of secondary windings. Each group has three identical windings. The phase shift between any two adjacent winding groups is 20° for the seven-level CHB drive. Since each of the secondary windings is connected to a three-phase diode rectifier, this configuration is essentially an 18-pulse separate type diode rectifier discussed in Chapter 3.

The power cell is composed of a three-phase diode rectifier, a dc capacitor, and a single-phase H-bridge inverter as shown in Fig. 12.4-1b. Each power cell is protected by fuses at the input and a bidirectional bypass switch S_{BP} at the output. The nominal output voltage of each power cell is typically 650 V (rms fundamental voltage). This design leads to the use of low-voltage components such as 1700 V IGBTs that are mass produced with a cost advantage over high-voltage ($\geq$3300 V) IGBTs. It is worth noting that the power cells should be insulated from each other and from ground at medium voltage levels even though they use low-voltage components.

Three power cells are cascaded at their ac output to form one line-to-neutral voltage of the three-phase system output. The inverter can produce seven distinct line-to-neutral voltage levels. The phase-shifted multicarrier sinusoidal modulation scheme presented in Chapter 7 is normally used in the CHB inverter. To boost the output voltage of each H-bridge, the 3rd harmonic injection method introduced in Chapter 6 can be adopted.

Table 12.4-1 summarizes the configuration of the multilevel CHB inverter fed drives for medium voltage applications. The topologies of the rectifier and inverter usually vary with the drive system operating voltages. For instance, with the utility/motor voltage of 4160 V, a 24-pulse diode rectifier and a nine-level CHB inverter can be selected. To reduce switching losses, the IGBT switching frequency $f_{sw,dev}$ is typically around 600 Hz. However, the equivalent inverter switching frequency $f_{sw,inv}$ is 4800 Hz, much higher than the device switching frequency $f_{sw,dev}$ due to the multilevel structure. The power rating of the drive is in the range of 0.2–14 MW [13].

The multilevel CHB inverter drive has a number of unique features:

- Modular construction for cost reduction and easy repair. The low-voltage power cells can be mass produced for the multilevel CHB inverters operating at various medium voltages. Defective power cells can be easily replaced, which minimizes the down time of the production line.
- Nearly sinusoidal output waveforms. The CHB inverter is able to produce ac voltage waveforms with small voltage steps. The inverter normally does not require any filters at its output. The motor is protected from high *dv/dt* stresses and has minimal harmonic power losses.
- Bypass function for improved system availability. The faulty power cells can be bypassed, and the drive can continue operating at reduced capacity with the remaining cells. Although bypassing the defective cells may cause three-phase unbalanced operation for the motor, it allows the process to continue. To produce a three-phase balanced line-to-line voltage with the faulty cells

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Table 12.4-1 Configurations of CHB Inverter Fed MV Drives Using Low-Voltage IGBTs

	Multipulse Diode Rectifier			Multilevel CHB Inverter					
Nominal Utility/Motor Voltage (V)	Rectifier Pulses	Secondary Windings	Transformer Secondary Cables	Power Cells	IGBTs (1700 V)	Voltage Levels (v_{AN})	Nominal Power Cell Output (V)	$f_{sw,dev}$ (Hz)	$f_{sw,inv}$ (Hz)
2300	18	9	27	9	36	7	650	600	3600
3300	18	9	27	9	36	7	650	600	3600
4160	24	12	36	12	48	9	650	600	4800
6600	36	18	54	18	72	13	650	600	7200

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bypassed, a neutral point shift method can be implemented [10–13]. This method is able to use all the healthy power cells to produce a three-phase balanced voltage with the highest possible magnitude.

- Provision for $N + 1$ redundancy. The drive system reliability can be improved by adding a redundant power cell to each of the inverter phase legs. When a power cell fails, it can be bypassed without causing reduction in the inverter output capacity.
- Nearly sinusoidal line current. This is mainly due to the use the multipulse diode rectifier.

There are a number of drawbacks for the multilevel CHB drive, including the following:

- High cost of phase-shifting transformer. The multiwinding transformer is the most expensive device in the CHB drive. Its secondary windings should be specially designed such that the symmetry of leakage inductances is preserved for harmonic current cancellation.
- Large number of cables. The multilevel CHB inverter drive normally requires 27–45 cables connecting the power cells to the transformer. It is, therefore, expensive to place the transformer away from the drive. With the transformer

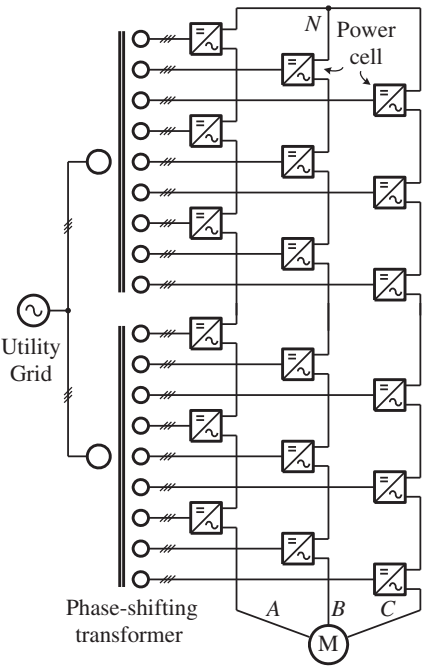

Figure 12.4-2 Topology of a 6600 V drive with two identical seven-level CHB inverters in cascade.

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installed inside the drive cabinet, the footprint of the drive increases, and so does the room cooling requirements.

- Large component count. The CHB inverter drive uses many low-voltage components, which potentially reduces reliability of the system.

12.4.2 CHB Inverter Drive for 6.6 kV/11.8 kV Motors

The operating voltage of the multilevel CHB inverter can be extended to 6600 V. A practical design for such a drive system is shown in Fig. 12.4-2, where two identical units of the seven-level CHB inverters are connected in cascade. The IGBTs used in the H-bridge power cells are typically rated at 1700 V, and each power cell produces a nominal voltage of 650 V. The power rating of the drive is in the range of 0.2–3.7 MW without IGBTs in series or parallel [9].

To drive the motors rated at 11.8 kV, the output voltage of each power cell can be increased to 1370 V. With five cells in cascade for the 11-level CHB inverter, its line-to-line voltages can reach 11.8 kV. High-voltage IGBTs should be used in this drive.

12.5 NPC/H-BRIDGE INVERTER FED DRIVES

Figure 12.5-1 shows the drive configuration using five-level NPC/H-bridge inverter [14]. The phase-shifting transformer has three identical groups of secondary

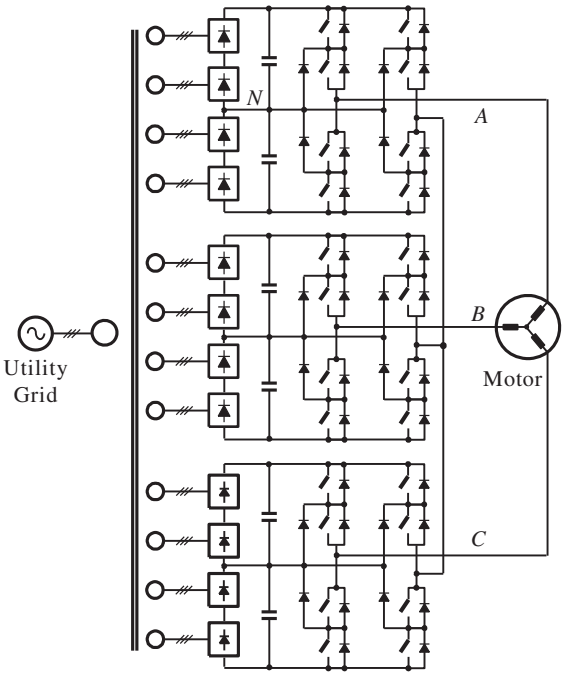

Figure 12.5-1 Five-level NPC/H-bridge inverter fed MV drive.

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windings. Each group of secondary windings feeds a 24-pulse diode rectifier. The phase shift between any two adjacent secondary windings in each group is 15°. The neutral point of the NPC/H-bridges is tied to the midpoint of the rectifiers to avoid inverter neutral voltage deviation. The inverter phase voltage v_{AN} is composed of five voltage levels while its line-to-line voltage v_{AB} has nine levels as discussed in Chapter 9.

The drive features very low ac line harmonic distortion, no switching devices in series, and low motor current THD. However, it requires a complex phase-shifting transformer with 12 secondary windings. The drive also requires a *dv/dt* filter at the inverter output. The power rating of the drive using high-voltage IGBTs is in the range of 0.2–4.8 MW.

12.6 ANPC INVERTER FED DRIVE

12.6.1 Three-Level ANPC Inverter Fed Drive

The configuration of a three-level ANPC inverter fed drive is shown in Fig. 12.6-1. Using the 3L-ANPC topology, as explained in Chapter 9, can make even distribution of power losses among the semiconductor switches, which leads to an increase in the inverter output power rating in comparison to the 3L-NPC inverter using the same switching devices. The neutral point *Z* of the dc bus is connected to the midpoint *X* of the two cascaded diode rectifiers. Since the dc output voltage of the diode rectifiers is equal and fixed, the voltage at the neutral point *Z* of the dc bus is also fixed. Thus, the neutral-point voltage control is no longer needed in this drive configuration.

The main specifications of the 3L-ANPC inverter fed drive are given in Table 12.6-1. The drive can operate at the motor voltages of 2300 V, 3300 V, and 4160 V. For the 2300 V drives, the ANPC inverter uses 18 pieces of 3300 V IGBTs without devices in series. For drives operating at higher voltages, two series-connected IGBTs can be used in each switch position in the inverter. The front-end converter can

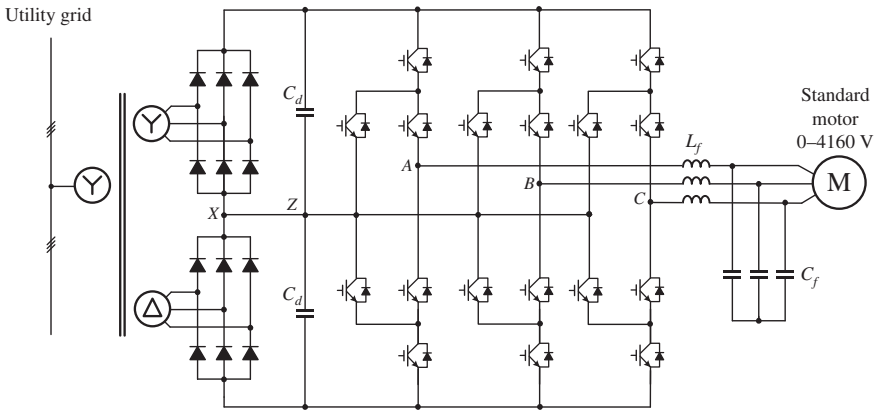

Figure 12.6-1 Three-level ANPC inverter fed drive with a 12-pulse diode rectifier.

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Table 12.6-1 Main Specifications for the 3L-ANPC Inverter Fed Drives

Drive System Specifications	Nominal input voltage	4160 V, 6000 V, 6600 V
	Output power rating	Air cooling: 0.3–2.0 MW Liquid cooling: 1.8–5.0 MW
	Output voltage rating	0–2300 V, 0–3300 V, 0–4160 V
	Output frequency	0–82.5 Hz
	Drive system efficiency	Typically >98%, excluding transformer
	Input power factor	>0.95
	Output waveform	Sinusoidal (with output filter)
	Motor type	Induction motor
Control Specifications	Control scheme	Direct torque control (DTC)
	Torque response time	10–20 ms
Power Converter Specifications	Rectifier type	Standard: 12-pulse diode rectifier Optional: 24-pulse diode rectifier
	Inverter type	3L-ANPC inverter
	Switching device and switching frequency	IGBT, 1000 Hz
	Modulation technique	Hysteresis modulation generated by DTC scheme

be either a 12- or 24-pulse diode rectifier, depending on applications and customer requirements. The direct torque control (DTC) scheme is used in the drive, which leads to the fast dynamic torque response. The power rating of the drive is in the range from 0.2 to 5 MW.

12.6.2 Five-Level ANPC Inverter Fed Drive

Similar to the three-level ANPC inverter, the five-level ANPC inverter introduced in Chapter 9 has also found practical applications in the MV drives [15]. There are two 5L-ANPC inverter fed drive configurations in the market. One has the same configuration as that shown in Fig. 12.6-1, where the 3L-ANPC inverter is replaced by a 5L-ANPC inverter. The 5L-ANPC inverter can produce a line-to-line voltage waveform with 9 voltage steps, which results in lower dv/dt and THD than the 3L-ANPC inverter. But this drive is not suitable for applications where four-quadrant operation is required since the power flow of the diode rectifier is in only one direction.

The other drive configuration uses a back-to-back converter topology, where a 5L-ANPC rectifier, instead of a 12-pulse diode rectifier, is used as the front-end converter. Both rectifier and inverter have identical converter topologies. This drive

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can be used in the areas where four-quadrant operation or frequent dynamic braking is required. The voltage and power rating of this drive can reach 6.9 kV and 2.0 MW [15].

12.7 MMC INVERTER FED DRIVE

An MV drive using modular multilevel converter (MMC) topology is shown in Fig. 12.7-1. This drive is developed and marketed for general applications, where four-quadrant operation is not required. A 12-pulse diode rectifier is used as a front end, but it can be replaced by 18- or 24-pulse diode rectifier depending on the grid-side harmonic requirements and the voltage ratings of the drive [16, 17].

The MMC employs submodules (SM) in cascade, and the number of half-bridge submodules per arm is dependent on the submodule voltage ratings and the output voltage of the converter. For the drives with rated voltage of 3.3–7.2 kV, 24, 30, or 36 low-voltage IGBT-based submodules can be used in the converter. This drive has been commercialized with power ratings up to 13.3 MW [16].

For drive applications where four-quadrant operation or high dynamic performance is required, the multipulse diode rectifier in Fig. 12.7-1 can be replaced with an MMC that has the identical structure as the inverter. This is a unique feature of the MMC topology since the converter requires only one dc supply (V_d), whereas the multilevel CHB inverter needs many isolated dc supplies for the cascaded power cells, which makes the four-quadrant operation much more complex and costly.

To achieve higher voltage operation with less number of cascaded submodules, a three-level submodules (3L-SM) shown in Fig. 12.7-1 instead of the half-bridge submodule can be used. This technology has been implemented in a commercial drive with the voltage rating up to 13.8 kV [17].

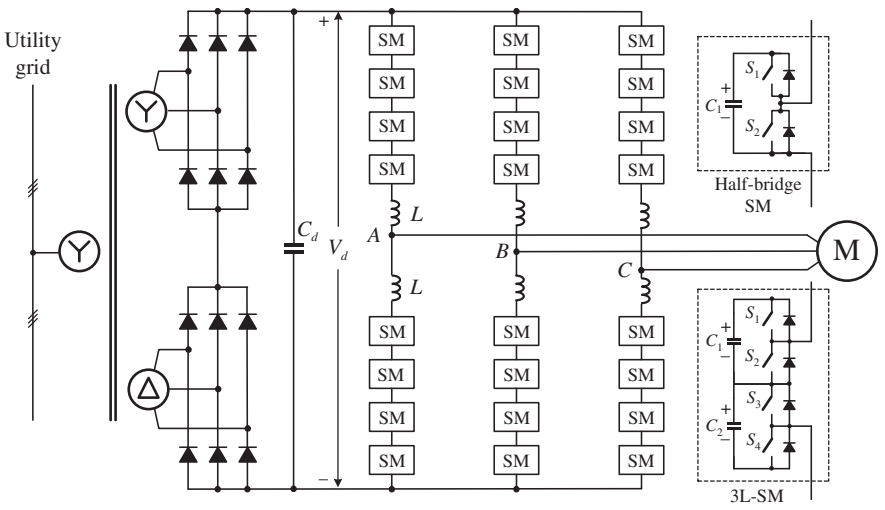

Figure 12.7-1 Modular multilevel converter fed drive with a 12-pulse diode rectifier.

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12.8 10 KV-CLASS DRIVES WITH MULTILEVEL CONVERTERS

As presented in the earlier chapters, a variety of multilevel converter topologies have been developed for MV drives. Most of the converters are suitable for drives with voltage rating from 2.3 to 7.2 kV. However, there is a market for the 10 kV-class drive, where the rated voltage of the drive is in the range of 10–13.8 kV. To develop such a drive, there are four possible methods that can be used to increase operating voltage of the drive: (1) increase the voltage levels of the multilevel converters (excluding multicell or multimodule converters), (2) put the switching devices in series, (3) put the power cells or submodules in cascade, and (4) combine some of the above methods.

In the first method, the operating voltage of the converter can be increased with the increase of the voltage levels. However, this method presents some limitations in the development of the 10 kV-class drive. For example, assuming the rated operating voltage of a three-level NPC converter is 4.16 kV, the operating voltage can be doubled if a six-level NPC converter is used. However, the number of components in the 6L-NPC converter is more than four times that in the 3L-NPC converter. It requires a total of 95 components, including 30 active switches, 60 clamping diodes, and 5 dc capacitors as shown in Table 8.6-2, whereas the 3L-NPC converter only requires 20 components, including 12 active switches, 6 clamping diodes, and 2 dc capacitors. The multilevel NPC converter is not a suitable topology in developing 10 kV-class drive due to the high cost.

The second method uses the switching devices in series to increase the operating voltage of the converter. To ensure equal voltage sharing among the switches in series, an equal static and dynamic voltage sharing circuit must be implemented. For transistor-based devices such as IGBTs, an active overvoltage clamping circuit through the IGBT gate driver presented in Section 2.3.3 can be implemented for each of the series-connected IGBTs. For the thyristor-based switches such as GCTs, a passive voltage equalization circuit in Fig. 2.3-1 can be used. In addition, the switching characteristics for the series-connected GCTs should be matched. The converter using this method to achieve 10 kV-class operation may have *dv*/*dt* issues. For example, in a 10 kV two-level converter using series-connected switches, the *dv*/*dt* in its output voltage could be higher than 10 kV/μs, which induces high bearing current and is also harmful to the stator winding insulation if it is not mitigated. This method may cause a reduction in converter reliability due to series-connected switches.

For the third method, multiple power cells or submodules are connected in cascade to raise the operating voltage of the converter for the 10 kV-class drive. There are no switches in series in the converter, and therefore the issues with equal voltage sharing for the series-connected switches are avoided. The modular structure of the converter provides a flexibility to scale up the converter operating voltage. The typical examples for the modular converter are the cascaded CHB converter and MMC topologies presented in the earlier chapters. They are a good candidate for use in 10 kV-class drives [13, 16, 17]. They also produce high-quality

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output voltage waveforms with low *dv*/*dt* and THD, but at the expense of high component count.

The 10 kV-class drive can also be realized by combining the first and second methods discussed above. For example, by adding series-connected switches to each of the switch positions in the three-level neutral point piloted (NPP) converter presented earlier, a 10 kV-class drive can be developed [19].

To summarize, the 10 kV-class drive can be implemented with different methods mentioned above. Many factors should be considered for the final selection of a converter topology for use in the 10 kV-class drive. These factors include the operating requirements and technical specifications of the drive, the selection of the switching devices, the reliability and protection of the converters, and the cost, efficiency, and size of the drive.

12.9 SUMMARY

This chapter presents a number of practical MV drive configurations using multilevel voltage source inverters, which include two-level VSI, three-level NPC inverter, multilevel CHB inverter, NPC/H-bridge inverter, ANPC inverter, and MMC topology. Most of the MV drives presented in this chapter employ a multipulse diode rectifier for cost reduction and line current harmonic mitigation. This type of low-cost drives can be used for general applications where the dynamic performance of the drive is not a top priority. For applications where a four-quadrant operation or high dynamic performance is required, the multipulse diode rectifier in the drive can be replaced by an active rectifier with the same converter topology as the inverter. The chapter ends with a discussion of the development of 10 kV-class drives.

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Chapter 13

Current Source Inverter Fed Drives

13.1 INTRODUCTION

The current source inverter (CSI) technology is well suited for medium voltage (MV) drives. The CSI fed drive features simple converter structure, motor-friendly waveforms, inherent four-quadrant operation capability, and reliable fuseless short-circuit protection, but it may have a slower dynamic response than the VSI drive. As indicated in Chapter 1, the majority (around 85%) of the installed MV drives are for applications where the drive dynamic performance may not be the top requirement, such as high-power fans, pumps, compressors, and ship propellers. Therefore, the CSI drive is perfectly fitted into this type of applications. The power rating of the PWM current source drives is normally in the range of 1–10 MW and can be further increased with parallel inverters. For a higher power rating up to 100 MW, the load-commutated inverter (LCI) is a preferred choice, with which the voltage source inverters (VSI) normally cannot compete in terms of the cost and efficiency of the system.

This chapter presents a number of CSI drives with various front-end converters such as single-bridge PWM rectifier, dual-bridge PWM rectifier, and phase-shifted SCR rectifiers. The advantages and limitations of these drives are analyzed and the main specifications are provided. A transformerless CSI drive for standard ac motors is introduced. The chapter ends with an introduction to LCI synchronous motor drives.

13.2 CSI DRIVES WITH PWM RECTIFIERS

13.2.1 CSI Drives with Single-Bridge PWM Rectifier

Figure 13.2-1 shows a typical CSI drive using a single-bridge PWM current source rectifier (CSR) as a front end. The rectifier and inverter have identical topology using

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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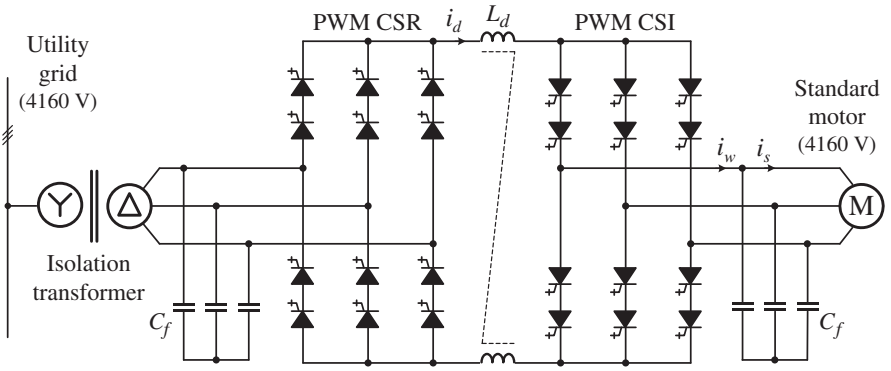

Figure 13.2-1 A typical 4160 V CSI drive with a PWM rectifier and inverter.

symmetrical GCTs. With the GCT voltage rating of 6000 V and two GCTs connected in series in each of the converter branches, the drive is capable of operating at the utility voltage of 4160 V (line-to-line). For higher voltage applications, the number of series-connected devices can be increased while the converter topology remains unchanged. For example, three GCTs can be in series for the 6600 V drives.

The PWM rectifier can use either the space vector modulation (SVM) or selective harmonic elimination (SHE) schemes. The SHE scheme is preferred due to its superior harmonic performance. With a switching frequency of 420 Hz and the utility frequency of 60 Hz, a maximum of three low-order current harmonics (the 5th, 7th, and 11th) can be eliminated. The other high-order harmonics can be attenuated by the filter capacitor, leading to a very low line current THD.

The dc current i_d can be adjusted by the rectifier delay angle or modulation index control presented in Chapter 11. The delay angle control produces a lagging power factor that can compensate the leading current produced by the filter capacitor, resulting in an improved power factor. Further, when the CSI drive is for fan or pump applications, its input power factor can be made near unity over a wide speed range [1]. This can be realized by using the SHE scheme with a delay angle control and properly selecting the line- and motor-side filter capacitors. This control scheme features simple design procedure, easy digital implementation, and reduced switching loss (due to the lack of bypass operation), and therefore is a preferred control scheme for the PWM CSI drive.

A typical switching pattern arrangement for the inverter is illustrated in Fig. 13.2-2, where f_{sw} is the GCT switching frequency, f_1 is the inverter fundamental output frequency, and N_p is the number of pulses per half cycle of the inverter PWM current i_w. The SHE scheme (without bypass pulses) is used at high inverter output frequencies while the trapezoidal pulse width modulation (TPWM) or space vector modulation (SVM) scheme is utilized when the motor operates at low speeds. As discussed in the previous chapter, the SHE scheme has a better harmonic performance than TPWM, but it is difficult, if not impossible, to eliminate more than four harmonics in i_w simultaneously. The switching frequency of the inverter is normally below 500 Hz in industrial CSI drives.

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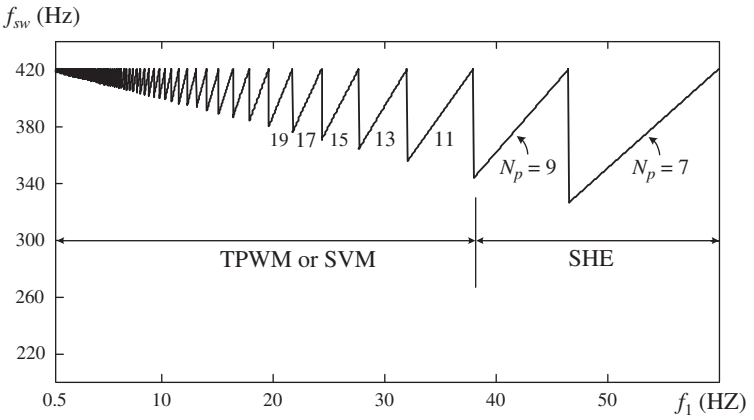

Figure 13.2-2 Switching frequency f_{sw} versus inverter output frequency f_1.

It is worth noting that the GCT device is capable of operating at higher switching frequencies. The main reasons for limiting the switching frequency is the GCT's high thermal resistance that prevents efficient heat transfer from the device to its heatsink. In addition, the device switching frequency should be kept low for switching loss reduction. Prior to the advent of the GCT device, GTO was the main switching device for the high-power CSI drive, where the device switching frequency was around 200 Hz [2].

Similar to the VSI drives, the rectification and inversion process in the CSI drive generates common-mode voltages. If not mitigated, the common-mode voltage would appear on the motor, causing premature failure of its winding insulation. The problem can be effectively solved by introducing an isolation transformer and grounding the neutral of the inverter filter capacitor as shown in Fig. 13.2-1. In doing so, the motor is not subject to any common-mode voltages. Therefore, the CSI drive is suitable for retrofit applications, where standard ac motors are usually used.

The CSI MV drive has the following features:

- Simple converter topology with low component count. The converter topology is simple and independent of operating voltages. Both converters use symmetrical GCT devices which do not require anti-parallel freewheeling diodes, leading to a minimum component count.
- Motor-friendly waveforms. The motor voltage and current waveforms are close to sinusoidal and does not contain any voltage steps with high *dv/dt*. These waveforms are compatible with standard ac motors without derating.
- High input power factor. The CSI drive using the PWM rectifier as a front end has a minimum input power factor of 98% over a wide speed range [3]. This is in contrast with the conventional CSI drive with an SCR rectifier, where the power factor varies with the drive operating conditions.
- Simple PWM scheme. The SHE and TPWM schemes are normally used in the current source converters. These schemes are much simpler than those for

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multilevel voltage source inverters. Only six gate signals are required for the six groups of synchronous switches per converter. The number of the gate signals does not vary with the number of the switches in series.

- Reliable fuseless short-circuit protection. In case of short circuit at the inverter outputs, the rate of rise of the dc current is limited by the dc choke, providing sufficient time for the drive controller to react. For a quick system shutdown, the rectifier can produce a negative dc voltage, forcing the dc current to fall quickly. The drive can be safely shut down at the moment when the dc current falls to zero. The drive does not need fuses for the short-circuit or over-current protection.
- $N + 1$ provision for high reliability. In certain applications where high system reliability is required, a redundant switching device ($N + 1$) can be added to each of the six converter branches. Since the GCT device is normally short circuited at failure, the CSI drive is able to operate continuously at its full capacity with a failed device.
- Long input and output cables. There is virtually no limit on the length of the cables connecting the transformer to the rectifier or connecting the inverter to the motor due to near sinusoidal input and output waveforms.
- Four-quadrant operation and regenerative braking capability. The power flow in the CSI drive is bidirectional. No additional components are required for four-quadrant operation and dynamic braking.

The main drawbacks of the CSI drive include

- Limited dynamic performance. This is mainly due to the use of a dc choke that limits the rate of dc current changes. The dynamic performance of the drive can be significantly improved when the inverter output current is directly regulated by modulation index instead of dc current adjustment through the rectifier. However, this method may cause additional conduction losses for the switching devices.
- Potential LC resonances. The line- and motor-side filter capacitors constitute resonant modes with the line and motor inductances. The line-side LC resonances can be effectively avoided by properly sizing the filter capacitor and line reactor such that the resultant LC resonant frequency is lower than that of the lowest harmonic produced by the rectifier. This approach can be equally applied to suppress the motor-side LC resonances. In addition, the active damping control discussed in Chapter 11 also provides an effective means for LC resonance suppression.

Figure 13.2-3 shows the picture of a 4160 V CSI drive system. The left cabinet contains an advanced digital controller for the drive system and the line- and motor-side filter capacitors. The middle cabinet houses two identical PWM converters, one for the rectifier and the other for the inverter. Each converter is composed of 12 pieces of 6000 V symmetrical GCTs. All the GCT devices in both converters are installed in

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Figure 13.2-3 Picture of a 4160 V drive with PWM GCT rectifier and inverter. Courtesy of Rockwell Automation Canada.

six power-cage modules designed for quick device assembly and replacement. The dc choke and air-cooling system are installed in the right cabinet.

The main specifications for the MV CSI drives are given in Table 13.2-1 [3]. They are divided into three groups. The drive system specifications include typical voltage/power ratings, efficiency, input power factor, line current THD, and ride-through capability. The control specifications include the control scheme, parameter tuning method, speed regulation, and bandwidth of the controllers. The converter type, modulation techniques, and switching devices are listed as the converter specifications.

13.2.2 CSI Drives for Custom Motors

In certain applications where a new motor is to be ordered, the total cost of the drive can be reduced by replacing the isolation transformer with a three-phase line reactor L_s as shown in Fig. 13.2-4. The motor in this case must be custom-made with enhanced insulation to withstand common-mode voltage stresses. The additional benefits of such a drive include reduced footprint and operating cost due to the elimination of the isolation transformer.

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Table 13.2-1 Main Specifications for the Medium Voltage CSI Drives

Drive System Specifications	Nominal input voltage	2300 V, 3300 V, 4160 V, 6600 V
	Output power rating	200–9000 HP (150–6700 kW)
	Output voltage rating	0–2300 V, 0–3300 V, 0–4160 V, 0–6600 V
	Output frequency	0.2–85 Hz
	Drive system efficiency	>96.0% (including transformer losses)
	Input power factor	Typically >0.98 (with PWM rectifier)
	Line current THD	Typically <5% (with PWM rectifier)
	Output waveform	Near sinusoidal current and voltage
	Motor type	Induction or synchronous
	Overload capability	150%—one minute (standard)
	Power loss ride through	5 cycles
	Cooling	Forced air or liquid
	Regenerative braking capability	Inherent. No additional hardware or software required
Control Specifications	Control scheme	Direct field oriented (vector) control
	Shaft encoder	Optional
	Parameter tuning method	Automatic self-tuning control
	Speed regulation	0.1% without shaft encoder 0.01% with shaft encoder
	Speed regulator bandwidth	5–25 radians/second
	Torque regulator bandwidth	15–50 radians/second
Power Converter Specifications	Rectifier type	6-pulse SCR, 18-pulse SCR or PWM GCT
	Number of rectifier switches per phase	2@2400 V, 4@3300 V/4160 V, 6@6600 V
	Inverter type	PWM current source
	Number of inverter switches per phase	2@2400 V, 4@3300 V/4160 V, 6@6600 V
	GCT/SCR peak off-state voltage rating	6000 V
	Modulation technique	Rectifier: SHE Inverter: SHE combined with TPWM
	GCT switching frequency	Rectifier: <500 Hz Inverter: <500 Hz
	Inverter/rectifier switch failure mode	Non-rupture, non-arc

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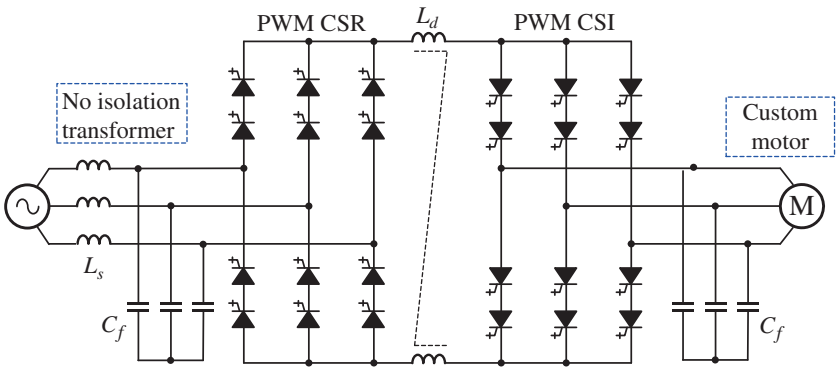

Figure 13.2-4 PWM CSI drive without isolation transformer.

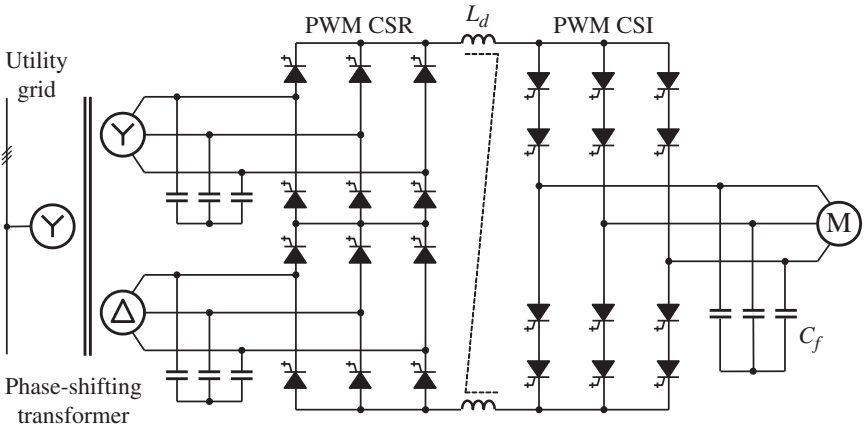

Figure 13.2-5 A 4160 V CSI drive with dual-bridge PWM rectifier for sinusoidal inputs.

13.2.3 CSI Drives with Dual-Bridge PWM Rectifier

Figure 13.2-5 illustrates a 4160 V CSI drive with a dual-bridge PWM rectifier. As discussed in Chapter 11, the rectifier uses SHE scheme to eliminate the 11th and 13th harmonics and phase-shifting transformer to cancel the 5th, 7th, 17th, and 19th harmonics. The other remaining high-order harmonics can be absorbed by the filter capacitors. As a result, the line current waveform is virtually sinusoidal [4, 5]. This drive is suitable for applications that require extremely low line current distortion.

13.3 TRANSFORMERLESS CSI DRIVE FOR STANDARD AC MOTORS

As discussed earlier, one of the major problems in both CSI- and VSI-based MV drives is the common-mode voltage, which would cause premature failure of motor

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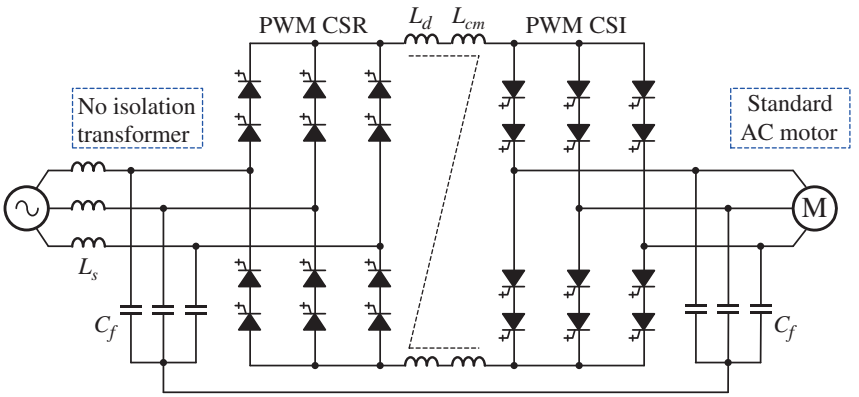

Figure 13.3-1 Transformerless CSI drive with an integrated dc choke.

winding insulation if not mitigated. The problem can be solved by introducing an isolation or phase-shifting transformer, but this solution brings a few drawbacks such as high manufacturing cost, increased drive size and weight, and high operating cost due to transformer losses. Another solution is to use a custom-made ac motor with an increased insulation level. This solution, however, cannot be used for retrofit applications where standard motors are already in use.

Figure 13.3-1 shows the schematic of a transformerless CSI drive, where an integrated dc choke with a single magnetic core is used [6]. The choke can provide two inductances, a differential inductance L_d that is inherently required by the CSI drive and a common-mode inductance L_{cm} that can block the common-mode voltage. The use of the integrated dc choke leads to the elimination of the isolation transformer, resulting in a significant reduction in manufacturing and operating costs. To ensure that the motor is not subject to any common-mode voltages, the neutral points of the line- and motor-side filter capacitors are connected together directly or through a small RL circuit. The detailed analysis of this drive is given in Chapter 17 (Transformerless MV Drives).

13.4 CSI DRIVE WITH MULTIPULSE SCR RECTIFIER

13.4.1 CSI Drive with 18-Pulse SCR Rectifier

The multipulse SCR rectifiers presented in Chapter 4 can also be used as a front end for the CSI drive. Among the various rectifier topologies, the 18-pulse SCR rectifier is a preferred choice due to its high performance to price ratio. For a given voltage and power rating, the 12-pulse rectifier has a cost advantage over the 18-pulse, but its line current THD may not meet the harmonic guidelines set by most standards. The 24-pulse rectifier has a better line current THD than the 18-pulse, but it costs more due to the increased number of SCR devices and complex phase-shifting transformer.

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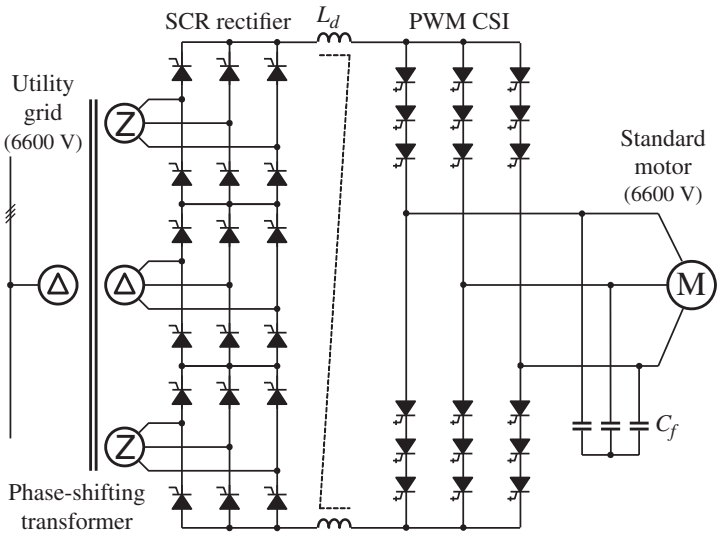

Figure 13.4-1 A typical 6600 V CSI drive with an 18-pulse SCR rectifier.

Figure 13.4-1 shows a typical CSI fed drive with an 18-pulse SCR rectifier. The rectifier has three units of 6-pulse rectifiers in cascade while the inverter uses three series-connected GCT devices per branch. With the voltage rating of 6000 V for SCRs and GCTs, the drive is capable of operating at the utility voltage of 6600 V (line-to-line). The CSI drive can meet the harmonic requirements set by IEEE standard 519-2014, and can also be used in retrofit applications with standard ac motors.

13.4.2 Low-Cost CSI Drive with 6-Pulse SCR Rectifier

Figure 13.4-2 shows a low-cost CSI drive with a 6-pulse SCR rectifier. The drive provides the most economical solution in certain applications. An optional harmonic

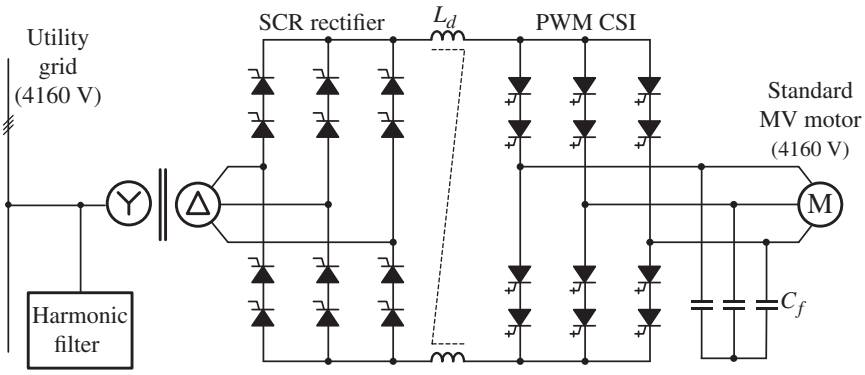

Figure 13.4-2 A low-cost CSI drive with 6-pulse SCR rectifier.

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filter can be used to reduce the line current THD. The installation cost can be further reduced for new motor applications where the isolation transformer can be replaced by a three-phase line reactor.

The low-cost CSI drive can also serve as a soft starter for motors requiring a high starting torque. The conventional SCR-based variable-voltage motor starters are incapable of producing a high starting torque due to the reduced starting voltage and the fixed supply frequency.

13.5 LCI DRIVES FOR SYNCHRONOUS MOTORS

The load commuted inverter (LCI) fed synchronous motor drives are often found in high-power applications with a power rating up to 100 MW. The LCI drive uses low-cost SCR thyristors in the rectifier and inverter, leading to a substantial cost reduction. The LCI drives are often used for high-power fans, pumps, compressors, and wind tunnels, where the drive dynamic performance is not the top requirement.

13.5.1 LCI Drives with 12-Pulse Input and 6-Pulse Output

Figure 13.5-1 illustrates the typical configuration of an LCI fed synchronous motor drive. It uses a 12-pulse SCR rectifier with a dc link choke to provide a controllable dc current to the inverter. The number of series connected SCRs depends on the voltage ratings of the SCRs and the utility supply [7, 8].

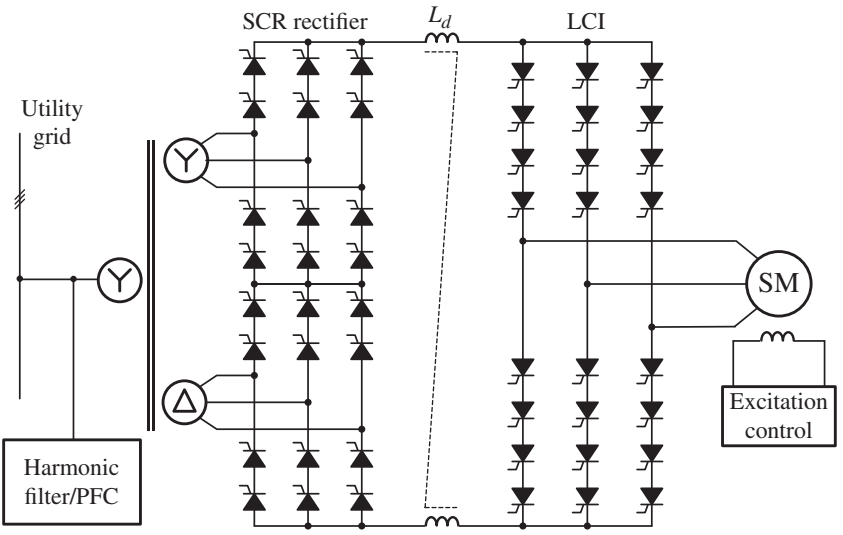

Figure 13.5-1 An LCI drive system with 12-pulse input and 6-pulse output.

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It is well known that the line current of the 12-pulse SCR rectifier does not contain the 5th or 7th harmonics, but it does have a large amount of the 11th and 13th harmonics. Therefore, a harmonic filter is often required for the LCI drive. The filter is of an LC series-resonant type, normally tuned at the 11th and 13th harmonics. By a proper selection of LC parameters, the turned filter can also serve as a power factor compensator (PFC).

The main features of the LCI fed synchronous drive include low cost, high efficiency, reliable operation, and inherent regenerative braking capability. The main drawbacks are high torque pulsation, slow transient response, and variable input power factor.

13.5.2 LCI Drives with 12-Pulse Input and 12-Pulse Output

The harmonic currents in the LCI drive with 6-pulse output cause harmonic torques and power losses in the synchronous motor. The harmonic torque may excite mechanical vibrations due to the mechanical resonant modes of the drive system. These problems can be mitigated by the LCI drive with a 12-pulse output as shown in Fig. 13.5-2.

The synchronous motor has two sets of separate stator windings with 30° electrical phase shift between them. The dominant harmonic torque produced by the 5th and 7th harmonic currents in the two stator windings can be cancelled by the six-phase motor. As a result, the torque pulsation is substantially reduced and the mechanical stress on the shaft train is also decreased.

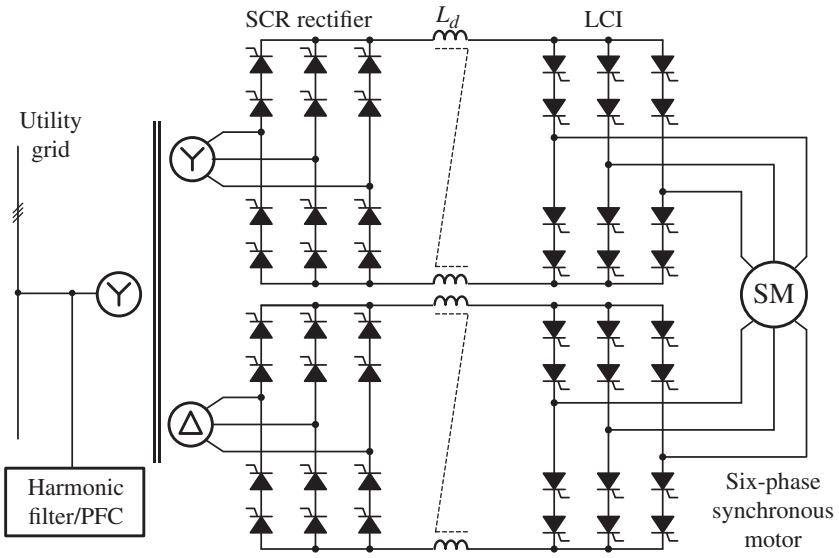

Figure 13.5-2 LCI drive with 12-pulse input and 12-pulse output.

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13.6 SUMMARY

This chapter focuses on a number of CSI drives with various front-end converters including single-bridge PWM rectifier, dual-bridge PWM rectifier, and phase-shifted SCR rectifiers. The CSI drives are widely used in industry where the dynamic performance of the drive is not the top requirement, such as high-power fans, pumps, compressors, and ship propellers. The advantages and limitations of the drives are analyzed and the main specifications are provided.

A transformerless PWM CSI drive for standard ac motors is also introduced. The drive uses an integrated dc choke instead of isolation or phase-shifting transformer to block the common-mode voltages, which leads to a significant reduction in manufacturing cost. The detailed analysis of this drive is given in Chapter 17 of this book.

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Chapter 14

Control of Induction Motor Drives

14.1 INTRODUCTION

The induction motor can be controlled by a variety of control schemes. The simplest control scheme is the volts per hertz (V/F) control, where the stator voltage of the induction motor is controlled to be proportional to its frequency when the motor speed is made adjustable. This control scheme can keep the stator magnetic flux approximately constant, typically at its rated value, when the motor operates in steady state. However, the stator flux cannot be controlled during the transients or dynamic adjustments of the motor speeds. The electromagnetic torque of the motor cannot be controlled either. Therefore, the drive system using the V/F control is not suitable for applications where high dynamic performance is required.

A number of advanced control schemes have been developed for the drives requiring high dynamic performance, among which the field oriented control (FOC) and direct torque control (DTC) have become the industrial standards and are widely used in high-power medium voltage (MV) drives. The basic principle of these schemes is to decouple the torque-producing and flux-producing components from the measured three-phase stator current such that the torque and magnetic flux of the induction motor can be controlled independently, the same way as the dc motor drive. In doing so, a superior dynamic performance is achieved. These two methods are also widely used in the drives where the dynamic performance of the drive is not a top priority since they can be cost-effectively implemented in real time by a digital controller.

This chapter focuses on the FOC and DTC schemes for induction motor drives. It starts with an introduction to reference frame transformation, followed by the dynamic model of the induction motor. Direct and indirect FOC schemes for the voltage source inverter (VSI) and current source inverter (CSI) fed drives are presented. The DTC for the MV drive is elaborated. Important concepts are illustrated with computer

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simulations. The chapter ends with a comparison between the FOC and DTC schemes for the induction motor drives.

14.2 REFERENCE FRAME TRANSFORMATION

The use of reference frame theory can simplify the analysis of electric machines and also provide a powerful tool for the digital implementation of sophisticated control schemes for the induction motor drives. A number of reference frames have been proposed over the years [1], of which the stationary and synchronous reference frames are most commonly used. The transformation of variables between the two frames is presented below.

14.2.1 *abc/dq* Frame Transformation

The transformation of the three-phase (*abc*-axis) variables of an induction motor to the equivalent two-phase (*dq*-axis) variables can be performed by the following *abc*/*dq* frame transformation:

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos\theta & \cos(\theta - 2\pi/3) & \cos(\theta - 4\pi/3) \\ -\sin\theta & -\sin(\theta - 2\pi/3) & -\sin(\theta - 4\pi/3) \end{bmatrix} \cdot \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \quad (14.2\text{-}1)$$

where x represents either current, voltage, or flux linkage, and θ is the angular displacement between the a-axis and d-axis of the three-phase and two-phase reference frames as shown in Fig. 14.2-1. The three-phase variables, x_a, x_b, and x_c, are in the **stationary reference frame** which does not rotate in space whereas the two-phase variables, x_d and x_q, are in the **synchronous reference frame** whose direct (d) and quadrature (q) axes rotate in space at the synchronous speed ω_e. Note that ω_e is the angular electrical (not mechanical) speed of the rotating magnetic field of the motor, given by

$$\omega_e = 2\pi f_s \quad (14.2\text{-}2)$$

where f_s is the frequency of the stator variables. The angle θ can be found from

$$\theta(t) = \int_0^t \omega_e(t)dt \quad (14.2\text{-}3)$$

Note that the transformation equation of (14.2-1) is valid only for a three-phase balanced system, in which

$$x_a + x_b + x_c = 0 \quad (14.2\text{-}4)$$

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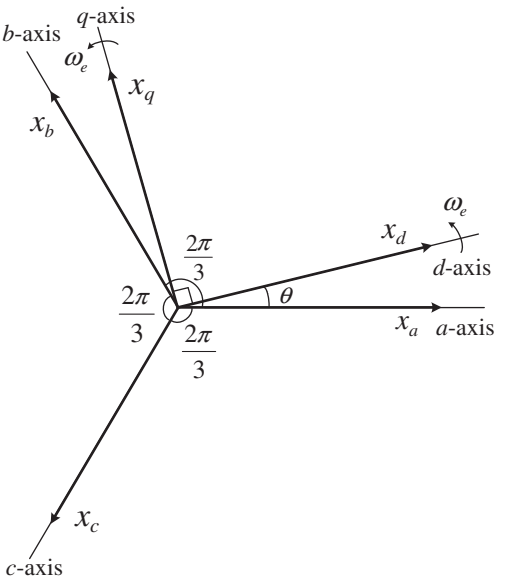

Figure 14.2-1 Variables in three-phase (*abc*) stationary frame and two-phase (*dq*) synchronous frame.

Similarly, the two-phase variables in the synchronous frame can be transformed back to the three-phase stationary frame by

$$\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \cos(\theta - 2\pi/3) & -\sin(\theta - 2\pi/3) \\ \cos(\theta - 4\pi/3) & -\sin(\theta - 4\pi/3) \end{bmatrix} \cdot \begin{bmatrix} x_d \\ x_q \end{bmatrix} \tag{14.2-5}$$

which is referred to as ***dq/abc* transformation**.

The relationship between a space vector and its phase variables is illustrated in Fig. 14.2-2a, where a current space vector $\vec{i}_s$ rotates at a certain speed ω in the stationary frame (refer to Chapter 6 for space vector definition). Its phase currents i_{as}, i_{bs}, and i_{cs} can be obtained by projecting $\vec{i}_s$ onto their corresponding *abc* axes. Since the three axes are stationary in space, each of the phase currents varies one cycle over time when $\vec{i}_s$ rotates one revolution in space. If the length (magnitude) and the rotating speed of $\vec{i}_s$ are constant, the waveforms of the phase currents are sinusoidal with a phase displacement of $2\pi/3$ between each other.

Figure 14.2-2b illustrates another case where the current vector $\vec{i}_s$ is in the *dq*-axis synchronous reference frame. Assuming that $\vec{i}_s$ rotates at the same speed as that of the *dq*-axis frame, the stator current angle ϕ, which is the angle between $\vec{i}_s$ and the *d*-axis, is constant. The resultant *dq*-axis current components, i_{ds} and i_{qs}, are of dc signals. As will be seen in the subsequent sections, this transformation can be utilized to simplify the simulation, design, and digital implementation of drive systems, where

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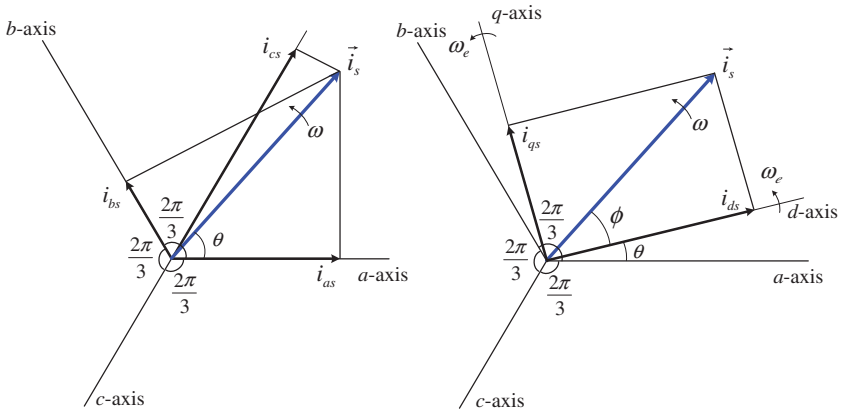

(a) *abc*-axis currents in the stationary frame (b) *dq*-axis currents in the synchronous frame

Figure 14.2-2 Decomposition of current vector $\vec{i}_s$.

any three-phase ac variables can be effectively represented by the two dc variables in the *dq*-axis frame.

14.2.2 *abc*/$\alpha\beta$ Stationary Transformation

With the rotating speed of the two-phase reference frame set at zero ($\omega_e = 0$), both two-phase and three-phase frames are stationary in space. The two-phase stationary frame is called $\alpha\beta$-axis reference frame. The transformation of the three-phase variables to the two-phase variables can be obtained by setting θ in (14.2-1) to zero, from which

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \cdot \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \tag{14.2-6}$$

where x_α and x_β are the variables on α- and β-axis of the two-phase reference frame, and the transformation is referred to as ***abc*/$\alpha\beta$ transformation**. Note that the α-axis variable can be expressed as

$$x_\alpha = \frac{2}{3}\left(x_a - \frac{1}{2}x_b - \frac{1}{2}x_c\right) = x_a \tag{14.2-7}$$

which is equal to the *a*-axis variable x_a. Similarly, the two-phase to three-phase stationary transformation, which is denoted ***$\alpha\beta$*/*abc* transformation**, can be performed by

$$\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \cdot \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} \tag{14.2-8}$$

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14.3 INDUCTION MOTOR DYNAMIC MODELS

There are two commonly used dynamic models for the induction motor. One is based on space vector theory and the other is derived from *dq*-axis theory. The space vector model features compact mathematical expressions and concise space vector diagram whereas the *dq*-axis model does not need to use complex numbers or variables. Both models are equally valid for the analysis of transient and steady-state performance of the induction motor. In what follows, the two models are presented and their relationship is revealed.

14.3.1 Space Vector Motor Model

It is assumed in the following analysis that the induction motor is three-phase symmetrical and its magnetic core is linear with a negligible core loss. The space vector model for an induction motor is generally composed of three sets of equations [2]. The first set is the voltage equations, given by

$$\begin{cases} \vec{v}_s = R_s\vec{i}_s + p\vec{\lambda}_s + j\omega\vec{\lambda}_s \\ \vec{v}_r = R_r\vec{i}_r + p\vec{\lambda}_r + j(\omega - \omega_r)\vec{\lambda}_r \end{cases} \tag{14.3-1}$$

where

$\vec{v}_s$, $\vec{v}_r$ – stator and rotor voltage vectors,
$\vec{i}_s$, $\vec{i}_r$ – stator and rotor current vectors,
$\vec{\lambda}_s$, $\vec{\lambda}_r$ – stator and rotor flux-linkage vectors,
R_s, R_r – stator and rotor winding resistance,
ω – rotating speed of an arbitrary reference frame,
ω_r – rotor angular speed (electrical), and
p – derivative operator ($p = d/dt$).

The terms $j\omega\vec{\lambda}_s$ and $j(\omega - \omega_r)\vec{\lambda}_r$ on the right-hand side of (14.3-1) are referred to as **speed voltages** which are induced by the rotation of the reference frame.

The second set is the flux-linkage equations

$$\begin{cases} \vec{\lambda}_s = L_s\vec{i}_s + L_m\vec{i}_r \\ \vec{\lambda}_r = L_r\vec{i}_r + L_m\vec{i}_s \end{cases} \tag{14.3-2}$$

where

$L_s = L_{ls} + L_m$ – stator self-inductance,
$L_r = L_{lr} + L_m$ – rotor self-inductance,
L_{ls}, L_{lr} – stator and rotor leakage inductances, and
L_m – magnetizing inductance.

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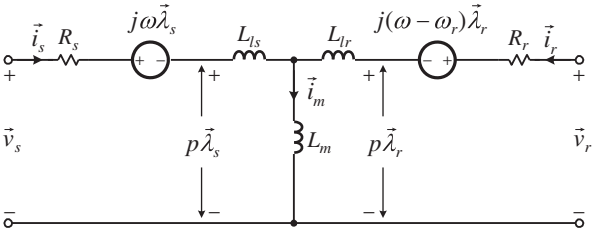

Figure 14.3-1 Space vector model of an induction motor in the arbitrary reference frame.

Note that all the rotor parameters and variables, such as R_r, L_{lr}, $\vec{i}_r$, and $\vec{\lambda}_r$, in the above questions are referred to the stator side.

The third set is the motion equation, given by

$$\begin{cases} \dfrac{J}{P}p\omega_r = T_e - T_L \\ T_e = \dfrac{3P}{2}\mathrm{Re}\left(j\,\vec{\lambda}_s\,\vec{i}_s^{\,*}\right) = -\dfrac{3P}{2}\,\mathrm{Re}\left(j\,\vec{\lambda}_r\,\vec{i}_r^{\,*}\right) \end{cases} \tag{14.3-3}$$

where

J – total moment of inertia of the rotor and load,
P – number of pole pairs,
T_L – load torque, and
T_e – electromagnetic torque.

The above equations constitute the space vector model of the induction motor whose schematic representation is given in Fig. 14.3-1. The motor model is in the arbitrary reference frame which rotates in space at the arbitrary speed of ω.

In the simulation and digital implementation of advanced control systems, the motor models in the synchronous and stationary reference frames are often employed. The synchronous-frame motor model can be easily obtained by setting the arbitrary speed ω in (14.3-1) to the synchronous speed ω_e. Figure 14.3-2a shows the equivalent circuit of a squirrel cage motor in the synchronous frame, where the rotor winding is shorted ($\vec{v}_r = 0$) and ω_{sl} is the angular slip frequency, given by

$$\omega_{sl} = \omega_e - \omega_r \tag{14.3-4}$$

To obtain the model in the stationary (stator) frame, we can set the arbitrary speed ω of the rotating reference frame to zero. The resultant equivalent circuit is shown in Fig. 14.3-2b.

14.3.2 *dq*-Axis Motor Model

The induction motor *dq*-axis model can be derived using the three-phase circuit theory and then transformed into the two-phase (*dq*-axis) frame [1]. Alternatively, it can also

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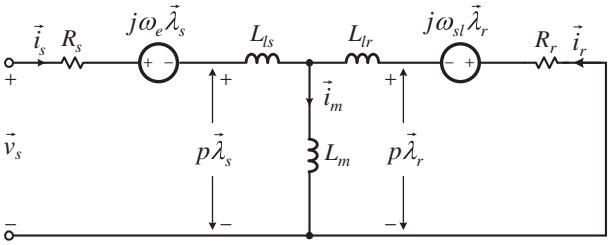
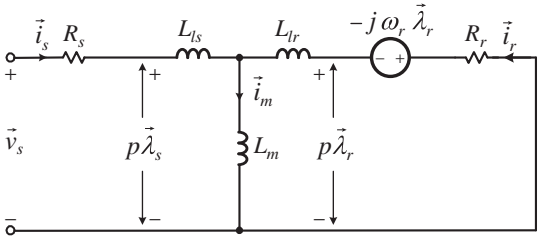


(a) Motor model in the synchronous frame



(b) Motor model in the stationary (stator) frame

Figure 14.3-2 Space vector models for a squirrel cage induction motor.

be obtained by decomposing the space vectors in the space vector motor model into the d- and q-axis components [2], that is,

$$\begin{cases} \vec{v}_s = v_{ds} + j\, v_{qs};\ \ \vec{i}_s = i_{ds} + j\, i_{qs};\ \ \vec{\lambda}_s = \lambda_{ds} + j\, \lambda_{qs} \\ \vec{v}_r = v_{dr} + j\, v_{qr};\ \ \vec{i}_r = i_{dr} + j\, i_{qr};\ \ \vec{\lambda}_r = \lambda_{dr} + j\, \lambda_{qr} \end{cases} \tag{14.3-5}$$

Substituting (14.3-5) to (14.3-1), the dq-axis voltage equations for the induction motor can be obtained:

$$\begin{cases} v_{ds} = R_s i_{ds} + p\lambda_{ds} - \omega\lambda_{qs} \\ v_{qs} = R_s i_{qs} + p\lambda_{qs} + \omega\lambda_{ds} \\ v_{dr} = R_r i_{dr} + p\lambda_{dr} - (\omega - \omega_r)\lambda_{qr} \\ v_{qr} = R_r i_{qr} + p\lambda_{qr} + (\omega - \omega_r)\lambda_{dr} \end{cases} \tag{14.3-6}$$

where the stator and rotor flux linkages can be calculated by

$$\begin{cases} \lambda_{ds} = L_{ls} i_{ds} + L_m (i_{ds} + i_{dr}) \\ \lambda_{qs} = L_{ls} i_{qs} + L_m (i_{qs} + i_{qr}) \\ \lambda_{dr} = L_{lr} i_{dr} + L_m (i_{ds} + i_{dr}) \\ \lambda_{qr} = L_{lr} i_{qr} + L_m (i_{qs} + i_{qr}) \end{cases} \tag{14.3-7}$$

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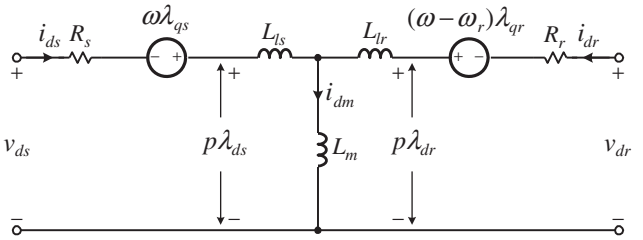
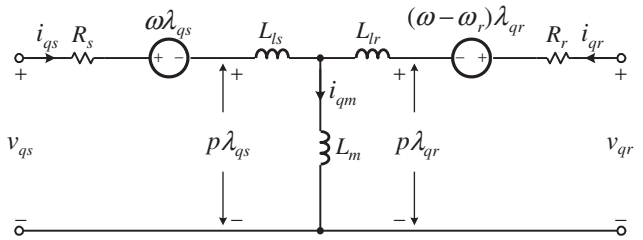


(a) d-axis circuit



(b) q-axis circuit

Figure 14.3-3 Induction motor dq-axis model in the arbitrary reference frame.

The electromagnetic torque can be expressed in a number of ways. Some of the commonly used expressions are

$$T_e = \begin{cases} \dfrac{3P}{2}(i_{qs}\lambda_{ds} - i_{ds}\lambda_{qs}) \\ \dfrac{3PL_m}{2}(i_{qs}i_{dr} - i_{ds}i_{qr}) \\ \dfrac{3PL_m}{2L_r}(i_{qs}\lambda_{dr} - i_{ds}\lambda_{qr}) \end{cases} \tag{14.3-8}$$

Equations (14.3-6) to (14.3-8) together with the motion equation of (14.3-3) represent the dq-axis model of the induction motor, whose equivalent circuit is shown in Fig. 14.3-3.

14.3.3 Induction Motor Transient Characteristics

It is instructive to study transient characteristics of the induction motor during free acceleration using the dynamic motor models. The motor under investigation is a low-power squirrel cage motor with the following parameters: $V_{LL} = 208$ V, $f_s = 60$ Hz, $Z_{base} = 15.4\ \Omega$, $R_s = 0.068$ pu, $R_r = 0.045$ pu, $L_{ls} = L_{lr} = 0.058$ pu, $L_m = 1.95$ pu, $P = 1$, and $J = 0.02\,\text{kg} \cdot \text{m}^2$. Figure 14.3-4 shows the block diagram for computer simulation with the motor model in the stationary frame ($\omega = 0$). The three-phase supply voltages v_{as}, v_{bs}, and v_{cs} are transformed to the $\alpha\beta$-axis stator voltages $v_{\alpha s}$

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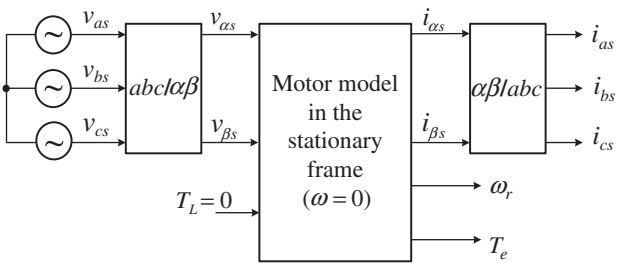

Figure 14.3-4 Block diagram for simulation of motor free acceleration using the stationary-frame motor model.

and $v_{\beta s}$ through the *abc*/$\alpha\beta$ transformation. The simulated $\alpha\beta$-axis stator currents $i_{\alpha s}$ and $i_{\beta s}$ are then converted to the three-phase currents i_{as}, i_{bs}, and i_{cs} by the $\alpha\beta$/*abc* transformation.

Figure 14.3-5a shows simulated transient waveforms of the stator current i_{as} and rotor speed n_r during motor free acceleration (the motor starts under the rated voltage and frequency without mechanical load). The rotor speed n_r in rpm relates to the angular electrical rotor speed ω_r by

$$n_r = \frac{30}{\pi P}\omega_r \tag{14.3-9}$$

The peak starting current is approximately 8.4 pu, which represents the rms starting current of 5.9 pu. The starting time is around 0.5 s due to the low moment of inertia and high starting current. The measured waveforms during the free acceleration are given in Fig. 14.3-5b, which match with the simulated results very well.

Figure 14.3-6 shows simulated and measured transient waveforms of the motor during a three-phase fault. The motor operates near its synchronous speed when its three-phase terminals are shorted. The maximum peak stator current is close to that during the free acceleration. The measured waveforms correlate closely with the simulated ones.

Figure 14.3-7 illustrates the simulation block diagram with the motor model in the synchronous frame. Using the *abc*/*dq* and *dq*/*abc* transformation blocks, the three-phase supply voltages v_{as}, v_{bs}, and v_{cs} in the stationary frame can be transformed to the *dq*-axis voltages v_{ds} and v_{qs} in the synchronous frame while the simulated *dq*-axis currents i_{ds} and i_{qs} in the synchronous frame can be converted to three-phase currents i_{as}, i_{bs}, and i_{cs} in the stationary frame. The angle θ in the transformation blocks can be obtained by the *abc*/$\alpha\beta$ transformation and $\tan^{-1}$ blocks shown in the figure or using equation (14.2-3) directly.

Figure 14.3-8a shows the simulated current waveforms for i_{as}, i_{ds}, and i_{qs} during motor free acceleration. The waveform for i_{as} in the stationary frame is exactly the same as that in Fig. 14.3-5. The *dq*-axis currents i_{ds} and i_{qs} are in the synchronous frame, and therefore are of dc signals in steady state. The amplitude of the stator

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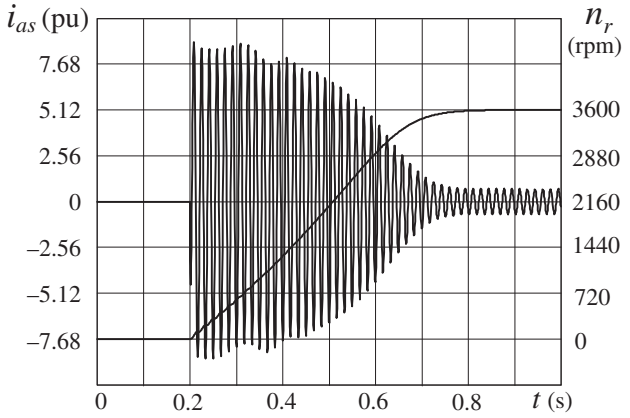
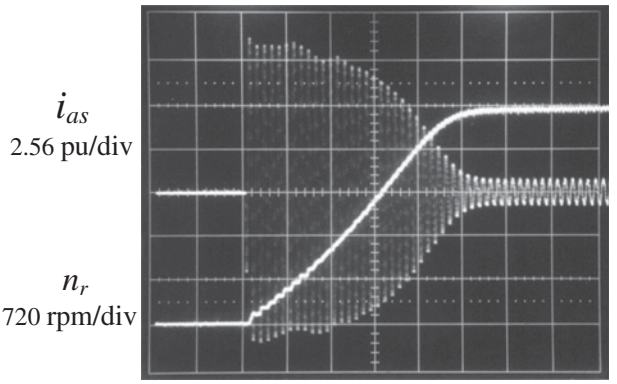


(a) Simulated waveforms



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(b) Measured waveforms

Figure 14.3-5 Waveforms of an induction motor during free acceleration.

current $\vec{i}_s$ can be obtained by $i_s = \sqrt{i_{ds}^2 + i_{qs}^2}$. The dq-axis voltages v_{ds} and v_{qs} in the synchronous frame are constant dc voltages, and therefore not shown in the figure.

The stator current angle ϕ, which is the angle between $\vec{i}_s$ and the d-axis as shown in Fig. 14.2-2b, will affect waveforms for i_{ds} and i_{qs}. The waveforms in Fig. 14.3-8a are obtained with $\phi = 0$ by adjusting the initial angle θ_0 such that the d-axis of the synchronous frame is aligned with $\vec{i}_s$. This leads to $i_{qs} = 0$ and $i_{ds} = i_s$ when the motor is in steady-state operation. If the q-axis of the synchronous frame is aligned with $\vec{i}_s$ ($\phi = 90°$), the steady-state dq-axis currents are $i_{qs} = i_s$ and $i_{ds} = 0$. However, the waveform for i_s is not affected by ϕ.

The torque response during the motor free acceleration is shown in Fig. 14.3-8b. Although it is obtained with the motor model in the synchronous frame, the response remains the same for the motor model in any other reference frames.

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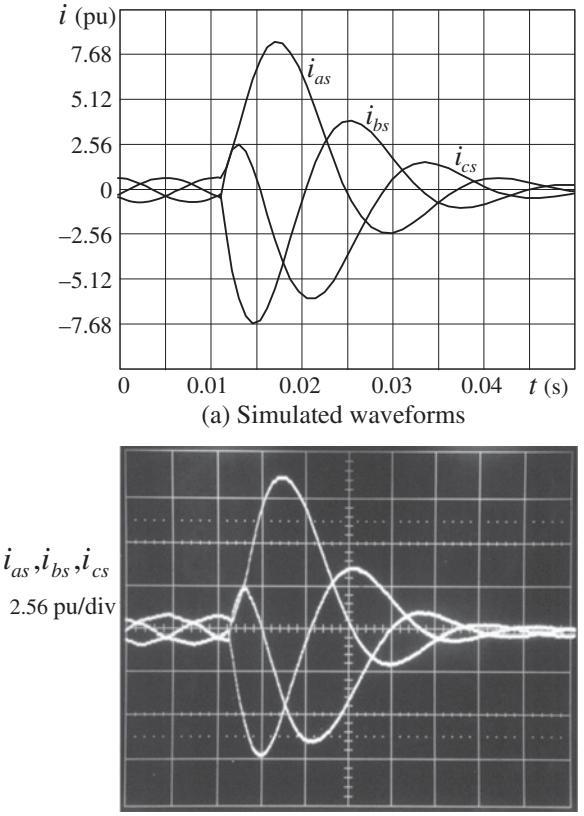

Figure 14.3-6 Stator current waveforms of an induction motor during a three-phase fault.

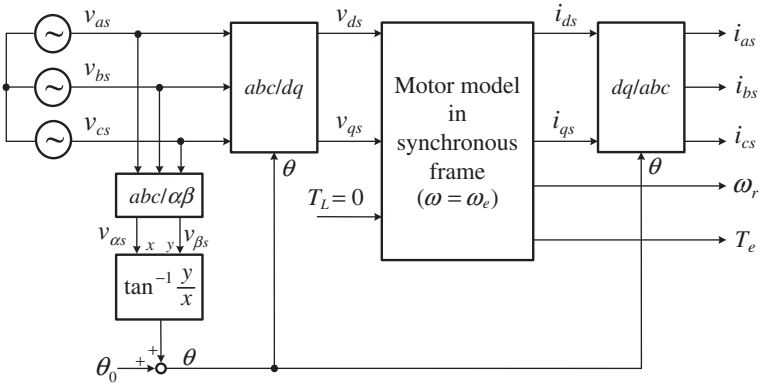

Figure 14.3-7 Block diagram for simulation of motor free acceleration using the synchronous-frame motor model.

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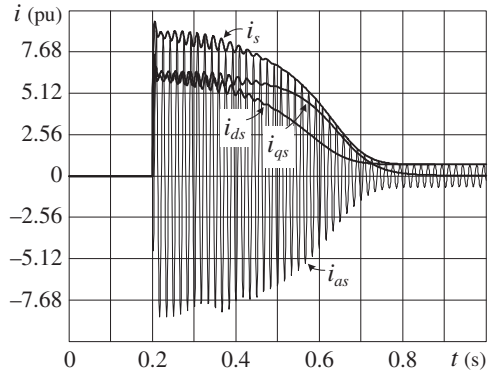
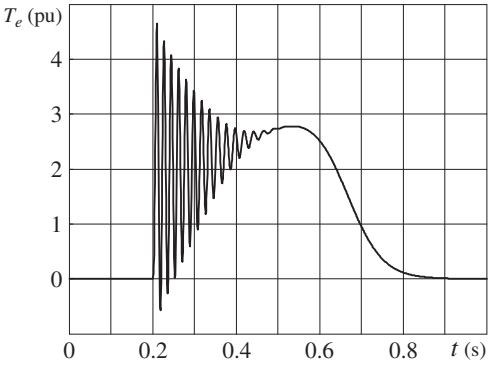


(a) i_{ds} and i_{qs} in the synchronous frame



(b) Electromagnetic Torque

Figure 14.3-8 Simulated waveforms for motor free acceleration using the synchronous-frame motor model.

14.4 PRINCIPLE OF FIELD ORIENTED CONTROL (FOC)

14.4.1 Field Orientation

It is well known that the dc motor drive has an excellent dynamic performance. This is mainly due to the decoupled (separate) control of stator magnetic field and electromagnetic torque of the motor. The torque is developed by the interaction of two perpendicular magnetic fields. One field is generated by the field current i_f in the stator winding and the other is produced by the armature (rotor) current i_a. The developed torque can be expressed as

$$T_e = K_a \lambda_f i_a \tag{14.4-1}$$

where K_a is an armature constant and λ_f is the flux produced by i_f. In high-performance dc drives, λ_f is normally kept constant by keeping i_f constant, and thus the torque T_e is proportional to and can be directly controlled by i_a.

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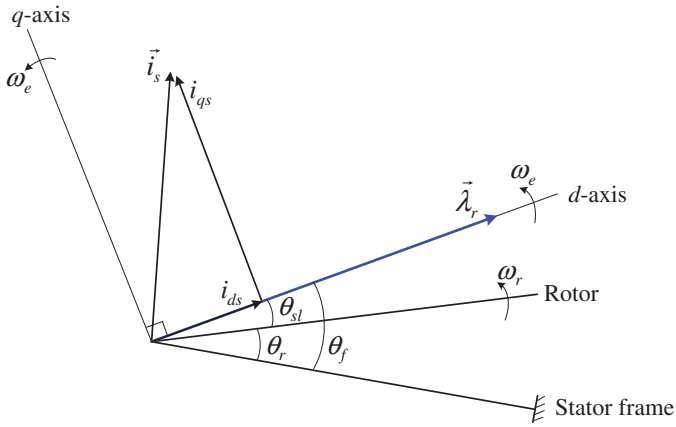

Figure 14.4-1 Rotor flux field orientation (the d-axis is aligned with $\vec{\lambda}_r$).

The FOC, also known as vector control, for induction motor emulates the dc motor control. Using a proper field orientation, the stator current can be decomposed into a flux-producing component and a torque-producing component. These two components are then controlled separately.

The field orientation can be generally classified into stator flux, air-gap flux, and rotor flux orientations [3, 4]. Since the **rotor flux orientation** is extensively used in ac drives, this scheme is to be analyzed in detail. Its operating principle can be easily applied to the other two field orientation schemes.

The rotor flux orientation is achieved by aligning the d-axis of the synchronous reference frame with the rotor flux vector $\vec{\lambda}_r$ as shown in Fig. 14.4-1. The resultant dq-axis rotor flux components are

$$\lambda_{qr} = 0 \quad \text{and} \quad \lambda_{dr} = \lambda_r \tag{14.4-2}$$

where λ_r is the magnitude of $\vec{\lambda}_r$. Substituting (14.4-2) into the last equation of (14.3-8) yields

$$T_e = K_T \, \lambda_{dr} \, i_{qs} = K_T \, \lambda_r \, i_{qs} \tag{14.4-3}$$

where $K_T = \frac{2PL_m}{3L_r}$. Equation (14.4-3) indicates that with the rotor field orientation the torque expression for the induction motor is similar to that of a dc motor. If λ_r can be kept constant during the motor operation, the developed torque can be directly controlled by the q-axis stator current i_{qs}.

The stator current vector $\vec{i}_s$ in Fig. 14.4-1 can be resolved into two components along the dq axes. The d-axis current i_{ds} is referred to as **flux-producing current** while the q-axis current i_{qs}, which is perpendicular to i_{ds}, is the **torque-producing current.** In the FOC, i_{ds} is normally kept at its rated value while i_{qs} is controlled independently. With the decoupled control for i_{ds} and i_{qs}, a high-performance drive can be realized.

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One of the key issues associated with the rotor flux oriented control is to accurately determine the **rotor flux angle** θ_f for field orientation. Various schemes can be used to find θ_f. For instance, it can be calculated from measured stator voltages and currents, or can be found from

$$\theta_f = \theta_r + \theta_{sl} \tag{14.4-4}$$

where θ_r and θ_{sl} are the measured rotor position angle and calculated slip angle, respectively.

14.4.2 General Block Diagram of FOC

Depending on how the rotor flux angle θ_f is obtained, the control schemes can be classified into direct and indirect FOCs. If θ_f is obtained by using flux-sensing devices embedded inside the motor or using measured motor terminal voltages and currents, the method is referred to as **direct FOC**. If the rotor flux angle θ_f is obtained from detected rotor position angle θ_r and calculated slip angle θ_{sl} as shown in (14.4-4), this scheme is known as **indirect FOC** [3–5].

A general block diagram of an induction motor drive with rotor flux oriented control is shown in Fig. 14.4-2. Since the essence of the FOC is the decoupled control of the rotor flux λ_r and electromagnetic torque T_e, these two variables are controlled separately. The torque reference T_e^* is generated by **speed controller** based on the reference speed ω_r^* and the detected or estimated rotor speed ω_r. The rotor flux reference λ_r^* is a function of ω_r^*. When the motor operates at or below its rated speed, λ_r^* is normally kept at its rated value. With the rated speed exceeded, λ_r^* should be weakened accordingly such that the stator voltage and the output power of the motor would not exceed their ratings.

The two references λ_r^* and T_e^* are then sent to **flux/torque controller**, where they are compared with calculated rotor flux λ_r and torque T_e for a closed-loop control.

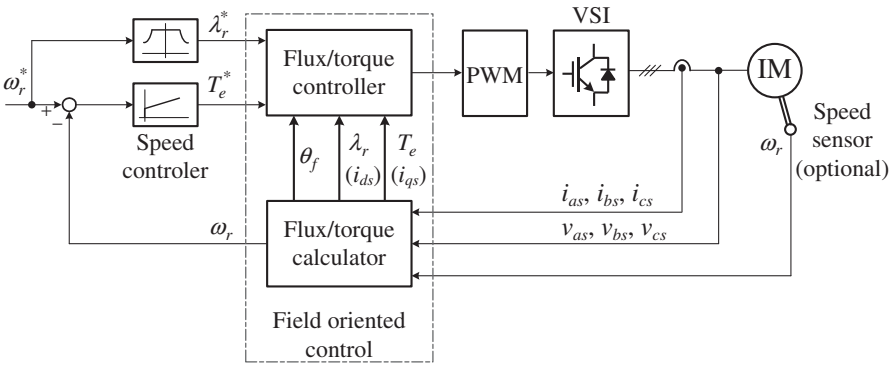

Figure 14.4-2 General block diagram of rotor flux FOC.

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The flux/torque controller generates reference signals for the PWM block which produces gate signals for the inverter to adjust its output voltage and frequency.

Based on the measured stator voltage and current variables and the motor model, the **flux/torque calculator** calculates (1) the rotor flux angle θ_f for field orientation, (2) the rotor flux magnitude λ_r or the flux-producing current i_{ds}, (3) the electromagnetic torque T_e or the torque-producing current i_{qs}, and (4) the rotor speed ω_r. Depending on drive system requirements and the type of the FOC scheme employed, the rotor speed ω_r may also be directly detected by a digital speed sensor. It is worth noting that the flux/torque calculator, also known as **flux/torque observer** or estimator in the literature, is the most important functional block in the FOC scheme.

14.5 DIRECT FIELD ORIENTED CONTROL

14.5.1 System Block Diagram

Figure 14.5-1 shows a typical block diagram of direct FOC for induction motor, where the rotor speed control is not shown for simplicity. There are three feedback control loops, one for the rotor flux linkage λ_r, one for the d-axis (flux-producing) stator current i_{ds}, and another for the q-axis (torque-producing) stator current i_{qs}.

For the rotor flux control, the calculated λ_r is compared with its reference λ_r^* to generate the d-axis stator current reference i_{ds}^* through flux controller. The q-axis stator current reference i_{qs}^* is generated according to the torque reference T_e^*. The feedback dq-axis stator currents i_{ds} and i_{qs} are compared with their references, and the errors are sent to current controllers to generate stator voltage references v_{ds}^* and v_{qs}^*. The dq-axis voltages in the synchronous frame are then transformed to the three-phase stator voltages v_{as}^*, v_{bs}^*, and v_{cs}^* in the stationary frame for the PWM block. Various PWM schemes can be used. If a carrier-based modulation scheme is

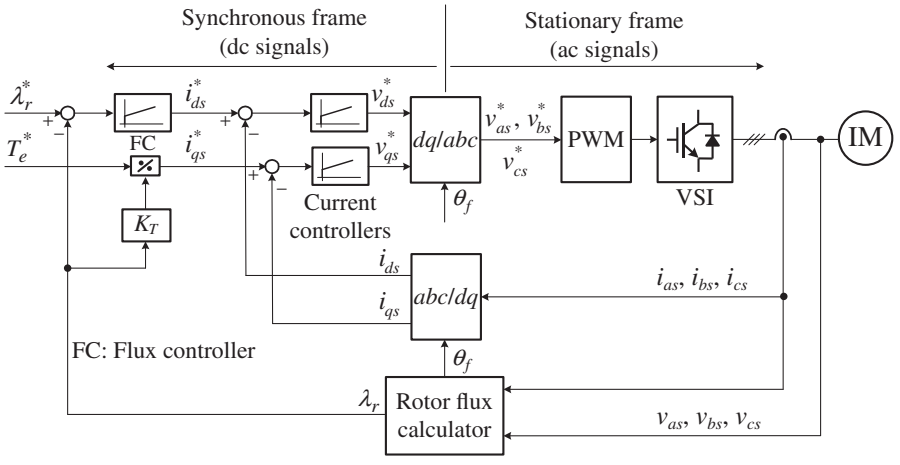

Figure 14.5-1 Direct field oriented control with rotor flux orientation.

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employed, v_{as}^*, v_{bs}^*, and v_{cs}^* are the modulating signals which are compared with a triangular carrier wave to generate PWM gatings for the switching devices in the inverter.

As shown in Fig. 14.5-1, the rotor flux angle θ_f is used in the *abc*/*dq* and *dq*/*abc* transformation blocks for field orientation. The variables on the left of the transformation blocks are all dc signals in the synchronous frame while those on the right side of the transformation blocks are all ac variables in the stationary frame.

14.5.2 Rotor Flux Calculator

Based on the stationary-frame motor model in Fig. 14.3-2b, the stator flux vector can be expressed as

$$\vec{\lambda}_s = \int \left(\vec{v}_s - R_s \vec{i}_s\right) dt \tag{14.5-1}$$

The rotor flux vector can be found from the flux-linkage equations (14.3-2)

$$\vec{\lambda}_r = L_r \frac{\vec{\lambda}_s - L_s \vec{i}_s}{L_m} + L_m \vec{i}_s = \frac{L_r}{L_m}\left(\vec{\lambda}_s - \sigma L_s \vec{i}_s\right) \tag{14.5-2}$$

where σ is the total leakage factor, defined by

$$\sigma = 1 - \frac{L_m^2}{L_s L_r} \tag{14.5-3}$$

Decomposing the rotor flux $\vec{\lambda}_r$ into the d- and q-axis components, we have

$$\begin{cases} \lambda_{dr} = \dfrac{L_r}{L_m}\left(\lambda_{ds} - \sigma L_s i_{ds}\right) \\ \lambda_{qr} = \dfrac{L_r}{L_m}\left(\lambda_{qs} - \sigma L_s i_{qs}\right) \end{cases} \tag{14.5-4}$$

from which the magnitude and angle of the rotor flux are

$$\begin{cases} \lambda_r = \sqrt{\lambda_{dr}^2 + \lambda_{qr}^2} \\ \theta_f = \tan^{-1}\dfrac{\lambda_{qr}}{\lambda_{dr}} \end{cases} \tag{14.5-5}$$

It is can be noted from (14.5-1) to (14.5-5) that

- The rotor flux magnitude λ_r and its angle θ_f can be identified based on measured stator voltage $\vec{v}_s$, stator current $\vec{i}_s$, and motor parameters (L_s, L_r, L_m, and R_s).

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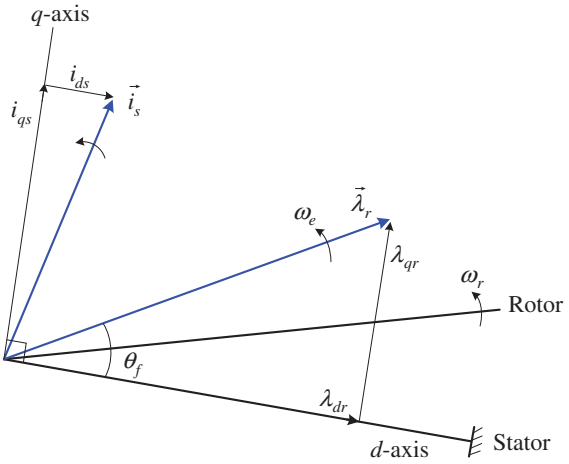

Figure 14.5-2 Vector diagram for $\vec{\lambda}_r$ and $\vec{i}_s$ used in rotor flux calculator.

- Since the stationary-frame motor model is used, all the variables such as λ_{dr}, λ_{qr}, i_{ds}, and i_{qs} (except λ_r and θ_f) are of ac signals. Neglecting the switching harmonics, they are sinusoidal in steady state.

Figure 14.5-2 shows the vector diagram for the rotor flux vector $\vec{\lambda}_r$ and stator current vector $\vec{i}_s$ used in the rotor flux calculator. When the two vectors rotate one revolution in space, their dq-axis components λ_{dr}, λ_{qr}, i_{ds}, and i_{qs} in the stationary (stator) frame vary one cycle over time.

Figure 14.5-3 shows the block diagram for the digital implementation of rotor flux calculator. Of the three stator voltages v_{as}, v_{bs}, and v_{cs}, only two voltages need to be measured and the third one can be found from $v_{as} + v_{bs} + v_{cs} = 0$. To reduce the number of voltage sensors, the stator voltages can also be reconstructed by using the inverter switching function and measured dc voltage. The stator voltages and currents

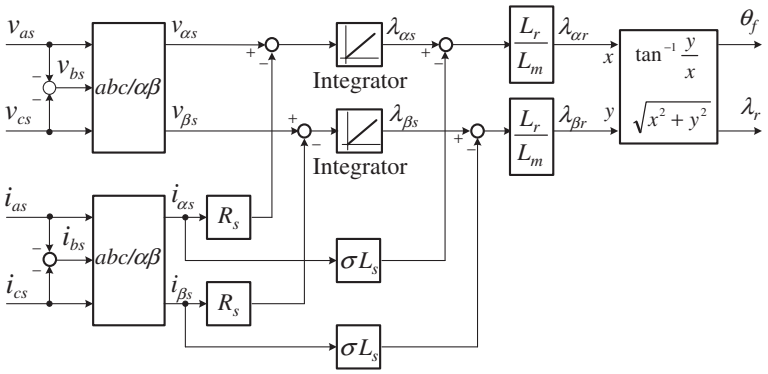

Figure 14.5-3 Block diagram for the rotor flux calculation.

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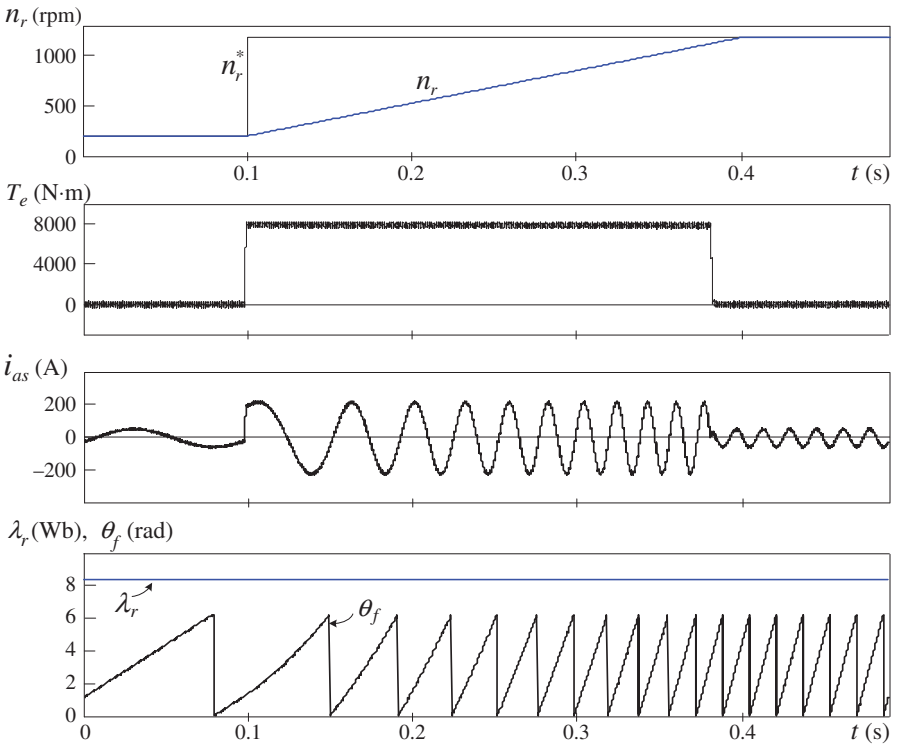

Figure 14.5-4 Simulated waveforms for an induction motor drive with direct FOC scheme.

are then transformed to two-phase variables through *abc/αβ* stationary transformation blocks. The other blocks are derived from equations (14.5-1) to (14.5-5). The output of the rotor flux calculator is the rotor flux angle θ_f and its amplitude λ_r.

Figure 14.5-4 shows the simulated waveforms for an induction motor drive using the direct FOC scheme. The block diagram of the drive system is shown in Fig. 14.5-1, to which a rotor speed control loop is added for simulation. The motor nameplate data and parameters are listed in Table 14.5-1. The inverter is controlled by a sinusoidal

Table 14.5-1 Motor Nameplate and Parameters

Motor Ratings		Motor Parameters	
Rated output power	1250 hp	Stator resistance, R_s	0.21 Ω
Rated line-to-line voltage	4160 V	Rotor resistance, R_r	0.146 Ω
Rated stator current	150 A	Stator leakage inductance, L_{ls}	5.2 mH
Rated speed	1189 rpm	Rotor leakage inductance, L_{lr}	5.2 mH
Rated torque	7490 N · m	Magnetizing inductance, L_m	155 mH
Rate stator flux linkage	9.0 Wb	Moment of inertia, J	22 kg · m^2
Rated rotor flux linkage	8.35 Wb		

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PWM scheme with switching frequency of 740 Hz. The rotor flux reference λ_r^* is set to its rated value of 8.35 Wb. The maximum torque is limited to its rated value of 7490 N · m during the transients by the speed controller.

The motor initially operates at a rotor speed of $n_r = 200$ rpm. The speed reference n_r^* has a step increase from 200 rpm to the rated rotor speed of 1189 rpm at the time instant of $t = 0.1$ s. The motor accelerates under the no-load condition while its torque is limited to the rated value. The stator current i_{as} rises to its rated value during the transient. When n_r reaches its reference of 1189 rpm around $t = 0.4$ s, the electromagnetic torque of the motor T_e drops to zero, and i_{as} reduces to a value that corresponds to the magnetizing current of the motor. Due to the decoupled flux and torque control by the FOC scheme, the rotor flux magnitude λ_r is kept constant during transients. The rotor flux angle θ_f is also given in the figure.

14.6 INDIRECT FIELD ORIENTED CONTROL

A digital rotor speed sensor is required for indirect field orientated control schemes. The rotor flux angle θ_f for field orientation is obtained from the measured rotor speed and calculated slip angle based on motor parameters. A typical block diagram of the indirect FOC is shown in Fig. 14.6-1. Since the rotor speed ω_r is directly measured, the rotor flux angle θ_f can be found from

$$\theta_f = \int (\omega_r + \omega_{sl})dt \tag{14.6-1}$$

where ω_{sl} is the angular slip frequency.

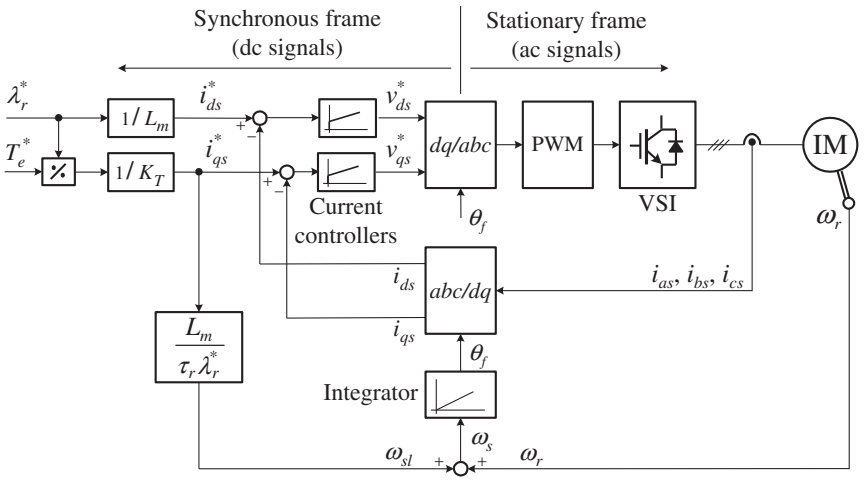

Figure 14.6-1 Indirect field oriented control with rotor flux orientation.

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The slip frequency ω_{sl} can be derived from the synchronous-frame motor model of Fig. 14.3-2a, from which

$$p\vec{\lambda}_r = -R_r \vec{i}_r - j\omega_{sl} \vec{\lambda}_r \tag{14.6-2}$$

Substituting the rotor current,

$$\vec{i}_r = \frac{1}{L_r}\left(\vec{\lambda}_r - L_m \vec{i}_s\right) \tag{14.6-3}$$

into (14.6-2) yields

$$p\vec{\lambda}_r = -\frac{R_r}{L_r}\left(\vec{\lambda}_r - L_m\vec{i}_s\right) - j\omega_{sl}\vec{\lambda}_r \tag{14.6-4}$$

from which

$$\vec{\lambda}_r(1 + \tau_r(p + j\omega_{sl})) = L_m\vec{i}_s \tag{14.6-5}$$

where τ_r is the rotor time constant, defined by

$$\tau_r = L_r/R_r \tag{14.6-6}$$

Decomposing (14.6-5) into the dq-axis components and taking into account the rotor flux orientation ($j\lambda_{qr} = 0$ and $\lambda_{dr} = \lambda_r$), we have

$$\begin{cases} \lambda_r(1 + p\tau_r) = L_m i_{ds} \\ \omega_{sl}\,\tau_r\,\lambda_r = L_m i_{qs} \end{cases} \tag{14.6-7}$$

from which

$$\omega_{sl} = \frac{L_m}{\tau_r\,\lambda_r}\,i_{qs} \tag{14.6-8}$$

As shown in Fig. 14.6-1, the rotor flux and torque are controlled by two feedback loops separately. Based on (14.6-7), the relationship between the rotor flux reference λ_r^* and d-axis current reference i_{ds}^* can be expressed as

$$i_{ds}^* = \frac{(1 + p\tau_r)}{L_m}\lambda_r^* \tag{14.6-9}$$

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Since λ_r^* is normally kept constant during operation ($p\lambda_r^* = 0$), (14.6-9) can be simplified to

$$i_{ds}^* = \frac{1}{L_m}\lambda_r^* \tag{14.6-10}$$

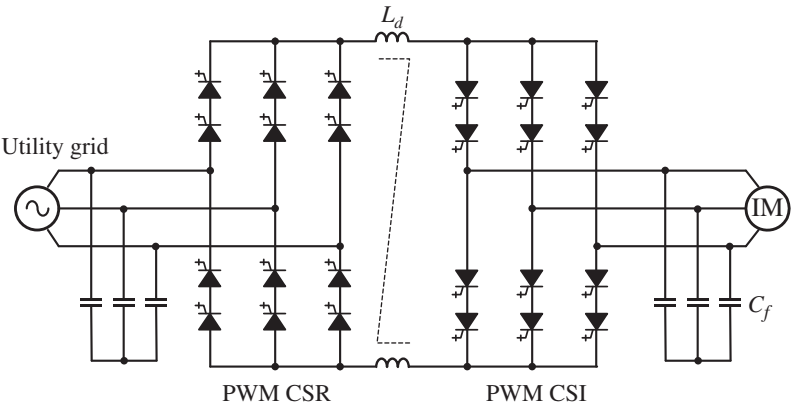
The q-axis current reference i_{qs}^* can be obtained from the torque equation of (14.4-3)

$$i_{qs}^* = \frac{1}{K_T \lambda_r^*} T_e^* \tag{14.6-11}$$

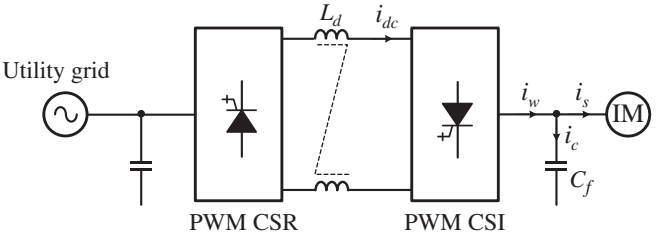
For a given λ_r^*, the torque-producing current i_{qs}^* is proportional to T_e^*.

14.7 FOC FOR CSI FED DRIVES

In VSI fed MV drives, both inverter output voltage and frequency can be controlled by its PWM scheme. However, this is not the case for the CSI fed drive shown in Fig. 14.7-1, where the inverter output frequency is controlled by its PWM scheme



(a) Schematic



(b) Block diagram

Figure 14.7-1 A PWM current source converter-based drive.

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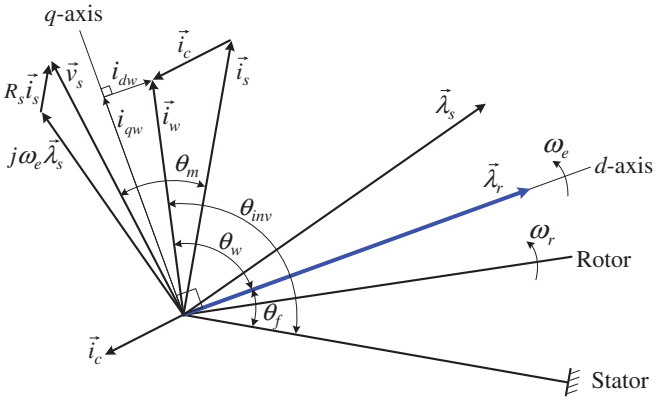

Figure 14.7-2 Vector diagram for a CSI fed drive with rotor flux orientation.

whereas the inverter output current i_w is adjusted by dc current i_{dc} of the rectifier. In addition, the stator current i_s is not directly controlled by i_w due to the filter capacitor C_f. Therefore, additional measures are required to maintain field orientation in CSI drives.

Figure 14.7-2 shows the space vector diagram for the CSI drive with rotor flux oriented control. The d-axis of the synchronous reference frame is aligned with the rotor flux vector $\vec{\lambda}_r$. The stator flux vector $\vec{\lambda}_s$ leads the rotor flux $\vec{\lambda}_r$ by a small angle due to the leakage inductances. The stator voltage $\vec{v}_s$ is the sum of the speed voltage $j\omega_e\vec{\lambda}_s$ in Fig. 14.3-2a and the stator resistance voltage drop $R_s\vec{i}_s$. The stator current $\vec{i}_s$ lags $\vec{v}_s$ by θ_m, which is the motor power factor angle. The capacitor current $\vec{i}_c$ leads $\vec{v}_s$ by $\pi/2$. The inverter PWM current $\vec{i}_w$ is a vector sum of $\vec{i}_s$ and $\vec{i}_c$, and its angle with respect to $\vec{\lambda}_r$ is θ_w. The inverter firing angle is

$$\theta_{inv} = \theta_w + \theta_f \tag{14.7-1}$$

where θ_f is the rotor flux angle for field orientation. When the drive operates in steady state, θ_w is constant since both $\vec{i}_w$ and $\vec{\lambda}_r$ are in the synchronous reference frame while θ_f and θ_{inv} vary periodically between zero and 2π.

Figure 14.7-3 shows a simplified block diagram for the CSI fed drive with direct FOC. The FOC scheme is implemented with three feedback control loops, one for the rotor speed ω_r, one for the rotor flux λ_r, and another for the dc current i_{dc}. The rotor speed ω_r is obtained by $\omega_r = \omega_e - \omega_{sl}$, where $\omega_e = d\theta_f/dt$ and ω_{sl} can be calculated by (14.6-8).

The q-axis (torque-producing) stator current reference i_{qs}^* and d-axis (flux-producing) current reference i_{ds}^* are generated in the same manner as that shown in Fig. 14.5-1. The dq-axis inverter PWM current references can be expressed as

$$\begin{cases} i_{dw}^* = i_{cd} + i_{ds}^* \\ i_{qw}^* = i_{cq} + i_{qs}^* \end{cases} \tag{14.7-2}$$

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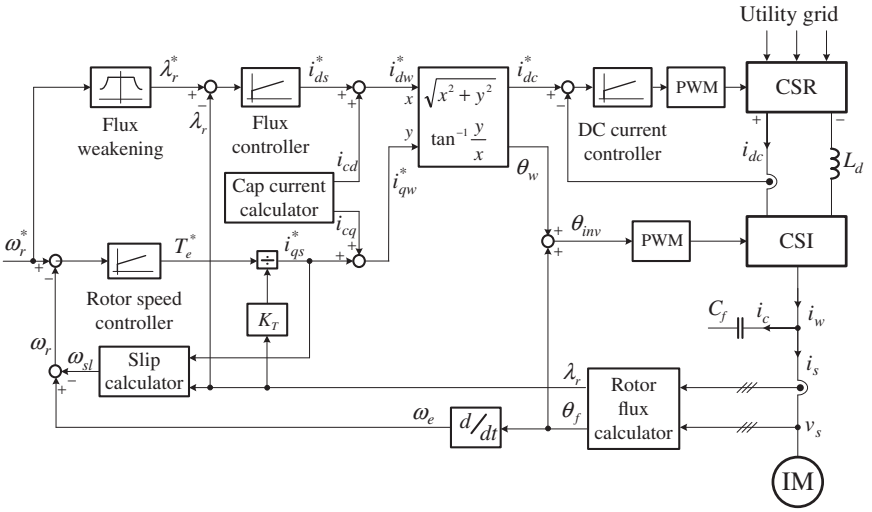

Figure 14.7-3 Simplified block diagram for a CSI fed drive using direct FOC scheme.

where i_{cd} and i_{cq} are the dq-axis capacitor currents, given by

$$\begin{cases} i_{cd} = (pv_{ds} - \omega_e v_{qs}) C_f \\ i_{cq} = (pv_{qs} + \omega_e v_{ds}) C_f \end{cases} \tag{14.7-3}$$

The first term on the right-hand side of the equation represents capacitor transient current and the second term is the steady-state current. To reduce the sensitivity and noise caused by the derivative terms (pv_{ds} and pv_{qs}), the effect of the capacitor transient response on the drive dynamic performance may be neglected. Equation (14.7-3) can then be simplified to

$$\begin{cases} i_{cd} = -\omega_e v_{qs} C_f \\ i_{cq} = \omega_e v_{ds} C_f \end{cases} \tag{14.7-4}$$

for use in the capacitor current calculator in Fig. 14.7-3.

Since the magnitude of i_w is proportional to the dc current, the dc current reference can be found from

$$i_{dc}^* = \sqrt{\left(i_{dw}^*\right)^2 + \left(i_{qw}^*\right)^2} \tag{14.7-5}$$

The inverter firing angle θ_{inv} is the sum of θ_f and θ_w, where θ_f can be obtained from the rotor flux calculator in Fig. 14.5-3 and θ_w can be determined by

$$\theta_w = \tan^{-1}\left(i_{qw}^* / i_{dw}^*\right) \tag{14.7-6}$$

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Various PWM schemes, such as SHE, TPWM, and SVM presented in Chapters 10 and 11, can be employed for the PWM blocks in Fig. 14.7-3.

14.8 DIRECT TORQUE CONTROL (DTC)

Direct torque control (DTC) is one of the advanced control schemes for ac drives [6–10]. It is characterized by simple control algorithm, easy digital implementation and robust operation [11, 12]. In this section, the principle of DTC scheme is introduced and simulation results are provided.

14.8.1 Principle of Direct Torque Control

The electromagnetic torque developed by an induction motor can be expressed in a number of ways, one of which is

$$T_e = \frac{3P}{2}\frac{L_m}{\sigma L_s L_r}\lambda_s \lambda_r \sin\theta_T \tag{14.8-1}$$

where θ_T is the angle between the stator flux vector $\vec{\lambda}_s$ and rotor flux vector $\vec{\lambda}_r$, often known as **torque angle**. This equation indicates that T_e can be directly controlled by θ_T.

The main variable to be controlled in the DTC scheme is the stator flux vector $\vec{\lambda}_s$. Referring to the induction motor model of Fig. 14.3-2b, $\vec{\lambda}_s$ relates the stator voltage vector $\vec{v}_s$ by

$$p\vec{\lambda}_s = \vec{v}_s - R_s\vec{i}_s \tag{14.8-2}$$

The equation shows that the derivative of $\vec{\lambda}_s$ reacts instantly to changes in $\vec{v}_s$. As discussed in Chapter 6, the stator voltage $\vec{v}_s$, which is the inverter load voltage, can be controlled by the reference vector $\vec{V}_{ref}$ in the space vector modulation. Since $\vec{V}_{ref}$ is synthesized by the stationary voltage vectors of the inverter, a proper selection of the stationary vectors can make the magnitude and angle of $\vec{\lambda}_s$ adjustable.

Figure 14.8-1 shows the principle of DTC for a two-level VSI fed induction motor drive. The dq-axis plane for the stator flux $\vec{\lambda}_s$ is divided into six sectors, I–VI. The stator flux $\vec{\lambda}_s$ in the figure falls into sector I, and its angle θ_s is referenced to the d-axis of the stationary reference frame. The rotor flux vector $\vec{\lambda}_r$ lags $\vec{\lambda}_s$ by θ_T.

Let us now examine the impact of the stationary voltage vectors $\vec{V}_0$ to $\vec{V}_6$ on $\vec{\lambda}_s$ and θ_T. Assume that $\vec{\lambda}_s$ and θ_T in Fig. 14.8-1 are the initial stator flux vector and torque angle. When voltage vector $\vec{V}_2$ is selected, the stator flux vector will become $\vec{\lambda}'_s = \vec{\lambda}_s + \vec{V}_2\Delta t$ after a short time interval Δt, leading to an increase in flux magnitude ($\lambda'_s > \lambda_s$) and torque angle ($\theta'_T > \theta_T$). If voltage vector $\vec{V}_5$ is selected, $\vec{\lambda}_s$ will change

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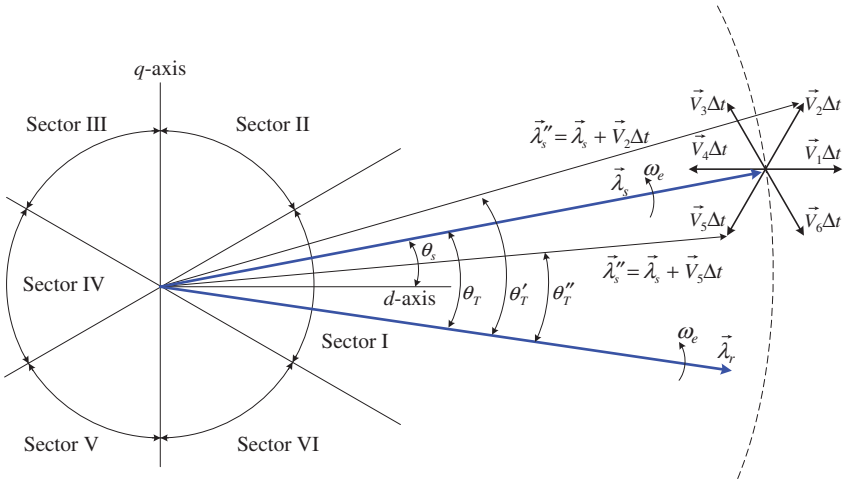

Figure 14.8-1 Principle of direct torque control.

to $\vec{\lambda}_s'' = \vec{\lambda}_s + \vec{V}_5\Delta t$, causing a decrease in flux magnitude ($\lambda_s'' < \lambda_s$) and torque angle $\theta_T'' < \theta_T$. Similarly, the selection of $\vec{V}_3$ and $\vec{V}_6$ can make one variable increase and the other decrease. Therefore, λ_s and θ_T can be controlled separately by proper selection of the inverter voltage vectors.

Note that the changes in $\vec{v}_s$ have much less impact on $\vec{\lambda}_r$ for a short time interval Δt due to the large rotor time constant. Therefore, it is assumed in the above analysis that the rotor flux vector $\vec{\lambda}_r$ is kept constant during Δt.

14.8.2 Switching Logic

Figure 14.8-2 shows a typical block diagram of a DTC-based induction motor drive, where the rotor speed feedback loop is not shown for simplicity. Similar to the FOC schemes, the stator flux and electromagnetic torque are controlled separately to achieve superior dynamic performance. The stator flux reference λ_s^* is compared with the calculated stator flux λ_s, and the error $\Delta\lambda_s$ is sent to flux comparator. The torque reference T_e^* is compared with the calculated torque T_e, and their difference ΔT_e is the input of torque comparator. The output of the flux and torque comparators (x_λ and x_T) are sent to switching logic unit for proper selection of the voltage vectors (switching states) of the inverter.

Both flux and torque comparators are of hysteresis (tolerance band) type, whose transfer characteristics are shown in Fig. 14.8-3. The flux comparator has two output levels ($x_\lambda = +1$ and -1) while the torque comparator has three output levels ($x_T = +1,\ 0,$ and -1), where "+1" requests an increase in λ_s or θ_T, "−1" demands a decrease in λ_s or θ_T, and "0" signifies no changes. The tolerance band for the flux and torque comparators is δ_λ and δ_T, respectively.

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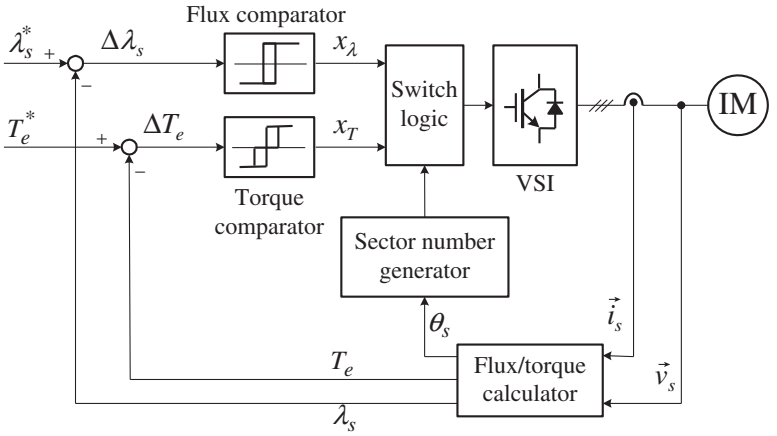
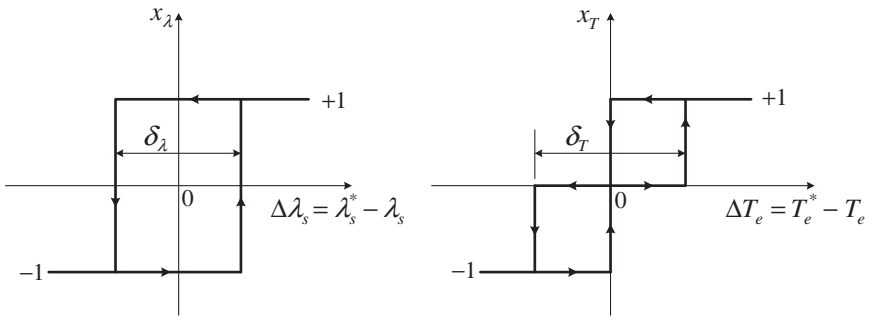


Figure 14.8-2 Block diagram of direct torque control scheme.

Table 14.8-1 gives the switching logic for the stator flux reference $\vec{\lambda}_s^*$ rotating in the counterclockwise direction. The input variables are x_λ, x_T, and the sector number, and the output variables are the inverter voltage vectors. The output of the comparators decides which voltage vector should be selected. Assuming that $\vec{\lambda}_s^*$ is in sector I, the comparator output of $x_\lambda = x_T = +1$ signifies an increase in λ_s and T_e. Voltage vector $\vec{V}_2$ can then be selected from the table. This selection will make both λ_s and θ_T increase as shown in Fig. 14.8-1.

When the output of the torque comparator x_T is zero (no need to adjust T_e), zero vector $\vec{V}_0$ can be selected. The alternative use of the switching states [OOO] and [PPP] for $\vec{V}_0$ in the switching table can help reduce the device switching frequency. For instance, when x_T changes between "+1" and "0" or between "0" and "−1," the zero states in the table ensures that only two switches are involved during the transition, one being turned on and the other being turned off.



(a) Flux comparator (b) Torque comparator

Figure 14.8-3 Characteristics of hysteresis comparators.

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Table 14.8-1 Switching Logic for $\vec{\lambda}_s^*$ Rotating in the Counterclockwise Direction

Comparator Output		Sector					
x_λ	x_T	I	II	III	IV	V	VI
+1	+1	$\vec{V}_2$ PPO	$\vec{V}_3$ OPO	$\vec{V}_4$ OPP	$\vec{V}_5$ OOP	$\vec{V}_6$ POP	$\vec{V}_1$ POO
	0	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO
	−1	$\vec{V}_6$ POP	$\vec{V}_1$ POO	$\vec{V}_2$ PPO	$\vec{V}_3$ OPO	$\vec{V}_4$ OPP	$\vec{V}_5$ OOP
−1	+1	$\vec{V}_3$ OPO	$\vec{V}_4$ OPP	$\vec{V}_5$ OOP	$\vec{V}_6$ POP	$\vec{V}_1$ POO	$\vec{V}_2$ PPO
	0	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP
	−1	$\vec{V}_5$ OOP	$\vec{V}_6$ POP	$\vec{V}_1$ POO	$\vec{V}_2$ PPO	$\vec{V}_3$ OPO	$\vec{V}_4$ OPP

The operation of the DTC can be further explained by the stator flux trajectory diagram of Fig. 14.8-4. Assume that the reference vector $\vec{\lambda}_s^*$ rotates in the counterclockwise direction during rotor speed acceleration and the output of the torque comparator is $x_T = +1$. When $\vec{\lambda}_s$ reaches the outer band limit at point a in sector II, the output of the flux comparator x_λ becomes "−1," and vector $\vec{V}_4$ is selected from

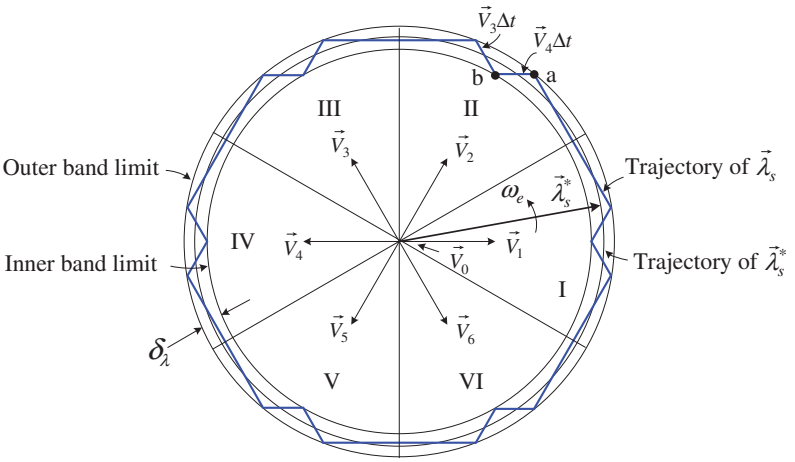

Figure 14.8-4 Trajectories of the stator flux $\vec{\lambda}_s$ and its reference $\vec{\lambda}_s^*$ with $x_T = 1$.

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Table 14.8-2 Switching Logic for the Stator Flux Rotating in the Clockwise Direction

Hysteresis Comparator		Sector					
x_λ	x_T	I	II	III	IV	V	VI
1	+1	$\vec{V}_6$ POP	$\vec{V}_5$ OOP	$\vec{V}_4$ OPP	$\vec{V}_3$ OPO	$\vec{V}_2$ PPO	$\vec{V}_1$ POO
	0	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO
	−1	$\vec{V}_2$ PPO	$\vec{V}_1$ POO	$\vec{V}_6$ POP	$\vec{V}_5$ OOP	$\vec{V}_4$ OPP	$\vec{V}_3$ OPO
−1	+1	$\vec{V}_5$ OOP	$\vec{V}_4$ OPP	$\vec{V}_3$ OPO	$\vec{V}_2$ PPO	$\vec{V}_1$ POO	$\vec{V}_6$ POP
	0	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP	$\vec{V}_0$ OOO	$\vec{V}_0$ PPP
	−1	$\vec{V}_3$ OPO	$\vec{V}_2$ PPO	$\vec{V}_1$ POO	$\vec{V}_6$ POP	$\vec{V}_5$ OOP	$\vec{V}_4$ OPP

the switching table, which will cause a decrease in λ_s. When $\vec{\lambda}_s$ hits the inner band limit at point b, x_λ becomes +1, and vector $\vec{V}_3$ is selected, making an increase in λ_s. The trajectory of $\vec{\lambda}_s$ in the figure is not very smooth due to the large width of the tolerance band δ_λ, which translates into a high stator flux ripple and low switching frequency. The quality of the stator flux waveform can be improved by reducing δ_λ, but it is achieved at the expense of an increase in the switching frequency.

The switching logic given in Table 14.8-1 is only valid for the motor rotating in the counterclockwise direction. When the motor operates in the clockwise direction, the switching logic in Table 14.8-2 can be used.

14.8.3 Stator Flux and Torque Calculation

The stator flux vector $\vec{\lambda}_s$ in the stationary frame can be expressed as

$$\begin{aligned}\vec{\lambda}_s &= \lambda_{ds} + j\lambda_{qs} \\ &= \int (v_{ds} - R_s i_{ds})dt + j\int (v_{qs} - R_s i_{qs})dt\end{aligned} \tag{14.8-3}$$

from which its magnitude and angle are

$$\begin{cases}\lambda_s = \sqrt{\lambda_{ds}^2 + \lambda_{qs}^2} \\ \theta_s = \tan^{-1}\left(\dfrac{\lambda_{qs}}{\lambda_{ds}}\right)\end{cases} \tag{14.8-4}$$

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where v_{ds}, v_{qs}, i_{ds}, and i_{qs} are the measured stator voltages and currents. The developed electromagnetic torque can be calculated by

$$T_e = \frac{3P}{2}(i_{qs}\lambda_{ds} - i_{ds}\lambda_{qs}) \tag{14.8-5}$$

The above equations illustrate that the stator flux and developed torque can be obtained by using measured stator voltages and currents. The only motor parameter required in the calculations is the stator resistance R_s. This is in contrast to the direct rotor flux FOC schemes, where almost all the motor parameters are needed.

14.8.4 DTC Drive Simulation

Figure 14.8-5 shows the simulated waveforms for an induction motor drive using the DTC scheme given in Fig. 14.8-2. The rotor speed feedback loop, based on which the

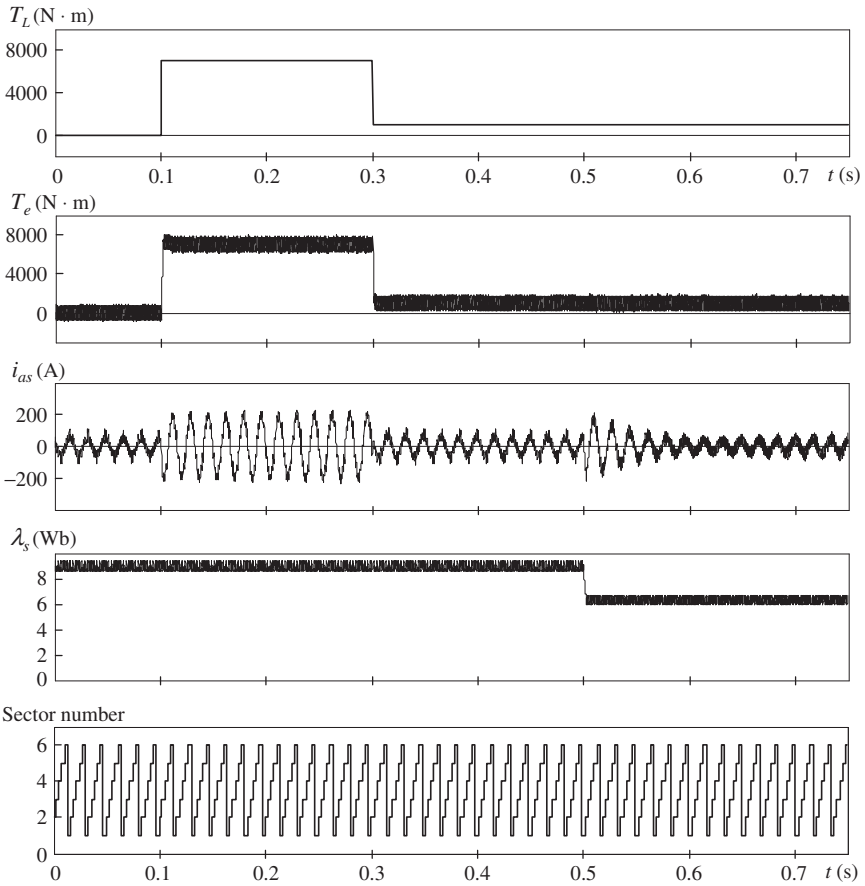

Figure 14.8-5 Simulated waveforms for a DTC drive operating at the rated rotor speed.

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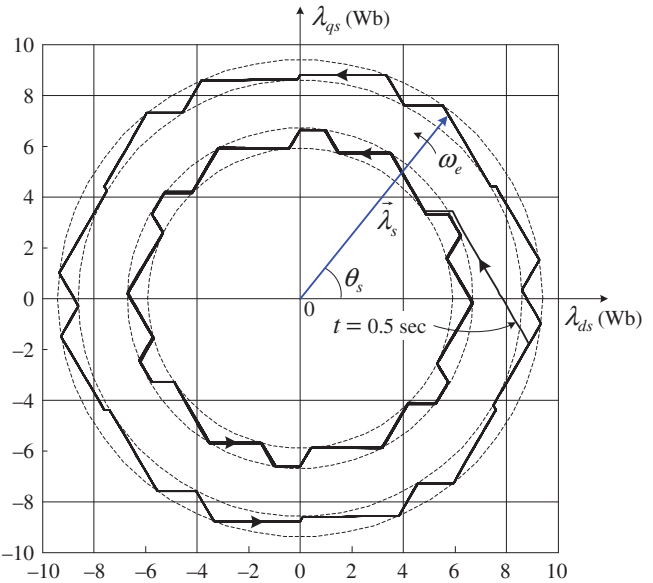

Figure 14.8-6 Trajectories of the stator flux λ_s in Fig. 14.8-5 for $0.35 \leq t \leq 0.75$ s.

torque reference T_e^* is generated, is not shown for simplicity. The motor parameters used in the simulation are given in Table 14.5-1.

The tolerance bands δ_T and δ_λ for the torque and flux comparators are adjusted such that the average switching frequency f_{sw} of the switching devices is around 800 Hz. The stator flux reference λ_s^* is set to its rated value of 9.0 Wb.

The motor operates at the rated speed of $n_r = 1189$ rpm under no-load conditions. Assuming the load torque T_L is suddenly increased to its rated value of 7490 N · m at $t = 0.1$ s and then decreased to 1000 N · m at $t = 0.3$ s, the generated torque T_e responds quickly. The torque ripple is set by the torque tolerance band δ_T. The stator current i_{as} varies with T_e accordingly.

Since the stator flux λ_s and the motor torque T_e are controlled independently, λ_s is kept constant during the sudden load torque changes. To demonstrate the effectiveness of the stator flux control, its reference λ_s^* has a step reduction from its rated value of 9.0–6.3 Wb at $t = 0.5$ s. The stator flux λ_s responds quickly while the stator current i_{as} is adjusted accordingly to keep T_e constant. The sector number obtained from the sector number generator in Fig. 14.8-2 is also shown in the figure.

Figure 14.8-6 shows the trajectories of the stator flux λ_s in Fig. 14.8-5 for $0.35\text{ s} \leq t \leq 0.75$ s. The outer and inner trajectories correspond to the steady-state operation before and after the stator flux reduction at $t = 0.5$ s.

14.8.5 Comparison Between DTC and FOC Schemes

Based on the analysis given in the preceding sections, the features and drawbacks for the DTC and rotor flux FOC schemes are summarized in Table 14.8-3.

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Table 14.8-3 Comparison Between DTC and FOC Schemes

Comparison	DTC	FOC
Field orientation (reference frame transformation)	Not required	Required
Control scheme	Simple	Complex
Stator current control	No	Yes
Motor parameters required	R_s	R_s, L_{ls}, L_{lr}, L_m, and R_r
Sensitivity to motor parameter variations	Not very sensitive	Sensitive
PWM scheme	Hysteresis band	Carrier based, SVM, or hysteresis band
Switching frequency	Variable	Defined (for carrier based and SVM)

14.9 SUMMARY

Two advanced control schemes, the FOC and DTC, for high-performance induction motor drives are presented in this chapter. There exist a variety of FOC schemes with many different variations and approaches. To make the subject matter easy to understand, this chapter focuses on the rotor flux oriented control schemes. The other reason for selecting this scheme for analysis lies in its simplicity and wide acceptance in the drive industry.

To facilitate the development of the FOC and DTC schemes, the *dq*-axis dynamic model for the induction motor is introduced. The algorithms for the rotor flux oriented control for VSI and CSI fed MV drives are presented. The operating principle of the DTC scheme is discussed. A comparison between the FOC and DTC schemes is provided. The important concepts are illustrated with computer simulations.

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Chapter 15

Control of Synchronous Motor Drives

15.1 INTRODUCTION

High-power variable-speed synchronous motor drives are widely used in industry. The power rating of a large synchronous motor drive can reach 100 MW while its voltage can be as high as 13.8 kV [1]. The synchronous motor drive is often used in applications where a high dynamic performance is required, such as rolling mills and mine hoists. When the synchronous motor drive is used for extruders, pumps, fans, and compressors, its variable-speed operation can provide considerable energy savings in comparison with the fixed-speed operation. Synchronous motor drives have also found applications in the propulsion system of large vessels [2].

The synchronous motors can be controlled by various methods, including zero *d*-axis current (ZDC) control, maximum torque per ampere (MTPA) control, direct torque control (DTC), and power factor control (PFC). In this chapter, the dynamic and steady-state models of the synchronous motor are introduced. The operating principle of the ZDC, MTPA, and DTC schemes for the voltage source converter (VSC) and current source converter (CSC) fed synchronous motor drives is elaborated. The characteristics and dynamic performance of the drives employing these control schemes are analyzed. The important concepts are illustrated by computer simulations.

15.2 MODELING OF SYNCHRONOUS MOTOR

15.2.1 Construction

The synchronous motor used in high-power medium voltage (MV) drives can be generally classified into two categories: wound rotor synchronous motor (WRSM) and permanent magnet synchronous motor (PMSM). In the WRSM, the rotor magnetic

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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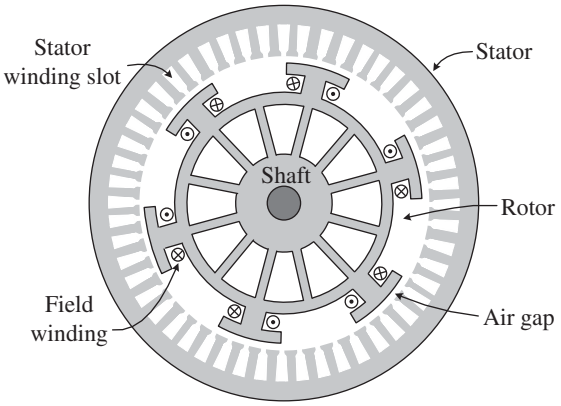

Figure 15.2-1 Salient-pole WRSM (six-pole configuration).

flux is generated by the current in the rotor field winding while the PMSM uses permanent magnets to produce the rotor flux. Depending on the shape of the rotor and the distribution of the air gap along the perimeter of the rotor, the synchronous motor can be classified into salient-pole and cylindrical (non-salient-pole) machines.

As its name indicates, the WRSM generates a rotor magnetic flux through a wound rotor configuration. Figure 15.2-1 illustrates a typical six-pole WRSM. The construction of the stator is similar to that of the induction motor. The rotor has a field winding, which is wound around the pole shoes. The rotor poles are positioned symmetrically on the rotor perimeter in a radial configuration around the shaft. The air gap is uneven in the salient-pole synchronous motors.

The rotor field winding requires direct current excitation. The rotor current can be supplied directly by brushes in contact with slip rings attached to the shaft, and the slip rings are electrically connected to the rotor winding. Alternatively, a brushless exciter that is physically attached to the shaft can be used. The exciter generates alternating currents that are converted into dc current for rotor winding by a diode rectifier mounted on the rotor shaft. The first option is simple, but the brushes and slip rings need regular maintenance. In contrast, the second option is more costly and complex, but requires little maintenance.

In PMSM, the magnetic flux of the rotor is generated by permanent magnets. Therefore, these motors are brushless. Given the lack of rotor windings, the power density of the motor can be increased, which can in turn reduce the size and weight of the motor. In addition, no rotor winding losses are incurred, thereby increasing the efficiency of the motor. The main drawback of these motors is that they are expensive and prone to demagnetization. Depending on how the permanent magnets are mounted on the rotor, the PMSM can be classified into surface-mounted and inset PM motors.

In the surface-mounted PMSM, permanent magnets are placed on the rotor surface. Figure 15.2-2 depicts such a motor, where eight magnets are mounted evenly on the surface of the rotor core and separated by non-iron materials placed between two adjacent magnets. Given that the permeability of magnets is close to that of

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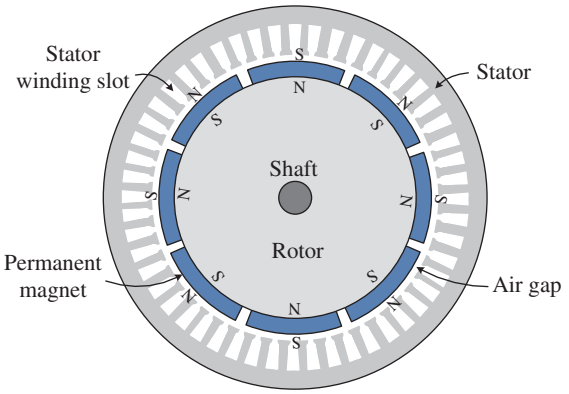

Figure 15.2-2 Surface-mounted PMSM (eight-pole configuration).

non-iron materials, the effective air gap between the rotor and the stator is uniformly distributed around the rotor surface. This type of motor is known as a cylindrical or non-salient-pole PMSM.

The main advantage of the surface-mounted PMSM is its simplicity and low construction cost in comparison with the inset PMSM. However, the magnets are subject to centrifugal forces that may detach them from the rotor at high rotational speeds. Therefore, the surface-mounted PMSM is mainly for low-speed applications, where the rotor speed is up to a few thousand revolutions per minute (rpm).

In the inset PMSM, the permanent magnets are inset into the rotor as shown in Fig. 15.2-3. Saliency is generated by the different permeabilities of the rotor core material and magnets. This configuration reduces the rotational stress of the magnets associated with the centrifugal forces in comparison with the surface-mounted PMSM. Therefore, this type of motors can operate at high rotor speeds.

15.2.2 Dynamic Model of Synchronous Motors (SM)

Figure 15.2-4 shows a general *dq*-axis model of a synchronous motor. To simplify the analysis, the motor is modeled in the rotor synchronous reference frame, where all

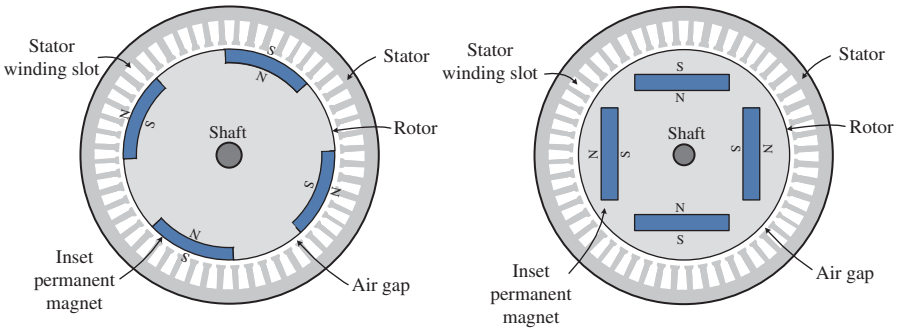

Figure 15.2-3 Inset PMSM (four-pole configuration).

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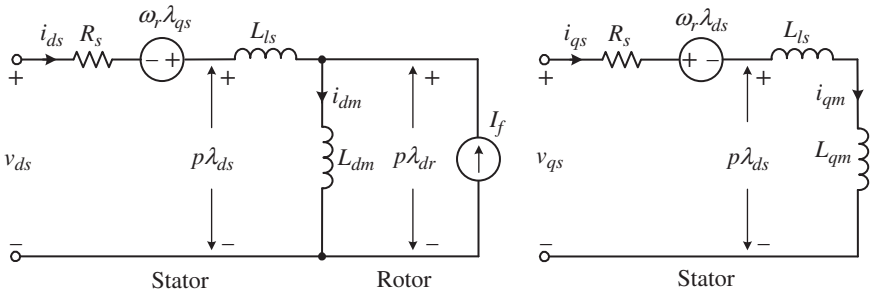

(a) *d*-axis circuit (b) *q*-axis circuit

Figure 15.2-4 General *dq*-axis model of a synchronous motor in the rotor synchronous reference frame.

the variables are of dc quantities. The stator circuit of the *dq*-axis model is essentially the same as that of the induction motor (IM) model shown in Fig. 14.3.3 with the following modifications:

- The speed of the arbitrary reference frame ω in the IM model is replaced with the rotor speed ω_r in the rotor synchronous frame.
- The magnetizing inductance L_m is replaced with the *dq*-axis magnetizing inductances L_{dm} and L_{qm} of the synchronous motor. In a cylindrical synchronous motor, the *dq*-axis magnetizing inductances are equal ($L_{dm} = L_{qm}$) whereas in the salient-pole motors, the *d*-axis magnetizing inductance is normally lower than the *q*-axis magnetizing inductance ($L_{dm} < L_{qm}$).

To model the rotor circuit, the field current in the rotor winding of a WRSM is represented by a constant current source I_f in the *d*-axis circuit as shown in 15.2-4a [2]. In the PMSM, the permanent magnets that replace the field winding in the WRSM can be modeled by an equivalent current source I_f with a fixed magnitude.

To simplify the synchronous motor model presented in Fig. 15.2-4, the following mathematical manipulations can be performed. The stator voltage equations for the motor can be expressed as

$$\begin{cases} v_{ds} = R_s i_{ds} - \omega_r \lambda_{qs} + p\lambda_{ds} \\ v_{qs} = R_s i_{qs} + \omega_r \lambda_{ds} + p\lambda_{qs} \end{cases} \tag{15.2-1}$$

where λ_{ds} and λ_{qs} are the *d*- and *q*-axis stator flux linkages, respectively, which are given by

$$\begin{cases} \lambda_{ds} = L_{ls} i_{ds} + L_{dm}(I_f + i_{ds}) = L_d i_{ds} + \lambda_r \\ \lambda_{qs} = (L_{ls} + L_{qm}) i_{qs} = L_q i_{qs} \end{cases} \tag{15.2-2}$$

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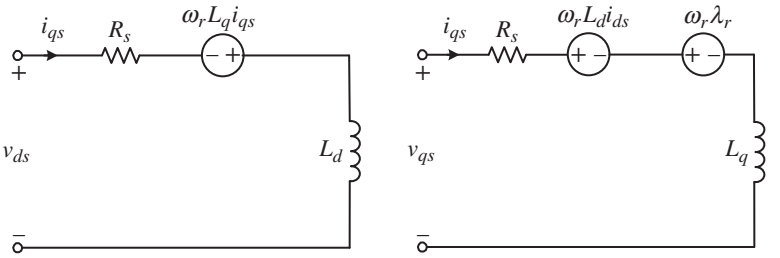

(a) d-axis circuit (b) q-axis circuit

Figure 15.2-5 Simplified dq-axis model of a synchronous motor in the rotor synchronous reference frame.

where λ_r is the rotor flux linkage, and L_d and L_q are the stator dq-axis self-inductances, respectively. These variables are defined by

$$\begin{cases} \lambda_r = L_{dm} I_f \\ L_d = L_{ls} + L_{dm} \\ L_q = L_{ls} + L_{qm} \end{cases} \tag{15.2-3}$$

Substituting Eq. (15.2-2) into (15.2-1) and considering $d\lambda_r/dt = 0$ for the constant field current I_f in the WRSM and the constant rotor flux linkage λ_r in the PMSM, we have

$$\begin{cases} v_{ds} = R_s i_{ds} - \omega_r L_q i_{qs} + L_d p i_{ds} \\ v_{qs} = R_s i_{qs} + \omega_r L_d i_{ds} + \omega_r \lambda_r + L_q p i_{qs} \end{cases} \tag{15.2-4}$$

Figure 15.2-5 shows a simplified model for the synchronous motors, which is derived from Eq. (15.2-4). It is noted that

- The simplified model is as accurate as the general model of Fig. 15.2-4 since no assumption is made during the derivation. The performance analysis based on the general and simplified models will give identical results.
- The synchronous motor model is valid for both WRSM and PMSM. For a given field current I_f in the WRSM, the rotor flux can be calculated by $\lambda_r = L_{dm} I_f$. For the PMSM, the rotor flux λ_r is produced by permanent magnets, and its value can be derived from the nameplate data and parameters of the motor.
- The SM model is also valid for both cylindrical and salient-pole motors. For a cylindrical motor, the dq-axis self-inductances, L_d and L_q, are equal whereas they are different in a salient-pole motor. The d-axis self-inductance of a salient-pole motor is usually lower than the q-axis self-inductance ($L_d < L_q$).

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The electromagnetic torque produced by a synchronous motor can be calculated with the same equation for the induction motor given in Eq. 14.3-8, that is,

$$T_e = \frac{3P}{2}(i_{qs}\lambda_{ds} - i_{ds}\lambda_{qs}) \tag{15.2-5}$$

Substituting Eq. (15.2-2) into (15.2-5), we obtain

$$T_e = \frac{3P}{2}[\lambda_r\, i_{qs} + (L_d - L_q)\; i_{ds} i_{qs}] \tag{15.2-6}$$

The rotor speed ω_r is governed by the motion equation, namely,

$$\omega_r = \frac{P}{JS}(T_e - T_l) \tag{15.2-7}$$

where P is the number of pole pairs, J is the moment of inertia of the rotor and its mechanical load, S is the Laplace operator, and T_l is the load torque.

To derive the synchronous motor model for computer simulation, Eq. (15.2-4) can be rearranged as follows:

$$\begin{cases} i_{ds} = \dfrac{1}{S}(v_{ds} - R_s i_{ds} + \omega_r L_q i_{qs})/L_d \\ i_{qs} = \dfrac{1}{S}(v_{qs} - R_s i_{qs} - \omega_r L_d i_{ds} - \omega_r \lambda_r)/L_q \end{cases} \tag{15.2-8}$$

where the differential operator p in Eq. (15.2-4) is replaced by Laplace operator S.

The block diagram for the dynamic simulation of a synchronous motor is shown in Fig. 15.2-6, which is derived from Eqs. (15.2-6) to (15.2-8). The input variables of the synchronous motor model are the dq-axis stator voltages v_{ds} and v_{qs}, the rotor flux linkage λ_r, and the mechanical load T_l. The output variables are the dq-axis stator currents i_{ds} and i_{qs}, the rotor speed ω_r, and the electromagnetic torque T_e of the motor.

15.2.3 Steady-State Equivalent Circuits

The steady-state model of a synchronous motor provides a useful tool to analyze the steady-state operation of the motor. The steady-state model can be developed from the dynamic model shown in Fig. 15.2-5. Considering that the dq-axis stator currents, i_{ds} and i_{qs}, in the rotor synchronous reference frame are of constant dc values in steady state, their derivatives in Eq. (15.2-4), pi_{ds} and pi_{qs}, become zero. Therefore,

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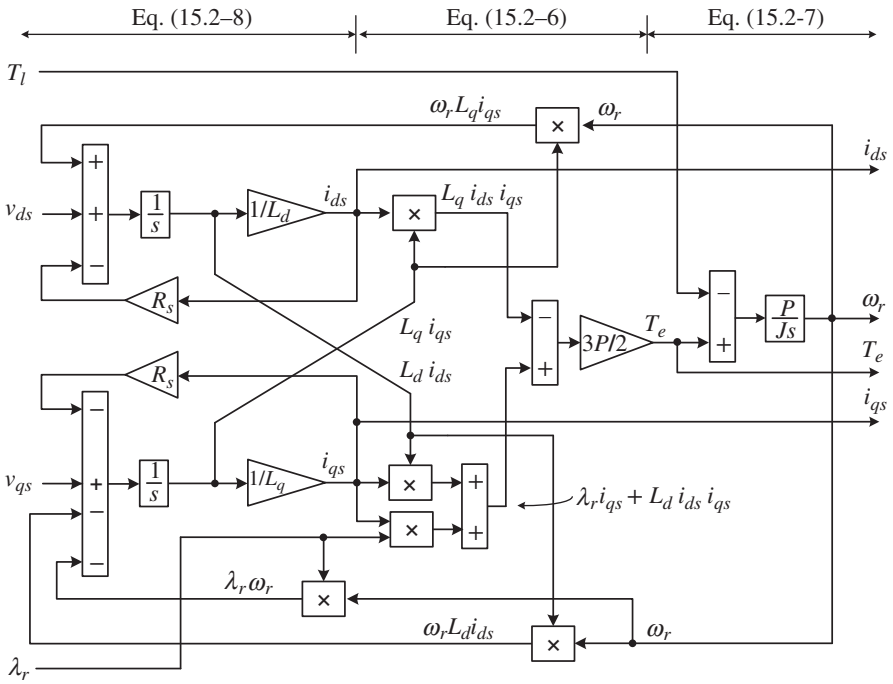
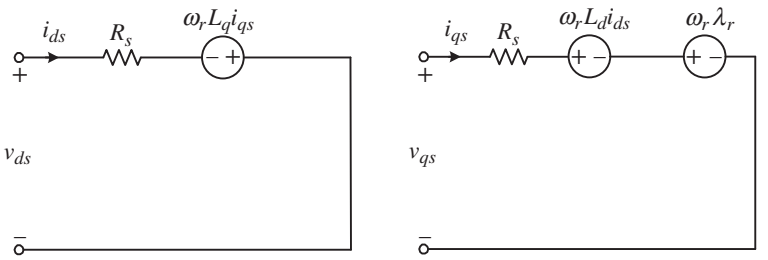


Figure 15.2-6 Block diagram for the dynamic simulation of synchronous motors.

the equations that describe the steady-state characteristics of the synchronous motor are given by

$$\begin{cases} v_{ds} = R_s i_{ds} - \omega_r L_q i_{qs} \\ v_{qs} = R_s i_{qs} + \omega_r L_d i_{ds} + \omega_r \lambda_r \end{cases} \tag{15.2-9}$$

Based on Eq. (15.2-9), a steady-state model for the synchronous motor can be derived, which is illustrated in Fig. 15.2-7 for steady-state analysis.



(a) *d*-axis circuit (b) *q*-axis circuit

Figure 15.2-7 Steady-state model of a synchronous motor.

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15.3 VSC FED SM DRIVE WITH ZERO *d*-AXIS CURRENT (ZDC) CONTROL

15.3.1 Introduction

The synchronous motor can be controlled by a number of methods to achieve different objectives [3]. For instance, the *d*-axis stator current of the motor can be set to zero during the operation to simplify control scheme design and implementation. In this section, the ZDC control scheme is introduced and analyzed.

15.3.2 Principle of ZDC Control

The ZDC control can be realized by resolving the three-phase stator current in the stationary reference frame into *dq*-axis components in the synchronous reference frame. The *d*-axis component i_{ds} is then kept at zero by the controller [4]. With the ZDC, the stator current i_s is equal to its *q*-axis component i_{qs}, that is,

$$\begin{cases} \vec{i}_s = i_{ds} + ji_{qs} = ji_{qs} \\ i_s = \sqrt{i_{ds}^2 + i_{qs}^2} = i_{qs} \end{cases} \quad \text{for } i_{ds} = 0 \tag{15.3-1}$$

where $\vec{i}_s$ is the space vector of the stator current, and i_s represents its magnitude, which is also the peak value of the three-phase stator current in the stationary reference frame.

The electromagnetic torque of the motor

$$T_e = \frac{3}{2}P(\lambda_r i_{qs} + (L_d - L_q)\, i_{ds} i_{qs}) \tag{15.3-2}$$

can then be simplified to

$$T_e = \frac{3}{2}P\lambda_r i_{qs} = \frac{3}{2}P\lambda_r i_s \tag{15.3-3}$$

The above equation indicates that when $i_{ds} = 0$, the motor torque is proportional to the stator current i_s. With a constant rotor flux linkage λ_r, the torque exhibits a linear relationship with the stator current. This is similar to torque production in a dc machine with a constant field current, where the electromagnetic torque is proportional to the armature current.

A space vector diagram for a synchronous motor with the ZDC control is illustrated in Fig. 15.3-1. The vector diagram is derived according to the SM steady-state model under the assumption that the stator winding resistance R_s is negligibly small and the rotor flux linkage vector $\vec{\lambda}_r$ is aligned with the *d*-axis of the synchronous reference frame. All the vectors in the diagram rotate in space at the rotor synchronous speed.

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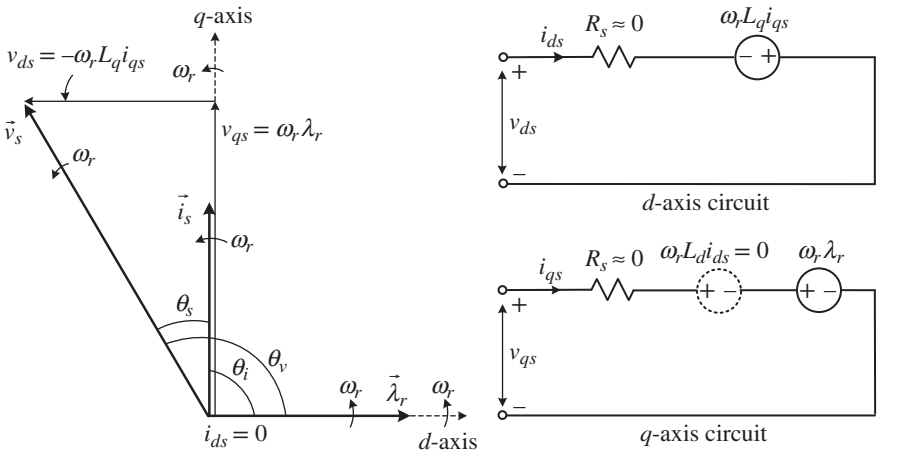

(a) Vector diagram (b) *dq*-axis steady-state model

Figure 15.3-1 Space vector diagram of a synchronous motor with ZDC control.

The stator current vector $\vec{i}_s$ is perpendicular to the rotor flux vector $\vec{\lambda}_r$. The magnitude of the stator voltage is given by

$$v_s = \sqrt{(v_{ds})^2 + (v_{qs})^2} = \sqrt{(-\omega_r L_q i_{qs})^2 + (\omega_r \lambda_r)^2} \tag{15.3-4}$$

The stator power factor angle is defined by

$$\theta_s = \theta_v - \theta_i \tag{15.3-5}$$

where θ_v and θ_i are the angles of the stator voltage and current vectors, given by

$$\begin{cases} \theta_v = \tan^{-1} \dfrac{v_{qs}}{v_{ds}} \\ \theta_i = \tan^{-1} \dfrac{i_{qs}}{i_{ds}} = \dfrac{\pi}{2} \end{cases} \tag{15.3-6}$$

With the ZDC scheme, the stator power factor angle can be calculated by

$$\theta_s = \theta_v - \theta_i = \left(\tan^{-1} \frac{v_{qs}}{v_{ds}} \right) - \frac{\pi}{2} \qquad \text{for } i_{ds} = 0 \tag{15.3-7}$$

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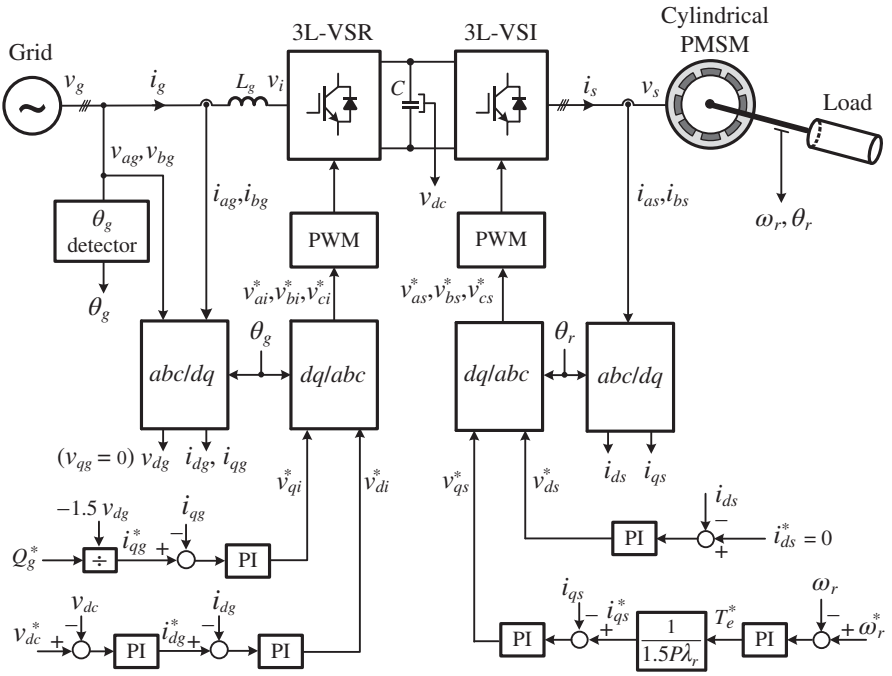

Figure 15.3-2 Control scheme of a synchronous motor drive with ZDC control.

15.3.3 Implementation of ZDC Control in VSC Fed SM Drive

In a synchronous motor drive, a number of system variables should be controlled: (1) the motor torque and speed, (2) the input power factor of the drive, and (3) the dc-link voltage of the power converters. The motor torque and speed are normally controlled by the inverter whereas the input power factor and the dc-link voltage are controlled by the rectifier. In this section, a block diagram for the SM drive with the ZDC control is presented, and its dynamic performance is analyzed.

The block diagram of the ZDC controlled SM drive is shown in Fig. 15.3-2, where a three-level (3L) voltage source rectifier (VSR) and a three-level voltage source inverter (VSI) are employed. For the SM drive at the megawatt levels, multilevel converters, such as the three-level NPC and three-level ANPC converters presented in Chapters 8 and 9, are usually used.

(a) Rectifier Control The main function of the VSR is to control input power factor of the drive through the reactive power control and the dc bus voltage v_{dc} based on grid voltage oriented control (VOC) scheme as shown in Fig. 15.3-2 [3, 4]. The grid voltages v_{ag} and v_{bg}, grid currents i_{ag} and i_{bg}, which are also the rectifier input currents, and the dc-link voltage v_{dc} are measured. In the measurement of three-phase

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variables, we only need to measure two of them since the third one can be calculated from $x_a + x_b + x_c = 0$ in a three-phase balanced system.

To realize the grid VOC, the grid voltage angle θ_g is detected. This angle is used for the transformation of variables from the *abc* stationary frame to the *dq* rotating reference frame through the *abc/dq* transformation or from the rotating frame back to the stationary frame through the *dq/abc* transformation as shown in Fig. 15.3-2. With the grid VOC, the speed of the *dq*-axis rotating frame is given by $\omega_g = 2\pi f_g$, where f_g is the grid frequency.

To find the grid voltage angle θ_g, the grid phase voltages, v_{ag} and v_{bg}, are measured and transformed to the $\alpha\beta$ frame through the *abc*/$\alpha\beta$ transformation presented in Chapter 14.

$$\begin{cases} v_{\alpha g} = \dfrac{2}{3}\left(v_{ag} - \dfrac{1}{2}v_{bg} - \dfrac{1}{2}v_{cg}\right) = v_{ag} \\ v_{\beta g} = \dfrac{2}{3}\left(\dfrac{\sqrt{3}}{2}v_{bg} - \dfrac{\sqrt{3}}{2}v_{cg}\right) = \dfrac{\sqrt{3}}{3}(v_{ag} + 2v_{bg}) \end{cases} \quad \text{for } v_{ag} + v_{bg} + v_{cg} = 0 \tag{15.3-8}$$

The grid voltage angle θ_g and its peak value v_g can then be found from

$$\begin{cases} \theta_g = \tan^{-1}\dfrac{v_{\beta g}}{v_{\alpha g}} \\ v_g = \sqrt{(v_{\alpha g})^2 + (v_{\beta g})^2} \end{cases} \tag{15.3-9}$$

There are three feedback control loops in the system: two inner current loops for the control of the *dq*-axis currents i_{dg} and i_{qg}, and one outer dc voltage feedback loop for the control of the dc-link voltage v_{dc}. The measured three-phase grid currents in the *abc* stationary frame i_{ag}, i_{bg}, and i_{cg} are transformed to the *dq*-axis currents i_{dg} and i_{qg} in the *dq* rotating frame, which represent the active and reactive components of the three-phase grid current in the VOC scheme, respectively.

With the VOC scheme, the grid voltage vector is aligned with the *d*-axis of the rotating reference frame. In doing so, the *q*-axis component of the grid voltage, v_{qg}, is equal to zero and its *d*-axis component is equal to the magnitude of the grid voltage, that is, $v_{qg} = 0$ and $v_{dg} = \sqrt{v_g^2 - v_{qg}^2} = v_g$. As a result, the calculation of the active and reactive grid power can be simplified as

$$\begin{cases} P_g = \dfrac{3}{2}(v_{dg}i_{dg} + v_{qg}i_{qg}) = \dfrac{3}{2}v_{dg}i_{dg} \\ Q_g = \dfrac{3}{2}(v_{qg}i_{dg} - v_{dg}i_{qg}) = -\dfrac{3}{2}v_{dg}i_{qg} \end{cases} \quad \text{for } v_{qg} = 0 \tag{15.3-10}$$

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from which the q-axis current reference can be calculated by

$$i_{qg}^{*} = \frac{Q_g^{*}}{-1.5 v_{dg}} \tag{15.3-11}$$

where Q_g^{*} is the reactive power reference, which can be set to zero for unity power factor operation, a positive value for leading power factor operation, or a negative value for lagging power factor operation. To control the reactive power accurately, the measured dq-axis current i_{qg} is compared with its reference i_{qg}^{*}, and the error is sent to the q-axis current PI controller to generate the q-axis voltage reference v_{qi}^{*} for the rectifier, as shown in Fig. 15.3-2.

To control the dc-link voltage tightly, the measured dc-link voltage v_{dc} is compared with its reference v_{dc}^{*}, and the error is sent to the dc-link voltage PI controller, which generates the d-axis current reference i_{dg}^{*}. Compared with the measured d-axis current i_{dg}, the difference between i_{dg}^{*} and i_{dg} is sent to the d-axis current PI controller, which generates the d-axis voltage reference v_{di}^{*} for the rectifier.

The dq-axis voltage references v_{di}^{*} and v_{qi}^{*} in the dq-axis rotating frame, which are of dc quantity, are then transformed into three-phase reference voltages v_{ai}^{*}, v_{bi}^{*}, and v_{ci}^{*} in the abc stationary frame via the dq/abc transformation, which are sent to the PWM generator. The three-phase reference voltages are sinusoidal waveforms, based on which a carrier-based modulation or space vector modulation (SVM) can be employed to generate gate signals for the switching devices in the rectifier, which controls its reactive power and dc-link voltage according to their respective references.

(b) Inverter Control The main function of the VSI is to control the torque and speed of the motor while its d-axis stator current is set to zero to achieve ZDC control. This is implemented by three feedback loops: two inner loops for the dq-axis stator currents i_{ds} and i_{qs}, and one outer loop for the rotor speed ω_r as shown in Fig. 15.3-2.

To control the speed of the motor, the rotor speed ω_r is detected, which is compared with its reference ω_r^{*}. The error is sent to the speed PI controller, which generates the torque reference T_e^{*}. The reference for the q-axis stator current i_{qs}^{*}, which is the torque-producing component of the stator current, is calculated according to the torque equation of (15.3-3). The d-axis stator current reference i_{ds}^{*} is set to zero.

The measured dq-axis stator currents, i_{ds} and i_{qs}, are then compared with their references i_{ds}^{*} and i_{qs}^{*}. The errors are sent to two PI controllers, which generate the dq-axis voltage references v_{ds}^{*} and v_{qs}^{*} for the inverter. These two references in the synchronous frame are then transformed into three-phase voltage references v_{as}^{*}, v_{bs}^{*}, and v_{cs}^{*} in the stationary frame via the dq/abc transformation. For the reference frame transformation between the dq-axis rotor synchronous frame and the abc stationary frame, the rotor position angle θ_r is needed, which is detected by an encoder mounted on the rotor shaft as shown in Fig. 15.3-2.

The three-phase voltage references v_{as}^{*}, v_{bs}^{*}, and v_{cs}^{*} are then sent to the PWM generator. Either carrier-based modulation or space vector modulation can be employed to generate gate signals for the switching devices in the inverter, which controls the d-axis stator current i_{ds} and the rotor speed ω_r according to their respective references.

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Table 15.3-1 System Parameters of a 2.45 MW VSC Fed Drive

PMSM	2.45 MW/4000 V/53.33 Hz/400 rpm/490 A Cylindrical machine
Control scheme	Zero *d*-axis current (ZDC) scheme
System input variables	*d*-axis stator current: $i_{ds}^* = 0$
	Rotor speed reference: ω_r^* (step input)
	Rectifier reactive power reference: $Q_g^* = 0$
	dc-link voltage reference: $v_{dc}^* = 7045$ V (3.05 pu)
Rectifier	Topology: three-level NPC converter
	Modulation scheme: SVM
	Switching frequency: 740 Hz
	Filter inductance L_g: 0.775 mH (0.045 pu)
Inverter	Topology: three-level NPC converter
	Modulation scheme: SVM
	Switching frequency: 740 Hz
	Harmonic filter: No
dc-link filter	Capacitor: 2500 μF (4.0 pu)
Grid voltage/frequency	4000 V/60 Hz

15.3.4 Transient Analysis

The dynamic performance of a synchronous motor drive using three-level VSCs with the ZDC scheme is investigated through computer simulation. A 2.45 MW cylindrical PMSM is used in the investigation and its parameters are given in Table A-1 at the end of this chapter. The drive system parameters are listed in Table 15.3-1.

Figure 15.3-3 shows the start-up transient response of the 2.45 MW SM drive with the ZDC scheme. A three-level neutral point clamped (3L-NPC) converter is used in the rectifier and inverter, and both converters are modulated by a conventional SVM scheme. The device switching frequency is 740 Hz, which leads to an equivalent switching frequency of 1480 Hz for both converters due to the three-level converter topology.

The rotor speed reference ω_r^* has a step input from zero to its rated value of 1 pu at $t = 0.05$ s. The motor accelerates under the no-load condition while its torque is limited to the rated value set by the saturation level of the speed PI controller. Due to the ZDC control scheme, the *d*-axis stator current i_{ds} is kept at zero while the *q*-axis i_{qs}, which is the torque-producing component of the stator current, is proportional to the torque T_e. The peak value of the phase-*a* stator current i_{as} is equal to i_{qs} with the ZDC scheme. The dc-link voltage v_{dc} is maintained around its reference value, which is 3.05 pu.

It is noted that when the drive is operating in steady state under no-load conditions, after the startup, the stator current i_{as} (and also i_{qs}) is very low. This is mainly due to

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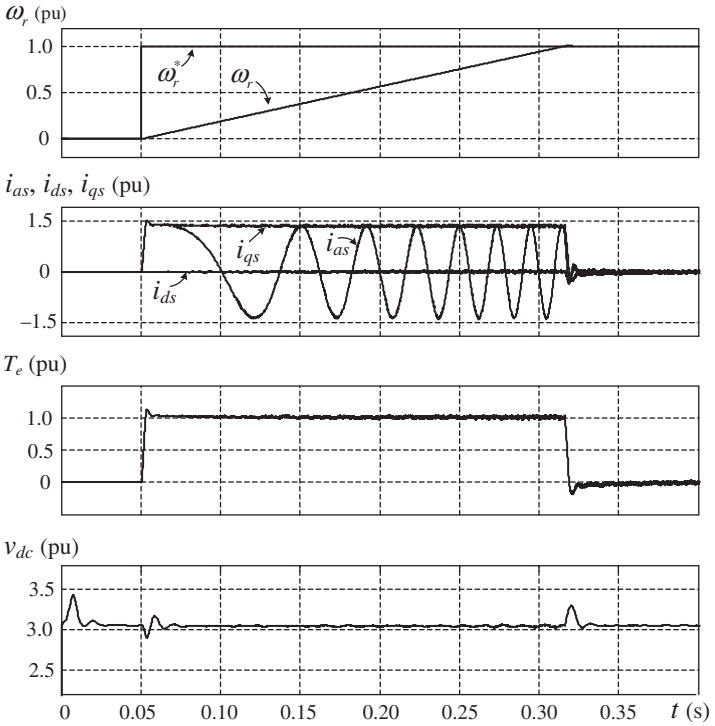

Figure 15.3-3 Start-up transient of a 2.45 MWSM drive with ZDC scheme.

the fact that the rotating field in the air gap of the PMSM is produced by the permanent magnets. A small stator current is provided to cover the stator winding losses and small rotational losses (windage and friction). This is not like the induction motors, where a much higher stator current is needed to produce a rotating field in the air gap even though the motor is running in steady state under no-load conditions. In large induction motors, the no-load stator current would be in the range of 0.25 to 0.3 pu, which corresponds to the magnetizing inductance of 4.0 to 3.0 pu, respectively.

It is also noted that in order to reduce the drive start-up time in the simulation, the moment of inertia of the motor is reduced on purpose such that the drive can start within a short period of time. In practice, it takes longer time to start, especially when the drive starts under the loaded conditions.

Figure 15.3-4 shows the transient response of the 2.45 MW drive to a step change in the load torque. The motor operates initially at its rated speed of 400 rpm (1 pu) under the no-load conditions. A step load torque of 1 pu is applied to the motor at $t =$ 0.1 s. The d-axis current i_{ds} is kept around zero by the ZDC scheme, and its q-axis current i_{qs} is proportional to the torque T_e. The motor speed ω_r^* is kept at the rated value of 1 pu except at the moment when the load torque is applied. The dc-link voltage is maintained around its reference value of 3.05 pu.

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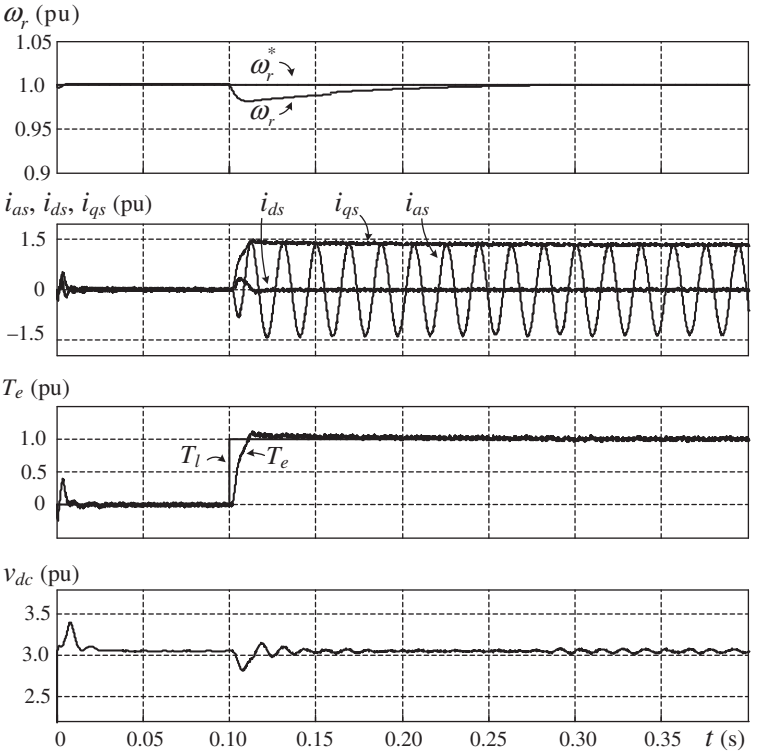

Figure 15.3-4 Response of a 2.45 MW SM drive to a step load torque change (ZDC scheme).

15.4 VSC FED SM DRIVE WITH MTPA CONTROL

15.4.1 Introduction

In addition to the ZDC control scheme presented in the preceding section, the synchronous motor can also be controlled by maximum torque per ampere (MTPA) scheme, where the motor torque can be produced by a minimum stator current [3–6]. With this approach, the power losses in the stator winding and also in the power converters can be minimized, which improves the efficiency of the drive system. In this section, the MTPA control method is presented and analyzed.

15.4.2 Principle of MTPA Control

As discussed earlier, the electromagnetic torque of a synchronous motor can be calculated by

$$T_e = \frac{3}{2}P(\lambda_r\, i_{qs} + (L_d - L_q)\; i_{ds} i_{qs}) \tag{15.4-1}$$

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The above equation indicates that the motor torque is a function of dq-axis stator currents i_{ds} and i_{qs}. By adjusting the ratio of i_{ds} to i_{qs}, it is possible to produce a given torque by a minimum stator current. For a given stator current

$$i_s = \sqrt{i_{ds}^2 + i_{qs}^2} \tag{15.4-2}$$

its d-axis component can be calculated by

$$i_{ds} = \sqrt{i_s^2 - i_{qs}^2} \tag{15.4-3}$$

Substituting the above equation into the torque equation of (15.4-1), the electromagnetic torque of the motor can be rewritten as

$$T_e = \frac{3}{2}P\left(\lambda_r\, i_{qs} + (L_d - L_q)\left(\sqrt{i_s^2 - i_{qs}^2}\right) i_{qs}\right) \tag{15.4-4}$$

In a cylindrical motor, the dq-axis self-inductances, L_d and L_q, are equal. The above torque equation is simplified to

$$T_e = \frac{3}{2}P\lambda_r i_{qs} \tag{15.4-5}$$

where the motor torque is proportional to the q-axis stator current i_{qs}, but it is not affected by the d-axis stator current i_{ds}. To achieve the MTPA control in a cylindrical motor drive, we can set the d-axis current i_{ds} to zero. In doing so, the motor torque is produced by a minimum stator current ($i_s = i_{qs}$). To summarize, the MTPA control for a cylindrical motor is essentially the ZDC control.

In a salient-pole motor, the principle of the MTPA scheme can be illustrated through the following derivations. By differentiating the torque equation of (15.4-4) with respect to i_{qs}, we have

$$\frac{dT_e}{di_{qs}} = \frac{3P}{2}\left(\lambda_r + (L_d - L_q)\, i_{ds} - (L_d - L_q)\, i_{qs}^2 \frac{1}{\sqrt{i_s^2 - i_{qs}^2}}\right) \tag{15.4-6}$$

To achieve the MTPA operation, the above equation is set to zero

$$\lambda_r + (L_d - L_q)\, i_{ds} - (L_d - L_q)\frac{i_{qs}^2}{i_{ds}} = 0 \tag{15.4-7}$$

from which the d-axis stator current is given by

$$i_{ds} = \frac{-\lambda_r}{2(L_d - L_q)} \pm \sqrt{\frac{\lambda_r^2}{4(L_d - L_q)^2} + i_{qs}^2} \qquad \text{for } L_d \neq L_q \tag{15.4-8}$$

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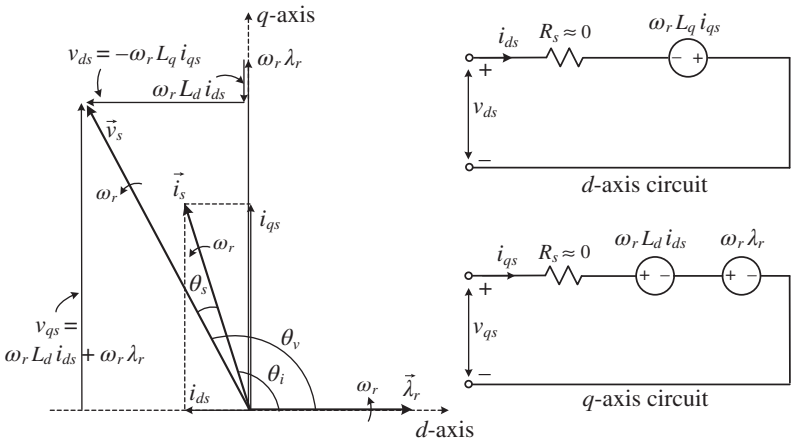

(a) Vector diagram (b) *dq*-axis steady-state model

Figure 15.4-1 Space vector diagram of a synchronous motor with MTPA control.

The *d*-axis stator current shown in the above equation has two possible values. The first term on the right side of the equation has a positive value because the *d*-axis self-inductance L_d of a salient-pole synchronous motor is normally lower than the *q*-axis self-inductance L_q. To minimize the *d*-axis stator current for the MTPA control, the second term should have a negative value, that is,

$$i_{ds} = \frac{-\lambda_r}{2(L_d - L_q)} - \sqrt{\frac{\lambda_r^2}{4(L_d - L_q)^2} + i_{qs}^2} \qquad \text{for } L_d \neq L_q \qquad (15.4\text{-}9)$$

The space vector diagram of a synchronous motor with the MTPA control is given in Fig. 15.4-1a, which is derived according to its *dq*-axis steady-state model given in Fig. 15.4-1b. The equations for the calculations of stator voltage angle θ_v, stator current angle θ_i, and stator power factor angle θ_s are the same as those given in the ZDC scheme, and therefore are not repeated here. The main difference between the vector diagrams of the two schemes is that the stator current angle θ_i for the ZDC scheme is equal to $\pi/2$ whereas it varies with the operating conditions in the MTPA scheme. It is noted that the *d*-axis stator current i_{ds} is negative, which is caused by the negative value of the *d*-axis stator voltage v_{ds}. Equation (15.4-9) also shows that the *d*-axis current is negative since the second term on the right side of the equation has a higher value than the first term.

Figure 15.4-2 shows an example of the variation of the stator current i_s with its *d*-axis current i_{ds} for a given torque of $T_e = 0.8$ pu. Different combinations of the *dq*-axis stator currents i_{ds} and i_{qs} can produce the same required torque, but the minimum stator current takes place at point *Q*, at which the MTPA operation is achieved.

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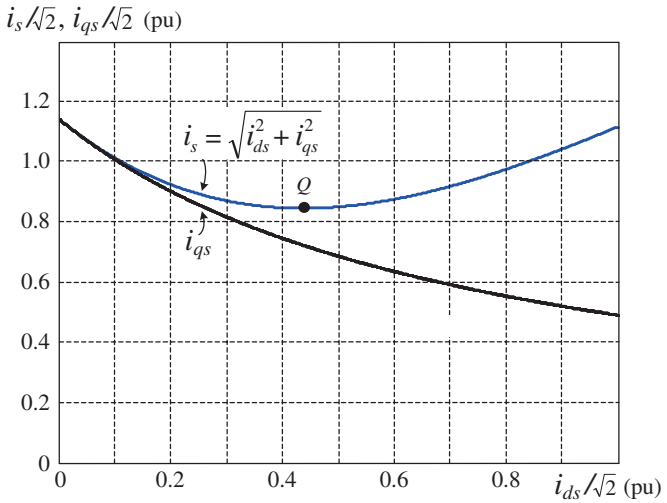

Figure 15.4-2 Trajectory of the stator current to produce a torque of $T_e = 0.8$ pu.

15.4.3 Implementation of MTPA Control in VSC Fed SM Drive

The block diagram of a synchronous motor drive with MTPA control is shown in Fig. 15.4-3. The power converters and controls are the same as those shown in Fig. 15.3-2 for the ZDC scheme except that the inverter is controlled by the MTPA scheme for a salient-pole synchronous motor. To implement the MTPA control, the dq-axis stator currents, i_{ds} and i_{qs}, need to be controlled according to their references, i_{ds}^* and i_{qs}^*, respectively.

The reference for the q-axis stator current can be calculated from the torque equation (15.4-1):

$$i_{qs}^* = \frac{2T_e^*}{3P(\lambda_r + (L_d - L_q))\, i_{ds}} \tag{15.4-10}$$

where T_e^* is the torque reference generated by the speed PI controller, i_{ds} is the measured d-axis stator current, and λ_r is the rotor flux linkage, which is a constant set by the permanent magnets in the PMSM or by the rotor field current in the WRSM.

The reference for the d-axis current i_{ds}^* can then be obtained from (15.4-9):

$$i_{ds}^* = \frac{-\lambda_r}{2(L_d - L_q)} - \sqrt{\frac{\lambda_r^2}{4(L_d - L_q)^2} + (i_{qs}^*)^2} \tag{15.4-11}$$

where i_{qs}^* is the q-axis stator current reference given by (15.4-10). Based on the above two equations, the block diagram for the MTPA control algorithm is shown in Fig. 15.4-3b.

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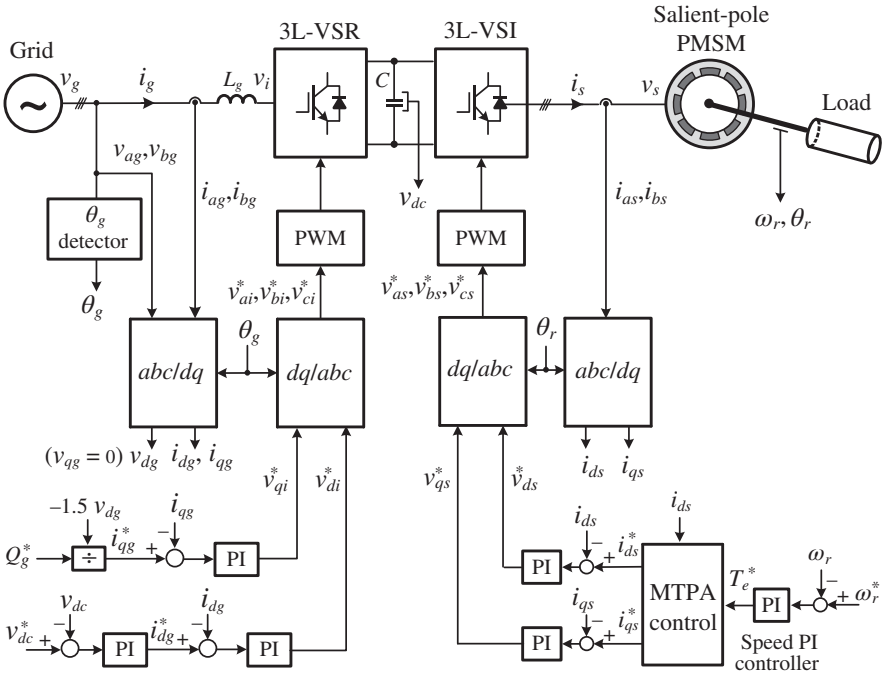
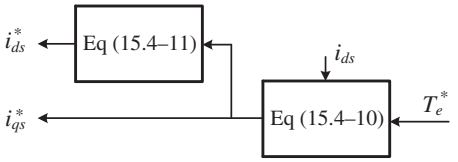


(a) System block diagram



(b) MTPA control

Figure 15.4-3 Block diagram of a salient-pole PMSM drive with MTPA control.

15.4.4 Transient Analysis

The dynamic performance of a synchronous motor drive with the MTPA scheme is investigated by computer simulation. The drive system parameters are the same as those given in Table 15.3-1 except that the *d*-axis current reference is no longer to be zero; it varies with the operating conditions of the drive. A salient-pole synchronous motor of 2.5 MW is used in the study, and its nameplate data and motor parameters are given in Table A-2 at the end of this chapter.

Figure 15.4-4 shows the dynamic response of the SM drive with MTPA control during the system start-up followed by a sudden change in load torque. The speed reference ω_r^* has a step increase from zero to the rated value at $t = 0.1$ s. The motor accelerates under the no-load condition, and its torque is limited to 1.3 pu by the speed PI controller. The rotor speed ω_r increases linearly with time due to

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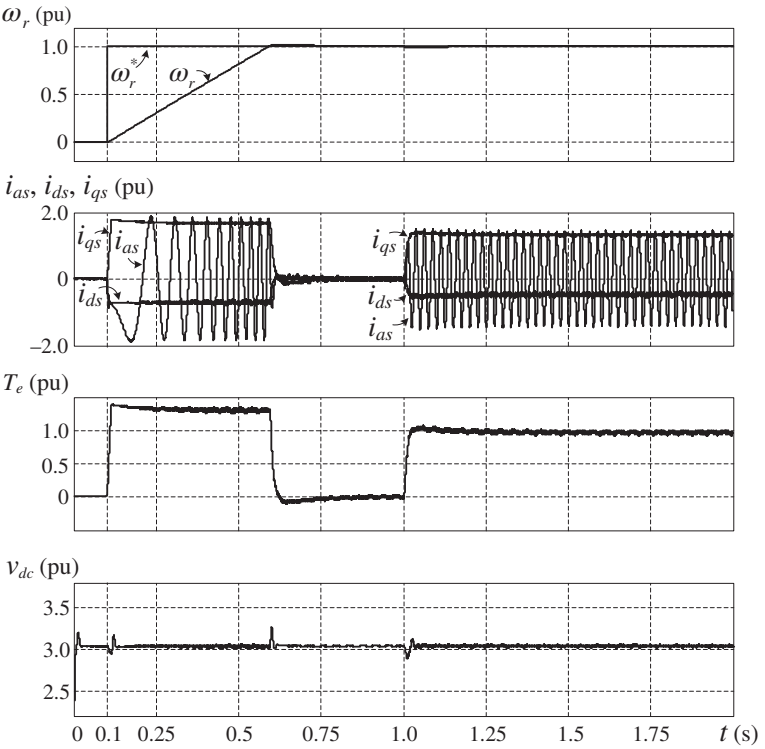

Figure 15.4-4 Transient response of a salient-pole SM drive with MTPA control.

the constant torque operation during the startup transient. When the rotor speed ω_r reaches its reference at approximately $t = 0.6$ s, its torque drops to zero.

At $t = 1.0$ s, a rated load torque is applied to the motor. The motor torque T_e responds quickly and increases to the rated value as well. Both dq-axis stator currents vary accordingly such that magnitude of the stator current is minimized by the MTPA scheme to minimize the stator winding and converter power losses. The dc-link voltage is maintained around the set value of 3.05 pu by the rectifier.

15.5 VSC FED SM DRIVE WITH DTC SCHEME

15.5.1 Introduction

The DTC is another advanced control scheme developed for motor drives for applications where a high dynamic performance is required [6, 7]. The principle of the DTC scheme was introduced in Chapter 14, but it was for induction motor (IM) drives using two-level VCSs (2L-VSC). In this section, the principle of DTC for synchronous motor drives using three-level NPC converters is introduced, and its dynamic performance is investigated.

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15.5.2 Principle of DTC

The electromagnetic torque of a synchronous motor can be expressed by

$$T_e = \frac{3P}{2}\lambda_s\lambda_r \sin\theta_T \tag{15.5-1}$$

where θ_T is the torque angle, which is the angle between the stator flux linkage vector $\vec{\lambda}_s$ and the rotor flux linkage vector $\vec{\lambda}_r$.To realize the DTC control, the magnitude of the stator flux λ_s is kept constant (normally at its rated value) by the controller, and the motor torque T_e can then be controlled by θ_T directly. This method is, therefore, referred to as the DTC. With the magnitude of the stator flux kept constant, the magnitude of the rotor flux λ_r is almost constant since the difference between the stator and rotor fluxes is only the leakage flux produced by the stator leakage inductance.

From the synchronous motor model developed in the earlier section, the stator flux vector $\vec{\lambda}_s$ can be mainly determined by the stator voltage:

$$p\vec{\lambda}_s = \vec{v}_s - R_s\vec{i}_s \tag{15.5-2}$$

where R_s is the stator winding resistance, which is quite small, especially in large synchronous motors. The above equation indicates that the derivative of $\vec{\lambda}_s$ reacts instantly to the changes in $\vec{v}_s$. As discussed in Chapter 14, the stator voltage $\vec{v}_s$ can be controlled by the reference vector $\vec{V}_{ref}$ in the space vector modulation scheme. Since the reference vector $\vec{V}_{ref}$ can be synthesized by the voltage vectors of the inverter, the magnitude and angle of the stator flux vector $\vec{\lambda}_s$ can be adjusted by proper selection of the inverter voltage vectors.

(a) Space Vector Diagram Figure 15.5-1 shows the space vector diagram of a three-level VSI for the DTC control. There are 19 space vectors, $\vec{V}_0$ to $\vec{V}_{18}$, which were defined in Table 8.3.1 in Chapter 8. To realize the DTC control, the space vector diagram is divided into 12 sectors from I to XXII. The stator flux vector $\vec{\lambda}_s$, rotor flux vector $\vec{\lambda}_r$, and torque angle θ_T are also shown in the figure. Both stator and rotor flux vectors are rotating in space at the synchronous speed ω_s.

Figure 15.5-2 illustrates the impact of the selection of the voltage vectors on $\vec{\lambda}_s$ and θ_T. Assuming the stator flux vector $\vec{\lambda}_s$ is in Sector I, selection of voltage vector $\vec{V}_7$ will make both λ_s and θ_T increase as shown in Fig. 15.5-2a, while selection of $\vec{V}_{12}$ will make λ_s increase and θ_T decrease as shown in Fig. 15.5-2b. Similarly, selection of $\vec{V}_{10}$ will make both λ_s and θ_T decrease whereas selection of $\vec{V}_9$ will make λ_s decrease and θ_T increase. It can be concluded from the above analysis that the magnitude of the stator flux λ_s and the torque angle θ_T can be fully adjusted by selection of proper voltage vectors of the inverter.

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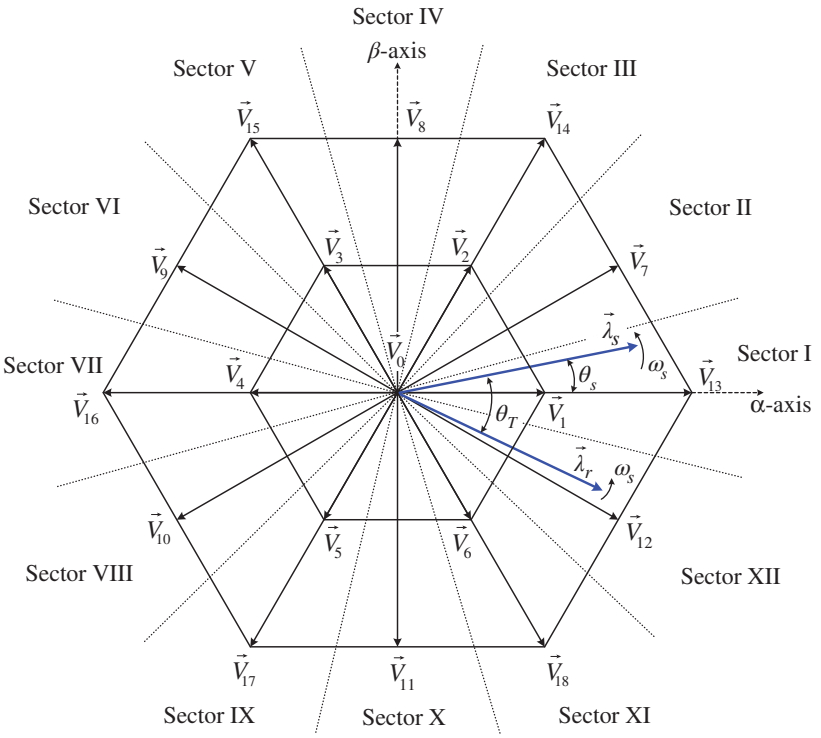

Figure 15.5-1 Space vector diagram of a three-level inverter for DTC control.

(b) Hysteresis Comparators for Stator Flux and Torque Figure 15.5-3 shows a simplified block diagram for the DTC scheme with a three-level VSI. The stator flux λ_s and electromagnetic torque T_e of the motor are controlled independently to achieve high dynamic performance. The stator flux reference λ_s^* is compared with the measured stator flux λ_s obtained from the flux/torque calculator, and the error $\Delta\lambda_s$ is sent to the flux comparator. The torque reference T_e^* is compared with the measured torque T_e from the flux/torque calculator and their difference ΔT_e is the input of the torque comparator. The output of the flux and torque comparators (x_λ and x_T) are sent to a switching logic unit to select the appropriate voltage vectors (switching states) of the inverter according to the sector that the stator flux reference $\vec{\lambda}_s^*$ falls into.

Both flux and torque comparators are of hysteresis type, and their transfer characteristics are given in Fig. 15.5-4. The hysteresis comparators are designed for three-level inverters. The flux comparator has two output levels, $x_\lambda = +1$ and -1, while the torque comparator has five output levels, $x_T = +2, +1, 0, -1$, and -2, where "+1" requests an increase in λ_s or θ_T, "+2" requests a large increase, "−1" demands a decrease, "−2" demands a large decrease, "0" signifies no changes, and δ_λ and δ_T are the tolerance bands for the stator flux and torque comparators, respectively. If the tolerance bands are reduced, the stator flux and motor torque will be more accurately

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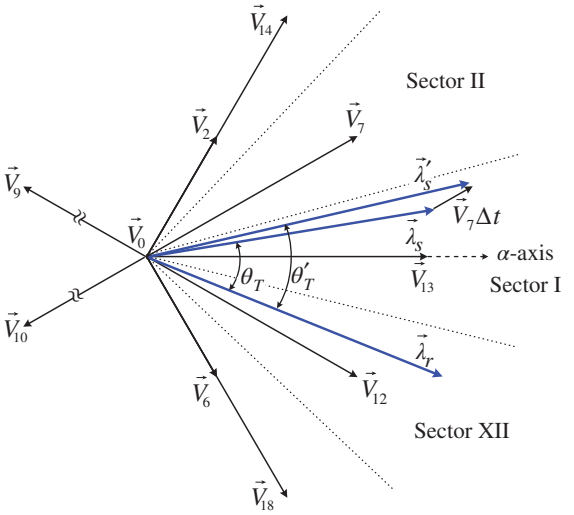
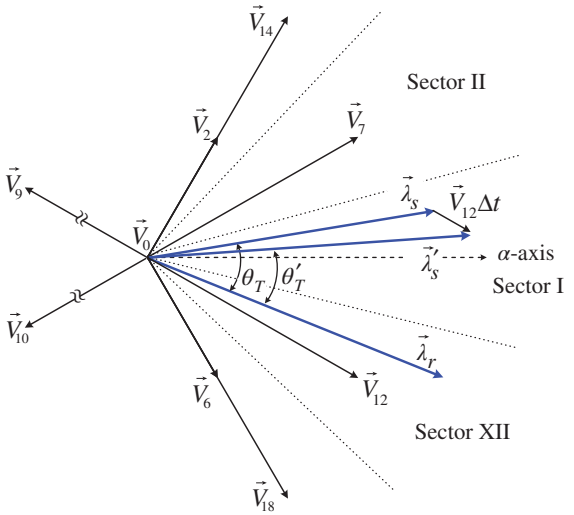


(a) Selection of Vector $\vec{V}_7$



(b) Selection of Vector $\vec{V}_{12}$

Figure 15.5-2 Impact of voltage vector selection on $\vec{\lambda}_s$ and θ_T.

controlled with less ripples but this is at the expense of higher switching frequencies for the inverter, and vice versa.

(c) Switching Logic Table 15.5-1 presents the switching logic for the stator flux reference $\vec{\lambda}_s^*$ that rotates in the counter clockwise direction in space [7, 8]. The input variables of the table are x_λ, x_T, and the sector number while its output indicates

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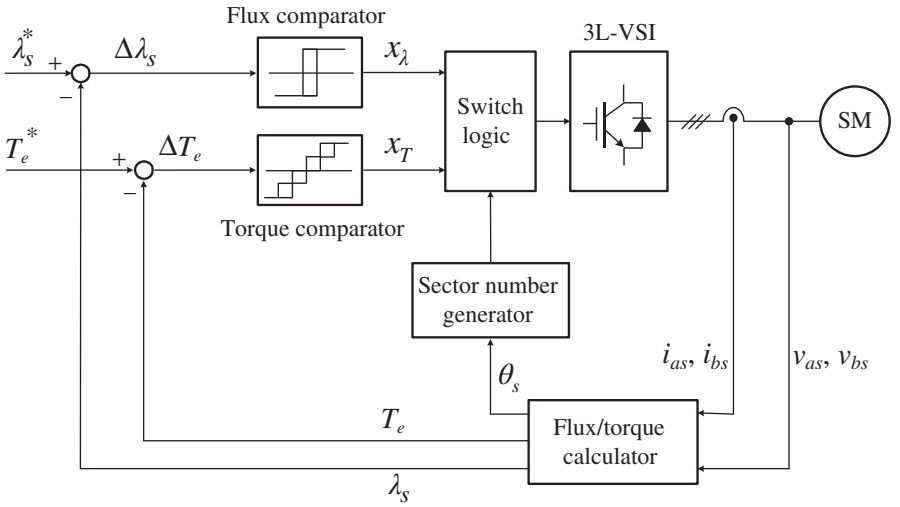
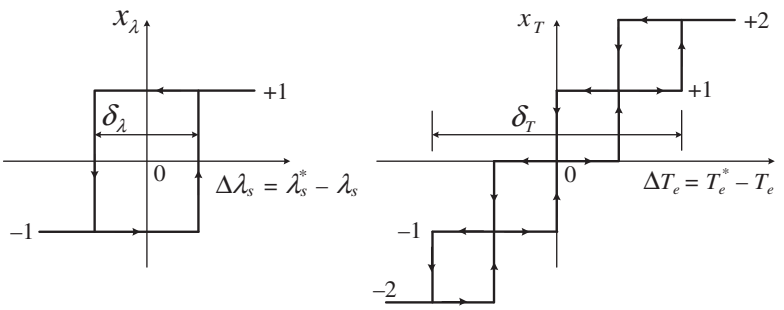


Figure 15.5-3 Simplified block diagram of the DTC scheme.

which inverter voltage vector should be selected according to the inputs. Assuming that the stator flux reference $\vec{\lambda}_s^*$ is in sector I and the outputs of the comparators are $x_\lambda = x_T = +1$, it demands an increase in λ_s and T_e. Voltage vector $\vec{V}_7$ can then be selected from the table, which will make both λ_s and θ_T (i.e.,T_e) increase as shown in Fig. 15.5-2a. Similarly, when $x_\lambda = +1$ and $x_T = -1$, selection of $\vec{V}_{12}$ will make λ_s increase and θ_T decrease as illustrated in Fig. 15.5-2b.

If the output of the stator flux comparator remains at $x_\lambda = +1$, but the output of the torque comparator is $x_T = +2$, it demands a large increase in T_e. Voltage vector $\vec{V}_{14}$ can be selected from the table. This vector is also shown in Fig. 15.5-2a, where it can be visualized that the selection of $\vec{V}_{14}$ will lead to a larger increase in torque angle than $\vec{V}_7$.



(a) Flux comparator (b) Torque comparator

Figure 15.5-4 Characteristics of the hysteresis comparators.

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Table 15.5-1 DTC Switching Logic for a Three-Level Voltage Source Inverter

Comparator Output		Sector Number											
x_λ	x_T	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
+1	+2	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$
	+1	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$
	0	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$
	−1	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$
	−2	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$
−1	+2	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$
	+1	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$
	0	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$	$\vec{V}_0$
	−1	$\vec{V}_{10}$	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$
	−2	$\vec{V}_{17}$	$\vec{V}_{11}$	$\vec{V}_{18}$	$\vec{V}_{12}$	$\vec{V}_{13}$	$\vec{V}_7$	$\vec{V}_{14}$	$\vec{V}_8$	$\vec{V}_{15}$	$\vec{V}_9$	$\vec{V}_{16}$	$\vec{V}_{10}$

When the output of the torque comparator x_T is zero (no need to adjust T_e), zero vector $\vec{V}_0$ can be selected from Table 15.5-1. The switching logic for all 12 sectors from I to XII is summarized in Table 15.5-1.

It is noted that the switching logic presented in Table 15.5-1 is not unique. For example, when $x_\lambda = +1$ and $x_T = \pm 1$, the selection of the small vectors $\vec{V}_2$ and $\vec{V}_6$ instead of the medium vectors $\vec{V}_7$ and $\vec{V}_{12}$ is also valid. However, the selection of the small vectors may slow down the system dynamic response due to the shorter length (lower voltage) of these vectors than the medium vectors, but it may result in lower harmonic distortion in the inverter output voltages.

(d) Stator Flux and Torque Calculator Similar to the DTC scheme for the induction motor drive presented in Chapter 14, the stator flux vector $\vec{\lambda}_s$ of the synchronous motor can be expressed as

$$\begin{aligned}\vec{\lambda}_s &= \lambda_{ds} + j\lambda_{qs} \\ &= \int (v_{ds} - R_s i_{ds})dt + j \int (v_{qs} - R_s i_{qs})dt\end{aligned} \tag{15.5-3}$$

from which the magnitude and angle of the stator flux linkage vector can be obtained by

$$\begin{cases} \lambda_s = \sqrt{\lambda_{ds}^2 + \lambda_{qs}^2} \\ \theta_s = \tan^{-1}\left(\dfrac{\lambda_{qs}}{\lambda_{ds}}\right) \end{cases} \tag{15.5-4}$$

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The *dq*-axis stator voltages and currents in Eq. (15.5-3); v_{ds}, v_{qs}, i_{ds}, and i_{qs}, can be calculated from the measured three-phase stator voltages and currents. The developed electromagnetic torque can be determined by

$$T_e = \frac{3P}{2}(i_{qs}\lambda_{ds} - i_{ds}\lambda_{qs}) \tag{15.5-5}$$

The above equations illustrate that the stator flux λ_s, the stator flux angle θ_s, and the developed torque T_e can all be obtained from the measured stator voltages and currents. The only motor parameter required in the calculations is the stator resistance R_s, which is easy to measure and its value does not vary with the operating conditions of the motor too much.

The stator flux vector $\vec{\lambda}_s$ of a synchronous motor can also be calculated from Eq. (15.2-2), which is

$$\vec{\lambda}_s = \lambda_{ds} + j\lambda_{qs} = (L_d i_{ds} + \lambda_r) + j(L_q i_{qs}) \tag{15.5-6}$$

Compared with Eq. (15.5-3), the above question does not required integrators to find the stator flux linkage, but need the motor *dq*-axis inductances and rotor flux linkage in the calculation.

15.5.3 Implementation of DTC Control in VSC Fed SM Drive

Figure 15.5-5 shows the block diagram of a synchronous motor drive with the DTC control scheme, where the rectifier with its control is omitted since it has been discussed in the earlier sections. The dc-link voltage is kept constant by the rectifier while the motor torque and speed are controlled by the inverter with the DTC scheme.

The control block diagram of Fig. 15.5-5 is essentially the same as that given in Fig. 15.5-3 except that a motor speed feedback control is added. As discussed earlier, the essence of the DTC control is to keep the stator flux λ_s constant and control the torque T_e independently. This is achieved by the stator flux and torque comparators, whose outputs are sent to the switching logic unit to generate switching signals for the inverter. The drive system with the DTC scheme has the following features:

- There is only one PI controller for the motor speed control, but no PI controllers are needed for the stator flux or torque control.
- There is no need for PWM schemes such as carrier-based or space vector modulation. The gate signals for the inverter are directly generated by the switching logic.
- There is no need for reference frame transformation between the *abc* stationary frame and the *dq* rotating frame.

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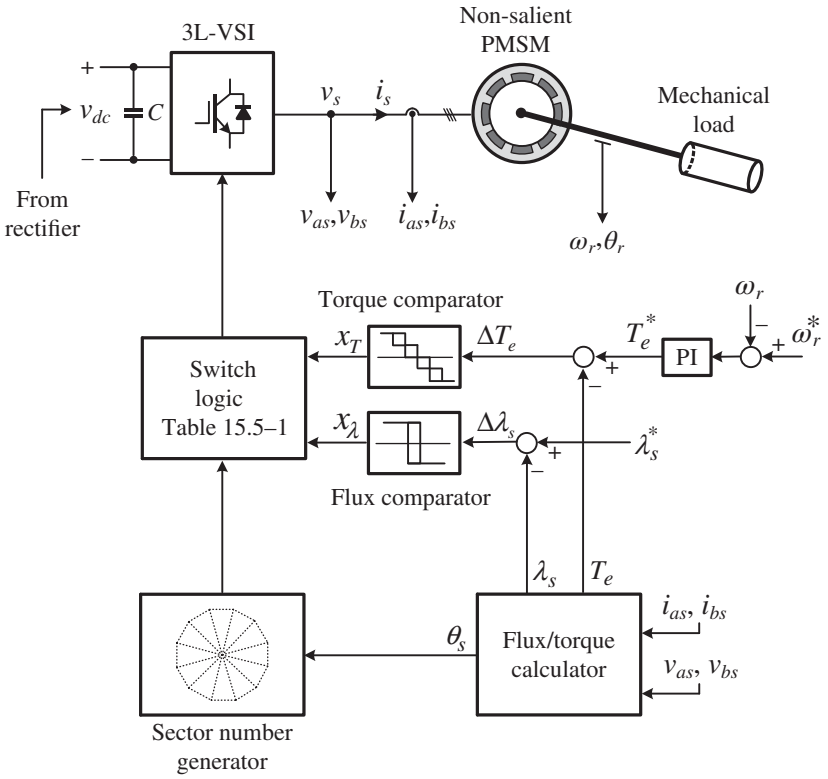

Figure 15.5-5 Block diagram of a synchronous motor drive with DTC control.

Therefore, the DTC scheme is simpler than the ZDC and MTPA schemes while its dynamic performance is comparable due to the decoupled stator flux and torque control.

15.5.4 Transient Analysis

The performance of the DTC scheme for a synchronous motor drive is investigated by computer simulation. The parameters of the drive system are given in Table 15.5-2. A 2.45 MW cylindrical PMSM is used in the investigation, and its parameters are listed in Table A-1 at the end of this chapter.

Figure 15.5-6 shows the dynamic response of the DTC controlled drive to the step changes in the load torque. The drive initially operates at its rated speed of $n_r = 400$ rpm under the no-load condition. The load torque T_l is suddenly increased to its rated value of 1.0 pu at $t = 0.2$ s, decreased to zero at $t = 0.4$ s, and finally increased to 0.5 pu at 0.6 s. It can be observed that the motor torque T_e responds quickly and follows the load torque profile closely. The torque ripple is set by the torque tolerance band δ_T. The phase-a stator current i_{as} varies with T_e accordingly.

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Table 15.5-2 System Parameters of a 2.45 MW drive with DTC Control

PMSM	2.45 MW/4000 V/53.33 Hz/400 rpm/490 A cylindrical machine
Control scheme	DTC with switching logic in Table 15.5-1
System input variables	Stator flux reference: $\lambda_s^* = 6.892$ Wb (rated) Rotor speed reference: $\omega_r^* = 400$ rpm (rated) Rectifier reactive power reference: $Q_g^* = 0$ dc-link voltage reference: $v_{dc}^* = 7045$ V (3.05 pu)
Rectifier	Topology: three-level NPC Modulation scheme: No Average witching frequency: 1000 Hz Filter inductance L_g: 0.775 mH (0.045 pu)
Inverter	Topology: three-level NPC Modulation scheme: No Averaging switching frequency: 1000 Hz Stator flux tolerance band: $\delta_\lambda = 0.8$ Wb (0.16 pu) Torque tolerance band: $\delta_T = 4000$ N · m (0.07 pu) Harmonic filter: No
dc-link filter	Capacitor: 2500 μF (4.0 pu)
Grid voltage/frequency	4000 V/60 Hz

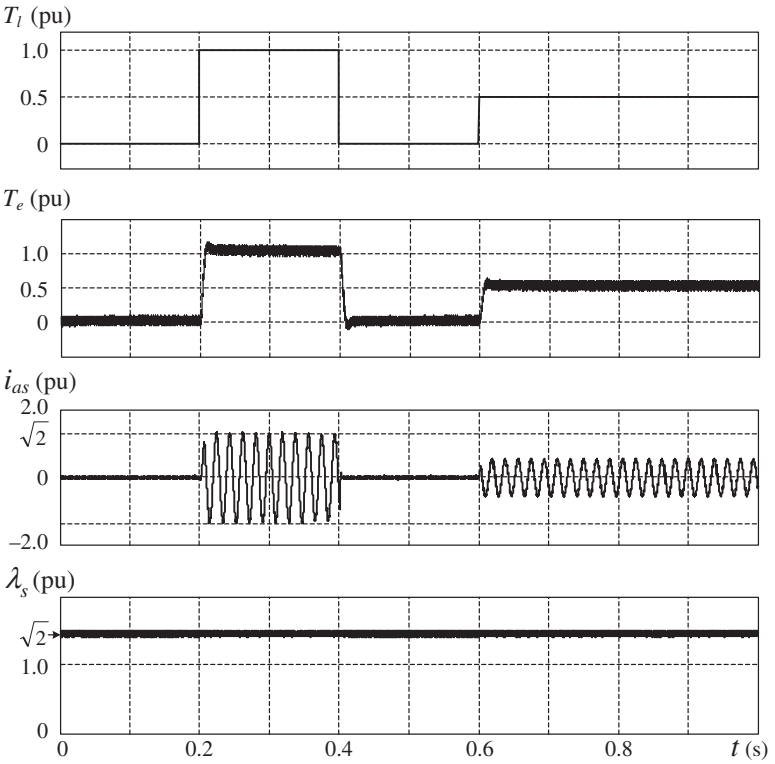

Figure 15.5-6 Dynamic response to the load torque change in a DTC controlled SM drive.

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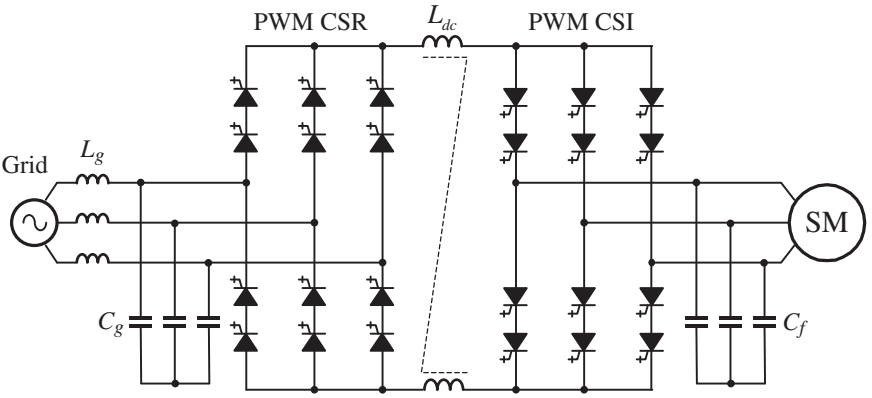
The stator flux λ_s is kept constant by the stator flux comparator. No obvious disturbances can be observed from the stator flux waveform during the sudden changes in the load torque. This is due to the fact that the stator flux is calculated by $\vec{\lambda}_s = \int (v_{ds} - R_s i_{ds})dt + j \int (v_{qs} - R_s i_{qs})dt$, where the stator winding resistance R_s is quite small, especially for large-size synchronous motors. In addition, the integral function in the stator flux calculation can be considered as a low-pass filter. As a result, a sudden change in the stator current due to the load torque change does not produce noticeable variations in the stator flux waveform.

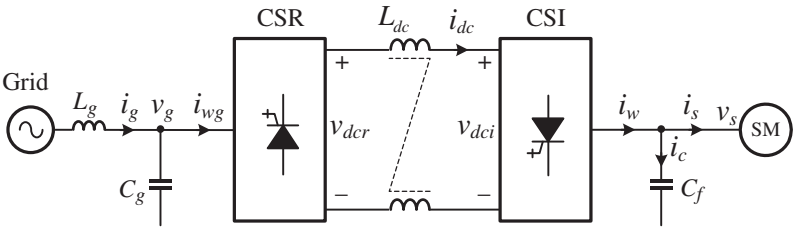
15.6 CONTROL OF CSC FED SM DRIVES

15.6.1 Introduction

The CSC fed synchronous motor drives have found various applications in industry. Figure 15.6-1 shows the typical configuration of a PWM CSC fed synchronous motor drive, where a current source rectifier (CSR) and a current source inverter (CSI) are used. The two converters are linked by a dc inductor L_{dc}, which makes the dc-link current smooth and continuous.



(a) Schematic diagram



(b) Simplified block diagram

Figure 15.6-1 Configuration of a PWM CSC fed synchronous motor drive.

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The CSC fed SM drive is characterized by a simple converter topology, inherent four-quadrant operation, and reliable fuseless short-circuit protection. In VSC fed MV drives, both inverter output voltage and frequency are controlled by the inverter PWM scheme. In contrast, in CSC fed drives, the magnitude of the inverter output current is controlled by the dc-link current i_{dc} produced by the rectifier whereas the output frequency of the inverter is controlled by the inverter PWM scheme. In addition, the stator current of the motor is not directly controlled by the inverter due to the presence of the filter capacitor C_f between the inverter and the motor. Therefore, additional measures are required to control the CSC fed drives.

The control schemes presented in the earlier sections for the VSC fed synchronous motor drives can also be applied to the CSC fed SM drives, including the ZDC control and MTPA control. This section presents both ZDC and MTPA control methods for the CSC fed synchronous motor drives.

15.6.2 CSC Fed SM Drive with ZDC Control

(a) Inverter Control To facilitate the analysis of the CSC fed synchronous motor drive with the ZDC control scheme, a space vector diagram for the drive is shown in Fig. 15.6-2. The rotor rotates in space at the synchronous speed ω_r. The rotor axis is aligned with the d-axis of the synchronous reference frame. With the ZDC control where the d-axis current i_{ds} is zero, the stator current vector $\vec{i}_s$ is aligned with the q-axis of the synchronous frame, and its magnitude is equal to the q-axis current i_{qs}. The stator voltage vector $\vec{v}_s$ leads its current vector $\vec{i}_s$ by θ_s, which is the stator power factor angle. The capacitor current $\vec{i}_c$ leads $\vec{v}_s$ by $\pi/2$. The inverter PWM current $\vec{i}_w$ is a vector sum of $\vec{i}_s$ and $\vec{i}_c$, and its angle with respect to the d-axis of the synchronous frame is θ_w, which can be calculated by $\theta_w = \tan^{-1}(i_{qw}/i_{dw})$. From Fig. 15.6-2, the inverter firing angle can be calculated by

$$\theta_{inv} = \theta_w + \theta_r \tag{15.6-1}$$

where θ_r is the rotor position angle.

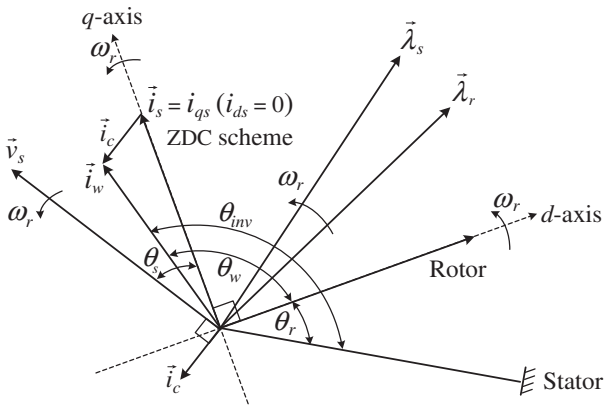

Figure 15.6-2 Vector diagram for a CSC fed SM drive with ZDC control.

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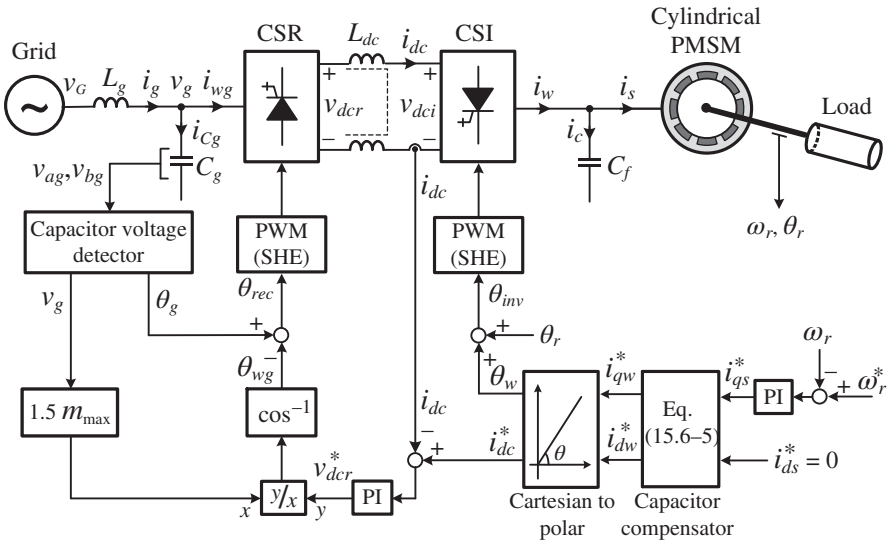

Figure 15.6-3 Block diagram of a CSC fed SM drive with ZDC scheme.

The block diagram of a CSC fed SM drive with the ZDC scheme is shown in Fig. 15.6-3, where the rectifier controls the dc-link current and the inverter controls the motor speed. To minimize the harmonic distortion in the rectifier input and inverter output currents, selective harmonic elimination (SHE) modulation scheme is used for both converters. The modulation index of the SHE scheme is fixed at its maximum value. The rectifier and the inverter are controlled by their firing angles θ_{rec} and θ_{inv}, respectively.

To achieve the ZDC control, the d-axis stator current reference i^*_{ds} is set to zero, and its q-axis stator current reference i^*_{qs} is set by the speed PI controller as shown in Fig. 15.6-3. The PI controller compares the measured rotor speed ω_r with its reference ω^*_r and makes the rotor speed follow its reference closely. The inverter output current i_w is a sum of the filter capacitor current i_c and stator current i_s. The dq-axis inverter output currents can be calculated by

$$\begin{cases} i_{dw} = i_{cd} + i_{ds} = -\omega_r C_f v_{qs} + i_{ds} \\ i_{qw} = i_{cq} + i_{qs} = \omega_r C_f v_{ds} + i_{qs} \end{cases} \tag{15.6-2}$$

where i_{cd} and i_{cq} are the dq-axis filter capacitor currents, given by

$$\begin{cases} i_{cd} = -\omega_r C_f v_{qs} \\ i_{cq} = \omega_r C_f v_{ds} \end{cases} \tag{15.6-3}$$

Note that the derivative terms associated with the capacitor current are not included in the above equation for simplicity since these terms have little impact on the dynamic performance of the drive.

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The *dq*-axis inverter output currents in Eq. (15.6-2) are a function of both *dq*-axis stator voltages and currents. By substituting the *dq*-axis stator voltage equations of (15.2-9) into (15.6-2), the inverter *dq*-axis output currents are a function of the stator current only:

$$\begin{cases} i_{dw} = -\omega_r^2 C_f \lambda_r - \omega_r^2 C_f L_d i_{ds} + i_{ds} \\ i_{qw} = -\omega_r^2 C_f L_q i_{qs} + i_{qs} \end{cases} \tag{15.6-4}$$

where the stator winding resistance is omitted for simplicity. Equation (15.6-4) can be used to serve as the inverter output current references i_{dw}^* and i_{qw}^* with the filter capacitor current taking into account:

$$\begin{cases} i_{dw}^* = -\omega_r^2 C_f \lambda_r + (1 - \omega_r^2 C_f L_d) i_{ds}^* \\ i_{qw}^* = (1 - \omega_r^2 C_f L_q)\, i_{qs}^* \end{cases} \tag{15.6-5}$$

Equation (15.6-5) is implemented in the control scheme of Fig. 15.6-3 as the capacitor compensator block.

With the cartesian to polar coordinates conversion, the inverter output current references i_{dw}^* and i_{qw}^* can be converted to the dc-link current reference i_{dc}^* and inverter output current angle θ_w, given by

$$\begin{cases} \theta_w = \tan^{-1}\left(\dfrac{i_{qw}^*}{i_{dw}^*}\right) \\ i_{dc}^* = \sqrt{(i_{dw}^*)^2 + (i_{qw}^*)^2} \end{cases} \tag{15.6-6}$$

The inverter firing angle θ_{inv} can then be calculated from Eq. (15.6-1), which is $\theta_{inv} = \theta_w + \theta_r$.

(b) Rectifier Control As shown in Fig. 15.6-3, the main function of the PWM CSR is to adjust the dc current i_{dc} through the firing angle control. The dc output voltage of the rectifier is given by (from Eq. (11.2-10))

$$v_{dcr} = 3/2 v_g \cos(\theta_{wg}) m \tag{15.6-7}$$

where v_g is the peak phase voltage of the grid-side capacitor, θ_{wg} is the angle of the rectifier input current i_{wg}, defined as $\theta_{wg} = \tan^{-1}(i_{qwg}/i_{dwg})$, and m_a is the amplitude modulation of the SHE scheme. From Eq. 15.6-7, the angle of the rectifier input current can be calculated by

$$\theta_{wg} = \cos^{-1}\frac{v_{dcr}}{1.5 m_{\max} v_g} \tag{15.6-8}$$

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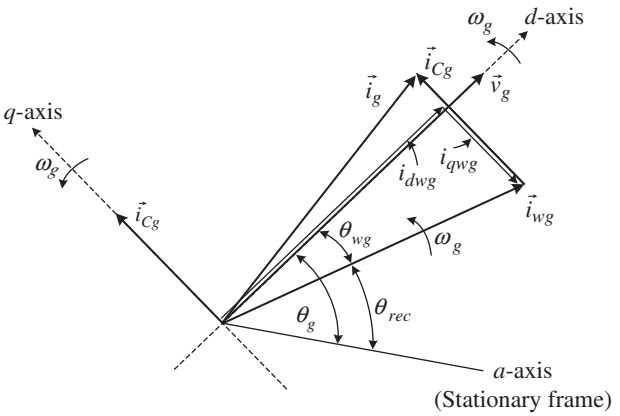

Figure 15.6-4 Vector diagram of the rectifier with firing angle control.

where $m_{\max}$ is the maximum modulation index, which is between 0.98 and 1.1, depending on the number of harmonics to be eliminated. The above equation is implemented in the rectifier control scheme shown in Fig. 15.6-3.

The relationship among the rectifier input current angle θ_{wg}, the grid-side capacitor voltage angle θ_g, and the rectifier firing angle θ_{rec} is illustrated in a vector diagram of Fig. 15.6-4. Similar to the grid VOC discussed in Section 15.3, the grid-side capacitor voltage vector $\vec{v}_g$, which is also the rectifier input voltage vector, is aligned with the d-axis of a rotating reference frame. The rotating reference frame rotates in space at a speed given by $\omega_g = 2\pi f_g$, where f_g is the frequency of the grid. With the delay angle control, the rectifier input current $\vec{i}_{wg}$ lags its input voltage $\vec{v}_g$ by θ_{wg}. The capacitor current $\vec{i}_{Cg}$ leads $\vec{v}_g$ by $\pi/2$. The grid current $\vec{i}_g$ is the sum of $\vec{i}_{wg}$ and $\vec{i}_{Cg}$ as shown in Fig. 15.6-4. The rectifier firing angle can then be obtained from

$$\theta_{rec} = \theta_g - \theta_{wg} \tag{15.6-9}$$

To detect the grid-side capacitor voltage, its phase voltages, v_{ag} and v_{bg}, are measured. The measured capacitor voltages are sent to the capacitor voltage detector, which calculates the magnitude of the grid-side capacitor voltage v_g, and its angle θ_g. The algorithm for the capacitor voltage detector is the same as that given in Eq. (15.3-9) and thus is not repeated here.

It is noted that with the firing angle control only (no modulation index control), the rectifier input power factor is not controllable. However, the lagging reactive component of the rectifier input current (i_{qwg}) compensates the leading reactive current (i_{Cg}) produced by the grid-side capacitor, which improves the overall input power factor of the drive system as shown in Fig. 15.6-4. For the full control of the rectifier input power factor, the control scheme for the PWM CSR presented in Chapter 11 can be considered.

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Table 15.6-1 System Parameters of a 2.45 MW CSC Fed Drive

PMSM	2.45 MW/4000 V/53.3 Hz/400 rpm/490 A cylindrical machine
Control scheme	Zero d-axis current (ZDC) control
System input variables	d-axis stator current reference: $i_{ds}^* = 0$ Rotor speed reference: ω_r^* (ramp function)
Rectifier	Topology: PWM current source rectifier Modulation scheme: SHE Modulation index: $m_a \approx 1$ Switching frequency: 420 Hz Harmonic filter: $L_g = 0.15$ pu, $C_g = 0.35$ pu
Inverter	Topology: PWM current source inverter Modulation scheme: SHE Modulation index: $m_a \approx 1$ Switching frequency: 420 Hz Harmonic filter: $C_f = 0.3$ pu
dc-link filter	dc choke: $L_{dc} = 1.3$ pu
Grid voltage/frequency	4000 V/60 Hz

15.6.3 Transient Analysis of a CSC Fed SM Drive with ZDC Control

The performance of a CSC fed SM drive controlled by the ZDC scheme is investigated by computer simulation. A cylindrical 2.45 MW synchronous motor is employed in the study, and its parameters are listed in Table A-1 at the end of this Chapter. The drive system parameters are given in Table 15.6-1.

Figure 15.6-5 shows the system dynamic response during the drive start up followed by a sudden change in the load torque. To start up the drive smoothly, the rotor speed reference ω_r^* is implemented with a ramp function, which increases from zero linearly with time and reaches the rated value (1.0 pu) at 1 s. The drive starts under no-load conditions. At $t = 1.5$ s, a rated load torque is applied. The rotor speed ω_r follows its reference closely, and is kept constant when the start up process is over except a small speed drop when the rated load torque is applied.

The d-axis stator current reference i_{ds}^* is set to zero while the q-axis stator current reference i_{qs}^* is given by the output of the speed PI controller. The measured stator dq-axis currents, i_{ds} and i_{qs}, follow their references closely except that they contain some current ripples caused by the switching harmonics in the three-phase stator current. The waveform of the phase-a stator current i_{as} is also given in the figure.

With the ZDC control, the motor electromagnetic torque T_e is proportional to the q-axis stator current i_{qs}, which can be verified by comparing the waveforms of T_e with i_{qs} in Fig. 15.6-5. The dc-link current i_{dc} is controlled by the rectifier, and its waveform is also given in the figure.

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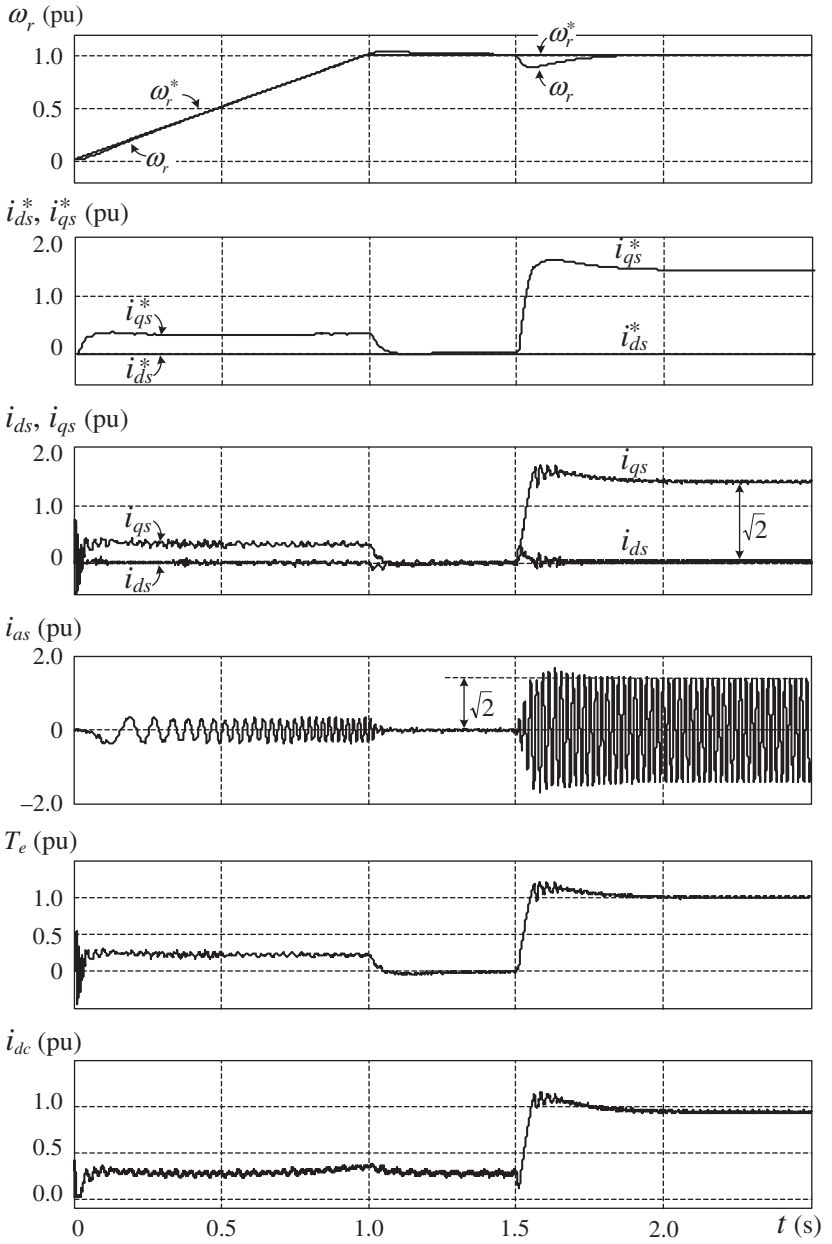

Figure 15.6-5 Transient response of a CSC fed SM drive with ZDC control.

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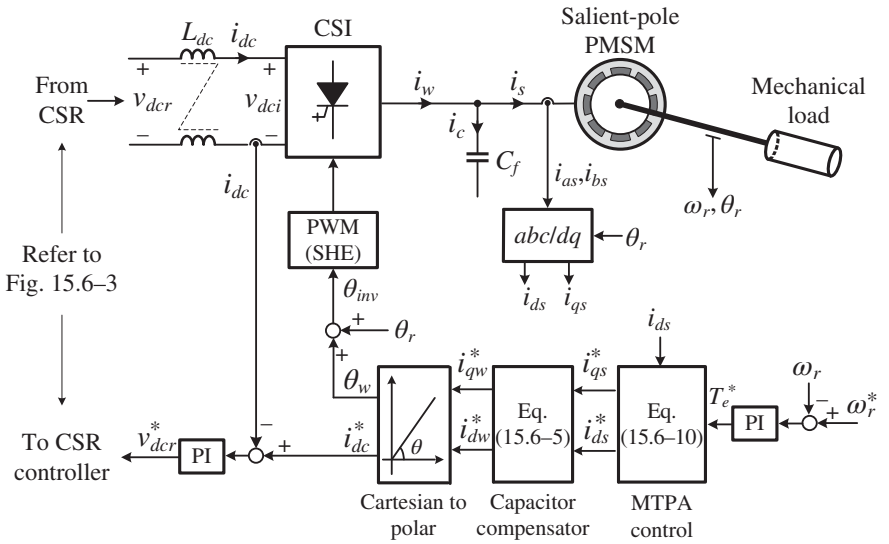

Figure 15.6-6 Block diagram of a CSC fed SM drive with MTPA scheme.

It is noted that when the drive is operating in steady state during 1.2 s $< t <$ 1.5 s under no-load conditions, the stator current i_{as} is quite small. This phenomenon has been explained in Section 15.3.4, and is not repeated here. However, the dc-link current i_{dc} is not significantly reduced during this period. It mainly flows through the filter capacitor C_f at the output of the inverter, but the average dc-link voltage is quite low. The dc-link power in this case is equal to the sum of the inverter power losses and motor power losses including copper, iron, and rotational losses.

15.6.4 CSC Fed SM Drive with MTPA Control

The MTPA control presented in the earlier sections for the VSC fed drive can be equally applied to the CSC fed drive. Figure 15.6-6 shows the block diagram of a CSC fed drive with the MTPA control. The block diagram is essentially the same as that for the CSC fed drive with the ZDC control in Fig. 15.6-3 except that an MTPA block is added to the system. The essence of the MTPA control is to adjust the dq-axis stator currents separately to produce a given torque with a minimal stator current. The algorithm for the MTPA control was developed earlier, and is given here for the purpose of convenience:

$$\begin{cases} i_{qs}^* = \dfrac{2T_e^*}{3P(\lambda_r + (L_d - L_q))\, i_{ds}} \\ i_{ds}^* = \dfrac{-\lambda_r}{2(L_d - L_q)} - \sqrt{\dfrac{\lambda_r^2}{4(L_d - L_q)^2} + (i_{qs}^*)^2} \end{cases} \tag{15.6-10}$$

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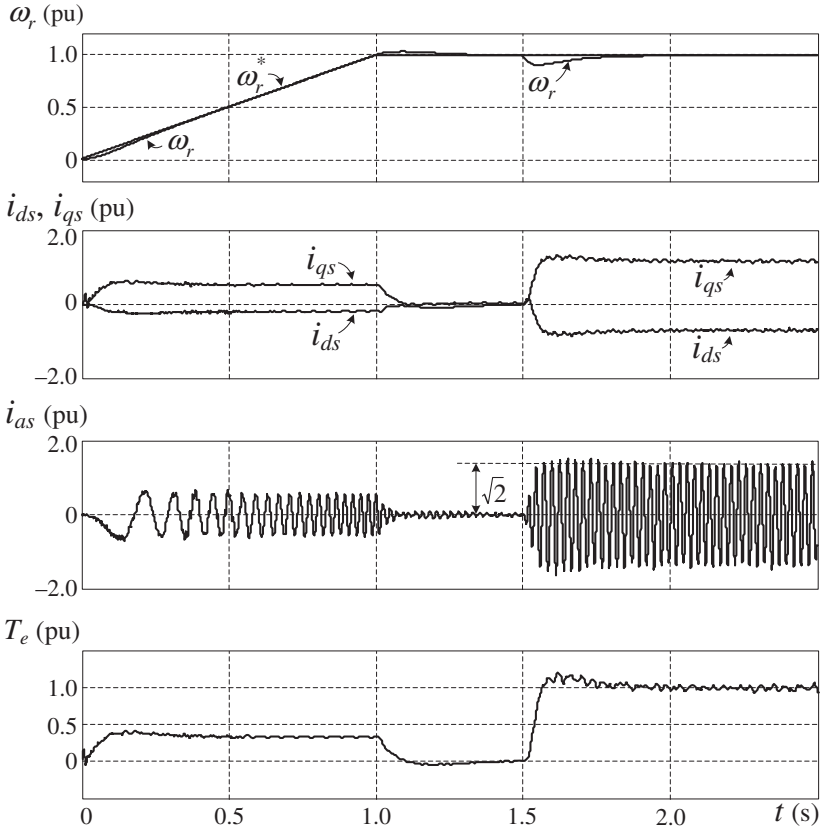

Figure 15.6-7 Transient response of a CSC fed SM drive with MTPA control.

The dynamic performance of the CSC fed SM drive with the MTPA control is investigated by computer simulation. The drive system parameters are the same as those given in Table 15.6-1 except that the *d*-axis current reference i_{ds}^* is no longer to be zero; it varies with the operating conditions of the drive. A 2.5 MW salient-pole synchronous motor is used in the study, and its parameters are given in Table A-2 at the end of this chapter.

Figure 15.6-7 shows the dynamic response of the drive during the system start up followed by a sudden change in load torque. To start up the drive smoothly, the rotor speed reference ω_r^* is increased linearly with time from zero to the rated value at $t = 1.0$ s. The rotor speed ω_r follows its reference closely. At $t = 1.5$ s, a rated load torque is applied to the motor, and the motor torque T_e increases accordingly. With the MTPA control, both *dq*-axis stator currents i_{ds} and i_{qs} vary accordingly such that the magnitude of the stator current i_s is minimized under any operating conditions. The motor torque T_e is produced by both i_{ds} and i_{qs}, and it is no longer proportional to i_{qs} as in the case of the ZDC control. The *d*-axis stator current i_{ds} has a negative value as discussed in Section 15.4.2.

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15.7 SUMMARY

This chapter focuses on the control schemes for the synchronous motor drives. The chapter starts with the dynamic and steady-state models for the synchronous motors in the rotor synchronous frame, which are generalized models suitable for both wound rotor and PMSMs. They are also applicable to salient-pole and cylindrical motors.

A number of control schemes are introduced for the synchronous motor drives, including the ZDC, MTPA control, and DTC. The operating principle of these schemes is analyzed, and their implementation in VSC and CSC fed synchronous motor drives is discussed. The drive system dynamic performance is illustrated with computer simulations. The synchronous motor drive with the ZDC control features linear torque–current characteristics and easy controller design. The drive with the MTPA scheme benefits from reduced power losses in the motor and power converters while the DTC controlled drive has the advantage of simple control scheme and easy implementation.

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APPENDIX

Table A-1 2.45 MW/4000 V/53.3 Hz Cylindrical PMSM

Motor Ratings		Motor Parameters	
Rated output power	2.45 MW	Stator resistance, R_s	24.21 mΩ
Rated line-to-line voltage	4000 V	d-axis synchronous inductance, L_d	9.816 mH
Rated stator current	490 A	q-axis synchronous inductance, L_q	9.816 mH
Rated speed	400 rpm	Rated stator frequency	53.33 Hz
Rated torque	58.5 kN · m	Rated power factor	0.716
Rated rotor flux linkage	4.97 Wb	Number of pole pairs	8
Moment of inertia	312 kg · m^2		

Table A-2 2.5 MW/4000 V/40 Hz Salient-Pole PMSM

Motor Ratings		**Motor Parameters**	
Rated output power	2.5 MW	Stator resistance, R_s	24.25 mΩ
Rated line-to-line voltage	4000 V	d-axis synchronous inductance, L_d	8.9995 mH
Rated stator current	485 A	q-axis synchronous inductance, L_q	21.8463 mH
Rated speed	400 rpm	Rated stator frequency	40 Hz
Rated torque	59.7 kN · m	Rated power factor	0.739
Rated rotor lux linkage	4.76 Wb	Number of pole pairs	6
Moment of inertia	420 kg · m^2		

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Part Six

Special Topics on MV Drives

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Chapter 16

Matrix Converter Fed MV Drives

16.1 INTRODUCTION

As presented in the earlier chapters, an ac–dc–ac converter topology is normally used in the MV drive, where a rectifier and an inverter are interconnected by a dc link. As an alternative to the dc-link based ac–dc–ac converters, ac–ac converter topologies, such as the classic matrix converter (MC), have been studied for applications in variable-speed drives [1–4]. However, the classic matrix converter cannot be directly used in the MV drive mainly due to the lack of switching devices with sufficiently high current and voltage ratings. Borrowing the idea from the multilevel cascaded H-bridge (CHB) converter topologies, multi-module cascaded matrix converters(CMCs) have been developed for the MV drive, which can reach the power rating of 6 MW and voltage rating of 6.6 kV using the low-voltage switching devices, such as low-voltage IGBTs available in the market [5, 6].

The multi-module CMC topology is composed of multiple units of three-phase to single-phase (3×1) matrix converter modules connected in cascade in each of the converter phases. One of the main features of the CMC topology is the direct power conversion from the ac input to ac output without the need of a dc link. In an ac–dc–ac voltage source converter (VSC) fed MV drive, a large dc-link capacitor is required. The dc-link capacitor is costly and bulky since it should be medium-voltage film capacitor instead of low-voltage electrolytic capacitors that are commonly used in low-voltage drives. The dc-link inductor in a current source converter (CSC) fed MV drive is also costly and bulky. The elimination of the dc link in the CMC helps to reduce the cost and size of the drive.

As a combination of direct power conversion and multi-module cascaded structure, the CMC has a number of other features, including modular design for cost reduction, four-quadrant operation capability, and nearly sinusoidal inputs and

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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outputs. However, with the absence of the dc link, the modulation scheme for the CMC is complicated. The CMC is also sensitive to input voltage disturbances, and has little fault ride-through capability [4, 7]. Despite its disadvantages, the multi-module CMC has found niche application in the MV drive.

In this chapter, the classic matrix converter is introduced. The 3 × 1 MC module, which is the building block for the CMC, is analyzed. The three-module and six-module CMC topologies for the MV drive are presented. Typical output voltage waveforms and input current waveforms for these converters are provided. A commercial MV drive employing the multi-module CMCs is introduced.

16.2 CLASSIC MATRIX CONVERTER (MC)

The classic MC is a single-stage converter that converts power directly from three-phase ac inputs with a fixed voltage and frequency to three-phase ac outputs with a variable voltage and frequency. This converter does not require dc-link components and employs bidirectional switches that allow any output phase to be connected to any input phase. The classic matrix converter is characterized by inherent four-quadrant operation capability, high power density, and a controllable input power factor [8, 9].

16.2.1 Classic MC Configuration

Figure 16.2-1 illustrates a simplified circuit diagram of a classic three-phase input and three-phase output (3 × 3) matrix converter, where nine bidirectional switches

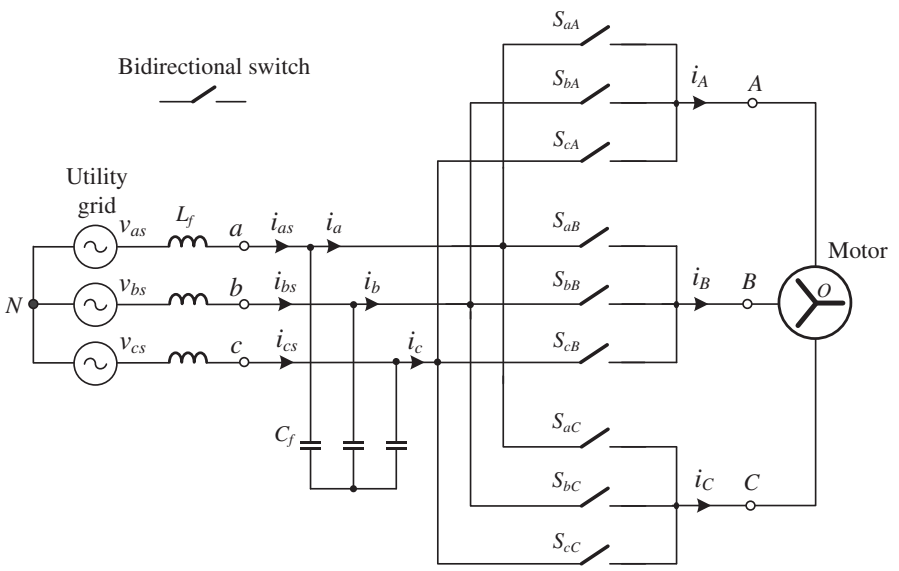

Figure 16.2-1 Classic three-phase input and three-phase output (3 × 3) matrix converter.

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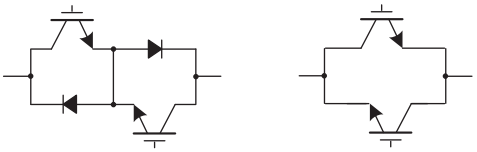

(a) Anti-series IGBTs (b) Anti-parallel RB-IGBTs

Figure 16.2-2 Configurations of a bidirectional switch.

are employed. These switches connect the output terminals A, B, and C directly to the input terminals a, b, and c of the converter such that the desired output voltages can be synthesized from the input voltages. The three-phase filter (L_f, C_f) on the converter input side reduces the high-frequency switching harmonics and also the voltage spikes that may be produced during the commutation of the switching devices in the MC converter [8].

Figure 16.2-2 shows two typical configurations of the bidirectional switch for use in the matrix converters. The bidirectional switch can be configured in practice by two IGBT–diode pairs connected in anti-series or two reverse-blocking (RB) IGBTs connected in anti-parallel as illustrated in Figs. 16.2-2a and 16.2-2b, respectively. The bidirectional switch can allow currents in both directions and block voltages of both polarities. The switch with the anti-series configuration has higher conduction losses since the two devices (one IGBT and one diode) are in the current path, whereas the anti-parallel configuration is more expensive due to the use of RB-IGBTs.

16.2.2 Switching Constraints and Waveform Synthesis

Similar to the VSC and CSCs, the matrix converter is subject to switching constraints. Any switching state that may cause short circuit of the input filter capacitor or interrupt the inductive load current is not allowed. Therefore, at any instant of time, three and only three switches in the converter must be in the ON states, one from the phase-A group of switches (S_{aA}, S_{bA}, S_{cA}), one from the phase-B group (S_{aB}, S_{bB}, S_{cB}), and another from the phase-C group (S_{aC}, S_{bC}, S_{cC}). If more than three switches are on, the filter capacitor C_f at the input terminals of the converter will be short circuited by the switches, causing excessively high current, whereas if one of the three conducted switches are turned off without turning on another one in the same switch group simultaneously, the inductive load (motor) current will be interrupted, causing destructively high voltage on the switches. The above stated switching constraints can be expressed by

$$\begin{cases} S_{aA} + S_{bA} + S_{cA} = 1 \\ S_{aB} + S_{bB} + S_{cB} = 1 \\ S_{aC} + S_{bC} + S_{cC} = 1 \end{cases} \tag{16.2-1}$$

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where

$$S_{jk} = \begin{cases} 1, & S_{jk} \text{ is closed} \\ 0, & S_{jk} \text{ is open} \end{cases} \quad j \in \{a,b,c\},\ k \in \{A,B,C\} \tag{16.2-2}$$

These constraints must be satisfied when a PWM switching scheme is developed for the matrix converter. A variety of switching schemes have been developed for the classic matrix converter [1, 9–11], including space vector modulation (SVM), carrier based sinusoidal modulations, and transfer function based modulation, but all these methods are quite complex due to the direct power conversion from the ac input to ac output without a dc link. However, the principle of the modulation scheme for the matrix converter is simple: the output voltage of the converter is synthesized from its input voltage, and the input current of the converter is constructed from its output current. The three-phase output voltage of the converter can be synthesized by

$$\begin{cases} v_{AN} = S_{aA}v_{aN} + S_{bA}v_{bN} + S_{cA}v_{cN} \\ v_{BN} = S_{aB}v_{aN} + S_{bB}v_{bN} + S_{cB}v_{cN} \\ v_{CN} = S_{aC}v_{aN} + S_{bC}v_{bN} + S_{cC}v_{cN} \end{cases} \tag{16.2-3}$$

from which the converter line-to-line output voltages can be calculated by

$$\begin{cases} v_{AB} = v_{AN} - v_{BN} \\ v_{BC} = v_{BN} - v_{CN} \\ v_{CA} = v_{CN} - v_{AN} \end{cases} \tag{16.2-4}$$

The three-phase input current of the converter can be constructed from

$$\begin{cases} i_a = S_{aA}i_A + S_{aB}i_B + S_{aC}i_C \\ i_b = S_{bA}i_A + S_{bB}i_B + S_{bC}i_C \\ i_c = S_{cA}i_A + S_{cB}i_B + S_{cC}i_C \end{cases} \tag{16.2-5}$$

Figure 16.2-3 illustrates how the output voltage of the matrix converter is synthesized from its input voltages, where v_{aN}, v_{bN}, and v_{cN} are the voltages at the converter input terminals a, b, and c with respect to the neutral point N of the utility grid, S_{aA}, S_{bA}, and S_{cA} are the switching status of the switches in the phase-A switch group, v_{AN} is the voltage at the phase-A output terminal with respect to the neutral of the grid, and v_{AN1} is the fundamental-frequency component of v_{AN}. The switching status shown in the figure satisfies the switching constraint, which is $S_{aA} + S_{bA} + S_{cA} = 1$.

The phase-A output voltage of the matrix converter can be synthesized by $v_{AN} = S_{aA}v_{aN} + S_{bA}v_{bN} + S_{cA}v_{cN}$. During period I of Fig. 16.2-3, S_{cA} is turned on ($S_{cA} = 1$), S_{aA} and S_{bA} are turned off ($S_{aA} = S_{bA} = 0$), and $v_{AN} = v_{cN}$. During period II, $S_{bA} = 1$ and $S_{aA} = S_{cA} = 0$, from which $v_{AN} = v_{bN}$. Following the same steps for all the switching

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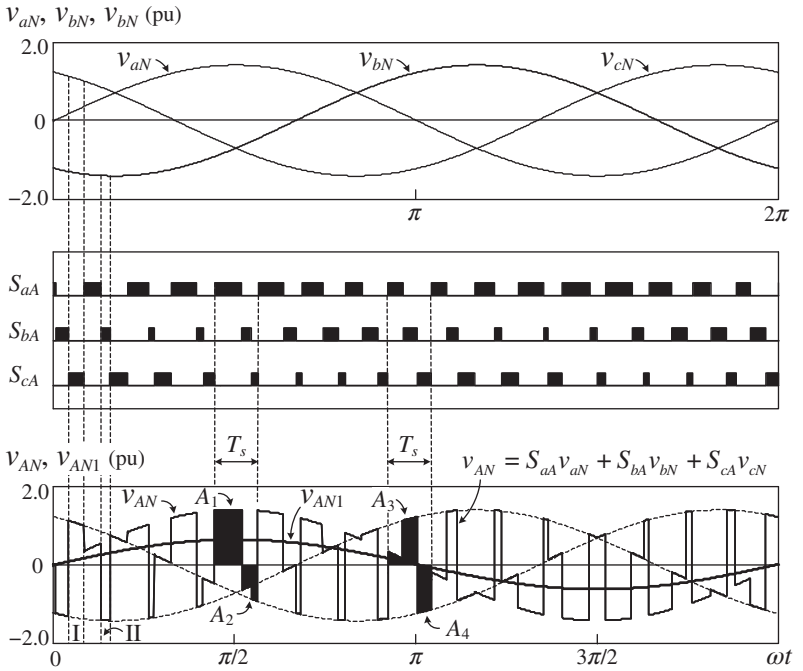

Figure 16.2-3 Synthesis of the output voltage in the classic 3 × 3 matrix converter.

periods, the waveforms of the converter output voltage v_{AN} can be constructed from its three-phase input voltage waveforms.

The fundamental component of the converter phase-*A* output voltage, v_{AN1}, is also illustrated in the figure, which can be considered as an average value of v_{AN} over switching period T_s. For example, when v_{AN1} reaches its peak at $\omega t = \pi/2$, area $A1$ (positive) is much greater than area $A2$ (negative), the average value of $A1$ and $A2$ over T_s, which represents the magnitude of v_{AN1}, is positive. When v_{AN1} decreases to zero at $\omega t = \pi$, the positive area $A3$ is close to the negative area $A4$, and the average value of $A3$ and $A4$ is close to zero.

Figure 16.2-4 illustrates how the input current of the matrix converter is synthesized from its output currents, where i_A, i_B, and i_C are the three-phase output current of the converter, S_{aA}, S_{aB}, and S_{aC} are the switching status of the switches connected to the input terminal a of the converter, i_a is the converter phase-a input current, and i_{as} is the phase-a supply current.

The phase-a input current of the matrix converter can be synthesized by $i_a = S_{aA}i_A + S_{aB}i_B + S_{aC}i_C$. During period I of Fig. 16.2-4, switches S_{aA}, S_{aB}, and S_{aC} are turned on ($S_{aA} = S_{aB} = S_{aC} = 1$), $i_a = i_A + i_B + i_C = 0$ due to the three-phase balanced current waveforms. During period II, all three switches are turned off ($S_{aA} = S_{aB} = S_{aC} = 0$), which leads to $i_a = 0$. During period III, $S_{aA} = 1$ and $S_{aB} = S_{aC} = 0$, from which $i_a = i_A$. During period IV, $S_{aA} = S_{aB} = 1$, $S_{aC} = 0$, and $i_a = i_A + i_B$. Following

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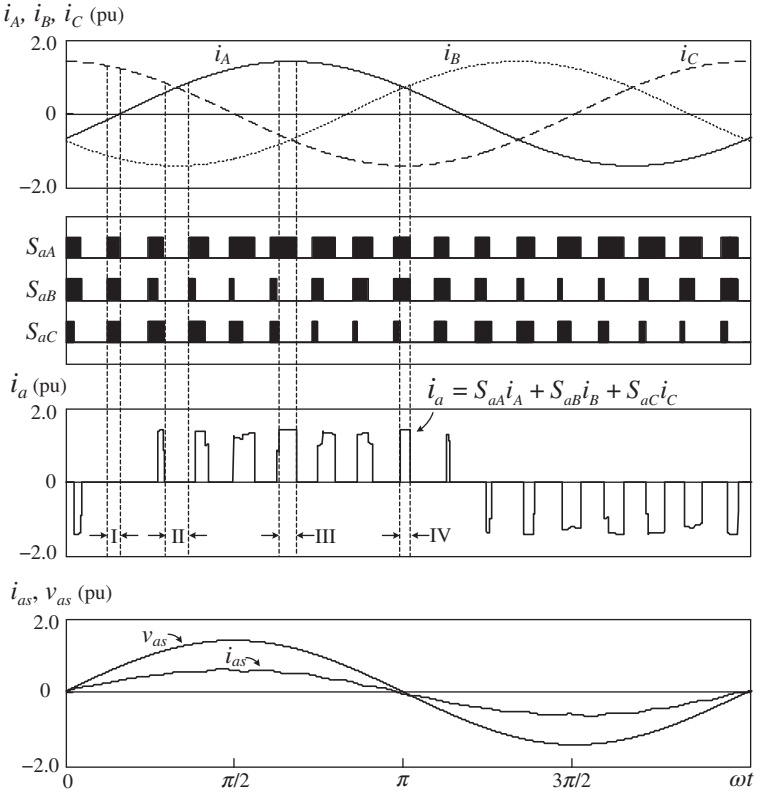

Figure 16.2-4 Synthesis of the input current in the classic 3 × 3 matrix converter.

the same steps for all the switching periods, the converter input current waveform can be constructed from its three-phase output current waveforms.

The waveforms of the phase-*a* supply current and voltage, i_{as} and v_{as}, are also shown in Fig. 16.2-4. The switching harmonics in i_a are attenuated by the filter inductor L_f and capacitor C_f, which makes i_{as} nearly sinusoidal.

Figure 16.2-5 shows the waveforms in the classic matrix converter using the modulation presented in [1]. The converter operates under the condition of switching frequency of 1 kHz, modulation index of 0.9, and output frequency of 90 Hz (f_o). The grid frequency f_g is 60 Hz, the input filter inductor and capacitor are 0.05 pu and 0.25 pu, and the load resistance and inductance are 0.5 pu and 0.35 pu, respectively.

The phase-*A* output voltage v_{AN} is synthesized from the input voltage of the converter, and its fundamental-frequency component v_{AN1} is also shown in the figure. The load current i_A is close to sinusoidal. The converter phase-*a* input current i_a is synthesized from the its output currents. The switching harmonics in i_a are filtered out by C_f and L_f, resulting in a sinusoidal supply current of i_{as}. The input power factor of the converter is controlled by the modulation scheme, which is close to unity.

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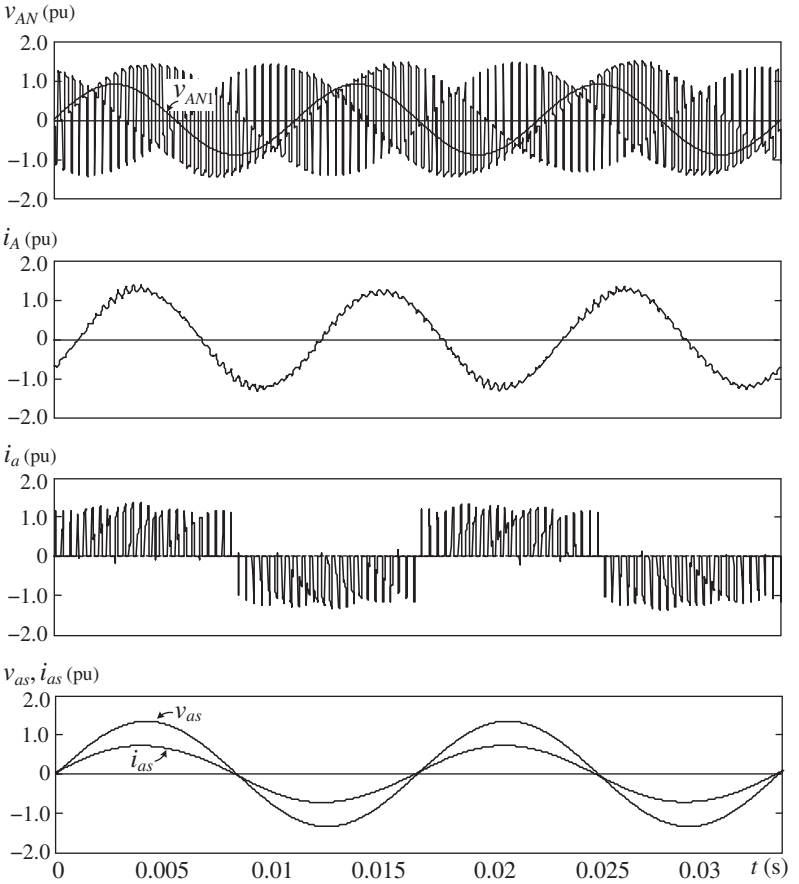

Figure 16.2-5 Waveforms in the classic 3 × 3 matrix converter ($f_g = 60$ Hz, $f_o = 90$ Hz).

16.3 THREE-MODULE MATRIX CONVERTER

To increase the voltage and power rating of the matrix converter, multi-module matrix converters were proposed for MV drives [5, 6, 12, 13]. Figure 16.3-1 shows a block diagram for three-module matrix converter, where the motor is connected to three units of three-phase to single-phase (3 × 1) MC modules. Each of the 3 × 1 MC modules requires a three-phase supply, which is provided by a secondary winding of a multi-winding transformer. To increase the voltage and power ratings of the drive further, two or more identical 3 × 1 MC modules can be connected in cascade in each of the converter phases. These CMCs will be discussed in the subsequent section.

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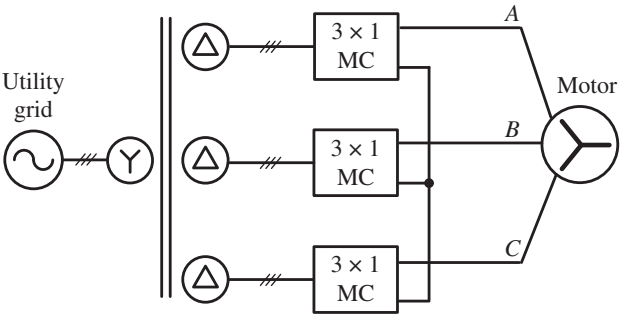

Figure 16.3-1 Block diagram for three-module matrix converter fed drive.

16.3.1 Three-Phase to Single-Phase (3 × 1) MC Module

The 3 × 1 MC module is the building block of the multi-module CMCs, and its circuit diagram is shown in Fig. 16.3-2. The MC module consists of six bidirectional switches, three in the upper group (S_{ap}, S_{bp}, S_{cp}) and three in the lower group (S_{aq}, S_{bq}, S_{cq}). A three-phase filter capacitor C_f is connected to the input of the module, which is used to suppress the switching harmonics and assist the commutation of the switching devices in the module.

The switching constraints discussed for the classic 3 × 3 matrix converter are also applicable to the 3 × 1 MC module. At any instant of time, two switches and only two in the converter must be in the ON states, one from the upper group of switches and the other from the lower group. If more than two switches are on, the filter capacitor C_f will be short circuited, causing excessively high current, whereas if one of the two conducted switches are turned off without turning on another one in

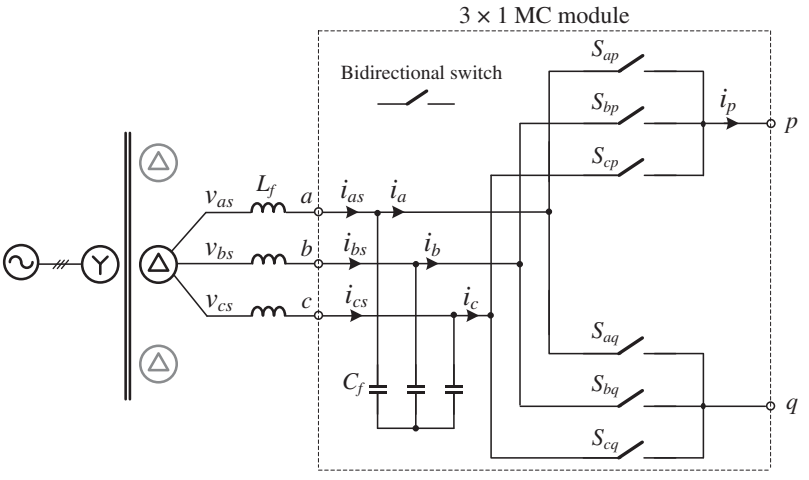

Figure 16.3-2 Circuit diagram of a 3 × 1 MC module.

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the same switch group simultaneously, the inductive load current will be interrupted, causing excessively high voltage on the switches. The switching constraints can be summarized as

$$\begin{cases} S_{ap} + S_{bp} + S_{cp} = 1 \\ S_{aq} + S_{bq} + S_{cq} = 1 \end{cases} \tag{16.3-1}$$

where

$$S_{jk} = \begin{cases} 1, & S_{jk} \text{ is closed} \\ 0, & S_{jk} \text{ is open} \end{cases} \qquad j \in \{a, b, c\}, k \in \{p, q\} \tag{16.3-2}$$

The output voltage v_{pq} of the 3×1 MC module can be synthesized from its input voltage v_a, v_b, and v_c:

$$v_{pq} = (S_{ap}v_a + S_{bp}v_b + S_{cp}v_c) - (S_{aq}v_a + S_{bq}v_b + S_{cq}v_c) \tag{16.3-3}$$

based on which the relationship between the switching status and the output voltage is summarized in Table 16.3-1.

The input currents i_a, i_b, and i_c of the 3×1 matrix converter can be found from its output current i_p:

$$\begin{cases} i_a = (S_{ap} - S_{aq})\, i_p \\ i_b = (S_{bp} - S_{bq})\, i_p \\ i_c = (S_{cp} - S_{cq})\, i_p \end{cases} \tag{16.3-4}$$

Table 16.3-1 Switching Status and Output Voltage of the 3×1 MC Module

Switch Status						Output Voltage
Sap	*Sbp*	*Scp*	*Saq*	*Sbq*	*Scq*	*vpq*
1	0	0	1	0	0	$v_a - v_a = 0$
0	1	0	0	1	0	$v_b - v_b = 0$
0	0	1	0	0	1	$v_c - v_c = 0$
1	0	0	0	1	0	$v_a - v_b = v_{ab}$
0	1	0	0	0	1	$v_b - v_c = v_{bc}$
0	0	1	1	0	0	$v_c - v_a = v_{ca}$
1	0	0	0	0	1	$v_a - v_c = v_{ac}$
0	1	0	1	0	0	$v_b - v_a = v_{ba}$
0	0	1	0	1	0	$v_c - v_b = v_{cb}$

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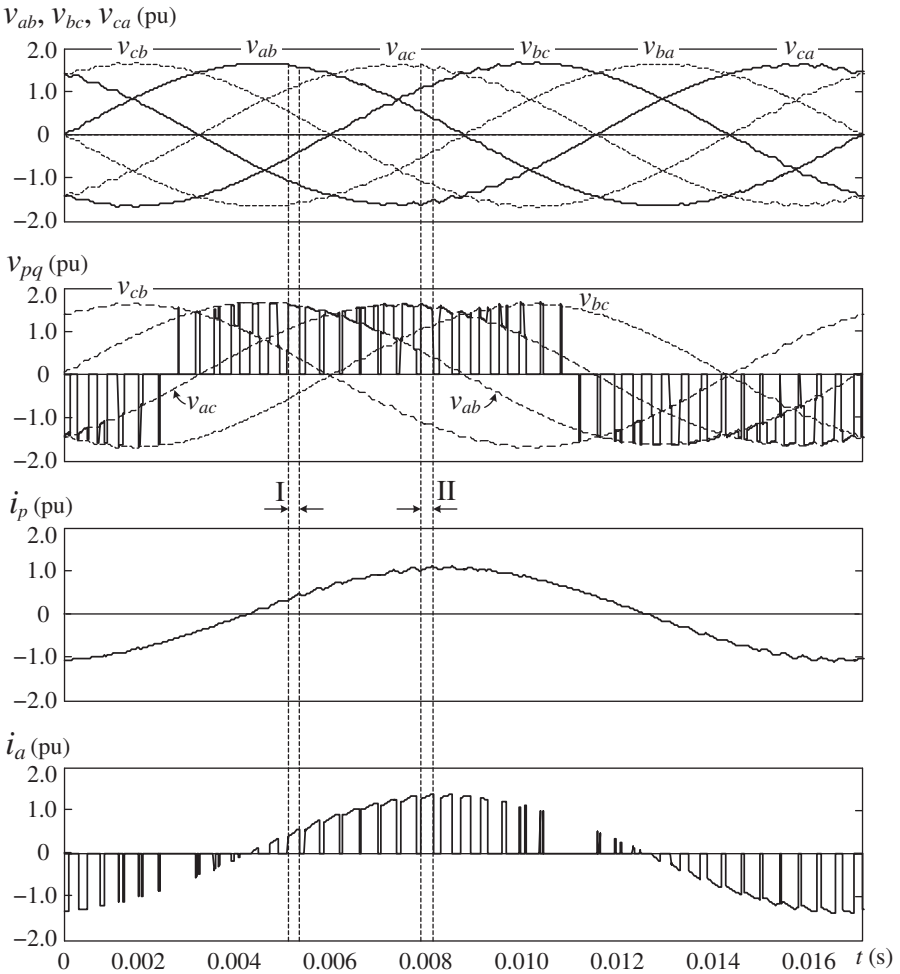

Figure 16.3-3 Synthesis of the output voltage and input current in the 3 × 1 MC module.

Figure 16.3-3 shows the typical output voltage and input current waveforms in the 3 × 1 matrix converter, where v_{ab}, v_{bc}, and v_{ca} are the line-to-line input voltages, v_{pq} is the output voltage, i_p is the output current, and i_a is the input current of the converter. To facilitate the analysis, the waveforms for v_{cb}, v_{ac}, and v_{ba} are also plotted. During period I, switches S_{ap} and S_{bq} are turned on and all the other switches are off (gate signals not shown). The converter output voltage v_{pq} is equal to v_{ab} according to Table 16.3-1. The converter input current $i_a = (S_{ap} - S_{aq})i_p = i_p$. Similarly, during period II, $S_{ap} = S_{cq} = 1$, all the other switches are off, $v_{pq} = v_{ac}$ and $i_a = i_p$. In summary, v_{pq} can be synthesized from the input line-to-line voltages of the converter and its input current is constructed from the output current of the converter.

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16.3.2 Three-Module MC Topology

Figure 16.3-4 shows the circuit diagram of the three-module matrix converter [12, 13], where each module is composed of a 3 × 1 matrix converter. A three-phase, multi-winding transformer is employed and each of the three secondary windings feeds a 3 × 1 MC module. A three-phase filter capacitor is connected to each of the MC modules to filter out the switching harmonics in the input current of the converter and absorb voltage spikes that may be generated during the commutation of the switching devices in the converter.

The leakage inductances of the transformer can serve as a filter inductance, and therefore a physical filter inductor between the transformer secondary winding and

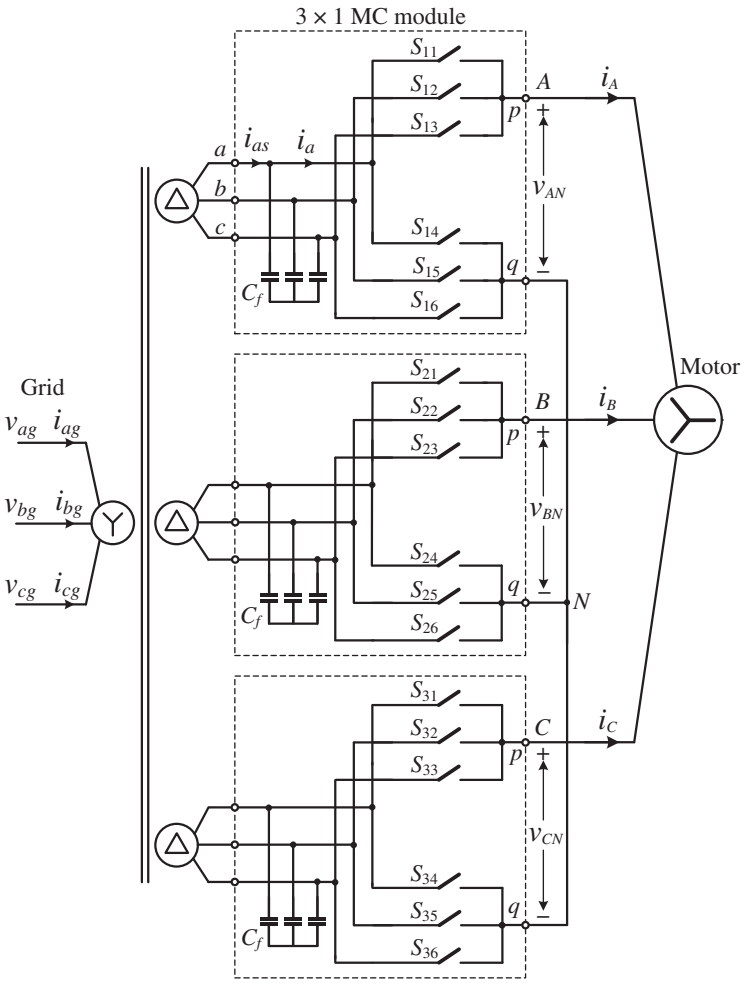

Figure 16.3-4 Circuit diagram of a three-module matrix converter.

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the MC module is normally not required. However, a three-phase filter may be added to the primary side of the transformer to reduce the line current THD for the grid code compliance. On the converter output side, the q terminals of the three modules are connected to form a neutral point N, and the p terminals are connected to the motor.

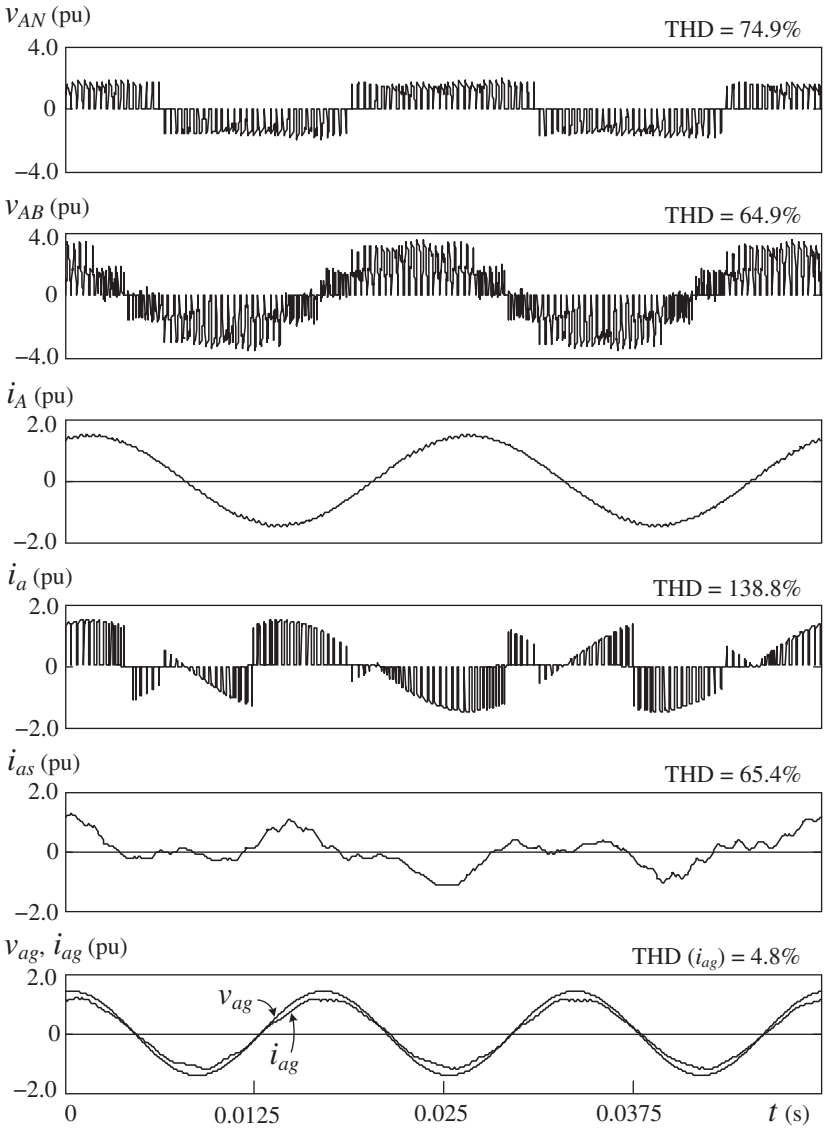

Figure 16.3-5 Simulated waveforms of a three-module matrix converter (f_g = 60 Hz, f_o = 40 Hz).

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16.3.3 Input and Output Waveforms

The main objective of the modulation scheme for the multi-modular matrix converter is to make the converter output voltage and frequency adjustable and to make the input current of the drive system, which is also the transformer primary current, close to sinusoidal with minimal harmonic distortion. The input power factor of matrix converter should be made adjustable as well.

A number of modulation schemes were proposed for the multi-modular matrix converters [12–15].One of them is called direct transfer function based modulation [12], where a mathematical model for the three-module matrix converter is established. This model relates the primary (input) currents of the transformer directly to the converter output currents and transformer secondary voltages by a transfer function. The transfer function contains a transfer matrix, which is derived to obtain a desired three-phase output voltage with variable magnitude and frequency while achieving sinusoidal input currents on the transformer primary side.

Another modulation scheme is referred to as indirect space vector based modulation, where an indirect circuit model for the three-module MC topologies is established [13]. This indirect circuit model is composed of a rectifier, a fictitious dc link, and an inverter, based on which the SVM schemes are applied separately to the rectifier and inverter to determine the switching states and dwell times. The switching states and dwell times for the rectifier and inverter are then converted and combined to produce the final gate signals for the three-module matrix converter.

Figure 16.3-5 shows the simulated waveforms in the three-module matrix converter using the direct transfer function based modulation scheme. The parameters for the simulation are given in Table 16.3-2. The waveform of the converter phase-*A* output voltage v_{AN} is similar to that given in Fig. 16.3-3 for the 3×1 MC module. The converter line-to-line voltage is obtained by $v_{AB} = v_{AN} - v_{BN}$. The phase-*A* output current i_A is close to sinusoidal due to the load inductance. The input current of the 3×1 module i_a is not periodical, which is mainly due to the difference between the grid and output frequencies. The transformer secondary current i_{as} contains much

Table 16.3-2 Simulation Parameters for a Three-Module Matrix Converter

Grid frequency (f_g)	60 Hz
Grid voltage (line-to-line)	1387 V
Transformer turns ratio (N_p/N_s)	3/2
Transformer leakage inductance (Referred to the secondary side)	0.6 mH (0.04 pu)
Input filter capacitance (C_f)	91 μF (0.2 pu)
Switching frequency	1.8 kHz
Modulation index	0.9
Output frequency (f_o)	40 Hz
Load resistance	4.3 Ω (0.75 pu)
Load inductance	7.7 mH (0.5 pu)

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less switching harmonics than i_a due to the filter capacitor C_f. However, the input current i_{ag} on the primary side is nearly sinusoidal, which is achieved by the direct transfer function based modulation scheme. The input power factor is adjusted to be unity.

16.4 MULTI-MODULE CASCADED MATRIX CONVERTER (CMC)

The three-module matrix converter discussed in the preceding section has higher voltage and power ratings than the conventional 3 × 3 matrix converter. To increase the voltage and power ratings of the converter further for the MV drives, CMC topologies can be used [6, 12, 13]. Similar to the multilevel CHB converters presented in Chapter 7, where multiple H-bridge power cells are connected in cascade, the CMC topologies use 3 × 1 MC modules in cascade.

Figure 16.4-1 shows the two configurations of the CMC topologies, where (a) is a six-module CMC topology with two modules in cascade per phase while (b) shows a 3*n*-module CMC topology with *n* modules in cascade per phase for the MV drives. Each module is composed of a 3 × 1 matrix converter. Number of modules in cascade

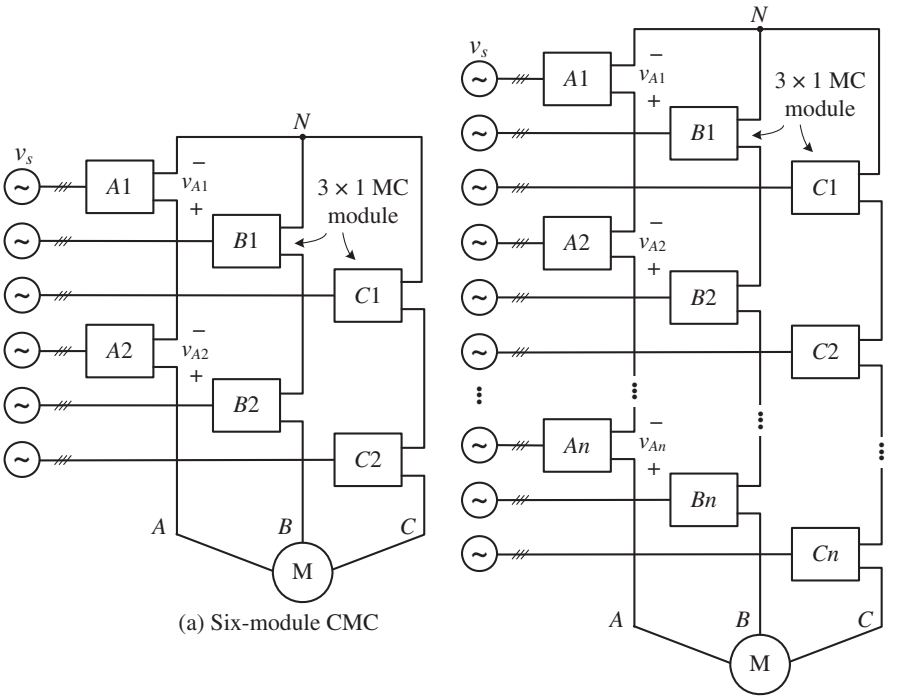

Figure 16.4-1 Multi-module CMC topologies.

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is dependent on the voltage rating of the module and the motor. It is reported that in a 6.6 kV commercial drive, six MC models rated at 635 V each are in cascade per phase, and the drive requires 18 MC modules [5, 6]. The CMC topology needs a separate three-phase power supply for each MC module. These power supplies can be provided by a transformer with multiple secondary windings.

16.4.1 Nine-Module CMC Topology

Figure 16.4-2 shows the block diagram of a nine-module CMC topology. A phase-shifting transformer with nine secondary windings is used to provide nine sets of three-phase balanced voltage supply for the 3 × 1 MC modules. The nine secondary windings are divided into three groups with a phase displacement of 20° between any two adjacent winding groups. The transformer is essentially the same as that used in the 18-pulse rectifiers presented in Chapter 3. The transformer can cancel four low-order dominant harmonics (the 5th, 7th, 11th, and 13th) in the line current of the secondary windings, which helps to make the primary current close to sinusoidal without any harmonic filters on the primary side of the transformer. In addition, the transformer provides electric isolation for the drive system.

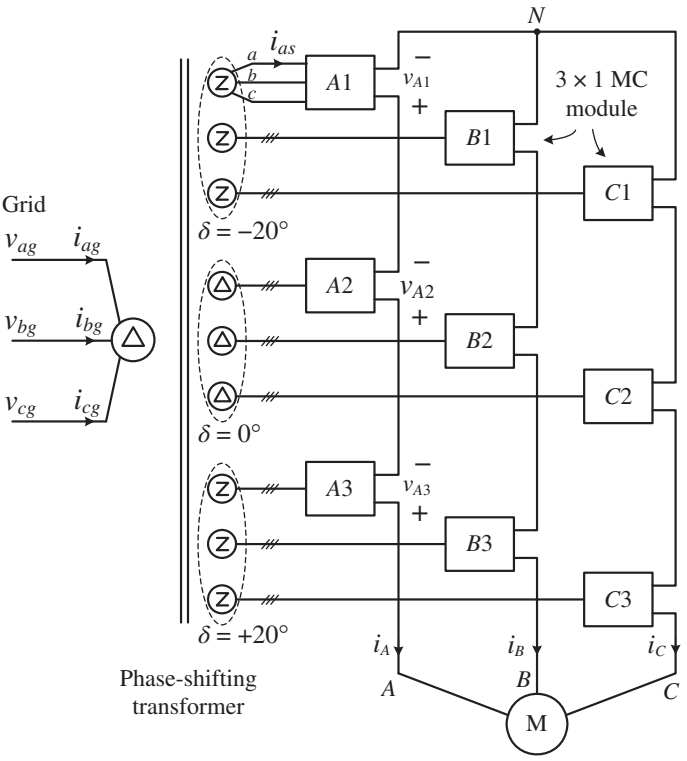

Figure 16.4-2 Block diagram of a nine-module CMC.

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16.4.2 Input and Output Waveforms

Figure 16.4-3 shows a set of typical waveforms of the nine-module CMC using the direct transfer function based modulation [12]. The parameters for the simulation are given in Table 16.4-1. The waveform of the output voltage v_{A1} of the MC module $A1$ is similar to v_{pq} in Fig. 16.3-3. With three MC modules in cascade per phase, the waveform of the phase-A output voltage v_{AN} is close to sinusoidal with switching harmonics superimposed. The converter line-to-line voltage v_{AB} has a THD of 20.5%,

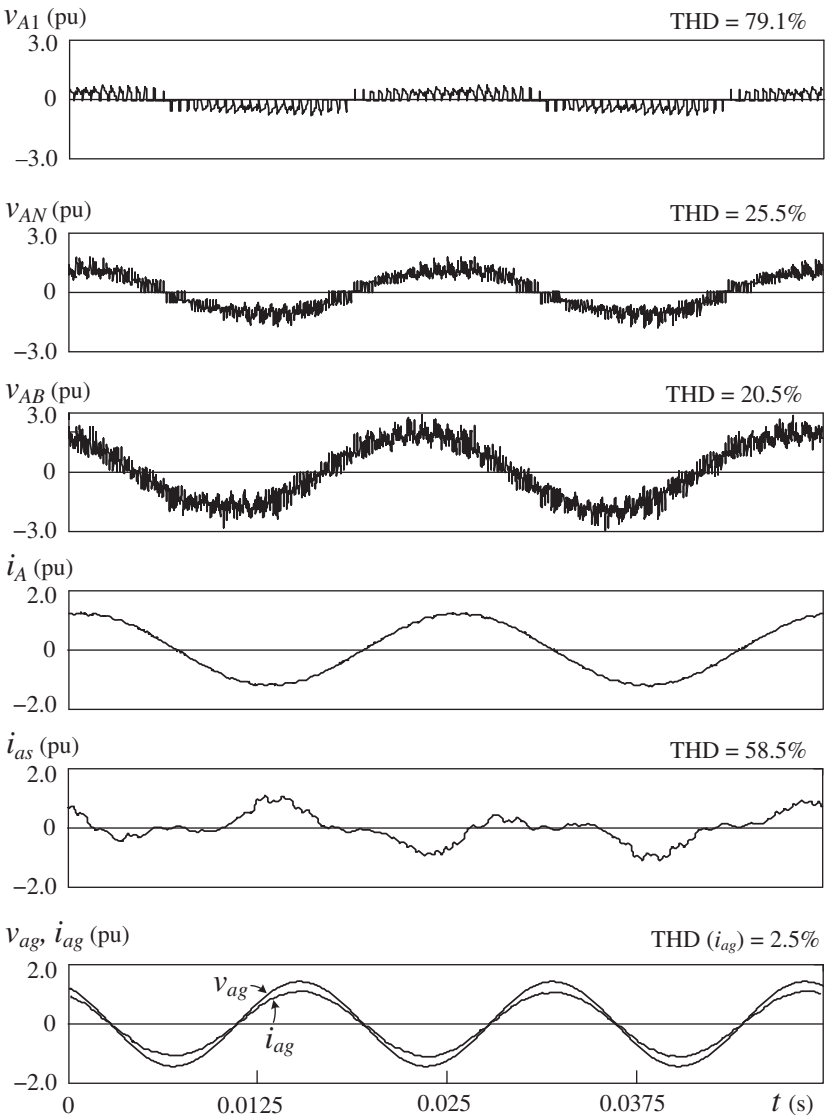

Figure 16.4-3 Simulated waveforms of a nine-module CMC at f_o = 40 Hz.

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Table 16.4-1 Simulation Parameters for a Nine-Module CMC

Grid frequency (f_g)	60 Hz
Grid voltage (line -to-line)	4160 V
Transformer turns ratio (N_p/N_s)	9/2
Transformer leakage inductance (Referred to the secondary side)	0.6 mH (0.04 pu)
Input filter capacitance (C_f)	91 μF (0.25 pu)
Switching frequency	1.8 kHz
Modulation index	0.9
Output frequency (f_o)	40 Hz
Load resistance	17.3 Ω (1.0 pu)
Load inductance	11.5 mH (0.25 pu)

which is much lower than 64.9% produced by the three-module MC shown in Fig. 16.3-5. The waveforms of the converter phase-*A* output current i_A, the transformer secondary current i_{as}, and the transformer primary current i_{ag} are all very close to those given in Fig. 16.3-5 for the three-module matrix converter, and thus are not discussed.

Figure 16.4-4 shows the simulated waveforms of the nine-module CMC operating at the output frequency of 80 Hz. The waveform of the converter line-to-line voltage v_{AB} is essentially the same as that in Fig. 16.4-3 except that its fundamental frequency is increased from 40 to 80 Hz. The input current of the drive i_{ag} has a THD of only 2.6%, and is adjusted by the modulation scheme to achieve unity power factor operation.

The measured waveforms from a laboratory nine-module CMC topology are illustrated in Fig. 16.4-5, where v_{AN} is the converter phase voltage, v_{AB} is the line-to-line voltage, and i_A is the phase-*A* output current. The configuration of the laboratory converter is the same as that shown in Fig. 16.4-2 with a phase-shifting transformer of nine secondary windings. The converter operates under the conditions of v_{AB} = 208 V, modulation index of $m_a = 0.9$, switching frequency of 1.8 kHz, and output

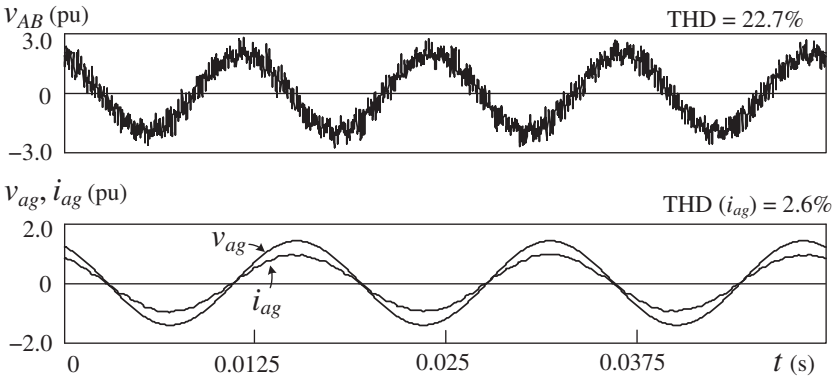

Figure 16.4-4 Simulated waveforms of a nine-module CMC at f_o = 80 Hz.

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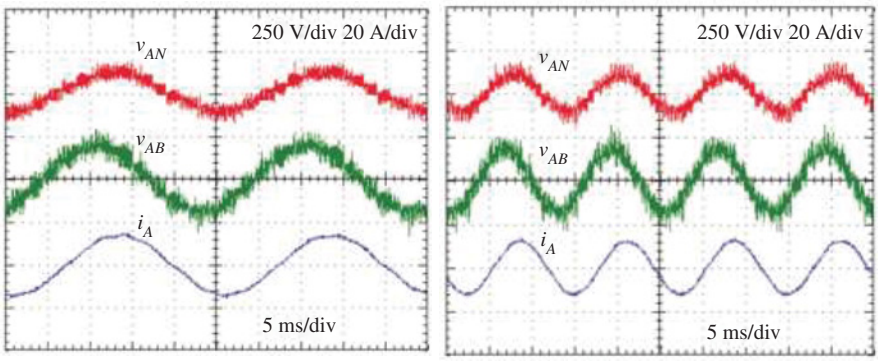

Figure 16.4-5 Experimental waveforms of a nine-module CMC [12]. (a) f_o = 40 Hz (b) f_o = 80 Hz

frequency of 40 Hz and 80 Hz. The measured waveforms are consistent with the simulated ones given in Figs. 16.4-3 and 16.4-4.

The multi-module CMC topology presented in this section has a number of features:

- Modular structure. The converter is composed of multiple units of identical 3 × 1 MC modules, which can be mass produced for the MV drives with different power and voltage ratings, leading to a reduction in manufacturing cost.
- No dc-link components. Unlike the VSC or CSC fed drives, in which a large dc capacitor or dc choke is required, the CMC fed drive does not need any dc-link components for cost reduction.
- High quality output voltage and input current waveforms. The converter output voltage waveform is produced by phase-shifted cascaded 3 × 1 MC modules, which leads to a lower THD and *dv/dt*. Its input current is nearly sinusoidal due to the use of phase-shifting transformer and a small filter at the inputs of each MC module.
- Medium voltage operation without switching devices in series. The low-voltage 3 × 1 MC modules are in cascade to produce high ac voltages at the converter output without semiconductor switches in series. The problems of equal voltage sharing for series-connected switches are eliminated.
- Four-quadrant operation. The MV drive using the CMC technology can operate in four quadrants without additional components.

However, the CMC has some drawbacks, including

- It needs a phase-shifting transformer to provide isolated ac supplies for the MC modules, which increases the manufacturing and operating cost of the drive.
- It has high component count. The CMC topology uses a large number of IGBTs. A nine-module CMC requires 108 IGBTs.

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16.5 MULTI-MODULE CMC FED MV DRIVE

Figure 16.5-1 shows a picture of a commercial 3.3 kV drive using a nine-module CMC topology [5]. The left cabinet contains a phase-shifting transformer with nine secondary windings and digital controllers, and the right cabinet houses nine units of 3 × 1 matrix converter modules. The modular design helps to reduce the manufacturing cost since the modules can be mass produced for drives with different voltage ratings. It also helps to reduce the downtime for repair when the modules are damaged and need to be replaced.

The configuration of the MV drive is illustrated in Fig. 16.5-2. The rated voltage of each 3 × 1 MC module is 635 V, at which inexpensive low-voltage IGBTs can be used. This helps to reduce the manufacturing cost of the drive. With three modules in cascade per phase, the phase voltage of the converter is 1905 V, and the line-to-line voltage of the drive reaches 3.3 kV, which is one of the standard medium voltages. The waveform of the line-to-line voltage v_{AB} is similar to that shown in Fig. 16.4-5. The motor current would be nearly sinusoidal due to the motor leakage inductances, and therefore no additional *LC* filter is required at the output of the drive.

The main specifications for the CMC fed MV drive are given in Table 16.5-1 [5].The drive has two voltage ratings of 3.3 kV and 6.6 kV. Both drives use the same 3 × 1 MC modules. For the 3.3 kV drive, nine MC modules are used in the drive with three modules in cascade per phase while in the 6.6 kV drive, 18 MC modules are

Figure 16.5-1 A commercial 3.3 kV nine-module CMC fed drive. Courtesy of Yaskawa.

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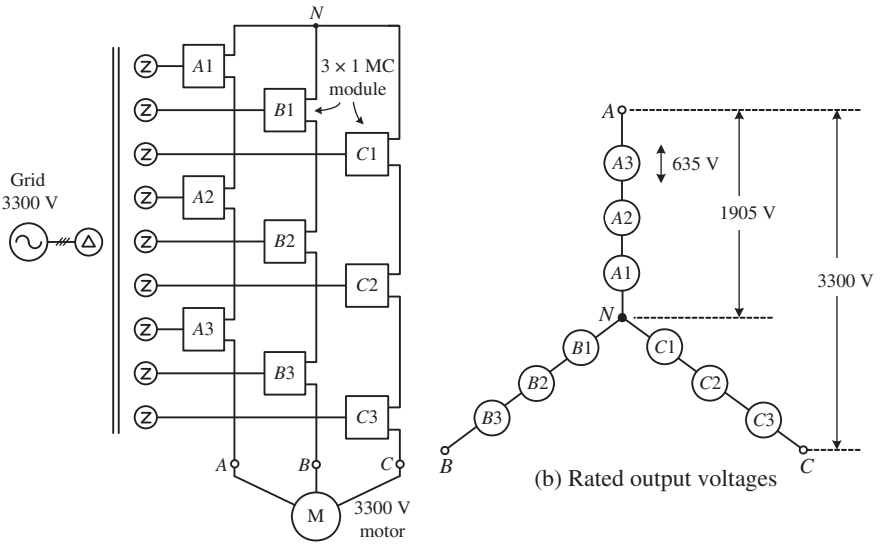

Figure 16.5-2 Configuration of the 3.3 kV CMC fed drive and its rated output voltages.

Table 16.5-1 Main Specifications of CMC Fed Drives

Drive System Specifications	Nominal input voltage	3.3 kV or 6.6 kV
	Output power rating	0.2–6.0 MVA
	Output voltage rating	0–3.3 kV (9-module CMC) 0–6.6 kV (18-module CMC)
	Output frequency	0–120 Hz
	Input power factor	>0.95
	Motor type	Induction motor
	Overload capability	Standard: 120% for 1 min Optional: 250% for 1 min
	Cooling	Forced air
	Regenerative braking capability	Yes
Control Specifications	Control scheme	Open-loop vector control, or flux vector control
	Frequency control accuracy	±0.5%
	Four-quadrant operation	Yes
Power Converter Specifications	Transformer	Phase-shifting transformer with six or nine secondary windings
	Converter type	Cascaded matrix converter with 3 × 1 MC modules of 635 V each
	Converter efficiency	~0.98%

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employed with six modules in cascade per phase. The output frequency of the drive could reach 120 Hz, which covers the speed range of most MV drives.

16.6 SUMMARY

This chapter presents a relatively new type of MV drives using a multi-module CMC topology. The matrix converter has a unique topology that performs direct ac to ac power conversion with a single power stage without a dc link that is normally required by the VSC and CSC fed drives. The CMC is constructed from multiple units of 3×1 matrix converter, which is derived from the conventional 3×3 matrix converter. The operating principle of the 3×1, 3×3, and multi-module CMCs is introduced. To illustrate the concept of direct ac to ac power conversion, simulation waveforms for these converters are provided. Finally, a commercial MV drive using multi-module CMC topology is introduced.

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Chapter 17

Transformerless MV Drives

17.1 INTRODUCTION

Both rectifier and inverter in a medium voltage (MV) drive produce common-mode (CM) voltages due to the switching action of the semiconductor switches in the converters. If not mitigated, the CM voltages will be superimposed on the motor phase (line-to-neutral) voltages, causing premature failure of the motor winding insulation. Traditionally, the problem is mitigated by introducing an isolation or phase-shifting transformer between the utility grid and the rectifier of the drive. In doing so, the CM voltages are blocked by the transformer, and will not appear on the motor. The insulation of the transformer in this case should be designed to take the additional voltage stress caused by the CM voltages.

Although this method provides an effective solution and it also has the side benefits of line current harmonic reduction by the leakage inductance of the isolation transformer or switching harmonic cancellation by the phase-shifting transformer, the cost and physical size of the drive are increased due to the use of the transformer. The operating cost of the drive is also increased due to the transformer power losses. With a typical transformer efficiency of 98% at the rated operating conditions, a 10 MW drive could have the power loss of 200 kW. The annual operating cost due to the transformer power loss is considerable since most of the MV drives operate in the field all year round.

This chapter focuses on the development of MV drives without the need of isolation or phase-shifting transformers. The common-mode voltage issue in the MV drive is analyzed, and mitigation methods for the reduction or elimination of CM voltages are discussed. The principle and realization of transformerless VSC and CSC fed MV drives are presented.

High-Power Converters and AC Drives, Second Edition. Bin Wu and Mehdi Narimani.
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17.2 COMMON-MODE VOLTAGE ISSUES AND CONVENTIONAL SOLUTION

17.2.1 Definition of CM Voltages

To analyze the common-mode voltages in a drive system, a simplified block diagram of the drive is shown in Fig. 17.2-1, where v_{cm1} and v_{cm2} are the CM voltages generated by the voltage source rectifier (VSR) and voltage source inverter (VSI), respectively. A differential-mode (DM) filter is required on the grid side to filter out the switching harmonics produced by the rectifier for the grid-code compliance. The DM filter on motor side is optional, depending on the inverter topology and drive system requirements. It may be a sinusoidal filter, after which a sinusoidal voltage waveform will be applied to the motor, or it may be a *dv*/*dt* filter for the reduction of *dv*/*dt* of the inverter output voltages. The DM filter may not be needed if the motor is driven by a multilevel inverter whose output voltage waveform is composed of a large number of small voltage steps with a low *dv*/*dt*.

It is assumed in the following analysis that the DM filters in the drive do not present any CM impedance. Depending on their configuration and design, a practical DM filter may present some CM impedance. For example, the leakage inductance of a three-phase inductor in the DM filter may be considered as the CM inductance as well, but its value is small and has little impact on CM voltage analysis.

The CM voltage generated by the rectifier, v_{cm1}, can be measured at the midpoint z of the dc link with respect to the neutral of the three-phase grid g, that is,

$$v_{cm1} = v_{zg} \tag{17.2-1}$$

The neutral of the three-phase grid g is normally grounded.

The CM voltage generated by the inverter can be measured at the neutral point of the motor stator winding with respect to the dc-link midpoint z:

$$v_{cm2} = v_{oz} \tag{17.2-2}$$

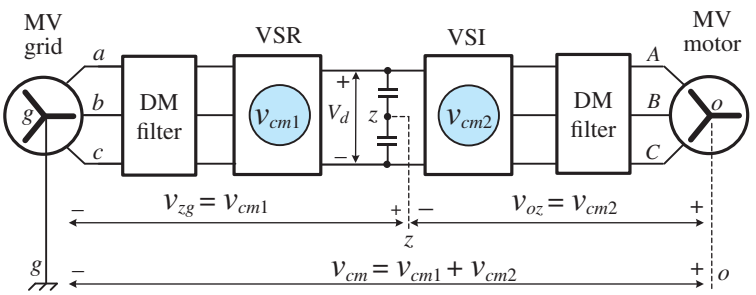

Figure 17.2-1 CM voltages in an MV drive.

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To calculate the CM voltage generated by the inverter, the voltages at the motor terminals A, B, and C with respect to the dc-link midpoint z can be expressed as

$$\begin{cases} v_{Az} = v_{Ao} + v_{oz} \\ v_{Bz} = v_{Bo} + v_{oz} \\ v_{Cz} = v_{Co} + v_{oz} \end{cases} \tag{17.2-3}$$

where v_{Ao}, v_{Bo}, and v_{Co} are the phase voltages of the stator winding, from which

$$(v_{Az} + v_{Bz} + v_{Cz}) = (v_{Ao} + v_{Bo} + v_{Co}) + 3v_{oz} \tag{17.2-4}$$

Considering a three-phase balanced system, where $v_{Ao} + v_{Bo} + v_{Co} = 0$, the above equation can be simplified to

$$v_{oz} = v_{cm2} = (v_{Az} + v_{Bz} + v_{Cz})/3 \tag{17.2-5}$$

Equation (17.2-5) can be used to calculate the CM voltage generated by the inverter. Similarly, the CM voltage generated by the rectifier can be calculated by

$$v_{zg} = v_{cm1} = (v_{za} + v_{zb} + v_{zc})/3 \tag{17.2-6}$$

where $v_{za,}$ v_{zb}, and v_{zc} are the voltages at the dc-link midpoint z with respect to the three-phase inputs of the rectifier, a, b, and c, respectively.

The total CM voltage in the drive is defined between the neutral point of the stator winding o and the system ground g, given by

$$v_{cm} = v_{og} = v_{oz} + v_{zg} = v_{cm2} + v_{cm1} \tag{17.2-7}$$

17.2.2 CM Voltage Waveforms

To investigate the CM voltage and its impact on the motor winding, the drive system of Fig. 17.2-1 with a two-level (2L) VSR and VSI are employed as an example. A carrier-based sinusoidal modulation with a switching frequency of 720 Hz and modulation index of 1.0 are used. The grid frequency is 60 Hz and the inverter output frequency is also 60 Hz.

Figure 17.2-2 shows the simulated waveforms of the CM voltages normalized to the dc-link voltage V_d, where (a) and (b) are the waveforms of the CM voltages generated by the rectifier and inverter, respectively, while (c) is the waveform of the total CM voltage v_{cm}. The magnitude of the CM voltage generated by the rectifier and inverter is half of the dc-link voltage ($0.5V_d$), respectively, and the magnitude of the total CM voltage v_{cm} is twice that of the rectifier or inverter.

Figure 17.2-3 illustrates the voltage waveforms on the stator winding, where (a) is the motor phase (line-to-neutral) voltage v_{Ao}, which is close to sinusoidal due to

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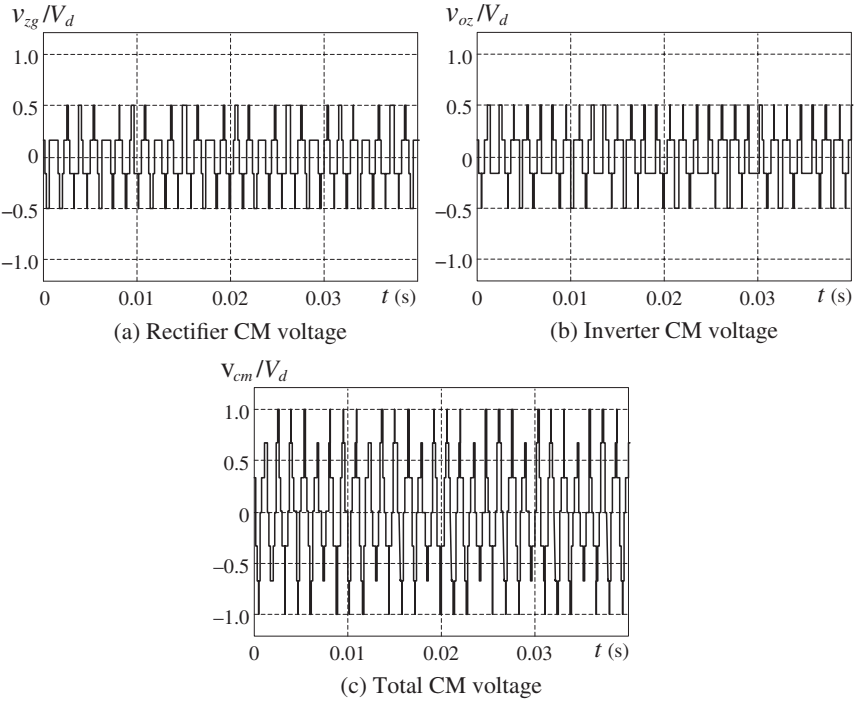

Figure 17.2-2 CM voltage waveforms in a 2L-VSC fed drive.

the motor-side DM filter and (b) shows the waveform of the motor line-to-ground voltage v_{Ag}, where the CM voltage v_{cm} is superimposed on motor phase voltage $v_{Ao,}$ and the peak of v_{Ag} is around three times of that of v_{Ao}. For the standard MV motors, the stator winding is not designed to withstand such a high voltage, and its insulation will deteriorate, resulting in premature failure of the stator winding insulation.

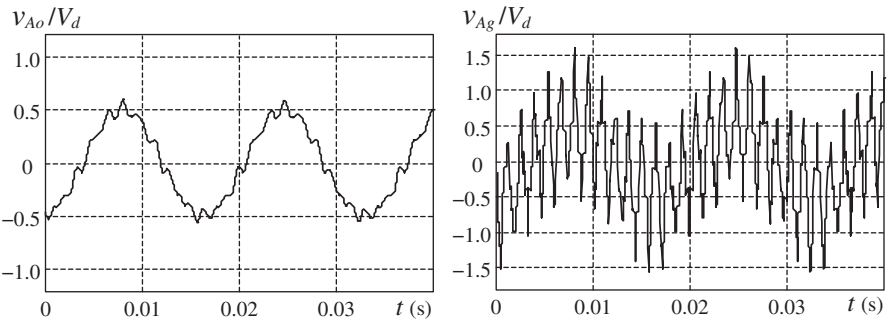

Figure 17.2-3 Voltage stress on stator winding in a 2L-VSC fed drive.

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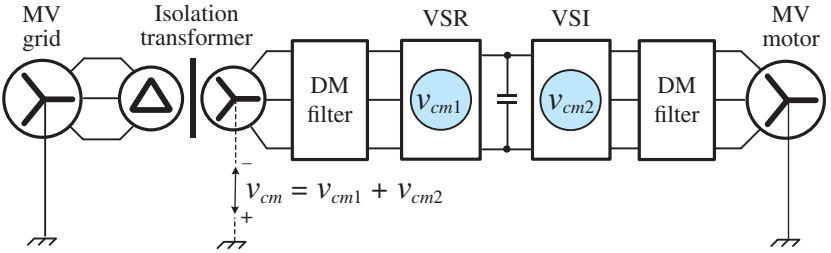

Figure 17.2-4 Conventional solution to the CM voltage issues in an MV drive.

17.2.3 Conventional Solution

The conventional solution to the CM voltage issue is to add an isolation transformer (or a phase-shifting transformer with a 12-pulse or 18-pulse rectifier) to the MV drives to block the CM voltages as shown in Fig. 17.2-4. With the neutral point of the stator winding grounded, the CM voltage v_{cm} produced by the rectifier and inverter is applied to the neutral of the transformer winding. It should be noted that introducing an isolation transformer to the drive does not eliminate the CM voltage, and the CM voltage is simply transferred from the motor to the transformer. Therefore, the transformer windings should be designed with higher level of voltage insulation to withstand the additional CM voltages.

However, the neutral point of the stator winding of an MV motor is normally not accessible in practice. A more practical solution is to leave the neutral point of both transformer and motor ungrounded (floating). In doing so, the distribution of the CM voltage between the transformer and motor is determined by the parasitic capacitances of the transformer and motor windings, C_T and C_M, with respect to ground as shown in Fig. 17.2-5. These are the CM capacitances, through which only the CM current caused by v_{cm} can flow.

According to the core and winding structure of the transformer and motor, the motor parasitic capacitance, C_M, is normally much higher than that of the transformer C_T. As a result, most CM voltage will appear on C_T, that is, on the transformer neutral to ground. The CM voltage on the motor neutral to ground will be negligibly small. For example, with a C_M of 60 nF and C_T of 0.5 nF in an MV drive [1], the CM voltage across C_M is less than 1%, and most of the CM voltage is applied to C_T.

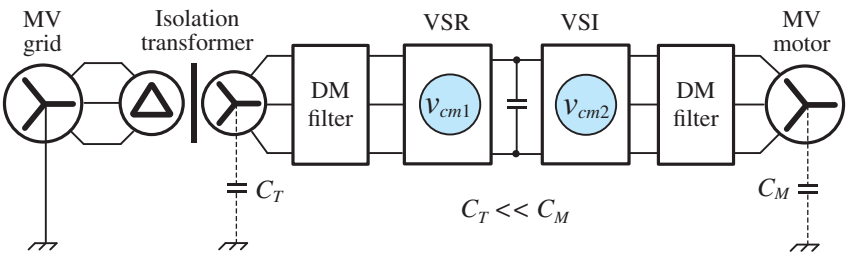

Figure 17.2-5 Parasitic CM capacitances of the transformer and motor in an MV drive.

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Another possible solution to the CM voltage issue is to design the MV motor with a higher level of voltage insulation to withstand the CM voltages. This approach is only valid for applications where a new motor is needed. It is not applicable for retrofit applications, where the standard MV motors have already been installed and operated in the field. The retrofit application is one of the main application areas of the MV drives for energy conservation as discussed in Chapter 1 of this book.

17.3 CM VOLTAGE REDUCTION IN MULTILEVEL VSC

The multilevel voltage source converters, such as neutral-point clamped (NPC) converters, have been widely used in the MV drives. As discussed in the earlier chapters, the number of redundant switching states in a multilevel converter increases exponentially with the number of voltage levels. It is, therefore, possible to reduce or even eliminate the CM voltages by selecting proper switching states that produce lower or no CM voltages [2–6]. In this section, the principle for CM voltage reduction/elimination methods is presented and analyzed.

17.3.1 Space Vector Modulation for CM Voltage Reduction

To facilitate the discussion of the principle of CM voltage reduction, a three-level (3L) NPC inverter shown in Fig. 17.3-1 is selected as an example.

The definition of the switching states for the 3L-NPC inverter is given in Table 17.3-1, where the switching state $[S_A]$ for inverter Phase A is defined with numbers [2], [1], and [0] instead of letters [P], [O], and [N] given in Table 8.2-1 of Chapter 8.

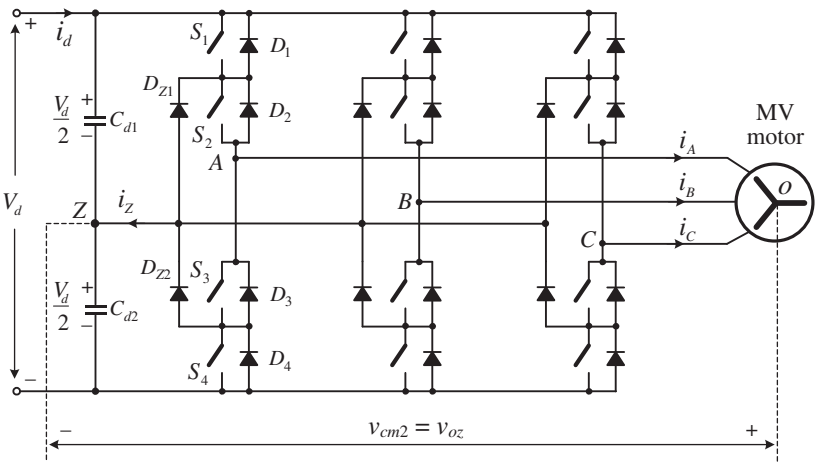

Figure 17.3-1 3L-NPC inverter as an example for CM voltage reduction.

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Table 17.3-1 Definition of Switching States in 3L-NPC Inverter

Switching State S_A	Device Switching Status (Phase A)				Inverter Terminal Voltage v_{Az}
	S_1	S_2	S_3	S_4	
2	On	On	Off	Off	$V_d/2$
1	Off	On	On	Off	0
0	Off	Off	On	On	$-V_d/2$

This arrangement is more convenient for the inverters with higher number of voltage levels to be discussed in this chapter.

Following the same procedures given in Chapter 8, all 19 space vectors (V_0 to V_{18}) in the 3L-NPC inverter can be determined and shown in its space vector diagram of Fig. 17.3-2. There are a total of 27 switching states, which are also shown in the figure. It is noted that the arrow in the space vector notation $\vec{V}$ is omitted in this chapter for simplicity since a large number of space vectors are presented in this chapter.

Referring to Table 17.3-1, the inverter terminal voltages with respect to the dc-link midpoint z can be calculated by

$$\begin{cases} v_{Az} = V_d(S_A - 1)/2 \\ v_{Bz} = V_d(S_B - 1)/2 \\ v_{Cz} = V_d(S_C - 1)/2 \end{cases} \tag{17.3-1}$$

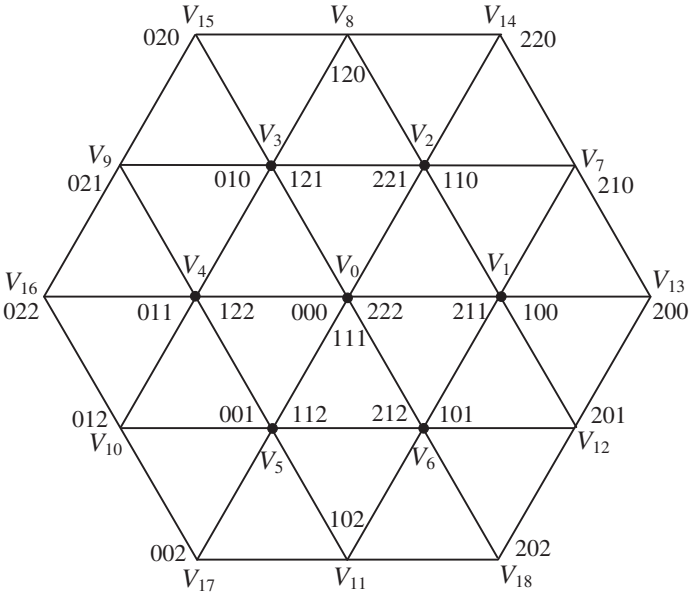

Figure 17.3-2 Space vector diagram of the 3L-NPC inverter.

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Table 17.3-2 CM Voltages Produced by the Switching State in a 3L-NPC Inverter

Voltage Vectors	Switching States	CMV	Voltage Vectors	Switching States	CMV
V_0	000	$-V_d/2$	V_7	210	0
	111	0	V_8	120	0
	222	$V_d/2$	V_9	021	0
V_1	100	$-V_d/3$	V_{10}	012	0
	211	$V_d/6$	V_{11}	102	0
V_2	110	$-V_d/6$	V_{12}	201	0
	221	$V_d/3$	V_{13}	200	$-V_d/6$
V_3	010	$-V_d/3$	V_{14}	220	$V_d/6$
	121	$V_d/6$	V_{15}	020	$-V_d/6$
V_4	122	$V_d/3$	V_{16}	022	$V_d/6$
	011	$-V_d/6$	V_{17}	002	$-V_d/6$
V_5	001	$-V_d/3$	V_{18}	202	$V_d/6$
	112	$V_d/6$			
V_6	101	$-V_d/6$			
	212	$V_d/3$			

where S_A, S_B, and S_C denote switching states of phase A, B, and C, respectively. Substituting Eq. (17.3-1) into Eq. (17.2-5), the inverter CM voltage produced by the switching state [S_A S_B S_C] can be calculated by

$$v_{cm2} = V_d(S_A + S_B + S_C - 3)/6 \tag{17.3-2}$$

For example, the CM voltage generated by switching state [200] is

$$v_{cm2} = V_d(2 + 0 + 0 - 3)/6 = -V_d/6 \tag{17.3-3}$$

Similarly, the CM voltage produced by all other switching states can be calculated and are summarized in Table 17.3-2. The CM voltages have a set of different magnitudes: 0, $\pm V_d/2$, $\pm V_d/3$, $\pm V_d/6$. The magnitude of the highest CM voltage is $V_d/2$, which is the half of the dc-link voltage V_d.

17.3.2 Reduction of CM Voltage Scheme 1 (RCM1)

The basic principle for the CM voltage reduction is that the switching states that generate high CM voltages are eliminated (not selected for use) when designing the

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Table 17.3-3 Elimination of the Switching States with RCM1 Scheme

Vectors	Switching States	CMV	Vectors	Switching States	CMV
V_0	~~000~~	~~$-V_d/2$~~	V_7	210	0
	111	0	V_8	120	0
	~~222~~	~~$V_d/2$~~	V_9	021	0
V_1	~~100~~	~~$-V_d/3$~~	V_{10}	012	0
	211	$V_d/6$	V_{11}	102	0
V_2	110	$-V_d/6$	V_{12}	201	0
	~~221~~	~~$V_d/3$~~	V_{13}	200	$-V_d/6$
V_3	~~010~~	~~$V_d/3$~~	V_{14}	220	$V_d/6$
	121	$V_d/6$	V_{15}	020	$-V_d/6$
V_4	~~122~~	~~$V_d/3$~~	V_{16}	022	$V_d/6$
	011	$-V_d/6$	V_{17}	002	$-V_d/6$
V_5	~~001~~	~~$-V_d/3$~~	V_{18}	202	$V_d/6$
	112	$V_d/6$			
V_6	101	$-V_d/6$			
	~~212~~	~~$V_d/3$~~			

SVM schemes. With the elimination of certain switching states, the range of the amplitude modulation index (m_a) may be affected. This leads to two CM voltage reduction schemes, defined as

- Reduction of CM voltage scheme 1(RCM1), where the modulation index is kept in its full range ($0 \leq m_a \leq 1$).
- Reduction of CM voltage scheme 2(RCM2), where magnitude of the CM voltages is reduced as much as possible, possibly to zero, but the modulation index range has to be reduced ($0 \leq m_a < 0.962$).

To develop the RCM1 scheme, let us take a 3L-NPC inverter as an example. To reduce the CM voltages, the switching states that generate high CM voltages, such as $\pm V_d/2$ and $\pm V_d/3$, are eliminated. As a result, the highest CM voltage produced by the switching states, this is, produced by the inverter, is only $V_d/6$ as shown in Table 17.3-3.

Figure 17.3-3 shows the space vector diagram for the 3L-NPC inverter with RCM1 scheme, where the switching states that produce the CM voltages of $\pm V_d/2$ and $\pm V_d/3$ are eliminated, but no switching states or voltage vectors on the outer hexagon of the space vector diagram are eliminated. This implies that the maximum length of the reference vector ($V_{ref,\max}$) that corresponds to the radius of the largest circle that can be inscribed within the hexagon can be achieved, which leads to the full range of the modulation index, that is, $0 \leq m_a \leq 1$.

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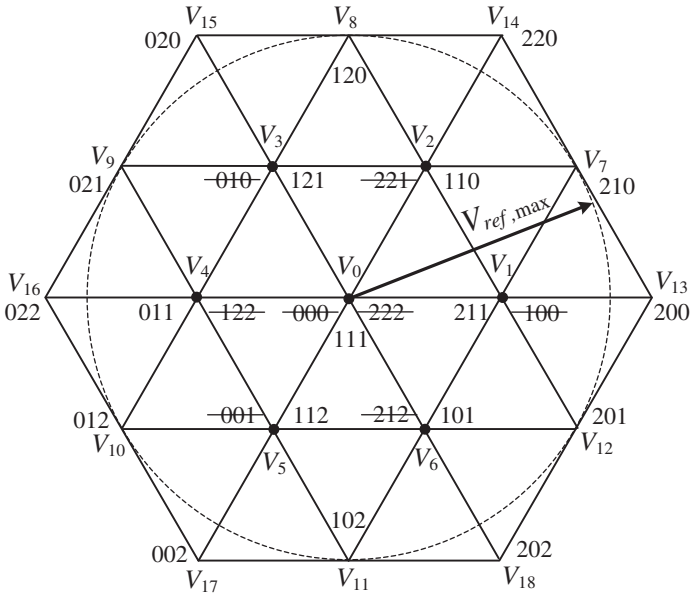
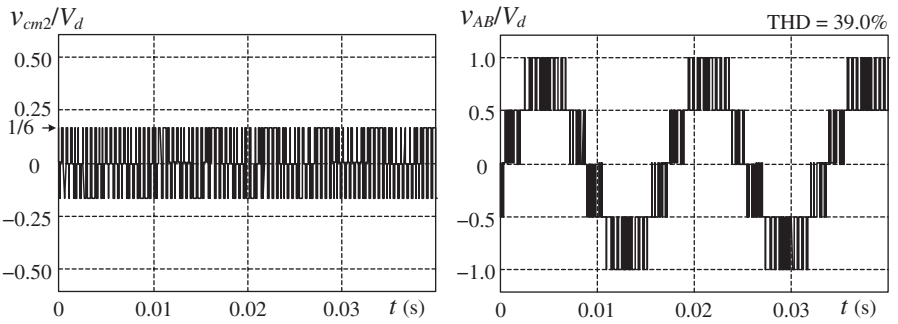


Figure 17.3-3 Space vector diagram for a 3L-NPC inverter with the RCM1 scheme.

Note that with the elimination of certain switching states for CM voltage reduction, the number of redundant switching states is reduced. For example, there are no redundant switching states in the space vector diagram of Fig. 17.3-3, where each voltage vector corresponds to only one switching state. As a result, the switching sequence design may not be as flexible as that in the conventional SVM scheme.

The waveforms of the CM voltage and output voltage of the inverter with RCM1 method are shown in Fig. 17.3-4. The inverter operates under the condition of the



(a) CM voltage generated by the inverter (b) Inverter line-to-line voltage

Figure 17.3-4 Waveforms of the CM and inverter output voltages in a 3L-NPC inverter with RCM1.

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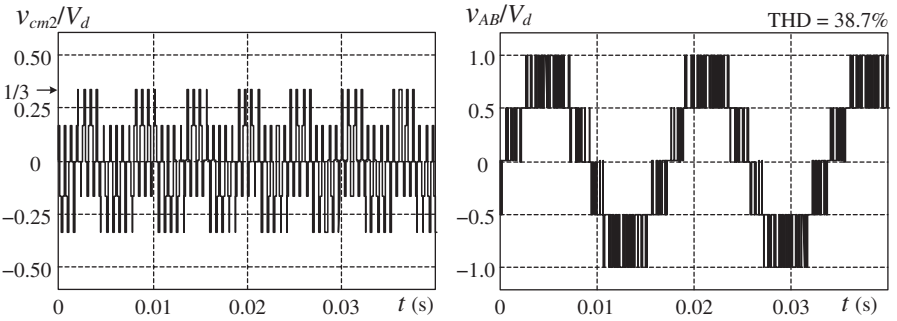

(a) CM voltage generated by the inverter (b) Inverter line-to-line voltage

Figure 17.3-5 Waveforms of the CM and inverter output voltages in a 3L-NPC inverter with C-SVM scheme.

fundamental frequency of 60 Hz, device switching frequency of 720 Hz, and modulation index of 0.8. It can be observed that the amplitude of the CM voltage generated by the inverter is only $V_d/6$. The line-to-line output voltage of the inverter is also shown in the figure with a THD of 39%.

Figure 17.3-5 shows the waveforms of the CM voltage and output voltage of the inverter with conventional space vector modulation (C-SVM) scheme. The inverter operates under the same conditions as those for the RCM1 scheme. The magnitude of the CM voltage produced by the C-SVM scheme is $V_d/3$, which is twice as high as that by the RCM1 scheme. It is noted that when designing the C-SVM scheme, the redundant switching states [000] and [222] associated with the zero vector V_0 are not used. Otherwise, the magnitude of the CM voltage produced by the C-SVM scheme could be $V_d/2$.

It is interesting to note that THD of the inverter line-to-line voltage with the RCM1 scheme (39%) is almost the same as that of C-SVM scheme (38.7%). This is primarily due to the fact that all the space vectors from V_0 to V_{18} are used in the design of both RCM1 and C-SVM switching patterns.

17.3.3 Reduction of CM Voltage Scheme 2 (RCM2)

To reduce the CM voltage as much as possible, the CM voltage reduction scheme 2 (RCM2) can be used, in which the magnitude of the CM voltages is minimized by eliminating more switching states than the RCM1 scheme. For odd-level inverters, such as three- and five-level inverters, it is possible to eliminate all the switching states that generate the CM voltage, which leads to a switching pattern that does not generate any CM voltages.

For the 3L-NPC inverter with RCM2 scheme, the switching states that produce the CM voltages of $\pm V_d/2$, $\pm V_d/3$, and $\pm V_d/6$ in Table 17.3-2 are all eliminated. As a result, the inverter does not generate any CM voltages. Fig. 17.3-6 shows the space

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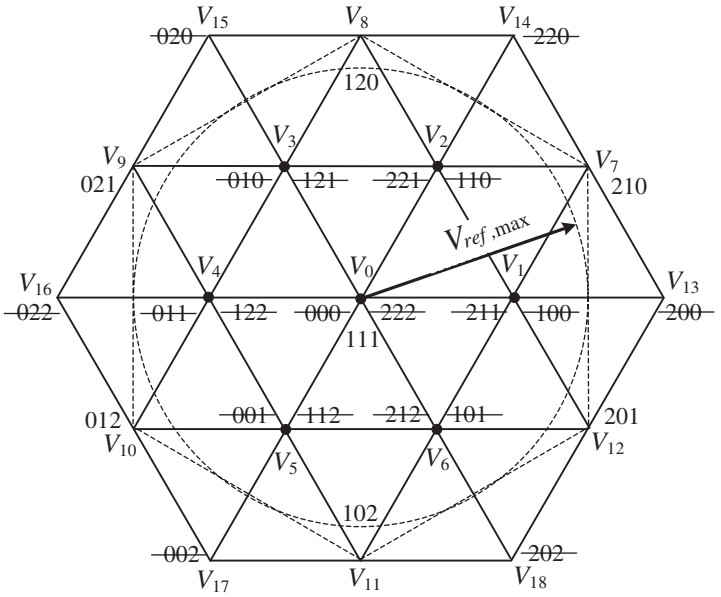

Figure 17.3-6 Space vector diagram of a 3L-NPC inverter with RCM2 scheme.

vector diagram for the 3L-NPC inverter with the RCM2 scheme. It can be observed that

- All the small vectors of V_1 to V_6 are not used in the switching pattern design since the switching states associated with the these vectors are all eliminated.
- All the large vectors of V_{13} to V_{18} on the outer hexagon are not used either.
- Only the zero vector V_0 and medium vectors of V_7 to V_{12} are used to develop the switching patterns.
- The maximum length of the reference vector ($V_{ref,\max}$) corresponds to the radius of the largest circle that can be inscribed within the new hexagon composed of the vectors of V_7 to V_{12}. This leads to the reduction of maximum modulation index, which is reduced from 1.0 to 0.866. This value can be derived from geometry of the outer and new hexagons (dashed lines).

The simulated waveforms for the CM and line-to-line voltages of the 3L-NPC inverter with the RCM2 scheme are shown in Fig. 17.3-7. The inverter operates under the condition of the fundamental frequency of 60 Hz, device switching frequency of 720 Hz, modulation index of 0.8, and a lagging load power factor of 0.9. The CM voltage is completely eliminated, but this is at the expense of higher line voltage THD, which is increased to 62.6% from 38.7% of the C-SVM scheme.

It should be noted that in the 3L-NPC inverter with RCM2 scheme, there are no redundant switching states anymore and only 7 switching states out of a total of 27 are used for switching pattern design. However, this limitation will not lead

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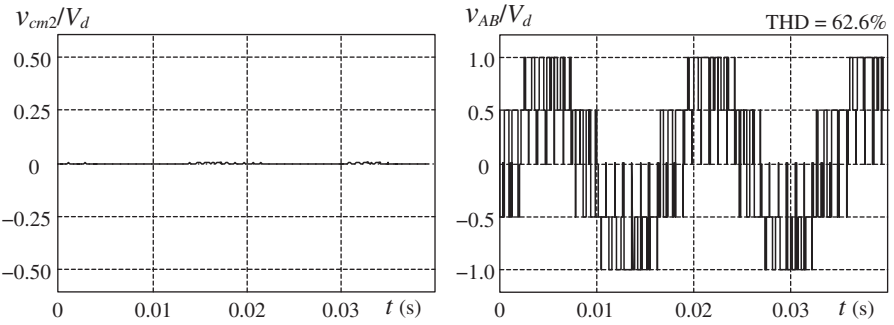

(a) CM voltage generated by the inverter (b) Inverter line-to-line voltage

Figure 17.3-7 Waveforms of the CM and inverter output voltage in a 3L-NPC inverter with RCM2 scheme.

to any switching transition that involves two voltage levels in the inverter phase voltages v_{Az}, v_{Bz}, and v_{Cz}. For example, for the given position of the reference vector V_{ref} in Fig. 17.3-6, the reference vector can be synthesized by three voltage vectors, V_0, V_7, and V_{12}, and their corresponding switching states are [111], [210], and [201], respectively. The switching transition for the switching devices in inverter phase A would take place among the switching states: [1], [2], and [2]. There is only one voltage level change between the switching states [1] and [2] during the switching transition, and the voltage level does not change from [2] to [2]. This implies that the requirement for the switching pattern design presented in Section 6.3.5 is satisfied.

A comparison of C-SVM, RCM1, and RCM2 schemes for the 3L-NPC inverter is given in Table 17.3-4 in terms of the range of modulation index m_a, magnitude of the CM voltages, and THD of the inverter line-to-line voltage at the modulation index of 0.8. With a properly designed switching pattern, the magnitude of the CM voltage produced by the C-SVM is $V_d/3$ (it could be $V_d/2$ if the redundant switching states [000] and [222] associated with the zero vector V_0 in Table 17.3-2 are used in the switching pattern design). The RCM1 scheme can reduce the CM voltage to $V_d/6$, whereas the RCM2 method can completely eliminate the CM voltages, but this is at the expense of reduced modulation index range and higher THD of the inverter line-to-line voltage. It is noted that the comparison is made based on the 3L-NPC inverter, but it is valid for all other three-level VSIs.

Table 17.3-4 Comparison of C-SVM, RCM1, and RCM2 Schemes for a 3L-VSI

Modulation Scheme	C-SVM	RCM1	RCM2
Range of modulation index (m_a)	0–1	0–1	0–0.866
Magnitude of CM voltages	$V_d/3$	$V_d/6$	0
THD of line-to-line voltage ($m_a = 0.8$)	38.7%	39%	62.6%

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17.3.4 CM Voltage Reduction in *n*-Level VSC

With the increase of the voltage levels, there are more redundant switching states, and it is therefore more flexible to eliminate switching states for the CM voltage reduction. Table 17.3-5 illustrates all the voltage vectors, switching states, and CM voltages for a four-level (4L) VSI. There are 37 voltage vectors and 64 switching states. The magnitude of the CM voltages generated by the switching state has five different levels: $\pm 9V_d/18$, $\pm 7V_d/18$, $\pm 5V_d/18$, $\pm 3V_d/18$, and $\pm V_d/18$.

It can be observed from Table 17.3-5 that there are no switching states that can produce zero CM voltage, which implies that the CM voltages cannot be completely

Table 17.3-5 Switching States and CM Voltages of a 4L-VSI

Vectors	Switching States	CMV	Vectors	Switching States	CMV	Vectors	Switching States	CMV
V_0	000	$-9V_d/18$	V_7	200	$-5V_d/18$	V_{18}	201	$-3V_d/18$
	111	$-3V_d/18$		311	$V_d/18$		312	$3V_d/18$
	222	$3V_d/18$	V_8	210	$-3V_d/18$	V_{19}	300	$-3V_d/18$
	333	$9V_d/18$		321	$3V_d/18$	V_{20}	310	$-V_d/18$
V_1	100	$-7V_d/18$	V_9	220	$-V_d/18$	V_{21}	320	$V_d/18$
	211	$-V_d/18$		331	$5V_d/18$	V_{22}	330	$3V_d/18$
	322	$5V_d/18$	V_{10}	120	$-3V_d/18$	V_{23}	230	$V_d/18$
V_2	110	$-5V_d/18$		231	$3V_d/18$	V_{24}	130	$-V_d/18$
	221	$V_d/18$	V_{11}	020	$-5V_d/18$	V_{25}	030	$-3V_d/18$
	332	$7V_d/18$		131	$V_d/18$	V_{26}	031	$-V_d/18$
V_3	010	$-7V_d/18$	V_{12}	021	$-3V_d/18$	V_{27}	032	$V_d/18$
	121	$-V_d/18$		132	$3V_d/18$	V_{28}	033	$3V_d/18$
	232	$5V_d/18$	V_{13}	022	$-V_d/18$	V_{29}	023	$3V_d/18$
V_4	011	$-5V_d/18$		133	$5V_d/18$	V_{30}	013	$-V_d/18$
	122	$V_d/18$	V_{14}	012	$-3V_d/18$	V_{31}	003	$-3V_d/18$
	233	$7V_d/18$		123	$3V_d/18$	V_{32}	103	$-V_d/18$
V_5	001	$-7V_d/18$	V_{15}	002	$-5V_d/18$	V_{33}	203	$V_d/18$
	112	$-V_d/18$		113	$V_d/18$	V_{34}	303	$3V_d/18$
	223	$5V_d/18$	V_{16}	102	$-3V_d/18$	V_{35}	302	$V_d/18$
V_6	101	$-5V_d/18$		213	$3V_d/18$	V_{36}	301	$-V_d/18$
	212	$V_d/18$	V_{17}	202	$-V_d/18$			
	323	$7V_d/18$		313	$5V_d/18$			

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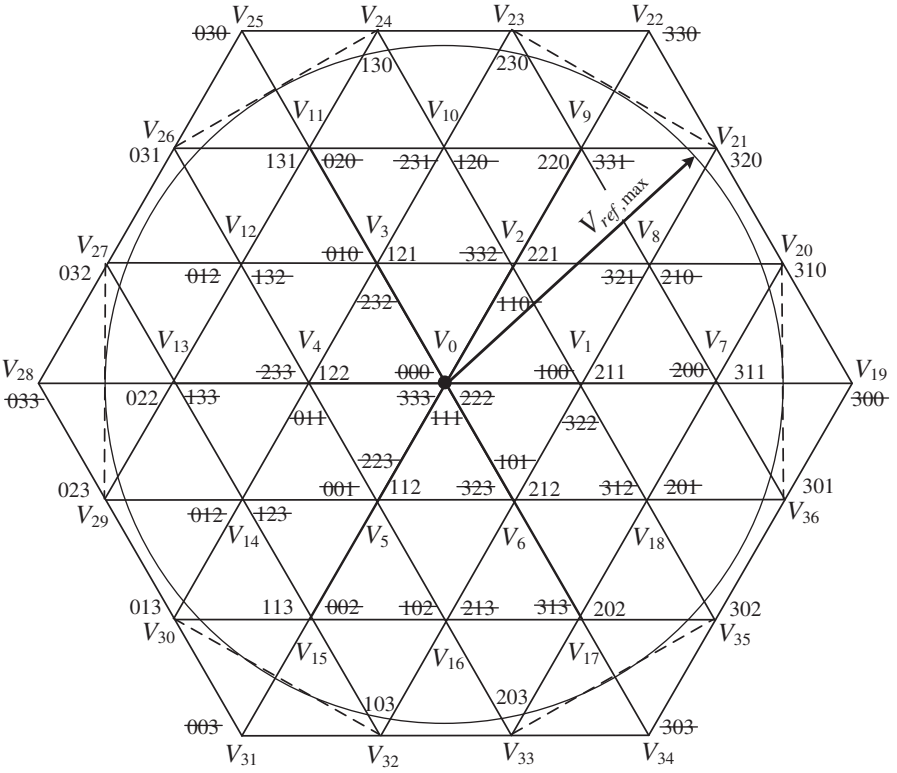

Figure 17.3-8 Space vector diagram of a 4L-VSI with RCM2 scheme.

eliminated in the four-level converter. This conclusion is valid for all even-level voltage source converters. With the RCM1 scheme for the 4L-VSC, we can eliminate the switching states that produce the CM voltages of $\pm 9V_d/18$, $\pm 7V_d/18$, and $\pm 5V_d/18$ while keep the full modulation index range ($0 \le m_a \le 1$). In doing so, the maximum magnitude of the CM voltage in the converter is $V_d/6$.

For the RCM2 method, we can eliminate all the switching states except those that produce the lowest CM voltage, which is $\pm V_d/18$. In this case, only 24 switching states out of a total of 64 are used in the switching pattern design. The resultant space vector diagram is shown in Fig. 17.3-8. It can be seen that six vectors (V_{19}, V_{22}, V_{25}, V_{28}, V_{31}, and V_{34}) on the outer hexagon of the space vector diagram cannot be used due to the elimination of their corresponding switching states. As a consequence, the maximum length of the reference vector, $V_{ref,\max}$, is reduced, which leads to the reduction of the modulation index to $0 \le m_a \le 0.9623$. The magnitude of the CM voltage generated by the inverter is reduced to $V_d/18$, which is only 5.6% of the dc-link voltage.

The simulated waveforms for the CM and line-to-line voltages in a 4L-VSI with RCM2 scheme are shown in Fig. 17.3-9. The inverter operates under the condition of the fundamental frequency of 60 Hz, device switching frequency of 720 Hz, modulation index of 0.8, and load power factor of 0.9. The magnitude of the CM

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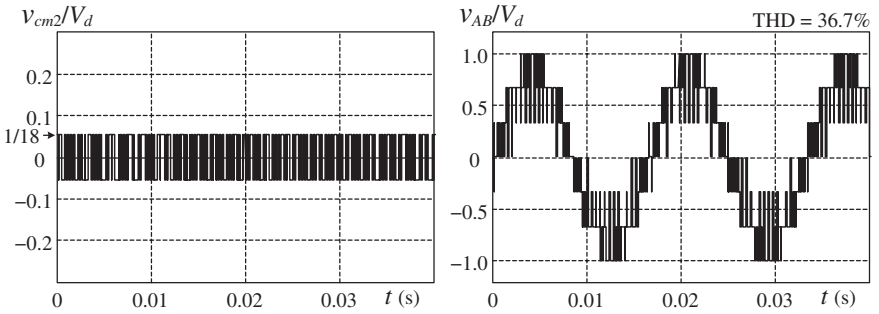
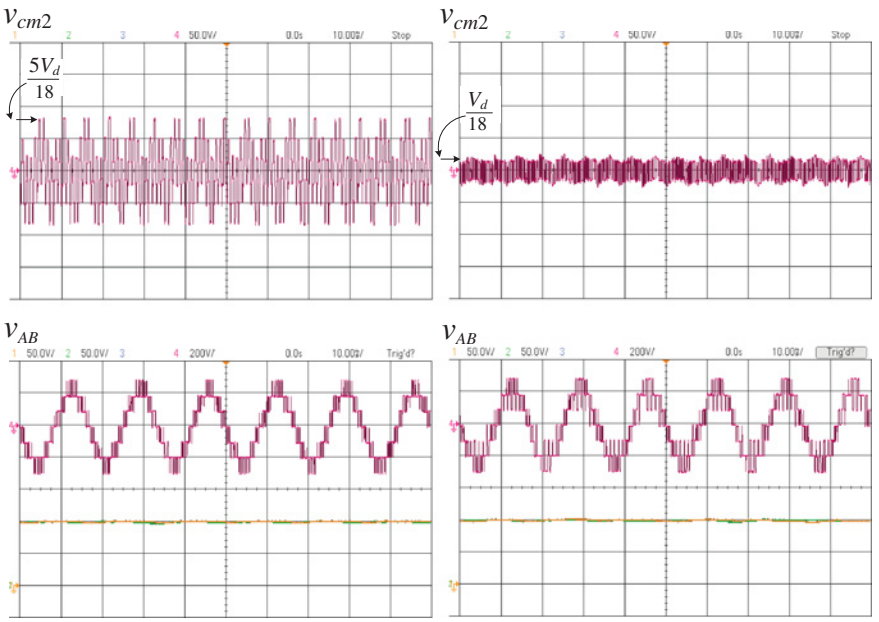


(a) CM voltage generated by the inverter (b) Inverter line-to-line voltage

Figure 17.3-9 Simulated waveforms in a 4L-VSI with RCM2 scheme.

voltage is only $V_d/18$, which is consistent with the theoretical analysis. However, the THD of the inverter line-to-line voltage is 36.7% in comparison to 24.8% produced by the 4L-VSI with the C-SVM scheme under the same operating conditions.

Figure 17.3-10 shows the measured waveforms of the CM and line-to-line voltages in a four-level VSI with the C-SVM and RCM2 schemes. The inverter operates with the dc-link voltage of 300 V, the fundamental output frequency of 60 Hz, modulation index of 0.8, and device switching frequency of 420 Hz. It can be observed that the magnitude of the CM voltage produced by the C-SVM is $5V_d/18$, which is five times that produced by the RCM2 scheme.



(a) Conventional SVM (b) RCM2 scheme

Figure 17.3-10 Measured waveforms in a 4L-VSI with C-SVM and RCM2 schemes.

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Table 17.3-6 Comparison of C-SVM, RCM1, and RCM2 Schemes for a 4L-VSI

Modulation Scheme	C-SVM	RCM1	RCM2
Range of modulation index (m_a)	0–1	0–1	0–0.962
Magnitude of CM voltages	$5V_d/18$	$3V_d/18$	$V_d/18$
THD of line-to-line voltage ($m_a = 0.8$)	25.1%	25.1%	37.2%

A comparison among conventional SVM, RCM1, and RCM2 schemes for the four-level VSIs is given in Table 17.3-6. With a properly designed switching pattern, the magnitude of the CM voltage produced by the C-SVM scheme is $5V_d/18$. It is noted that the magnitude of the CM voltage by the C-SVM scheme could be $7V_d/18$ if the switching states of vectors V_1 to V_6 that produce CM voltage of $7V_d/18$ in Table 17.3-5 are used, and it could be further increased to $9V_d/18$ if the redundant switching states [000] and [333] associated with the zero vector V_0 are used in the switching pattern design. The RCM1 method can reduce the CM voltage to $3V_d/18$ whereas the RCM2 method can further reduce the CM voltage to $V_d/18$ as summarized in Table 17.3-6.

The THD of the line-to-line voltage of 4L- and 5L-VSI modulated by the C-SVM, RCM1, and RCM2 schemes versus the modulation index m_a are given in Fig. 17.3-11. The RCM1 has the same THD profile as the C-SVM, which is due to the fact that all the voltage vectors in the space vector diagram are used in the switching pattern

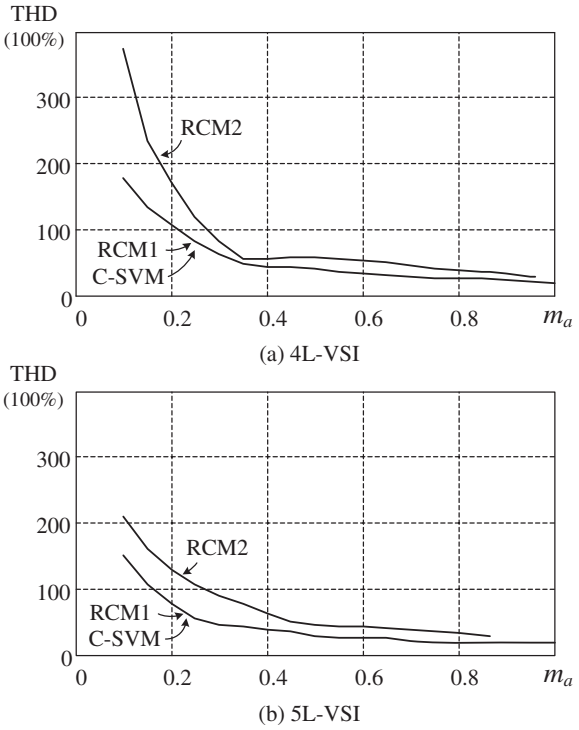

Figure 17.3-11 Comparison of THD of the output voltage in 4L- and 5L-VSI with C-SVM, RCM1, and RCM2 schemes.

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Table 17.3-7 CM Voltages in *n*-Level VSI with RCM1 and RCM2 Schemes

CM Voltage Reduction Schemes	Voltage Levels	Magnitude of CMV	m_a
RCM1	3-level	$V_d/6$	0–1.0
	4-level	$V_d/6$	0–1.0
	5-level	$V_d/12$	0–1.0
	6-level	$V_d/10$	0–1.0
	7-level	$V_d/9$	0–1.0
RCM2	3-level	0	0–0.866
	4-level	$V_d/18$	0–0.962
	5-level	0	0–0.866
	6-level	$V_d/30$	0–0.923
	7-level	0	0–0.866

design with the RCM1 and C-SVM schemes. The THD of the line-to-line voltage of the inverter using the RCM2 scheme is higher than that produced by the C-SVM and RCM1 schemes since some of the voltage vectors in the space vector diagram are not selected for use in the switching pattern design for CM voltage minimization.

A summary of the CM voltages in *n*-level VSIs using RCM1 and RCM2 schemes is given in Table 17.3-7. As discussed earlier, the RCM1 scheme can reduce the magnitude of the CM voltage to a certain extent while keeping the modulation index m_a at its full range. The RCM2 scheme can reduce the magnitude of the CM voltages further or even eliminate the CM voltage, but this is at the expense of reduced modulation index range and increased THD of the inverter line-to-line voltages.

It should be noted that the analysis and discussion presented in this section is based on multilevel VSIs, but they are also equally applicable to the multilevel VSRs.

17.4 TRANSFORMERLESS DRIVES WITH MULTILEVEL VSC

To develop a multilevel VSC fed MV drive without requiring isolation transformers, three methods can be generally employed: (1) elimination of the CM voltages with the RCM2 scheme, (2) suppression of CM voltage by CM filters, and (3) combined method of the CM filters and RCM1/RCM2 schemes. The principle of these methods is introduced, and their advantages and drawbacks are discussed.

17.4.1 Elimination of CM Voltages by Switching Scheme in Multilevel VSC

As discussed earlier, the CM voltages in the odd-level voltage source converters can be eliminated with the RCM2 scheme, which leads to an MV drive without the need

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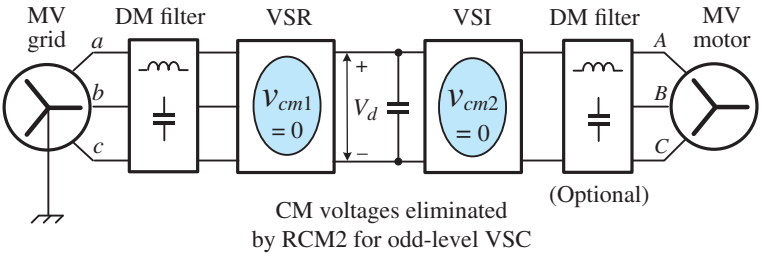

Figure 17.4-1 Transformerless MV drive with CM voltage eliminated by RCM2.

of any transformers or CM filters to block the CM voltages. Fig. 17.4-1 shows a simplified block diagram of a transformerless drive, where odd-level, such as 5- or 7-level, VSR and VSI are used.

To meet the grid code, a differential-mode filter is normally required on the grid side to filter out the switching harmonics produced by the rectifier. The motor-side DM filter is optional, depending on the voltage levels of the inverter and THD requirement on the motor voltage. The leakage inductances of the stator and rotor windings can also help to reduce the motor current THD.

The elimination of the isolation transformer in the MV drive helps to reduce the manufacturing cost, which is in addition to the reduction of the physical size and operating cost. However, there are some limitation and disadvantages associated with this method.

- This method is not applicable to the MV drives using even-level converters, such as 4L or 6L converters.
- The THD of the rectifier input current and inverter output voltage is increased. Large-size DM filters may be required, especially on the grid side to meet the grid code.
- The use of RCM2 scheme for the rectifier will cause a 15.5% increase in the dc-link voltage V_d due to the maximum modulation index of 0.866. This is due to the fact that the rectifier can be considered as a boost converter, and its dc voltage increases with the decrease of the modulation index. As a result, the voltage ratings for all the devices in the drive may increase, including the dc capacitors and semiconductor switches. This may lead to the increase of the manufacturing cost of the drive.

17.4.2 Suppression of CM Voltage by CM Filters

To mitigate the CM voltage issues in an MV drive, a CM filter can be introduced to the system. Figure 17.4-2a shows a simplified block diagram of such a drive, where a three-phase CM inductor L_{cm} is used as a CM filter to block the CM voltages produced by the converters and also to limit the CM current.

The CM inductor can be placed in different locations in the drive. It can be on the grid side as shown in the figure, in the dc link, or on the motor side. A CM resistor

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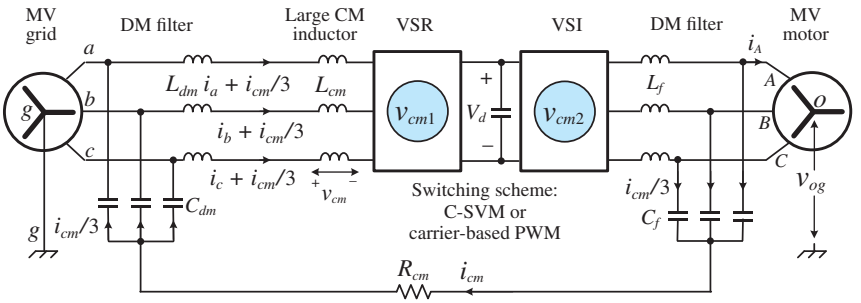
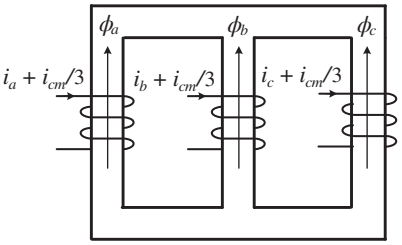
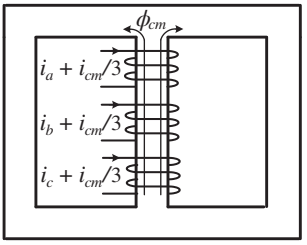


(a) System block diagram



(b) Structure of DM inductor L_{dm}



(c) Structure of CM inductor L_{cm}

Figure 17.4-2 Transformerless MV drive with a large CM inductor.

R_{cm} is also added in the CM current path. The main purpose of adding this resistor is to provide a resistive damping to damp out possible LC oscillations in the CM circuits, but it is not for CM voltage reduction since its resistance is much lower than the reactance of the CM inductor.

Two DM filters are used to filter out the switching harmonics produced by the converters. The neutral points of the three-phase DM filter capacitors are connected to provide a path for the CM current i_{cm} caused by the CM voltages and also to ensure that most of the CM voltages are applied to and blocked by the CM inductor. To reduce the CM current to an acceptable level, a large CM inductance $L_{cm}$is required.

Figure 17.4-2b shows the structure of the DM inductor L_{dm}. The magnetic core has three legs with a winding on each leg. Both three-phase current (i_a, i_b, and i_c) and CM current ($i_{cm}/3$) flow through the three-phase winding. The three-phase current generates a three-phase magnetic flux (ϕ_a, ϕ_b, and ϕ_c) in the magnetic core, which produces a DM inductance L_{dm}. However, the CM currents in the three-phase winding are all in-phase with the same magnitude and wave shape, and therefore the magnetic flux produced by the CM currents are cancelled in the magnetic core. As a result, the DM inductor does not produce CM inductance, which is the ideal case. In practice, the DM inductor has some leakage inductance, which can be considered as a CM inductance, but its value is small.

Figure 17.4-2c shows the structure of the CM inductor L_{cm}. The magnetic core has similar structure to that for the DM inductor, but three windings are all on the center leg. Both three-phase current (i_a, i_b, and i_c) and CM current ($i_{cm}/3$) flow through the

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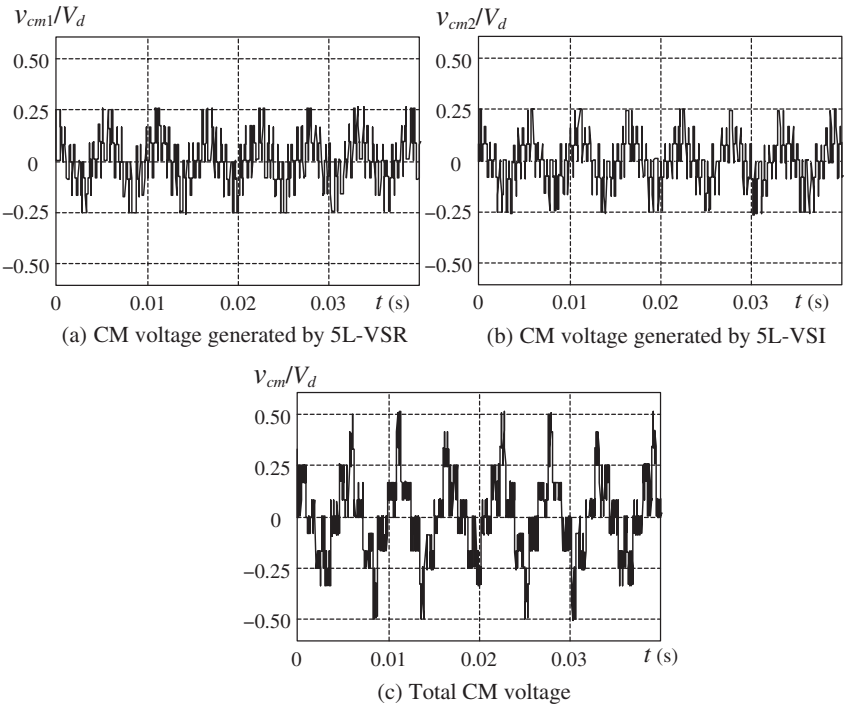
winding. For a three-phase balanced system, the sum of the three-phase currents at any time instant is zero, and therefore the magnetic fluxes produced by the three-phase currents in the CM inductor are cancelled. The CM current i_{cm} generates the CM flux ϕ_{cm} in the magnetic core, which produces the CM inductance L_{cm}.

To investigate the transformerless drive system given in Fig. 17.4-2 further, let us consider a 6 kV/5 MW drive fed by a five-level VSR and inverter. Both converters are modulated by the conventional SVM scheme with a modulation index of 0.95. The switching frequency of the switchers in both converters is 720 Hz. The rectifier is connected to an MV grid of 6 kV/60 Hz, and the inverter output frequency is also 60 Hz. The grid-side DM filter is composed of a three-phase inductor L_{dm} of 0.15 pu (2.87 mH) and a three-phase capacitor C_{dm} of 0.2 pu (74 μF), respectively. The motor-side DM filter is composed of three-phase inductor L_f of 0.03 pu (0.57 mH) and a three-phase capacitor C_f of 0.05 pu (18 μF), respectively. To suppress the CM voltage/current effectively, a large CM inductance of 28 pu (535 mH) is selected to keep the CM current around 0.03 pu, and the CM voltage on motor is around 0.1 pu. The CM damping resistance R_{cm} is 0.5 pu (3.6 Ω).

The waveforms of the CM voltages, v_{cm1} and v_{cm2}, generated by the rectifier and inverter are shown in Figs. 17.4-3a and 17.4-3b, respectively, and their magnitude is $0.25V_d$. The magnitude of the total CM voltage, v_{cm}, is twice that of the rectifier or inverter as shown in Fig. 17.4-3c.



(a) CM voltage generated by 5L-VSR

(b) CM voltage generated by 5L-VSI

(c) Total CM voltage

Figure 17.4-3 CM voltage waveforms in a transformerless drive with a large-size CM inductance.

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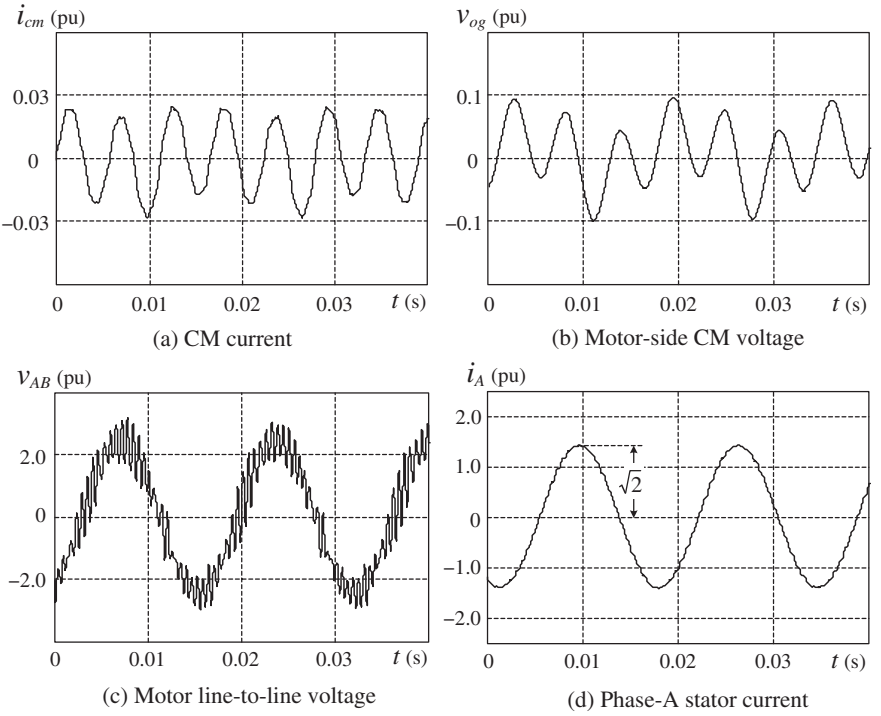

Figure 17.4-4 CM current and motor-side CM voltage waveforms in a transformerless drive with a large-size CM inductance.

Figures 17.4-4a and 17.4-4b show the waveforms of the CM current i_{cm} and motor-side CM voltage v_{og} in the transformerless drive of Fig. 17.4-2. The magnitude of the CM current is limited to around 0.03 pu mainly by the large CM inductance (28 pu). The CM voltage applied to the motor v_{og}, which is the voltage of the stator winding neutral o with respect to ground g, is only around 0.1 pu. The frequency of the dominant harmonic in the CM voltage is 180 Hz, which is three times the inverter fundamental output frequency of 60 Hz. Therefore, the CM voltage is also referred to as the zero-sequence voltage. It is should be noted that

- The three-phase DM filter capacitor C_f also acts as a CM filter capacitor and its equivalent CM capacitance is $3C_f$. Since its impedance is much lower than that of the large CM inductor, a small CM voltage will appear on C_f. This also applies to the DM filter capacitor, C_{dm}, on the grid side.
- For a given total CM voltage v_{cm}, the magnitude of the CM current i_{cm} and motor CM voltage v_{og} can be adjusted by the value of L_{cm}, C_{dm}, and C_f.
- The waveform of v_{og} does not contain high *dv/dt* switching transients, which is due to L_{cm} and C_f.

The waveforms of motor line-to-line voltage v_{AB} and phase-*A* stator current i_A are shown in Figs. 17.4-4c and 17.4-4d, respectively.

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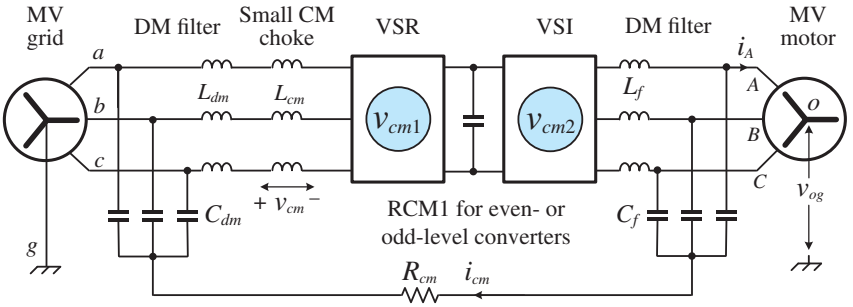

Figure 17.4-5 Example of combined method (small L_{cm} + RCM1) for transformerless drive.

17.4.3 Combined Method of CM Filters and CM Voltage Reduction Schemes

To minimize the size and cost of the CM filters, the method of using CM filters can be combined with CM voltage reduction schemes presented in this chapter as shown in Fig. 17.4-5. With the reduction of CM voltages, a smaller CM inductance is needed to block the CM voltages. This method can be applied to both even- and odd-level converters.

Figure 17.4-6 illustrates the simulated CM voltage waveforms for the MV drive of Fig. 17.4-5 using five-level rectifier and inverter modulated by the RCM1 scheme.

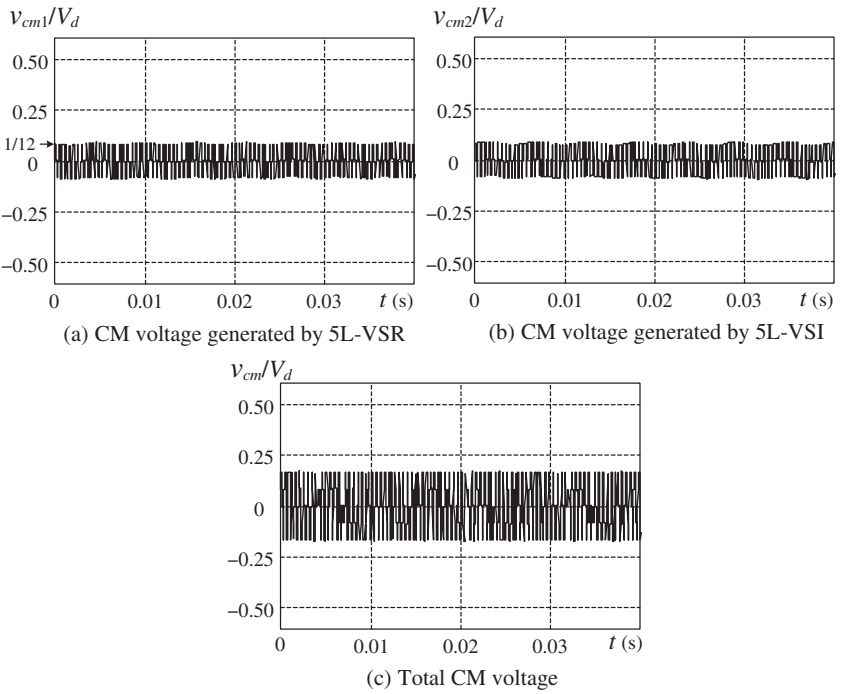

Figure 17.4-6 Waveforms in the transformerless drive with the combined CM voltage mitigation method.

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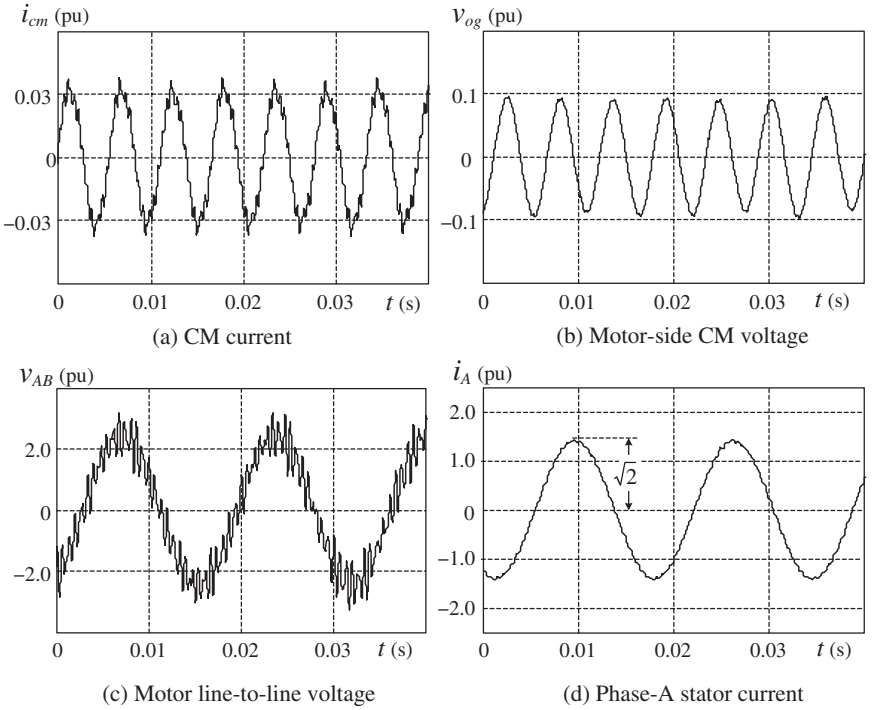

Figure 17.4-7 Waveforms in the transformerless drive with a combined CM voltage mitigation method.

The converters operate under the same conditions as those given in the preceding section. The magnitude of the CM voltages, v_{cm1} and v_{cm2}, generated by the five-level rectifier and inverter is $V_d/12$ ($0.083V_d$), respectively, while the total CM voltage has a magnitude of $0.166V_d$.

To reduce the CM current to around 0.03 pu, the CM inductance L_{cm} of 8 pu is selected. The other parameters of the drive system remain the same as those in the previous case. The simulated waveforms for the CM current i_{cm} and motor CM voltage v_{og} are shown in Figs. 17.4-7a and 17.4-7b, respectively. The peak value of i_{cm} is around 0.03 pu and v_{og} is about 0.1 pu. Compared with the previous case, where the CM inductance is 28 pu, the CM inductance is reduced to 8 pu, which leads to the cost savings in the MV drive. The waveforms of motor line-to-line voltage v_{AB} and phase-*A* stator current i_A are also shown in Figs. 17.4-7c and 17.4-7d, respectively.

17.5 TRANSFORMERLESS CSI FED DRIVES

17.5.1 Conventional Solution

Like the voltage source converters, the current source rectifier (CSR) and current source inverter (CSI) also generate CM voltages due to the switching action of the

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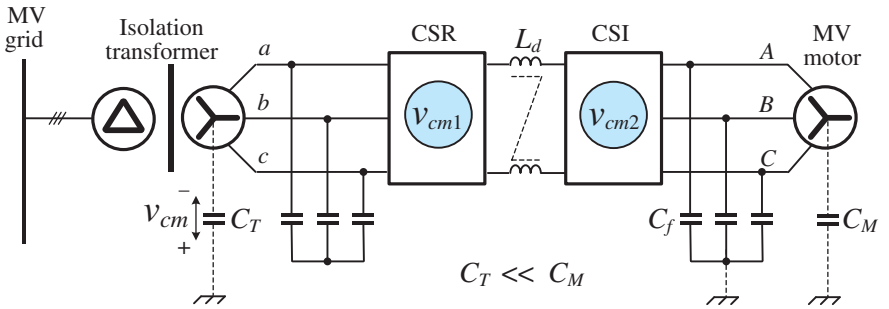

Figure 17.5-1 Conventional solution to CM voltage issues in a CSI drive.

semiconductor switches in the converters. If not mitigated, the CM voltages will be superimposed on the motor phase voltages, which increase the voltage stress on the motor winding and causes premature failure of winding insulation.

The conventional solution to the CM voltage issue in the CSI fed MV drive is to introduce an isolation transformer to the system as shown in Fig. 17.5-1. As discussed earlier in the preceding section, the parasitic capacitance of the motor winding to ground, C_M, is much higher than that of the transformer winding to ground C_T. As a result, most of the CM voltage v_{cm} produced by the rectifier and inverter will appear on the transformer, and motor is protected from damage by the CM voltages.

Alternatively, the neutral point of the motor-side filter capacitor C_f can be grounded. In a three-phase balanced system, the potential of the neutral of the stator winding is equal to that of the neutral of C_f. Grounding the neutral of C_f equivalently makes the neutral of the stator winding grounded. As a result, all the CM voltages are transferred to the transformer, and thus the motor is not subject to CM voltages.

However, the use of the isolation transformer not only increases the manufacturing cost and physical size of the drive, it also increases the operating cost of the drive due to the transformer power losses as discussed earlier.

17.5.2 Integrated DC Choke for Transformerless CSI Fed Drives

Figure 17.5-2 shows the schematic of a transformerless CSI fed drive, where an integrated dc choke is used [7]. The choke can provide two inductances, a differential inductance L_d that is inherently required by the CSI drive and a common-mode inductance L_{cm} that can block the CM voltage. The use of the integrated dc choke leads to the elimination of the isolation transformer, resulting in a significant reduction in manufacturing and operating costs. To ensure that the motor is not subject to CM voltages, the neutral points of the grid- and motor-side filter capacitors are connected together.

Figure 17.5-3 shows the concept of two integrated dc chokes, one with four coils and the other with two coils. Both designs use a single magnetic core of three legs. For the four-coil design in Fig. 17.5-3a, the choke has two air gaps of equal length placed in the outer legs. The differential coils x_1 and x_2 are wound over the outer legs while the common-mode coils y_1 and y_2 are placed on the center leg. The number of

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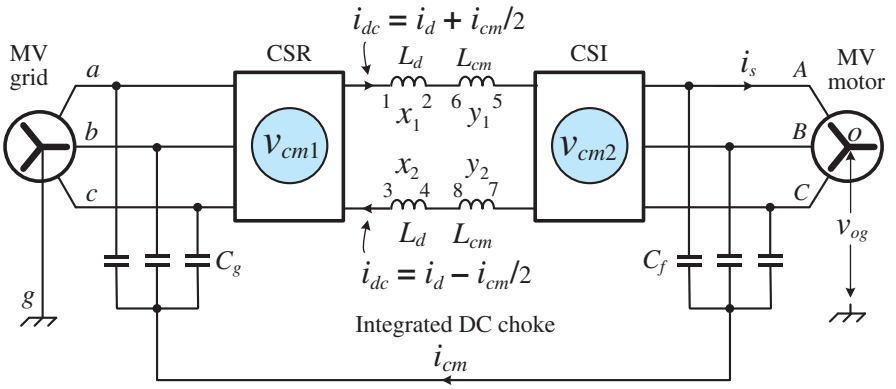
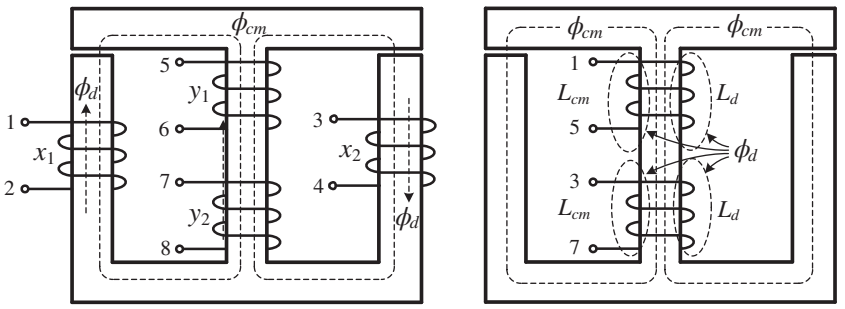


Figure 17.5-2 Integrated DC choke for CM voltage suppression.

turns of the differential and CM coils can be adjusted separately to produce required differential and CM inductances.

The connection of the integrated choke in the dc link of the drive is shown in Fig. 17.5-2, where the differential coil x_1 is in series with the CM coil y_1 in the positive dc bus while coils x_2 and y_2 are in series in the negative bus.

The dc current i_{dc} contains two components, a differential current i_d and a CM current i_{cm}. The differential current i_d flows through the positive dc bus and returns to the rectifier from the negative bus. The CM current i_{cm} is equally split in the dc-link circuit, one half ($i_{cm}/2$) in the positive dc bus and the other half in the negative dc bus due to the symmetrical arrangement of the choke and converters. The CM current i_{cm} flows from the rectifier through the dc link, inverter, motor-side filter capacitor, grid-side filter capacitor, and returns to the rectifier. Its magnitude is mainly limited by the CM inductance of the choke. Assuming the same number of turns for all four coils, the ratio of L_{cm} to L_d can reach 2.25 [7, 8]. The turns ratio of the differential and CM coils can be further adjusted to achieve different ratios of the differential to CM inductance.



(a) Four-coil structure (b) Two-coil structure

Figure 17.5-3 Concept of the integrated DC choke.

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For the integrated chock with two-coil design in Fig. 17.5-3b, the magnetic core has similar structure to that of the four-coil design, but it has only two coils. The two coils are placed on the center legs for the production of the CM magnetic flux ϕ_{cm}. When the CM current flows through these coils, the choke produces a high CM inductance. The differential inductance L_d is produced by the leakage flux ϕ_d, and no separate coils are needed for L_d. The geometry of the integrated choke can be designed in such way that it produces high leakage fluxes, which leads to a sufficiently high differential inductance to smooth the dc-link (differential) current. The integrated dc choke with the two-coil design is more cost effective than the four-coil design, and therefore it has been implemented in the commercial MV drives.

The concept of the integrated choke with four-coil design is verified on a low-power laboratory CSI fed drive with the differential inductance L_d of 1.0 pu and CM inductance L_{cm} of 4.8 pu. A six-pulse SCR rectifier is connected to a three-phase utility supply of 208 V/60 Hz. The inverter operates at a switching frequency of 180 Hz with the motor-side filter capacitor of $C_f = 0.31$ pu. The motor operates near rated speed under no-load conditions, at which the CM voltage reaches its highest level. The fundamental frequency of the drive is 57.7 Hz at which the effect of the beat frequency (the frequency difference between the grid and the inverter) can be observed.

Figure 17.5-4 shows the measured waveforms of the drive with the integrated dc choke installed, where v_{AB} is the motor line-to-line voltage, i_s is the stator current, i_{dc} is the dc current, v_{Ag} is the motor line-to-ground voltage, v_{og} is the motor neutral-to-ground voltage, which is also the motor-side CM voltage, and i_{cm} is the CM current.

Figure 17.5-4a shows the waveforms of the drive with $L_d = 1.0$ pu and $L_{cm} = 0$ (the CM coils y_1 and y_2 are not used in the testing). The neutral point of the motor-side filter capacitor C_f is left open to avoid excessive CM current since $L_{cm} = 0$. The motor neutral-to-ground voltage v_{og} contains triplen harmonics, which are of zero-sequence components with the 3rd and 6th harmonics being dominant. The motor line-to-ground voltage v_{Ag} is composed of two components: the motor phase voltage v_{Ao} and neutral-to-ground voltage v_{og}. Obviously, with v_{og} superimposed on v_{Ao}, the motor line-to-ground voltage v_{Ag} is severely distorted with high peaks, causing damages to the motor insulation system.

With the CM coils and the capacitor neutrals connected as shown in Fig. 17.5-2, the measured waveforms are shown in Fig. 17.5-4b. Comparing with Fig. 17.5-4a, we can make the following observations:

- The motor line-to-line voltage v_{AB}, stator current i_s and dc current i_{dc} are essentially the same as those in Fig. 17.5-4a. It implies that the CM inductance provided by the integrated dc choke does not affect the operation of the drive.
- The waveform the motor line-to-ground v_{Ag} is close to sinusoidal, free of CM voltages. The CM voltage is fully blocked by the integrated dc choke. This observation can be verified by the waveform of the CM voltage v_{og}, which is essentially zero.
- There is a small amount of CM current i_{cm}. To reduce the CM current further, the number of turns of the CM coils can be increased.

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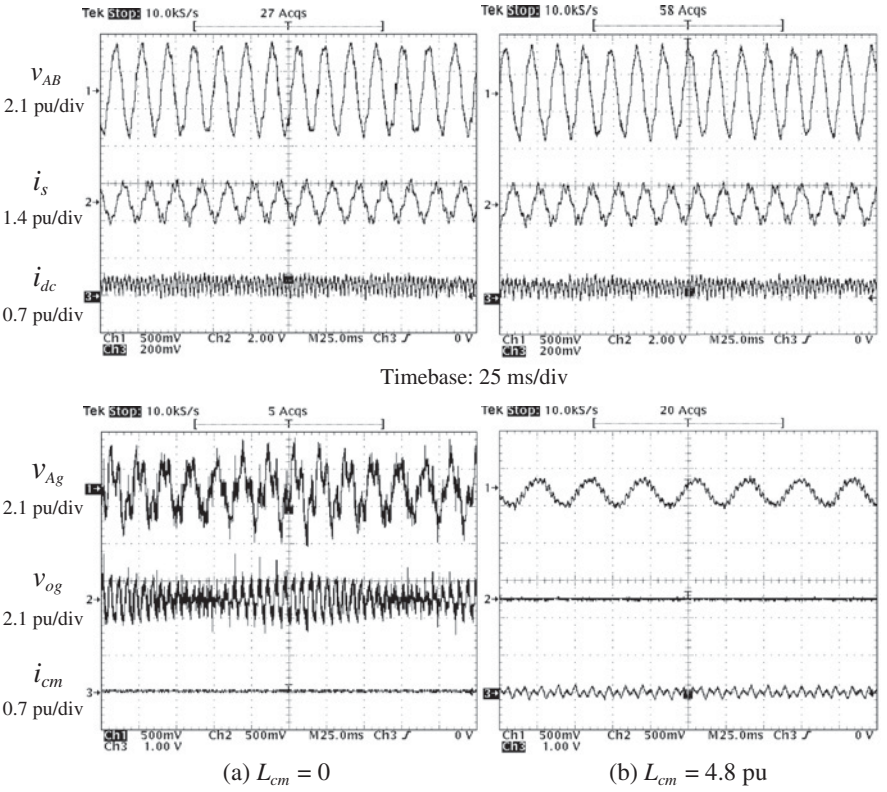

Figure 17.5-4 Measured waveforms in transformerless CSI drives.

The transformerless CSI MV drive using the integrated dc choke of Fig. 17.5-3b has been in production since 2003. A picture of such a drive is given in Fig. 13.2-3 of Chapter 13. This is the world's first transformerless MV drives for standard MV motors. A few thousands of them are currently operating in the field.

17.6 SUMMARY

The common-mode (CM) voltage generated by the power converters is a major issue in developing transformerless medium-voltage (MV) drives. In this chapter, the CM voltages in an MV drive are defined, the mechanism of the CM voltage generation is analyzed, and the principle of the CM voltage reduction by switching schemes is discussed. Two switching schemes for the CM voltage reduction, RCM1 and RCM2 schemes, for the multilevel voltage source converters are elaborated. Three types of VSI fed transformerless drives are presented. The CSI-fed transformerless drive using integrated dc choke is discussed. The key concepts for the development of transformerless drives are illustrated with computer simulations and experiments.

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